图纸履历 **PLAN HISTORY**

日期 DATE	版本 REVISION	标记 MARK	说 明 DESCRIPTION	设绘 DESIGNED	校核 CHKD	审定 APPRD
2008-10-17	A		FOR APPROVAL			

船号 HULL NO.	BHSC2008407-1/2		34,800吨散货船 34,800DWT BULK CARRIER	CSDC CODE	DC081210,101,001JS
船级社 CLASS.	BV				
设绘 DESIGNED			**装载手册（初步）** **PRELIMINARY LOADING MANUAL-TYPICAL LOADING**	详细设计	DETAIL DESIGN
校对 CHECKED				版本 REV.	A
审核 REVIEWED				比例 SCALE	N.A.
标检 STANDARDIZED				重量 WEIGHT	N.A.
审定 APPROVED				页数 PAGE	1/340

	船 厂 BUILDER	

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1. GENERAL

1.1. General Description

Name of ship …….....

Type of ship …….. Bulk Carrier

Owner …….....

Designer …….....CSDC

Builder…….....……Builder's number…….....

Class……...……...BV 1+HULL+MACH Bulk Carrier CSR BC-A, (holds 2,4 may be empty) ESP Unrestricted navigation, AUT-UMS, VeriSTAR-HULL, GRAB[20], IN WATER SURVEY, MOMSHAFT

Navigation area…….....Unrestricted Area

Flag state……...……

Port of registry……...……

Official number……..……

IMO number……...……..…….

Call sign……....……

Date of keel laid……....……

Date of launch……....……

Date of delivery……...……..……

Class Number…….....……..

(Note: The blank item will be filled in final booklet.)

1.2. Principal Dimension

Length overall ……..….....179.88 m

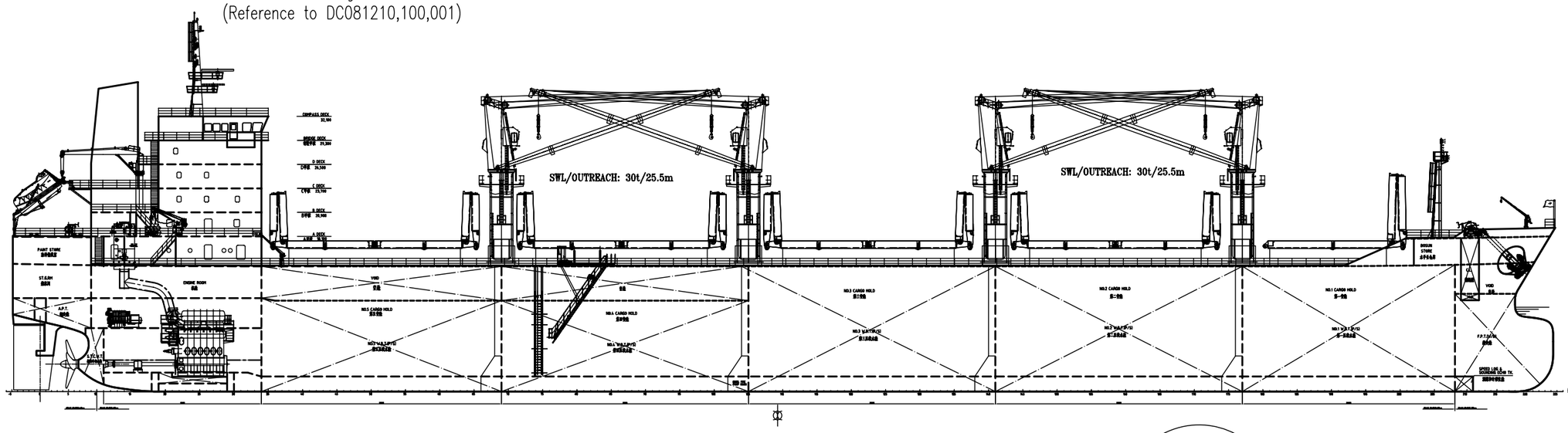
Length between perpendiculars ……..…….....172.00 m

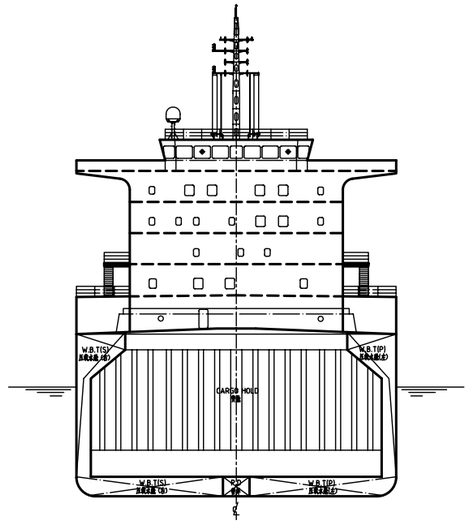
Breadth moulded ……...…….....28.80 m

Depth moulded ……...…….....14.60 m

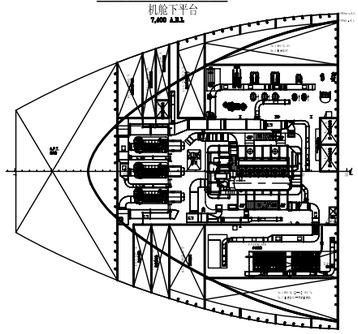
Designed draft moulded ……...…….….....9.83 m

Scantling draft moulded .....….....10.27 m

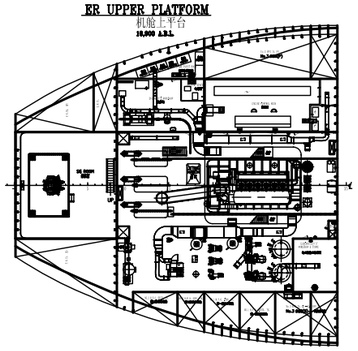




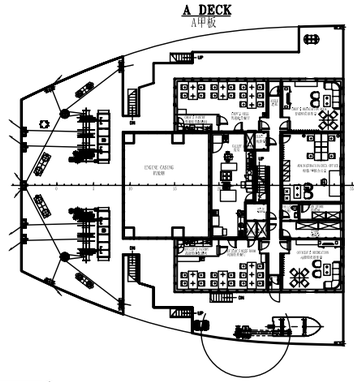
ER LOWER PLATFORM
机舱下平台



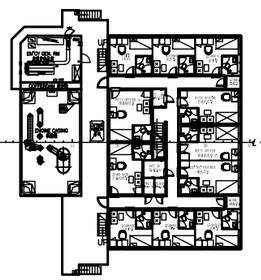
ER UPPER PLATFORM
机舱上平台



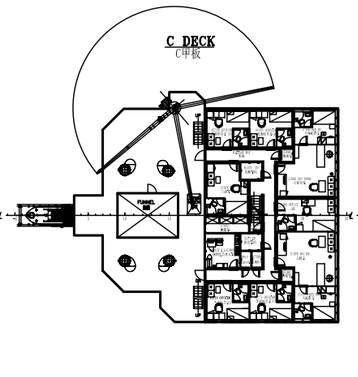
A DECK
A甲板



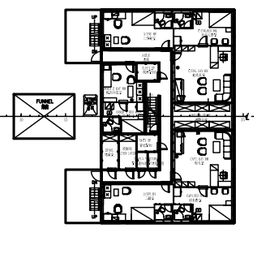
B DECK
B甲板



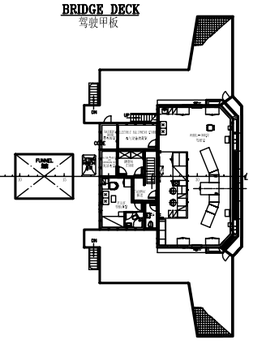
C DECK
C甲板



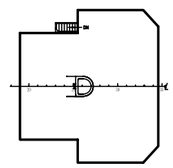
D DECK
D甲板



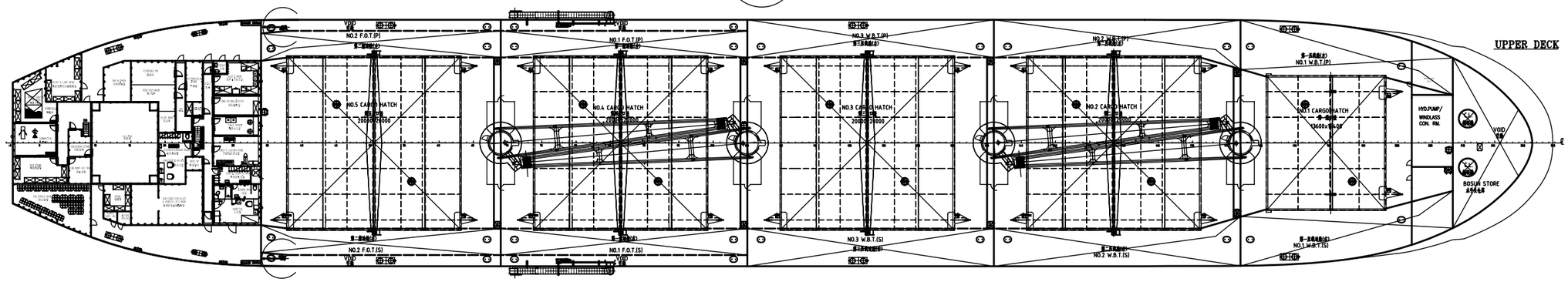
BRIDGE DECK
驾驶甲板



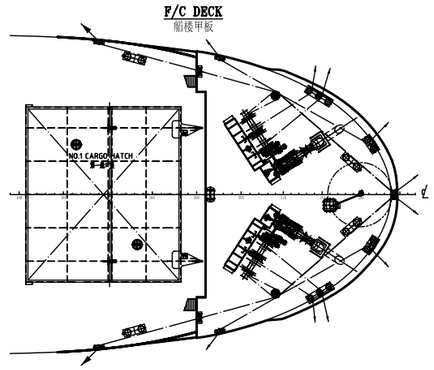
COMPASS DECK
罗经甲板



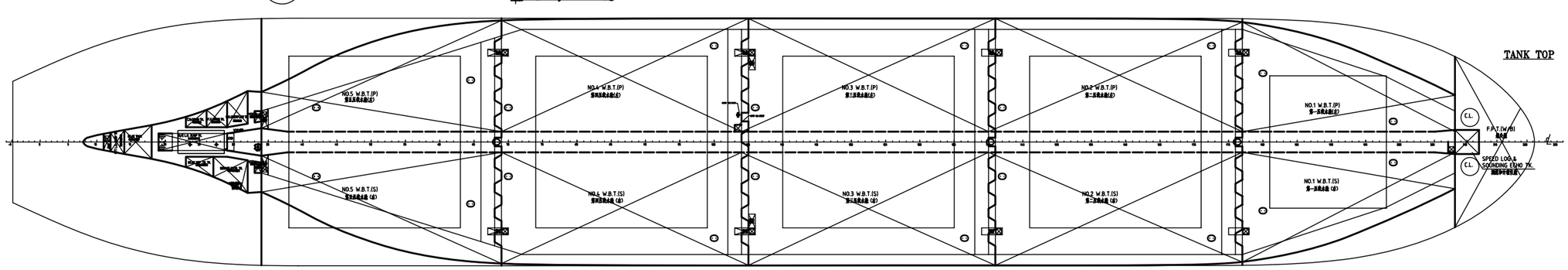
UPPER DECK



F/C DECK
船楼甲板



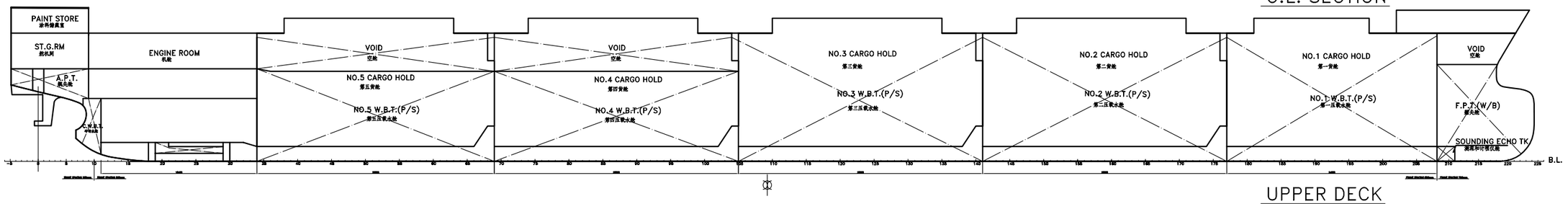
TANK TOP



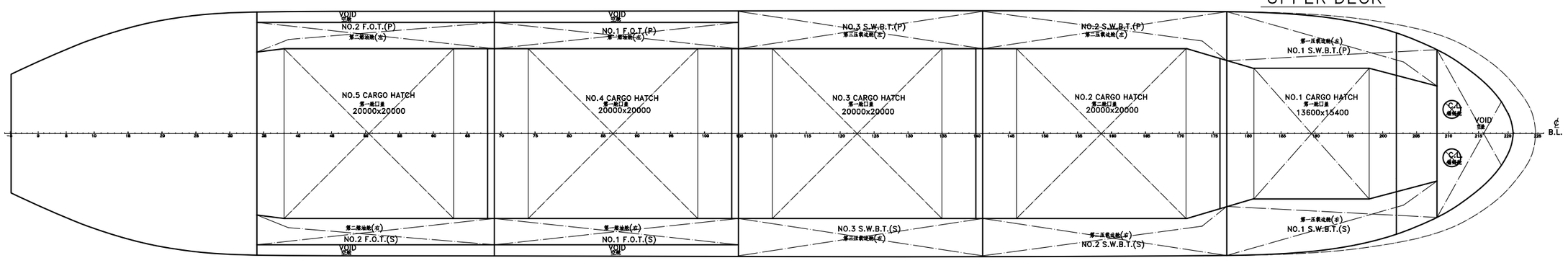
PRINCIPAL PARTICULARS
主尺度

LENGTH O.A. 总长	~179.88m	SERVICE SPEED 服务航速	ABOUT 14.0 KNOTS
LENGTH B.P. 垂线间长	172.00m	COMPLEMENT 定员	29
BREADTH MLD. 型宽	28.80m	CLASSIFICATION 船级	
DEPTH MLD. 型深	14.60m		BV, I-HULL-MACH BULK CARRIER CSR BC-A,
DESIGN DRAFT 设计吃水	9.830m		INOLD 2,4, MAYBE EMPTY) ESP, UNRESTRICTED NAVIGATION,
SCANTLING DRAFT 结构吃水	10.270m		AUT-UMS, VeriSTAR-HULL, GRAB(20),
DEADWEIGHT at 10.30m 载重量 (10.30m)	~35,000T		INWATERSURVEY, MONSHAFT
MAIN ENGINE 主机	1 SET OF MAN-B&W 6S42ME MARK 7		
SMCR	6480kW×136.0 rpm(100% OF MCR)		
CSR	5832kW×131.2 rpm(90% OF SMCR)		

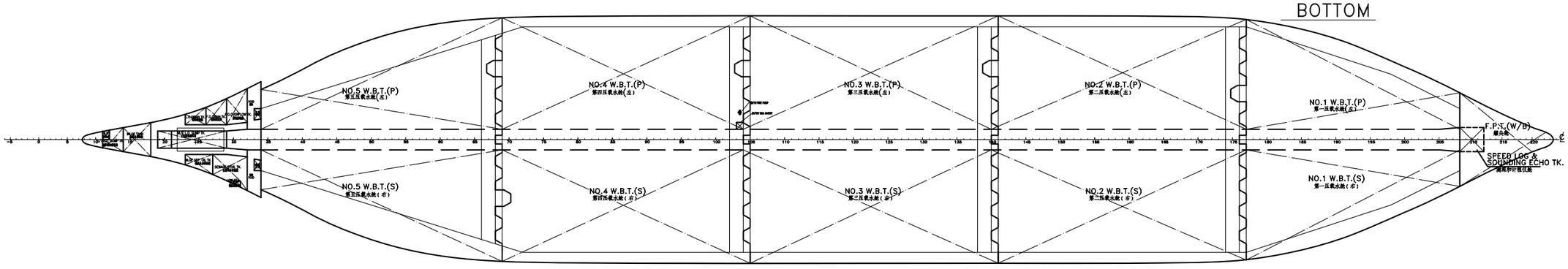
C.L. SECTION



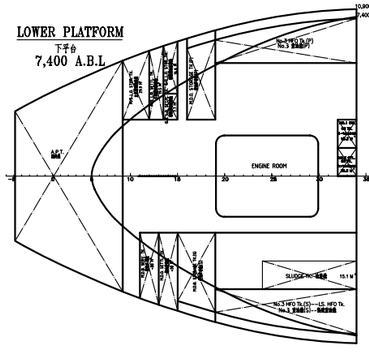
UPPER DECK



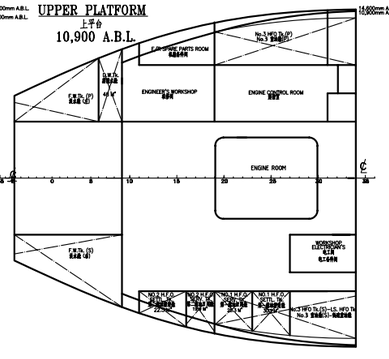
BOTTOM



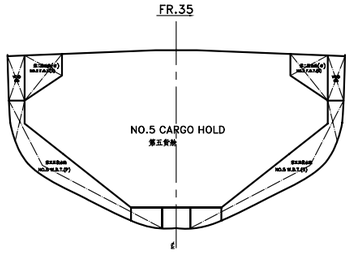
LOWER PLATFORM
下平台
7,400 A.B.L.



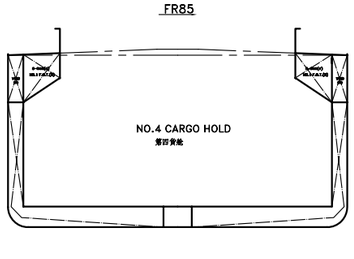
UPPER PLATFORM
上平台
10,900 A.B.L.



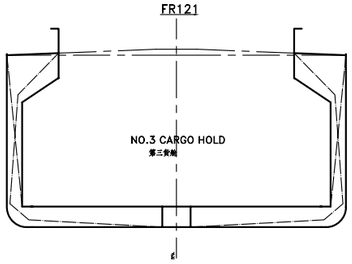
FR.35



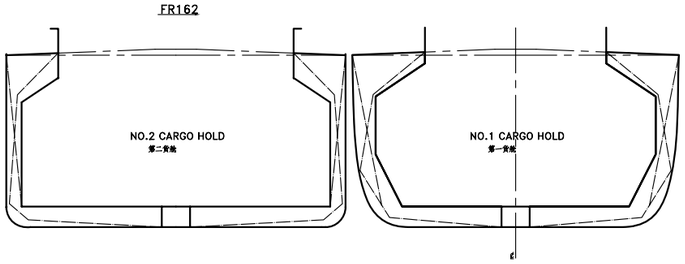
FR85



FR121

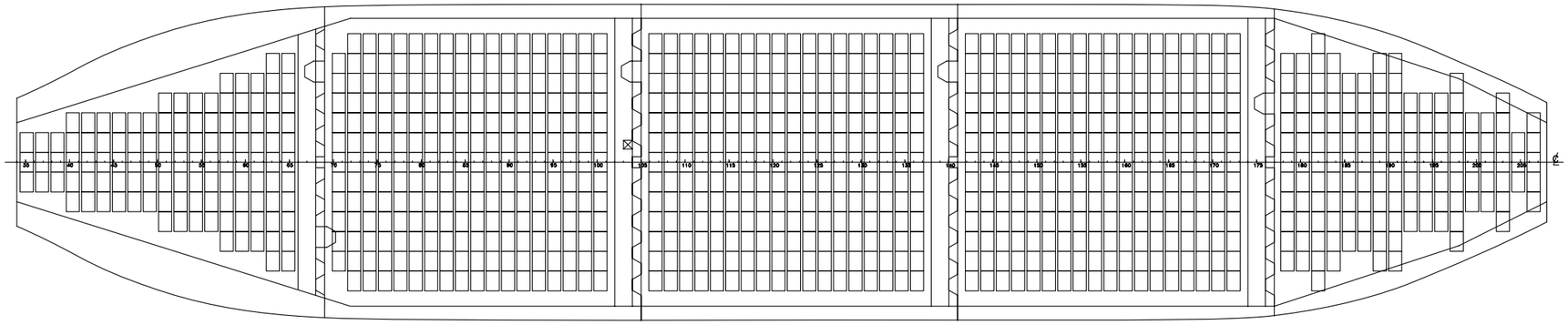


FR162



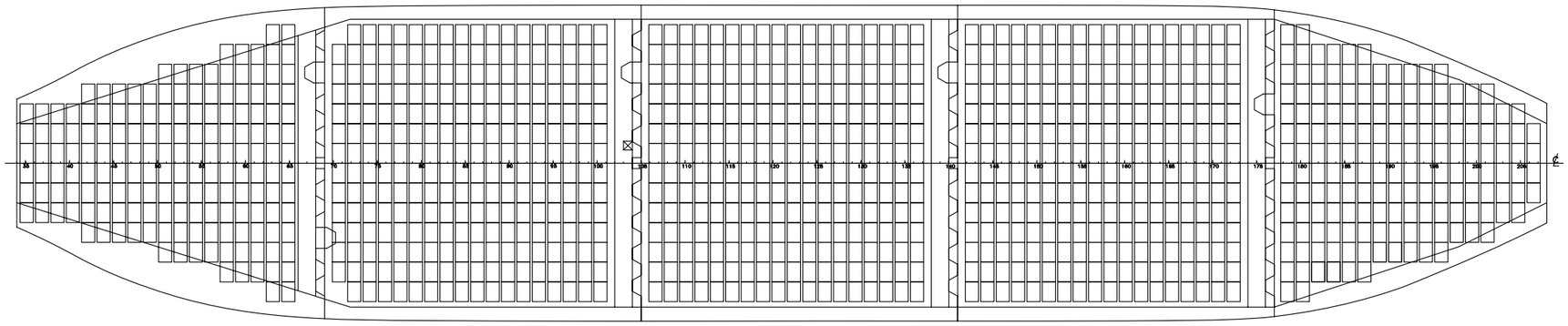
STEEL COIL ARRANGEMENT (TIER 2)

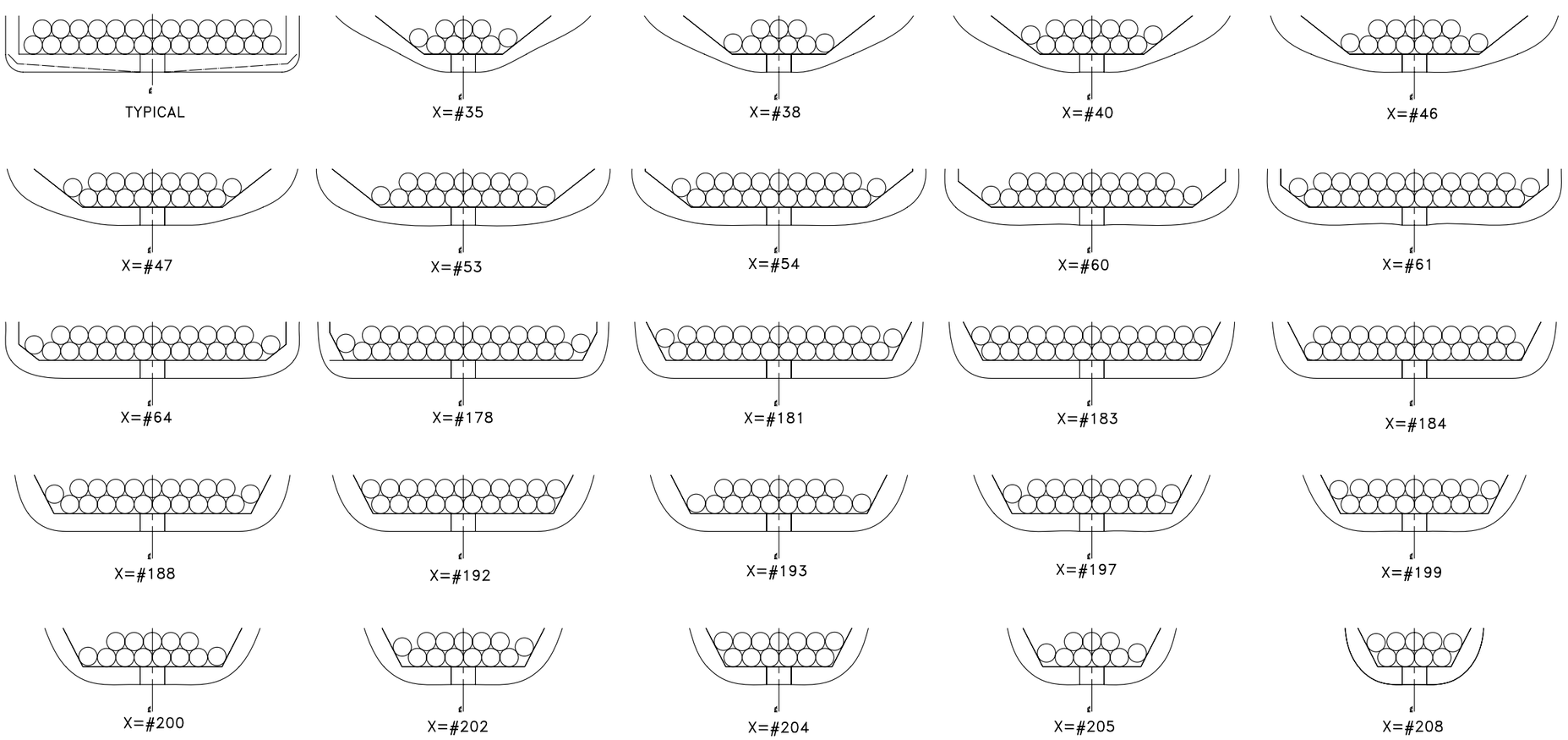
90 BAYS, BAY SPACE 200mm,(20t 1200X1800 LXD)



STEEL COIL ARRANGEMENT (TIER 1)

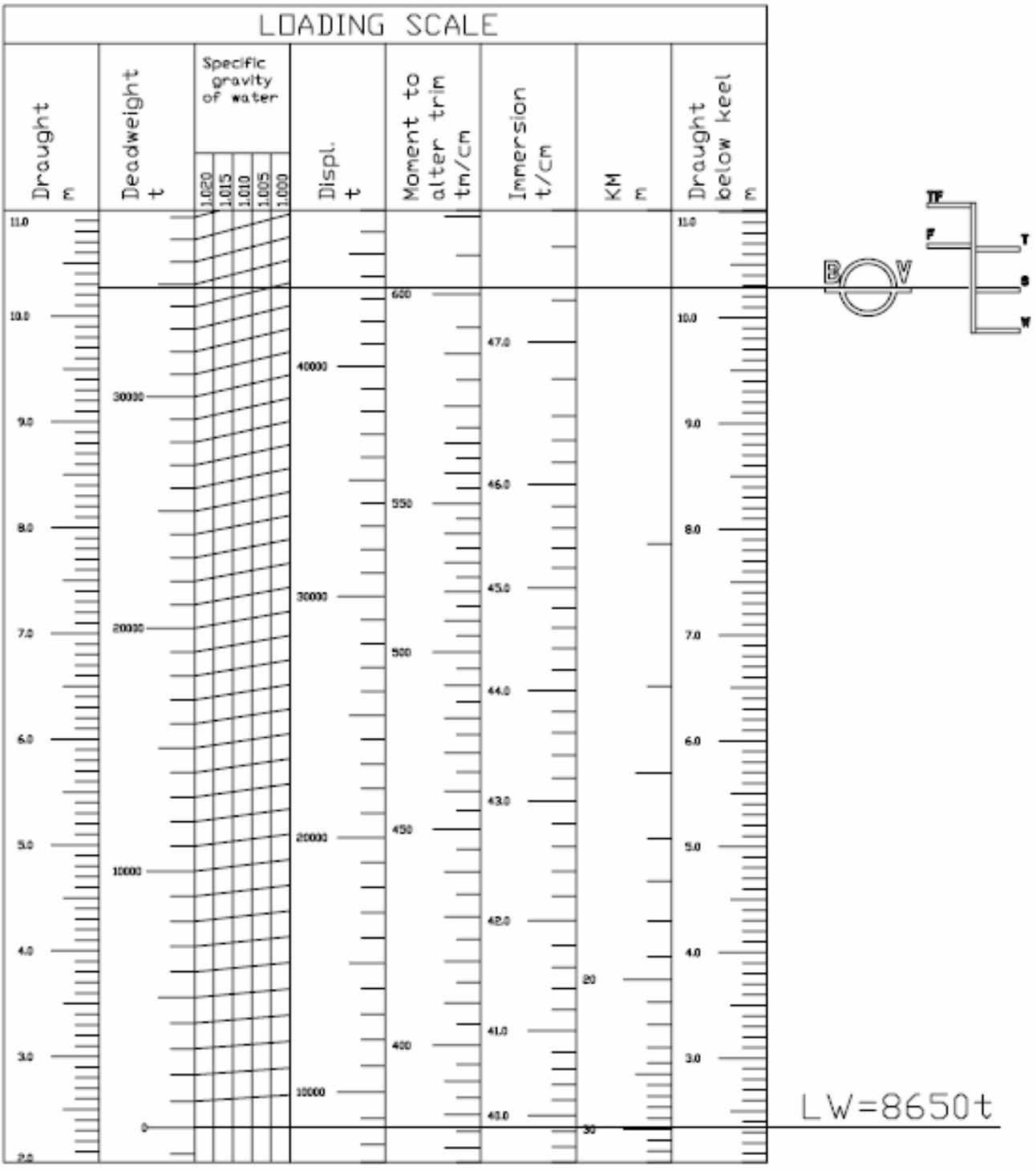
90 BAYS, BAY SPACE 200mm,(20t 1200X1800 LXD)





1.6. Freeboard Mark and Deadweight Scale

Mark	Description	Freeboard (m)	Draft(moulded) (m)
S	Summer	4356	10.270
F	Summer fresh water	4132	10.494
T	Tropic	4142	10.484
TF	Tropic fresh water	3918	10.708
W	Winter	4570	10.056



2. MASTER'S INSTRUCTION

2.1. General

(1) This booklet is made for the guidance to the master so as to have the safe operation and proper loading
of the vessel.

(2) This booklet present the necessary data and instructions to arrange cargoes, fuel oil, ballast water etc. in such a way that the vessel can be free from creation of unacceptable stress in hull structure and maintain sufficient stability under each loading condition during service.

(3) The master of the vessel should read the whole of the booklet and be well familiar with the characteristic of the vessel before placing it into service. Any ignorance of the instructions in this booklet may jeopardize the safety of crew and vessel.

(4) It should be noted that however this booklet can not cover unusual condition or contingencies.

(5) Calculation in the booklet is obtained from NAPA program of NAPA Oy. Helsinki. Finland. Napa Release 2006.1.

(6) Coordinate system in the booklet is right-handed. Origin of the coordinate system is located at the intersection of frame0 (AP), centerline of ship (CL) and the base line (BL).

	Positive value (+)	Negative value (-)	Remark
x	Forward	Aftward	All parameter in x direction, suck as CGX, LCB ...
y	Port side	Starboard side	All parameter in y direction, suck as CGY ...
z	Upward	—	All parameter in z direction, suck as CGZ, KM ...
trim	By bow	By stern	
heel	Port side	Starboard side	Such as heeling angle ...

(7) SI units to be adopted in the booklet.

2.2. Stability Guidance

The loading conditions given in this booklet are more or less fictitious, just to illustrate the features of the vessel representing the whole range of operation. Of course they comply with regulations are required for the ship. In general those are:

— Freeboard (see chapter ‘1.6 Freeboard Mark and Deadweight Scale’)

— Longitudinal strength (see chapter ‘2.4 Loading Criteria and Instruction’)

— Intact and damage stability criteria (see chapter ‘3. Stability Data’)

An existing loading condition that differs from those in this booklet should be investigated whether these criterions are met. This can be done with the aid of loading computer easily and get reliable results. However if the computer fails or what ever happens that precludes usage of which, then the following is optional:

— Checking of freeboard can be done by reading of draft marks;

— Longitudinal strength and immersion anger of openings can not be determined with aid of this manual;

— for checking stability there are two possibilities: The easiest way is to determine the center of gravity, correcting it by the effect of free surface if any and see whether this value is in stable zone of diagrams in chapter ‘3.6 Stability Limiting Curve’. This diagram contains of intact, damage and wind age stability. Alternatively-in doubt or whether otherwise required-a curve of righting arms can be calculated and drawn to check intact stability criteria(see chapter ‘2.5.2 Trim and Stability Calculation’). Damage stability can be checked via required KG_{min} values(see chapter ‘3.6 Stability Limiting Curve’).

2.3. General Precautions against Capsizing

(1) Compliance with the stability criteria indicated hereinafter does not ensure immunity against capsizing regardless of the circumstance or preventing the master from his responsibility. Master should therefore have prudence and good seamanship in the season of the year, weather forecast and the navigational zone and take the appropriate action to speed and course according to prevailing circumstances as well as ICLL66and Protocol of1988 for damage stability.

(2) Master should ensure that the cargo stowed in the ship is compliant with the criteria. If necessary the weight of cargo should be limited to the extend that ballast weight may be required.

(3) Before a voyage care Master should ensure that the cargo and sizeable pieces of equipment have been properly loaded to minimize the possibility of both longitudinal and lateral shifting while at sea, on the effect of acceleration caused by rolling and pitching.

(4) In all loading conditions, the initial metacentric height should be corrected for the free surface of each kind of tank which having larger free surface, and selecting the maxim free surface moment of these tanks. The correction of stability curves by the method of increasing the gravity center of ship, the increasing value is the corrective value for sum of free surface of tanks.

(5) Weather tight closing devices such as doors, hatches, etc. should be kept close during navigation. No modification or removal is allowed for doorsills, hatch comings, air piping, ventilators and other piping systems without authorization by the classification society.

(6) Master should note that stability can be adversely affected by such influence as beam wind on ships with large wind age area, water trapped on deck and in deck cargo, rolling characteristics and following seas.

(7) Master should be aware of the possibility that the carriage of certain bulk cargo may adversely affect the stability of the ship. Attention is drawn to the recommendations made in the IMO’s code of safe practice for solid bulk cargo.

2.4. Loading Criteria and Instruction

The standard loading conditions described in this booklet fully warrant all of trim, stability and longitudinal strength. In case to make the loading condition different from which in this booklet, the master should comply with the criteria of stability and longitudinal strength in any condition. It is recommended to comply with the requirement as follows:

(1) Designed allowable local loading on inner bottom

Cargo hold	1	2	3	4	5
Maximum uniform loading(t/m^2)	22	22	22	22	22

(2) Stability

It is necessary to comply with the requirement of stability criteria of IMO document about intact stability shown in '2.6 Stability Criteria Evaluation'. To keep necessary righting lever and metacentric height, care should be taken on the following points.

a) Cargo weight distribution in vertical direction.

b) Filling up proper ballast water in tanks.

c) Minimizing free surface due to ballast water and oil.

In each loading condition, GM value due to free surface of liquids in tanks should be higher than GM limit curve obtained from the intact stability criteria. Besides intact stability, keeping the vessel at a satisfactory safety level in view of damage stability, the GM value should also be higher than GM limit curve obtained from the damage stability criteria. The two GM limit curves are shown in "3.6 Stability Limiting Curves".

(3) Damage stability for SOLAS74 Chapter XII

Additional safety measures for bulk carriers of "Amendments to the annex to the International Convention for the Safety of Life at Sea.1974".

The following criteria in condition of equilibrium after flooding have been considered:

1) The final waterline after flooding, taking into account signage, heel and trim is below the lower edge of any opening through flooding may take place.
2) The angle of heel due to unsymmetrical flooding does not exceed 15°. If no part of the deck is immersed an angle of heel up to 17° may be accepted.
3) The metacentric height in the flooded condition is positive.
4) When any part of the deck outside the compartment assumed flooded in a particular case of damage is immersed or in any case where the margin of stability in the flooded condition may be considered, the residual stability is to be investigated:

   a) The righting lever curve should have a minimum range of 20° beyond the position of equilibrium with a maximum righting lever of at least 0.1m within this range.

   b) The area under the righting lever curve within this range should not be less than 0.0175mrad.

(4) Longitudinal strength

The scantlings are suitable for maximum permissible still water bending moment(BM) of 1,070,000 KN·m and -950,000KN·m amid ship in sea-going conditions. And the scantlings are suitable for a maximum permissible bending moment(BM) of 1,360,000 KN·m and -1,310,000 KN·m amid ship in flooded conditions.

In any loading condition and flooded condition the BM and SF value should not be more than the maximum permissible values. The details of designed permissible SWBM and SWSF are as follows:

			FR9	FR25	FR34	FR52	FR69	FR87	FR105	FR123	FR141	FR159	FR177	FR192	FR208	FR214	FR220
BM (KN.m)	**INTACT CONDITION**	**HOGGING**	40000	215000	**360000**	780000	**1070000**	1070000	**1070000**	1070000	**1070000**	1070000	**1070000**	220000	**43000**	15000	3000
		SAGGING	-100	-31000	**-950000**	-210000	**-950000**	-950000	**-950000**	-950000	**-950000**	-950000	**-950000**	-66000	**-9000**	-5000	-1000
	FLOOD CONDITION	**HOGGING**	40000	220000	**380000**	880000	**1360000**	1360000	**1360000**	1360000	**1360000**	1250000	**848000**	250000	**50000**	15000	5000
		SAGGING	-20000	-150000	**-320000**	-750000	**-1310000**	-1310000	**-1310000**	-1310000	**-1310000**	-950000	**-820000**	-200000	**-22000**	-5000	-5000
SF (KN)	**INTACT CONDITION**	**POSITIVE**	10000	20000	**25000**	20000	**48000**	6500	**10000**	4000	**50000**	25000	**10000**	5000	**5000**	800	-400
		NEGATIVE	-250	-6500	**-13000**	-10000	**-13000**	-30000	**-55000**	-10000	**-13000**	-15000	**-45000**	-18000	**-10000**	-4000	-1200
	FLOOD CONDITION	**POSITIVE**	10000	21000	**27000**	24000	**62000**	20000	**35000**	25000	**62760**	32000	**40000**	18000	**10000**	2000	0
		NEGATIVE	-5000	-20000	**-29000**	-36000	**-43000**	-33000	**-65260**	-25000	**-36000**	-22000	**-60000**	-22000	**-10000**	-5000	-1500

NOTE: the BM & SF in oter frames to be attained by linearity interpolation.

(5) Local strength

Based on the design loading criteria for local strength, hold mass curve via maximum allowable and minimum required mass of cargo and double bottom capability of each hold as a function of the draught at mid-hold position are shown. In any loading condition, mass of cargo in each hold should be in the allowable area of the hold mass curve.

(6) Minimum draft forward

The scantling is approved for operation at draft forward not less than 3.8 meters. In the opinion of the master, sea conditions are likely to cause regular slamming; other appropriation measures such as a change in speed, heeling or an increase in the draught forward may also need to be taken.

(7) Bridge visibility

The view of the sea surface from the conning position in wheel house should not be obscured more than two shop lengths, forward of the bow to 10° on each side irrespective of the ship's draft trim.

(8) Propeller size, location and immersion

Propeller diameter D = 5.3m;

Height of the shaft line from the base line = 3.2m;

To keep efficiency of propeller, draft aft should be deep enough to take sufficient propeller immersion.

For the heavy ballast condition the propeller immersion I/D is to be at least 60% where:

I: the distance from propeller centerline to waterline

D: propeller diameter.

2.5. Calculation Method for Loading Condition

2.5.1 Displacement Calculation from Draught Reading

Hydrostatic data and Cross Curves of stability are calculated for even keel (trim=0). However it may be used also for the moderately heeled and trimmed ship. The inaccuracy arising is believed to be in reasonable limits.

On condition that the displacement would be determined by reading of draft marks, it becomes necessary to calculate an equivalent draft from the measured ones. This is the draft at LCF; it best fits a slightly heeled and trimmed ship.

(1) Read the draught at aft, midship and fore draught marks(T_{f0} T_{a0} T_{m0})

For all marks maximum keel thickness of 20 mm is relevant.

Mark	Absolute position	Position rel. to perpend/midship
AFT2	#0+1.0	1.0 m fwd. of A.P. (Rudder horn)
AFT1	#13	9.0 m fwd. of A.P.
MID	#108	1.0 m aft. of Lpp/2
FWD	#215-0.2	2.3 m aft of F.P.

(2) Calculate the draught at "FP" (T_f), "AP"(T_a) and midship(T_m)

Readings at fore, aft, mid and mark AFT1 then:

$$T_f = T_{f0} - 2.3 \times (T_{a0} - T_{f0}) / 160.7$$

$$T_a = T_{a0} + 9.0 \times (T_{a0} - T_{f0}) / 160.7$$

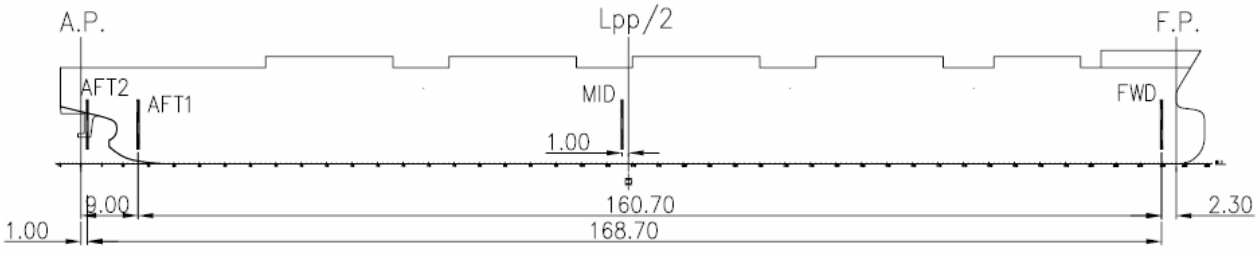
$$T_m = T_{m0} - 1.0 \times (T_{a0} - T_{f0}) / 160.7$$

Readings at fore,aft, mid and mark AFT2 then:

$$T_f = T_{f0} - 2.3 \times (T_{a0} - T_{f0}) / 168.7$$

$$T_a = T_{a0} + 1.0 \times (T_{a0} - T_{f0}) / 168.7$$

$$T_m = T_{m0} - 1.0 \times (T_{a0} - T_{f0}) / 168.7$$



(3) Trim(T_r), draught at midship without deflection(T_M), mean draft(T) and deflection(Δd) are obtained as follows:

$T_M = (T_a + T_f) / 2$ in meter

$\Delta d = T_m - T_M$ in meter

$T = (T_a + 6 \times T_m + T_f) / 8$ in meter

$T_r = T_a - T_f$ in meter

Where:

T_m draught at midship

$T_r < 0$ trim by the stem

$T_r > 0$ trim by the bow

$\Delta d > 0$ sagging condition

$\Delta d < 0$ hogging condition

(4) Displacement(D1), tons per cm immersion(TCP) and center of floating(LCA) corresponding to "T" are obtained from "3.2 Hydrostatic Data".

(5) Displacement correction due to trim:

$D2 = -100 \times TCP \times t_r \times (LCA - Lpp/2) / Lpp$ in tons

(6) Displacement correction due to specific gravity of seawater (D3):

When the specific gravity (S.G.) of seawater differs from standard of 1.025 displacement should also be corrected by following formula:

$D3 = (D1+D2) \times (S.G. - 1.025) / 1.025$ in tons

(7) Final displacement (Disp) will be determined as follows:

$Disp = D1+D2+D3$ in tons

2.5.2 Trim and stability calculation

This calculation is made with the attached blank sheets of "Trim and Stability Calculation Sheet 1, 2 and 3".

At the first stage the weight, the position of center of gravity and moment distribution are to be calculated by "Trim and Stability Calculation Sheet 1" and on the next stage trim and stability calculations are performed by "Trim and Stability Calculation Sheet 2 and 3".

(1) Put the weight of cargo, oil, water ballast etc. into column (1) and put the center of gravity of each loading weight of length from AP(CGX), breadth from centerline(CGY), and height above base line(CGZ) into column (2)、(4) and (6) respectively.

CGX, CGY and CGZ corresponding to loading weight for each tank are obtained from "4.2 Tanks Capacity Table".

(2) Multiply the value in column (1) and (2) ((1) and (4), (1) and (6)), put the results into column (3) ((5) (7)) as the longitudinal moment, transverse moment and vertical moment of each loading weight.

(3) Put "Frs. Mom" (the product of specific gravity of liquid and moment of inertia of free surface of each tank) into column(8). Moment of inertia corresponding to loading weight of each tank is also to be obtained from "4.2 Tanks Capacity Table".

(4) The total weight, longitudinal moment, transverse moment, vertical moment and free surface moment of inertia are to be put into each column in the "Trim and Stability Calculation Sheet 2"

Then sum up the columns (1)', (3)', (5)' and (7)' respectively and the results of grand total are to be written in the bottom line. These figures mean displacement, longitudinal moment from AP, transverse moment from centerline, vertical moment above base line and total of free surface effect of liquid in tanks respectively.

(5) The center of gravity corresponding to the calculated condition can be obtained as follows.

CGX =Σ(3)' /Σ(1)' in meter

CGY =Σ(5)' /Σ(1)' in meter

CGZ(KG) =Σ(7)' /Σ(1)' in meter

(6) Trim and draught corresponding to the displacement can be calculated as follows.

Trim: T_r = Disp×(CGX－LCB) / (100×MCT) in meter

Draught at AP: T_a = do－T_r×LCA / Lpp in meter

Draught at FP: T_f = do＋T_r×(Lpp－LCA) / Lpp in meter

Draught mean: dm = (T_f-T_a) / 2 in meter

Where:

T = draught corresponding to the displacement obtained from"3.2 Hydrostatic Data"

Notice:

T, MCT, LCB and LCA correspond to the corrected displacement (D'=D×1.025/S.G.) in"3.2 Hydrostatic Data" when the specific gravity (S.G.) of seawater differs from standard of 1.025.

(7) Ship's initial metacentric height GM can be calculated as follows:

GM = GM_o＋GM_{corr} in meter

GM_o= KMT－KG in meter

GM_{corr} = －Σ(Frs.mom) / Disp in meter

Where:

GM	initial metacentric height with correction due to free surface effect
GM_o	initial metacentric height without correction due to free surface effect
KMT	transverse metacentric height corresponding to displacement in "3.2 Hydrostatic Data"
KG	vertical position of center of gravity of the ship above base line (CGZ) obtained from the result of "Trim and Stability Calculation Sheet 2"
GM_{corr}	free surface correction to the initial stability
Σ(Frs.mom)	total of free surface moment obtained from the result of "Trim and Stability Calculation Sheet 2"

(8) Ship's heeling angle can be calculated as follows.

Heeling angle = arctg (Σ(Frs.mom) / (Disp/GM)) = arctg (CGY/GM) in degree

(9) Ship's statically stability curve (righting lever vs heeling angle can be obtained by following method.

$GZ = GZ_o - KG' \times \sin\theta$ in meter

$KG' = KG + KG_{corr}$ in meter

$KG_{corr} = \Sigma(Frs.mom) / Disp$ in meter

Where:

GZ	righting lever
GZ_o	righting lever when center of gravity of the ship is assumed on the base line (shown in "3.3 Cross Curve Data")
KG	vertical position of center of gravity of the ship above base line (CGZ) obtained from the result of "Trim and Stability Calculation Sheet 2"
KG'	vertical position of center of gravity of the ship above base line with correction due to free surface effect
KG_{corr}	increase of height of center of gravity of the ship due to free surface
Σ(Frs.mom)	total of free surface moment obtained from the result of "Trim and Stability Calculation Sheet 2"

Stability curve calculation to be performed using "Trim and Stability Calculation Sheet 3"

Plot point (GZ vs θ). GZ in ordinate and θ in abscissa, and connect these points by a fair curve. Thus the statically stability curve is obtained.

(10) About free surface correction

For all loading conditions, the initial metacentric height and the righting lever curve should be corrected due to free surface of liquids in tanks.

For calculating the free surface effects in tanks containing consumable liquids, it should be assumed that each type of liquids at least one transverse pair or a single center line tank has a free surface and the tank or combination of tanks taken into account should be those where the effects of free surface is the greatest.

Where water ballast tanks are to be filled or discharged during the course of a voyage, the free surface effects should be calculated to take account of the most onerous transitory stage relating to such operations. It assumes only one ballast tank or pair of tanks is allowed to have a free surface at any moment.

Trim and Stability Calculation Sheet 1

(Detail of deadweight distribution)

Loading condition:_____

		Weight (t) (1)	L.C.G. (m) (2)	L.Momont (t-m) (3)=(1)*(2)	T.C.G. (m) (4)	T.Moment (t-m) (5)=(1)*(4)	V.C.G. (m) (6)	V.Moment (t-m) (7)=(1)*(6)	Frs.Mom (t-m) (8)
CARGO									
CH1	NO.1 CARGO HOLD								
CH2	NO.2 CARGO HOLD								
CH3	NO.3 CARGO HOLD								
CH4	NO.4 CARGO HOLD								
CH5	NO.5 CARGO HOLD								
TOTAL									
HEAVY FUEL OIL									
FOT1P	NO.1 FULL OIL TANK(P)								
FOT1S	NO.1 FULL OIL TANK(S)								
FOT2P	NO.2 FULL OIL TANK(P)								
FOT2S	NO.2 FULL OIL TANK(S)								
HFOTKP	H.F.O. TK. (P)								
HFOTKS	H.F.O. TK. (S)								
NO1_HFO_SERV_TK	NO.1 H.F.O. SERV. TK.								
NO1_HFO_SETTL_TK	NO.1 H.F.O. SETTL. TK.								
NO2_HFO_SERV_TK	NO.2 H.F.O. SERV. TK.								
NO2_HFO_SETTL_TK	NO.2 H.F.O. SETTL. TK.								
TOTAL									
DIESEL OIL									
MDOSTORTKP	M.D.O. STORAGE TK.(P)								
MDOSTORTKS	M.D.O. STORAGE TK.(S)								
MDOSERVTK	M.D.O. SERV. TK.								
MDOSETTLTK	M.D.O. SETTL. TK.								
TOTAL									
LUBRICATING OIL									
MELOSTORTK	M/E L.O. STORAGE TK.								
MELOSEETTLTK	M/E L.O. SETTL. TK.								
GELOSTORTK	G/E L.O. STOR. TK.								
GELOSEETTLTK	G/E L.O. SETTL. TK.								
NO1CYLOTK	No.1 Cyl LO Tk.								
NO2CYLOTK	No.2 Cyl LO Tk.								
STLOSUTK	S/T L.O. SUMP TK.								
MELOSUMPTK	M/E L.O. SUMP TK.								
TOTAL									
FRESH WATER									
FWTP	F.W.TK. (P)								
FWTS	F.W.TK. (S)								
DWT	D.W.TK.								
CWT	COOLING WATER TANK								
TOTAL									
WATER BALLAST									
FPT	F.P.T.								
WBT1P	NO.1 W.B.T.(P)								
WBT1S	NO.1 W.B.T.(S)								
WBT2P	NO.2 W.B.T.(P)								
WBT2S	NO.2 W.B.T.(S)								
WBT3P	NO.3 W.B.T.(P)								
WBT3S	NO.3 W.B.T.(S)								
WBT4P	NO.4 W.B.T.(P)								
WBT4S	NO.4 W.B.T.(S)								
WBT5P	NO.5 W.B.T.(P)								
WBT5S	NO.5 W.B.T.(S)								
APT	APT								
TOTAL									
MISCELLANEOUS									
BT	BILGE TANK								
FOOVERFLOW_TK	F.O.OVERFLOW TK.								
FODRAIN_TK	F.O.DRAIN TK.								
LODRAIN_TK	L.O.DRAIN TK.								
BILGE_SEP_OIL_TK	BILGE SEP. OIL TK.								
SEWAGE_STO_TK	SEWAGE STOR. TK.								
SLUDGE_TK	SLUDGE TK.								
TOTAL									

Trim and Stability Calculation Sheet 2

(Summary of deadweight distribution)

Loading condition:_____

	Weight (t) (1)	CGX (m) (2)	L.Momont (t-m) (3)=(1) × (2)	CGY (m) (4)	T.Moment (t-m) (5)=(1) × (4)	CGZ (m) (6)	V.Moment (t-m) (7)=(1) × (6)	Frs.mom (t-m) (8)
Crew & Effect								
Provision 10%								
Spare								
Cargo								
Ballast Water								
Fresh Water								
Heavy Fuel Oil								
Diesel Oil								
Lubrication Oil								
Miscellaneous								
DEADWEIGHT								
LIGHTWEIGHT								
DISPLACEMENT								

Trim and Stability Calculation Sheet 3

(Stability curve)

Loading condition:_____

θ (0)	Sinθ (1)	Disp (t) (2)	KG (m) (3)	Σ(Frs.mom) (t-m) (4)	ΣGMcorr (m) (5)=(4)/(2)	KG (m) (6)=(3)+(5)	GZo (m) (7)	GZ (m) (8)=(7)-(6) ×(1)
10	0.17365							
12	0.20791							
20	0.34202							
30	0.50000							
40	0.64279							
50	0.76604							
60	0.86603							
70	0.93969							

2.6. Stability Criteria Evaluation

The following stability criteria (1) and (2) comply with the requirement of IMO Resolution A.749 (18).

(1) General Stability Requirements (IMO Resolution A.749(18))

The stability curves are to comply with the following requirements in Fig.1

a) $A_1 \geq 0.055$m-rad

The area under the righting lever curve (GZ curve) up to $\theta=30^\circ$ angle of heel should not be less than 0.055 meter-radians.

b) $A_1+A_2 \geq 0.09$m-rad

The area under the righting lever curve(GZ curve)up to $\theta=40^\circ$ or the angle of flooding θ_f if this angle is less than $\theta=40^\circ$ should not be less than 0.09 meter-radians.

c) $A_2 \geq 0.03$m-rad

The area under the righting lever curve(GZ curve)between the angles of heel 30° and 40° or between 30° and θ_f if this angle is less than 40° should not be less than 0.03 meter-radians.

d) GZ30≥0.2m

The righting lever GZ should be at least 0.20 m at an angle of heel equal or greater than 30°.

e) $\theta_{max} \geq 25^\circ$

The maximum righting arm should occur at an angle of heel preferably exceeding 30° but not less than 25°

f) GM≥0.15m

Initial metacentric height GM is to be not less than 0.15m.

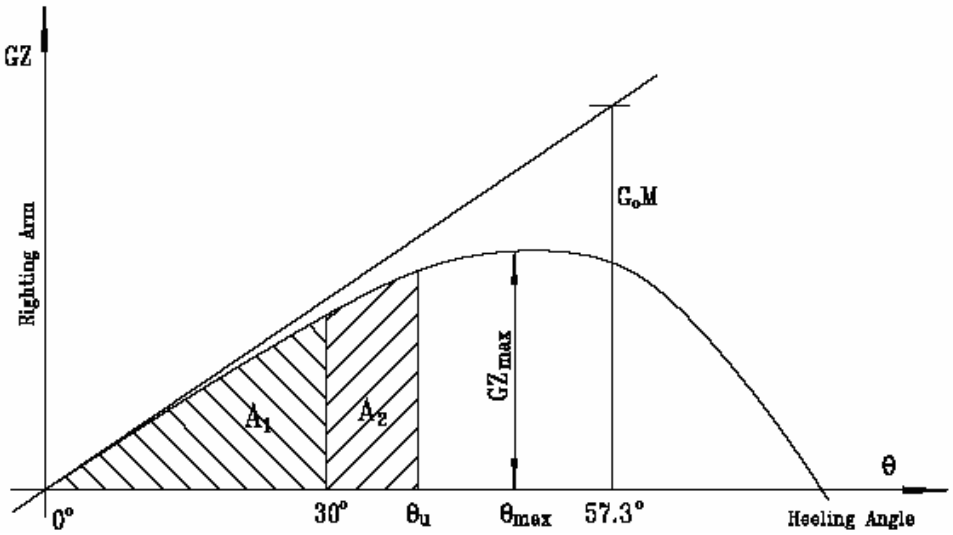


Fig.1 Stability Curve (General Stability Requirement)

Where:

A_1= Area under stability curve between 0° and 30° (m-rad)

A_2= Area under stability curve between 30° and θ_f (m-rad)

θ_f = Heeling angle (θ) to be taken of whichever is less flooding angle in relevant loading condition or 40°

GZ_{max}= Maximum righting lever (m)

θ_{max} = Heeling angle at which righting arm reaches maximum (θ)

GM = Initial metancentric height corrected by free surface effect (m)

(2) Stability Requirement in Wind and Waves (IMO Resolution A.749(18))

Stability Curves and wind-heeling moment lever curves of ships are to comply with the following requirement in Fig.2

a). Heeling angle caused by steady wind is to be less than 16° or an angle corresponding to 80% of immersing angle of deck edge whichever is less. In general deck edge means the point of intersection between the continuation of top of freeboard deck at ship's of lowest point of freeboard deck and the outside of side shell.

b). Area"b" is not less than area "a".

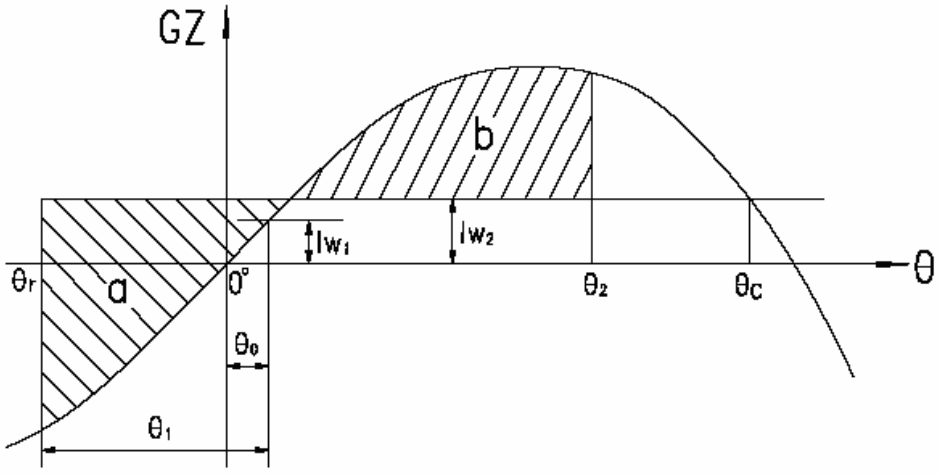


Fig.2 Stability and Wind-heeling Moment Lever Curve
(Stability Requirement in Wind and Waves)

Where:

A_1 = Area under stability curve between 0° and 30° (m-rad)

l_{w1} = Heeling moment lever caused by steady wind (m) given by the follow in formula:

$$l_{w1} = P \times A \times Z / 1000 / g / \triangle$$

l_{w2} = Heeling moment lever caused by gust (m) given by the following formula:

$$l_{w2} = 1.5 \times l_{w1}$$

P = 504 N/m^2. The value of P used for ships in restricted service may be reduced subject to the approval of the administration

A = Projected lateral area of the portion of the ship and deck cargo above the waterline (m^2)

Z =Vertical distance from the center of A to the center of the underwater lateral area of

approximately to a point at one half the draught (m)

△= Displacement (t)

B = Mounded breadth of the ship(m)

θ_r = Angle of rolling stop motion. In general, it may be given by the formula(θ_0-θ_1)

θ_c = Angle of second intercept between wind heeling lever l_{w2} and GZ curves

θ_0 = Angle of heel under action of steady wind

θ_1 = Angle of roll to windward due to wave action

θ_2 = Angle of down flooding, θ_c or 50°, whichever is less

a = Area encircled by stability curve, l_{w2} and θ_r (m-rad)

b = Area encircled by stability curve, l_{w2} and θ_2(m-rad)

d = Mean moulded draught of the ship (m)

The angle of roll for ships with anti-rolling devices should be determined without taking into account the operation of these devices.

$\theta_1 = 109\times k\times x_1\times x_2\times (rs)^{1/2}$ in degree

Where:

x_1 = Values obtained from Table1 according to the value of B/d. In case the value of B/d becomes intermediate value are to be determined by interpolation.

x_2 = Values obtained from Table2 according to C_b. In case C_b becomes intermediate values are to be determined by interpolation.

C_b = Block coefficient given by the following formula: C_b=△/ (1.025LBd)

L = Length of the ship at waterline (m)

k = Factors determined as follows:

a) For round-bilged ships having neither bilge keels nor bar keels:1.0

b) For ships with sharp bilges: 0.7

c) For ships with bilge keels and/or bar keels:

Values obtained from Table3 according to the value of $100A_k$ / (LB).

In case $100A_k$/(LB) become intermediate values are to be determined by interpolation

A_k = Total area of bilge keels, or area of the lateral of the bar keel, or sum of these areas (m^2).

r = Values obtained from the following formula. However, the value of r need not be taken over than 1.0 : r = 0.73+0.6 OG/d

with:

OG = Distance between the center of gravity and the waterline (m) (+ if center of gravity is above the waterline, - if it is below)

s = Values obtained from Table4 according to the value of T. In case T becomes intermediate values are to be determined by interpolation.

T = Rolling period (see) obtained from the following formula:

$T = [2B/ (GM)^{1/2}]\times(0.373+0.023B/d-0.043L/100)$ in second

GM = initial metacentric height corrected for free surface effect (m)

Table 1: values of factor x_1

B/d	≤2.4	2.5	2.6	2.7	2.8	2.9	3.0	3.1	3.2	3.3	3.4	≥3.5
x_1	1.0	0.98	0.96	0.95	0.93	0.91	0.90	0.88	0.86	0.84	0.82	0.80

Table 2: values of factor x_2

C_b	≤0.45	0.50	0.55	0.60	0.65	≥0.70
x_2	0.75	0.82	0.89	0.95	0.97	1.00

Table 3: vakues of factor k

$100A_k$ / (LB)	0	1.0	1.5	2.0	2.5	3.0	3.5	≥4.0
k	1.0	0.98	0.95	0.88	0.79	0.74	0.72	0.70

Table 4: values of factor s

T	≤6	7	8	12	14	16	18	≥20
s	0.100	0.098	0.093	0.065	0.053	0.044	0.038	0.035

(3) Method of Stability Evaluation

a) Refer to "2.5.2 Trim and stability calculation (1)-(7)".
Calculate the displacement(Disp), draught corresponding to the displacement, initial metacentric height with correction due to free surface effect (GM) and etc. in actual loading condition.

b) Refer to "2.5.2 Trim and stability calculation (9)".
Draw the statically stability curve in actual loading condition.

c) Refer to "2.6 (1) General Stability Requirements (IMO Resolution A.749(18))".
Fill up "IMO A.749(18) Stability Criteria Calculation Sheet 1".
Check if the requirements of General Stability Requirements (A.749(18)) are satisfied or not.

d) Refer to "3.4 Wind Profile Curves".
Find out the projected lateral area above the water line (A) and vertical distance (Z) from the center of (A) to the center of under water lateral area. In case △ becomes intermediate values are to be determined by interpolation. Calculate lw1 and l_{w2} using "IMO A.749(18) Stability Criteria Calculation Sheet 2" and draw them in the statically stability curve.

e) Refer to "2.6(2) Stability Requirement in Wind and Waves (IMO Resolution A.749 (18))".
Find out θ_0, θ_w, θ_c(θ_w: Angle of heel under action of gust).
Find out θ_2 to be taken of whichever is least down flooding angle θ_c or 50°.

f) Refer to "2.6 (2) Stability Requirement in Wind and Waves (IMO Resolution A.749(18))".
Calculate θ_1 by "IMO A.749(18) Stability Criteria Calculation Sheet 2" and draw it in the statically stability curve.

g) Fill up "IMO A.749(18) Stability Criteria Calculation Sheet 3"

h) Check whether Stability Requirement in Wind and Waves (A.749(18)) are satisfied or not.

IMO A.749(18) Stability Criteria Calculation Sheet 1

Loading condition:_____

Displacement=_____t EM=_____m Σ(Frs.mom)=_____tm

KG=_____m KG_{corr}=_____m KG'=_____m

Gm_o =_____m GM_{corr}=_____m GM=_____m (≥0.15m)

	θ	0°	10°	20°	30°	40°
	Sinθ	0.0000	0.1736	0.3420	0.5000	0.6428
(1)	KG'×Sinθ					
(2)	GZ_o					
(3)	GZ=(2)-(1)					
(4)	Coeff	1	3	3	1	—
(5)	(3) × (4)					—
(6)	Coeff	1	4	2	4	1
(7)	(3) × (6)					

(8)	Σ(5) =
(9)	Σ(7) =

(10)	(8)×π / 48 =
(11)	(9) ×π / 54 =

Area (0°~30°) = (10) =	⩾ 0.055 m-rad
Area (0°~40°) = (11) – (12) =	⩾ 0.09 m-rad
Area (30°~40°) = (11) – (10) – (12) =	⩾ 0.03 m-rad

In case down flooding angel (θ_f) is 40°or over (12) is taken 0.

In case down flooding angel (θ_f) is less than 40°, the correction (12) is taken as follows:

θ	40°	θ_f
GZ	a	b
(12)	ΔAREA = (a+b) × (40-θU) × π /360 =	

The following criteria is found by using GZ curve.

GZ_{30}° =	⩾ 0.2 m
θ_{max} =	⩾ 25°

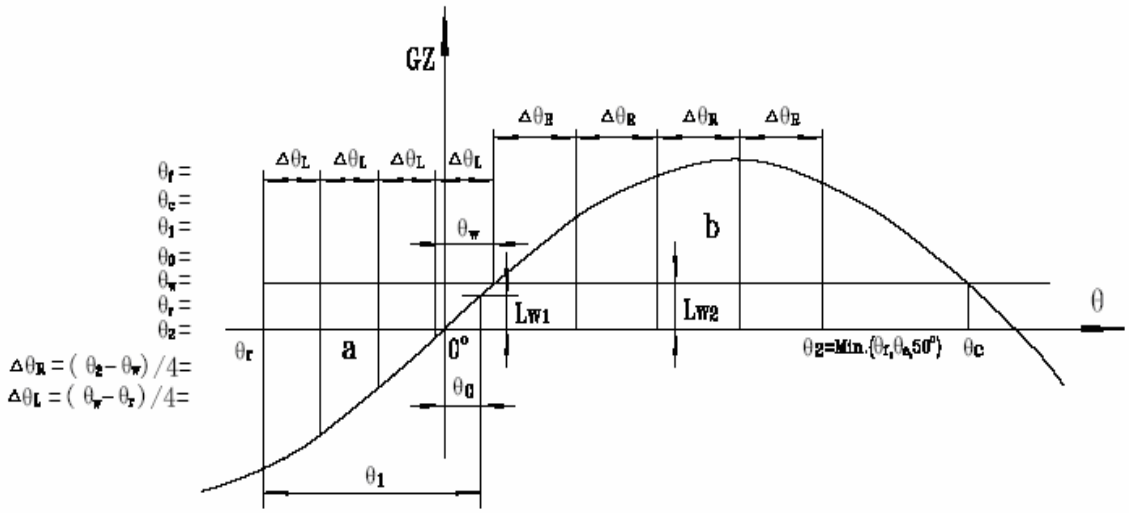
IMO A.749(18) Stability Criteria Calculation Sheet 2

Loading condition:_____

Displacement(ext.)	Disp(△)	=
Draught(ext.)	T_0	=
Draught(mld.)	T	=
Metacentric height(corrected by free surface effect)	GM	=
Center of gravity	KG	=
Center of gravity(corrected by free surface effect)	KG’	=
Water line length	L	=
Breadth (mld.)	B	=
Depth (mld.)	D	=
Bilge keel area	A_k	=
Projected lateral area	A	=
Vertical distance	Z	=
P×A×Z / 1000/ g /△	1_{w2}	=
$1.5\times1_{w1}$	1_{w1}	=
Flooding angle	θ_f	=
Min.(θ_f θ_c 50°)	θ_2	=
B/d		=
x_1		=
C_b=△/ (1.025LBd)		=
x_2		=
		=
100Ak / (LB)		=
k (from table 3)		=
OG=KG-d		=
r = 0.73+0.6 OG/d		=
Rolling period T= [2B/ (GM)1/2]×(0.373+0.023B/d-0.043L/100)		=
s (from table 4)		=
$\theta1 = 109\times k\times x_1\times x_2\times(rs)^{1/2}$		=
80% of immersing angle of deck edge ($0.8\theta_d$)=[arctan(2×(D’-d))×0.8		=
D’ (from top of keel to top of freeboard deck at ship’s side)		=
B’ (from outside of side shell to outside of another side shell)		=

IMO A.749(18)Stability Criteria Calculation Sheet 3

Loading condition:_____



Area“b”：

(1)	(deg.)	θ_W	$\theta_W+\Delta\theta_R$	$\theta_W+2\theta_R$	$\theta_W+3\Delta\theta_R$	$\theta_W+4\Delta\theta_R$
(2)	GZ	—				
(3)	$(GZ-1_{w2})$	0				
(4)	Simpson’s F	1	4	2	4	1
(5)	(3) × (4)	0				

$\Sigma(5)$ = _____

$b = 1/3 \times \Delta\theta_R \times \Sigma(5) \times \pi/180$ = _____

Area “a”：

(6)	(deg.)	θ_W	$\theta_R-\Delta\theta_L$	$\theta_W-2\Delta\theta_L$	$\theta_W-3\Delta\theta_L$	$\theta_W-4\Delta\theta_L$
(7)	GZ	—				
(8)	1_{w2}	—				
(9)	$(1_{w2} - GZ)$	0				
(10)	Simpson’s F	1	4	2	4	1
(11)	(4) × (5)	0				

$\Sigma(11)$ = _____

$a = 1/3 \times \Delta\theta_L \times \Sigma(11) \times \pi/180$ = _____

Criteria：

b = _____>a = _____

θ_0=_____<Min.(16°, $0.8\theta_d$)

3. STABILITY DATA

3.1. Bonjean Data

Station Number:

Stat No .	X(m)	Stat No .	X(m)	Stat No .	X(m)
0	0	**6**	51.6	**15**	129
0.5	4.3	**7**	60.2	**16**	137.6
1	8.6	**8**	68.8	**17**	146.2
1.5	12.9	**9**	77.4	**18**	154.8
2	17.2	**10**	86	**18.5**	159.1
2.5	21.5	**11**	94.6	**19**	163.4
3	25.8	**12**	103.2	**19.5**	167.7
4	34.4	**13**	111.8	**20**	172
5	43	**14**	120.4	**20.25**	174.15

Explanation of symbols:

T draught, molded m

The calculations are based on the following dimensions:

REFERENCE LENGTH	172.000 M
BREADTH MOULDED	28.800 M
DISTANCE FROM FRAME NO. 0 TO REF. POINT	86.000 M

T	X AREA	0.000 MOM BL	X AREA	4.300 MOM BL	X AREA	8.600 MOM BL	X AREA	12.900 MOM BL	T
1.00	0.00	0.00	0.00	0.00	0.59	0.45	2.07	1.21	1.00
1.50	0.00	0.00	0.00	0.00	1.53	1.64	3.67	3.22	1.50
2.00	0.00	0.00	0.00	0.00	2.65	3.61	5.49	6.41	2.00
2.50	0.00	0.00	0.00	0.00	3.89	6.39	7.47	10.88	2.50
3.00	0.00	0.00	0.00	0.00	5.16	9.88	9.59	16.72	3.00
3.50	0.00	0.00	0.00	0.00	6.42	14.00	11.87	24.12	3.50
4.00	0.00	0.00	0.00	0.00	7.69	18.75	14.33	33.34	4.00
4.50	0.00	0.00	0.00	0.00	8.98	24.23	17.05	44.91	4.50
5.00	0.00	0.00	0.00	0.00	10.36	30.79	20.14	59.63	5.00
5.50	0.00	0.00	0.00	0.00	11.94	39.09	23.79	78.84	5.50
6.00	0.00	0.00	0.00	0.00	13.89	50.35	28.36	105.14	6.00
6.50	0.00	0.00	0.00	0.00	16.56	67.05	34.35	142.64	6.50
7.00	0.00	0.00	0.00	0.00	20.78	95.63	41.89	193.61	7.00
7.50	0.00	0.00	1.73	12.70	26.92	140.23	50.69	257.47	7.50
8.00	0.00	0.00	6.54	50.10	34.43	198.47	60.39	332.66	8.00
8.50	1.63	13.63	12.88	102.40	42.88	268.26	70.75	418.17	8.50
9.00	6.13	53.07	20.11	165.70	52.04	348.41	81.63	513.34	9.00
9.50	11.67	104.40	27.98	238.59	61.73	438.06	92.91	617.68	9.50
10.00	17.89	165.00	36.35	320.16	71.84	536.67	104.51	730.83	10.00
10.50	24.60	233.83	45.10	409.83	82.30	643.85	116.38	852.47	10.50
11.00	31.70	310.19	54.15	507.14	93.03	759.24	128.47	982.55	11.00
11.50	39.11	393.53	63.45	611.86	103.99	882.58	140.77	1120.88	11.50
12.00	46.76	483.44	72.98	723.76	115.15	1013.66	153.24	1267.38	12.00
12.50	54.61	579.61	82.68	842.62	126.48	1152.50	165.85	1421.89	12.50
13.00	62.63	681.83	92.54	968.34	137.97	1298.93	178.58	1584.24	13.00

T	X AREA	17.200 MOM BL	X AREA	21.500 MOM BL	X AREA	25.800 MOM BL	X AREA	34.400 MOM BL	T
1.00	3.44	1.92	4.80	2.67	7.01	3.88	12.47	6.82	1.00
1.50	5.88	4.97	8.27	7.03	12.01	10.17	21.22	17.82	1.50
2.00	8.63	9.81	12.35	14.20	17.90	20.51	31.29	35.51	2.00
2.50	11.69	16.69	17.04	24.77	24.69	35.82	42.47	60.70	2.50
3.00	15.05	25.95	22.36	39.44	32.41	57.09	54.52	93.87	3.00
3.50	18.76	38.03	28.39	59.07	41.12	85.44	67.22	135.16	3.50
4.00	22.91	53.60	35.24	84.78	50.81	121.82	80.40	184.61	4.00
4.50	27.63	73.70	43.02	117.92	61.42	166.95	93.95	242.18	4.50
5.00	33.16	100.00	51.86	159.95	72.79	221.00	107.76	307.79	5.00
5.50	39.81	134.96	61.73	211.80	84.78	283.95	121.77	381.39	5.50
6.00	47.77	180.81	72.49	273.68	97.25	355.67	135.94	462.86	6.00
6.50	56.95	238.21	83.96	345.40	110.10	436.04	150.22	552.13	6.50
7.00	67.10	306.79	95.99	426.60	123.27	524.92	164.58	649.04	7.00
7.50	77.99	385.72	108.45	516.99	136.68	622.14	178.98	753.44	7.50
8.00	89.41	474.31	121.26	616.30	150.29	727.67	193.40	865.15	8.00
8.50	101.27	572.11	134.36	724.33	164.08	841.41	207.81	984.10	8.50
9.00	113.46	678.84	147.68	840.92	178.00	963.20	222.23	1110.26	9.00
9.50	125.94	794.28	161.19	965.88	192.02	1092.91	236.65	1243.62	9.50
10.00	138.67	918.34	174.84	1098.98	206.12	1230.43	251.07	1384.20	10.00
10.50	151.60	1050.90	188.60	1240.06	220.29	1375.63	265.49	1531.98	10.50
11.00	164.70	1191.76	202.46	1388.98	234.50	1528.41	279.90	1686.98	11.00
11.50	177.94	1340.75	216.38	1545.60	248.75	1688.72	294.32	1849.18	11.50
12.00	191.30	1497.74	230.35	1709.83	263.03	1856.48	308.74	2018.59	12.00
12.50	204.76	1662.62	244.38	1881.62	277.33	2031.68	323.16	2195.21	12.50
13.00	218.30	1835.26	258.44	2060.91	291.65	2214.26	337.58	2379.04	13.00

T	X AREA	43.000 MOM BL	X AREA	51.600 MOM BL	X AREA	60.200 MOM BL	X AREA	68.800 MOM BL	T
1.00	19.94	10.52	25.00	12.78	27.06	13.62	27.97	13.96	1.00
1.50	32.01	25.65	38.64	29.85	41.24	31.36	42.29	31.86	1.50
2.00	44.87	48.19	52.68	54.43	55.60	56.49	56.70	57.08	2.00
2.50	58.29	78.40	66.95	86.55	70.01	88.93	71.12	89.52	2.50
3.00	72.10	116.41	81.33	126.10	84.43	128.58	85.54	129.17	3.00
3.50	86.20	162.22	95.75	172.95	98.85	175.44	99.96	176.03	3.50
4.00	100.48	215.78	110.16	227.02	113.27	229.50	114.38	230.10	4.00
4.50	114.86	276.92	124.58	288.30	127.68	290.78	128.79	291.37	4.50
5.00	129.28	345.40	139.00	356.78	142.10	359.27	143.21	359.86	5.00
5.50	143.70	421.09	153.42	432.48	156.52	434.96	157.63	435.55	5.50
6.00	158.12	504.00	167.84	515.38	170.94	517.86	172.05	518.46	6.00
6.50	172.53	594.11	182.25	605.49	185.36	607.98	186.47	608.57	6.50
7.00	186.95	691.43	196.67	702.81	199.77	705.30	200.88	705.89	7.00
7.50	201.37	795.96	211.09	807.34	214.19	809.83	215.30	810.42	7.50
8.00	215.79	907.70	225.51	919.08	228.61	921.57	229.72	922.16	8.00
8.50	230.21	1026.65	239.93	1038.03	243.03	1040.52	244.14	1041.11	8.50
9.00	244.62	1152.81	254.34	1164.19	257.45	1166.67	258.56	1167.27	9.00
9.50	259.04	1286.17	268.76	1297.56	271.86	1300.04	272.97	1300.63	9.50
10.00	273.46	1426.75	283.18	1438.13	286.28	1440.62	287.39	1441.21	10.00
10.50	287.88	1574.53	297.60	1585.92	300.70	1588.40	301.81	1588.99	10.50
11.00	302.30	1729.53	312.02	1740.91	315.12	1743.39	316.23	1743.99	11.00
11.50	316.71	1891.73	326.43	1903.11	329.54	1905.60	330.65	1906.19	11.50
12.00	331.13	2061.14	340.85	2072.52	343.95	2075.01	345.06	2075.60	12.00
12.50	345.55	2237.76	355.27	2249.14	358.37	2251.63	359.48	2252.22	12.50
13.00	359.97	2421.59	369.69	2432.97	372.79	2435.46	373.90	2436.05	13.00

T	X AREA	77.400 MOM BL	X AREA	86.000 MOM BL	X AREA	94.600 MOM BL	X AREA	103.200 MOM BL	T
1.00	28.10	14.01	28.10	14.01	28.10	14.01	28.10	14.01	1.00
1.50	42.43	31.94	42.43	31.94	42.43	31.94	42.43	31.94	1.50
2.00	56.85	57.16	56.85	57.16	56.85	57.16	56.85	57.16	2.00
2.50	71.26	89.60	71.26	89.60	71.26	89.60	71.26	89.60	2.50
3.00	85.68	129.25	85.68	129.25	85.68	129.25	85.68	129.25	3.00
3.50	100.10	176.11	100.10	176.11	100.10	176.11	100.10	176.11	3.50
4.00	114.52	230.17	114.52	230.17	114.52	230.17	114.52	230.17	4.00
4.50	128.94	291.45	128.94	291.45	128.94	291.45	128.94	291.45	4.50
5.00	143.35	359.93	143.35	359.93	143.35	359.93	143.35	359.93	5.00
5.50	157.77	435.63	157.77	435.63	157.77	435.63	157.77	435.63	5.50
6.00	172.19	518.53	172.19	518.53	172.19	518.53	172.19	518.53	6.00
6.50	186.61	608.65	186.61	608.65	186.61	608.65	186.61	608.65	6.50
7.00	201.03	705.97	201.03	705.97	201.03	705.97	201.03	705.97	7.00
7.50	215.44	810.50	215.44	810.50	215.44	810.50	215.44	810.50	7.50
8.00	229.86	922.24	229.86	922.24	229.86	922.24	229.86	922.24	8.00
8.50	244.28	1041.18	244.28	1041.18	244.28	1041.18	244.28	1041.18	8.50
9.00	258.70	1167.34	258.70	1167.34	258.70	1167.34	258.70	1167.34	9.00
9.50	273.12	1300.71	273.12	1300.71	273.12	1300.71	273.12	1300.71	9.50
10.00	287.53	1441.28	287.53	1441.28	287.53	1441.28	287.53	1441.28	10.00
10.50	301.95	1589.07	301.95	1589.07	301.95	1589.07	301.95	1589.07	10.50
11.00	316.37	1744.06	316.37	1744.06	316.37	1744.06	316.37	1744.06	11.00
11.50	330.79	1906.26	330.79	1906.26	330.79	1906.26	330.79	1906.26	11.50
12.00	345.21	2075.68	345.21	2075.68	345.21	2075.68	345.21	2075.68	12.00
12.50	359.62	2252.30	359.62	2252.30	359.62	2252.30	359.62	2252.30	12.50
13.00	374.04	2436.13	374.04	2436.13	374.04	2436.13	374.04	2436.13	13.00

T	X AREA	111.800 MOM BL	X AREA	120.400 MOM BL	X AREA	129.000 MOM BL	X AREA	137.600 MOM BL	T
1.00	28.10	14.01	28.05	13.99	27.54	13.74	26.29	13.17	1.00
1.50	42.43	31.94	42.38	31.90	41.69	31.43	39.96	30.28	1.50
2.00	56.85	57.16	56.78	57.12	55.99	56.48	53.90	54.67	2.00
2.50	71.26	89.60	71.20	89.56	70.38	88.84	68.00	86.41	2.50
3.00	85.68	129.25	85.62	129.21	84.79	128.49	82.20	125.46	3.00
3.50	100.10	176.11	100.04	176.06	99.22	175.36	96.44	171.75	3.50
4.00	114.52	230.17	114.46	230.13	113.63	229.43	110.71	225.24	4.00
4.50	128.94	291.45	128.87	291.41	128.05	290.69	124.99	285.92	4.50
5.00	143.35	359.93	143.29	359.89	142.46	359.15	139.27	353.75	5.00
5.50	157.77	435.63	157.71	435.59	156.87	434.82	153.54	428.71	5.50
6.00	172.19	518.53	172.13	518.49	171.29	517.71	167.82	510.83	6.00
6.50	186.61	608.65	186.55	608.60	185.71	607.81	182.11	600.10	6.50
7.00	201.03	705.97	200.96	705.93	200.12	705.12	196.40	696.55	7.00
7.50	215.44	810.50	215.38	810.46	214.54	809.65	210.69	800.16	7.50
8.00	229.86	922.24	229.80	922.20	228.96	921.39	224.98	910.94	8.00
8.50	244.28	1041.18	244.22	1041.14	243.38	1040.34	239.28	1028.90	8.50
9.00	258.70	1167.34	258.64	1167.30	257.79	1166.50	253.58	1154.04	9.00
9.50	273.12	1300.71	273.05	1300.67	272.21	1299.87	267.89	1286.36	9.50
10.00	287.53	1441.28	287.47	1441.24	286.63	1440.44	282.20	1425.87	10.00
10.50	301.95	1589.07	301.89	1589.03	301.05	1588.23	296.51	1572.56	10.50
11.00	316.37	1744.06	316.31	1744.02	315.47	1743.22	310.82	1726.45	11.00
11.50	330.79	1906.26	330.73	1906.22	329.88	1905.42	325.14	1887.54	11.50
12.00	345.21	2075.68	345.14	2075.64	344.30	2074.83	339.46	2055.82	12.00
12.50	359.62	2252.30	359.56	2252.26	358.72	2251.45	353.79	2231.31	12.50
13.00	374.04	2436.13	373.98	2436.08	373.14	2435.28	368.12	2414.00	13.00

T	X AREA	146.200 MOM BL	X AREA	154.800 MOM BL	X AREA	159.100 MOM BL	X AREA	163.400 MOM BL	T
1.00	23.29	11.80	17.09	8.82	13.07	6.84	9.18	4.91	1.00
1.50	35.80	27.46	26.77	20.94	20.76	16.48	14.90	12.08	1.50
2.00	48.69	50.03	36.95	38.77	28.96	30.84	21.10	22.94	2.00
2.50	61.85	79.65	47.52	62.57	37.57	50.22	27.67	37.74	2.50
3.00	75.22	116.42	58.43	92.58	46.50	74.81	34.54	56.65	3.00
3.50	88.75	160.40	69.62	128.96	55.72	104.77	41.66	79.79	3.50
4.00	102.41	211.61	81.06	171.86	65.18	140.25	48.98	107.23	4.00
4.50	116.15	270.03	92.71	221.41	74.85	181.37	56.45	139.00	4.50
5.00	129.97	335.67	104.56	277.70	84.72	228.25	64.05	175.10	5.00
5.50	143.84	408.52	116.58	340.79	94.77	281.01	71.75	215.52	5.50
6.00	157.77	488.57	128.74	410.73	104.98	339.75	79.53	260.28	6.00
6.50	171.73	575.86	141.03	487.56	115.35	404.54	87.39	309.42	6.50
7.00	185.73	670.35	153.44	571.34	125.86	475.49	95.33	363.03	7.00
7.50	199.76	772.05	165.96	662.09	136.50	552.67	103.36	421.24	7.50
8.00	213.81	880.94	178.57	759.82	147.28	636.19	111.50	484.27	8.00
8.50	227.88	997.04	191.26	864.55	158.18	726.12	119.74	552.33	8.50
9.00	241.97	1120.31	204.03	976.29	169.20	822.55	128.13	625.70	9.00
9.50	256.07	1250.76	216.87	1095.03	180.33	925.54	136.66	704.61	9.50
10.00	270.19	1388.41	229.77	1220.79	191.58	1035.20	145.35	789.32	10.00
10.50	284.32	1533.27	242.73	1353.63	202.93	1151.57	154.20	880.09	10.50
11.00	298.47	1685.31	255.74	1493.55	214.39	1274.75	163.23	977.12	11.00
11.50	312.62	1844.51	268.81	1640.52	225.95	1404.85	172.42	1080.60	11.50
12.00	326.78	2010.90	281.92	1794.58	237.62	1541.94	181.80	1190.73	12.00
12.50	340.95	2184.47	295.07	1955.75	249.38	1686.05	191.34	1307.70	12.50
13.00	355.12	2365.23	308.27	2124.02	261.24	1837.27	201.07	1431.67	13.00

T	X AREA	167.700 MOM BL	X AREA	172.000 MOM BL	T
1.00	5.53	3.04	2.62	1.50	1.00
1.50	9.24	7.69	4.53	3.89	1.50
2.00	13.33	14.87	6.67	7.64	2.00
2.50	17.72	24.76	8.97	12.84	2.50
3.00	22.36	37.51	11.41	19.55	3.00
3.50	27.18	53.18	13.95	27.80	3.50
4.00	32.14	71.80	16.56	37.61	4.00
4.50	37.21	93.34	19.24	48.97	4.50
5.00	42.34	117.74	21.96	61.91	5.00
5.50	47.52	144.90	24.71	76.37	5.50
6.00	52.70	174.72	27.48	92.28	6.00
6.50	57.88	207.06	30.23	109.48	6.50
7.00	63.02	241.80	32.93	127.68	7.00
7.50	68.14	278.88	35.53	146.53	7.50
8.00	73.22	318.25	37.98	165.49	8.00
8.50	78.29	360.08	39.57	178.48	8.50
9.00	83.39	404.76	39.62	178.90	9.00
9.50	88.58	452.77	39.66	179.36	9.50
10.00	93.91	504.68	39.68	179.46	10.00
10.50	99.41	561.12	39.54	178.10	10.50
11.00	105.14	622.69	39.58	178.52	11.00
11.50	111.12	689.94	39.61	178.84	11.50
12.00	117.36	763.35	39.67	179.57	12.00
12.50	123.88	843.21	39.72	180.10	12.50
13.00	130.67	929.80	39.75	180.50	13.00

3.2. Hydrostatic Data

MAIN CHARACTERISTICS OF THE VESSEL:

Length between perpendiculars	172.00 m
Bread moulded	28.80 m
Design draught	9.83 m
Scantling draught	10.27 m
x-coordinate of after perpendicular	0.00 m
x-coordinate of reference point	86.00 m
x-coordinate of midship section	86.00 m
x-coordinate of building frame 0	0.00 m
Thickness of keelplate	0.020 m
Mean thickness of shell plating	0.018 m
Trim	0.000 m
Heel	0.000 deg.
Density of water	1.025 ton/m^3
Shell thickness used in the calculation	18.00 mm
x-coordinate of aft end of DWL	-3.20 m
x-coordinate of fore end of DWL	172.02 m

EXPLANATION OF SYMBOLS:

T	Draught moulded	m
DISP	Total displacement	t
VOLM	Volume moulded	m^3
WLA	Waterline area	m^2
WSA	Wetted surface area	m^2
LCB	Longitudinal center of buoyancy	m
VCB	Vertical center of buoyancy	m
LCA	Longitudinal center of flotation	m
KMT	Transversal metacentric height	m
KML	Longitudinal metacentric height	m
MCT	Moment to change trim	tm/cm
TCP	Change of displacement/change of draught	t/cm
CB	Block coeffecient	
CW	Waterplane coefficient	
CP	Prismatic coefficient	
CM	Midship section coefficient	

COORDINATE SYSTEM:

Righthanded origin of coordinate system is located at the intersection of frame0(AP), center line of ship(CL) and the base line(BL).

HYDROSTATIC TABLES:

T m	DISP t	VOLM m3	WLA m2	WSA m2	LCB m	VCB m	LCA m
2.000	7341.2	7087.4	3844.8	4155	95.140	1.029	93.89
2.100	7736.2	7472.1	3860.4	4196	95.073	1.081	93.79
2.200	8132.8	7858.2	3875.5	4236	95.007	1.133	93.70
2.300	8530.9	8245.9	3890.1	4276	94.942	1.185	93.61
2.400	8930.5	8635.1	3904.0	4316	94.880	1.237	93.53
2.500	9331.5	9025.6	3917.4	4355	94.818	1.289	93.44
2.600	9733.8	9417.4	3930.1	4394	94.758	1.341	93.35
2.700	10137.4	9810.5	3942.6	4433	94.699	1.393	93.27
2.800	10542.3	10204.8	3954.6	4472	94.641	1.446	93.18
2.900	10948.4	10600.3	3966.0	4511	94.584	1.498	93.10
3.000	11355.7	10996.9	3977.2	4549	94.528	1.550	93.02
3.100	11764.1	11394.7	3988.0	4588	94.472	1.602	92.94
3.200	12173.6	11793.5	3998.3	4626	94.417	1.654	92.85
3.300	12584.2	12193.4	4008.6	4664	94.363	1.706	92.77
3.400	12995.8	12594.3	4018.6	4702	94.310	1.758	92.69
3.500	13408.4	12996.1	4028.5	4740	94.257	1.810	92.60
3.600	13822.1	13399.0	4038.0	4778	94.204	1.862	92.52
3.700	14236.7	13802.9	4047.4	4816	94.152	1.914	92.43
3.800	14652.3	14207.7	4056.6	4854	94.100	1.966	92.34
3.900	15068.9	14613.4	4065.6	4892	94.049	2.018	92.26
4.000	15486.4	15020.0	4074.5	4930	93.997	2.070	92.17
4.100	15904.7	15427.5	4083.2	4967	93.946	2.122	92.09
4.200	16324.0	15835.9	4091.8	5005	93.896	2.174	92.00
4.300	16744.2	16245.1	4100.3	5043	93.845	2.226	91.92
4.400	17165.3	16655.2	4108.9	5080	93.795	2.278	91.82
4.500	17587.2	17066.2	4117.1	5118	93.744	2.330	91.73
4.600	18010.1	17478.0	4125.3	5156	93.694	2.383	91.64
4.700	18433.7	17890.7	4133.5	5193	93.644	2.435	91.55
4.800	18858.2	18304.2	4141.7	5231	93.593	2.487	91.45
4.900	19283.6	18718.5	4149.9	5269	93.543	2.539	91.36
5.000	19709.9	19133.7	4158.3	5307	93.492	2.591	91.26
5.100	20137.0	19549.7	4166.7	5345	93.442	2.643	91.15
5.200	20565.0	19966.6	4175.2	5383	93.391	2.695	91.04
5.300	20994.0	20384.4	4183.6	5421	93.339	2.747	90.94
5.400	21423.8	20803.0	4192.3	5460	93.288	2.799	90.82
5.500	21854.5	21222.5	4201.3	5499	93.236	2.852	90.69
5.600	22286.3	21643.0	4210.7	5538	93.183	2.904	90.56
5.700	22719.0	22064.4	4220.1	5578	93.129	2.956	90.43
5.800	23152.7	22486.8	4229.4	5618	93.075	3.008	90.29
5.900	23587.4	22910.2	4239.0	5657	93.020	3.061	90.15
6.000	24023.1	23334.6	4248.6	5697	92.964	3.113	90.01
6.100	24459.9	23760.0	4258.0	5737	92.907	3.165	89.87
6.200	24897.6	24186.3	4267.5	5777	92.850	3.218	89.73
6.300	25336.4	24613.7	4277.2	5817	92.792	3.270	89.58
6.400	25776.2	25042.0	4286.9	5858	92.733	3.323	89.44
6.500	26217.1	25471.4	4296.7	5898	92.673	3.375	89.28

T m	DISP t	VOLM m3	WLA m2	WSA m2	LCB m	VCB m	LCA m
6.600	26659.1	25901.8	4307.2	5939	92.612	3.428	89.12
6.700	27102.2	26333.4	4317.3	5980	92.551	3.480	88.96
6.800	27546.3	26766.0	4327.0	6021	92.488	3.533	88.80
6.900	27991.6	27199.6	4336.9	6062	92.424	3.586	88.64
7.000	28437.9	27634.3	4346.6	6103	92.360	3.638	88.49
7.100	28885.4	28070.1	4355.7	6143	92.295	3.691	88.34
7.200	29333.8	28506.9	4364.3	6184	92.229	3.744	88.20
7.300	29783.3	28944.6	4372.9	6224	92.162	3.797	88.05
7.400	30233.7	29383.3	4380.6	6263	92.095	3.849	87.93
7.500	30685.0	29822.9	4388.0	6302	92.028	3.902	87.81
7.600	31136.1	30262.5	4407.6	6353	91.963	3.955	87.45
7.700	31588.4	30702.9	4419.2	6396	91.897	4.008	87.25
7.800	32041.8	31144.5	4430.1	6439	91.830	4.061	87.06
7.900	32496.2	31587.1	4440.6	6481	91.763	4.114	86.88
8.000	32951.6	32030.7	4450.1	6522	91.695	4.167	86.72
8.100	33408.0	32475.1	4459.0	6563	91.626	4.220	86.56
8.200	33865.1	32920.4	4467.1	6603	91.558	4.273	86.42
8.300	34323.1	33366.5	4475.1	6643	91.489	4.326	86.27
8.400	34785.1	33816.3	4482.8	6681	91.428	4.379	86.13
8.500	35242.7	34262.2	4490.0	6722	91.354	4.433	85.98
8.600	35701.7	34709.2	4497.0	6762	91.282	4.486	85.83
8.700	36162.1	35157.9	4504.7	6802	91.212	4.539	85.70
8.800	36623.4	35607.3	4512.5	6841	91.144	4.593	85.57
8.900	37085.1	36057.1	4520.9	6880	91.076	4.646	85.43
9.000	37553.2	36512.6	4544.3	6933	90.993	4.700	84.99
9.100	38019.5	36966.8	4553.3	6974	90.919	4.754	84.83
9.200	38486.6	37421.8	4560.0	7013	90.844	4.807	84.72
9.300	38954.3	37877.4	4566.2	7051	90.770	4.860	84.62
9.400	39422.6	38333.6	4571.8	7089	90.696	4.914	84.53
9.500	39891.5	38790.4	4576.9	7127	90.623	4.967	84.45
9.600	40360.9	39247.6	4581.9	7164	90.551	5.020	84.38
9.700	40830.8	39705.4	4586.6	7201	90.480	5.074	84.31
9.800	41301.1	40163.6	4591.1	7238	90.409	5.127	84.25
9.830	41442.3	40301.2	4592.4	7250	90.388	5.143	84.24
9.900	41771.9	40622.3	4595.4	7275	90.339	5.180	84.20
10.000	42243.2	41081.3	4599.8	7312	90.271	5.233	84.14
10.100	42714.9	41540.9	4604.1	7349	90.203	5.287	84.09
10.200	43187.0	42000.8	4608.2	7386	90.135	5.340	84.05
10.270	43517.7	42323.0	4610.9	7411	90.089	5.377	84.02
10.300	43659.5	42461.1	4612.1	7422	90.069	5.393	84.01
10.400	44132.3	42921.8	4616.0	7458	90.004	5.446	83.97
10.500	44605.6	43382.9	4619.9	7495	89.940	5.499	83.94
10.600	45079.3	43844.4	4623.6	7531	89.877	5.552	83.92
10.700	45553.3	44306.2	4627.4	7567	89.814	5.605	83.89
10.800	46027.8	44768.4	4631.2	7603	89.753	5.658	83.87
10.900	46502.6	45231.0	4634.9	7639	89.693	5.711	83.85

T m	DISP t	VOLM m3	WLA m2	WSA m2	LCB m	VCB m	LCA m
11.000	46977.8	45693.9	4638.5	7675	89.633	5.764	83.83
11.100	47453.4	46157.3	4642.0	7711	89.575	5.817	83.81
11.200	47929.3	46620.9	4645.5	7747	89.517	5.870	83.80
11.300	48405.5	47084.9	4648.8	7783	89.461	5.923	83.79
11.400	48882.2	47549.3	4652.2	7819	89.406	5.976	83.78
11.500	49359.1	48014.0	4655.6	7855	89.351	6.029	83.77
11.600	49836.5	48479.0	4659.0	7891	89.298	6.082	83.77
11.700	50314.2	48944.4	4662.4	7927	89.245	6.135	83.76
11.800	50792.2	49410.2	4665.6	7963	89.193	6.188	83.76
11.900	51270.6	49876.2	4668.8	7999	89.143	6.240	83.76
12.000	51749.3	50342.6	4671.9	8035	89.093	6.293	83.76
12.100	52228.3	50809.3	4675.2	8071	89.044	6.346	83.77
12.200	52707.7	51276.3	4678.4	8106	88.996	6.399	83.77
12.300	53187.4	51743.7	4681.7	8142	88.949	6.452	83.78
12.400	53667.4	52211.4	4684.9	8178	88.903	6.504	83.78
12.500	54147.8	52679.4	4688.0	8214	88.857	6.557	83.79
12.600	54628.5	53147.7	4691.1	8250	88.813	6.610	83.80
12.700	55109.5	53616.3	4694.1	8286	88.769	6.662	83.81
12.800	55590.8	54085.2	4697.1	8322	88.726	6.715	83.83
12.900	56072.4	54554.4	4700.1	8358	88.684	6.768	83.84
13.000	56554.3	55023.9	4703.1	8394	88.643	6.821	83.85
13.100	57036.5	55493.7	4706.1	8430	88.603	6.873	83.86
13.200	57519.0	55963.8	4709.0	8466	88.563	6.926	83.87
13.300	58001.8	56434.2	4711.8	8502	88.524	6.979	83.89
13.400	58485.0	56904.9	4714.7	8538	88.486	7.031	83.90
13.500	58968.3	57375.9	4717.4	8573	88.448	7.084	83.91
13.600	59452.0	57847.1	4720.1	8609	88.411	7.136	83.93
13.700	59936.0	58318.6	4722.8	8645	88.375	7.189	83.94
13.800	60420.2	58790.4	4725.5	8681	88.340	7.242	83.95
13.900	60904.7	59262.4	4728.2	8717	88.305	7.294	83.97
14.000	61389.5	59734.7	4730.8	8753	88.270	7.347	83.98
14.100	61874.5	60207.3	4733.5	8788	88.237	7.399	84.00
14.200	62359.8	60680.1	4736.0	8824	88.204	7.452	84.01
14.300	62845.4	61153.2	4738.7	8860	88.172	7.504	84.03
14.400	63331.3	61626.6	4741.4	8896	88.140	7.557	84.04
14.500	63817.4	62100.2	4744.1	8932	88.109	7.609	84.06
14.600	64303.8	62574.1	4746.8	8968	88.078	7.662	84.08

T m	KMT m	KML m	MCT tm/cm	CB	TCP t/cm	CP	CM	CW
2.000	32.989	858.170	365.8	0.7154	39.4	0.7321	0.977	0.7747
2.100	31.616	822.473	369.4	0.7183	39.6	0.7343	0.978	0.7779
2.200	30.368	789.873	372.9	0.7211	39.7	0.7364	0.979	0.7809
2.300	29.232	759.985	376.4	0.7238	39.9	0.7385	0.980	0.7839
2.400	28.188	732.386	379.6	0.7263	40.0	0.7405	0.981	0.7867

T m	KMT m	KML m	MCT tm/cm	CB	TCP t/cm	CP	CM	CW
2.500	27.228	706.905	382.8	0.7288	40.2	0.7424	0.982	0.7894
2.600	26.341	683.192	385.9	0.7312	40.3	0.7443	0.982	0.7920
2.700	25.523	661.225	388.9	0.7335	40.4	0.7462	0.983	0.7945
2.800	24.764	640.738	391.8	0.7357	40.5	0.7480	0.984	0.7969
2.900	24.056	621.534	394.7	0.7379	40.7	0.7497	0.984	0.7992
3.000	23.399	603.550	397.4	0.7400	40.8	0.7515	0.985	0.8015
3.100	22.785	586.645	400.1	0.7420	40.9	0.7532	0.985	0.8037
3.200	22.212	570.718	402.8	0.7440	41.0	0.7548	0.986	0.8058
3.300	21.675	555.761	405.4	0.7459	41.1	0.7564	0.986	0.8078
3.400	21.173	541.660	407.9	0.7478	41.2	0.7580	0.987	0.8099
3.500	20.704	528.358	410.5	0.7496	41.3	0.7595	0.987	0.8119
3.600	20.261	515.728	412.9	0.7514	41.4	0.7610	0.987	0.8138
3.700	19.844	503.797	415.4	0.7531	41.5	0.7625	0.988	0.8157
3.800	19.451	492.444	417.8	0.7548	41.6	0.7640	0.988	0.8176
3.900	19.081	481.651	420.2	0.7564	41.7	0.7654	0.988	0.8194
4.000	18.731	471.444	422.6	0.7580	41.8	0.7668	0.989	0.8212
4.100	18.400	461.674	424.9	0.7596	41.9	0.7682	0.989	0.8229
4.200	18.086	452.402	427.3	0.7612	41.9	0.7695	0.989	0.8247
4.300	17.789	443.558	429.6	0.7627	42.0	0.7709	0.989	0.8264
4.400	17.508	435.150	432.0	0.7641	42.1	0.7722	0.990	0.8281
4.500	17.241	427.070	434.3	0.7656	42.2	0.7735	0.990	0.8298
4.600	16.986	419.339	436.6	0.7670	42.3	0.7747	0.990	0.8314
4.700	16.745	411.984	438.9	0.7684	42.4	0.7760	0.990	0.8331
4.800	16.516	404.940	441.3	0.7698	42.5	0.7772	0.990	0.8347
4.900	16.299	398.229	443.6	0.7712	42.5	0.7785	0.991	0.8364
5.000	16.092	391.847	446.1	0.7725	42.6	0.7797	0.991	0.8381
5.100	15.895	385.760	448.5	0.7738	42.7	0.7808	0.991	0.8398
5.200	15.707	379.948	451.1	0.7751	42.8	0.7820	0.991	0.8415
5.300	15.528	374.341	453.6	0.7764	42.9	0.7832	0.991	0.8432
5.400	15.357	369.044	456.2	0.7777	43.0	0.7843	0.992	0.8449
5.500	15.194	364.050	458.9	0.7790	43.1	0.7855	0.992	0.8467
5.600	15.039	359.363	461.9	0.7802	43.2	0.7866	0.992	0.8486
5.700	14.891	354.820	464.8	0.7814	43.3	0.7878	0.992	0.8505
5.800	14.747	350.465	467.7	0.7827	43.4	0.7889	0.992	0.8524
5.900	14.614	346.306	470.7	0.7839	43.4	0.7900	0.992	0.8543
6.000	14.488	342.318	473.8	0.7851	43.5	0.7911	0.992	0.8563
6.100	14.367	338.413	476.8	0.7863	43.6	0.7923	0.992	0.8582
6.200	14.252	334.656	479.8	0.7875	43.7	0.7934	0.993	0.8600
6.300	14.144	331.093	482.9	0.7887	43.8	0.7945	0.993	0.8620
6.400	14.040	327.642	486.0	0.7899	43.9	0.7956	0.993	0.8640
6.500	13.940	324.363	489.3	0.7911	44.0	0.7967	0.993	0.8659
6.600	13.847	321.340	492.7	0.7923	44.1	0.7978	0.993	0.8680
6.700	13.757	318.309	496.1	0.7934	44.3	0.7989	0.993	0.8701
6.800	13.671	315.302	499.3	0.7946	44.4	0.8000	0.993	0.8720
6.900	13.590	312.434	502.6	0.7958	44.5	0.8011	0.993	0.8740
7.000	13.513	309.611	505.9	0.7969	44.6	0.8022	0.993	0.8760

T m	KMT m	KML m	MCT tm/cm	CB	TCP t/cm	CP	CM	CW
7.100	13.439	306.736	508.9	0.7981	44.6	0.8033	0.994	0.8779
7.200	13.368	303.832	511.8	0.7993	44.7	0.8044	0.994	0.8796
7.300	13.301	301.006	514.6	0.8004	44.8	0.8055	0.994	0.8813
7.400	13.236	298.063	517.2	0.8016	44.9	0.8066	0.994	0.8829
7.500	13.173	295.171	519.6	0.8027	45.0	0.8077	0.994	0.8844
7.600	13.115	295.108	527.1	0.8038	45.2	0.8087	0.994	0.8882
7.700	13.058	293.258	531.2	0.8049	45.3	0.8098	0.994	0.8905
7.800	13.004	291.314	535.1	0.8061	45.4	0.8108	0.994	0.8928
7.900	12.955	289.312	538.8	0.8072	45.5	0.8119	0.994	0.8949
8.000	12.906	287.168	542.2	0.8083	45.6	0.8129	0.994	0.8969
8.100	12.859	284.950	545.3	0.8094	45.7	0.8140	0.994	0.8987
8.200	12.814	282.647	548.1	0.8105	45.8	0.8150	0.994	0.9003
8.300	12.772	280.374	550.9	0.8115	45.9	0.8160	0.994	0.9019
8.400	12.731	278.074	553.5	0.8127	45.9	0.8171	0.995	0.9035
8.500	12.693	275.750	555.9	0.8137	46.0	0.8181	0.995	0.9049
8.600	12.657	273.465	558.3	0.8148	46.1	0.8191	0.995	0.9063
8.700	12.622	271.369	561.0	0.8158	46.2	0.8201	0.995	0.9079
8.800	12.589	269.370	563.8	0.8168	46.3	0.8211	0.995	0.9095
8.900	12.558	267.539	566.8	0.8179	46.3	0.8221	0.995	0.9112
9.000	12.531	269.059	577.2	0.8190	46.6	0.8232	0.995	0.9153
9.100	12.503	267.464	580.7	0.8201	46.7	0.8242	0.995	0.9178
9.200	12.475	265.455	583.2	0.8211	46.7	0.8252	0.995	0.9191
9.300	12.450	263.367	585.5	0.8222	46.8	0.8263	0.995	0.9204
9.400	12.426	261.221	587.5	0.8232	46.9	0.8273	0.995	0.9215
9.500	12.403	259.041	589.3	0.8243	46.9	0.8283	0.995	0.9226
9.600	12.382	256.881	591.0	0.8253	47.0	0.8293	0.995	0.9236
9.700	12.362	254.744	592.7	0.8263	47.0	0.8303	0.995	0.9246
9.800	12.343	252.597	594.2	0.8273	47.1	0.8312	0.995	0.9255
9.830	12.338	251.955	594.7	0.8276	47.1	0.8315	0.995	0.9257
9.900	12.326	250.475	595.7	0.8283	47.1	0.8322	0.995	0.9264
10.000	12.310	248.420	597.3	0.8293	47.1	0.8331	0.995	0.9272
10.100	12.296	246.382	598.7	0.8303	47.2	0.8341	0.995	0.9281
10.200	12.282	244.364	600.2	0.8313	47.2	0.8350	0.996	0.9289
10.270	12.273	242.959	601.1	0.8319	47.3	0.8357	0.996	0.9295
10.300	12.270	242.369	601.5	0.8322	47.3	0.8359	0.996	0.9297
10.400	12.258	240.427	602.9	0.8332	47.3	0.8368	0.996	0.9305
10.500	12.248	238.513	604.3	0.8341	47.4	0.8377	0.996	0.9313
10.600	12.238	236.628	605.6	0.8350	47.4	0.8386	0.996	0.9321
10.700	12.230	234.780	607.0	0.8359	47.4	0.8395	0.996	0.9328
10.800	12.223	232.980	608.3	0.8368	47.5	0.8404	0.996	0.9336
10.900	12.217	231.201	609.6	0.8377	47.5	0.8412	0.996	0.9344
11.000	12.212	229.443	610.9	0.8386	47.5	0.8421	0.996	0.9351
11.100	12.207	227.715	612.2	0.8395	47.6	0.8429	0.996	0.9358
11.200	12.203	226.024	613.5	0.8403	47.6	0.8438	0.996	0.9365
11.300	12.201	224.350	614.7	0.8412	47.7	0.8446	0.996	0.9372
11.400	12.199	222.707	615.9	0.8420	47.7	0.8454	0.996	0.9379

T m	KMT m	KML m	MCT tm/cm	CB	TCP t/cm	CP	CM	CW
11.500	12.198	221.102	617.2	0.8428	47.7	0.8462	0.996	0.9385
11.600	12.198	219.532	618.5	0.8437	47.8	0.8470	0.996	0.9392
11.700	12.199	217.987	619.7	0.8445	47.8	0.8478	0.996	0.9399
11.800	12.201	216.447	620.9	0.8453	47.8	0.8486	0.996	0.9406
11.900	12.203	214.933	622.1	0.8461	47.9	0.8494	0.996	0.9412
12.000	12.206	213.446	623.3	0.8469	47.9	0.8501	0.996	0.9418
12.100	12.211	212.002	624.5	0.8477	47.9	0.8509	0.996	0.9425
12.200	12.216	210.590	625.7	0.8485	48.0	0.8517	0.996	0.9431
12.300	12.221	209.197	626.9	0.8492	48.0	0.8524	0.996	0.9438
12.400	12.228	207.828	628.2	0.8500	48.0	0.8532	0.996	0.9444
12.500	12.235	206.469	629.3	0.8508	48.1	0.8539	0.996	0.9451
12.600	12.242	205.135	630.5	0.8515	48.1	0.8546	0.996	0.9457
12.700	12.250	203.817	631.7	0.8523	48.1	0.8553	0.996	0.9463
12.800	12.259	202.521	632.8	0.8530	48.1	0.8561	0.996	0.9469
12.900	12.269	201.246	634.0	0.8537	48.2	0.8568	0.996	0.9475
13.000	12.279	199.989	635.1	0.8545	48.2	0.8575	0.996	0.9481
13.100	12.290	198.752	636.3	0.8552	48.2	0.8582	0.997	0.9487
13.200	12.302	197.531	637.4	0.8559	48.3	0.8589	0.997	0.9493
13.300	12.314	196.325	638.5	0.8566	48.3	0.8595	0.997	0.9499
13.400	12.326	195.133	639.6	0.8573	48.3	0.8602	0.997	0.9505
13.500	12.340	193.957	640.7	0.8580	48.4	0.8609	0.997	0.9510
13.600	12.354	192.790	641.7	0.8587	48.4	0.8616	0.997	0.9516
13.700	12.368	191.641	642.8	0.8593	48.4	0.8622	0.997	0.9521
13.800	12.383	190.510	643.8	0.8600	48.4	0.8629	0.997	0.9526
13.900	12.398	189.397	644.8	0.8607	48.5	0.8635	0.997	0.9532
14.000	12.415	188.301	645.9	0.8613	48.5	0.8642	0.997	0.9537
14.100	12.431	187.218	646.9	0.8620	48.5	0.8648	0.997	0.9543
14.200	12.448	186.146	647.9	0.8627	48.5	0.8654	0.997	0.9548
14.300	12.466	185.099	648.9	0.8633	48.6	0.8661	0.997	0.9553
14.400	12.484	184.077	650.0	0.8639	48.6	0.8667	0.997	0.9558
14.500	12.502	183.072	651.0	0.8646	48.6	0.8673	0.997	0.9564
14.600	12.521	182.084	652.1	0.8652	48.7	0.8679	0.997	0.9569

---------- total displacement t ----------

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
2.000	6801.4	6897.3	7022.6	7094.3	7171.6	7253.6	7341.2	7433.8
2.100	7194.5	7293.1	7419.1	7490.7	7567.7	7649.2	7736.2	7828.0
2.200	7590.2	7690.9	7817.3	7888.8	7965.0	8046.5	8132.8	8223.8
2.300	7988.2	8090.5	8217.1	8288.5	8364.2	8445.2	8530.9	8621.1
2.400	8388.4	8491.8	8618.5	8689.6	8765.0	8845.5	8930.5	9019.9
2.500	8790.6	8894.8	9021.4	9091.7	9167.1	9247.1	9331.5	9420.0
2.600	9194.7	9299.3	9425.8	9495.7	9570.7	9650.1	9733.8	9821.6
2.700	9600.4	9705.4	9831.0	9901.1	9975.6	10054.5	10137.4	10224.4
2.800	10007.9	10113.0	10238.0	10307.7	10381.8	10460.1	10542.3	10628.5
2.900	10417.0	10521.9	10646.4	10715.7	10789.3	10866.9	10948.4	11033.7
3.000	10827.6	10932.3	11056.0	11124.9	11197.9	11274.9	11355.7	11440.2
3.100	11239.7	11344.0	11466.9	11535.3	11607.7	11684.0	11764.1	11847.8
3.200	11653.3	11757.0	11879.1	11946.9	12018.7	12094.3	12173.6	12256.4
3.300	12068.3	12170.8	12292.4	12359.6	12430.7	12505.7	12584.2	12666.2
3.400	12484.7	12586.3	12706.8	12773.4	12843.9	12918.1	12995.8	13076.9
3.500	12902.5	13003.0	13122.4	13188.4	13258.1	13331.5	13408.4	13488.7
3.600	13321.6	13420.9	13539.1	13604.4	13673.4	13746.0	13822.1	13901.5
3.700	13742.1	13840.1	13956.9	14021.5	14089.7	14161.5	14236.7	14315.2
3.800	14163.4	14260.4	14375.8	14439.5	14507.0	14578.0	14652.3	14729.9
3.900	14586.6	14681.9	14795.7	14858.7	14925.3	14995.4	15068.9	15145.5
4.000	15011.2	15104.6	15216.7	15278.8	15344.6	15413.8	15486.4	15562.1
4.100	15437.1	15528.4	15638.7	15699.9	15764.8	15833.1	15904.7	15979.5
4.200	15864.5	15953.4	16061.7	16122.0	16185.9	16253.3	16324.0	16397.9
4.300	16293.2	16379.6	16485.8	16545.0	16608.0	16674.5	16744.2	16817.1
4.400	16723.3	16807.1	16910.8	16969.1	17031.0	17096.5	17165.3	17237.2
4.500	17154.7	17235.7	17337.0	17394.1	17455.0	17519.5	17587.2	17658.1
4.600	17587.5	17665.6	17764.1	17820.1	17879.9	17943.3	18010.1	18079.9
4.700	18021.7	18096.8	18192.4	18247.1	18305.7	18368.0	18433.7	18502.5
4.800	18457.3	18529.1	18621.8	18675.1	18732.5	18793.7	18858.2	18926.0
4.900	18894.3	18962.7	19052.2	19104.1	19160.3	19220.2	19283.6	19350.3
5.000	19332.8	19397.5	19483.9	19534.2	19589.0	19647.6	19709.9	19775.4
5.100	19772.6	19833.5	19916.6	19965.4	20018.7	20076.0	20137.0	20201.4
5.200	20213.7	20270.8	20350.5	20397.7	20449.4	20505.3	20565.0	20628.2
5.300	20656.2	20709.5	20785.6	20831.1	20881.2	20935.6	20994.0	21055.8
5.400	21100.0	21149.4	21221.7	21265.6	21314.1	21366.9	21423.8	21484.3
5.500	21545.0	21590.6	21659.1	21701.2	21748.0	21799.2	21854.5	21913.7
5.600	21991.0	22033.1	22097.6	22137.9	22183.0	22232.5	22286.3	22344.0
5.700	22437.4	22476.9	22537.3	22575.7	22619.0	22666.9	22719.0	22775.1
5.800	22885.3	22921.9	22978.3	23014.7	23056.2	23102.3	23152.7	23207.2
5.900	23334.5	23368.1	23420.5	23454.8	23494.4	23538.7	23587.4	23640.2
6.000	23785.1	23815.4	23863.8	23896.1	23933.7	23976.2	24023.1	24074.3
6.100	24236.8	24263.3	24308.4	24338.7	24374.1	24414.7	24459.9	24509.3
6.200	24689.7	24711.9	24754.2	24782.3	24815.8	24854.3	24897.6	24945.2
6.300	25143.7	25161.8	25201.1	25227.2	25258.6	25295.0	25336.4	25382.2
6.400	25598.7	25613.1	25649.1	25673.2	25702.5	25736.9	25776.2	25820.1
6.500	26054.6	26065.5	26098.2	26120.4	26147.5	26179.8	26217.1	26259.1
6.600	26511.6	26519.0	26547.5	26568.6	26593.7	26623.9	26659.1	26699.1
6.700	26969.3	26973.5	26997.8	27017.9	27041.1	27069.1	27102.2	27140.0
6.800	27427.9	27429.1	27449.3	27468.0	27489.4	27515.4	27546.3	27582.1
6.900	27887.1	27885.6	27901.9	27918.1	27938.8	27962.8	27991.6	28025.2
7.000	28353.2	28343.0	28355.7	28369.7	28389.2	28411.3	28437.9	28469.4
7.100	28817.4	28801.2	28810.5	28822.3	28839.7	28860.7	28885.4	28914.7
7.200	29282.4	29260.3	29266.3	29276.1	29291.1	29311.2	29333.8	29361.0
7.300	29748.1	29720.2	29723.0	29730.9	29743.7	29762.1	29783.3	29808.4
7.400	30214.6	30180.5	30180.7	30186.7	30197.5	30213.4	30233.7	30256.7

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
7.500	30681.7	30647.6	30639.1	30643.5	30652.2	30665.9	30685.0	30706.1
7.600	31149.4	31112.8	31098.4	31101.1	31108.0	31119.5	31136.1	31156.4
7.700	31617.6	31578.8	31558.4	31559.6	31564.7	31574.1	31588.4	31607.6
7.800	32086.5	32045.6	32019.1	32018.8	32022.2	32029.7	32041.8	32059.1
7.900	32555.8	32512.9	32480.3	32478.8	32480.5	32486.2	32496.2	32511.1
8.000	33025.6	32980.9	32948.0	32939.5	32939.7	32943.6	32951.6	32964.2
8.100	33495.9	33449.5	33413.9	33400.7	33399.6	33401.8	33408.0	33422.0
8.200	33966.6	33918.5	33880.6	33862.3	33860.2	33860.8	33865.1	33875.4
8.300	34437.7	34388.1	34348.0	34332.4	34321.4	34320.5	34323.1	34330.3
8.400	34909.3	34858.1	34815.9	34798.9	34783.0	34781.0	34785.1	34786.2
8.500	35381.2	35328.6	35284.4	35266.1	35251.1	35242.0	35242.7	35243.8
8.600	35853.5	35799.5	35753.4	35733.9	35717.4	35703.6	35701.7	35702.2
8.700	36326.2	36270.8	36222.9	36202.3	36184.3	36167.9	36162.1	36161.3
8.800	36799.2	36742.5	36692.9	36671.2	36651.9	36636.6	36623.4	36621.2
8.900	37272.5	37214.5	37163.3	37140.6	37123.7	37102.1	37085.1	37081.8
9.000	37746.1	37686.9	37634.1	37610.4	37589.8	37569.3	37553.2	37543.1
9.100	38220.0	38159.6	38105.3	38080.7	38058.3	38037.2	38019.5	38004.8
9.200	38694.1	38632.7	38576.8	38554.2	38527.4	38505.6	38486.6	38466.9
9.300	39168.6	39106.0	39048.7	39023.4	38997.4	38974.6	38954.3	38937.3
9.400	39643.3	39579.6	39524.7	39493.5	39467.7	39444.0	39422.6	39404.2
9.500	40118.2	40053.5	39995.1	39965.1	39938.5	39913.9	39891.5	39871.8
9.600	40593.4	40527.6	40466.8	40437.1	40409.7	40384.2	40360.9	40340.1
9.700	41068.7	41002.0	40939.1	40909.4	40881.3	40855.0	40830.8	40809.0
9.800	41544.3	41476.6	41412.4	41382.1	41353.2	41326.2	41301.1	41278.4
9.830	41687.0	41619.0	41554.4	41523.9	41494.9	41467.6	41442.3	41419.4
9.900	42020.1	41951.4	41885.9	41855.0	41825.5	41797.7	41771.9	41748.4
10.000	42496.1	42428.6	42359.7	42328.2	42298.2	42269.7	42243.2	42218.8
10.100	42972.2	42902.3	42833.8	42801.8	42771.1	42742.1	42714.9	42689.8
10.200	43448.5	43376.8	43308.2	43275.6	43244.4	43214.8	43187.0	43161.2
10.270	43782.1	43709.6	43640.4	43607.4	43575.9	43545.9	43517.7	43491.4
10.300	43925.0	43852.3	43782.8	43749.7	43718.0	43687.9	43659.5	43633.0
10.400	44401.6	44327.9	44257.6	44224.1	44191.9	44161.3	44132.3	44105.4
10.500	44881.2	44803.8	44732.7	44698.8	44666.2	44635.0	44605.6	44578.2
10.600	45356.2	45279.9	45208.0	45173.7	45140.7	45109.1	45079.3	45051.4
10.700	45832.2	45756.1	45683.6	45648.9	45615.5	45583.6	45553.3	45525.1
10.800	46309.2	46232.6	46159.4	46124.4	46090.6	46058.3	46027.8	45999.2
10.900	46786.4	46709.2	46635.4	46600.1	46566.0	46533.5	46502.6	46473.7
11.000	47263.8	47186.0	47111.6	47076.0	47041.7	47008.9	46977.8	46948.6
11.100	47741.3	47663.0	47588.1	47552.3	47517.7	47484.7	47453.4	47423.9
11.200	48218.9	48140.2	48064.8	48028.8	47994.0	47960.8	47929.3	47899.6
11.300	48696.7	48617.5	48541.8	48505.5	48470.6	48437.2	48405.5	48375.8
11.400	49174.7	49095.1	49018.9	48982.5	48947.5	48914.0	48882.2	48852.2
11.500	49652.8	49572.8	49496.3	49459.8	49424.7	49391.0	49359.1	49329.1
11.600	50131.0	50050.7	49974.0	49937.3	49902.1	49868.5	49836.5	49806.4
11.700	50609.4	50528.8	50451.8	50415.2	50379.9	50346.2	50314.2	50284.0
11.800	51087.9	51007.0	50930.0	50893.3	50858.0	50824.2	50792.2	50762.0
11.900	51566.7	51485.5	51408.4	51371.6	51336.3	51302.6	51270.6	51240.3
12.000	52045.5	51964.2	51887.0	51850.3	51815.0	51781.3	51749.3	51719.1
12.100	52524.5	52443.1	52365.8	52329.2	52294.0	52260.3	52228.3	52198.1
12.200	53003.7	52922.1	52845.0	52808.4	52773.2	52739.6	52707.7	52677.5
12.300	53483.1	53401.4	53324.4	53287.8	53252.8	53219.3	53187.4	53157.3
12.400	53962.6	53881.0	53804.0	53767.6	53732.6	53699.2	53667.4	53637.4
12.500	54442.3	54360.7	54283.9	54247.6	54212.7	54179.4	54147.8	54117.8
12.600	54922.2	54840.7	54764.1	54727.9	54693.2	54660.0	54628.5	54598.6
12.700	55402.2	55320.9	55244.6	55208.5	55173.9	55140.9	55109.5	55079.7
12.800	55882.5	55801.4	55725.3	55689.3	55654.9	55622.0	55590.8	55561.2
12.900	56363.0	56282.1	56206.3	56170.5	56136.2	56103.5	56072.4	56042.9

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
13.000	56843.7	56763.0	56687.5	56651.9	56617.8	56585.2	56554.3	56525.0
13.100	57324.7	57244.2	57169.0	57133.6	57099.7	57067.3	57036.5	57007.3
13.200	57805.8	57725.6	57650.8	57615.6	57581.9	57549.7	57519.0	57490.0
13.300	58287.2	58207.3	58132.9	58097.8	58064.3	58032.3	58001.8	57973.0
13.400	58768.8	58689.2	58615.2	58580.4	58547.0	58515.2	58485.0	58456.2
13.500	59250.6	59171.4	59097.8	59063.2	59030.1	58998.4	58968.3	58939.8
13.600	59732.7	59653.9	59580.7	59546.3	59513.3	59481.9	59452.0	59423.6
13.700	60215.1	60136.6	60063.8	60029.6	59996.9	59965.7	59936.0	59907.8
13.800	60697.6	60619.6	60547.2	60513.3	60480.8	60449.7	60420.2	60392.1
13.900	61180.4	61102.9	61030.9	60997.2	60964.9	60934.1	60904.7	60876.8
14.000	61663.5	61586.4	61514.9	61481.4	61449.3	61418.7	61389.5	61361.8
14.100	62146.8	62070.1	61999.1	61965.8	61933.9	61903.5	61874.5	61847.0
14.200	62630.4	62554.2	62483.6	62450.5	62418.9	62388.6	62359.8	62332.5
14.300	63114.2	63038.4	62968.4	62935.5	62904.0	62874.0	62845.4	62818.3
14.400	63598.3	63523.0	63453.4	63420.7	63389.5	63359.7	63331.3	63304.4
14.500	64082.6	64007.8	63938.7	63906.2	63875.2	63845.6	63817.4	63790.8
14.600	64567.2	64492.9	64424.2	64392.0	64361.2	64331.7	64303.8	64277.4

---------- volume moulded m3 ----------

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
2.000	6561.6	6654.6	6776.7	6846.6	6922.0	7002.0	7087.4	7177.8
2.100	6944.2	7040.0	7162.8	7232.6	7307.7	7387.2	7472.1	7561.6
2.200	7329.4	7427.4	7550.5	7620.3	7694.6	7774.0	7858.2	7947.0
2.300	7717.0	7816.5	7939.9	8009.5	8083.4	8162.4	8245.9	8333.9
2.400	8106.6	8207.2	8330.8	8400.1	8473.6	8552.1	8635.1	8722.3
2.500	8498.2	8599.6	8723.1	8791.7	8865.3	8943.3	9025.6	9112.0
2.600	8891.6	8993.6	9116.9	9185.1	9258.3	9335.7	9417.4	9503.0
2.700	9286.8	9389.1	9512.0	9579.9	9652.6	9729.5	9810.5	9895.3
2.800	9683.6	9786.0	9908.0	9975.9	10048.2	10124.5	10204.8	10288.8
2.900	10082.0	10184.3	10305.6	10373.2	10445.0	10520.7	10600.3	10683.5
3.000	10481.8	10583.9	10704.6	10771.7	10843.0	10918.1	10996.9	11079.3
3.100	10883.2	10984.9	11104.8	11171.4	11242.1	11316.6	11394.7	11476.3
3.200	11285.9	11387.1	11506.2	11572.3	11642.3	11716.1	11793.5	11874.3
3.300	11690.1	11790.1	11908.7	11974.3	12043.7	12116.8	12193.4	12273.3
3.400	12095.6	12194.7	12312.4	12377.3	12446.1	12518.4	12594.3	12673.4
3.500	12502.5	12600.6	12717.1	12781.4	12849.5	12921.1	12996.1	13074.5
3.600	12910.7	13007.6	13123.0	13186.6	13254.0	13324.8	13399.0	13476.5
3.700	13320.2	13415.9	13529.9	13592.8	13659.4	13729.5	13802.9	13879.4
3.800	13730.6	13825.2	13937.8	14000.0	14065.9	14135.1	14207.7	14283.3
3.900	14142.7	14235.7	14346.8	14408.3	14473.3	14541.7	14613.4	14688.1
4.000	14556.1	14647.4	14756.8	14817.4	14881.6	14949.2	15020.0	15093.8
4.100	14971.0	15060.2	15167.8	15227.6	15290.9	15357.6	15427.5	15500.4
4.200	15387.2	15474.1	15579.8	15638.7	15701.1	15766.9	15835.9	15907.9
4.300	15804.7	15889.2	15992.9	16050.7	16112.2	16177.1	16245.1	16316.2
4.400	16223.6	16305.5	16406.9	16463.7	16524.2	16588.1	16655.2	16725.4
4.500	16643.7	16723.0	16821.9	16877.7	16937.2	17000.1	17066.2	17135.4
4.600	17065.2	17141.7	17238.0	17292.6	17351.0	17412.9	17478.0	17546.2
4.700	17488.0	17561.6	17655.1	17708.5	17765.8	17826.6	17890.7	17957.8
4.800	17912.3	17982.6	18073.2	18125.3	18181.4	18241.2	18304.2	18370.3
4.900	18337.8	18404.9	18492.5	18543.2	18598.1	18656.6	18718.5	18783.6
5.000	18764.8	18828.3	18912.9	18962.1	19015.6	19072.9	19133.7	19197.7
5.100	19193.1	19253.0	19334.3	19382.1	19434.1	19490.1	19549.7	19612.6
5.200	19622.7	19678.9	19756.9	19803.1	19853.6	19908.3	19966.6	20028.3
5.300	20053.7	20106.0	20180.6	20225.2	20274.2	20327.4	20384.4	20444.8
5.400	20485.9	20534.5	20605.4	20648.4	20695.8	20747.4	20803.0	20862.1
5.500	20919.3	20964.2	21031.3	21072.6	21118.4	21168.5	21222.5	21280.3
5.600	21353.9	21395.1	21458.4	21497.9	21542.0	21590.5	21643.0	21699.4
5.700	21788.5	21827.3	21886.7	21924.3	21966.8	22013.5	22064.4	22119.3
5.800	22224.7	22260.7	22316.1	22351.8	22392.5	22437.6	22486.8	22540.1
5.900	22662.3	22695.2	22746.8	22780.5	22819.2	22862.6	22910.2	22961.9
6.000	23101.1	23130.9	23178.6	23210.3	23247.1	23288.7	23334.6	23384.6
6.100	23541.1	23567.4	23611.6	23641.3	23676.1	23715.8	23760.0	23808.3
6.200	23982.2	24004.2	24045.7	24073.4	24106.1	24143.9	24186.3	24232.9
6.300	24424.3	24442.4	24481.0	24506.6	24537.4	24573.1	24613.7	24658.5
6.400	24867.5	24881.9	24917.3	24941.0	24969.7	25003.4	25042.0	25085.0
6.500	25311.6	25322.5	25354.7	25376.5	25403.2	25434.9	25471.4	25512.5
6.600	25756.7	25764.2	25792.6	25813.0	25837.7	25867.4	25901.8	25941.0
6.700	26202.6	26206.9	26231.0	26250.6	26273.4	26300.9	26333.4	26370.5
6.800	26649.4	26650.7	26670.7	26689.2	26710.1	26735.6	26766.0	26801.1
6.900	27096.8	27095.3	27111.6	27127.6	27147.8	27171.4	27199.6	27232.6
7.000	27550.1	27540.8	27553.5	27567.3	27586.5	27608.1	27634.3	27665.3
7.100	28002.3	27987.2	27996.5	28008.2	28025.4	28045.9	28070.1	28098.9
7.200	28455.2	28434.4	28440.4	28450.2	28465.0	28484.6	28506.9	28533.6
7.300	28908.9	28882.4	28885.3	28893.2	28905.9	28924.1	28944.6	28969.3
7.400	29363.3	29330.9	29331.1	29337.1	29347.8	29363.5	29383.3	29406.0

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
7.500	29818.3	29785.3	29777.6	29782.0	29790.7	29804.2	29822.9	29843.6
7.600	30274.0	30238.4	30225.0	30227.8	30234.6	30246.0	30262.5	30282.2
7.700	30730.1	30692.4	30673.1	30674.3	30679.4	30688.8	30702.9	30721.7
7.800	31186.8	31147.0	31121.9	31121.7	31125.1	31132.5	31144.5	31161.7
7.900	31644.0	31602.3	31571.2	31569.7	31571.5	31577.2	31587.1	31601.8
8.000	32101.7	32058.2	32026.3	32018.5	32018.8	32022.7	32030.7	32043.1
8.100	32559.8	32514.6	32480.1	32467.9	32466.8	32469.0	32475.1	32489.0
8.200	33018.4	32971.6	32934.7	32917.6	32915.4	32916.1	32920.4	32930.0
8.300	33477.4	33429.0	33389.9	33374.9	33364.7	33363.9	33366.5	33373.5
8.400	33936.7	33886.9	33845.8	33829.3	33814.5	33812.4	33816.3	33817.7
8.500	34396.5	34345.2	34302.2	34284.4	34269.9	34261.6	34262.2	34263.4
8.600	34856.6	34804.0	34759.1	34740.1	34724.0	34714.9	34709.2	34709.9
8.700	35317.1	35263.1	35216.4	35196.4	35178.9	35163.1	35157.9	35157.2
8.800	35777.9	35722.6	35674.3	35653.2	35634.4	35619.5	35607.3	35605.2
8.900	36239.0	36182.4	36132.5	36110.4	36093.8	36073.0	36057.1	36053.9
9.000	36700.4	36642.6	36591.2	36568.1	36548.5	36528.1	36512.6	36503.3
9.100	37162.0	37103.2	37050.2	37026.2	37004.3	36983.9	36966.8	36953.1
9.200	37624.0	37564.0	37509.6	37487.1	37461.4	37440.2	37421.8	37403.3
9.300	38086.2	38025.1	37969.3	37944.5	37919.2	37897.0	37877.4	37860.9
9.400	38548.6	38486.5	38432.9	38402.6	38377.5	38354.3	38333.6	38315.7
9.500	39011.3	38948.2	38891.1	38862.0	38836.1	38812.1	38790.4	38771.3
9.600	39474.2	39410.1	39350.6	39321.8	39295.1	39270.3	39247.6	39227.5
9.700	39937.4	39872.2	39810.9	39781.9	39754.5	39728.9	39705.4	39684.2
9.800	40400.7	40334.6	40272.0	40242.4	40214.3	40188.0	40163.6	40141.5
9.830	40539.7	40473.3	40410.3	40380.6	40352.3	40325.7	40301.2	40278.8
9.900	40864.2	40800.9	40733.3	40703.1	40674.4	40647.4	40622.3	40599.3
10.000	41327.9	41261.5	41194.9	41164.2	41134.9	41107.2	41081.3	41057.6
10.100	41791.8	41723.4	41656.8	41625.5	41595.7	41567.4	41540.9	41516.4
10.200	42255.8	42185.8	42118.9	42087.1	42056.8	42027.9	42000.8	41975.7
10.270	42580.8	42510.1	42442.5	42410.4	42379.7	42350.5	42323.0	42297.4
10.300	42720.0	42649.1	42581.3	42549.0	42518.2	42488.8	42461.1	42435.4
10.400	43184.4	43112.5	43043.9	43011.2	42979.9	42950.0	42921.8	42895.6
10.500	43651.1	43576.1	43506.7	43473.7	43441.9	43411.5	43382.9	43356.2
10.600	44114.2	44039.9	43969.8	43936.4	43904.2	43873.4	43844.4	43817.2
10.700	44578.0	44503.9	44433.1	44399.3	44366.8	44335.6	44306.2	44278.7
10.800	45042.8	44968.1	44896.7	44862.6	44829.7	44798.2	44768.4	44740.6
10.900	45507.7	45432.4	45360.5	45326.0	45292.8	45261.1	45231.0	45202.8
11.000	45972.8	45896.9	45824.4	45789.7	45756.3	45724.3	45693.9	45665.5
11.100	46438.0	46361.7	46288.7	46253.7	46220.0	46187.8	46157.3	46128.6
11.200	46903.4	46826.5	46753.1	46717.9	46684.0	46651.6	46620.9	46592.0
11.300	47368.9	47291.6	47217.7	47182.4	47148.4	47115.8	47084.9	47055.9
11.400	47834.5	47756.9	47682.6	47647.1	47613.0	47580.3	47549.3	47520.1
11.500	48300.3	48222.3	48147.7	48112.1	48077.9	48045.1	48014.0	47984.7
11.600	48766.2	48687.9	48613.1	48577.4	48543.0	48510.2	48479.0	48449.7
11.700	49232.3	49153.7	49078.7	49042.9	49008.5	48975.6	48944.4	48915.0
11.800	49698.6	49619.7	49544.5	49508.7	49474.3	49441.4	49410.2	49380.7
11.900	50165.0	50085.8	50010.6	49974.8	49940.3	49907.4	49876.2	49846.7
12.000	50631.5	50552.2	50476.9	50441.1	50406.7	50373.8	50342.6	50313.1
12.100	51098.2	51018.7	50943.4	50907.7	50873.3	50840.5	50809.3	50779.9
12.200	51565.0	51485.5	51410.2	51374.5	51340.2	51307.5	51276.3	51246.9
12.300	52032.1	51952.5	51877.3	51841.6	51807.4	51774.8	51743.7	51714.3
12.400	52499.2	52419.7	52344.6	52309.1	52274.9	52242.4	52211.4	52182.1
12.500	52966.6	52887.1	52812.2	52776.7	52742.7	52710.3	52679.4	52650.2
12.600	53434.1	53354.7	53280.0	53244.7	53210.8	53178.4	53147.7	53118.6
12.700	53901.9	53822.5	53748.1	53712.9	53679.1	53646.9	53616.3	53587.3
12.800	54369.8	54290.6	54216.4	54181.4	54147.8	54115.7	54085.2	54056.3
12.900	54837.9	54759.0	54685.0	54650.1	54616.7	54584.8	54554.4	54525.7

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
13.000	55306.3	55227.5	55153.9	55119.2	55085.9	55054.1	55023.9	54995.4
13.100	55774.8	55696.3	55623.0	55588.5	55555.4	55523.8	55493.7	55465.3
13.200	56243.6	56165.4	56092.4	56058.1	56025.1	55993.7	55963.8	55935.5
13.300	56712.6	56634.7	56562.1	56527.9	56495.2	56464.0	56434.2	56406.1
13.400	57181.8	57104.2	57032.0	56998.0	56965.5	56934.5	56904.9	56876.9
13.500	57651.3	57574.0	57502.2	57468.4	57436.1	57405.2	57375.9	57348.0
13.600	58121.0	58044.1	57972.7	57939.1	57906.9	57876.3	57847.1	57819.4
13.700	58590.9	58514.4	58443.4	58410.0	58378.1	58347.6	58318.6	58291.1
13.800	59061.0	58984.9	58914.3	58881.2	58849.5	58819.2	58790.4	58763.0
13.900	59531.5	59455.7	59385.6	59352.6	59321.1	59291.1	59262.4	59235.3
14.000	60002.1	59926.8	59857.1	59824.4	59793.1	59763.2	59734.7	59707.7
14.100	60473.0	60398.1	60328.9	60296.4	60265.3	60235.6	60207.3	60180.5
14.200	60944.1	60869.7	60800.9	60768.6	60737.7	60708.2	60680.1	60653.5
14.300	61415.5	61341.5	61273.2	61241.1	61210.4	61181.1	61153.2	61126.8
14.400	61887.1	61813.6	61745.7	61713.9	61683.4	61654.3	61626.6	61600.4
14.500	62359.0	62286.0	62218.6	62186.9	62156.6	62127.7	62100.2	62074.2
14.600	62831.1	62758.6	62691.6	62660.2	62630.1	62601.4	62574.1	62548.4

---------- long. centre of buoy. m ----------

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
2.000	73.776	79.278	84.747	87.428	90.059	92.629	95.140	97.584
2.100	74.634	79.909	85.133	87.694	90.209	92.667	95.073	97.416
2.200	75.416	80.479	85.480	87.932	90.337	92.698	95.007	97.259
2.300	76.130	80.996	85.794	88.146	90.454	92.722	94.942	97.111
2.400	76.784	81.466	86.078	88.339	90.558	92.741	94.880	96.970
2.500	77.383	81.895	86.336	88.508	90.651	92.755	94.818	96.837
2.600	77.934	82.289	86.570	88.665	90.733	92.765	94.758	96.710
2.700	78.441	82.650	86.781	88.808	90.806	92.771	94.699	96.588
2.800	78.909	82.982	86.977	88.938	90.872	92.774	94.641	96.472
2.900	79.343	83.288	87.156	89.057	90.930	92.773	94.584	96.360
3.000	79.745	83.571	87.321	89.165	90.981	92.770	94.528	96.253
3.100	80.117	83.833	87.473	89.263	91.027	92.764	94.472	96.150
3.200	80.464	84.076	87.613	89.352	91.067	92.756	94.417	96.050
3.300	80.787	84.298	87.742	89.434	91.102	92.746	94.363	95.953
3.400	81.088	84.507	87.861	89.508	91.133	92.734	94.310	95.859
3.500	81.369	84.702	87.971	89.576	91.159	92.720	94.257	95.768
3.600	81.632	84.883	88.072	89.637	91.182	92.705	94.204	95.680
3.700	81.877	85.052	88.165	89.693	91.201	92.688	94.152	95.594
3.800	82.103	85.210	88.250	89.743	91.216	92.669	94.100	95.509
3.900	82.317	85.357	88.329	89.789	91.229	92.649	94.049	95.427
4.000	82.517	85.493	88.402	89.830	91.239	92.628	93.997	95.347
4.100	82.704	85.621	88.469	89.866	91.245	92.606	93.946	95.268
4.200	82.878	85.739	88.530	89.899	91.249	92.582	93.896	95.190
4.300	83.041	85.849	88.585	89.927	91.251	92.557	93.845	95.114
4.400	83.194	85.950	88.636	89.952	91.251	92.531	93.795	95.040
4.500	83.336	86.044	88.682	89.973	91.248	92.505	93.744	94.966
4.600	83.468	86.131	88.723	89.991	91.243	92.477	93.694	94.894
4.700	83.591	86.211	88.759	90.006	91.236	92.448	93.644	94.822
4.800	83.706	86.284	88.791	90.018	91.226	92.418	93.593	94.752
4.900	83.812	86.351	88.819	90.026	91.215	92.387	93.543	94.682
5.000	83.910	86.413	88.843	90.031	91.202	92.355	93.492	94.613
5.100	84.000	86.468	88.863	90.034	91.187	92.322	93.442	94.545
5.200	84.084	86.518	88.879	90.033	91.169	92.288	93.391	94.477
5.300	84.161	86.563	88.891	90.029	91.150	92.253	93.339	94.410
5.400	84.231	86.602	88.901	90.023	91.128	92.216	93.288	94.343
5.500	84.296	86.637	88.906	90.014	91.105	92.179	93.236	94.276
5.600	84.357	86.667	88.909	90.003	91.079	92.139	93.183	94.210
5.700	84.416	86.693	88.908	89.989	91.052	92.099	93.129	94.143
5.800	84.468	86.715	88.904	89.973	91.023	92.057	93.075	94.076
5.900	84.515	86.733	88.897	89.954	90.992	92.014	93.020	94.009
6.000	84.558	86.748	88.888	89.932	90.960	91.970	92.964	93.942
6.100	84.596	86.761	88.875	89.909	90.925	91.924	92.907	93.874
6.200	84.630	86.773	88.860	89.883	90.889	91.878	92.850	93.807
6.300	84.661	86.781	88.843	89.855	90.851	91.830	92.792	93.738
6.400	84.689	86.785	88.823	89.825	90.811	91.780	92.733	93.670
6.500	84.713	86.787	88.802	89.793	90.769	91.730	92.673	93.600
6.600	84.735	86.786	88.781	89.760	90.726	91.678	92.612	93.531
6.700	84.754	86.782	88.759	89.725	90.682	91.624	92.551	93.461
6.800	84.771	86.776	88.734	89.689	90.636	91.570	92.488	93.390
6.900	84.786	86.769	88.707	89.655	90.589	91.514	92.424	93.319
7.000	84.780	86.759	88.679	89.618	90.541	91.458	92.360	93.247
7.100	84.782	86.748	88.648	89.580	90.495	91.400	92.295	93.174
7.200	84.782	86.736	88.617	89.540	90.448	91.342	92.229	93.101
7.300	84.780	86.722	88.584	89.499	90.400	91.285	92.162	93.027
7.400	84.778	86.708	88.550	89.456	90.350	91.229	92.095	92.952

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
7.500	84.774	86.675	88.515	89.413	90.299	91.171	92.028	92.878
7.600	84.769	86.649	88.480	89.369	90.247	91.112	91.963	92.803
7.700	84.763	86.623	88.443	89.324	90.194	91.052	91.897	92.728
7.800	84.757	86.596	88.407	89.278	90.141	90.992	91.830	92.654
7.900	84.750	86.568	88.370	89.233	90.087	90.931	91.763	92.581
8.000	84.742	86.540	88.316	89.186	90.033	90.869	91.695	92.508
8.100	84.734	86.512	88.270	89.140	89.978	90.808	91.626	92.443
8.200	84.726	86.483	88.223	89.094	89.924	90.746	91.558	92.363
8.300	84.716	86.454	88.176	89.027	89.869	90.684	91.489	92.285
8.400	84.707	86.425	88.129	88.972	89.815	90.621	91.428	92.207
8.500	84.697	86.396	88.082	88.917	89.745	90.559	91.354	92.132
8.600	84.687	86.368	88.035	88.862	89.683	90.498	91.282	92.056
8.700	84.676	86.338	87.989	88.808	89.620	90.442	91.212	91.981
8.800	84.666	86.310	87.942	88.753	89.558	90.358	91.144	91.906
8.900	84.655	86.281	87.896	88.699	89.504	90.287	91.076	91.831
9.000	84.644	86.252	87.850	88.645	89.436	90.217	90.993	91.757
9.100	84.632	86.223	87.805	88.591	89.374	90.149	90.919	91.684
9.200	84.621	86.194	87.760	88.544	89.311	90.080	90.844	91.611
9.300	84.609	86.166	87.715	88.487	89.251	90.013	90.770	91.521
9.400	84.597	86.138	87.679	88.432	89.191	89.946	90.696	91.441
9.500	84.585	86.109	87.630	88.380	89.131	89.879	90.623	91.363
9.600	84.574	86.081	87.584	88.329	89.073	89.814	90.551	91.284
9.700	84.561	86.054	87.539	88.278	89.015	89.749	90.480	91.207
9.800	84.549	86.026	87.496	88.227	88.957	89.685	90.409	91.130
9.830	84.546	86.018	87.483	88.212	88.940	89.666	90.388	91.107
9.900	84.537	85.998	87.453	88.178	88.901	89.621	90.339	91.054
10.000	84.525	85.976	87.411	88.129	88.845	89.559	90.271	90.980
10.100	84.512	85.945	87.369	88.080	88.789	89.497	90.203	90.906
10.200	84.500	85.916	87.328	88.032	88.735	89.436	90.135	90.833
10.270	84.491	85.897	87.299	87.999	88.697	89.394	90.089	90.782
10.300	84.488	85.889	87.287	87.985	88.681	89.376	90.069	90.761
10.400	84.475	85.863	87.247	87.938	88.628	89.317	90.004	90.690
10.500	84.468	85.836	87.207	87.892	88.576	89.258	89.940	90.620
10.600	84.452	85.810	87.168	87.847	88.524	89.201	89.877	90.551
10.700	84.437	85.784	87.130	87.802	88.473	89.144	89.814	90.483
10.800	84.424	85.759	87.092	87.758	88.423	89.088	89.753	90.417
10.900	84.412	85.733	87.054	87.714	88.374	89.033	89.693	90.351
11.000	84.399	85.708	87.017	87.672	88.326	88.979	89.633	90.287
11.100	84.386	85.684	86.981	87.629	88.278	88.926	89.575	90.223
11.200	84.374	85.659	86.945	87.588	88.231	88.874	89.517	90.161
11.300	84.361	85.635	86.910	87.547	88.185	88.823	89.461	90.099
11.400	84.349	85.611	86.875	87.507	88.139	88.772	89.406	90.039
11.500	84.337	85.588	86.841	87.468	88.095	88.723	89.351	89.980
11.600	84.325	85.565	86.807	87.429	88.051	88.674	89.298	89.921
11.700	84.312	85.542	86.774	87.391	88.008	88.626	89.245	89.864
11.800	84.301	85.520	86.741	87.353	87.966	88.579	89.193	89.808
11.900	84.289	85.498	86.710	87.316	87.924	88.533	89.143	89.753
12.000	84.277	85.476	86.678	87.280	87.884	88.488	89.093	89.699
12.100	84.266	85.455	86.647	87.245	87.844	88.443	89.044	89.645
12.200	84.254	85.434	86.617	87.210	87.805	88.400	88.996	89.593
12.300	84.243	85.413	86.588	87.176	87.766	88.357	88.949	89.542
12.400	84.232	85.393	86.559	87.143	87.728	88.315	88.903	89.491
12.500	84.221	85.374	86.530	87.110	87.691	88.274	88.857	89.442
12.600	84.211	85.354	86.502	87.078	87.655	88.233	88.813	89.393
12.700	84.200	85.335	86.475	87.047	87.620	88.194	88.769	89.346
12.800	84.190	85.317	86.448	87.016	87.585	88.155	88.726	89.299
12.900	84.180	85.298	86.422	86.986	87.551	88.117	88.684	89.253

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
13.000	84.170	85.281	86.396	86.956	87.517	88.079	88.643	89.208
13.100	84.161	85.263	86.371	86.927	87.484	88.043	88.603	89.163
13.200	84.151	85.246	86.347	86.899	87.452	88.007	88.563	89.120
13.300	84.142	85.230	86.322	86.871	87.421	87.972	88.524	89.077
13.400	84.133	85.213	86.299	86.844	87.390	87.937	88.486	89.035
13.500	84.125	85.197	86.276	86.817	87.359	87.903	88.448	88.994
13.600	84.116	85.182	86.253	86.791	87.330	87.870	88.411	88.954
13.700	84.108	85.167	86.231	86.765	87.301	87.837	88.375	88.914
13.800	84.100	85.152	86.209	86.740	87.272	87.805	88.340	88.875
13.900	84.093	85.138	86.188	86.716	87.244	87.774	88.305	88.837
14.000	84.085	85.124	86.168	86.691	87.217	87.743	88.270	88.799
14.100	84.078	85.110	86.147	86.668	87.190	87.713	88.237	88.762
14.200	84.071	85.096	86.127	86.645	87.163	87.683	88.204	88.726
14.300	84.064	85.083	86.108	86.622	87.138	87.654	88.172	88.690
14.400	84.058	85.071	86.089	86.600	87.112	87.626	88.140	88.655
14.500	84.052	85.058	86.070	86.578	87.087	87.598	88.109	88.621
14.600	84.046	85.046	86.052	86.557	87.063	87.570	88.078	88.588

---------- transv. metac. height m ----------

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
2.000	35.599	35.226	34.616	34.246	33.854	33.434	32.989	32.522
2.100	34.073	33.690	33.108	32.770	32.406	32.023	31.616	31.194
2.200	32.686	32.294	31.742	31.429	31.094	30.740	30.368	29.987
2.300	31.409	31.013	30.499	30.205	29.897	29.569	29.232	28.879
2.400	30.236	29.846	29.360	29.088	28.799	28.498	28.188	27.866
2.500	29.156	28.777	28.315	28.062	27.794	27.515	27.228	26.932
2.600	28.159	27.794	27.356	27.117	26.868	26.611	26.341	26.067
2.700	27.241	26.886	26.469	26.246	26.013	25.773	25.523	25.267
2.800	26.389	26.047	25.650	25.439	25.222	24.995	24.764	24.524
2.900	25.596	25.267	24.891	24.692	24.486	24.275	24.056	23.836
3.000	24.860	24.545	24.189	23.997	23.803	23.601	23.399	23.192
3.100	24.175	23.873	23.532	23.352	23.164	22.976	22.785	22.592
3.200	23.534	23.243	22.919	22.750	22.570	22.392	22.212	22.030
3.300	22.933	22.656	22.349	22.183	22.017	21.846	21.675	21.505
3.400	22.373	22.105	21.812	21.655	21.498	21.335	21.173	21.012
3.500	21.849	21.588	21.306	21.160	21.009	20.857	20.704	20.547
3.600	21.352	21.103	20.833	20.694	20.551	20.407	20.261	20.113
3.700	20.886	20.645	20.388	20.254	20.121	19.984	19.844	19.705
3.800	20.449	20.215	19.968	19.841	19.714	19.583	19.451	19.320
3.900	20.039	19.809	19.572	19.452	19.330	19.206	19.081	18.957
4.000	19.650	19.426	19.198	19.084	18.966	18.848	18.731	18.614
4.100	19.278	19.064	18.846	18.735	18.624	18.510	18.400	18.289
4.200	18.933	18.723	18.513	18.405	18.298	18.192	18.086	17.982
4.300	18.606	18.399	18.195	18.093	17.990	17.890	17.789	17.689
4.400	18.296	18.094	17.896	17.797	17.699	17.603	17.508	17.411
4.500	18.003	17.802	17.612	17.517	17.424	17.332	17.241	17.149
4.600	17.727	17.529	17.342	17.251	17.162	17.074	16.986	16.901
4.700	17.464	17.268	17.087	17.000	16.914	16.829	16.745	16.664
4.800	17.215	17.022	16.845	16.760	16.678	16.596	16.516	16.438
4.900	16.978	16.790	16.616	16.533	16.453	16.375	16.299	16.225
5.000	16.754	16.568	16.394	16.318	16.239	16.165	16.092	16.021
5.100	16.539	16.358	16.188	16.113	16.037	15.965	15.895	15.826
5.200	16.335	16.158	15.993	15.914	15.845	15.774	15.707	15.641
5.300	16.143	15.970	15.807	15.730	15.661	15.594	15.528	15.464
5.400	15.958	15.791	15.630	15.555	15.483	15.421	15.357	15.296
5.500	15.781	15.619	15.462	15.388	15.319	15.256	15.194	15.135
5.600	15.613	15.455	15.303	15.230	15.163	15.096	15.039	14.981
5.700	15.454	15.301	15.151	15.081	15.013	14.948	14.891	14.835
5.800	15.301	15.152	15.008	14.938	14.870	14.807	14.747	14.697
5.900	15.156	15.010	14.869	14.802	14.735	14.672	14.614	14.564
6.000	15.018	14.875	14.738	14.672	14.607	14.545	14.488	14.437
6.100	14.884	14.746	14.613	14.546	14.484	14.424	14.367	14.313
6.200	14.756	14.623	14.493	14.428	14.368	14.309	14.252	14.200
6.300	14.634	14.506	14.378	14.316	14.256	14.199	14.144	14.091
6.400	14.517	14.393	14.268	14.209	14.150	14.094	14.040	13.989
6.500	14.405	14.284	14.164	14.106	14.049	13.993	13.940	13.892
6.600	14.298	14.180	14.064	14.008	13.952	13.897	13.847	13.798
6.700	14.196	14.081	13.969	13.914	13.859	13.807	13.757	13.710
6.800	14.097	13.987	13.877	13.823	13.771	13.721	13.671	13.625
6.900	14.002	13.895	13.789	13.738	13.687	13.638	13.590	13.545
7.000	13.915	13.808	13.705	13.655	13.606	13.559	13.513	13.467
7.100	13.827	13.724	13.626	13.576	13.529	13.484	13.439	13.395
7.200	13.743	13.644	13.549	13.501	13.456	13.412	13.368	13.326
7.300	13.663	13.567	13.476	13.430	13.385	13.342	13.301	13.260
7.400	13.586	13.494	13.405	13.362	13.318	13.276	13.236	13.196

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
7.500	13.512	13.428	13.337	13.296	13.254	13.212	13.173	13.137
7.600	13.440	13.359	13.273	13.232	13.193	13.152	13.115	13.079
7.700	13.372	13.294	13.211	13.172	13.134	13.096	13.058	13.023
7.800	13.307	13.231	13.152	13.114	13.077	13.041	13.004	12.971
7.900	13.245	13.172	13.096	13.059	13.023	12.988	12.955	12.920
8.000	13.185	13.114	13.046	13.006	12.972	12.938	12.906	12.873
8.100	13.128	13.060	12.993	12.956	12.922	12.890	12.859	12.828
8.200	13.073	13.007	12.943	12.908	12.876	12.845	12.814	12.785
8.300	13.021	12.957	12.895	12.865	12.831	12.801	12.772	12.744
8.400	12.972	12.909	12.849	12.820	12.789	12.760	12.731	12.704
8.500	12.925	12.863	12.806	12.779	12.751	12.720	12.693	12.667
8.600	12.879	12.820	12.764	12.738	12.712	12.682	12.657	12.632
8.700	12.836	12.778	12.725	12.700	12.675	12.646	12.622	12.599
8.800	12.795	12.739	12.688	12.663	12.639	12.615	12.589	12.567
8.900	12.756	12.702	12.652	12.628	12.604	12.583	12.558	12.536
9.000	12.718	12.666	12.617	12.595	12.573	12.552	12.531	12.507
9.100	12.683	12.632	12.585	12.563	12.542	12.522	12.503	12.480
9.200	12.650	12.601	12.554	12.532	12.514	12.494	12.475	12.454
9.300	12.619	12.570	12.526	12.505	12.486	12.468	12.450	12.433
9.400	12.589	12.541	12.498	12.479	12.460	12.443	12.426	12.410
9.500	12.560	12.514	12.473	12.454	12.436	12.419	12.403	12.388
9.600	12.534	12.489	12.449	12.431	12.413	12.397	12.382	12.368
9.700	12.509	12.465	12.426	12.409	12.392	12.376	12.362	12.349
9.800	12.485	12.442	12.405	12.388	12.372	12.357	12.343	12.331
9.830	12.478	12.436	12.398	12.382	12.366	12.352	12.338	12.326
9.900	12.463	12.421	12.385	12.368	12.354	12.339	12.326	12.314
10.000	12.442	12.401	12.366	12.350	12.336	12.323	12.310	12.299
10.100	12.423	12.383	12.349	12.333	12.320	12.307	12.296	12.284
10.200	12.405	12.366	12.333	12.318	12.305	12.293	12.282	12.272
10.270	12.394	12.355	12.323	12.308	12.295	12.283	12.273	12.264
10.300	12.389	12.351	12.318	12.304	12.291	12.280	12.270	12.260
10.400	12.374	12.336	12.305	12.291	12.279	12.268	12.258	12.250
10.500	12.360	12.323	12.292	12.280	12.268	12.257	12.248	12.240
10.600	12.347	12.311	12.281	12.269	12.258	12.247	12.238	12.231
10.700	12.336	12.300	12.272	12.259	12.248	12.239	12.230	12.223
10.800	12.326	12.291	12.263	12.251	12.240	12.231	12.223	12.216
10.900	12.317	12.283	12.255	12.243	12.233	12.225	12.217	12.210
11.000	12.309	12.275	12.248	12.237	12.227	12.219	12.212	12.205
11.100	12.302	12.269	12.243	12.232	12.222	12.214	12.207	12.201
11.200	12.296	12.264	12.238	12.227	12.218	12.210	12.203	12.198
11.300	12.291	12.260	12.234	12.224	12.215	12.207	12.201	12.196
11.400	12.287	12.256	12.232	12.221	12.213	12.205	12.199	12.194
11.500	12.284	12.254	12.230	12.220	12.211	12.204	12.198	12.194
11.600	12.283	12.253	12.229	12.219	12.211	12.204	12.198	12.194
11.700	12.282	12.252	12.229	12.219	12.211	12.204	12.199	12.195
11.800	12.282	12.253	12.230	12.220	12.212	12.206	12.201	12.197
11.900	12.283	12.254	12.231	12.222	12.214	12.208	12.203	12.200
12.000	12.285	12.256	12.233	12.225	12.217	12.211	12.206	12.204
12.100	12.287	12.259	12.237	12.228	12.221	12.215	12.211	12.208
12.200	12.291	12.263	12.241	12.232	12.225	12.220	12.216	12.213
12.300	12.295	12.267	12.246	12.237	12.230	12.225	12.221	12.218
12.400	12.300	12.272	12.251	12.243	12.236	12.231	12.228	12.225
12.500	12.306	12.278	12.258	12.249	12.243	12.238	12.235	12.232
12.600	12.312	12.285	12.265	12.257	12.250	12.245	12.242	12.240
12.700	12.319	12.293	12.272	12.265	12.258	12.254	12.250	12.249
12.800	12.327	12.301	12.281	12.273	12.267	12.262	12.259	12.258
12.900	12.336	12.310	12.290	12.283	12.277	12.272	12.269	12.267

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
13.000	12.346	12.319	12.300	12.292	12.287	12.282	12.279	12.278
13.100	12.356	12.330	12.310	12.303	12.297	12.293	12.290	12.289
13.200	12.366	12.340	12.321	12.314	12.308	12.304	12.302	12.300
13.300	12.378	12.352	12.333	12.326	12.320	12.316	12.314	12.313
13.400	12.390	12.364	12.345	12.338	12.333	12.329	12.326	12.326
13.500	12.402	12.377	12.358	12.351	12.346	12.342	12.340	12.339
13.600	12.415	12.390	12.372	12.365	12.359	12.356	12.354	12.353
13.700	12.429	12.404	12.386	12.379	12.374	12.370	12.368	12.368
13.800	12.444	12.419	12.401	12.394	12.389	12.385	12.383	12.383
13.900	12.459	12.434	12.416	12.409	12.404	12.400	12.398	12.398
14.000	12.474	12.450	12.432	12.425	12.420	12.416	12.415	12.414
14.100	12.490	12.466	12.448	12.441	12.436	12.433	12.431	12.431
14.200	12.507	12.483	12.465	12.458	12.453	12.450	12.448	12.448
14.300	12.524	12.500	12.482	12.476	12.471	12.467	12.466	12.466
14.400	12.541	12.517	12.500	12.494	12.489	12.486	12.484	12.484
14.500	12.559	12.536	12.518	12.512	12.507	12.504	12.502	12.502
14.600	12.578	12.554	12.537	12.531	12.526	12.523	12.521	12.521

---------- long. metac. height m ----------

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
2.000	904.589	915.068	904.447	895.274	884.411	871.667	858.170	842.888
2.100	871.371	876.433	865.328	856.740	846.488	835.011	822.473	808.691
2.200	839.357	840.910	829.684	821.571	811.744	801.458	789.873	777.272
2.300	810.027	808.126	797.087	789.356	780.367	770.716	759.985	748.273
2.400	781.866	778.120	767.098	759.790	751.441	742.351	732.386	721.566
2.500	755.395	750.385	739.557	732.353	724.810	716.229	706.905	696.798
2.600	730.695	724.743	714.051	707.232	700.084	692.054	683.192	673.775
2.700	707.537	701.043	690.136	684.043	677.128	669.548	661.225	652.332
2.800	685.930	678.937	668.314	662.415	655.734	648.521	640.738	632.328
2.900	665.572	658.252	647.962	642.213	635.847	628.910	621.534	613.632
3.000	646.594	639.037	629.029	623.338	617.215	610.528	603.550	596.036
3.100	628.782	621.041	611.236	605.690	599.687	593.386	586.645	579.504
3.200	612.229	604.118	594.508	589.140	583.288	577.227	570.718	563.956
3.300	596.519	588.092	578.792	573.495	567.932	561.989	555.761	549.338
3.400	581.850	573.297	563.998	558.801	553.430	547.645	541.660	535.458
3.500	568.032	559.315	549.999	544.964	539.660	534.124	528.358	522.344
3.600	554.895	546.197	536.853	531.914	526.713	521.328	515.728	509.969
3.700	542.739	533.715	524.411	519.483	514.425	509.188	503.797	498.228
3.800	531.110	521.996	512.635	507.792	502.794	497.685	492.444	487.080
3.900	520.504	510.833	501.432	496.694	491.737	486.740	481.651	476.524
4.000	510.368	500.322	490.807	486.113	481.274	476.356	471.444	466.443
4.100	500.937	490.463	480.814	476.058	471.326	466.474	461.674	456.829
4.200	491.906	481.167	471.354	466.488	461.822	457.132	452.402	447.682
4.300	483.212	472.401	462.269	457.467	452.788	448.213	443.558	438.875
4.400	474.902	464.083	453.702	448.909	444.195	439.684	435.150	430.526
4.500	467.070	456.228	445.567	440.723	436.066	431.554	427.070	422.560
4.600	459.686	448.799	437.931	433.002	428.342	423.785	419.339	414.969
4.700	452.728	441.619	430.715	425.643	420.960	416.390	411.984	407.675
4.800	446.161	434.686	423.892	418.715	413.956	409.355	404.940	400.683
4.900	439.820	428.162	417.394	412.194	407.239	402.640	398.229	394.028
5.000	433.681	421.920	411.219	406.000	400.897	396.271	391.847	387.616
5.100	427.659	416.041	405.319	400.061	394.938	390.162	385.760	381.482
5.200	421.851	410.494	399.625	394.384	389.301	384.386	379.948	375.639
5.300	416.186	405.229	394.132	389.000	383.893	378.965	374.341	370.062
5.400	410.513	400.131	388.876	383.790	378.707	373.829	369.044	364.751
5.500	404.955	395.065	383.926	378.751	373.818	368.865	364.050	359.616
5.600	401.470	390.204	379.224	373.950	369.062	364.105	359.363	354.762
5.700	398.450	385.517	374.825	369.432	364.415	359.592	354.820	350.169
5.800	394.519	380.734	370.566	365.097	359.995	355.231	350.465	345.891
5.900	390.503	376.026	366.291	361.028	355.813	350.977	346.306	341.734
6.000	386.338	371.387	362.206	357.108	351.807	346.915	342.318	337.730
6.100	382.103	369.137	358.233	353.164	348.034	343.062	338.413	333.871
6.200	377.865	366.524	354.264	349.397	344.456	339.363	334.656	330.187
6.300	373.678	363.257	350.258	345.751	340.854	335.850	331.093	326.591
6.400	369.498	359.909	346.359	342.149	337.373	332.561	327.642	323.109
6.500	365.441	356.444	342.428	338.460	334.011	329.262	324.363	319.812
6.600	361.459	352.863	341.103	334.858	330.710	326.022	321.340	316.608
6.700	357.618	349.308	338.813	331.240	327.310	322.920	318.309	313.546
6.800	353.916	345.788	336.073	328.837	323.970	319.877	315.302	310.730
6.900	350.410	342.235	333.246	327.519	320.645	316.762	312.434	307.915
7.000	352.328	338.830	330.350	325.090	317.283	313.663	309.611	305.104
7.100	348.659	335.422	327.363	322.557	316.760	310.607	306.736	302.429
7.200	344.832	332.162	324.360	319.925	314.656	307.458	303.832	299.814
7.300	340.990	329.006	321.358	317.190	312.335	306.073	301.006	297.179
7.400	337.152	326.035	318.340	314.446	309.970	304.640	298.063	294.480

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
7.500	333.348	327.927	315.421	311.648	307.495	302.554	295.171	291.851
7.600	329.515	324.862	312.509	308.841	304.952	300.446	295.108	289.135
7.700	325.759	321.603	309.740	306.097	302.319	298.201	293.258	286.439
7.800	322.063	318.313	307.025	303.364	299.728	295.827	291.314	285.716
7.900	318.400	314.994	304.488	300.708	297.151	293.426	289.312	284.317
8.000	314.785	311.669	306.292	298.151	294.601	291.019	287.168	282.535
8.100	311.260	308.335	303.687	295.701	292.084	288.618	284.950	280.716
8.200	307.777	305.074	300.883	293.454	289.692	286.256	282.647	278.648
8.300	304.354	301.832	298.037	295.159	287.341	283.878	280.374	276.494
8.400	300.999	298.607	295.117	292.634	285.147	281.572	278.074	274.304
8.500	297.702	295.421	292.183	290.014	286.813	279.308	275.750	272.313
8.600	294.431	292.293	289.258	287.307	284.653	277.123	273.465	270.284
8.700	291.234	289.197	286.395	284.569	282.230	275.096	271.369	268.296
8.800	288.093	286.156	283.530	281.826	279.690	276.613	269.370	266.305
8.900	284.995	283.174	280.674	279.077	277.084	274.391	267.539	264.395
9.000	281.941	280.221	277.854	276.350	274.399	272.156	269.059	262.512
9.100	278.954	277.304	275.072	273.598	271.728	269.846	267.464	260.750
9.200	276.035	274.457	272.312	270.855	269.143	267.514	265.455	259.165
9.300	273.158	271.644	269.561	268.104	266.659	265.169	263.367	260.825
9.400	270.314	268.866	266.868	265.388	264.185	262.822	261.221	259.144
9.500	267.525	266.133	264.114	262.874	261.758	260.516	259.041	257.288
9.600	264.791	263.454	261.412	260.398	259.366	258.187	256.881	255.333
9.700	262.104	260.817	258.790	257.957	256.995	255.914	254.744	253.367
9.800	259.461	258.184	256.331	255.534	254.661	253.683	252.597	251.380
9.830	258.675	257.400	255.598	254.818	253.966	253.021	251.955	250.792
9.900	256.862	255.585	253.899	253.155	252.364	251.479	250.475	249.410
10.000	254.309	252.977	251.516	250.829	250.090	249.303	248.420	247.453
10.100	251.801	250.413	249.186	248.540	247.862	247.149	246.382	245.507
10.200	249.328	247.916	246.884	246.291	245.679	245.040	244.364	243.609
10.270	247.617	246.278	245.297	244.746	244.179	243.586	242.959	242.312
10.300	246.888	245.580	244.619	244.095	243.543	242.967	242.369	241.758
10.400	244.481	243.287	242.394	241.937	241.430	240.925	240.427	239.896
10.500	242.065	241.038	240.216	239.812	239.384	238.934	238.513	238.053
10.600	239.651	238.829	238.098	237.718	237.362	236.991	236.628	236.249
10.700	237.306	236.667	236.019	235.679	235.381	235.086	234.780	234.465
10.800	235.137	234.551	233.970	233.689	233.435	233.220	232.980	232.699
10.900	233.012	232.479	231.962	231.744	231.539	231.378	231.201	230.974
11.000	230.923	230.444	230.007	229.832	229.688	229.569	229.443	229.283
11.100	228.882	228.452	228.088	227.970	227.883	227.799	227.715	227.611
11.200	226.881	226.501	226.214	226.141	226.108	226.055	226.024	225.962
11.300	224.923	224.588	224.389	224.362	224.363	224.349	224.350	224.335
11.400	222.992	222.711	222.614	222.612	222.647	222.692	222.707	222.735
11.500	221.107	220.892	220.873	220.900	220.962	221.051	221.102	221.157
11.600	219.268	219.111	219.162	219.238	219.318	219.426	219.532	219.604
11.700	217.465	217.369	217.481	217.601	217.710	217.838	217.987	218.071
11.800	215.699	215.669	215.844	215.981	216.132	216.285	216.447	216.572
11.900	213.974	214.008	214.239	214.406	214.589	214.759	214.933	215.124
12.000	212.297	212.387	212.662	212.856	213.066	213.264	213.446	213.674
12.100	210.652	210.799	211.129	211.335	211.569	211.794	212.002	212.239
12.200	209.044	209.244	209.629	209.848	210.095	210.345	210.590	210.830
12.300	207.477	207.732	208.151	208.393	208.651	208.912	209.197	209.454
12.400	205.946	206.249	206.701	206.966	207.229	207.522	207.828	208.100
12.500	204.451	204.791	205.283	205.565	205.832	206.152	206.469	206.763
12.600	202.990	203.372	203.894	204.182	204.478	204.805	205.135	205.447
12.700	201.558	201.984	202.527	202.821	203.153	203.478	203.817	204.147
12.800	200.166	200.618	201.185	201.489	201.844	202.181	202.521	202.851
12.900	198.805	199.282	199.868	200.199	200.551	200.907	201.246	201.571

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
13.000	197.465	197.973	198.572	198.925	199.282	199.650	199.989	200.312
13.100	196.160	196.691	197.315	197.673	198.039	198.406	198.752	199.074
13.200	194.883	195.430	196.086	196.440	196.812	197.181	197.531	197.853
13.300	193.628	194.197	194.872	195.233	195.605	195.973	196.325	196.647
13.400	192.403	192.984	193.676	194.049	194.418	194.780	195.133	195.455
13.500	191.202	191.789	192.500	192.881	193.249	193.605	193.957	194.278
13.600	190.020	190.630	191.351	191.729	192.097	192.449	192.790	193.114
13.700	188.857	189.497	190.221	190.594	190.960	191.312	191.641	191.966
13.800	187.720	188.382	189.109	189.476	189.838	190.186	190.510	190.832
13.900	186.601	187.280	188.011	188.374	188.732	189.074	189.397	189.712
14.000	185.500	186.198	186.931	187.290	187.642	187.977	188.301	188.610
14.100	184.432	185.137	185.865	186.220	186.563	186.893	187.218	187.543
14.200	183.387	184.093	184.816	185.166	185.500	185.827	186.146	186.494
14.300	182.355	183.066	183.782	184.123	184.451	184.775	185.099	185.462
14.400	181.339	182.053	182.763	183.095	183.418	183.737	184.077	184.444
14.500	180.341	181.056	181.756	182.081	182.400	182.710	183.072	183.441
14.600	179.366	180.072	180.756	181.080	181.395	181.721	182.084	182.453

---------- long. centre of flotatio m ----------

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
2.000	89.45	90.94	92.02	92.51	92.98	93.43	93.89	94.33
2.100	89.63	90.97	92.00	92.47	92.92	93.36	93.79	94.22
2.200	89.75	90.99	91.98	92.43	92.86	93.28	93.70	94.11
2.300	89.86	91.00	91.95	92.38	92.80	93.21	93.61	94.01
2.400	89.93	91.01	91.92	92.33	92.74	93.14	93.53	93.91
2.500	89.98	91.00	91.88	92.28	92.68	93.06	93.44	93.81
2.600	90.01	90.99	91.84	92.23	92.62	92.99	93.35	93.71
2.700	90.02	90.97	91.79	92.18	92.56	92.92	93.27	93.61
2.800	90.03	90.95	91.75	92.13	92.49	92.84	93.18	93.52
2.900	90.03	90.92	91.70	92.07	92.43	92.77	93.10	93.42
3.000	90.02	90.88	91.65	92.02	92.36	92.69	93.02	93.33
3.100	90.00	90.85	91.60	91.96	92.30	92.62	92.94	93.24
3.200	89.97	90.80	91.55	91.89	92.22	92.55	92.85	93.15
3.300	89.94	90.75	91.49	91.83	92.15	92.47	92.77	93.06
3.400	89.90	90.70	91.43	91.76	92.08	92.39	92.69	92.98
3.500	89.85	90.65	91.37	91.69	92.01	92.31	92.60	92.89
3.600	89.80	90.60	91.30	91.63	91.93	92.23	92.52	92.79
3.700	89.73	90.54	91.23	91.55	91.86	92.15	92.43	92.71
3.800	89.65	90.47	91.16	91.48	91.78	92.07	92.34	92.62
3.900	89.57	90.39	91.09	91.40	91.70	91.98	92.26	92.53
4.000	89.49	90.32	91.01	91.32	91.62	91.90	92.17	92.44
4.100	89.39	90.23	90.93	91.24	91.53	91.82	92.09	92.35
4.200	89.28	90.14	90.84	91.16	91.45	91.73	92.00	92.26
4.300	89.18	90.04	90.76	91.07	91.37	91.64	91.92	92.17
4.400	89.07	89.93	90.66	90.98	91.28	91.56	91.82	92.08
4.500	88.96	89.82	90.56	90.89	91.19	91.47	91.73	91.99
4.600	88.84	89.69	90.46	90.79	91.09	91.38	91.64	91.89
4.700	88.71	89.58	90.34	90.68	91.00	91.28	91.55	91.80
4.800	88.57	89.45	90.22	90.58	90.89	91.18	91.45	91.71
4.900	88.44	89.32	90.10	90.46	90.79	91.08	91.36	91.61
5.000	88.30	89.19	89.97	90.34	90.67	90.97	91.26	91.52
5.100	88.16	89.05	89.84	90.21	90.55	90.87	91.15	91.42
5.200	88.03	88.91	89.71	90.08	90.43	90.75	91.04	91.31
5.300	87.89	88.76	89.58	89.94	90.30	90.63	90.94	91.21
5.400	87.77	88.61	89.44	89.81	90.17	90.50	90.82	91.10
5.500	87.65	88.47	89.30	89.68	90.03	90.37	90.69	90.99
5.600	87.42	88.32	89.15	89.54	89.89	90.24	90.56	90.87
5.700	87.17	88.18	89.00	89.40	89.76	90.10	90.43	90.74
5.800	86.99	88.05	88.84	89.25	89.62	89.96	90.29	90.61
5.900	86.81	87.92	88.69	89.09	89.47	89.82	90.15	90.48
6.000	86.65	87.80	88.54	88.93	89.32	89.68	90.01	90.34
6.100	86.50	87.53	88.39	88.78	89.16	89.53	89.87	90.20
6.200	86.36	87.29	88.24	88.62	89.00	89.38	89.73	90.06
6.300	86.23	87.10	88.11	88.47	88.85	89.23	89.58	89.91
6.400	86.10	86.93	87.98	88.32	88.70	89.07	89.44	89.77
6.500	85.97	86.76	87.86	88.19	88.54	88.91	89.28	89.62
6.600	85.85	86.61	87.56	88.05	88.39	88.76	89.12	89.47
6.700	85.72	86.47	87.33	87.93	88.25	88.60	88.96	89.32
6.800	85.60	86.33	87.15	87.72	88.11	88.44	88.80	89.15
6.900	85.46	86.20	86.97	87.43	87.98	88.30	88.64	88.99
7.000	84.95	86.06	86.80	87.24	87.86	88.16	88.49	88.84
7.100	84.84	85.94	86.65	87.05	87.53	88.03	88.34	88.67
7.200	84.74	85.81	86.50	86.88	87.32	87.90	88.20	88.51
7.300	84.65	85.68	86.36	86.73	87.14	87.65	88.05	88.36
7.400	84.57	85.53	86.23	86.57	86.96	87.39	87.93	88.22

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
7.500	84.49	85.04	86.10	86.43	86.79	87.20	87.81	88.08
7.600	84.43	84.91	85.97	86.30	86.63	87.01	87.45	87.95
7.700	84.36	84.81	85.84	86.16	86.49	86.84	87.25	87.82
7.800	84.30	84.72	85.70	86.03	86.35	86.68	87.06	87.53
7.900	84.24	84.63	85.56	85.89	86.21	86.53	86.88	87.30
8.000	84.19	84.55	85.07	85.76	86.07	86.39	86.72	87.10
8.100	84.13	84.48	84.94	85.62	85.94	86.25	86.56	86.91
8.200	84.08	84.41	84.83	85.47	85.80	86.11	86.42	86.73
8.300	84.04	84.34	84.73	84.99	85.67	85.98	86.27	86.57
8.400	83.99	84.28	84.64	84.87	85.52	85.84	86.13	86.41
8.500	83.95	84.22	84.55	84.76	85.05	85.70	85.98	86.27
8.600	83.90	84.16	84.48	84.67	84.91	85.56	85.83	86.13
8.700	83.86	84.11	84.40	84.58	84.79	85.40	85.70	86.00
8.800	83.82	84.06	84.33	84.50	84.68	84.92	85.57	85.87
8.900	83.78	84.00	84.27	84.42	84.59	84.78	85.43	85.74
9.000	83.75	83.96	84.20	84.34	84.49	84.67	84.99	85.62
9.100	83.71	83.91	84.14	84.27	84.40	84.58	84.83	85.48
9.200	83.67	83.86	84.08	84.20	84.32	84.50	84.72	85.34
9.300	83.64	83.82	84.02	84.12	84.25	84.42	84.62	84.90
9.400	83.61	83.78	83.96	84.05	84.19	84.34	84.53	84.77
9.500	83.58	83.74	83.90	83.99	84.13	84.27	84.45	84.67
9.600	83.55	83.69	83.84	83.94	84.07	84.21	84.38	84.58
9.700	83.52	83.65	83.79	83.89	84.01	84.15	84.31	84.50
9.800	83.49	83.61	83.74	83.85	83.96	84.10	84.25	84.44
9.830	83.48	83.60	83.73	83.83	83.95	84.08	84.24	84.42
9.900	83.46	83.57	83.70	83.80	83.91	84.04	84.20	84.37
10.000	83.43	83.53	83.66	83.76	83.87	83.99	84.14	84.31
10.100	83.40	83.48	83.63	83.72	83.83	83.95	84.09	84.26
10.200	83.37	83.44	83.59	83.68	83.79	83.91	84.05	84.21
10.270	83.35	83.42	83.57	83.66	83.76	83.88	84.02	84.18
10.300	83.34	83.41	83.56	83.65	83.75	83.87	84.01	84.17
10.400	83.31	83.39	83.53	83.62	83.72	83.84	83.97	84.13
10.500	83.28	83.36	83.50	83.59	83.69	83.81	83.94	84.10
10.600	83.24	83.34	83.48	83.56	83.66	83.78	83.92	84.07
10.700	83.20	83.31	83.45	83.54	83.64	83.75	83.89	84.04
10.800	83.18	83.29	83.43	83.52	83.62	83.73	83.87	84.01
10.900	83.16	83.27	83.41	83.50	83.60	83.71	83.85	83.99
11.000	83.15	83.25	83.39	83.48	83.58	83.70	83.83	83.97
11.100	83.13	83.23	83.38	83.46	83.57	83.68	83.81	83.95
11.200	83.11	83.22	83.36	83.45	83.56	83.67	83.80	83.94
11.300	83.10	83.20	83.35	83.44	83.55	83.66	83.79	83.93
11.400	83.08	83.19	83.34	83.44	83.54	83.65	83.78	83.92
11.500	83.07	83.18	83.33	83.43	83.53	83.65	83.77	83.91
11.600	83.06	83.17	83.33	83.42	83.53	83.64	83.77	83.90
11.700	83.05	83.17	83.33	83.42	83.53	83.64	83.76	83.90
11.800	83.04	83.16	83.32	83.42	83.53	83.64	83.76	83.89
11.900	83.03	83.16	83.33	83.42	83.53	83.64	83.76	83.89
12.000	83.02	83.16	83.33	83.42	83.53	83.64	83.76	83.89
12.100	83.02	83.16	83.33	83.43	83.53	83.65	83.77	83.90
12.200	83.01	83.16	83.33	83.43	83.54	83.65	83.77	83.90
12.300	83.01	83.16	83.34	83.44	83.54	83.65	83.78	83.91
12.400	83.01	83.16	83.34	83.45	83.55	83.66	83.78	83.91
12.500	83.01	83.17	83.35	83.45	83.56	83.67	83.79	83.92
12.600	83.02	83.18	83.36	83.46	83.57	83.68	83.80	83.93
12.700	83.02	83.18	83.37	83.47	83.58	83.70	83.81	83.94
12.800	83.02	83.19	83.38	83.48	83.59	83.71	83.83	83.95
12.900	83.03	83.20	83.39	83.50	83.61	83.72	83.84	83.96

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
13.000	83.04	83.21	83.40	83.51	83.62	83.73	83.85	83.97
13.100	83.05	83.22	83.42	83.52	83.63	83.75	83.86	83.98
13.200	83.05	83.23	83.43	83.54	83.65	83.76	83.87	83.99
13.300	83.06	83.25	83.45	83.55	83.66	83.77	83.89	84.00
13.400	83.08	83.26	83.46	83.57	83.68	83.79	83.90	84.01
13.500	83.09	83.27	83.48	83.59	83.70	83.80	83.91	84.02
13.600	83.10	83.29	83.50	83.60	83.71	83.82	83.93	84.03
13.700	83.11	83.31	83.51	83.62	83.73	83.83	83.94	84.05
13.800	83.13	83.32	83.53	83.64	83.74	83.85	83.95	84.06
13.900	83.14	83.34	83.55	83.65	83.76	83.86	83.97	84.07
14.000	83.16	83.35	83.56	83.67	83.77	83.88	83.98	84.09
14.100	83.17	83.37	83.58	83.69	83.79	83.89	84.00	84.10
14.200	83.19	83.39	83.60	83.70	83.81	83.91	84.01	84.12
14.300	83.21	83.41	83.62	83.72	83.82	83.92	84.03	84.13
14.400	83.22	83.43	83.63	83.74	83.84	83.94	84.04	84.15
14.500	83.24	83.45	83.65	83.75	83.85	83.95	84.06	84.17
14.600	83.26	83.47	83.67	83.77	83.87	83.97	84.08	84.19

---------- immersion/cm t/cm ----------

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
2.000	39.1	39.5	39.5	39.5	39.5	39.5	39.4	39.3
2.100	39.4	39.7	39.7	39.7	39.7	39.6	39.6	39.5
2.200	39.7	39.8	39.9	39.9	39.8	39.8	39.7	39.6
2.300	39.9	40.0	40.0	40.0	40.0	39.9	39.9	39.8
2.400	40.1	40.2	40.2	40.2	40.1	40.1	40.0	39.9
2.500	40.3	40.4	40.3	40.3	40.3	40.2	40.2	40.1
2.600	40.5	40.5	40.5	40.4	40.4	40.4	40.3	40.2
2.700	40.6	40.7	40.6	40.6	40.5	40.5	40.4	40.3
2.800	40.8	40.8	40.7	40.7	40.7	40.6	40.5	40.5
2.900	40.9	40.9	40.9	40.8	40.8	40.7	40.7	40.6
3.000	41.1	41.1	41.0	41.0	40.9	40.8	40.8	40.7
3.100	41.2	41.2	41.1	41.1	41.0	40.9	40.9	40.8
3.200	41.4	41.3	41.2	41.2	41.1	41.1	41.0	40.9
3.300	41.5	41.4	41.4	41.3	41.2	41.2	41.1	41.0
3.400	41.7	41.6	41.5	41.4	41.3	41.3	41.2	41.1
3.500	41.8	41.7	41.6	41.5	41.4	41.4	41.3	41.2
3.600	41.9	41.8	41.7	41.6	41.5	41.5	41.4	41.3
3.700	42.1	41.9	41.8	41.7	41.6	41.6	41.5	41.4
3.800	42.2	42.0	41.9	41.8	41.7	41.7	41.6	41.5
3.900	42.3	42.2	42.0	41.9	41.8	41.8	41.7	41.6
4.000	42.5	42.3	42.1	42.0	41.9	41.9	41.8	41.7
4.100	42.6	42.4	42.2	42.1	42.0	41.9	41.9	41.8
4.200	42.7	42.5	42.3	42.2	42.1	42.0	41.9	41.9
4.300	42.9	42.6	42.4	42.3	42.2	42.1	42.0	41.9
4.400	43.0	42.7	42.5	42.4	42.3	42.2	42.1	42.0
4.500	43.1	42.9	42.6	42.5	42.4	42.3	42.2	42.1
4.600	43.3	43.0	42.7	42.6	42.5	42.4	42.3	42.2
4.700	43.4	43.1	42.8	42.7	42.6	42.5	42.4	42.3
4.800	43.5	43.2	42.9	42.8	42.7	42.6	42.5	42.3
4.900	43.7	43.3	43.0	42.9	42.8	42.6	42.5	42.4
5.000	43.8	43.5	43.2	43.0	42.9	42.7	42.6	42.5
5.100	43.9	43.6	43.3	43.1	43.0	42.8	42.7	42.6
5.200	44.1	43.7	43.4	43.2	43.1	42.9	42.8	42.7
5.300	44.2	43.8	43.5	43.3	43.2	43.0	42.9	42.8
5.400	44.3	43.9	43.6	43.4	43.3	43.1	43.0	42.8
5.500	44.4	44.1	43.7	43.5	43.4	43.2	43.1	42.9
5.600	44.6	44.2	43.8	43.6	43.5	43.3	43.2	43.0
5.700	44.7	44.3	43.9	43.7	43.6	43.4	43.3	43.1
5.800	44.9	44.4	44.1	43.9	43.7	43.5	43.4	43.2
5.900	45.0	44.5	44.2	44.0	43.8	43.6	43.4	43.3
6.000	45.2	44.6	44.3	44.1	43.9	43.7	43.5	43.4
6.100	45.3	44.8	44.4	44.2	44.0	43.8	43.6	43.5
6.200	45.4	44.9	44.5	44.3	44.1	43.9	43.7	43.6
6.300	45.5	45.1	44.6	44.4	44.2	44.0	43.8	43.7
6.400	45.6	45.2	44.7	44.5	44.3	44.1	43.9	43.8
6.500	45.7	45.3	44.8	44.6	44.4	44.2	44.0	43.9
6.600	45.8	45.4	45.0	44.7	44.5	44.3	44.1	44.0
6.700	45.9	45.5	45.1	44.8	44.6	44.4	44.3	44.1
6.800	46.0	45.6	45.2	44.9	44.7	44.5	44.4	44.2
6.900	46.1	45.7	45.3	45.1	44.8	44.6	44.5	44.3
7.000	46.4	45.8	45.5	45.2	44.9	44.7	44.6	44.4
7.100	46.5	45.9	45.6	45.3	45.1	44.8	44.6	44.5
7.200	46.6	46.0	45.7	45.5	45.2	44.9	44.7	44.6
7.300	46.6	46.1	45.8	45.6	45.3	45.1	44.8	44.6
7.400	46.7	46.2	45.8	45.7	45.4	45.2	44.9	44.7

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
7.500	46.8	46.5	45.9	45.8	45.6	45.3	45.0	44.8
7.600	46.8	46.6	46.0	45.8	45.6	45.4	45.2	44.9
7.700	46.9	46.6	46.1	45.9	45.7	45.5	45.3	45.0
7.800	46.9	46.7	46.2	46.0	45.8	45.6	45.4	45.1
7.900	47.0	46.8	46.3	46.1	45.9	45.7	45.5	45.3
8.000	47.0	46.8	46.6	46.2	46.0	45.8	45.6	45.4
8.100	47.1	46.9	46.6	46.3	46.1	45.9	45.7	45.5
8.200	47.1	46.9	46.7	46.4	46.2	46.0	45.8	45.6
8.300	47.1	47.0	46.8	46.6	46.2	46.1	45.9	45.7
8.400	47.2	47.0	46.8	46.7	46.3	46.1	45.9	45.7
8.500	47.2	47.1	46.9	46.8	46.6	46.2	46.0	45.8
8.600	47.3	47.1	46.9	46.8	46.7	46.3	46.1	45.9
8.700	47.3	47.2	47.0	46.9	46.7	46.4	46.2	46.0
8.800	47.3	47.2	47.0	46.9	46.8	46.6	46.3	46.1
8.900	47.4	47.2	47.1	47.0	46.8	46.7	46.3	46.2
9.000	47.4	47.3	47.1	47.0	46.9	46.8	46.6	46.2
9.100	47.4	47.3	47.1	47.0	46.9	46.8	46.7	46.3
9.200	47.4	47.3	47.2	47.1	47.0	46.9	46.7	46.4
9.300	47.5	47.4	47.2	47.1	47.0	46.9	46.8	46.7
9.400	47.5	47.4	47.2	47.1	47.1	47.0	46.9	46.7
9.500	47.5	47.4	47.3	47.2	47.1	47.0	46.9	46.8
9.600	47.5	47.4	47.3	47.2	47.1	47.1	47.0	46.9
9.700	47.6	47.5	47.3	47.2	47.2	47.1	47.0	46.9
9.800	47.6	47.5	47.3	47.3	47.2	47.1	47.1	47.0
9.830	47.6	47.5	47.4	47.3	47.2	47.2	47.1	47.0
9.900	47.6	47.5	47.4	47.3	47.2	47.2	47.1	47.0
10.000	47.6	47.5	47.4	47.3	47.3	47.2	47.1	47.1
10.100	47.6	47.5	47.4	47.4	47.3	47.3	47.2	47.1
10.200	47.7	47.5	47.5	47.4	47.3	47.3	47.2	47.2
10.270	47.7	47.6	47.5	47.4	47.4	47.3	47.3	47.2
10.300	47.7	47.6	47.5	47.4	47.4	47.3	47.3	47.2
10.400	47.7	47.6	47.5	47.5	47.4	47.4	47.3	47.3
10.500	47.7	47.6	47.5	47.5	47.4	47.4	47.4	47.3
10.600	47.7	47.6	47.5	47.5	47.5	47.4	47.4	47.4
10.700	47.7	47.6	47.6	47.5	47.5	47.5	47.4	47.4
10.800	47.7	47.7	47.6	47.6	47.5	47.5	47.5	47.4
10.900	47.7	47.7	47.6	47.6	47.6	47.5	47.5	47.5
11.000	47.8	47.7	47.6	47.6	47.6	47.6	47.5	47.5
11.100	47.8	47.7	47.7	47.6	47.6	47.6	47.6	47.6
11.200	47.8	47.7	47.7	47.7	47.7	47.6	47.6	47.6
11.300	47.8	47.8	47.7	47.7	47.7	47.7	47.7	47.6
11.400	47.8	47.8	47.7	47.7	47.7	47.7	47.7	47.7
11.500	47.8	47.8	47.8	47.7	47.7	47.7	47.7	47.7
11.600	47.8	47.8	47.8	47.8	47.8	47.8	47.8	47.7
11.700	47.9	47.8	47.8	47.8	47.8	47.8	47.8	47.8
11.800	47.9	47.8	47.8	47.8	47.8	47.8	47.8	47.8
11.900	47.9	47.9	47.9	47.9	47.9	47.9	47.9	47.9
12.000	47.9	47.9	47.9	47.9	47.9	47.9	47.9	47.9
12.100	47.9	47.9	47.9	47.9	47.9	47.9	47.9	47.9
12.200	47.9	47.9	47.9	47.9	47.9	47.9	48.0	48.0
12.300	48.0	48.0	48.0	48.0	48.0	48.0	48.0	48.0
12.400	48.0	48.0	48.0	48.0	48.0	48.0	48.0	48.0
12.500	48.0	48.0	48.0	48.0	48.0	48.0	48.1	48.1
12.600	48.0	48.0	48.0	48.0	48.1	48.1	48.1	48.1
12.700	48.0	48.0	48.1	48.1	48.1	48.1	48.1	48.1
12.800	48.1	48.1	48.1	48.1	48.1	48.1	48.1	48.2
12.900	48.1	48.1	48.1	48.1	48.1	48.2	48.2	48.2

trim draught	-4.000	-3.000	-2.000	-1.500	-1.000	-0.500	0.000	0.500
13.000	48.1	48.1	48.1	48.2	48.2	48.2	48.2	48.2
13.100	48.1	48.1	48.2	48.2	48.2	48.2	48.2	48.3
13.200	48.1	48.2	48.2	48.2	48.2	48.3	48.3	48.3
13.300	48.2	48.2	48.2	48.2	48.3	48.3	48.3	48.3
13.400	48.2	48.2	48.2	48.3	48.3	48.3	48.3	48.3
13.500	48.2	48.2	48.3	48.3	48.3	48.3	48.4	48.4
13.600	48.2	48.3	48.3	48.3	48.3	48.4	48.4	48.4
13.700	48.3	48.3	48.3	48.4	48.4	48.4	48.4	48.4
13.800	48.3	48.3	48.4	48.4	48.4	48.4	48.4	48.5
13.900	48.3	48.3	48.4	48.4	48.4	48.4	48.5	48.5
14.000	48.3	48.4	48.4	48.4	48.5	48.5	48.5	48.5
14.100	48.4	48.4	48.4	48.5	48.5	48.5	48.5	48.5
14.200	48.4	48.4	48.5	48.5	48.5	48.5	48.5	48.6
14.300	48.4	48.4	48.5	48.5	48.5	48.6	48.6	48.6
14.400	48.4	48.5	48.5	48.5	48.6	48.6	48.6	48.6
14.500	48.5	48.5	48.5	48.6	48.6	48.6	48.6	48.7
14.600	48.5	48.5	48.6	48.6	48.6	48.6	48.7	48.7

3.3. Cross Curve Data

NOTE:

The stability lever is calculated assuming that the centre of gravity is on base line.

the calculation is based on stability data (including poop, forecastle and hatch coaming).

Expiation of symbols:

Trim	(+) trim by head, (-) trim by stern	m
Heel	Heeling angles	deg.
Draught (T)	Draught moulded	m
KN	Stability righting lever	m

trim 0.000 m

	KN								
draught	5.0	10.0	20.0	30.0	40.0	50.0	60.0	70.0	75.0
2.000	2.851	5.419	8.133	9.422	10.006	10.166	10.104	9.779	9.438
2.050	2.793	5.333	8.073	9.385	9.993	10.176	10.135	9.801	9.454
2.100	2.737	5.248	8.013	9.348	9.980	10.185	10.164	9.821	9.468
2.150	2.684	5.166	7.955	9.311	9.967	10.195	10.192	9.839	9.481
2.200	2.633	5.084	7.897	9.275	9.955	10.204	10.219	9.856	9.493
2.250	2.584	5.004	7.841	9.239	9.942	10.212	10.244	9.871	9.505
2.300	2.537	4.926	7.785	9.203	9.929	10.221	10.268	9.885	9.515
2.350	2.492	4.849	7.731	9.168	9.916	10.230	10.289	9.898	9.524
2.400	2.449	4.775	7.677	9.134	9.904	10.239	10.309	9.910	9.532
2.450	2.407	4.702	7.624	9.100	9.891	10.247	10.328	9.921	9.540
2.500	2.368	4.632	7.572	9.066	9.879	10.255	10.345	9.930	9.547
2.550	2.329	4.564	7.521	9.033	9.866	10.264	10.360	9.939	9.553
2.600	2.292	4.499	7.470	9.000	9.854	10.272	10.374	9.946	9.559
2.650	2.256	4.435	7.420	8.968	9.842	10.280	10.386	9.952	9.565
2.700	2.222	4.373	7.371	8.936	9.830	10.288	10.397	9.957	9.570
2.750	2.189	4.313	7.323	8.905	9.818	10.295	10.408	9.962	9.574
2.800	2.157	4.255	7.275	8.874	9.806	10.303	10.416	9.966	9.578
2.850	2.126	4.199	7.228	8.843	9.794	10.311	10.424	9.971	9.582
2.900	2.096	4.145	7.181	8.813	9.782	10.318	10.431	9.975	9.584
2.950	2.067	4.092	7.136	8.783	9.770	10.325	10.437	9.979	9.587
3.000	2.039	4.041	7.090	8.753	9.758	10.332	10.442	9.981	9.588
3.050	2.012	3.991	7.046	8.723	9.746	10.339	10.445	9.984	9.590
3.100	1.986	3.943	7.001	8.695	9.735	10.345	10.448	9.986	9.592
3.150	1.961	3.896	6.958	8.667	9.723	10.350	10.450	9.987	9.593
3.200	1.937	3.850	6.915	8.639	9.712	10.355	10.452	9.988	9.595
3.250	1.913	3.806	6.872	8.612	9.701	10.359	10.452	9.988	9.596
3.300	1.890	3.763	6.830	8.585	9.690	10.362	10.452	9.987	9.598
3.350	1.868	3.721	6.788	8.558	9.679	10.364	10.452	9.986	9.600
3.400	1.847	3.681	6.747	8.532	9.668	10.366	10.450	9.985	9.603
3.450	1.826	3.641	6.706	8.506	9.657	10.366	10.448	9.983	9.605
3.500	1.806	3.603	6.666	8.480	9.646	10.367	10.446	9.982	9.607
3.550	1.786	3.565	6.626	8.455	9.635	10.366	10.443	9.980	9.610
3.600	1.767	3.529	6.587	8.430	9.624	10.364	10.440	9.978	9.612
3.650	1.749	3.493	6.548	8.405	9.614	10.362	10.436	9.977	9.615
3.700	1.731	3.459	6.509	8.380	9.603	10.359	10.432	9.975	9.617
3.750	1.714	3.425	6.471	8.356	9.593	10.355	10.427	9.973	9.619
3.800	1.697	3.393	6.434	8.332	9.583	10.351	10.422	9.972	9.622
3.850	1.680	3.361	6.396	8.309	9.573	10.346	10.416	9.971	9.624
3.900	1.665	3.330	6.359	8.286	9.563	10.341	10.410	9.970	9.627
3.950	1.649	3.300	6.323	8.263	9.553	10.335	10.404	9.968	9.629
4.000	1.634	3.271	6.287	8.240	9.543	10.328	10.397	9.967	9.631
4.050	1.619	3.242	6.251	8.218	9.533	10.321	10.390	9.966	9.634
4.100	1.605	3.214	6.215	8.196	9.524	10.313	10.382	9.965	9.636
4.150	1.591	3.187	6.180	8.174	9.514	10.305	10.375	9.964	9.638
4.200	1.578	3.161	6.145	8.152	9.505	10.296	10.366	9.963	9.640
4.250	1.565	3.135	6.110	8.131	9.496	10.287	10.358	9.962	9.642
4.300	1.552	3.110	6.076	8.109	9.487	10.277	10.349	9.961	9.644
4.350	1.540	3.085	6.042	8.088	9.478	10.267	10.340	9.960	9.646
4.400	1.528	3.061	6.009	8.067	9.470	10.256	10.331	9.959	9.648
4.450	1.516	3.038	5.975	8.046	9.461	10.245	10.322	9.958	9.650
4.500	1.504	3.016	5.943	8.026	9.452	10.234	10.312	9.956	9.652
4.550	1.493	2.993	5.910	8.006	9.443	10.222	10.303	9.955	9.654
4.600	1.482	2.972	5.878	7.986	9.434	10.210	10.293	9.954	9.656
4.650	1.472	2.951	5.846	7.967	9.424	10.197	10.284	9.952	9.657
4.700	1.461	2.931	5.814	7.947	9.415	10.185	10.275	9.951	9.659
4.750	1.451	2.911	5.782	7.928	9.405	10.172	10.265	9.949	9.660
4.800	1.441	2.891	5.751	7.909	9.395	10.159	10.256	9.948	9.662
4.850	1.432	2.872	5.721	7.890	9.384	10.145	10.247	9.946	9.663
4.900	1.422	2.854	5.690	7.871	9.374	10.131	10.238	9.945	9.664
4.950	1.413	2.836	5.660	7.852	9.363	10.117	10.229	9.943	9.665
5.000	1.405	2.818	5.631	7.834	9.351	10.103	10.221	9.942	9.666
5.050	1.396	2.801	5.602	7.816	9.340	10.088	10.212	9.940	9.666
5.100	1.387	2.784	5.574	7.798	9.328	10.074	10.203	9.938	9.666

	KN								
draught	5.0	10.0	20.0	30.0	40.0	50.0	60.0	70.0	75.0
5.150	1.379	2.768	5.546	7.780	9.316	10.059	10.194	9.936	9.667
5.200	1.371	2.752	5.519	7.762	9.303	10.044	10.186	9.934	9.666
5.250	1.363	2.736	5.492	7.744	9.290	10.029	10.177	9.932	9.666
5.300	1.356	2.721	5.465	7.727	9.277	10.013	10.168	9.929	9.665
5.350	1.348	2.706	5.439	7.710	9.264	9.998	10.160	9.927	9.664
5.400	1.341	2.691	5.413	7.693	9.250	9.982	10.151	9.925	9.663
5.450	1.334	2.677	5.388	7.676	9.237	9.966	10.142	9.922	9.662
5.500	1.327	2.663	5.363	7.659	9.222	9.950	10.134	9.919	9.661
5.550	1.320	2.650	5.339	7.642	9.208	9.934	10.125	9.916	9.659
5.600	1.313	2.636	5.315	7.626	9.193	9.919	10.117	9.912	9.658
5.650	1.307	2.624	5.292	7.609	9.178	9.903	10.108	9.909	9.656
5.700	1.301	2.611	5.269	7.593	9.163	9.887	10.100	9.905	9.654
5.750	1.295	2.599	5.246	7.577	9.148	9.871	10.091	9.901	9.652
5.800	1.289	2.587	5.224	7.561	9.132	9.855	10.083	9.897	9.649
5.850	1.283	2.575	5.202	7.545	9.117	9.839	10.074	9.893	9.647
5.900	1.277	2.563	5.180	7.529	9.101	9.824	10.066	9.889	9.644
5.950	1.271	2.552	5.159	7.514	9.085	9.808	10.057	9.885	9.642
6.000	1.266	2.541	5.138	7.498	9.068	9.792	10.048	9.880	9.639
6.050	1.261	2.531	5.118	7.483	9.052	9.777	10.039	9.875	9.636
6.100	1.255	2.520	5.098	7.467	9.035	9.761	10.031	9.870	9.633
6.150	1.250	2.510	5.079	7.452	9.018	9.746	10.022	9.865	9.630
6.200	1.245	2.500	5.059	7.437	9.002	9.731	10.013	9.860	9.627
6.250	1.241	2.490	5.041	7.422	8.984	9.716	10.004	9.855	9.623
6.300	1.236	2.481	5.022	7.407	8.967	9.702	9.994	9.850	9.619
6.350	1.231	2.471	5.004	7.393	8.950	9.687	9.985	9.844	9.616
6.400	1.227	2.462	4.986	7.378	8.932	9.673	9.976	9.838	9.612
6.450	1.222	2.453	4.969	7.364	8.914	9.658	9.967	9.833	9.608
6.500	1.218	2.445	4.951	7.349	8.897	9.644	9.957	9.827	9.604
6.550	1.214	2.436	4.935	7.335	8.879	9.630	9.948	9.821	9.600
6.600	1.210	2.428	4.918	7.320	8.860	9.616	9.938	9.815	9.596
6.650	1.206	2.420	4.902	7.306	8.842	9.602	9.929	9.808	9.591
6.700	1.202	2.412	4.886	7.291	8.824	9.588	9.919	9.802	9.587
6.750	1.198	2.404	4.871	7.277	8.805	9.574	9.909	9.795	9.582
6.800	1.194	2.397	4.855	7.262	8.787	9.561	9.899	9.789	9.577
6.850	1.191	2.389	4.840	7.247	8.768	9.548	9.890	9.782	9.573
6.900	1.187	2.382	4.826	7.232	8.749	9.534	9.880	9.775	9.568
6.950	1.184	2.375	4.811	7.217	8.730	9.521	9.869	9.768	9.563
7.000	1.180	2.368	4.797	7.202	8.711	9.508	9.859	9.761	9.558
7.050	1.177	2.362	4.784	7.186	8.692	9.495	9.849	9.754	9.552
7.100	1.174	2.355	4.770	7.171	8.673	9.481	9.839	9.747	9.547
7.150	1.171	2.349	4.757	7.155	8.654	9.468	9.828	9.740	9.542
7.200	1.168	2.342	4.744	7.139	8.634	9.455	9.818	9.732	9.536
7.250	1.165	2.336	4.732	7.122	8.615	9.442	9.807	9.724	9.531
7.300	1.162	2.330	4.719	7.106	8.596	9.429	9.797	9.717	9.525
7.350	1.159	2.325	4.707	7.089	8.577	9.416	9.786	9.709	9.520
7.400	1.156	2.319	4.695	7.072	8.558	9.403	9.775	9.701	9.514
7.450	1.153	2.313	4.684	7.055	8.539	9.390	9.764	9.693	9.508
7.500	1.151	2.308	4.672	7.038	8.520	9.377	9.753	9.685	9.502
7.550	1.148	2.303	4.661	7.020	8.501	9.363	9.742	9.677	9.496
7.600	1.146	2.297	4.650	7.002	8.483	9.350	9.731	9.668	9.490
7.650	1.143	2.292	4.640	6.985	8.464	9.337	9.720	9.660	9.484
7.700	1.141	2.287	4.629	6.967	8.445	9.323	9.709	9.652	9.478
7.750	1.138	2.283	4.619	6.949	8.427	9.310	9.698	9.644	9.471
7.800	1.136	2.278	4.609	6.931	8.408	9.297	9.686	9.635	9.465
7.850	1.134	2.273	4.600	6.912	8.390	9.283	9.675	9.627	9.459
7.900	1.132	2.269	4.590	6.894	8.372	9.270	9.663	9.618	9.452
7.950	1.129	2.264	4.581	6.875	8.354	9.256	9.652	9.609	9.445
8.000	1.127	2.260	4.572	6.857	8.335	9.243	9.640	9.601	9.439
8.050	1.125	2.256	4.563	6.838	8.318	9.229	9.628	9.592	9.432
8.100	1.123	2.252	4.554	6.819	8.300	9.215	9.617	9.583	9.425
8.150	1.121	2.248	4.546	6.800	8.282	9.201	9.605	9.574	9.418
8.200	1.119	2.244	4.538	6.781	8.265	9.188	9.593	9.565	9.411
8.250	1.117	2.240	4.530	6.762	8.247	9.174	9.581	9.556	9.404
8.300	1.116	2.236	4.522	6.743	8.230	9.160	9.569	9.547	9.397
8.350	1.114	2.233	4.514	6.724	8.213	9.146	9.557	9.538	9.390
8.400	1.112	2.229	4.506	6.705	8.196	9.132	9.545	9.529	9.382

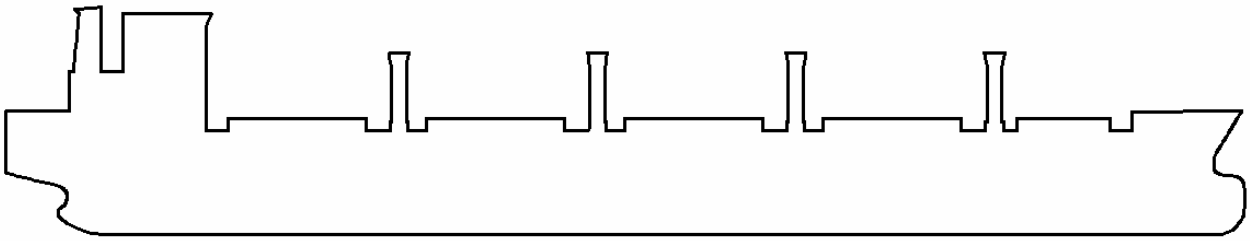
	KN								
draught	5.0	10.0	20.0	30.0	40.0	50.0	60.0	70.0	75.0
8.450	1.110	2.226	4.499	6.686	8.178	9.118	9.533	9.520	9.375
8.500	1.109	2.222	4.492	6.667	8.162	9.104	9.520	9.511	9.368
8.550	1.107	2.219	4.485	6.647	8.145	9.089	9.508	9.502	9.361
8.600	1.105	2.216	4.478	6.628	8.128	9.075	9.496	9.492	9.354
8.650	1.104	2.213	4.472	6.609	8.111	9.060	9.483	9.483	9.346
8.700	1.102	2.210	4.465	6.589	8.094	9.046	9.471	9.474	9.339
8.750	1.101	2.207	4.459	6.570	8.077	9.031	9.459	9.464	9.332
8.800	1.100	2.204	4.453	6.550	8.060	9.016	9.446	9.455	9.324
8.850	1.098	2.201	4.447	6.531	8.043	9.001	9.433	9.445	9.317
8.900	1.097	2.198	4.441	6.511	8.026	8.986	9.421	9.436	9.310
8.950	1.095	2.196	4.435	6.492	8.009	8.971	9.408	9.427	9.303
9.000	1.094	2.193	4.429	6.472	7.992	8.956	9.395	9.417	9.295
9.050	1.093	2.190	4.424	6.452	7.975	8.941	9.382	9.408	9.288
9.100	1.092	2.188	4.419	6.433	7.958	8.925	9.369	9.398	9.281
9.150	1.090	2.185	4.414	6.414	7.940	8.910	9.356	9.389	9.273
9.200	1.089	2.183	4.409	6.394	7.923	8.894	9.343	9.380	9.266
9.250	1.088	2.181	4.404	6.375	7.906	8.878	9.330	9.370	9.259
9.300	1.087	2.179	4.399	6.356	7.888	8.863	9.317	9.361	9.252
9.350	1.086	2.176	4.395	6.338	7.871	8.847	9.303	9.351	9.244
9.400	1.085	2.174	4.390	6.319	7.853	8.831	9.290	9.342	9.237
9.450	1.084	2.172	4.386	6.300	7.836	8.815	9.276	9.332	9.230
9.500	1.083	2.170	4.381	6.282	7.818	8.798	9.263	9.323	9.223
9.550	1.082	2.168	4.377	6.264	7.801	8.782	9.249	9.313	9.216
9.600	1.081	2.167	4.372	6.246	7.783	8.766	9.236	9.304	9.208
9.650	1.080	2.165	4.367	6.228	7.765	8.749	9.222	9.294	9.201
9.700	1.079	2.163	4.362	6.211	7.748	8.733	9.208	9.285	9.194
9.750	1.079	2.161	4.357	6.194	7.730	8.716	9.194	9.275	9.187
9.800	1.078	2.160	4.351	6.177	7.712	8.699	9.180	9.266	9.179
9.830	1.077	2.159	4.348	6.166	7.701	8.689	9.172	9.260	9.175
9.850	1.077	2.158	4.345	6.160	7.694	8.682	9.166	9.256	9.172
9.900	1.076	2.157	4.339	6.143	7.676	8.665	9.152	9.246	9.165
9.950	1.076	2.155	4.333	6.126	7.657	8.648	9.138	9.237	9.158
10.000	1.075	2.154	4.327	6.110	7.639	8.631	9.124	9.227	9.150
10.050	1.074	2.152	4.321	6.094	7.621	8.614	9.110	9.217	9.143
10.100	1.074	2.151	4.314	6.078	7.603	8.596	9.095	9.207	9.136
10.150	1.073	2.150	4.307	6.063	7.584	8.579	9.081	9.198	9.128
10.200	1.072	2.149	4.300	6.047	7.566	8.561	9.067	9.188	9.121
10.250	1.072	2.147	4.293	6.032	7.547	8.543	9.052	9.178	9.114
10.270	1.072	2.147	4.290	6.026	7.540	8.536	9.047	9.174	9.111
10.300	1.071	2.146	4.285	6.017	7.529	8.526	9.038	9.168	9.106
10.350	1.071	2.145	4.278	6.003	7.510	8.508	9.024	9.158	9.099
10.400	1.070	2.144	4.270	5.988	7.492	8.490	9.009	9.149	9.091
10.450	1.070	2.143	4.262	5.974	7.473	8.472	8.995	9.139	9.084
10.500	1.069	2.142	4.254	5.960	7.454	8.454	8.980	9.129	9.077
10.550	1.069	2.141	4.245	5.946	7.436	8.435	8.965	9.119	9.069
10.600	1.068	2.141	4.237	5.932	7.417	8.417	8.951	9.109	9.062
10.650	1.068	2.140	4.228	5.918	7.398	8.399	8.936	9.099	9.054
10.700	1.068	2.139	4.219	5.905	7.379	8.380	8.921	9.088	9.047
10.750	1.067	2.138	4.210	5.892	7.360	8.362	8.906	9.078	9.039
10.800	1.067	2.138	4.201	5.879	7.341	8.343	8.891	9.068	9.032
10.850	1.067	2.137	4.191	5.865	7.322	8.324	8.876	9.058	9.024
10.900	1.067	2.137	4.181	5.852	7.303	8.306	8.861	9.048	9.016
10.950	1.066	2.136	4.172	5.839	7.284	8.287	8.846	9.038	9.009
11.000	1.066	2.136	4.162	5.826	7.265	8.268	8.831	9.027	9.001
11.050	1.066	2.135	4.151	5.813	7.246	8.249	8.816	9.017	8.994
11.100	1.066	2.135	4.141	5.800	7.227	8.230	8.801	9.007	8.986
11.150	1.065	2.134	4.131	5.787	7.208	8.211	8.786	8.996	8.978
11.200	1.065	2.134	4.120	5.774	7.188	8.191	8.770	8.986	8.971
11.250	1.065	2.134	4.110	5.761	7.169	8.172	8.755	8.976	8.963
11.300	1.065	2.134	4.100	5.749	7.150	8.153	8.740	8.965	8.955
11.350	1.065	2.133	4.089	5.736	7.131	8.133	8.724	8.955	8.947
11.400	1.065	2.133	4.079	5.723	7.112	8.114	8.709	8.944	8.940
11.450	1.065	2.133	4.069	5.710	7.092	8.095	8.693	8.933	8.932
11.500	1.065	2.133	4.059	5.697	7.073	8.075	8.678	8.923	8.924
11.550	1.065	2.133	4.049	5.684	7.054	8.055	8.662	8.912	8.916
11.600	1.065	2.133	4.039	5.672	7.035	8.036	8.646	8.902	8.908

	KN								
draught	5.0	10.0	20.0	30.0	40.0	50.0	60.0	70.0	75.0
11.650	1.065	2.133	4.029	5.659	7.015	8.016	8.631	8.891	8.900
11.700	1.065	2.133	4.019	5.646	6.996	7.996	8.615	8.880	8.893
11.750	1.065	2.133	4.010	5.633	6.977	7.977	8.599	8.869	8.885
11.800	1.065	2.133	4.000	5.621	6.958	7.957	8.583	8.859	8.877
11.850	1.065	2.133	3.990	5.608	6.939	7.937	8.567	8.848	8.869
11.900	1.065	2.134	3.981	5.595	6.920	7.917	8.551	8.837	8.861
11.950	1.065	2.134	3.971	5.582	6.901	7.897	8.535	8.826	8.853
12.000	1.066	2.134	3.962	5.570	6.882	7.878	8.519	8.815	8.845

3.4. Wind Profile Curve

Explanation of symbols:

T	Draught moulded	m
Trim	(+) trim by head, (-) trim by stern	m
Coef	Weighting coefficient of profile curve (part)	
Area	Area of lateral profile above wl	m^2
CGZ	Vertical center of area of lateral profile above wl	m
Lever	Vertical center of area of total lateral profile	m
MOM	Moment of area above water line from T/2	m^3



TR=0, COEF=1.00

LCOND	AREA m2	ZCG m	MOM m3
T=2	2923.1	11.164	29710.8
T=2.5	2837.6	11.433	28894.4
T=3	2751.7	11.704	28077.7
T=3.5	2665.7	11.976	27260.8
T=4	2579.8	12.250	26444.0
T=4.5	2494.2	12.525	25627.8
T=5	2408.7	12.801	24812.0
T=5.5	2323.3	13.079	23996.2
T=6	2237.9	13.358	23180.5
T=6.5	2152.4	13.641	22364.6
T=7	2066.7	13.926	21548.0
T=7.5	1980.4	14.217	20729.2
T=8	1893.2	14.515	19907.2
T=8.5	1805.2	14.820	19081.9
T=9	1717.4	15.131	18257.2
T=9.5	1629.7	15.447	17433.0
T=9.83	1571.8	15.660	16889.4
T=10	1542.0	15.771	16609.4
T=10.27	1494.7	15.949	16164.7

3.5. Flooding Point and Flooding Angle Curve

Flooding point:

Open	**Text**	**Otype**	**Fr.**	**Y (m)**	**Z (m)**
OWIT1	Window above A deck	unprotected	#15	8.2	19.10
OWIT2	Window above A deck	unprotected	#15	-8.2	19.10
OWIT3	Window above A deck	unprotected	#30	9.6	19.10
OWIT4	Window above A deck	unprotected	#30	-9.6	19.10

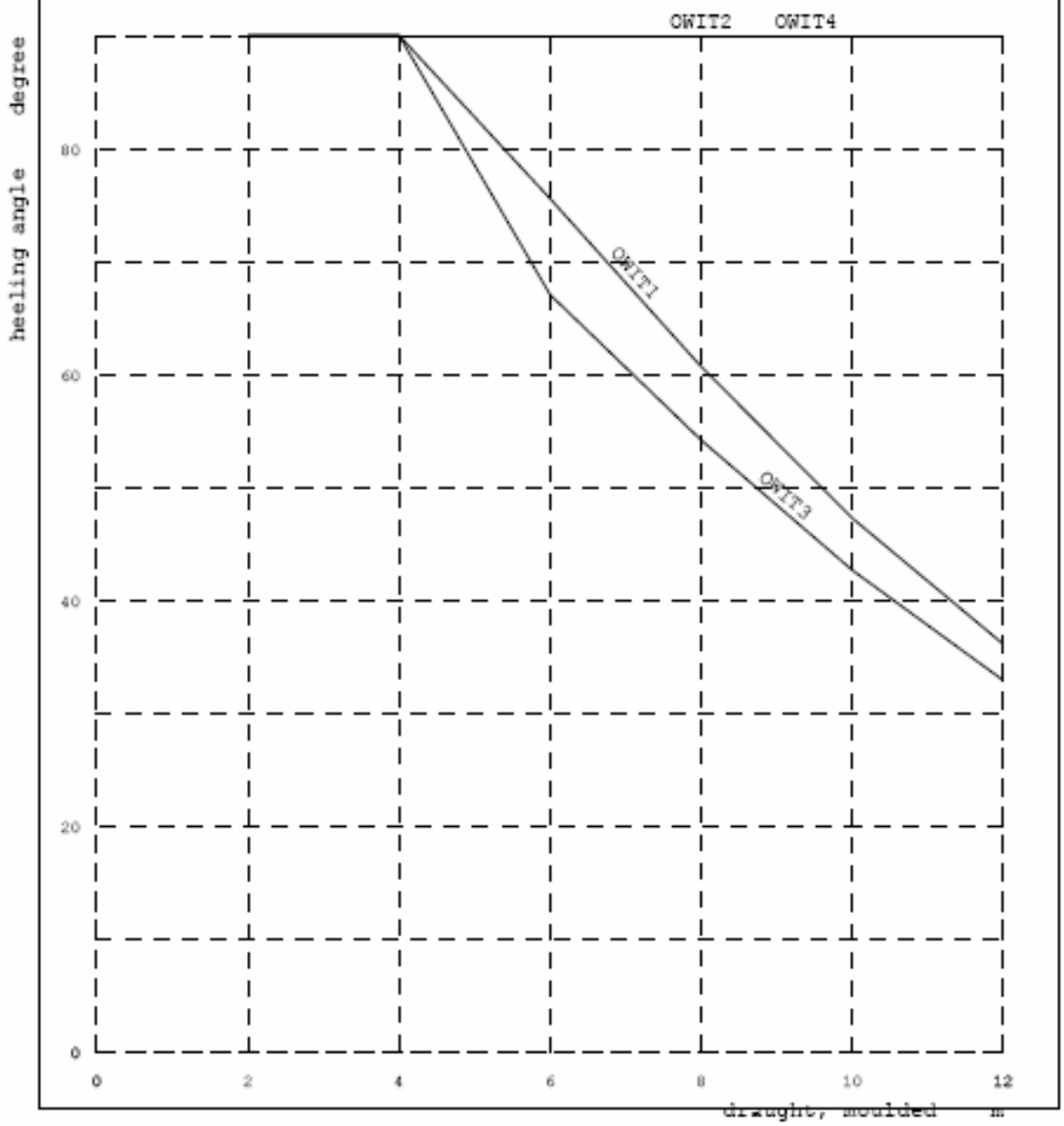
Explanation of symbols:

Open	Name of the opening	
Text	Description of the opening	
Otype	Type of the opening	
Fr.	Frame number	#
Y	y-coordinate of the opening	m
Z	z-coordinate of the opening	m

OPENING IMMERSION

BC_35K_WK_LD/A
DATE 2008-10-29
TIME 15:50
USER ADMI

STABHULL
RHO=1.025



3.6. Stability Limiting Curve

Please note: The following limiting curves calculated for zero trim: Whereas KG's of all loading cases presented in the booklet is calculated for the corresponding present trim, because after trim improving stability, KG's of loading cases can be higher than the same value read off the limiting curve.

Calculated with the following constant values:

Mean thickness of shell:	0.018 m
Thickness of keel plate:	0.020 m
Trim:	0.000 m
Depth (IMO wind criteria)	14.600 m
Density of sea water:	1.025 t/m^3

Stability criteria according to:

- IMO Resolution A.749(18)
- Minimum area under lever arm curve up to 30 deg. 0.055 mrad
- Minimum area under lever arm curve up to 40 deg. 0.090 mrad
- Minimum area under lever arm curve 30 to 40 deg. 0.030 mrad
- GZ greater than 0.20m at an angle of heel greater than 30 deg.
- Maximum righting arm at an angle greater than 25 deg.
- Initial GM greater than 0.15m

- IMO Resolution A.749(18) (Weather Criterion)
- - Area b / Area a is greater than 1.0
- - Area of bilge keels and bar keel 18.66 m2
- - Gust wind pressure 1.5*0.0514 t/m2 (acc. to reg.)
- -Angle of roll acc. to regulation
- - Heeling angle, steady wind less 16 deg.
- - Heeling angle, steady wind, residual freeboard greater 20%

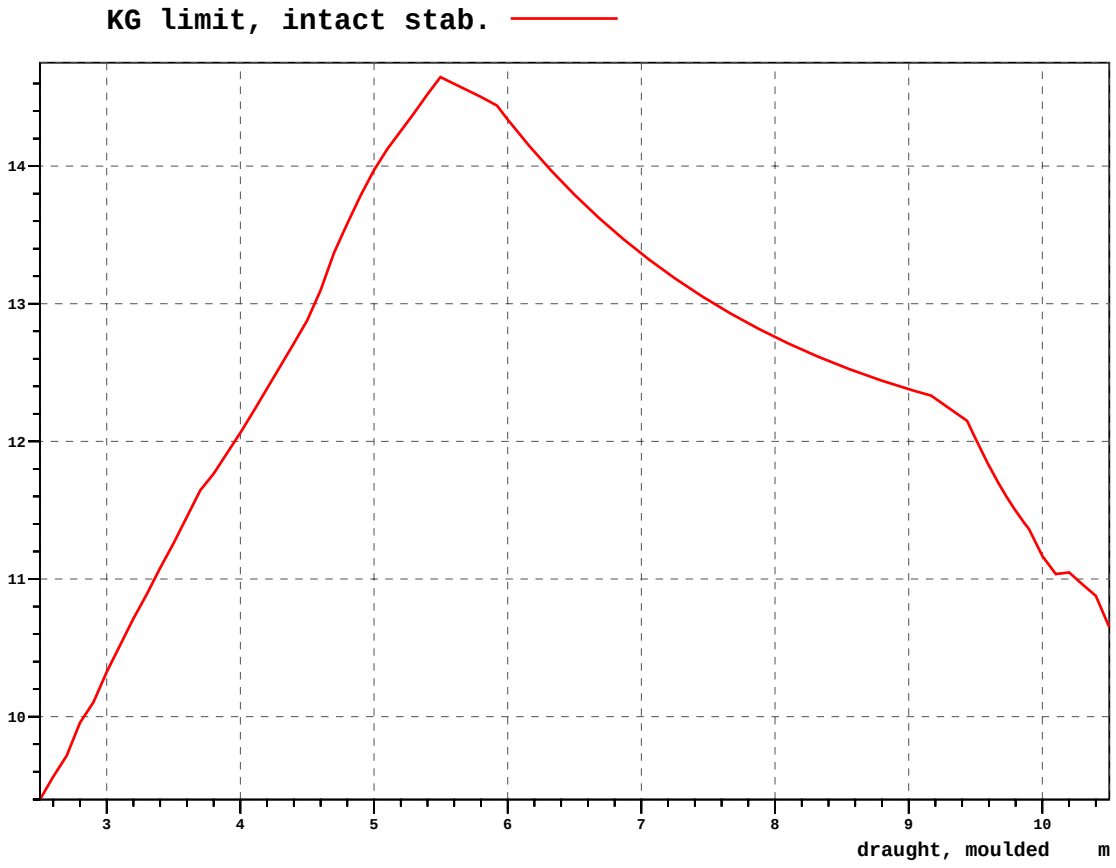
Explanation of symbols:

T	Draught	m
TR	(+) trim by head, (-) trim by stern	m
MINGM	Minimum GM in compliance with the criteria	m
MAXKG	Maximum KG in compliance with the criteria	m
DCRI	Name of the determining criterion	

LIMIT CURVE

T m	TR m	MINGM m	MAXKG m	DCRI
2.500	0.000	17.820	9.398	MAXGZ25
2.600	0.000	16.768	9.564	MAXGZ25
2.700	0.000	15.799	9.717	MAXGZ25
2.800	0.000	14.801	9.956	MAXGZ25
2.900	0.000	13.946	10.105	MAXGZ25
3.000	0.000	13.068	10.324	MAXGZ25
3.100	0.000	12.260	10.517	MAXGZ25
3.200	0.000	11.492	10.713	MAXGZ25
3.300	0.000	10.780	10.890	MAXGZ25
3.400	0.000	10.088	11.080	MAXGZ25
3.500	0.000	9.435	11.261	MAXGZ25
3.600	0.000	8.799	11.453	MAXGZ25
3.700	0.000	8.188	11.645	MAXGZ25
3.800	0.000	7.674	11.765	MAXGZ25
3.900	0.000	7.155	11.915	MAXGZ25
4.000	0.000	6.656	12.063	MAXGZ25
4.100	0.000	6.164	12.222	MAXGZ25
4.200	0.000	5.689	12.384	MAXGZ25
4.300	0.000	5.227	12.546	MAXGZ25
4.400	0.000	4.780	12.711	MAXGZ25
4.500	0.000	4.343	12.880	MAXGZ25
4.600	0.000	3.870	13.098	MAXGZ25
4.700	0.000	3.358	13.369	MAXGZ25
4.800	0.000	2.916	13.583	MAXGZ25
4.900	0.000	2.493	13.788	MAXGZ25
5.000	0.000	2.103	13.971	MAXGZ25
5.100	0.000	1.753	14.125	MAXGZ25
5.200	0.000	1.435	14.256	MAXGZ25
5.300	0.000	1.128	14.386	MAXGZ25
5.400	0.000	0.824	14.522	MAXGZ25
5.497	0.000	0.544	14.646	
5.500	0.000	0.541	14.644	AREA3040
5.600	0.000	0.434	14.598	AREA3040
5.700	0.000	0.335	14.550	AREA3040
5.800	0.000	0.246	14.502	AREA3040
5.900	0.000	0.165	14.450	AREA3040
5.920	0.000	0.150	14.440	
6.000	0.000	0.150	14.338	GM0.15
6.100	0.000	0.150	14.218	GM0.15
6.200	0.000	0.150	14.104	GM0.15
6.300	0.000	0.150	13.995	GM0.15
6.400	0.000	0.150	13.890	GM0.15
6.500	0.000	0.150	13.792	GM0.15
6.600	0.000	0.150	13.698	GM0.15
6.700	0.000	0.150	13.608	GM0.15
6.800	0.000	0.150	13.523	GM0.15
6.900	0.000	0.150	13.442	GM0.15
7.000	0.000	0.150	13.363	GM0.15
7.100	0.000	0.150	13.289	GM0.15
7.200	0.000	0.150	13.219	GM0.15
7.300	0.000	0.150	13.151	GM0.15
7.400	0.000	0.150	13.088	GM0.15
7.500	0.000	0.150	13.026	GM0.15
7.600	0.000	0.150	12.967	GM0.15
7.700	0.000	0.150	12.911	GM0.15
7.800	0.000	0.150	12.858	GM0.15
7.900	0.000	0.150	12.807	GM0.15
8.000	0.000	0.150	12.758	GM0.15
8.100	0.000	0.150	12.711	GM0.15
8.200	0.000	0.150	12.667	GM0.15
8.300	0.000	0.150	12.625	GM0.15
8.400	0.000	0.150	12.584	GM0.15
8.500	0.000	0.150	12.545	GM0.15
8.600	0.000	0.150	12.509	GM0.15
8.700	0.000	0.150	12.474	GM0.15

T m	TR m	MINGM m	MAXKG m	DCRI
8.800	0.000	0.150	12.441	GM0.15
8.900	0.000	0.150	12.409	GM0.15
9.000	0.000	0.150	12.379	GM0.15
9.100	0.000	0.150	12.351	GM0.15
9.167	0.000	0.150	12.333	
9.200	0.000	0.164	12.310	AREA3040
9.300	0.000	0.207	12.242	AREA3040
9.400	0.000	0.251	12.174	AREA3040
9.437	0.000	0.267	12.149	
9.500	0.000	0.382	12.020	MAXGZ25
9.600	0.000	0.556	11.826	MAXGZ25
9.700	0.000	0.715	11.646	MAXGZ25
9.800	0.000	0.846	11.497	MAXGZ25
9.900	0.000	0.964	11.362	MAXGZ25
10.000	0.000	1.142	11.168	MAXGZ25
10.100	0.000	1.258	11.036	MAXGZ25
10.200	0.000	1.233	11.047	MAXGZ25
10.300	0.000	1.308	10.960	MAXGZ25
10.400	0.000	1.379	10.878	MAXGZ25
10.500	0.000	1.596	10.650	MAXGZ25



Used intact stability criteria:

1. Criterion: AREA30, 'Area under GZ curve up to 30 deg.'
   Area under the GZ curve >0.055 mrad
   Range: 0, 30

2. Criterion: AREA40, 'Area under GZ curve up to 40 deg.'
   Area under the GZ curve >0.09 mrad
   Range: 0, MIN (40, FA)

3. Criterion: AREA3040, 'Area under GZ curve between 30 and 40 deg.'
   Area under the GZ curve >0.03 mrad
   Range: 30, MIN(40, FA)

4. Criterion: GZ0.2, 'Min. GZ > 0.2'
   Maximum GZ > 0.2m
   Range: 30, -

5. Criterion: MAXGZ25, 'Max. GZ at an angle > 25 deg.'
   Position of the maximum of the GZ curve > 25 deg.

6. Criterion: GM0.15, 'GM > 0.15m'
   Minimum GM > 0.15m

4. TANK CAPACITY DATA

Explanation of symbols:

FRMIN	frame number of aft end	#
FRMAX	frame number of fore end	#
ZMIN	height of lowest point	m
ZMAX	height of highest point	m
T	draught, moulded	m
H	height from tank bottom	m
FILL	filling degree	m^3
VOLM	volume moulded	m^3
VNET	net volume	m^3
WEIGHT	weight of load	t
CGX	x-coordinate of center of gravity	m
CGY	y-coordinate of center of gravity	m
CGZ	z-coordinate of center of gravity	m
IYMAX	max. IY of surface	m^4
IMAX	max. mom. of inertia of surface	m^4
FSM	free surf. mom/Max	tm

4.1 Tanks Location Table

NAME	DES	FRMIN #	FRMAX #	ZMIN m	ZMAX m
CAPACITY OF Grain Load (RHO=1)					
CH1	NO.1 CARGO HOLD	177	208	1.75	16.85
CH2	NO.2 CARGO HOLD	141	177	1.75	16.85
CH3	NO.3 CARGO HOLD	105	141	1.75	16.85
CH4	NO.4 CARGO HOLD	69	105	1.75	16.85
CH5	NO.5 CARGO HOLD	34	69	1.75	16.85
SUBTOTAL					
TOTAL					
NAME	DES	FRMIN #	FRMAX #	ZMIN m	ZMAX m
CAPACITY OF Ballast Water (RHO=1.025)					
FPT	FORE PEAK TK.	208	224	-0.00	11.40
WBT1P	NO.1 WATER BALLAST TK. (P)	177	208	0.00	14.98
WBT1S	NO.1 WATER BALLAST TK. (S)	177	208	0.00	14.98
WBT2P	NO.2 WATER BALLAST TK. (P)	141	177	0.00	14.83
WBT2S	NO.2 WATER BALLAST TK. (S)	141	177	0.00	14.83
WBT3P	NO.3 WATER BALLAST TK. (P)	105	141	0.00	14.77
WBT3S	NO.3 WATER BALLAST TK. (S)	105	141	0.00	14.77
WBT4P	NO.4 WATER BALLAST TK. (P)	69	105	0.00	10.60
WBT4S	NO.4 WATER BALLAST TK. (S)	69	105	0.00	10.60
WBT5P	NO.5 W.B.T.(P)	34	69	0.00	10.84
WBT5S	NO.5 WATER BALLAST TK. (S)	34	69	0.00	10.84
APT	AFT PEAK TK.	-5	9	6.48	10.90
SUBTOTAL					
CAPACITY OF Diesel Oil (RHO=0.86)					
MDOSTORTKP	M.D.O. STOR. TK. (P)	16	19	6.47	10.90
MDOSTORTKS	M.D.O. STOR. TK. (S)	15	19	6.47	10.90
MDOSERVTK	M.D.O. SERV. TK.	11	13	6.83	10.90
MDOSETTLTK	M.D.O. SETTL. TK.	13	15	6.47	10.90
SUBTOTAL					
CAPACITY OF Fresh Water (RHO=1)					
FWTP	FRESH WATER TK. (P)	-5	6	10.90	14.98
FWTS	FRESH WATER TK. (S)	-5	9	10.90	14.98
CWT	COOLING WATER TK.	7	11	0.82	7.40
DWT	DISTILLED WATER TK.	6	9	10.90	14.98
SUBTOTAL					

NAME	DES	FRMIN #	FRMAX #	ZMIN m	ZMAX m
CAPACITY OF Heavy Fuel Oil (RHO=0.98)					
FOT1P	NO.1 FUEL OIL TK. (P)	69	105	10.60	14.77
FOT1S	NO.1 FUEL OIL TK. (S)	69	105	10.60	14.77
FOT2P	NO.2 FUEL OIL TK. (P)	34	69	10.60	14.79
FOT2S	NO.2 FUEL OIL TK. (S)	34	69	10.60	14.79
HFOTKP	H.F.O. TK. (P)	19	34	4.62	14.79
HFOTKS	H.F.O. TK. (S)	19	34	4.62	14.79
NO1_HFO_SERV_TK	NO.1 H.F.O. SERV. TK.	19	23	10.90	13.80
NO1_HFO_SETTL_TK	NO.1 H.F.O. SETTL. TK.	23	27	10.90	13.80
NO2_HFO_SERV_TK	NO.2 H.F.O. SERV. TK.	16	19	10.90	13.80
NO2_HFO_SETTL_TK	NO.2 H.F.O. SETTL. TK.	11	16	10.90	13.80
SUBTOTAL					
CAPACITY OF Lubricating Oil (RHO=0.9)					
MELOSTORTK	M/E L.O. STOR. TK.	9	12	6.99	10.90
MELOSETTLTK	M/E L.O. SETTL. TK.	12	14	6.75	10.90
GELOSTORTK	G/E L.O. STOR. TK.	14	15	6.93	10.90
GELOSETTLTK	G/E L.O. SETTL. TK.	14	15	6.47	10.90
NO1CYLOTK	NO.1 CYL L.O. TK.	32	34	8.20	10.90
NO2CYLOTK	NO.2 CYL L.O. TK.	32	34	8.20	10.90
STLOSUTK	S/T L.O. SUMP TK.	11	14	0.18	2.20
MELOSUMPTK	M/E L.O. SUMP TK.	19	29	0.60	2.20
SUBTOTAL					
CAPACITY OF Miscellaneous (RHO=1)					
BT	BILGE TK.	14	18	0.00	2.20
BILGE_SEP_OIL_TK	BILGE SEP. OIL TK.	23	27	0.23	2.20
SEWAGE_STO_TK	SEWAGE STOR. TK.	27	32	0.12	2.21
LODRAIN_TK	L.O. DRAIN TK.	23	26	0.27	2.20
FODRAIN_TK	F.O. DRAIN TK.	26	29	0.13	2.20
FOOVERFLOW_TK	F.O. OVERFLOW TK.	29	32	0.12	2.20
SLUDGE_TK	SLUDGE TK.	24	34	7.40	8.20
SUBTOTAL					
TOTAL					

NAME	DES	FRMIN #	FRMAX #	ZMIN m	ZMAX m
CAPACITY OF Void (RHO=1.025)					
BW	BILGE WELL	11	12	1.60	2.20
BWP	BILGE WELL (P)	33	34	0.11	2.20
BWS	BILGE WELL (S)	33	34	0.11	2.20
LOW_SEA_CH	LOW SEA CHEST	29	31	1.74	3.70
HIGH_SEA_CH	HIGH SEA CHEST	29	31	2.80	4.75
CLP	CHAIN LOCKER (P)	209	212	11.40	18.06
CLS	CHAIN LOCKER (S)	209	212	11.40	18.06
SG	STEER GEAR ROOM	-5	9	10.90	15.10
TUNNEL	TUNNEL	34	208	0.00	1.75
V_DB	VOID OF DOUBLE BOTTOM	18	34	0.00	2.21
V_FPT	VOID SPACE ABOVE FPT	208	221	11.40	15.10
V_LSTOOL1	NO.1 LOWER STOOL	174	177	1.75	4.15
V_LSTOOL2	NO.2 LOWER STOOL	138	141	1.75	4.15
V_LSTOOL3	NO.3 LOWER STOOL	102	105	1.75	4.15
V_LSTOOL4	NO.4 LOWER STOOL	66	69	1.75	4.15
USTOOL1P	NO.1 UPPER STOOL (P)	176	177	12.67	15.10
USTOOL1S	NO.1 UPPER STOOL (S)	176	177	12.67	15.10
USTOOL2P	NO.2 UPPER STOOL (P)	140	141	12.67	15.10
USTOOL2S	NO.2 UPPER STOOL (S)	140	141	12.67	15.10
USTOOL3P	NO.3 UPPER STOOL (P)	104	105	12.67	15.10
USTOOL3S	NO.3 UPPER STOOL (S)	104	105	12.67	15.10
USTOOL4P	NO.4 UPPER STOOL (P)	68	69	12.67	15.10
USTOOL4S	NO.4 UPPER STOOL (S)	68	69	12.67	15.10
V_COFF1P	COFFERDAM BESIDE FOT1P	69	105	10.60	14.65
V_COFF1S	COFFERDAM BESIDE FOT1S	69	105	10.60	14.65
V_COFF2P	COFFERDAM BESIDE FOT2P	34	69	10.60	14.66
V_COFF2S	COFFERDAM BESIDE FOT2S	34	69	10.60	14.66
V_COFF3P	COFFERDAM BESIDE MDOSTORTKP	16	19	5.61	10.90
V_COFF3S	COFFERDAM BESIDE MDOSTORTKS	15	19	5.61	10.90
V_COFF4P	COFFERDAM ABOVE HFOTKP	31	34	13.80	14.79
V_COFF4S	COFFERDAM ABOVE HFOTKS	31	34	13.80	14.79
V_COFF5P	COFFERDAM BESIDE HFOTKP	19	34	3.72	14.68
V_COFF5S	COFFERDAM BESIDE HFOTKS	19	34	3.72	14.68
V_COFF6S	COFFERDAM ABOVE HFOTKS	12	27	13.80	14.79
SUBTOTAL					
TOTAL					

4.2 Tanks Capacity Table

Capacity of Grain Load, RHO=1, Reduction=1, FILL=1

NAME	VOLM m3	VNET m3	CGX m	CGY m	CGZ m	IYMAX m4	FSM tm
CH1	6666.2	6652.8	151.18	0.00	8.68	21042	21042.1
CH2	9981.7	9961.7	125.57	0.00	8.67	43163	43163.3
CH3	9994.1	9974.1	96.78	0.00	8.67	43163	43163.3
CH4	9993.7	9973.8	67.98	0.00	8.67	43163	43163.3
CH5	9013.2	8995.2	40.05	0.00	9.09	41805	41804.6
TOTAL	45648.9	45557.6	93.51	0.00	8.76		192336.7

Capacity of Ballast Water, RHO=1.025, Reduction=0.02, FILL=1

NAME	VOLM m3	VNET m3	CGX m	CGY m	CGZ m	IYMAX m4	FSM tm
FPT	770.7	755.3	168.53	0.00	5.89	883	904.9
WBT1P	1204.0	1179.9	151.92	9.30	7.03	2309	2366.8
WBT1S	1204.0	1179.9	151.92	-9.30	7.03	2309	2366.8
WBT2P	1388.7	1360.9	125.72	10.47	5.87	5350	5484.1
WBT2S	1388.7	1360.9	125.72	-10.47	5.87	5350	5484.1
WBT3P	1403.0	1374.9	97.00	10.53	5.79	5520	5657.8
WBT3S	1403.0	1374.9	97.00	-10.53	5.79	5520	5657.8
WBT4P	968.5	949.1	68.29	9.66	2.70	5478	5615.0
WBT4S	968.5	949.1	68.29	-9.66	2.70	5478	5615.0
WBT5P	925.1	906.6	39.87	9.38	3.43	3153	3231.6
WBT5S	925.1	906.6	39.87	-9.38	3.43	3153	3231.6
APT	337.7	331.0	2.28	0.00	9.53	3072	3148.4
TOTAL	12886.8	12629.1	102.73	0.00	5.34		48764.2

Capacity of Fresh Water, RHO=1, Reduction=0.02, FILL=1

NAME	VOLM m3	VNET m3	CGX m	CGY m	CGZ m	IYMAX m4	FSM tm
FWTP	95.6	93.7	0.94	6.60	13.03	46	45.7
FWTS	137.2	134.4	2.17	-6.85	13.01	84	84.3
CWT	25.2	24.7	6.47	0.00	5.10	181	181.0
DWT	41.6	40.7	4.97	7.43	12.96	33	32.6
TOTAL	299.6	293.6	2.53	0.00	12.34		343.6

Capacity of Heavy Fuel Oil, RHO=0.98, Reduction=0.02, FILL=0.98

NAME	VOLM m3	VNET m3	CGX m	CGY m	CGZ m	IYMAX m4	FSM tm
FOT1P	274.6	269.1	68.20	11.71	13.12	71	70.1
FOT1S	274.6	269.1	68.20	-11.71	13.12	71	70.1
FOT2P	265.9	260.5	39.85	11.70	13.13	70	68.9
FOT2S	265.9	260.5	39.85	-11.70	13.13	70	68.9
HFOTKP	263.6	258.3	20.57	11.15	11.11	46	45.3
HFOTKS	187.4	183.6	21.94	-11.14	10.40	35	34.5

NAME	VOLM m3	VNET m3	CGX m	CGY m	CGZ m	IYMAX m4	FSM tm
NO1_HFO_SERV_TK	25.8	25.3	15.46	-11.00	12.41	8	7.5
NO1_HFO_SETTL_TK	30.8	30.2	18.63	-11.27	12.38	12	11.3
NO2_HFO_SERV_TK	20.3	19.9	12.64	-11.07	12.41	7	6.8
NO2_HFO_SETTL_TK	22.9	22.4	9.62	-10.64	12.46	5	4.9
TOTAL	1631.7	1599.0	42.67	-0.15	12.44		388.2

Capacity of Lubricating Oil, RHO=0.9, Reduction=0.02, FILL=0.98

NAME	VOLM m3	VNET m3	CGX m	CGY m	CGZ m	IYMAX m4	FSM tm
MELOSTORTK	30.5	29.9	7.12	6.92	9.45	30	27.2
MELOSETTLTK	19.9	19.5	8.81	7.18	9.32	22	20.1
GELOSTORTK	14.9	14.6	10.02	8.29	9.42	12	10.5
GELOSETTLTK	7.8	7.7	10.01	5.59	8.86	0	0.4
NO1CYLOTK	10.4	10.2	25.00	3.60	9.55	2	1.7
NO2CYLOTK	10.4	10.2	25.00	1.20	9.55	2	1.7
STLOSUTK	6.2	6.1	8.89	-0.00	1.42	2	2.0
MELOSUMPTK	16.9	16.6	17.42	0.00	1.32	5	4.7
TOTAL	117.0	114.7	12.72	4.88	7.80		68.1

Capacity of Diesel Oil, RHO=0.85, Reduction=0.02, FILL=0.98

NAME	VOLM m3	VNET m3	CGX m	CGY m	CGZ m	IYMAX m4	FSM tm
MDOSTORTKP	41.5	40.7	12.66	7.28	9.32	52	44.3
MDOSTORTKS	52.9	51.9	12.30	-7.23	9.35	66	56.0
MDOSERVTK	24.8	24.3	8.23	-7.09	9.37	27	22.7
MDOSETTLTK	29.7	29.1	9.82	-7.33	9.24	35	30.0
TOTAL	148.9	145.9	11.23	-3.18	9.32		152.9

Capacity of Miscellaneous, RHO=1, Reduction=0.02, FILL=1

NAME	VOLM m3	VNET m3	CGX m	CGY m	CGZ m	IYMAX m4	FSM tm
BT	16.4	16.1	11.52	0.00	1.32	10	10.0
BILGE_SEP_OIL_TK	5.8	5.7	18.80	-2.35	1.54	2	1.7
SEWAGE_STO_TK	13.9	13.6	22.36	-2.93	1.47	10	10.3
LODRAIN_TK	3.9	3.8	18.32	2.29	1.56	1	0.9
FODRAIN_TK	6.6	6.5	20.69	2.59	1.49	3	3.4
FOOVERFLOW_TK	9.2	9.0	23.04	3.09	1.47	8	8.1
SLUDGE_TK	15.4	15.1	21.80	-8.40	7.80	9	9.2
TOTAL	71.2	69.8	19.16	-1.81	2.81		43.7

Capacity of Void, RHO=1.025, Reduction=0.02, FILL=1

NAME	VOLM m3	VNET m3	CGX m	CGY m	CGZ m	IYMAX m4	FSM tm
BW	1.0	1.0	7.81	0.00	1.92	1	0.8
BWP	1.9	1.9	25.39	2.94	1.26	0	0.1
BWS	1.9	1.9	25.39	-2.94	1.26	0	0.1
LOW_SEA_CH	3.5	3.4	22.69	-5.68	3.13	3	2.8
HIGH_SEA_CH	4.1	4.0	22.69	7.46	4.20	4	4.5
CLP	22.6	22.1	166.75	2.85	14.72	1	1.0
SG	365.2	357.9	1.37	0.00	12.98	674	690.7
TUNNEL	584.6	572.9	95.40	0.00	0.88	160	164.4
V_DB	90.2	88.4	20.78	0.00	1.12	307	314.9
V_FPT	297.5	291.5	168.20	0.00	13.36	1891	1938.5
V_LSTOOL1	96.9	95.0	139.34	0.00	2.77	2906	2978.8
V_LSTOOL2	100.6	98.6	110.53	0.00	2.75	3597	3686.9
V_LSTOOL3	100.6	98.6	81.74	0.00	2.75	3597	3686.9
V_LSTOOL4	98.6	96.6	52.94	0.00	2.76	2809	2878.8
USTOOL1P	16.5	16.1	139.80	4.35	13.82	62	63.3
USTOOL1S	16.5	16.1	139.80	-4.35	13.82	62	63.3
USTOOL2P	18.3	18.0	111.00	4.87	13.82	67	68.3
USTOOL2S	18.3	18.0	111.00	-4.87	13.82	67	68.3
USTOOL3P	18.3	18.0	82.20	4.87	13.82	67	68.3
USTOOL3S	18.3	18.0	82.20	-4.87	13.82	67	68.3
USTOOL4P	18.3	18.0	53.40	4.87	13.82	67	68.3
USTOOL4S	18.3	18.0	53.40	-4.87	13.82	67	68.3
V_COFF1P	150.7	147.7	68.20	13.75	12.61	5	5.4
V_COFF1S	150.7	147.7	68.20	-13.75	12.61	5	5.4
V_COFF2P	147.1	144.2	39.75	13.74	12.62	5	5.5
V_COFF2S	147.1	144.2	39.75	-13.74	12.62	5	5.5
V_COFF3P	17.3	17.0	12.62	8.71	8.17	4	4.5
V_COFF3S	22.7	22.3	12.23	-8.64	8.23	7	6.7
V_COFF4P	8.5	8.3	24.60	11.49	14.26	12	12.1
V_COFF4S	8.5	8.3	24.60	-11.49	14.26	12	12.1
V_COFF5P	92.4	90.6	20.28	12.40	9.77	15	15.6
V_COFF5S	92.3	90.4	20.28	-12.40	9.77	15	15.1
V_COFF6S	34.6	33.9	14.55	-11.14	14.27	34	34.8
TOTAL	2784.2	2728.5	72.32	-0.13	8.32		17008.3

5. TYPICAL LOADING CONDITION

5.1. Summary of Loading Condition

CASE	DESCRIPTION
LD00	LIGHT SHIP
LD01A	NORMAL BALLAST CONDITION,DEPARTURE
LD01B	NORMAL BALLAST CONDITION,FPT FULL,HALFWAY
LD01C	NORMAL BALLAST CONDITION,FPT EMPTY,HALFWAY
LD01D	NORMAL BALLAST CONDITION,ARRIVAL
LD01E	NORMAL BALLAST CONDITION FOR IACS,DEPARTURE
LD01F	NORMAL BALLAST CONDITION FOR IACS,ARRIVAL
LD02	DOCKING WITH 10% BUNKER
LD03A	HEAVY BALLAST CONDITION,DEPARTURE
LD03B	HEAVY BALLAST CONDITION,WBT2 FULL,HALFWAY
LD03C	HEAVY BALLAST CONDITION,WBT2 EMPTY,HALFWAY
LD03D	HEAVY BALLAST CONDITION,ARRIVAL
LD03E	HEAVY BALLAST CONDITION FOR IACS,DEPARTURE
LD03F	HEAVY BALLAST CONDITION FOR IACS,ARRIVAL
LD04A	HOMOGENEOUSLY LIGHT CARGO CONDITION,DEPRTURE
LD04B	HOMOGENEOUSLY LIGHT CARGO CONDITION,HALFWAY
LD04C	HOMOGENEOUSLY LIGHT CARGO CONDITION,ARRIVAL
LD05A	HOMOGENEOUSLY HEAVY CARGO CONDITION,DEPARTURE
LD05B	HOMOGENEOUSLY HEAVY CARGO CONDITION,HALFWAY
LD05C	HOMOGENEOUSLY HEAVY CARGO CONDITION,ARRIVAL
LD06A	ALTERNATIVE HEAVY CARGO CONDITION,DEPARTURE
LD06B	ALTERNATIVE HEAVY CARGO CONDITION,HALFWAY
LD06C	ALTERNATIVE HEAVY CARGO CONDITION,ARRIVAL
LD07	PROPELLER SHAFT SEAL MAINTENANCE WITH 10% BUNKER,DOCKING
LD08A	STEEL COIL CONDITION,DEPARTURE
LD08B	STEEL COIL CONDITION,HALFWAY
LD08C	STEEL COIL CONDITION,ARRIVAL

LD09A	MULTI-PORT HEAVY CARGO LOADING,DEPARTURE
LD09B	MULTI-PORT HEAVY CARGO LOADING,NO.2WBT2 FULL,FOLLOWED
LD09C	MULTI-PORT HEAVY CARGO LOADING,NO.2 WBT2 EMPTY,FOLLOWED
LD09D	MULTI-PORT HEAVY CARGO LOADING,ARRIVAL
LD09E	MULTI-PORT HEAVY CARGO LOADING,FOR IACS,DEPARTURE
LD09F	MULTI-PORT HEAVY CARGO LOADING,FOR IACS,ARRIVAL
LD10A	HOMOGENEOUSLY LIGHT CARGO CONDITION,SHORT VOYAGE,DEPARTURE
LD10B	HOMOGENEOUSLY LIGHT CARGO CONDITION,SHORT VOYAGE,ARRIVAL
LD11A	HOMOGENEOUSLY HEAVY CARGO CONDITION,SHORT VOYAGE,DEPARTURE
LD11B	HOMOGENEOUSLY HEAVY CARGO CONDITION,SHORT VOYAGE,ARRIVAL
LD12A	ALTERNATIVE HEAVY CARGO CONDITION,SHORT VOYAGE,DEPARTURE
LD12B	ALTERNATIVE HEAVY CARGO CONDITION,SHORT VOYAGE,ARRIVAL

Note:
"For IACS" included in the names of some load cases means the load case is only for CSR strength calculation not a real condition.
For example: LD01E,LD01F

CASE	DISP t	DWIT t	CARGO t	BUNKER t	T m	TR m	GM m	GMCORR m
LD00	8650.0	0.0	0.0	0.0	2.452	-3.544	19.277	0.000
LD01A	23697.4	15047.4	0.0	2059.0	5.971	-2.434	7.058	0.065
LD01B	22738.7	14088.7	0.0	1109.3	5.735	-1.351	7.482	0.067
LD01C	21964.5	13314.5	0.0	1109.3	5.582	-2.638	7.847	0.070
LD01D	21175.6	12525.6	0.0	327.4	5.384	-1.727	8.185	0.072
LD01E	22923.3	14273.3	0.0	2059.0	5.810	-3.676	7.405	0.067
LD01F	21949.8	13299.8	0.0	327.4	5.532	-0.411	7.809	0.070
LD02	11318.0	2668.0	0.0	221.5	3.000	-0.280	13.950	0.026
LD03A	33920.9	25270.9	0.0	2059.0	8.215	-0.532	4.504	0.336
LD03B	32962.1	24312.1	0.0	1109.3	8.004	0.383	4.702	0.346
LD03C	30172.2	21522.2	0.0	1109.3	7.400	-1.644	4.984	0.378
LD03D	29383.3	20733.3	0.0	327.4	7.223	-0.876	5.125	0.388
LD03E	31124.0	22474.0	0.0	2052.1	7.605	-2.524	4.772	0.366
LD03F	32173.2	23523.2	0.0	327.4	7.823	1.182	4.822	0.354
LD04A	43604.2	34954.2	32851.6	2059.0	10.267	-1.640	3.092	0.015
LD04B	42645.5	33995.5	32851.6	1109.3	10.075	-0.844	3.198	0.015
LD04C	41856.6	33206.6	32851.6	327.4	9.916	-0.184	3.258	0.015
LD05A	43608.8	34958.8	32856.1	2059.0	10.268	-1.661	7.018	0.015
LD05B	42650.0	34000.0	32856.1	1109.3	10.076	-0.878	7.212	0.015
LD05C	41861.1	33211.1	32856.1	327.4	9.916	-0.229	7.349	0.015
LD06A	43696.8	35046.8	32944.2	2059.0	10.268	-2.911	6.025	0.015
LD06B	42738.0	34088.0	32944.2	1109.3	10.078	-2.134	6.192	0.015
LD06C	41949.1	33299.1	32944.2	327.4	9.920	-1.492	6.308	0.015
LD07	12149.2	3499.2	0.0	278.5	3.143	1.202	12.731	0.065
LD08A	43071.9	34421.9	0.0	2059.0	10.153	-1.781	8.466	0.015
LD08B	42113.1	33463.1	0.0	1109.3	9.961	-1.001	8.697	0.015
LD08C	41324.2	32674.2	0.0	327.4	9.801	-0.355	8.864	0.016
LD09A	32413.2	23763.2	18870.7	2059.0	7.881	-2.647	6.656	0.351
LD09B	31454.4	22804.4	18870.7	1109.3	7.678	-1.794	6.927	0.362
LD09C	43537.1	34887.1	33553.2	1109.3	10.268	-0.496	6.929	0.273
LD09D	42558.1	33908.1	33553.2	327.4	10.065	-0.117	7.065	0.268
LD09E	29623.3	20973.3	18870.7	2059.0	7.267	-4.622	7.184	0.385
LD09F	45348.0	36698.0	33553.2	327.4	10.675	1.806	6.932	0.251
LD10A	43574.3	34924.3	33462.1	1088.3	10.267	-1.187	3.174	0.018
LD10B	42806.3	34156.3	33462.1	327.4	10.113	-0.556	3.234	0.018
LD11A	43559.8	34909.8	33786.9	1088.3	10.270	-0.702	7.172	0.018
LD11B	42791.9	34141.9	33786.9	327.4	10.115	-0.079	7.304	0.018
LD12A	43632.6	34982.6	33859.6	1088.3	10.268	-2.000	6.117	0.018
LD12B	42864.6	34214.6	33859.6	327.4	10.115	-1.381	6.229	0.018

CASE	XBMMX m	BMMAX tm	XBMMN m	BMMIN tm	XSHMX m	SHMAX t	XSHMN m	SHMIN t
LD00	71.4	57820	176.5	-1	25.8	1595	135.4	-875
LD01A	74.6	93140	176.5	-39	25.8	2196	138.6	-1533
LD01B	72.8	84904	176.9	-11	25.8	1968	138.6	-1400
LD01C	68.2	73569	176.9	-31	25.8	1850	138.4	-1099
LD01D	68.2	66546	176.9	-20	25.8	1658	138.6	-988
LD01E	68.2	80972	176.5	-69	25.8	2058	136.6	-1226
LD01F	72.8	77424	184.2	-0	27.4	1764	138.6	-1283
LD02	86.6	88327	-3.3	0	25.8	1943	138.6	-2127
LD03A	43.2	42550	98.3	-72617	111.4	3327	82.6	-4165
LD03B	43.2	37702	98.0	-78381	111.4	3432	82.6	-4190
LD03C	41.4	34634	96.0	-73678	111.4	4333	82.6	-3914
LD03D	41.4	30588	96.0	-78709	111.4	4422	82.6	-3934
LD03E	43.2	38834	96.3	-68329	111.4	4236	82.6	-3884
LD03F	41.4	33126	98.0	-83573	111.4	3528	82.6	-4204
LD04A	17.0	2446	75.3	-9812	165.0	267	29.0	-397
LD04B	15.4	1695	82.6	-13823	141.0	318	29.0	-496
LD04C	14.2	1132	89.4	-17395	141.0	385	25.8	-586
LD05A	16.8	2233	70.9	-15365	138.0	302	37.6	-661
LD05B	15.4	1563	73.1	-19156	138.0	384	37.1	-731
LD05C	13.8	1065	75.6	-22392	138.0	453	36.6	-790
LD06A	123.1	73338	32.1	-6675	53.8	4198	140.2	-3922
LD06B	123.1	70758	32.8	-8731	53.8	4171	140.2	-3843
LD06C	123.1	68539	33.1	-10455	53.8	4148	140.2	-3779
LD07	93.4	102141	-3.2	-0	29.0	1973	138.6	-2403
LD08A	25.8	8899	64.6	-8016	137.8	1197	40.8	-745
LD08B	23.4	7613	64.6	-11608	137.8	1279	39.0	-809
LD08C	21.8	6645	66.4	-14632	137.8	1346	39.0	-864
LD09A	70.0	86419	176.9	-33	53.8	2887	82.6	-2090
LD09B	70.0	81623	176.9	-19	53.8	2855	82.6	-2128
LD09C	15.0	1546	68.2	-18056	137.8	340	36.6	-728
LD09D	13.0	866	75.4	-24712	138.0	487	36.0	-845
LD09E	70.0	81643	176.5	-56	53.8	2827	140.2	-2830
LD09F	15.4	1646	120.4	-37921	141.0	1374	111.4	-789
LD10A	19.4	5573	100.8	-9846	5.9	433	29.0	-414
LD10B	18.6	4799	97.7	-13121	5.9	392	29.0	-481
LD11A	14.2	1257	73.1	-22568	138.0	434	36.6	-813
LD11B	12.6	806	75.4	-25491	138.0	498	36.0	-864
LD12A	123.1	71286	33.1	-10561	53.8	4254	140.2	-3911
LD12B	123.1	69205	33.3	-12192	53.8	4237	140.2	-3850

5.2 Bunker at Departure, Halfway and Arrival

LOADING CONDITION TEMP1: BUNKER AT DEPARTURE

DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	20.2	98.0	8.23	-7.08	9.34	22.22
M.D.O. SETTL. TK.	24.2	98.0	9.82	-7.32	9.21	29.35
M.D.O. STOR. TK. (P)	33.9	98.0	12.66	7.26	9.29	43.37
M.D.O. STOR. TK. (S)	43.2	98.0	12.30	-7.22	9.32	54.86
SUBTOTAL	121.6		11.23	-3.17	9.29	149.80
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	40.7	100.0	4.97	7.43	12.96	31.97
FRESH WATER TK. (P)	93.7	100.0	0.94	6.60	13.03	44.74
FRESH WATER TK. (S)	134.4	100.0	2.17	-6.85	13.01	82.65
SUBTOTAL	293.6		2.53	0.00	12.34	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	258.4	98.0	68.20	11.73	13.08	68.67
NO.1 FUEL OIL TK. (S)	258.4	98.0	68.20	-11.73	13.08	68.67
NO.2 FUEL OIL TK. (P)	250.2	98.0	39.86	11.71	13.09	0.00
NO.2 FUEL OIL TK. (S)	250.2	98.0	39.86	-11.71	13.09	0.00
H.F.O. TK. (P)	248.1	98.0	20.60	11.15	11.04	44.35
H.F.O. TK. (S)	176.3	98.0	21.94	-11.14	10.32	33.83
NO.1 H.F.O. SERV. TK.	24.3	98.0	15.46	-11.00	12.38	7.35
NO.1 H.F.O. SETTL. T.	29.0	98.0	18.63	-11.27	12.36	11.05
NO.2 H.F.O. SERV. TK.	19.1	98.0	12.64	-11.07	12.39	0.00
NO.2 H.F.O. SETTL. T.	21.5	98.0	9.62	-10.64	12.43	0.00
SUBTOTAL	1535.7		42.68	-0.15	12.40	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	6.8	98.0	10.01	5.59	8.82	0.36
G/E L.O. STOR. TK.	12.9	98.0	10.02	8.28	9.39	10.28
M/E L.O. SETTL. TK.	17.2	98.0	8.81	7.17	9.29	19.67
M/E L.O. STOR. TK.	26.4	98.0	7.12	6.90	9.42	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	101.1		12.73	4.87	7.77	56.99

DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	0.6	10.0	18.98	-1.96	0.68	1.69
BILGE TK.	1.6	10.0	11.66	0.00	0.32	9.85
F.O. DRAIN TK.	0.6	10.0	20.77	2.04	0.56	3.33
F.O. OVERFLOW TK.	0.9	10.0	23.05	2.35	0.50	7.97
L.O. DRAIN TK.	0.4	10.0	18.43	1.94	0.72	0.91
SEWAGE STOR. TK.	1.4	10.0	22.44	-2.26	0.52	10.07
SLUDGE TK.	1.5	10.0	21.80	-8.40	7.44	9.03
SUBTOTAL	7.0		19.24	-1.81	1.99	42.86
TOTAL	2059.0		33.55	-0.07	11.94	642.92

LOADING CONDITION TEMP2: BUNKER AT HALFWAY

DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	10.3	50.0	8.24	-6.66	8.53	22.22
M.D.O. SETTL. TK.	12.4	50.0	9.83	-6.86	8.34	29.35
M.D.O. STOR. TK. (P)	17.3	50.0	12.68	6.80	8.45	43.37
M.D.O. STOR. TK. (S)	22.0	50.0	12.35	-6.76	8.50	54.86
SUBTOTAL	62.0		11.26	-2.98	8.46	149.80
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	20.4	50.0	4.98	7.32	11.97	31.97
FRESH WATER TK. (P)	46.9	50.0	1.02	6.48	12.05	44.74
FRESH WATER TK. (S)	67.2	50.0	2.26	-6.75	12.02	82.65
SUBTOTAL	159.2		2.90	-0.00	10.95	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	131.9	50.0	68.20	11.90	12.29	68.67
NO.1 FUEL OIL TK. (S)	131.9	50.0	68.20	-11.90	12.29	68.67
NO.2 FUEL OIL TK. (P)	127.7	50.0	39.94	11.88	12.30	0.00
NO.2 FUEL OIL TK. (S)	127.7	50.0	39.94	-11.88	12.30	0.00
H.F.O. TK. (P)	126.6	50.0	21.26	10.98	9.26	44.35
H.F.O. TK. (S)	90.0	50.0	21.60	-10.89	8.63	33.83
NO.1 H.F.O. SERV. TK.	12.4	50.0	15.47	-10.92	11.69	7.35
NO.1 H.F.O. SETTL. T.	14.8	50.0	18.64	-11.21	11.66	11.05
NO.2 H.F.O. SERV. TK.	9.7	50.0	12.65	-10.98	11.70	0.00
NO.2 H.F.O. SETTL. T.	11.0	50.0	9.65	-10.54	11.75	0.00
SUBTOTAL	783.5		42.77	-0.15	11.35	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	4.1	60.0	10.01	5.58	8.05	0.36
G/E L.O. STOR. TK.	7.9	60.0	10.02	8.01	8.78	10.28
M/E L.O. SETTL. TK.	10.5	60.0	8.82	6.85	8.65	19.67
M/E L.O. STOR. TK.	16.1	60.0	7.14	6.62	8.83	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	76.6		14.10	4.02	6.96	56.99

DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	2.3	40.0	18.84	-2.15	1.07	1.69
BILGE TK.	6.4	40.0	11.55	0.00	0.73	9.85
F.O. DRAIN TK.	2.6	40.0	20.71	2.30	0.98	3.33
F.O. OVERFLOW TK.	3.6	40.0	23.05	2.70	0.93	7.97
L.O. DRAIN TK.	1.5	40.0	18.35	2.11	1.09	0.91
SEWAGE STOR. TK.	5.4	40.0	22.38	-2.58	0.94	10.07
SLUDGE TK.	6.0	40.0	21.80	-8.40	7.56	9.03
SUBTOTAL	27.9		19.18	-1.81	2.34	42.86
TOTAL	1109.3		32.71	-0.04	10.60	642.92

LOADING CONDITION TEMP3: BUNKER AT ARRIVAL

DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	2.1	10.0	8.28	-5.84	7.59	22.22
M.D.O. SETTL. TK.	2.5	10.0	9.88	-5.95	7.33	29.35
M.D.O. STOR. TK. (P)	3.5	10.0	12.81	5.91	7.45	43.37
M.D.O. STOR. TK. (S)	4.4	10.0	12.58	-5.91	7.52	54.86
SUBTOTAL	12.4		11.39	-2.61	7.47	149.80
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	4.1	10.0	4.98	7.15	11.13	31.97
FRESH WATER TK. (P)	9.4	10.0	1.14	6.32	11.15	44.74
FRESH WATER TK. (S)	13.4	10.0	2.40	-6.59	11.14	82.65
SUBTOTAL	51.6		4.32	-0.00	8.25	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	26.4	10.0	68.20	12.64	11.22	68.67
NO.1 FUEL OIL TK. (S)	26.4	10.0	68.20	-12.64	11.22	68.67
NO.2 FUEL OIL TK. (P)	25.5	10.0	40.23	12.62	11.23	0.00
NO.2 FUEL OIL TK. (S)	25.5	10.0	40.23	-12.62	11.23	0.00
H.F.O. TK. (P)	25.3	10.0	23.08	10.63	7.06	44.35
H.F.O. TK. (S)	18.0	10.0	23.44	-10.57	6.80	33.83
NO.1 H.F.O. SERV. TK.	2.5	10.0	15.49	-10.82	11.07	7.35
NO.1 H.F.O. SETTL. T.	3.0	10.0	18.65	-11.14	11.06	11.05
NO.2 H.F.O. SERV. TK.	1.9	10.0	12.65	-10.87	11.07	0.00
NO.2 H.F.O. SETTL. T.	2.2	10.0	9.70	-10.42	11.09	0.00
SUBTOTAL	156.7		43.37	-0.16	10.03	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	1.4	20.0	10.03	5.53	7.22	0.36
G/E L.O. STOR. TK.	2.6	20.0	10.04	7.48	7.96	10.28
M/E L.O. SETTL. TK.	3.5	20.0	8.83	6.21	7.80	19.67
M/E L.O. STOR. TK.	5.4	20.0	7.19	6.05	8.05	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	50.8		16.95	2.45	5.88	56.99

DES	MASS t	FILL %	XM m	YM m	ZM m	FRSM tm
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	4.5	80.0	18.81	-2.29	1.40	1.69
BILGE TK.	12.9	80.0	11.52	0.00	1.14	9.85
F.O. DRAIN TK.	5.2	80.0	20.70	2.51	1.34	3.33
F.O. OVERFLOW TK.	7.2	80.0	23.04	2.98	1.31	7.97
L.O. DRAIN TK.	3.1	80.0	18.32	2.24	1.42	0.91
SEWAGE STOR. TK.	10.9	80.0	22.37	-2.83	1.31	10.07
SLUDGE TK.	12.0	80.0	21.80	-8.40	7.72	9.03
SUBTOTAL	55.8		19.16	-1.81	2.67	42.86
TOTAL	327.4		27.78	-0.10	7.76	642.92

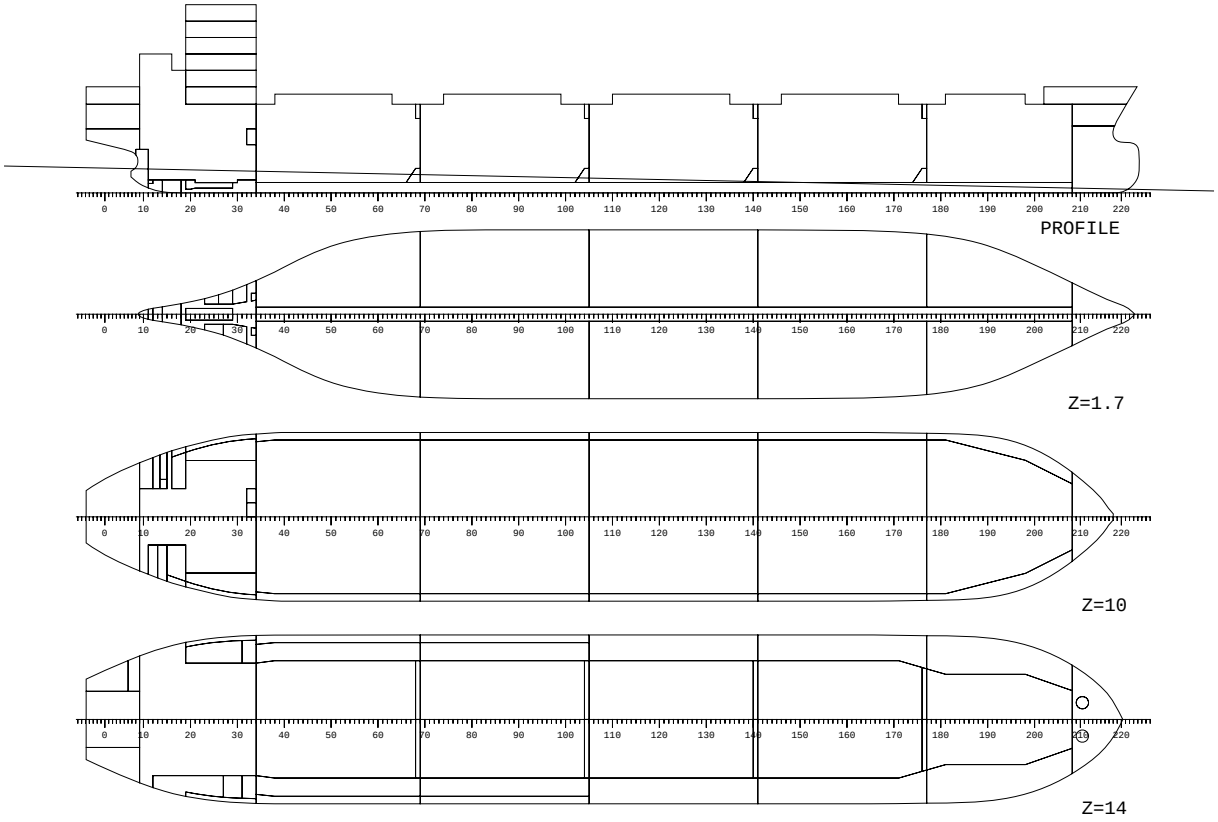
5.3 BALLAST WATER

BALLAST

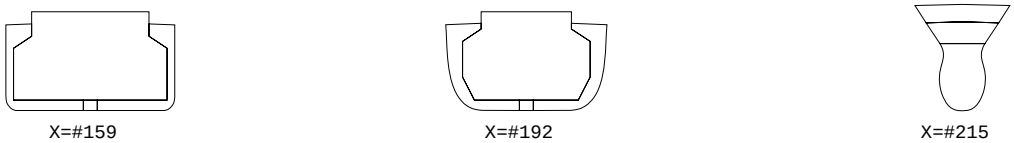
DES	MASS t	FILL %	XM m	YM m	ZM m
Ballast Water, (Dens = 1.025)					
AFT PEAK TK.	339.3	100.0	2.28	0.00	9.53
FORE PEAK TK.	774.2	100.0	168.53	0.00	5.89
NO.1 WATER BALLAST TK. (P)	1209.4	100.0	151.92	9.30	7.03
NO.1 WATER BALLAST TK. (S)	1209.4	100.0	151.92	-9.30	7.03
NO.2 WATER BALLAST TK. (P)	1394.9	100.0	125.72	10.47	5.87
NO.2 WATER BALLAST TK. (S)	1394.9	100.0	125.72	-10.47	5.87
NO.3 WATER BALLAST TK. (P)	1409.3	100.0	97.00	10.53	5.79
NO.3 WATER BALLAST TK. (S)	1409.3	100.0	97.00	-10.53	5.79
NO.4 WATER BALLAST TK. (P)	972.8	100.0	68.29	9.66	2.70
NO.4 WATER BALLAST TK. (S)	972.8	100.0	68.29	-9.66	2.70
NO.5 W.B.T.(P)	929.2	100.0	39.87	9.38	3.43
NO.5 WATER BALLAST TK. (S)	929.2	100.0	39.87	-9.38	3.43
SUBTOTAL	12944.8		102.73	0.00	5.34
TOTAL	12944.8		102.73	0.00	5.34

5.4. Detail Loading Condition

LOADING CASE 'LD00 -- LIGHT SHIP




X=#5 X=#52 X=#87 X=#123


X=#159 X=#192 X=#215

NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	0.0	0.00	0.00	0.00	0.00
Diesel Oil	DO	0.0	0.00	0.00	0.00	0.00
Fresh Water	FW	0.0	0.00	0.00	0.00	0.00
Grain Load	GRAIN	0.0	0.00	0.00	0.00	0.00
Heavy Fuel Oil	HFO	0.0	0.00	0.00	0.00	0.00
Lubricating Oil	LO	0.0	0.00	0.00	0.00	0.00
Miscellaneous	MIS	0.0	0.00	0.00	0.00	0.00
Deadweight		0.0	0.00	0.00	0.00	0.00
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		0.0	0.00	0.00	0.00	
Total weight		8650.0	79.30	0.00	10.20	

F L O A T I N G P O S I T I O N

Draught moulded	2.452	m	KM	29.48 m
Trim	-3.544	m	KG	10.20 m
Heel, PS=+	0.0	deg		
TA	4.224	m	GM0	19.28 m
TF	0.680	m	GMCORR	0.00 m
Trimming moment	-134996	tonm	GM	19.28 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	1.693	mrad	OK
AREA40	Area under GZ curve .	0.090	2.348	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.655	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.042	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	22.463	deg	NOT MET
GM0.15	GM > 0.15 m	0.150	19.277	m	OK
IMOWEATH.	IMO weather criterion	1.000	0.979		NOT MET
VISIBILI.		359.760	316.322		OK
PROPIMMER	Propeller should imm.	50.000	19.701	%	NOT MET

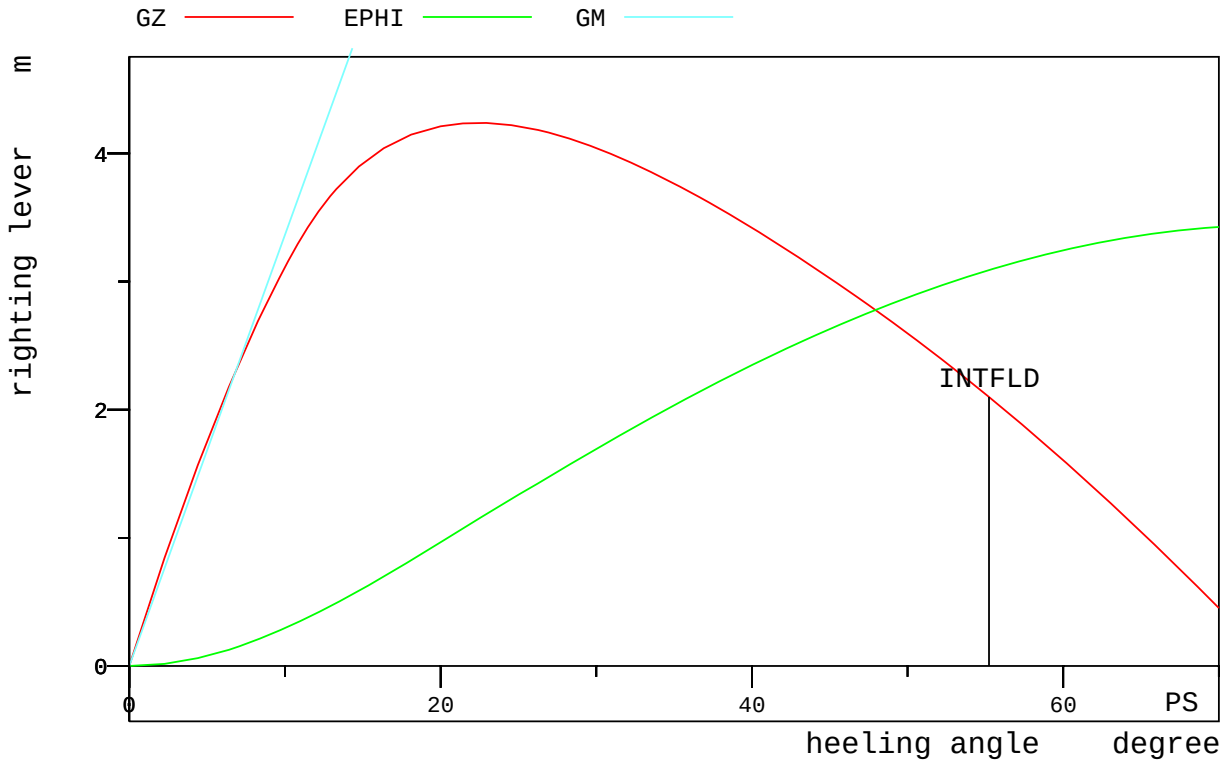
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-874.9 t	(14.6%)	POSITION:	135.4 m	171
SHEAR FORCE (MAX)	1595.0 t	(26.6%)		25.8 m	34
SAGGING MOMENT	-1.5 tm	(0.0%)		176.5 m	224
HOGGING MOMENT	57819.9 tm	(53.0%)		71.4 m	91

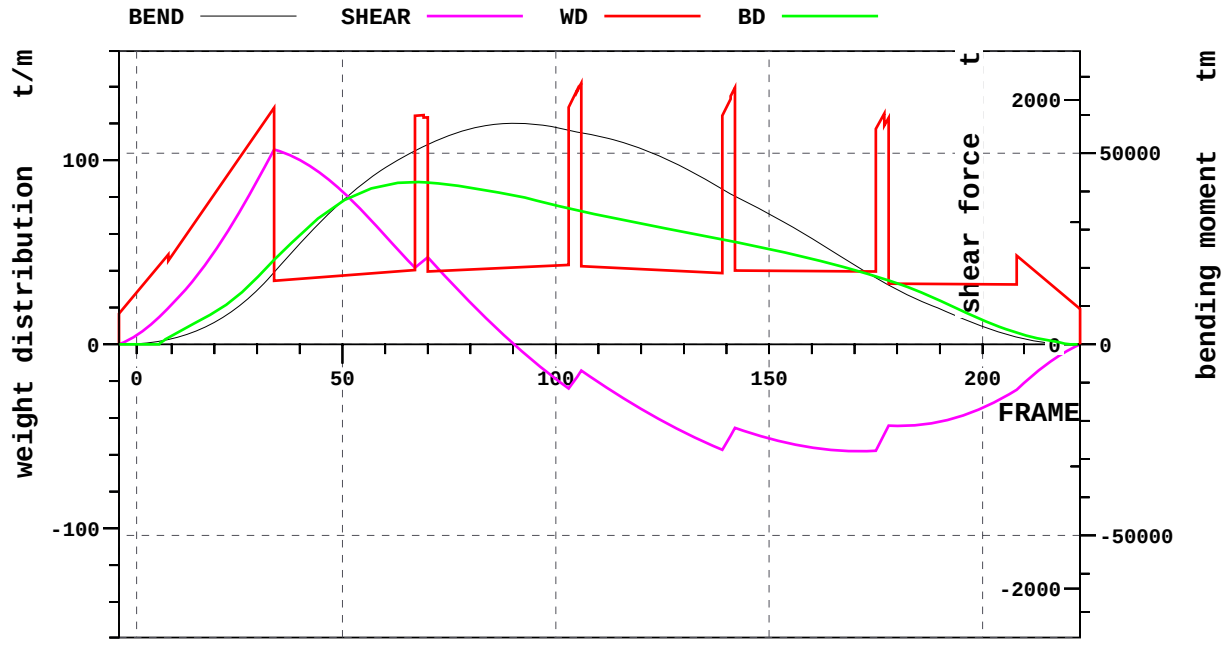
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.740)						
NO.1 CARGO HOLD	0.0	0.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
SUBTOTAL	0.0		0.00	0.00	0.00	0.0
TOTAL	0.0		0.00	0.00	0.00	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	0.0	0.0	0.00	0.00	0.00	0.00
M.D.O. SETTL. TK.	0.0	0.0	0.00	0.00	0.00	0.00
M.D.O. STOR. TK. (P)	0.0	0.0	12.66	7.28	9.32	0.00
M.D.O. STOR. TK. (S)	0.0	0.0	12.30	-7.23	9.35	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	0.00
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	0.0	0.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	0.0	0.0	4.97	7.43	12.96	0.00
FRESH WATER TK. (P)	0.0	0.0	0.94	6.60	13.03	0.00
FRESH WATER TK. (S)	0.0	0.0	2.17	-6.85	13.01	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	0.00

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	0.0	0.0	68.20	11.71	13.12	0.00
NO.1 FUEL OIL TK. (S)	0.0	0.0	68.20	-11.71	13.12	0.00
NO.2 FUEL OIL TK. (P)	0.0	0.0	39.85	11.70	13.13	0.00
NO.2 FUEL OIL TK. (S)	0.0	0.0	39.85	-11.70	13.13	0.00
H.F.O. TK. (P)	0.0	0.0	20.57	11.15	11.11	0.00
H.F.O. TK. (S)	0.0	0.0	21.94	-11.14	10.40	0.00
NO.1 H.F.O. SERV. TK.	0.0	0.0	15.46	-11.00	12.41	0.00
NO.1 H.F.O. SETTL. T.	0.0	0.0	18.63	-11.27	12.38	0.00
NO.2 H.F.O. SERV. TK.	0.0	0.0	12.64	-11.07	12.41	0.00
NO.2 H.F.O. SETTL. T.	0.0	0.0	9.62	-10.64	12.46	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	0.00
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	0.0	0.0	10.01	5.59	8.86	0.00
G/E L.O. STOR. TK.	0.0	0.0	10.02	8.29	9.42	0.00
M/E L.O. SETTL. TK.	0.0	0.0	8.81	7.18	9.32	0.00
M/E L.O. STOR. TK.	0.0	0.0	7.12	6.92	9.45	0.00
M/E L.O. SUMP TK.	0.0	0.0	17.42	0.00	1.32	0.00
NO.1 CYL L.O. TK.	0.0	0.0	25.00	3.60	9.55	0.00
NO.2 CYL L.O. TK.	0.0	0.0	25.00	1.20	9.55	0.00
S/T L.O. SUMP TK.	0.0	0.0	8.89	-0.00	1.42	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	0.00
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	0.0	0.0	18.80	-2.35	1.54	0.00
BILGE TK.	0.0	0.0	11.52	0.00	1.32	0.00
F.O. DRAIN TK.	0.0	0.0	20.69	2.59	1.49	0.00
F.O. OVERFLOW TK.	0.0	0.0	23.04	3.09	1.47	0.00
L.O. DRAIN TK.	0.0	0.0	18.32	2.29	1.56	0.00
SEWAGE STOR. TK.	0.0	0.0	22.36	-2.93	1.47	0.00
SLUDGE TK.	0.0	0.0	21.80	-8.40	7.80	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	0.00
TOTAL	0.0		0.00	0.00	0.00	0.00

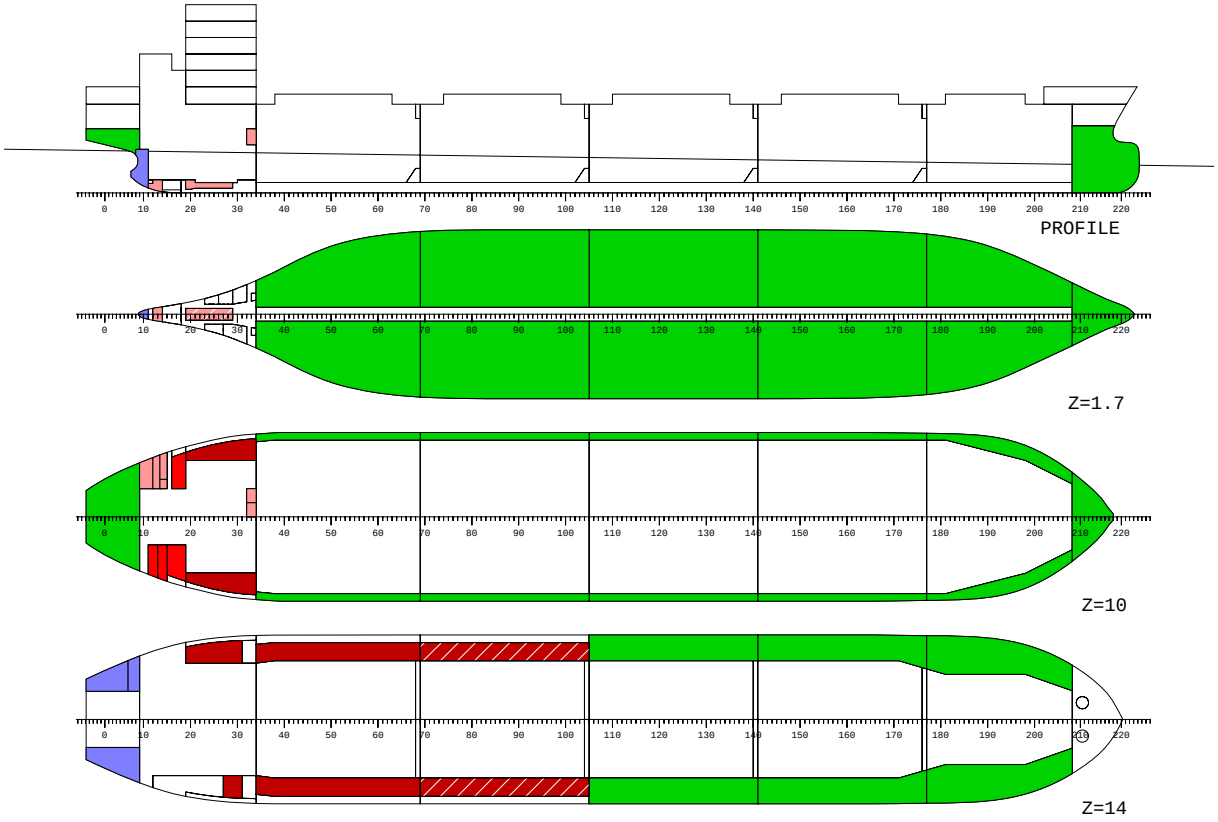


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	0.000	0.00	0.000	0.0	0.000
10.0	-0.234	3.11	0.294	0.0	0.000
12.0	-0.489	3.52	0.410	0.0	0.000
20.0	-2.381	4.21	0.963	0.0	0.000
30.0	-5.597	4.04	1.693	0.0	0.000
40.0	-8.972	3.42	2.348	0.0	0.000
50.0	-12.173	2.59	2.875	0.0	0.000
60.0	-15.089	1.61	3.244	0.0	0.000
70.0	-17.663	0.45	3.425	0.0	0.000

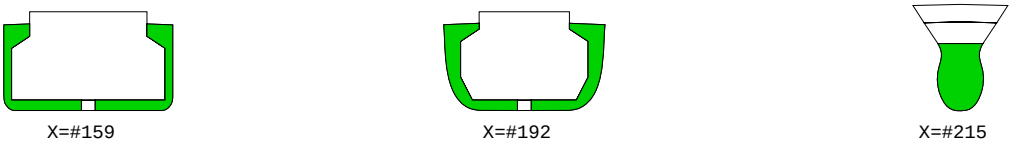


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	146	75	28.14	0.00
5.94	9.00	1242	299	48.79	3.59
18.60	25.00	9353	1033	98.36	25.62
25.80	34.00	18828	1595	128.48	45.88
40.20	52.00	39470	1183	37.72	79.96
53.80	69.00	51567	685	124.50	87.98
68.20	87.00	57660	103	41.42	82.28
82.60	105.00	55494	-267	138.20	73.75
97.00	123.00	49846	-589	40.55	64.97
111.40	141.00	39294	-748	133.31	56.20
125.80	159.00	28348	-839	39.83	46.94
140.20	177.00	16097	-735	125.39	35.47
152.20	192.00	8311	-618	32.78	21.67
165.00	208.00	1864	-373	32.53	6.39
169.20	214.00	684	-211	37.96	3.07
173.40	220.00	112	-82	27.72	0.80

LOADING CASE 'LD01A -- NORMAL BALLAST CONDITION, DEPARTURE







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	12944.8	102.73	0.00	5.34	886.83
Diesel Oil	DO	121.6	11.23	-3.17	9.29	149.80
Fresh Water	FW	293.6	2.53	0.00	12.34	159.36
Grain Load	GRAIN	0.0	0.00	0.00	0.00	0.00
Heavy Fuel Oil	HFO	1535.7	42.68	-0.15	12.40	233.92
Lubricating Oil	LO	101.1	12.73	4.87	7.77	56.99
	MASS	43.6	53.17	0.00	20.19	0.00
Miscellaneous	MIS	7.0	19.24	-1.81	1.99	42.86
Deadweight		15047.4	93.12	-0.01	6.28	1529.76
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		15047.4	93.12	-0.01	6.28	
Total weight		23697.4	88.07	-0.01	7.71	

F L O A T I N G P O S I T I O N

Draught moulded	5.971	m	KM	14.84 m
Trim	-2.434	m	KG	7.71 m
Heel, PS=+	-0.0	deg		
TA	7.188	m	GM0	7.12 m
TF	4.754	m	GMCORR	-0.06 m
Trimming moment	-120860	tonm	GM	7.06 m
Heel	-0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.997	mrad	OK
AREA40	Area under GZ curve .	0.090	1.694	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.697	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.154	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	40.433	deg	OK
GM0.15	GM > 0.15 m	0.150	7.058	m	OK
IMOWEATH.	IMO weather criterion	1.000	3.449		OK
VISIBILI.		359.760	221.760		OK
PROPIMMER	Propeller should imm.	50.000	75.626	%	OK

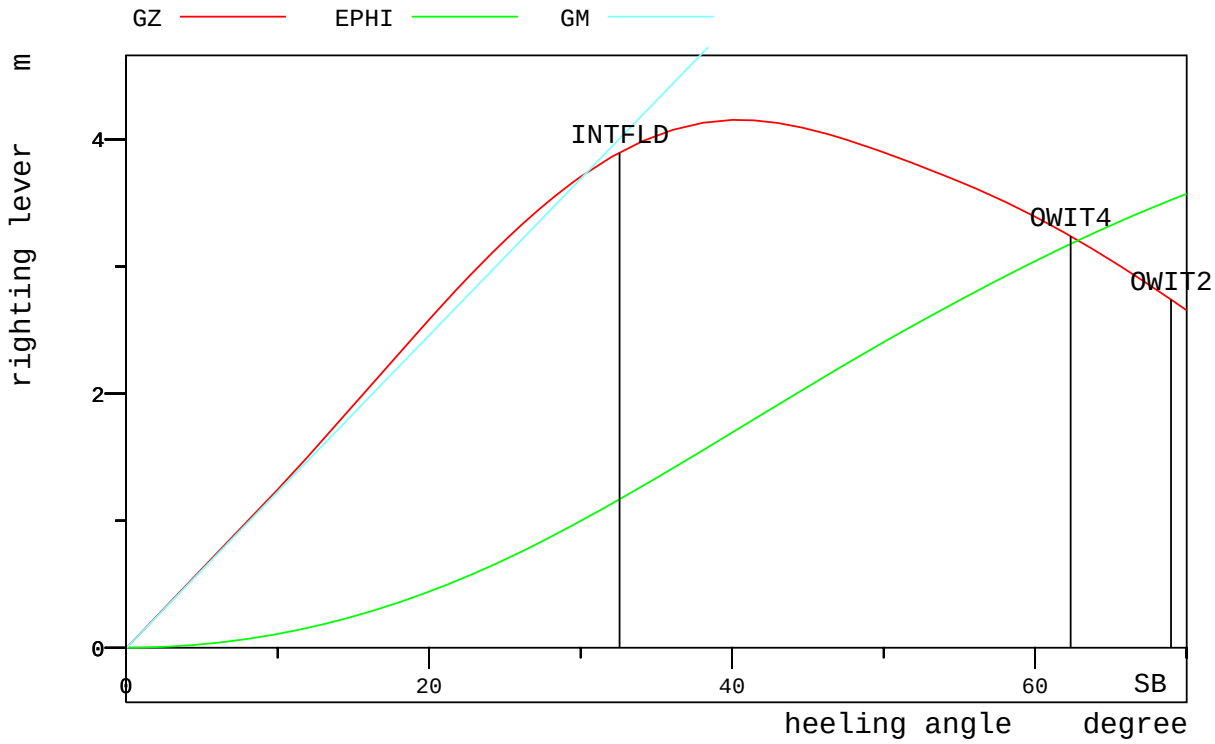
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN,CORR)	-1533.3 t	(25.6%)	POSITION:	138.6 m	175
SHEAR FORCE (MAX,CORR)	2196.2 t	(36.6%)		25.8 m	34
SAGGING MOMENT	-39.2 tm	(0.9%)		176.5 m	224
HOGGING MOMENT	93139.7 tm	(85.4%)		74.6 m	95

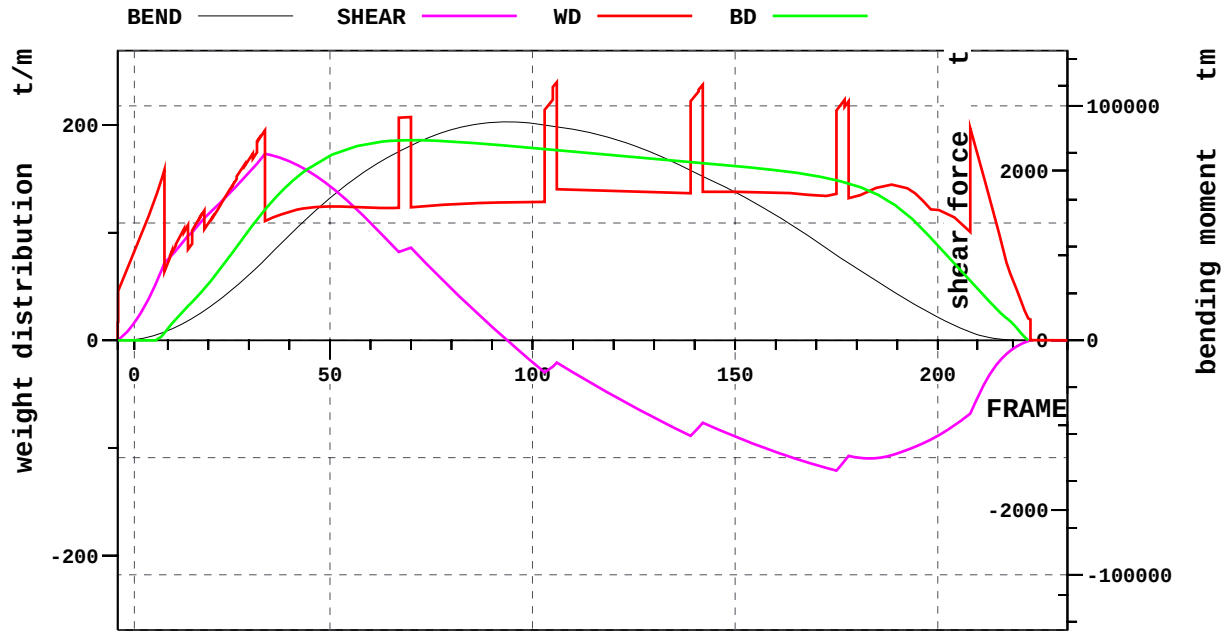
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.740)						
NO.1 CARGO HOLD	0.0	0.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
SUBTOTAL	0.0		0.00	0.00	0.00	0.0
, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	18.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	43.6		53.17	0.00	20.19	0.0
TOTAL	43.6		53.17	0.00	20.19	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	339.3	100.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	774.2	100.0	168.53	0.00	5.89	886.83
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	929.2	100.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	929.2	100.0	39.87	-9.38	3.43	0.00
SUBTOTAL	12944.8		102.73	0.00	5.34	886.83
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	20.2	98.0	8.23	-7.08	9.34	22.22
M.D.O. SETTL. TK.	24.2	98.0	9.82	-7.32	9.21	29.35
M.D.O. STOR. TK. (P)	33.9	98.0	12.66	7.26	9.29	43.37
M.D.O. STOR. TK. (S)	43.2	98.0	12.30	-7.22	9.32	54.86
SUBTOTAL	121.6		11.23	-3.17	9.29	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	40.7	100.0	4.97	7.43	12.96	31.97
FRESH WATER TK. (P)	93.7	100.0	0.94	6.60	13.03	44.74
FRESH WATER TK. (S)	134.4	100.0	2.17	-6.85	13.01	82.65
SUBTOTAL	293.6		2.53	0.00	12.34	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	258.4	98.0	68.20	11.73	13.08	68.67
NO.1 FUEL OIL TK. (S)	258.4	98.0	68.20	-11.73	13.08	68.67
NO.2 FUEL OIL TK. (P)	250.2	98.0	39.86	11.71	13.09	0.00
NO.2 FUEL OIL TK. (S)	250.2	98.0	39.86	-11.71	13.09	0.00
H.F.O. TK. (P)	248.1	98.0	20.60	11.15	11.04	44.35
H.F.O. TK. (S)	176.3	98.0	21.94	-11.14	10.32	33.83
NO.1 H.F.O. SERV. TK.	24.3	98.0	15.46	-11.00	12.38	7.35
NO.1 H.F.O. SETTL. T.	29.0	98.0	18.63	-11.27	12.36	11.05
NO.2 H.F.O. SERV. TK.	19.1	98.0	12.64	-11.07	12.39	0.00
NO.2 H.F.O. SETTL. T.	21.5	98.0	9.62	-10.64	12.43	0.00
SUBTOTAL	1535.7		42.68	-0.15	12.40	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	6.8	98.0	10.01	5.59	8.82	0.36
G/E L.O. STOR. TK.	12.9	98.0	10.02	8.28	9.39	10.28
M/E L.O. SETTL. TK.	17.2	98.0	8.81	7.17	9.29	19.67
M/E L.O. STOR. TK.	26.4	98.0	7.12	6.90	9.42	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	101.1		12.73	4.87	7.77	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	0.6	10.0	18.98	-1.96	0.68	1.69
BILGE TK.	1.6	10.0	11.66	0.00	0.32	9.85
F.O. DRAIN TK.	0.6	10.0	20.77	2.04	0.56	3.33
F.O. OVERFLOW TK.	0.9	10.0	23.05	2.35	0.50	7.97
L.O. DRAIN TK.	0.4	10.0	18.43	1.94	0.72	0.91
SEWAGE STOR. TK.	1.4	10.0	22.44	-2.26	0.52	10.07
SLUDGE TK.	1.5	10.0	21.80	-8.40	7.44	9.03
SUBTOTAL	7.0		19.24	-1.81	1.99	42.86
TOTAL	15003.8		93.24	-0.01	6.24	1529.76

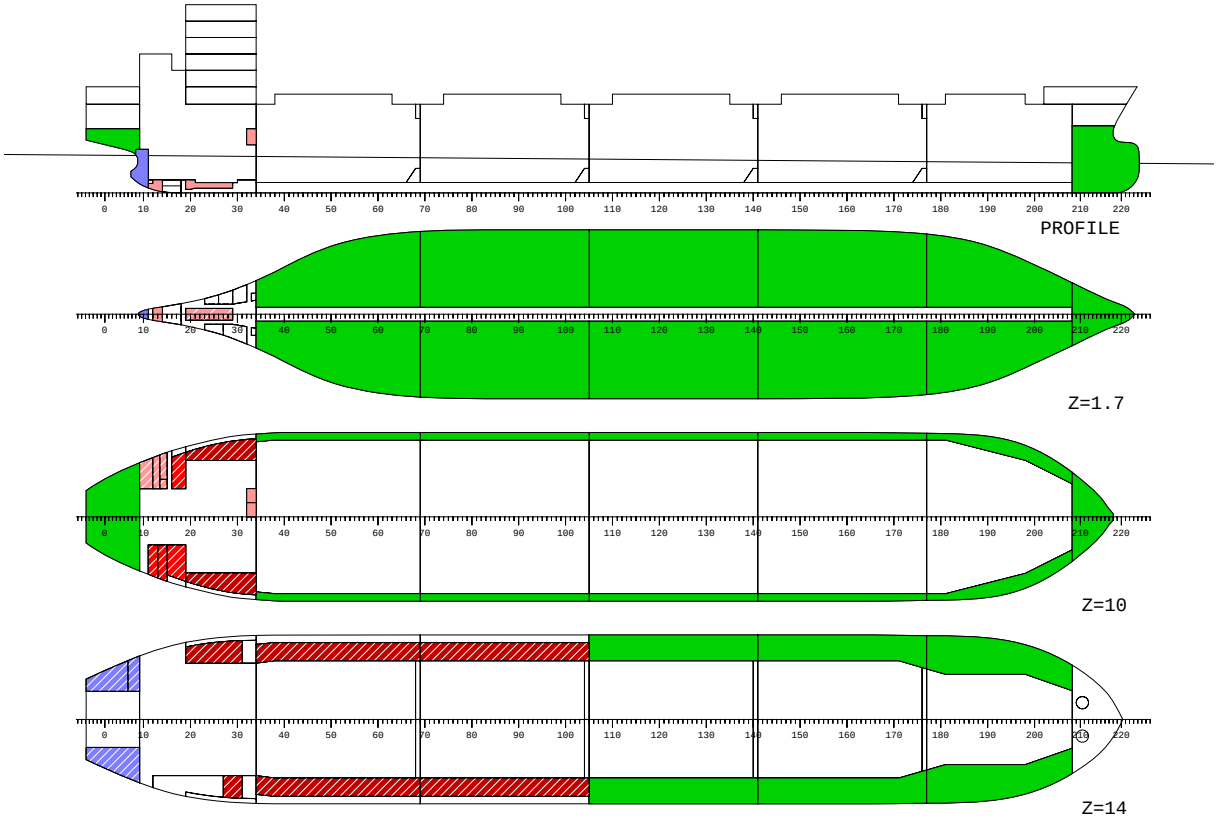


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.006	-0.01	0.000	0.0	0.000
10.0	0.018	1.24	0.108	265.6	0.011
12.0	0.036	1.50	0.156	318.1	0.013
20.0	0.168	2.58	0.441	523.2	0.022
30.0	0.178	3.71	0.997	764.9	0.032
40.0	-0.383	4.15	1.694	983.3	0.041
50.0	-1.509	3.90	2.403	1171.9	0.049
60.0	-2.721	3.39	3.042	1324.8	0.056
70.0	-3.978	2.66	3.573	1437.5	0.061

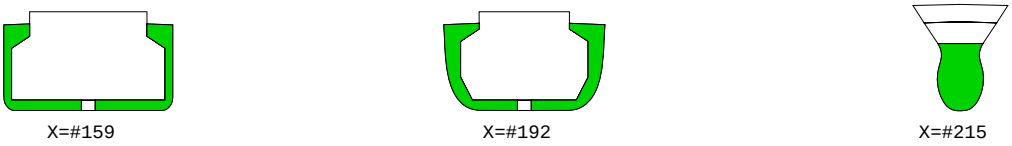


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	332	209	83.32	0.00
5.94	9.00	3482	911	64.05	6.91
18.60	25.00	20335	1720	138.31	76.69
25.80	34.00	34429	2196	110.86	121.59
40.20	52.00	63505	1738	124.27	174.42
53.80	69.00	82114	1075	207.11	185.88
68.20	87.00	92351	291	127.68	182.93
82.60	105.00	91246	-311	223.51	177.15
97.00	123.00	84280	-731	138.53	171.12
111.40	141.00	70702	-1028	232.89	165.10
125.80	159.00	54470	-1297	137.33	158.58
140.20	177.00	33978	-1420	223.55	146.74
152.20	192.00	17624	-1299	141.72	117.92
165.00	208.00	3506	-864	100.69	55.43
169.20	214.00	1011	-374	125.45	34.04
173.40	220.00	103	-103	56.15	15.43

LOADING CASE 'LD01B -- NORMAL BALLAST CONDITION, FPT FULL, HALFWAY







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	12944.8	102.73	0.00	5.34	886.83
Diesel Oil	DO	62.0	11.26	-2.98	8.46	149.80
Fresh Water	FW	159.2	2.90	-0.00	10.95	159.36
Grain Load	GRAIN	0.0	0.00	0.00	0.00	0.00
Heavy Fuel Oil	HFO	783.5	42.79	-0.15	11.15	233.92
Lubricating Oil	LO	76.6	14.10	4.02	6.96	56.99
	MASS	34.6	63.09	0.00	20.29	0.00
Miscellaneous	MIS	27.9	19.18	-1.81	2.34	42.86
Deadweight		14088.7	97.12	-0.00	5.78	1529.76
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		14088.7	97.12	-0.00	5.78	
Total weight		22738.7	90.34	-0.00	7.46	

F L O A T I N G P O S I T I O N

Draught moulded	5.735	m	KM	15.01 m
Trim	-1.351	m	KG	7.46 m
Heel, PS=+	0.0	deg		
TA	6.411	m	GM0	7.55 m
TF	5.060	m	GMCORR	-0.07 m
Trimming moment	-63868	tonm	GM	7.48 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	1.052	mrad	OK
AREA40	Area under GZ curve .	0.090	1.782	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.729	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.367	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	41.178	deg	OK
GM0.15	GM > 0.15 m	0.150	7.482	m	OK
IMOWEATH.	IMO weather criterion	1.000	3.372		OK
VISIBILI.		359.760	197.478		OK
PROPIMMER	Propeller should imm.	50.000	60.958	%	OK

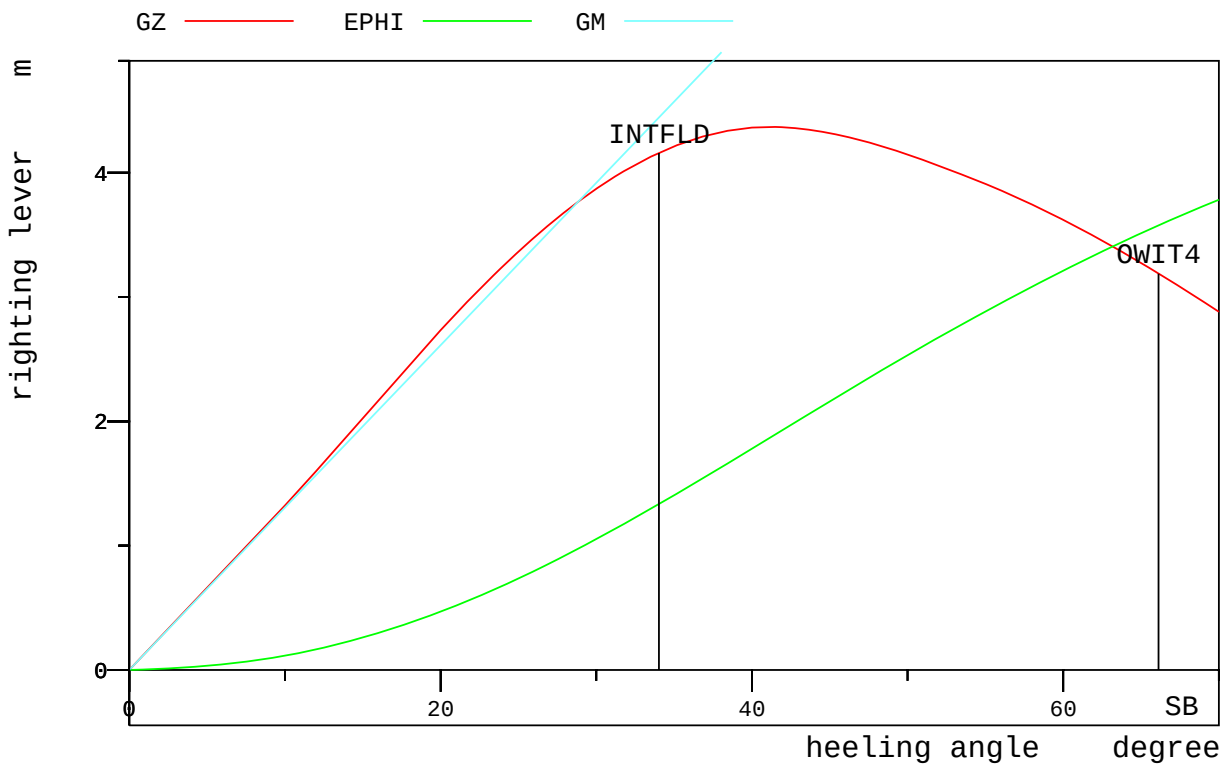
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN,CORR)	-1400.1 t	(23.3%)	POSITION:	138.6 m	175
SHEAR FORCE (MAX,CORR)	1968.2 t	(32.8%)		25.8 m	34
SAGGING MOMENT	-11.0 tm	(0.3%)		176.9 m	225
HOGGING MOMENT	84903.6 tm	(77.8%)		72.8 m	93

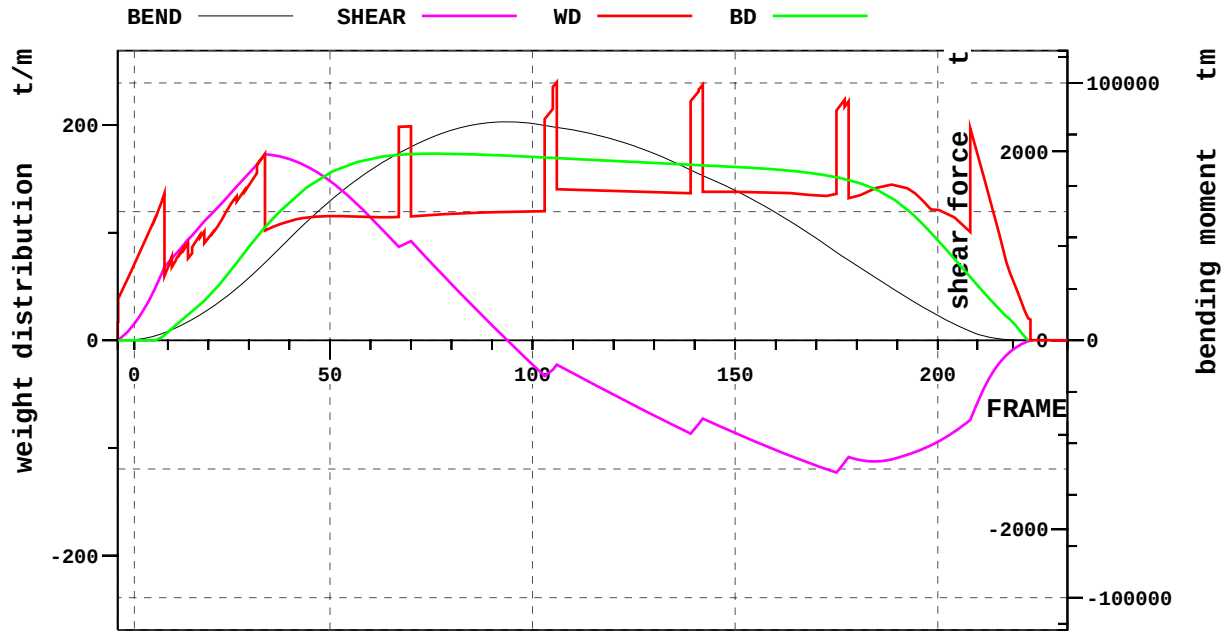
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.740)						
NO.1 CARGO HOLD	0.0	0.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
SUBTOTAL	0.0		0.00	0.00	0.00	0.0
, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	9.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	34.6		63.09	0.00	20.29	0.0
TOTAL	34.6		63.09	0.00	20.29	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	339.3	100.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	774.2	100.0	168.53	0.00	5.89	886.83
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	929.2	100.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	929.2	100.0	39.87	-9.38	3.43	0.00
SUBTOTAL	12944.8		102.73	0.00	5.34	886.83
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	10.3	50.0	8.24	-6.66	8.53	22.22
M.D.O. SETTL. TK.	12.4	50.0	9.83	-6.86	8.34	29.35
M.D.O. STOR. TK. (P)	17.3	50.0	12.68	6.80	8.45	43.37
M.D.O. STOR. TK. (S)	22.0	50.0	12.35	-6.76	8.50	54.86
SUBTOTAL	62.0		11.26	-2.98	8.46	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	20.4	50.0	4.98	7.32	11.97	31.97
FRESH WATER TK. (P)	46.9	50.0	1.02	6.48	12.05	44.74
FRESH WATER TK. (S)	67.2	50.0	2.26	-6.75	12.02	82.65
SUBTOTAL	159.2		2.90	-0.00	10.95	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	131.9	50.0	68.20	12.07	11.98	68.67
NO.1 FUEL OIL TK. (S)	131.9	50.0	68.20	-12.07	11.98	68.67
NO.2 FUEL OIL TK. (P)	127.7	50.0	40.00	12.05	12.00	0.00
NO.2 FUEL OIL TK. (S)	127.7	50.0	40.00	-12.05	12.00	0.00
H.F.O. TK. (P)	126.6	50.0	21.26	10.98	9.26	44.35
H.F.O. TK. (S)	90.0	50.0	21.60	-10.89	8.63	33.83
NO.1 H.F.O. SERV. TK.	12.4	50.0	15.47	-10.92	11.69	7.35
NO.1 H.F.O. SETTL. T.	14.8	50.0	18.64	-11.21	11.66	11.05
NO.2 H.F.O. SERV. TK.	9.7	50.0	12.65	-10.98	11.70	0.00
NO.2 H.F.O. SETTL. T.	11.0	50.0	9.65	-10.54	11.75	0.00
SUBTOTAL	783.5		42.79	-0.15	11.15	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	4.1	60.0	10.01	5.58	8.05	0.36
G/E L.O. STOR. TK.	7.9	60.0	10.02	8.01	8.78	10.28
M/E L.O. SETTL. TK.	10.5	60.0	8.82	6.85	8.65	19.67
M/E L.O. STOR. TK.	16.1	60.0	7.14	6.62	8.83	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	76.6		14.10	4.02	6.96	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	2.3	40.0	18.84	-2.15	1.07	1.69
BILGE TK.	6.4	40.0	11.55	0.00	0.73	9.85
F.O. DRAIN TK.	2.6	40.0	20.71	2.30	0.98	3.33
F.O. OVERFLOW TK.	3.6	40.0	23.05	2.70	0.93	7.97
L.O. DRAIN TK.	1.5	40.0	18.35	2.11	1.09	0.91
SEWAGE STOR. TK.	5.4	40.0	22.38	-2.58	0.94	10.07
SLUDGE TK.	6.0	40.0	21.80	-8.40	7.56	9.03
SUBTOTAL	27.9		19.18	-1.81	2.34	42.86
TOTAL	14054.1		97.20	-0.00	5.74	1529.76

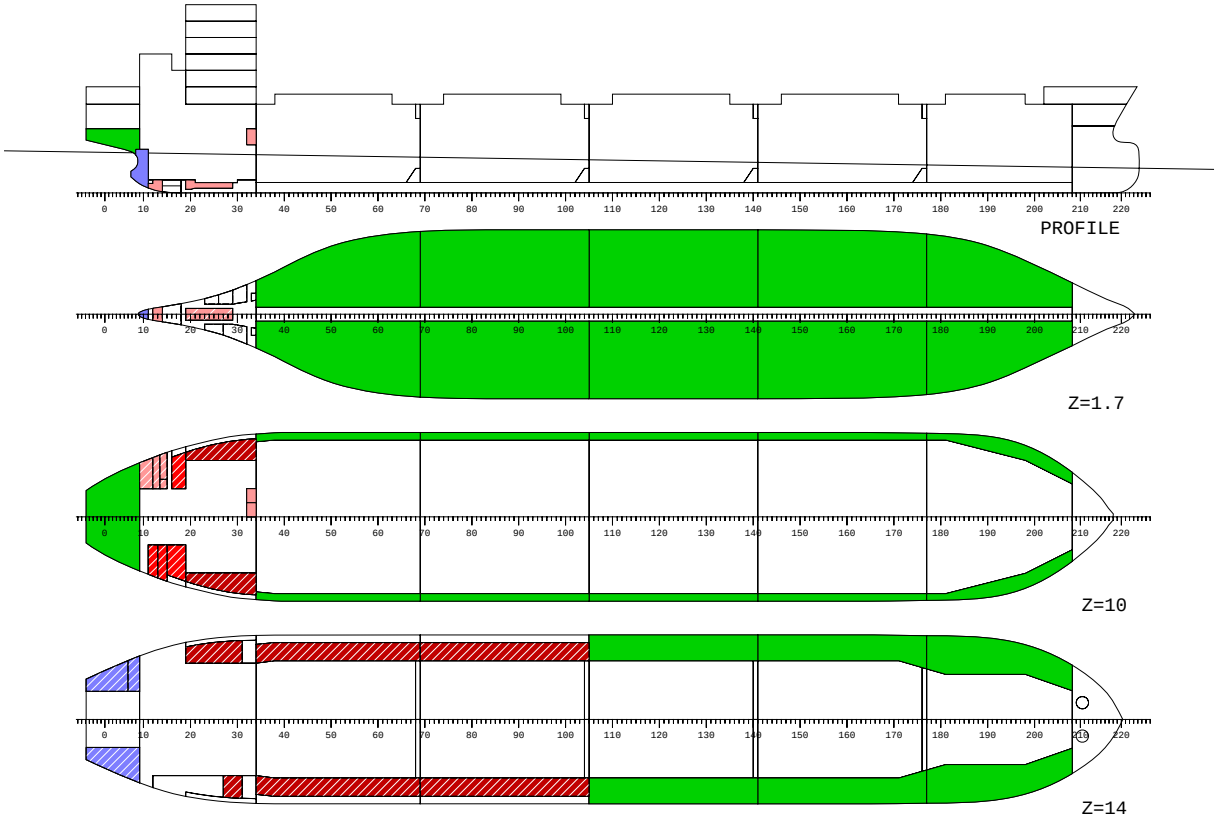


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.002	-0.00	0.000	0.0	0.000
10.0	0.024	1.32	0.115	265.6	0.012
12.0	0.043	1.60	0.166	318.1	0.014
20.0	0.174	2.73	0.469	523.2	0.023
30.0	0.129	3.87	1.052	764.9	0.034
40.0	-0.447	4.36	1.782	983.3	0.043
50.0	-1.586	4.15	2.531	1171.9	0.052
60.0	-2.860	3.62	3.212	1324.8	0.058
70.0	-4.150	2.88	3.782	1437.5	0.063

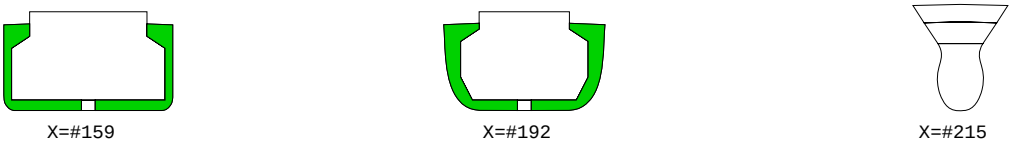


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	271	177	70.71	0.00
5.94	9.00	2933	778	98.31	4.75
18.60	25.00	17640	1531	122.69	62.40
25.80	34.00	30217	1968	102.06	105.16
40.20	52.00	56765	1619	115.35	158.92
53.80	69.00	74240	1031	198.65	172.91
68.20	87.00	84138	288	118.67	172.64
82.60	105.00	82984	-312	214.99	169.54
97.00	123.00	76645	-642	138.53	166.20
111.40	141.00	64715	-887	232.89	162.85
125.80	159.00	50598	-1143	137.33	159.01
140.20	177.00	32130	-1292	223.55	149.81
152.20	192.00	17003	-1218	141.72	122.65
165.00	208.00	3484	-841	100.69	59.02
169.20	214.00	1042	-364	125.45	36.58
173.40	220.00	141	-101	56.15	16.72

LOADING CASE 'LD01C -- NORMAL BALLAST CONDITION, FPT EMPTY, HALFWAY







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	12170.6	98.54	0.00	5.30	886.83
Diesel Oil	DO	62.0	11.26	-2.98	8.46	149.80
Fresh Water	FW	159.2	2.90	-0.00	10.95	159.36
Grain Load	GRAIN	0.0	0.00	0.00	0.00	0.00
Heavy Fuel Oil	HFO	783.5	42.79	-0.15	11.15	233.92
Lubricating Oil	LO	76.6	14.10	4.02	6.96	56.99
	MASS	34.6	63.09	0.00	20.29	0.00
Miscellaneous	MIS	27.9	19.18	-1.81	2.34	42.86
Deadweight		13314.5	92.97	-0.00	5.77	1529.76
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		13314.5	92.97	-0.00	5.77	
Total weight		21964.5	87.58	-0.00	7.52	

F L O A T I N G P O S I T I O N

Draught moulded	5.582	m	KM	15.43 m
Trim	-2.638	m	KG	7.52 m
Heel, PS=+	0.0	deg		
TA	6.901	m	GM0	7.92 m
TF	4.263	m	GMCORR	-0.07 m
Trimming moment	-127510	tonm	GM	7.85 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	1.090	mrad	OK
AREA40	Area under GZ curve .	0.090	1.829	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.739	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.410	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	41.077	deg	OK
GM0.15	GM > 0.15 m	0.150	7.847	m	OK
IMOWEATH.	IMO weather criterion	1.000	3.177		OK
VISIBILI.		359.760	233.660		OK
PROPIMMER	Propeller should imm.	50.000	70.205	%	OK

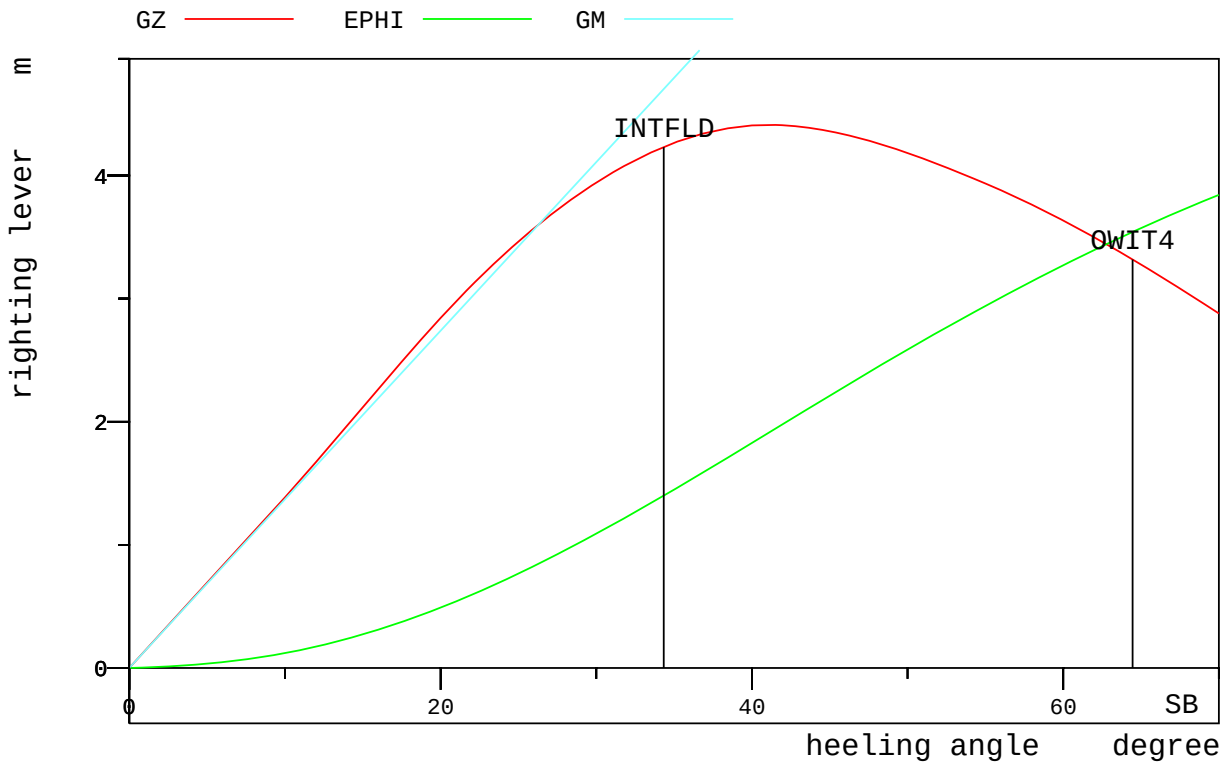
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN,CORR)	-1099.3 t	(18.3%)	POSITION:	138.4 m	175
SHEAR FORCE (MAX,CORR)	1850.2 t	(30.8%)		25.8 m	34
SAGGING MOMENT	-31.4 tm	(0.9%)		176.9 m	225
HOGGING MOMENT	73569.0 tm	(67.4%)		68.2 m	87

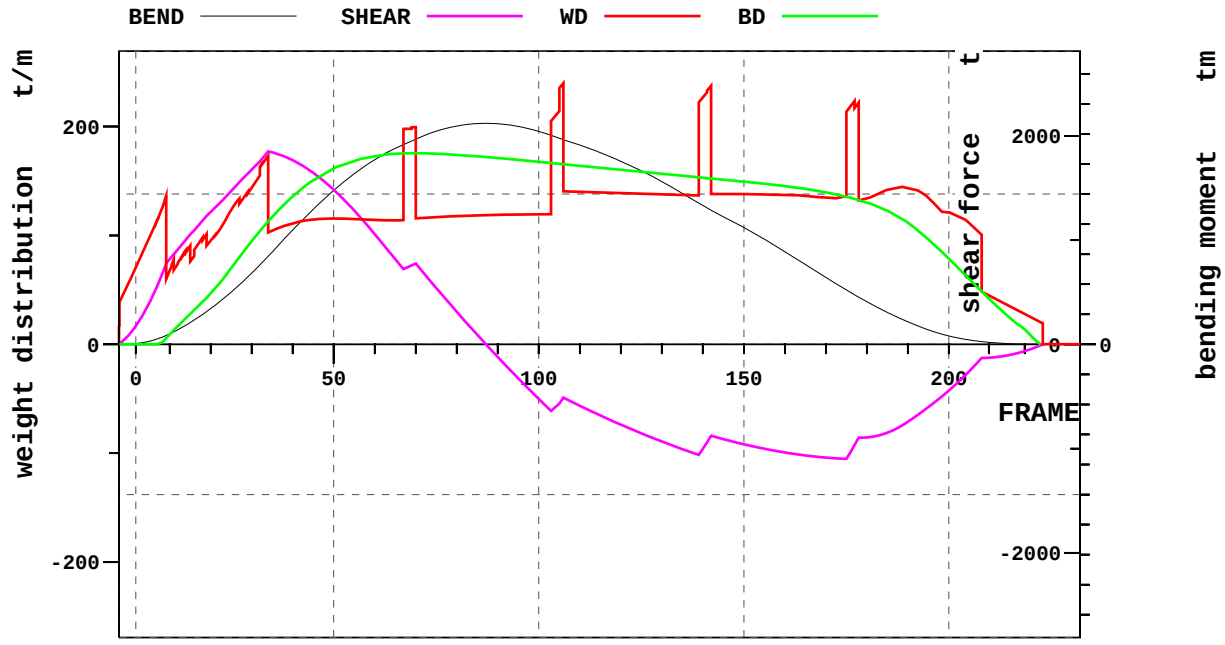
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.740)						
NO.1 CARGO HOLD	0.0	0.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
SUBTOTAL	0.0		0.00	0.00	0.00	0.0
, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	9.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	34.6		63.09	0.00	20.29	0.0
TOTAL	34.6		63.09	0.00	20.29	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	339.3	100.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	0.00	0.00	0.00	886.83
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	929.2	100.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	929.2	100.0	39.87	-9.38	3.43	0.00
SUBTOTAL	12170.6		98.54	0.00	5.30	886.83
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	10.3	50.0	8.24	-6.66	8.53	22.22
M.D.O. SETTL. TK.	12.4	50.0	9.83	-6.86	8.34	29.35
M.D.O. STOR. TK. (P)	17.3	50.0	12.68	6.80	8.45	43.37
M.D.O. STOR. TK. (S)	22.0	50.0	12.35	-6.76	8.50	54.86
SUBTOTAL	62.0		11.26	-2.98	8.46	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	20.4	50.0	4.98	7.32	11.97	31.97
FRESH WATER TK. (P)	46.9	50.0	1.02	6.48	12.05	44.74
FRESH WATER TK. (S)	67.2	50.0	2.26	-6.75	12.02	82.65
SUBTOTAL	159.2		2.90	-0.00	10.95	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	131.9	50.0	68.20	12.07	11.98	68.67
NO.1 FUEL OIL TK. (S)	131.9	50.0	68.20	-12.07	11.98	68.67
NO.2 FUEL OIL TK. (P)	127.7	50.0	40.00	12.05	12.00	0.00
NO.2 FUEL OIL TK. (S)	127.7	50.0	40.00	-12.05	12.00	0.00
H.F.O. TK. (P)	126.6	50.0	21.26	10.98	9.26	44.35
H.F.O. TK. (S)	90.0	50.0	21.60	-10.89	8.63	33.83
NO.1 H.F.O. SERV. TK.	12.4	50.0	15.47	-10.92	11.69	7.35
NO.1 H.F.O. SETTL. T.	14.8	50.0	18.64	-11.21	11.66	11.05
NO.2 H.F.O. SERV. TK.	9.7	50.0	12.65	-10.98	11.70	0.00
NO.2 H.F.O. SETTL. T.	11.0	50.0	9.65	-10.54	11.75	0.00
SUBTOTAL	783.5		42.79	-0.15	11.15	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	4.1	60.0	10.01	5.58	8.05	0.36
G/E L.O. STOR. TK.	7.9	60.0	10.02	8.01	8.78	10.28
M/E L.O. SETTL. TK.	10.5	60.0	8.82	6.85	8.65	19.67
M/E L.O. STOR. TK.	16.1	60.0	7.14	6.62	8.83	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	76.6		14.10	4.02	6.96	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	2.3	40.0	18.84	-2.15	1.07	1.69
BILGE TK.	6.4	40.0	11.55	0.00	0.73	9.85
F.O. DRAIN TK.	2.6	40.0	20.71	2.30	0.98	3.33
F.O. OVERFLOW TK.	3.6	40.0	23.05	2.70	0.93	7.97
L.O. DRAIN TK.	1.5	40.0	18.35	2.11	1.09	0.91
SEWAGE STOR. TK.	5.4	40.0	22.38	-2.58	0.94	10.07
SLUDGE TK.	6.0	40.0	21.80	-8.40	7.56	9.03
SUBTOTAL	27.9		19.18	-1.81	2.34	42.86
TOTAL	13279.9		93.05	-0.00	5.73	1529.76

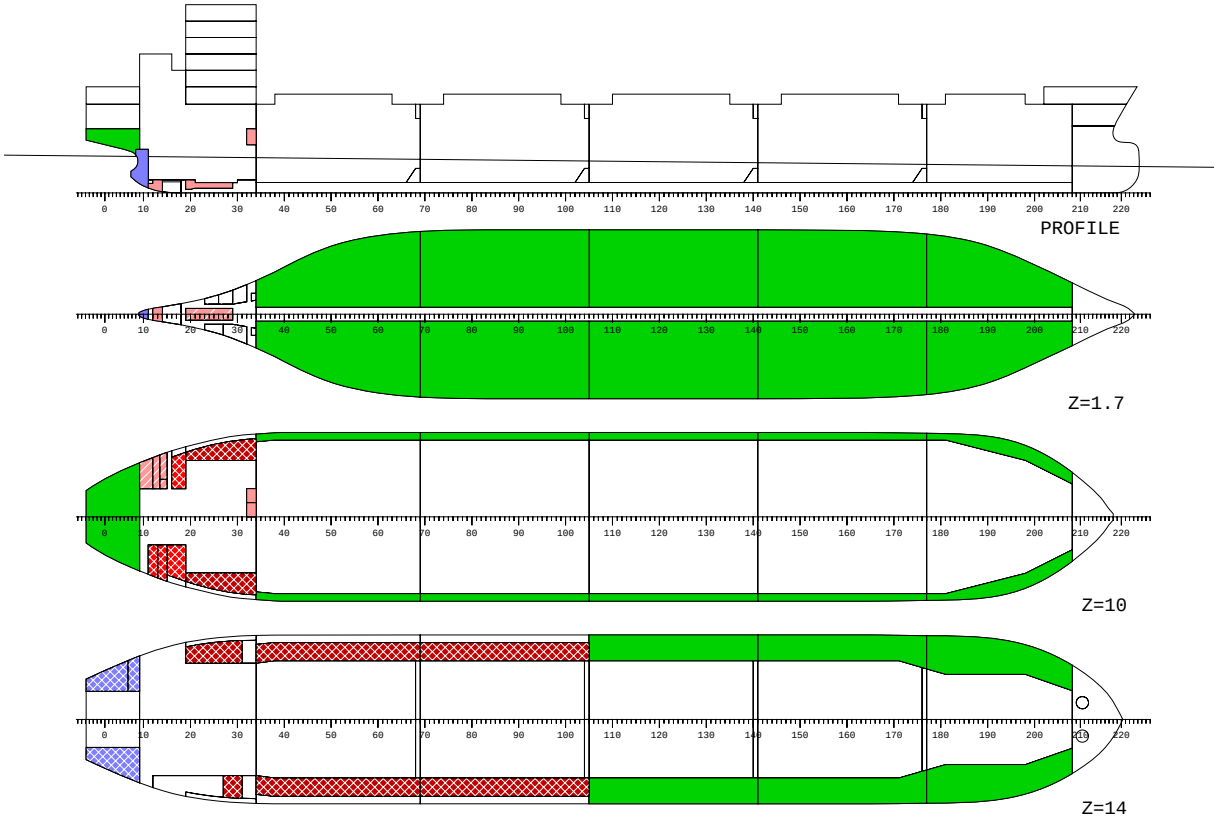


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.002	-0.00	0.000	0.0	0.000
10.0	0.023	1.39	0.121	265.6	0.012
12.0	0.041	1.67	0.174	318.1	0.014
20.0	0.160	2.84	0.490	523.2	0.024
30.0	0.019	3.94	1.090	764.9	0.035
40.0	-0.637	4.41	1.829	983.3	0.045
50.0	-1.829	4.18	2.586	1171.9	0.053
60.0	-3.162	3.63	3.271	1324.8	0.060
70.0	-4.494	2.88	3.842	1437.5	0.065

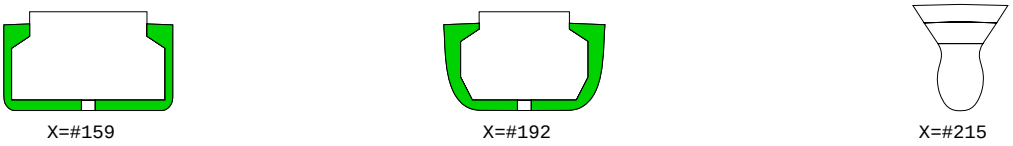


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	281	177	70.78	0.00
5.94	9.00	2960	778	98.23	5.54
18.60	25.00	17398	1470	122.83	69.81
25.80	34.00	29354	1850	137.86	113.04
40.20	52.00	53523	1406	115.33	164.53
53.80	69.00	67688	757	199.26	175.51
68.20	87.00	73569	4	118.67	172.05
82.60	105.00	68523	-569	214.35	165.77
97.00	123.00	59030	-821	138.53	159.24
111.40	141.00	45389	-943	232.89	152.71
125.80	159.00	31665	-1030	137.33	145.68
140.20	177.00	16351	-965	223.55	133.55
152.20	192.00	6388	-689	141.72	105.77
165.00	208.00	938	-130	100.69	48.84
169.20	214.00	435	-113	38.08	29.77
173.40	220.00	75	-64	27.85	13.40

LOADING CASE 'LD01D -- NORMAL BALLAST CONDITION, ARRIVAL







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	12170.6	98.54	0.00	5.30	886.83
Diesel Oil	DO	12.4	11.39	-2.61	7.47	149.80
Fresh Water	FW	51.6	4.32	-0.00	8.25	159.36
Grain Load	GRAIN	0.0	0.00	0.00	0.00	0.00
Heavy Fuel Oil	HFO	156.7	43.37	-0.16	10.03	233.92
Lubricating Oil	LO	50.8	16.95	2.45	5.88	56.99
	MASS	27.6	75.87	0.00	20.42	0.00
Miscellaneous	MIS	55.8	19.16	-1.81	2.67	42.86
Deadweight		12525.6	96.64	-0.00	5.40	1529.76
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		12525.6	96.64	-0.00	5.40	
Total weight		21175.6	89.56	-0.00	7.36	

F L O A T I N G P O S I T I O N

Draught moulded	5.384	m	KM	15.62 m
Trim	-1.727	m	KG	7.36 m
Heel, PS=+	0.0	deg		
TA	6.248	m	GM0	8.26 m
TF	4.521	m	GMCORR	-0.07 m
Trimming moment	-80437	tonm	GM	8.18 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	1.131	mrad	OK
AREA40	Area under GZ curve .	0.090	1.890	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.759	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.547	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	41.727	deg	OK
GM0.15	GM > 0.15 m	0.150	8.185	m	OK
IMOWEATH.	IMO weather criterion	1.000	3.087		OK
VISIBILI.		359.760	211.678		OK
PROPIMMER	Propeller should imm.	50.000	57.887	%	OK

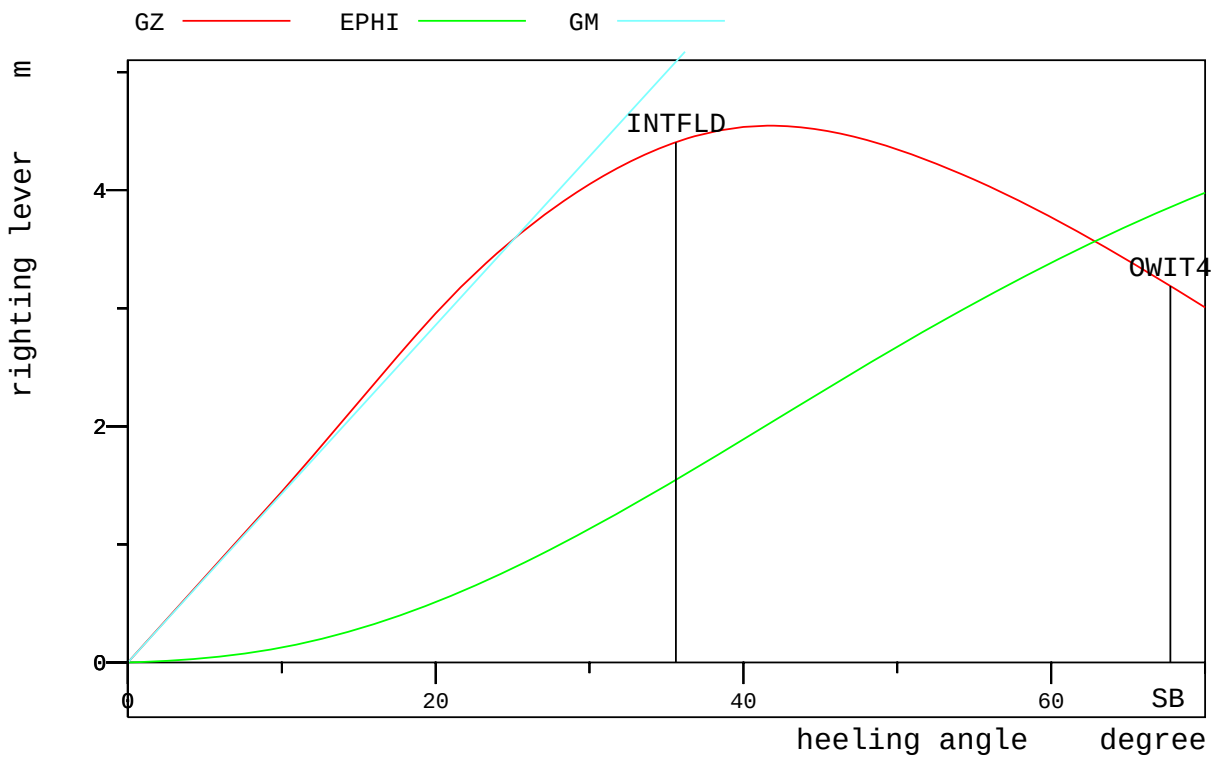
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-987.9 t	(16.5%)	POSITION:	138.6 m	175
SHEAR FORCE (MAX)	1657.7 t	(27.6%)		25.8 m	34
SAGGING MOMENT	-19.9 tm	(0.6%)		176.9 m	225
HOGGING MOMENT	66546.4 tm	(61.0%)		68.2 m	87

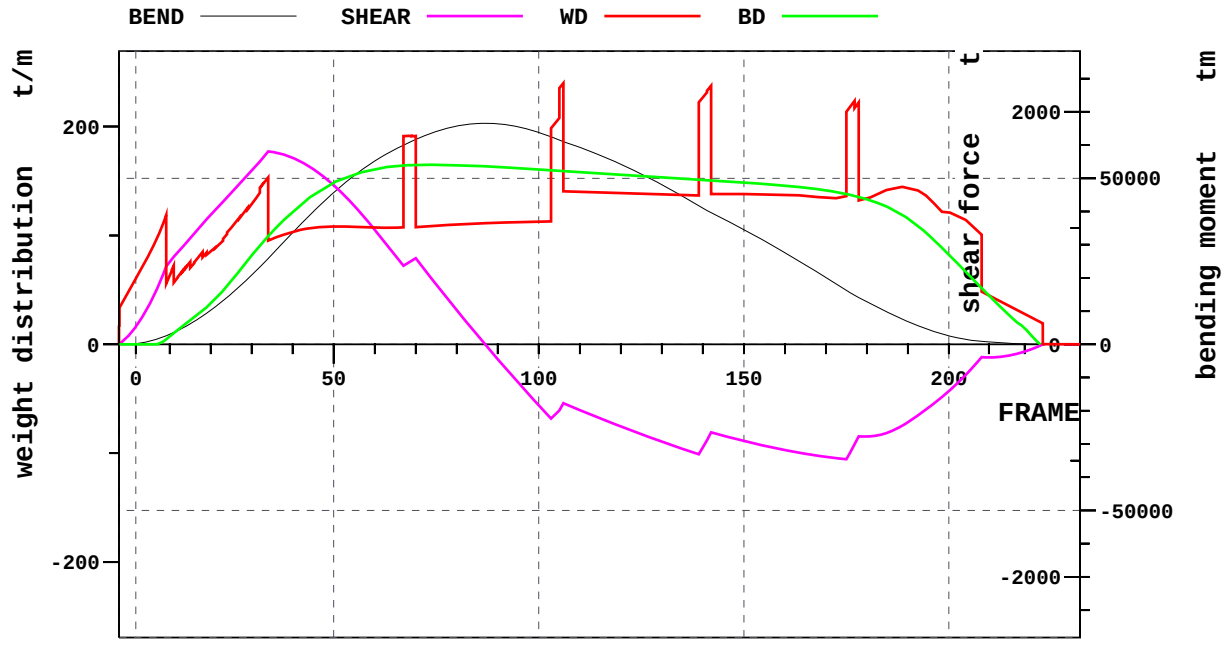
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.740)						
NO.1 CARGO HOLD	0.0	0.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
SUBTOTAL	0.0		0.00	0.00	0.00	0.0
, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 10%	2.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	99.00	0.00	19.20	0.0
SUBTOTAL	27.6		75.87	0.00	20.42	0.0
TOTAL	27.6		75.87	0.00	20.42	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	339.3	100.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	0.00	0.00	0.00	886.83
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	929.2	100.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	929.2	100.0	39.87	-9.38	3.43	0.00
SUBTOTAL	12170.6		98.54	0.00	5.30	886.83
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	2.1	10.0	8.28	-5.84	7.59	22.22
M.D.O. SETTL. TK.	2.5	10.0	9.88	-5.95	7.33	29.35
M.D.O. STOR. TK. (P)	3.5	10.0	12.81	5.91	7.45	43.37
M.D.O. STOR. TK. (S)	4.4	10.0	12.58	-5.91	7.52	54.86
SUBTOTAL	12.4		11.39	-2.61	7.47	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	4.1	10.0	4.98	7.15	11.13	31.97
FRESH WATER TK. (P)	9.4	10.0	1.14	6.32	11.15	44.74
FRESH WATER TK. (S)	13.4	10.0	2.40	-6.59	11.14	82.65
SUBTOTAL	51.6		4.32	-0.00	8.25	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	26.4	10.0	68.20	12.64	11.22	68.67
NO.1 FUEL OIL TK. (S)	26.4	10.0	68.20	-12.64	11.22	68.67
NO.2 FUEL OIL TK. (P)	25.5	10.0	40.23	12.62	11.23	0.00
NO.2 FUEL OIL TK. (S)	25.5	10.0	40.23	-12.62	11.23	0.00
H.F.O. TK. (P)	25.3	10.0	23.08	10.63	7.06	44.35
H.F.O. TK. (S)	18.0	10.0	23.44	-10.57	6.80	33.83
NO.1 H.F.O. SERV. TK.	2.5	10.0	15.49	-10.82	11.07	7.35
NO.1 H.F.O. SETTL. T.	3.0	10.0	18.65	-11.14	11.06	11.05
NO.2 H.F.O. SERV. TK.	1.9	10.0	12.65	-10.87	11.07	0.00
NO.2 H.F.O. SETTL. T.	2.2	10.0	9.70	-10.42	11.09	0.00
SUBTOTAL	156.7		43.37	-0.16	10.03	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	1.4	20.0	10.03	5.53	7.22	0.36
G/E L.O. STOR. TK.	2.6	20.0	10.04	7.48	7.96	10.28
M/E L.O. SETTL. TK.	3.5	20.0	8.83	6.21	7.80	19.67
M/E L.O. STOR. TK.	5.4	20.0	7.19	6.05	8.05	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	50.8		16.95	2.45	5.88	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	4.5	80.0	18.81	-2.29	1.40	1.69
BILGE TK.	12.9	80.0	11.52	0.00	1.14	9.85
F.O. DRAIN TK.	5.2	80.0	20.70	2.51	1.34	3.33
F.O. OVERFLOW TK.	7.2	80.0	23.04	2.98	1.31	7.97
L.O. DRAIN TK.	3.1	80.0	18.32	2.24	1.42	0.91
SEWAGE STOR. TK.	10.9	80.0	22.37	-2.83	1.31	10.07
SLUDGE TK.	12.0	80.0	21.80	-8.40	7.72	9.03
SUBTOTAL	55.8		19.16	-1.81	2.67	42.86
TOTAL	12498.0		96.69	-0.00	5.37	1529.76

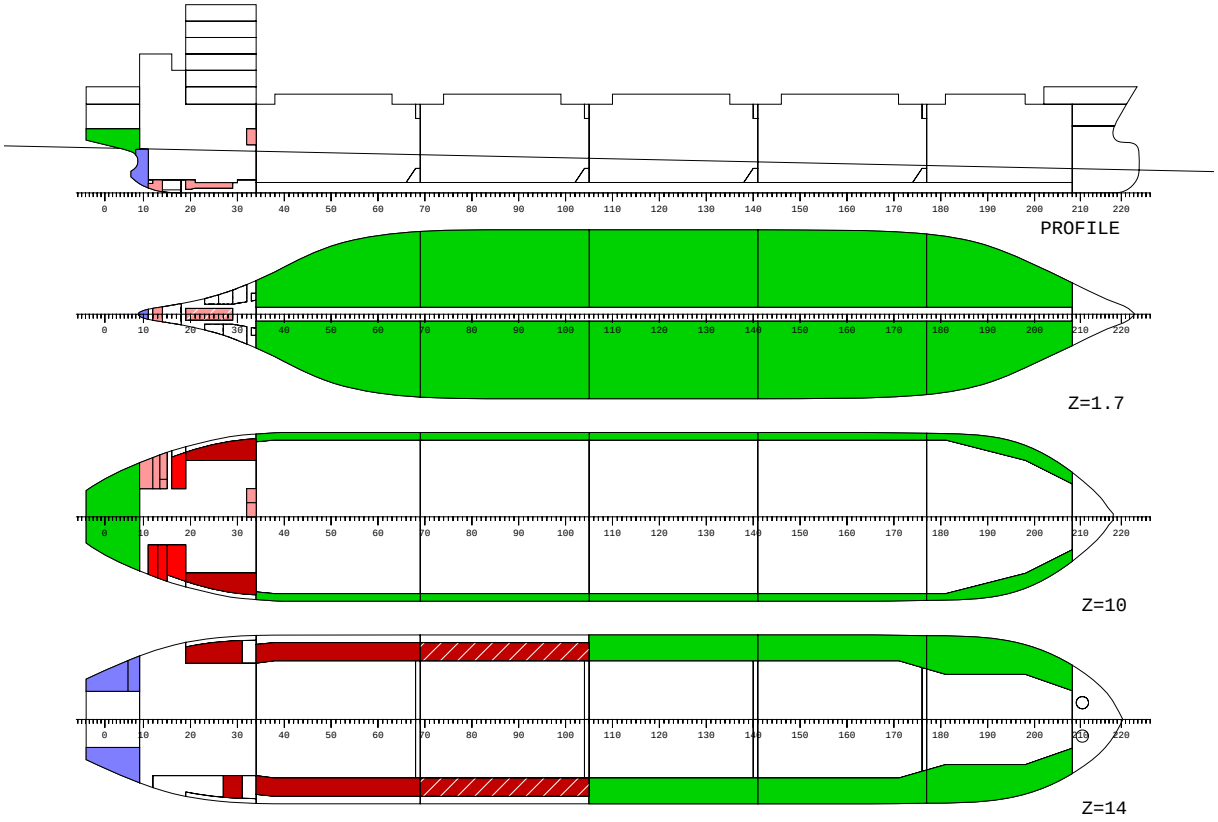


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.002	0.00	0.000	0.0	0.000
10.0	0.025	1.45	0.126	265.6	0.013
12.0	0.044	1.75	0.182	318.1	0.015
20.0	0.157	2.96	0.511	523.2	0.025
30.0	-0.042	4.05	1.131	764.9	0.036
40.0	-0.725	4.54	1.890	983.3	0.046
50.0	-1.925	4.35	2.673	1171.9	0.055
60.0	-3.318	3.77	3.385	1324.8	0.063
70.0	-4.684	3.01	3.978	1437.5	0.068

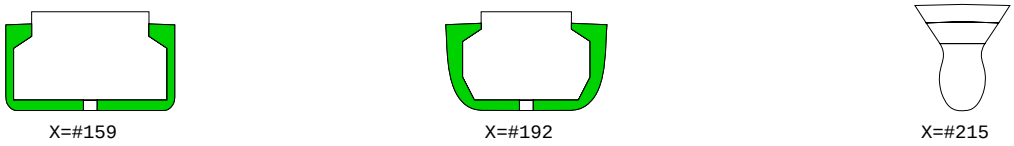


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	240	153	60.82	0.00
5.94	9.00	2542	671	87.24	4.61
18.60	25.00	15160	1302	107.02	58.30
25.80	34.00	25814	1658	95.30	99.43
40.20	52.00	47817	1299	108.01	151.52
53.80	69.00	60999	719	191.53	164.63
68.20	87.00	66546	-5	111.33	163.42
82.60	105.00	61434	-569	207.89	159.39
97.00	123.00	52497	-746	138.53	155.12
111.40	141.00	40266	-825	232.89	150.84
125.80	159.00	28338	-901	137.33	146.07
140.20	177.00	14741	-858	223.55	136.15
152.20	192.00	5821	-622	141.72	109.72
165.00	208.00	889	-110	100.70	51.83
169.20	214.00	437	-104	38.08	31.89
173.40	220.00	90	-62	27.85	14.49

LOADING CASE 'LD01E -- NORMAL BALLAST CONDITION, FOR IACS, DEPARTURE







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	12170.6	98.54	0.00	5.30	886.83
Diesel Oil	DO	121.6	11.23	-3.17	9.29	149.80
Fresh Water	FW	293.6	2.53	0.00	12.34	159.36
Grain Load	GRAIN	0.0	0.00	0.00	0.00	0.00
Heavy Fuel Oil	HFO	1535.7	42.68	-0.15	12.40	233.92
Lubricating Oil	LO	101.1	12.73	4.87	7.77	56.99
	MASS	43.6	53.17	0.00	20.19	0.00
Miscellaneous	MIS	7.0	19.24	-1.81	1.99	42.86
Deadweight		14273.3	89.03	-0.01	6.31	1529.76
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		14273.3	89.03	-0.01	6.31	
Total weight		22923.3	85.36	-0.01	7.78	

F L O A T I N G P O S I T I O N

Draught moulded	5.810	m	KM	15.25 m
Trim	-3.676	m	KG	7.78 m
Heel, PS=+	-0.0	deg		
TA	7.648	m	GM0	7.47 m
TF	3.972	m	GMCORR	-0.07 m
Trimming moment	-188025	tonm	GM	7.41 m
Heel	-0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	1.032	mrad	OK
AREA40	Area under GZ curve .	0.090	1.737	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.705	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.187	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	40.345	deg	OK
GM0.15	GM > 0.15 m	0.150	7.405	m	OK
IMOWEATH.	IMO weather criterion	1.000	3.255		OK
VISIBILI.		359.760	263.581		OK
PROPIMMER	Propeller should imm.	50.000	84.296	%	OK

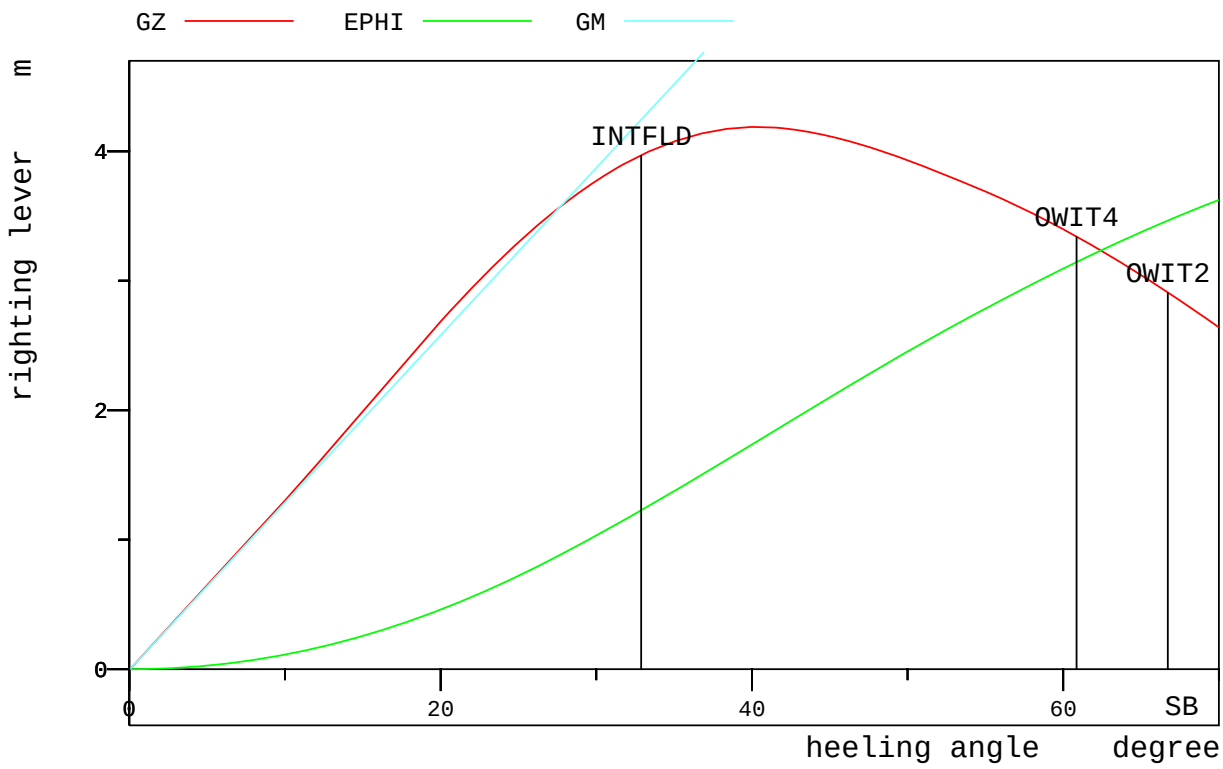
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN,CORR)	-1225.8 t	(20.4%)	POSITION:	136.6 m	172
SHEAR FORCE (MAX,CORR)	2057.7 t	(34.3%)		25.8 m	34
SAGGING MOMENT	-68.6 tm	(1.6%)		176.5 m	224
HOGGING MOMENT	80972.1 tm	(74.2%)		68.2 m	87

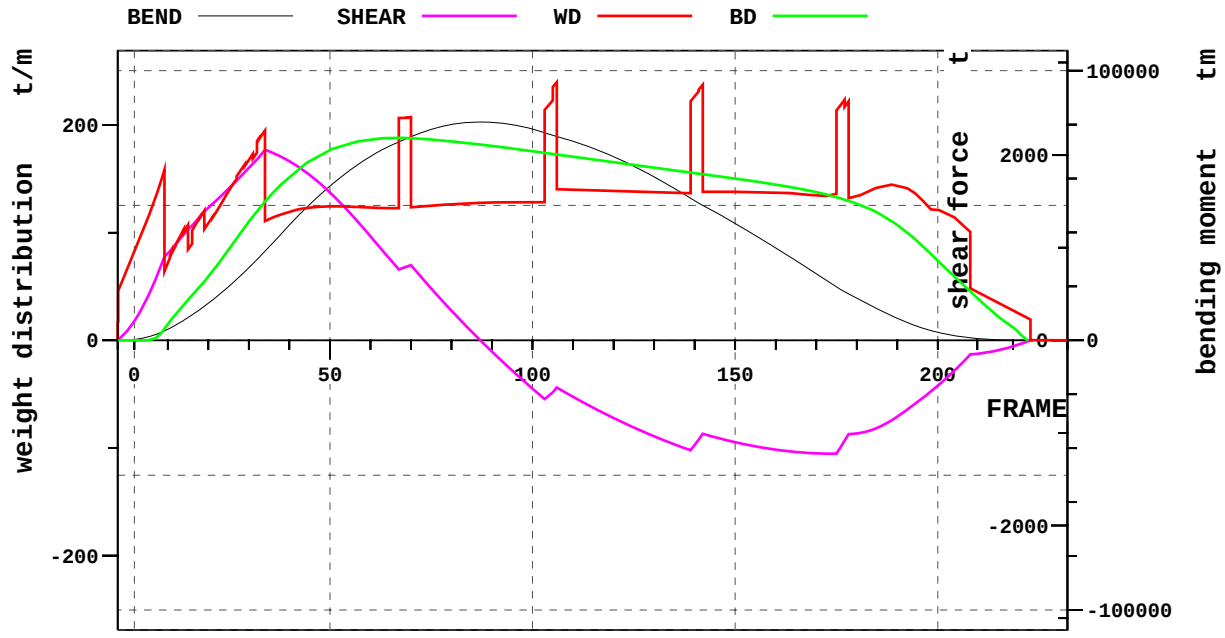
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.740)						
NO.1 CARGO HOLD	0.0	0.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
SUBTOTAL	0.0		0.00	0.00	0.00	0.0
, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	18.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	43.6		53.17	0.00	20.19	0.0
TOTAL	43.6		53.17	0.00	20.19	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	339.3	100.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	168.53	0.00	5.89	886.83
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	929.2	100.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	929.2	100.0	39.87	-9.38	3.43	0.00
SUBTOTAL	12170.6		98.54	0.00	5.30	886.83
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	20.2	98.0	8.23	-7.08	9.34	22.22
M.D.O. SETTL. TK.	24.2	98.0	9.82	-7.32	9.21	29.35
M.D.O. STOR. TK. (P)	33.9	98.0	12.66	7.26	9.29	43.37
M.D.O. STOR. TK. (S)	43.2	98.0	12.30	-7.22	9.32	54.86
SUBTOTAL	121.6		11.23	-3.17	9.29	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	40.7	100.0	4.97	7.43	12.96	31.97
FRESH WATER TK. (P)	93.7	100.0	0.94	6.60	13.03	44.74
FRESH WATER TK. (S)	134.4	100.0	2.17	-6.85	13.01	82.65
SUBTOTAL	293.6		2.53	0.00	12.34	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	258.4	98.0	68.20	11.73	13.08	68.67
NO.1 FUEL OIL TK. (S)	258.4	98.0	68.20	-11.73	13.08	68.67
NO.2 FUEL OIL TK. (P)	250.2	98.0	39.86	11.71	13.09	0.00
NO.2 FUEL OIL TK. (S)	250.2	98.0	39.86	-11.71	13.09	0.00
H.F.O. TK. (P)	248.1	98.0	20.60	11.15	11.04	44.35
H.F.O. TK. (S)	176.3	98.0	21.94	-11.14	10.32	33.83
NO.1 H.F.O. SERV. TK.	24.3	98.0	15.46	-11.00	12.38	7.35
NO.1 H.F.O. SETTL. T.	29.0	98.0	18.63	-11.27	12.36	11.05
NO.2 H.F.O. SERV. TK.	19.1	98.0	12.64	-11.07	12.39	0.00
NO.2 H.F.O. SETTL. T.	21.5	98.0	9.62	-10.64	12.43	0.00
SUBTOTAL	1535.7		42.68	-0.15	12.40	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	6.8	98.0	10.01	5.59	8.82	0.36
G/E L.O. STOR. TK.	12.9	98.0	10.02	8.28	9.39	10.28
M/E L.O. SETTL. TK.	17.2	98.0	8.81	7.17	9.29	19.67
M/E L.O. STOR. TK.	26.4	98.0	7.12	6.90	9.42	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	101.1		12.73	4.87	7.77	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	0.6	10.0	18.98	-1.96	0.68	1.69
BILGE TK.	1.6	10.0	11.66	0.00	0.32	9.85
F.O. DRAIN TK.	0.6	10.0	20.77	2.04	0.56	3.33
F.O. OVERFLOW TK.	0.9	10.0	23.05	2.35	0.50	7.97
L.O. DRAIN TK.	0.4	10.0	18.43	1.94	0.72	0.91
SEWAGE STOR. TK.	1.4	10.0	22.44	-2.26	0.52	10.07
SLUDGE TK.	1.5	10.0	21.80	-8.40	7.44	9.03
SUBTOTAL	7.0		19.24	-1.81	1.99	42.86
TOTAL	14229.7		89.14	-0.01	6.26	1529.76

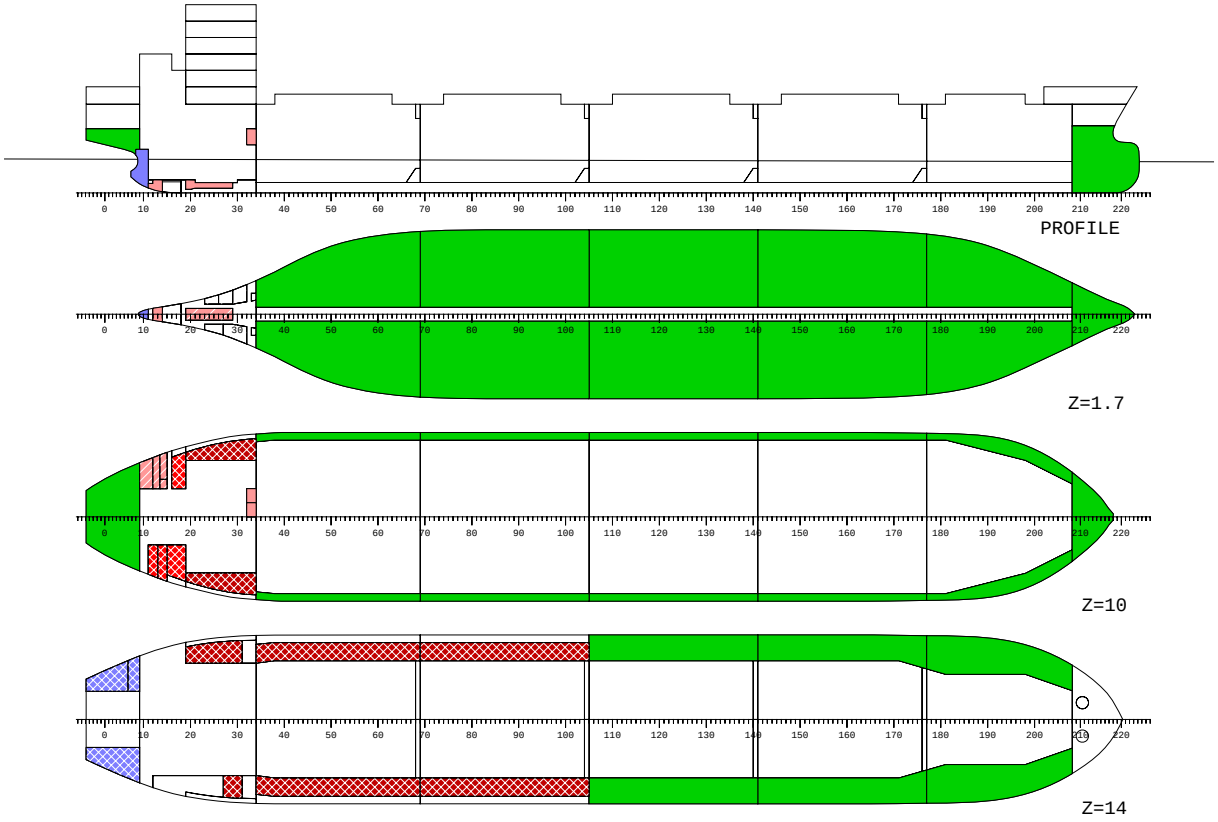


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.006	-0.01	0.000	0.0	0.000
10.0	0.017	1.30	0.113	265.6	0.012
12.0	0.034	1.57	0.163	318.1	0.014
20.0	0.152	2.68	0.461	523.2	0.023
30.0	0.070	3.77	1.032	764.9	0.033
40.0	-0.573	4.19	1.737	983.3	0.043
50.0	-1.743	3.93	2.452	1171.9	0.051
60.0	-3.016	3.40	3.094	1324.8	0.058
70.0	-4.318	2.64	3.624	1437.5	0.063

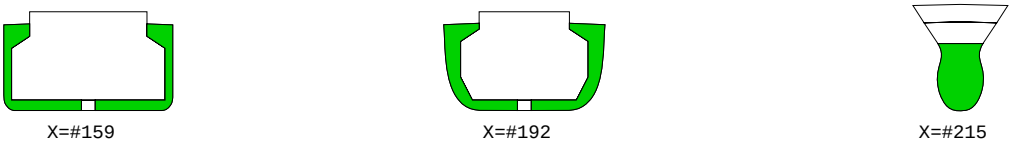


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	344	209	83.32	0.01
5.94	9.00	3509	904	64.09	10.59
18.60	25.00	19887	1637	138.37	84.19
25.80	34.00	33209	2058	110.86	129.03
40.20	52.00	59639	1508	124.40	179.43
53.80	69.00	74804	796	206.81	187.99
68.20	87.00	80972	5	127.81	181.96
82.60	105.00	76009	-562	223.20	173.11
97.00	123.00	66013	-902	138.53	164.01
111.40	141.00	50880	-1073	232.89	154.91
125.80	159.00	35207	-1174	137.33	145.31
140.20	177.00	18021	-1085	223.55	130.66
152.20	192.00	6927	-763	141.72	101.35
165.00	208.00	935	-151	100.69	45.49
169.20	214.00	388	-122	38.08	27.40
173.40	220.00	25	-66	27.85	12.18

LOADING CASE 'LD01F -- NORMAL BALLAST CONDITION, FOR IACS, ARRIVAL







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	12944.8	102.73	0.00	5.34	886.83
Diesel Oil	DO	12.4	11.39	-2.61	7.47	149.80
Fresh Water	FW	51.6	4.32	-0.00	8.25	159.36
Grain Load	GRAIN	0.0	0.00	0.00	0.00	0.00
Heavy Fuel Oil	HFO	156.7	43.37	-0.16	10.03	233.92
Lubricating Oil	LO	50.8	16.95	2.45	5.88	56.99
	MASS	27.6	75.87	0.00	20.42	0.00
Miscellaneous	MIS	55.8	19.16	-1.81	2.67	42.86
Deadweight		13299.8	100.83	-0.00	5.43	1529.76
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		13299.8	100.83	-0.00	5.43	
Total weight		21949.8	92.34	-0.00	7.31	

F L O A T I N G P O S I T I O N

Draught moulded	5.532	m	KM	15.19 m
Trim	-0.411	m	KG	7.31 m
Heel, PS=+	0.0	deg		
TA	5.738	m	GM0	7.88 m
TF	5.326	m	GMCORR	-0.07 m
Trimming moment	-18834	tonm	GM	7.81 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	1.091	mrad	OK
AREA40	Area under GZ curve .	0.090	1.840	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.748	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.496	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	41.758	deg	OK
GM0.15	GM > 0.15 m	0.150	7.809	m	OK
IMOWEATH.	IMO weather criterion	1.000	3.282		OK
VISIBILI.		359.760	179.614		OK
PROPIMMER	Propeller should imm.	50.000	48.259	%	NOT MET

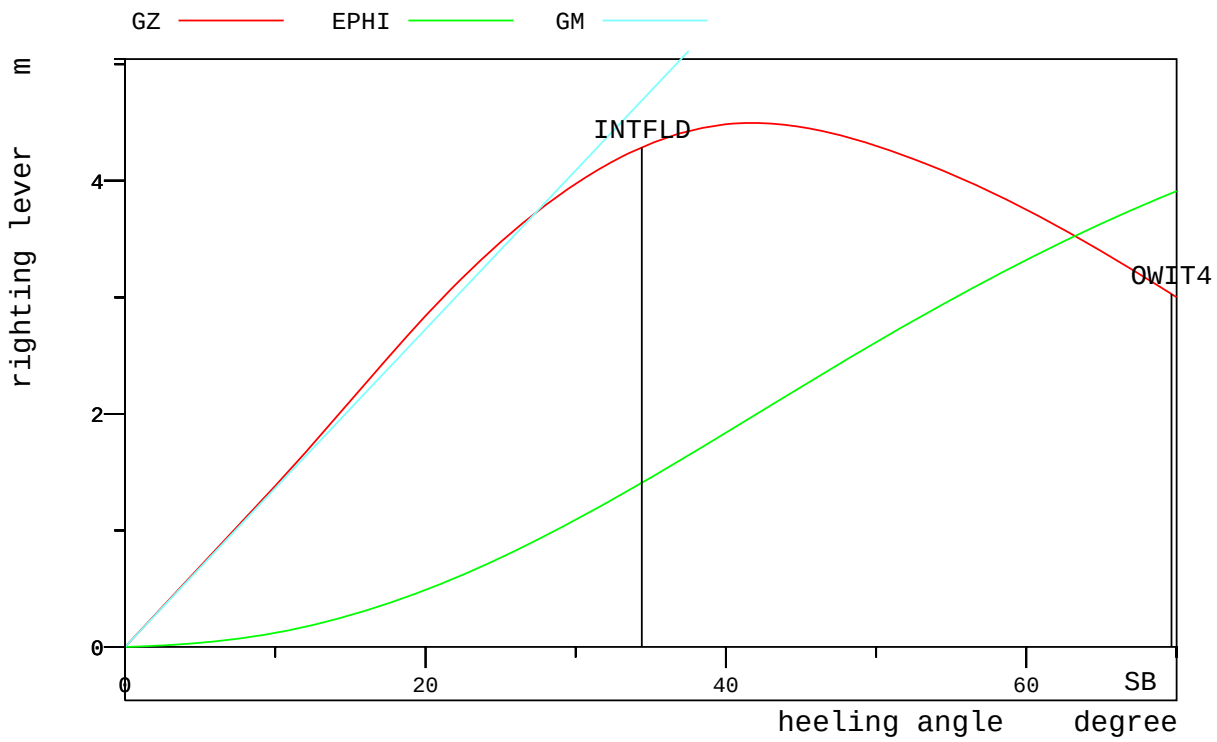
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-1283.0 t	(21.4%)	POSITION:	138.6 m	175
SHEAR FORCE (MAX)	1764.0 t	(29.4%)		27.4 m	36
SAGGING MOMENT	-				
HOGGING MOMENT	77424.0 tm	(71.0%)		72.8 m	93

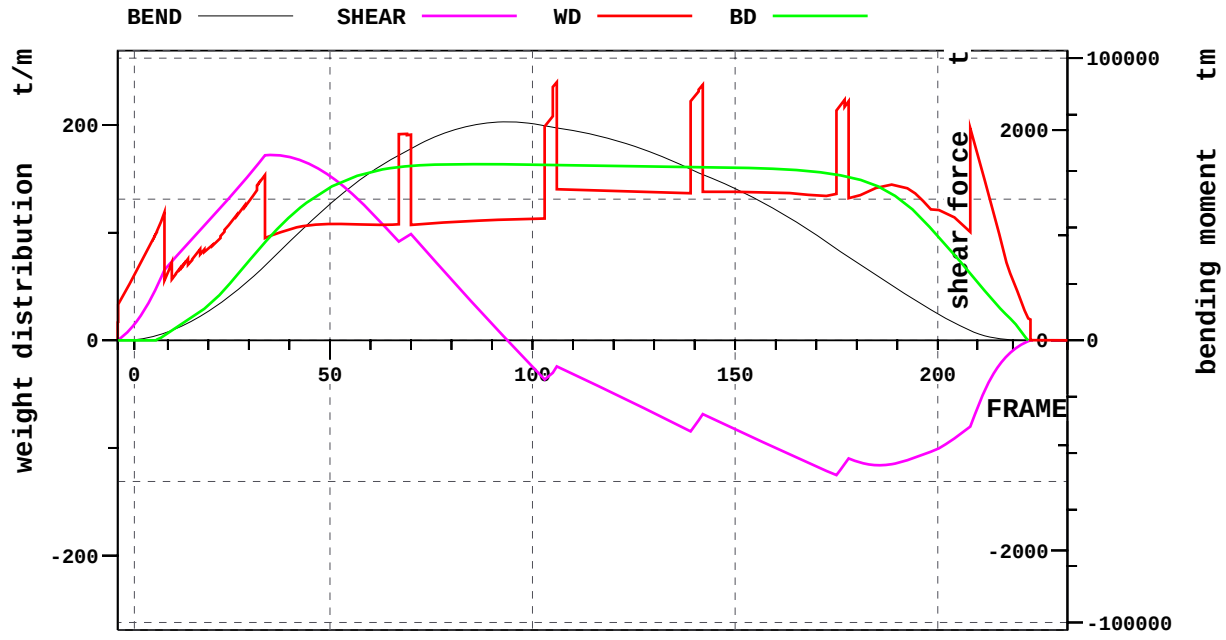
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.740)						
NO.1 CARGO HOLD	0.0	0.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
SUBTOTAL	0.0		0.00	0.00	0.00	0.0
, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 10%	2.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	99.00	0.00	19.20	0.0
SUBTOTAL	27.6		75.87	0.00	20.42	0.0
TOTAL	27.6		75.87	0.00	20.42	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	339.3	100.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	774.2	100.0	168.53	0.00	5.89	886.83
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	929.2	100.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	929.2	100.0	39.87	-9.38	3.43	0.00
SUBTOTAL	12944.8		102.73	0.00	5.34	886.83
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	2.1	10.0	8.28	-5.84	7.59	22.22
M.D.O. SETTL. TK.	2.5	10.0	9.88	-5.95	7.33	29.35
M.D.O. STOR. TK. (P)	3.5	10.0	12.81	5.91	7.45	43.37
M.D.O. STOR. TK. (S)	4.4	10.0	12.58	-5.91	7.52	54.86
SUBTOTAL	12.4		11.39	-2.61	7.47	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	4.1	10.0	4.98	7.15	11.13	31.97
FRESH WATER TK. (P)	9.4	10.0	1.14	6.32	11.15	44.74
FRESH WATER TK. (S)	13.4	10.0	2.40	-6.59	11.14	82.65
SUBTOTAL	51.6		4.32	-0.00	8.25	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	26.4	10.0	68.20	12.64	11.22	68.67
NO.1 FUEL OIL TK. (S)	26.4	10.0	68.20	-12.64	11.22	68.67
NO.2 FUEL OIL TK. (P)	25.5	10.0	40.23	12.62	11.23	0.00
NO.2 FUEL OIL TK. (S)	25.5	10.0	40.23	-12.62	11.23	0.00
H.F.O. TK. (P)	25.3	10.0	23.08	10.63	7.06	44.35
H.F.O. TK. (S)	18.0	10.0	23.44	-10.57	6.80	33.83
NO.1 H.F.O. SERV. TK.	2.5	10.0	15.49	-10.82	11.07	7.35
NO.1 H.F.O. SETTL. T.	3.0	10.0	18.65	-11.14	11.06	11.05
NO.2 H.F.O. SERV. TK.	1.9	10.0	12.65	-10.87	11.07	0.00
NO.2 H.F.O. SETTL. T.	2.2	10.0	9.70	-10.42	11.09	0.00
SUBTOTAL	156.7		43.37	-0.16	10.03	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	1.4	20.0	10.03	5.53	7.22	0.36
G/E L.O. STOR. TK.	2.6	20.0	10.04	7.48	7.96	10.28
M/E L.O. SETTL. TK.	3.5	20.0	8.83	6.21	7.80	19.67
M/E L.O. STOR. TK.	5.4	20.0	7.19	6.05	8.05	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	50.8		16.95	2.45	5.88	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	4.5	80.0	18.81	-2.29	1.40	1.69
BILGE TK.	12.9	80.0	11.52	0.00	1.14	9.85
F.O. DRAIN TK.	5.2	80.0	20.70	2.51	1.34	3.33
F.O. OVERFLOW TK.	7.2	80.0	23.04	2.98	1.31	7.97
L.O. DRAIN TK.	3.1	80.0	18.32	2.24	1.42	0.91
SEWAGE STOR. TK.	10.9	80.0	22.37	-2.83	1.31	10.07
SLUDGE TK.	12.0	80.0	21.80	-8.40	7.72	9.03
SUBTOTAL	55.8		19.16	-1.81	2.67	42.86
TOTAL	13272.2		100.88	-0.00	5.40	1529.76

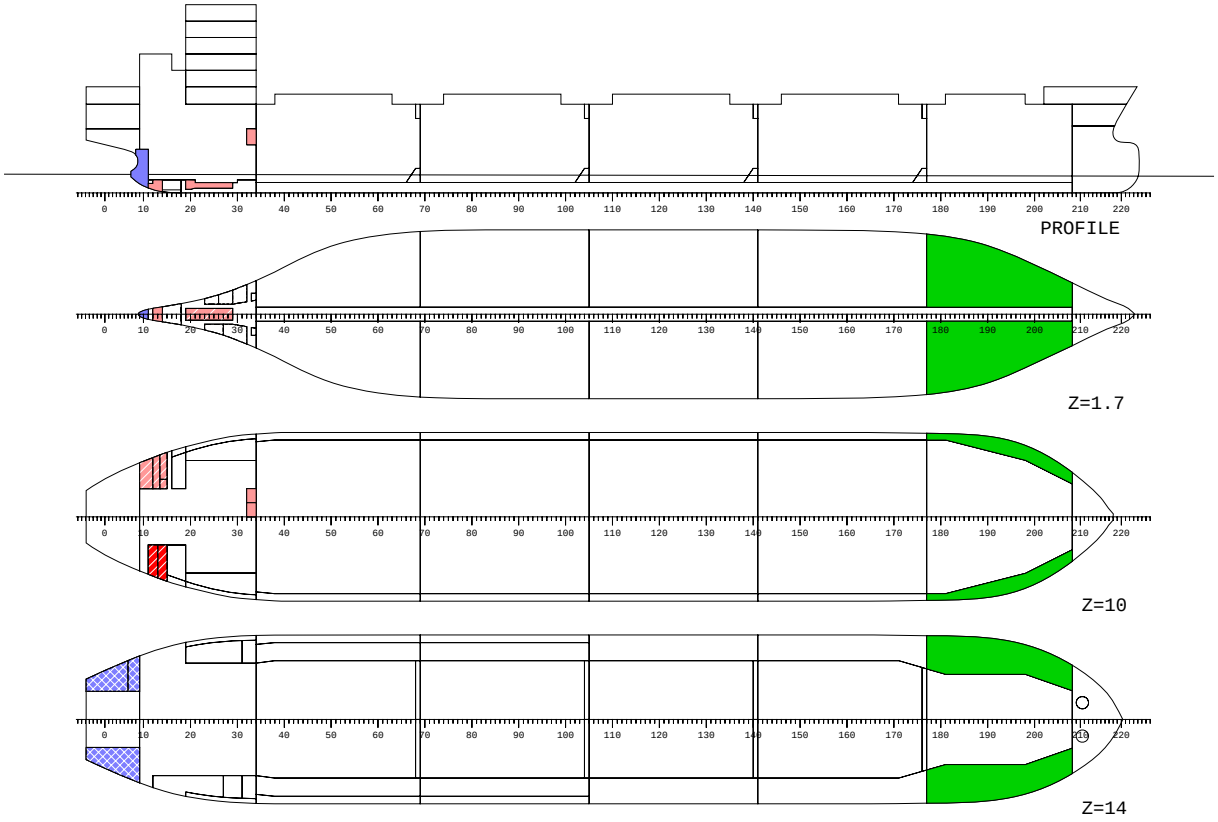


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.002	0.00	0.000	0.0	0.000
10.0	0.025	1.38	0.121	265.6	0.012
12.0	0.044	1.67	0.174	318.1	0.014
20.0	0.170	2.84	0.489	523.2	0.024
30.0	0.069	3.97	1.091	764.9	0.035
40.0	-0.535	4.49	1.840	983.3	0.045
50.0	-1.682	4.30	2.614	1171.9	0.053
60.0	-3.011	3.75	3.320	1324.8	0.060
70.0	-4.337	3.00	3.912	1437.5	0.065

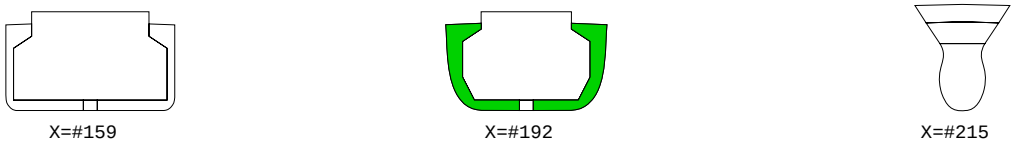


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	229	152	60.76	0.00
5.94	9.00	2511	671	55.65	4.34
18.60	25.00	15308	1351	106.89	51.36
25.80	34.00	26484	1762	95.02	91.38
40.20	52.00	50713	1506	108.02	145.51
53.80	69.00	67132	989	191.86	161.70
68.20	87.00	76683	281	111.34	163.76
82.60	105.00	75506	-309	208.23	162.99
97.00	123.00	69782	-561	138.53	161.97
111.40	141.00	59344	-762	232.89	160.95
125.80	159.00	47114	-1008	137.33	159.43
140.20	177.00	30446	-1179	223.55	152.52
152.20	192.00	16415	-1148	141.72	126.80
165.00	208.00	3439	-821	100.70	62.18
169.20	214.00	1048	-355	125.45	38.80
173.40	220.00	161	-99	56.15	17.85

LOADING CASE 'LD02 -- DOCKING WITH 10%' BUNKER







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	2418.8	151.92	0.00	7.03	0.00
Diesel Oil	DO	18.1	9.11	-6.63	8.22	51.57
Fresh Water	FW	51.6	4.32	-0.00	8.25	159.36
Grain Load	GRAIN	0.0	0.00	0.00	0.00	0.00
Heavy Fuel Oil	HFO	93.9	14.53	-11.01	12.39	29.91
Lubricating Oil	LO	50.8	16.95	2.45	5.88	9.80
	MASS	27.6	75.29	0.00	20.42	0.00
Miscellaneous	MIS	7.0	19.24	-1.81	1.99	42.86
Deadweight		2668.0	139.54	-0.39	7.35	293.50
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		2668.0	139.54	-0.39	7.35	
Total weight		11318.0	93.50	-0.09	9.53	

F L O A T I N G P O S I T I O N

Draught moulded	3.000	m	KM	23.50 m
Trim	-0.280	m	KG	9.53 m
Heel, PS=+	-0.4	deg		
TA	3.140	m	GM0	13.98 m
TF	2.860	m	GMCORR	-0.03 m
Trimming moment	-11081	tonm	GM	13.95 m
Heel	-0.4	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	1.430	mrad	OK
AREA40	Area under GZ curve .	0.090	2.086	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.656	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	3.910	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	26.725	deg	OK
GM0.15	GM > 0.15 m	0.150	13.950	m	OK
IMOWEATH.	IMO weather criterion	1.000	1.164		OK
VISIBILI.		359.760	209.614		OK
PROPIMMER	Propeller should imm.	50.000	0.000	%	NOT MET

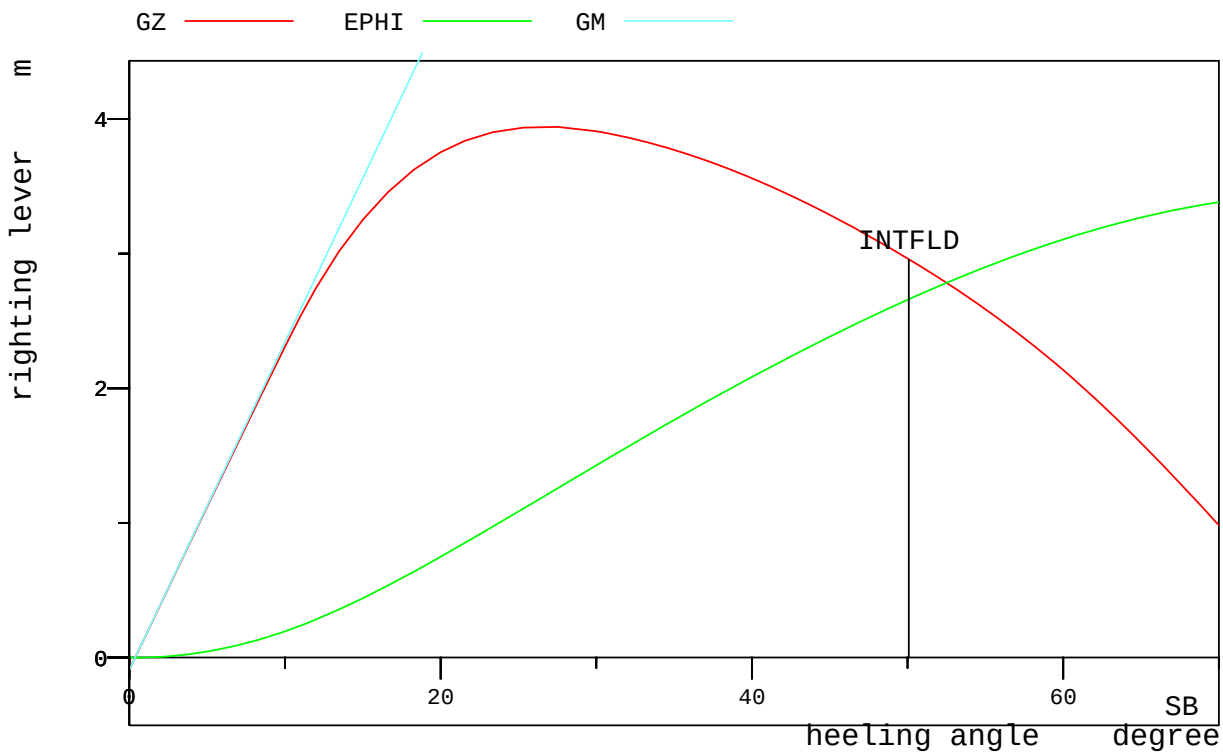
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-2127.3 t	(35.5%)	POSITION:	138.6 m	175
SHEAR FORCE (MAX)	1942.7 t	(32.4%)		25.8 m	34
SAGGING MOMENT	-				
HOGGING MOMENT	88327.5 tm	(81.0%)		86.6 m	110

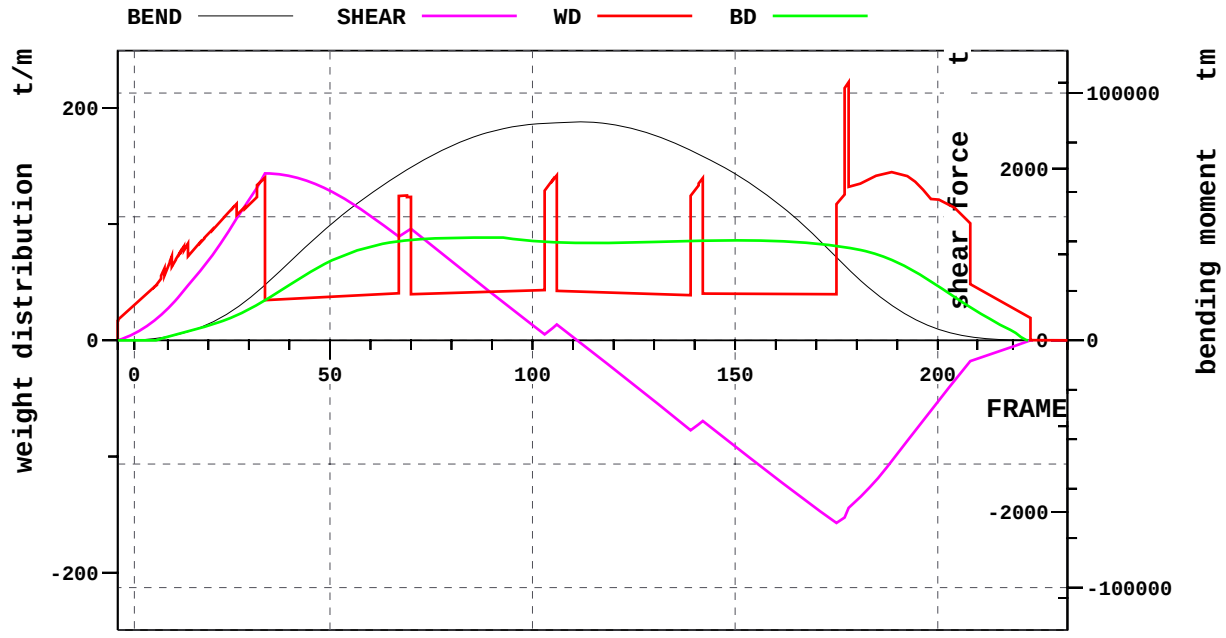
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.740)						
NO.1 CARGO HOLD	0.0	0.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
SUBTOTAL	0.0		0.00	0.00	0.00	0.0
, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	2.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	27.6		75.29	0.00	20.42	0.0
TOTAL	27.6		75.29	0.00	20.42	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	0.00	0.00	0.00	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	0.00	0.00	0.00	0.00
NO.2 WATER BALLAST T.	0.0	0.0	0.00	0.00	0.00	0.00
NO.3 WATER BALLAST T.	0.0	0.0	0.00	0.00	0.00	0.00
NO.3 WATER BALLAST T.	0.0	0.0	0.00	0.00	0.00	0.00
NO.4 WATER BALLAST T.	0.0	0.0	0.00	0.00	0.00	0.00
NO.4 WATER BALLAST T.	0.0	0.0	0.00	0.00	0.00	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	2418.8		151.92	0.00	7.03	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	8.3	40.0	8.24	-6.53	8.34	22.22
M.D.O. SETTL. TK.	9.9	40.0	9.84	-6.72	8.13	29.35
M.D.O. STOR. TK. (P)	0.0	0.0	12.66	7.28	9.32	0.00
M.D.O. STOR. TK. (S)	0.0	0.0	12.30	-7.23	9.35	0.00
SUBTOTAL	18.1		9.11	-6.63	8.22	51.57

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	4.1	10.0	4.98	7.15	11.13	31.97
FRESH WATER TK. (P)	9.4	10.0	1.14	6.32	11.15	44.74
FRESH WATER TK. (S)	13.4	10.0	2.40	-6.59	11.14	82.65
SUBTOTAL	51.6		4.32	-0.00	8.25	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	0.0	0.0	68.20	11.71	13.12	0.00
NO.1 FUEL OIL TK. (S)	0.0	0.0	68.20	-11.71	13.12	0.00
NO.2 FUEL OIL TK. (P)	0.0	0.0	39.85	11.70	13.13	0.00
NO.2 FUEL OIL TK. (S)	0.0	0.0	39.85	-11.70	13.13	0.00
H.F.O. TK. (P)	0.0	0.0	20.57	11.15	11.11	0.00
H.F.O. TK. (S)	0.0	0.0	21.94	-11.14	10.40	0.00
NO.1 H.F.O. SERV. TK.	24.3	98.0	15.46	-11.00	12.38	7.35
NO.1 H.F.O. SETTL. T.	29.0	98.0	18.63	-11.27	12.36	11.05
NO.2 H.F.O. SERV. TK.	19.1	98.0	12.64	-11.07	12.39	6.67
NO.2 H.F.O. SETTL. T.	21.5	98.0	9.62	-10.64	12.44	4.84
SUBTOTAL	93.9		14.53	-11.01	12.39	29.91
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	1.4	20.0	10.03	5.53	7.22	0.00
G/E L.O. STOR. TK.	2.6	20.0	10.04	7.48	7.96	0.00
M/E L.O. SETTL. TK.	3.5	20.0	8.83	6.21	7.80	0.00
M/E L.O. STOR. TK.	5.4	20.0	7.19	6.05	8.05	0.00
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	4.63
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	1.63
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	1.63
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	1.91
SUBTOTAL	50.8		16.95	2.45	5.88	9.80
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	0.6	10.0	18.98	-1.96	0.68	1.69
BILGE TK.	1.6	10.0	11.66	0.00	0.32	9.85
F.O. DRAIN TK.	0.6	10.0	20.77	2.04	0.56	3.33
F.O. OVERFLOW TK.	0.9	10.0	23.05	2.35	0.50	7.97
L.O. DRAIN TK.	0.4	10.0	18.43	1.94	0.72	0.91
SEWAGE STOR. TK.	1.4	10.0	22.44	-2.26	0.52	10.07
SLUDGE TK.	1.5	10.0	21.80	-8.40	7.44	9.03
SUBTOTAL	7.0		19.24	-1.81	1.99	42.86
TOTAL	2640.4		140.22	-0.40	7.21	293.50

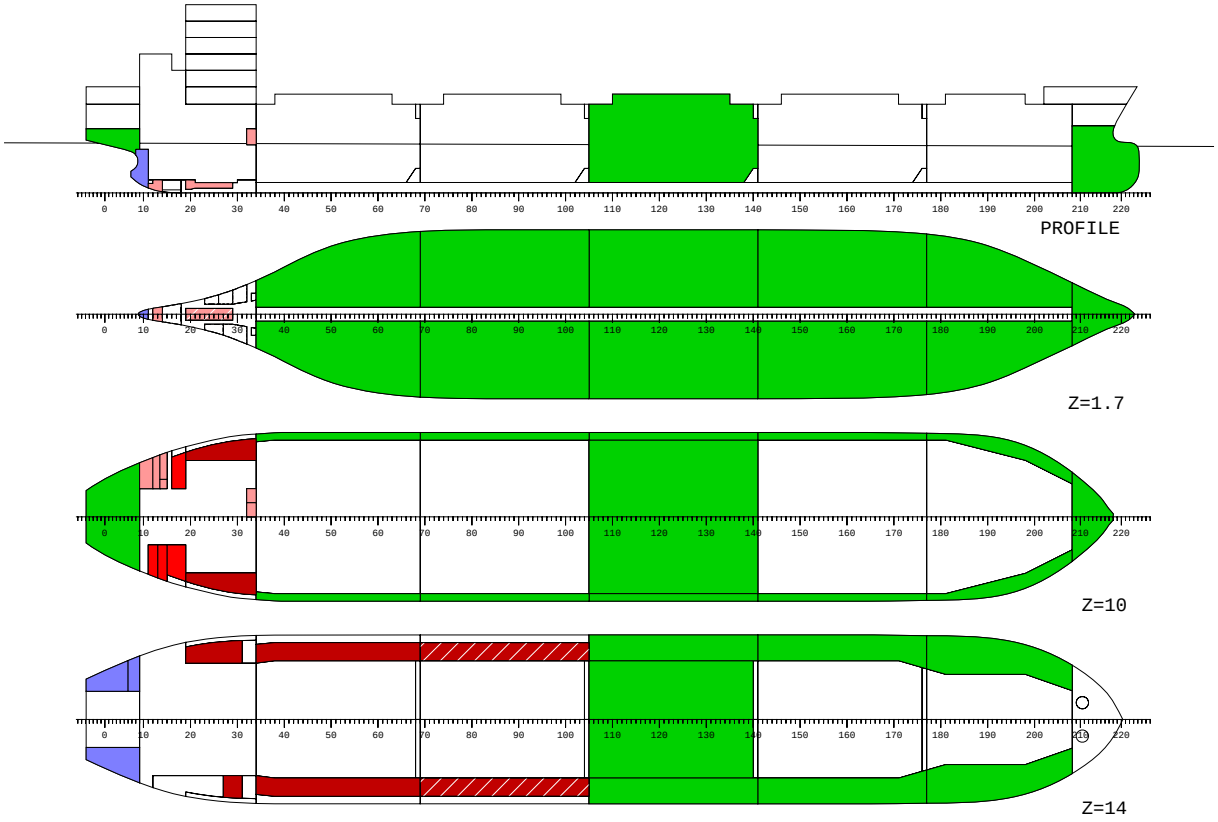


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.092	-0.09	0.000	0.0	0.000
10.0	-0.113	2.31	0.195	51.0	0.005
12.0	-0.154	2.75	0.283	61.0	0.005
20.0	-1.017	3.75	0.749	100.4	0.009
30.0	-3.065	3.91	1.429	146.7	0.013
40.0	-5.407	3.56	2.085	188.7	0.017
50.0	-7.722	2.96	2.658	224.8	0.020
60.0	-9.946	2.14	3.107	254.2	0.022
70.0	-12.126	0.98	3.383	275.8	0.024

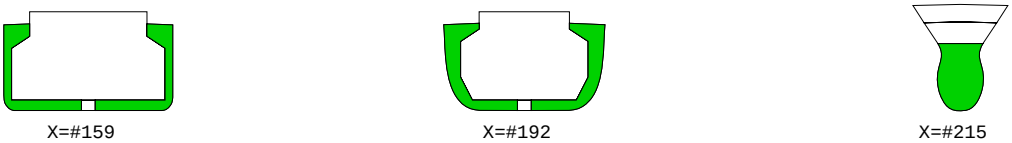


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	120	79	30.58	0.00
5.94	9.00	1263	333	55.64	2.36
18.60	25.00	10983	1269	110.17	18.65
25.80	34.00	22486	1943	34.62	34.90
40.20	52.00	49203	1690	37.83	70.89
53.80	69.00	68720	1270	124.61	86.08
68.20	87.00	83040	658	41.53	88.46
82.60	105.00	87843	143	138.31	88.02
97.00	123.00	86220	-442	40.66	87.32
111.40	141.00	75093	-979	133.42	86.63
125.80	159.00	57276	-1565	39.94	85.44
140.20	177.00	30234	-2062	125.51	80.17
152.20	192.00	10677	-1202	141.72	63.50
165.00	208.00	1470	-242	100.69	29.16
169.20	214.00	628	-158	38.08	17.68
173.40	220.00	140	-73	27.85	7.87

LOADING CASE 'LD03A -- HEAVY BALLAST CONDITION, DEPARTURE







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	12944.8	102.73	0.00	5.34	10748.87
Diesel Oil	DO	121.6	11.23	-3.17	9.29	149.80
Fresh Water	FW	293.6	2.53	0.00	12.34	159.36
Grain Load	GRAIN	10223.4	96.78	0.00	8.67	0.00
Heavy Fuel Oil	HFO	1535.7	42.68	-0.15	12.40	233.92
Lubricating Oil	LO	101.1	12.73	4.87	7.77	56.99
	MASS	43.6	53.17	0.00	20.19	0.00
Miscellaneous	MIS	7.0	19.24	-1.81	1.99	42.86
Deadweight		25270.9	94.60	-0.01	7.25	11391.80
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		25270.9	94.60	-0.01	7.25	
Total weight		33920.9	90.70	-0.00	8.00	

F L O A T I N G P O S I T I O N

Draught moulded	8.215	m	KM	12.84 m
Trim	-0.532	m	KG	8.00 m
Heel, PS=+	-0.0	deg		
TA	8.481	m	GM0	4.84 m
TF	7.949	m	GMCORR	-0.34 m
Trimming moment	-29594	tonm	GM	4.50 m
Heel	-0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.663	mrad	OK
AREA40	Area under GZ curve .	0.090	1.156	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.493	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	2.911	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	40.554	deg	OK
GM0.15	GM > 0.15 m	0.150	4.504	m	OK
IMOWEATH.	IMO weather criterion	1.000	5.324		OK
VISIBILI.		359.760	146.834		OK
PROPIMMER	Propeller should imm.	50.000	100.016	%	OK

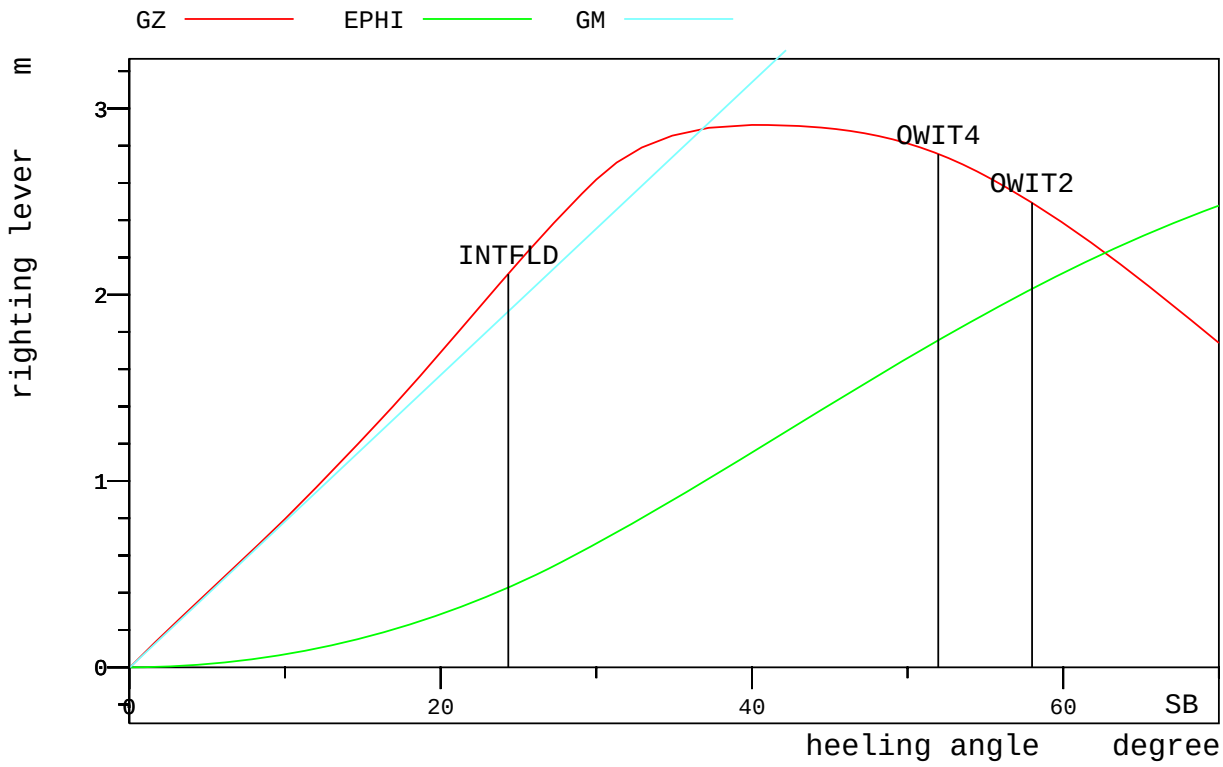
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN,CORR)	-4165.1 t	(69.4%)	POSITION:	82.6 m	105
SHEAR FORCE (MAX,CORR)	3327.5 t	(55.5%)		111.4 m	141
SAGGING MOMENT	-72617.4 tm	(75.0%)		98.3 m	125
HOGGING MOMENT	42549.8 tm	(46.1%)		43.2 m	56

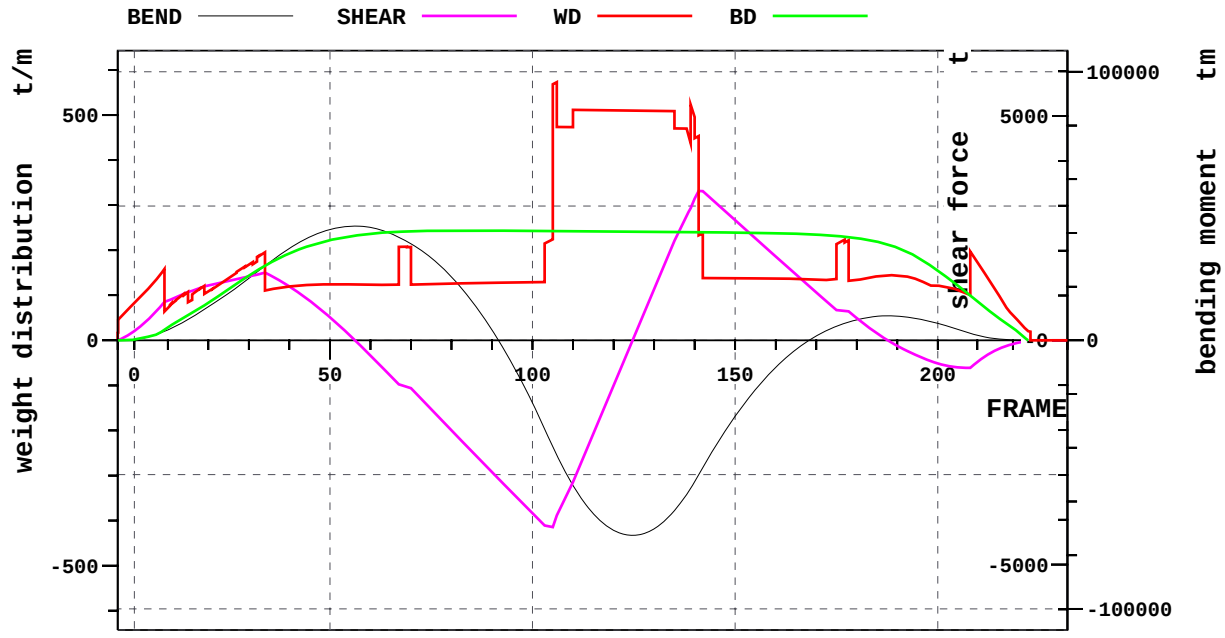
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.740)						
NO.1 CARGO HOLD	0.0	0.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	10223.4	100.0	96.78	0.00	8.67	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
SUBTOTAL	10223.4		96.78	0.00	8.67	0.0
, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	18.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	43.6		53.17	0.00	20.19	0.0
TOTAL	10267.0		96.60	0.00	8.72	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	339.3	100.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	774.2	100.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	10.47	5.87	5374.44
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	-10.47	5.87	5374.44
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	929.2	100.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	929.2	100.0	39.87	-9.38	3.43	0.00
SUBTOTAL	12944.8		102.73	0.00	5.34	10748.87
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	20.2	98.0	8.23	-7.08	9.34	22.22
M.D.O. SETTL. TK.	24.2	98.0	9.82	-7.32	9.21	29.35
M.D.O. STOR. TK. (P)	33.9	98.0	12.66	7.26	9.29	43.37
M.D.O. STOR. TK. (S)	43.2	98.0	12.30	-7.22	9.32	54.86
SUBTOTAL	121.6		11.23	-3.17	9.29	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	40.7	100.0	4.97	7.43	12.96	31.97
FRESH WATER TK. (P)	93.7	100.0	0.94	6.60	13.03	44.74
FRESH WATER TK. (S)	134.4	100.0	2.17	-6.85	13.01	82.65
SUBTOTAL	293.6		2.53	0.00	12.34	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	258.4	98.0	68.20	11.73	13.08	68.67
NO.1 FUEL OIL TK. (S)	258.4	98.0	68.20	-11.73	13.08	68.67
NO.2 FUEL OIL TK. (P)	250.2	98.0	39.86	11.71	13.09	0.00
NO.2 FUEL OIL TK. (S)	250.2	98.0	39.86	-11.71	13.09	0.00
H.F.O. TK. (P)	248.1	98.0	20.60	11.15	11.04	44.35
H.F.O. TK. (S)	176.3	98.0	21.94	-11.14	10.32	33.83
NO.1 H.F.O. SERV. TK.	24.3	98.0	15.46	-11.00	12.38	7.35
NO.1 H.F.O. SETTL. T.	29.0	98.0	18.63	-11.27	12.36	11.05
NO.2 H.F.O. SERV. TK.	19.1	98.0	12.64	-11.07	12.39	0.00
NO.2 H.F.O. SETTL. T.	21.5	98.0	9.62	-10.64	12.43	0.00
SUBTOTAL	1535.7		42.68	-0.15	12.40	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	6.8	98.0	10.01	5.59	8.82	0.36
G/E L.O. STOR. TK.	12.9	98.0	10.02	8.28	9.39	10.28
M/E L.O. SETTL. TK.	17.2	98.0	8.81	7.17	9.29	19.67
M/E L.O. STOR. TK.	26.4	98.0	7.12	6.90	9.42	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	101.1		12.73	4.87	7.77	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	0.6	10.0	18.98	-1.96	0.68	1.69
BILGE TK.	1.6	10.0	11.66	0.00	0.32	9.85
F.O. DRAIN TK.	0.6	10.0	20.77	2.04	0.56	3.33
F.O. OVERFLOW TK.	0.9	10.0	23.05	2.35	0.50	7.97
L.O. DRAIN TK.	0.4	10.0	18.43	1.94	0.72	0.91
SEWAGE STOR. TK.	1.4	10.0	22.44	-2.26	0.52	10.07
SLUDGE TK.	1.5	10.0	21.80	-8.40	7.44	9.03
SUBTOTAL	7.0		19.24	-1.81	1.99	42.86
TOTAL	15003.8		93.24	-0.01	6.24	11391.80

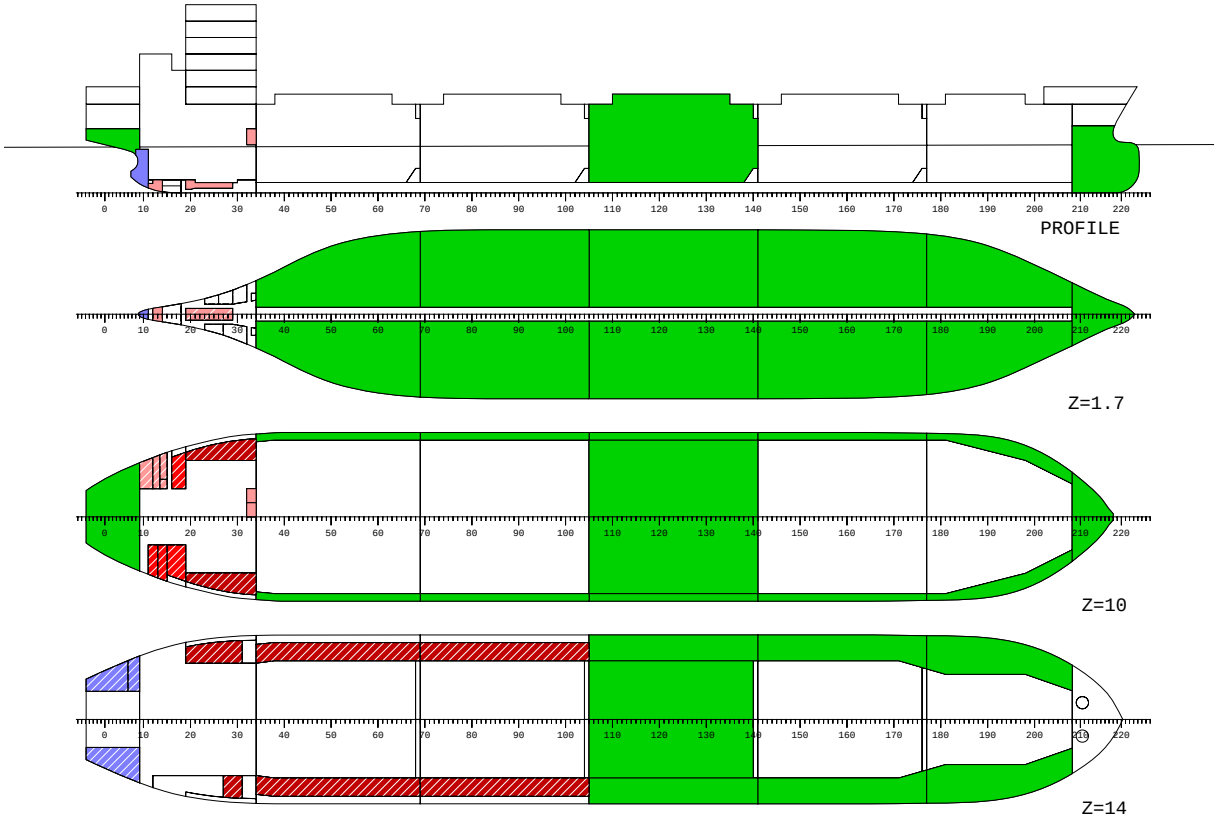


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.004	-0.00	0.000	0.0	0.000
10.0	0.014	0.80	0.069	1978.2	0.058
12.0	0.028	0.96	0.100	2368.5	0.070
20.0	0.150	1.69	0.284	3896.2	0.115
30.0	0.367	2.62	0.663	5695.9	0.168
40.0	0.016	2.91	1.156	7322.5	0.216
50.0	-0.637	2.81	1.659	8726.6	0.257
60.0	-1.516	2.38	2.117	9865.6	0.291
70.0	-2.491	1.74	2.478	10704.8	0.316

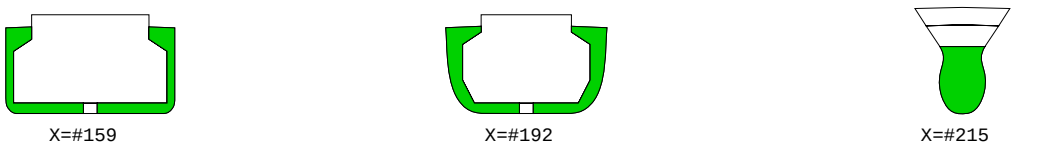


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	316	207	83.32	1.69
5.94	9.00	3315	856	111.25	23.32
18.60	25.00	17522	1324	138.22	112.97
25.80	34.00	27719	1510	110.65	165.44
40.20	52.00	41920	353	124.10	225.79
53.80	69.00	36763	-1037	207.79	241.69
68.20	87.00	10566	-2661	127.47	243.45
82.60	105.00	-39542	-4165	224.20	242.38
97.00	123.00	-72377	-361	510.17	241.06
111.40	141.00	-50123	3327	343.03	239.74
125.80	159.00	-11605	1949	137.33	237.93
140.20	177.00	6216	655	223.55	229.68
152.20	192.00	8725	-215	141.72	197.52
165.00	208.00	2611	-612	100.69	99.26
169.20	214.00	818	-271	125.45	61.55
173.40	220.00	128	-84	56.15	27.63

LOADING CASE 'LD03B -- HEAVY BALLAST CONDITION, WBT2 FULL, HALFWAY







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	12944.8	102.73	0.00	5.34	10748.87
Diesel Oil	DO	62.0	11.26	-2.98	8.46	149.80
Fresh Water	FW	159.2	2.90	-0.00	10.95	159.36
Grain Load	GRAIN	10223.4	96.78	0.00	8.67	0.00
Heavy Fuel Oil	HFO	783.5	42.80	-0.15	11.14	233.92
Lubricating Oil	LO	76.6	14.10	4.02	6.96	56.99
	MASS	34.6	63.09	0.00	20.29	0.00
Miscellaneous	MIS	27.9	19.18	-1.81	2.34	42.86
Deadweight		24312.1	96.98	-0.00	6.99	11391.80
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		24312.1	96.98	-0.00	6.99	
Total weight		32962.1	92.34	-0.00	7.83	

F L O A T I N G P O S I T I O N

Draught moulded	8.004	m	KM	12.88 m
Trim	0.383	m	KG	7.83 m
Heel, PS=+	0.0	deg		
TA	7.812	m	GM0	5.05 m
TF	8.196	m	GMCORR	-0.35 m
Trimming moment	20361	tonm	GM	4.70 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.693	mrad	OK
AREA40	Area under GZ curve .	0.090	1.211	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.518	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	3.065	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	40.720	deg	OK
GM0.15	GM > 0.15 m	0.150	4.702	m	OK
IMOWEATH.	IMO weather criterion	1.000	5.312		OK
VISIBILI.		359.760	133.882		OK
PROPIMMER	Propeller should imm.	50.000	87.403	%	OK

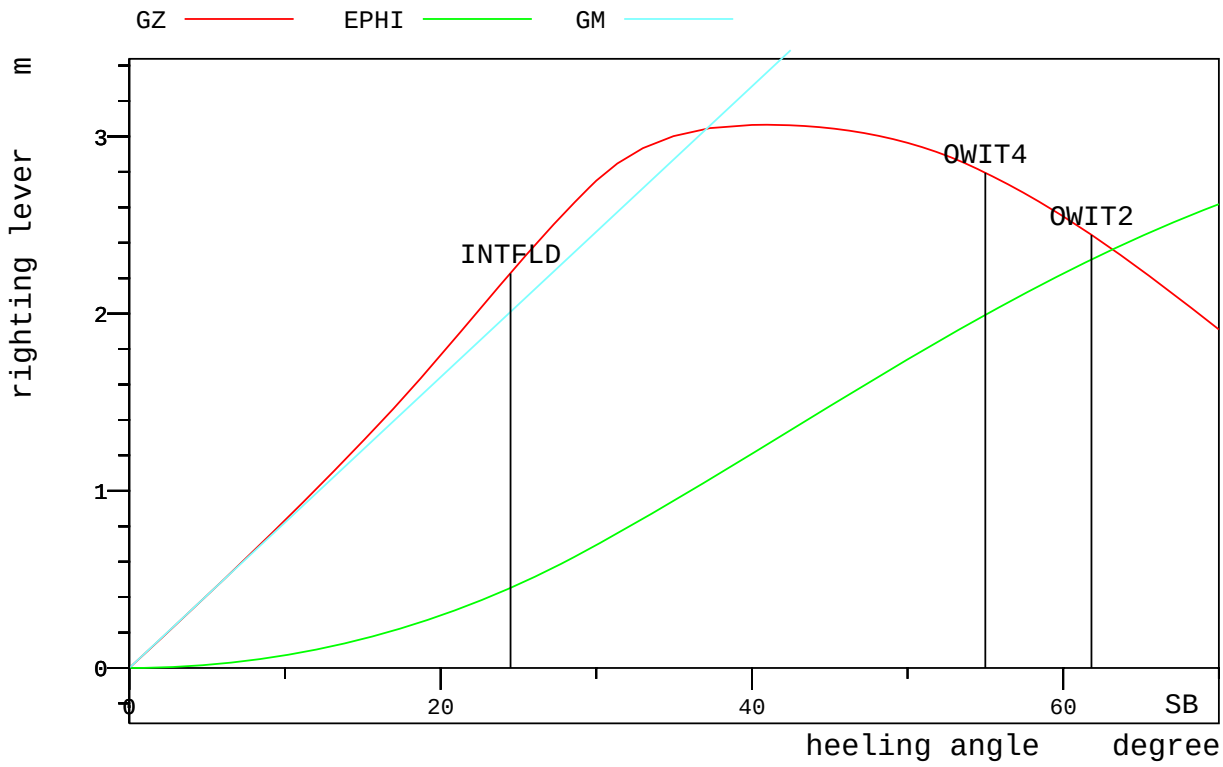
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-4189.7 t	(69.8%)	POSITION:	82.6 m	105
SHEAR FORCE (MAX)	3432.2 t	(57.2%)		111.4 m	141
SAGGING MOMENT	-78381.2 tm	(80.9%)		98.0 m	124
HOGGING MOMENT	37702.0 tm	(40.8%)		43.2 m	56

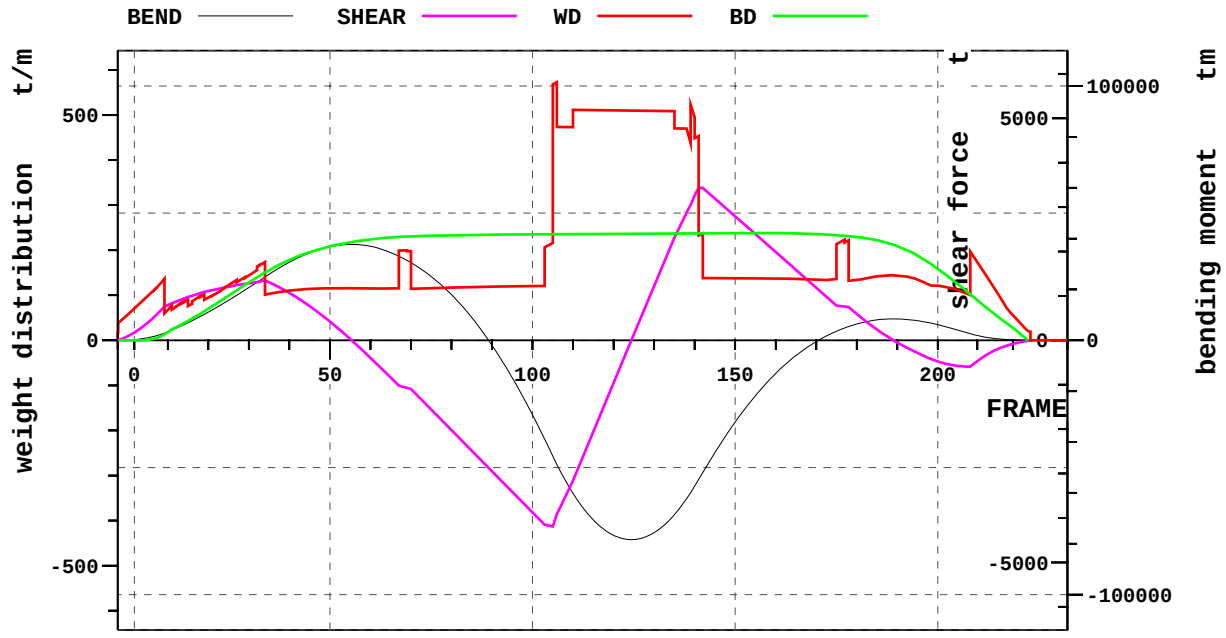
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.740)						
NO.1 CARGO HOLD	0.0	0.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	10223.4	100.0	96.78	0.00	8.67	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
SUBTOTAL	10223.4		96.78	0.00	8.67	0.0
, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	9.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	34.6		63.09	0.00	20.29	0.0
TOTAL	10258.0		96.67	0.00	8.71	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	339.3	100.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	774.2	100.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	10.47	5.87	5374.44
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	-10.47	5.87	5374.44
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	929.2	100.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	929.2	100.0	39.87	-9.38	3.43	0.00
SUBTOTAL	12944.8		102.73	0.00	5.34	10748.87
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	10.3	50.0	8.24	-6.66	8.53	22.22
M.D.O. SETTL. TK.	12.4	50.0	9.83	-6.86	8.34	29.35
M.D.O. STOR. TK. (P)	17.3	50.0	12.68	6.80	8.45	43.37
M.D.O. STOR. TK. (S)	22.0	50.0	12.35	-6.76	8.50	54.86
SUBTOTAL	62.0		11.26	-2.98	8.46	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	20.4	50.0	4.98	7.32	11.97	31.97
FRESH WATER TK. (P)	46.9	50.0	1.02	6.48	12.05	44.74
FRESH WATER TK. (S)	67.2	50.0	2.26	-6.75	12.02	82.65
SUBTOTAL	159.2		2.90	-0.00	10.95	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	131.9	50.0	68.20	12.07	11.98	68.67
NO.1 FUEL OIL TK. (S)	131.9	50.0	68.20	-12.07	11.98	68.67
NO.2 FUEL OIL TK. (P)	127.7	50.0	40.00	12.05	12.00	0.00
NO.2 FUEL OIL TK. (S)	127.7	50.0	40.00	-12.05	12.00	0.00
H.F.O. TK. (P)	126.6	50.0	21.27	10.98	9.24	44.35
H.F.O. TK. (S)	90.0	50.0	21.60	-10.89	8.63	33.83
NO.1 H.F.O. SERV. TK.	12.4	50.0	15.47	-10.92	11.69	7.35
NO.1 H.F.O. SETTL. T.	14.8	50.0	18.64	-11.21	11.66	11.05
NO.2 H.F.O. SERV. TK.	9.7	50.0	12.65	-10.98	11.70	0.00
NO.2 H.F.O. SETTL. T.	11.0	50.0	9.65	-10.54	11.75	0.00
SUBTOTAL	783.5		42.80	-0.15	11.14	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	4.1	60.0	10.01	5.58	8.05	0.36
G/E L.O. STOR. TK.	7.9	60.0	10.02	8.01	8.78	10.28
M/E L.O. SETTL. TK.	10.5	60.0	8.82	6.85	8.65	19.67
M/E L.O. STOR. TK.	16.1	60.0	7.14	6.62	8.83	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	76.6		14.10	4.02	6.96	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	2.3	40.0	18.84	-2.15	1.07	1.69
BILGE TK.	6.4	40.0	11.55	0.00	0.73	9.85
F.O. DRAIN TK.	2.6	40.0	20.71	2.30	0.98	3.33
F.O. OVERFLOW TK.	3.6	40.0	23.05	2.70	0.93	7.97
L.O. DRAIN TK.	1.5	40.0	18.35	2.11	1.09	0.91
SEWAGE STOR. TK.	5.4	40.0	22.38	-2.58	0.94	10.07
SLUDGE TK.	6.0	40.0	21.80	-8.40	7.56	9.03
SUBTOTAL	27.9		19.18	-1.81	2.34	42.86
TOTAL	14054.1		97.20	-0.00	5.74	11391.80

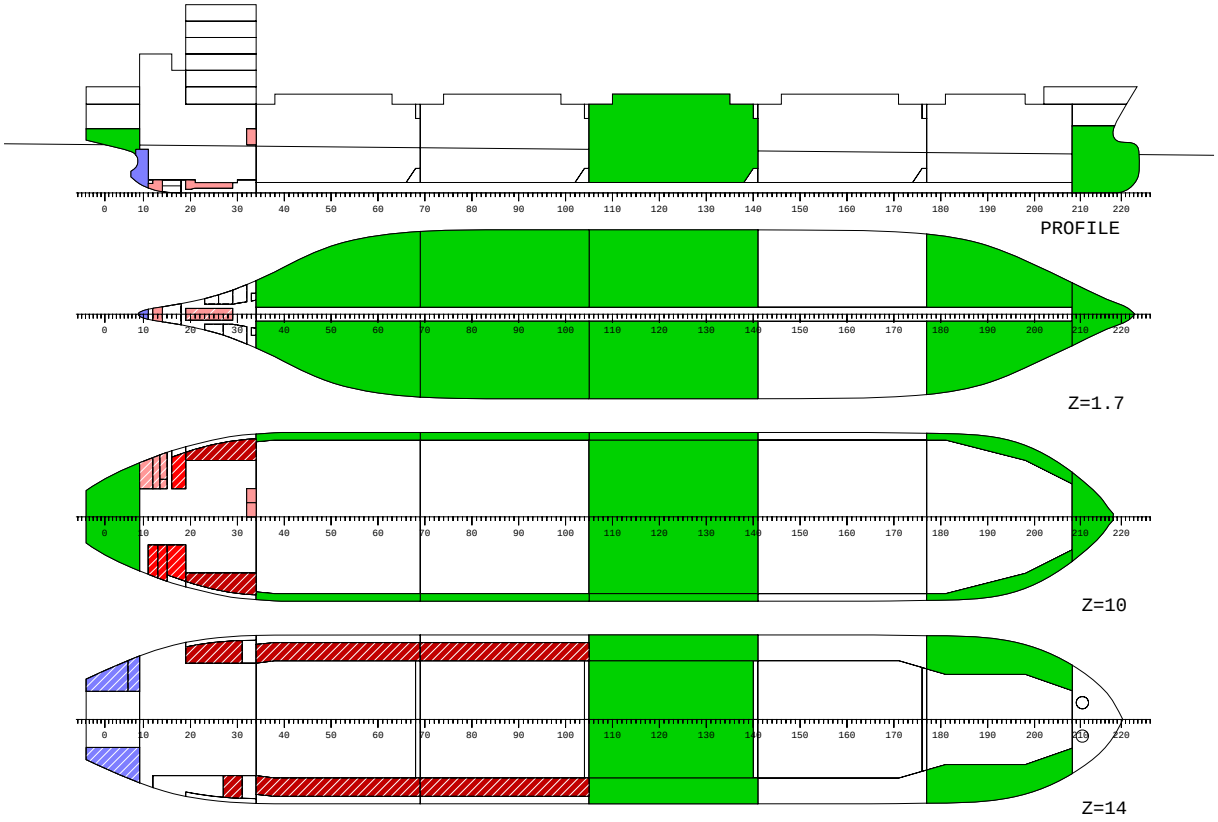


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.001	-0.00	0.000	0.0	0.000
10.0	0.017	0.83	0.072	1978.2	0.060
12.0	0.031	1.01	0.104	2368.5	0.072
20.0	0.156	1.76	0.296	3896.2	0.118
30.0	0.399	2.75	0.693	5695.9	0.173
40.0	0.042	3.06	1.211	7322.5	0.222
50.0	-0.638	2.96	1.741	8726.6	0.265
60.0	-1.524	2.55	2.226	9865.6	0.299
70.0	-2.508	1.91	2.617	10704.8	0.325

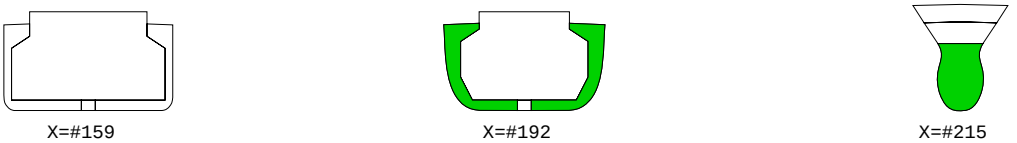


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	258	176	70.62	0.12
5.94	9.00	2868	760	98.43	14.21
18.60	25.00	15719	1207	122.51	98.73
25.80	34.00	24904	1348	101.19	150.46
40.20	52.00	37311	269	115.37	212.36
53.80	69.00	31354	-1068	199.49	230.40
68.20	87.00	4810	-2676	118.67	234.42
82.60	105.00	-45597	-4190	215.86	235.62
97.00	123.00	-78211	-305	510.17	236.57
111.40	141.00	-54805	3432	343.03	237.52
125.80	159.00	-14676	2070	137.33	237.97
140.20	177.00	4747	758	223.55	231.96
152.20	192.00	8251	-150	141.72	201.38
165.00	208.00	2613	-595	100.69	102.30
169.20	214.00	853	-265	125.45	63.46
173.40	220.00	164	-83	56.15	28.34

LOADING CASE 'LD03C -- HEAVY BALLAST CONDITION, WBT2 EMPTY, HALFWAY







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	10154.9	96.41	0.00	5.19	10748.87
Diesel Oil	DO	62.0	11.26	-2.98	8.46	149.80
Fresh Water	FW	159.2	2.90	-0.00	10.95	159.36
Grain Load	GRAIN	10223.4	96.78	0.00	8.67	0.00
Heavy Fuel Oil	HFO	783.5	42.80	-0.15	11.14	233.92
Lubricating Oil	LO	76.6	14.10	4.02	6.96	56.99
	MASS	34.6	63.09	0.00	20.29	0.00
Miscellaneous	MIS	27.9	19.18	-1.81	2.34	42.86
Deadweight		21522.2	93.25	-0.00	7.14	11391.80
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		21522.2	93.25	-0.00	7.14	
Total weight		30172.2	89.25	-0.00	8.02	

F L O A T I N G P O S I T I O N

Draught moulded	7.400	m	KM	13.38 m
Trim	-1.644	m	KG	8.02 m
Heel, PS=+	0.0	deg		
TA	8.222	m	GM0	5.36 m
TF	6.577	m	GMCORR	-0.38 m
Trimming moment	-89178	tonm	GM	4.98 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.735	mrad	OK
AREA40	Area under GZ curve .	0.090	1.283	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.548	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	3.210	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	38.469	deg	OK
GM0.15	GM > 0.15 m	0.150	4.984	m	OK
IMOWEATH.	IMO weather criterion	1.000	4.749		OK
VISIBILI.		359.760	180.615		OK
PROPIMMER	Propeller should imm.	50.000	95.128	%	OK

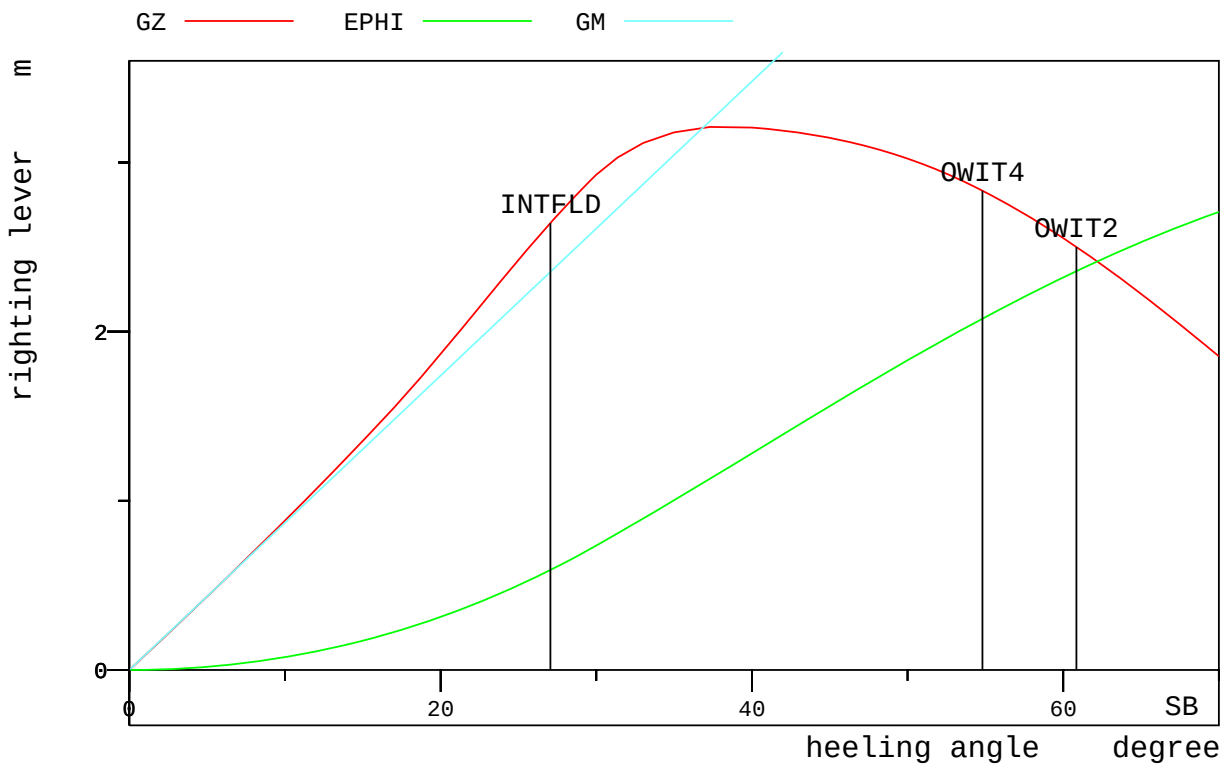
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-3913.8 t	(65.2%)	POSITION:	82.6 m	105
SHEAR FORCE (MAX)	4333.2 t	(72.2%)		111.4 m	141
SAGGING MOMENT	-73677.6 tm	(76.1%)		96.0 m	122
HOGGING MOMENT	34634.3 tm	(39.0%)		41.4 m	53

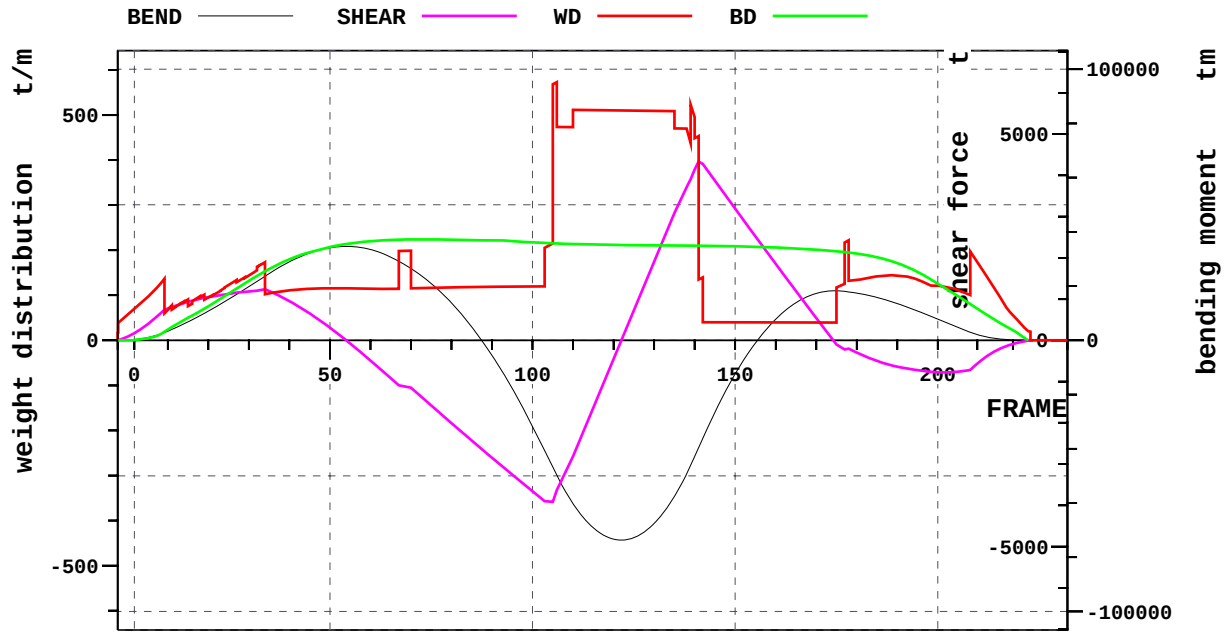
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.740)						
NO.1 CARGO HOLD	0.0	0.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	10223.4	100.0	96.78	0.00	8.67	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
SUBTOTAL	10223.4		96.78	0.00	8.67	0.0
, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	9.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	34.6		63.09	0.00	20.29	0.0
TOTAL	10258.0		96.67	0.00	8.71	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	339.3	100.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	774.2	100.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	0.00	0.00	0.00	5374.44
NO.2 WATER BALLAST T.	0.0	0.0	0.00	0.00	0.00	5374.44
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	929.2	100.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	929.2	100.0	39.87	-9.38	3.43	0.00
SUBTOTAL	10154.9		96.41	0.00	5.19	10748.87
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	10.3	50.0	8.24	-6.66	8.53	22.22
M.D.O. SETTL. TK.	12.4	50.0	9.83	-6.86	8.34	29.35
M.D.O. STOR. TK. (P)	17.3	50.0	12.68	6.80	8.45	43.37
M.D.O. STOR. TK. (S)	22.0	50.0	12.35	-6.76	8.50	54.86
SUBTOTAL	62.0		11.26	-2.98	8.46	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	20.4	50.0	4.98	7.32	11.97	31.97
FRESH WATER TK. (P)	46.9	50.0	1.02	6.48	12.05	44.74
FRESH WATER TK. (S)	67.2	50.0	2.26	-6.75	12.02	82.65
SUBTOTAL	159.2		2.90	-0.00	10.95	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	131.9	50.0	68.20	12.07	11.98	68.67
NO.1 FUEL OIL TK. (S)	131.9	50.0	68.20	-12.07	11.98	68.67
NO.2 FUEL OIL TK. (P)	127.7	50.0	40.00	12.05	12.00	0.00
NO.2 FUEL OIL TK. (S)	127.7	50.0	40.00	-12.05	12.00	0.00
H.F.O. TK. (P)	126.6	50.0	21.27	10.98	9.24	44.35
H.F.O. TK. (S)	90.0	50.0	21.60	-10.89	8.63	33.83
NO.1 H.F.O. SERV. TK.	12.4	50.0	15.47	-10.92	11.69	7.35
NO.1 H.F.O. SETTL. T.	14.8	50.0	18.64	-11.21	11.66	11.05
NO.2 H.F.O. SERV. TK.	9.7	50.0	12.65	-10.98	11.70	0.00
NO.2 H.F.O. SETTL. T.	11.0	50.0	9.65	-10.54	11.75	0.00
SUBTOTAL	783.5		42.80	-0.15	11.14	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	4.1	60.0	10.01	5.58	8.05	0.36
G/E L.O. STOR. TK.	7.9	60.0	10.02	8.01	8.78	10.28
M/E L.O. SETTL. TK.	10.5	60.0	8.82	6.85	8.65	19.67
M/E L.O. STOR. TK.	16.1	60.0	7.14	6.62	8.83	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	76.6		14.10	4.02	6.96	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	2.3	40.0	18.84	-2.15	1.07	1.69
BILGE TK.	6.4	40.0	11.55	0.00	0.73	9.85
F.O. DRAIN TK.	2.6	40.0	20.71	2.30	0.98	3.33
F.O. OVERFLOW TK.	3.6	40.0	23.05	2.70	0.93	7.97
L.O. DRAIN TK.	1.5	40.0	18.35	2.11	1.09	0.91
SEWAGE STOR. TK.	5.4	40.0	22.38	-2.58	0.94	10.07
SLUDGE TK.	6.0	40.0	21.80	-8.40	7.56	9.03
SUBTOTAL	27.9		19.18	-1.81	2.34	42.86
TOTAL	11264.2		90.14	-0.00	5.71	11391.80

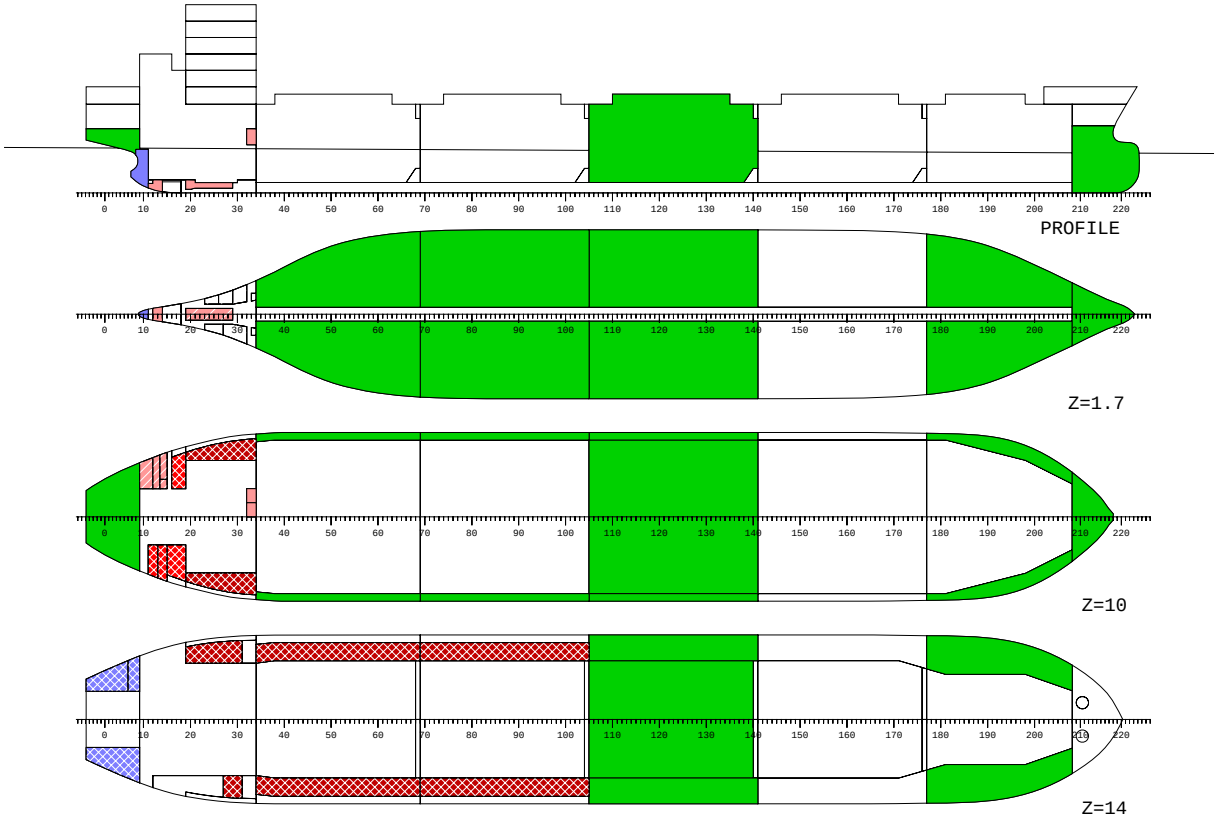


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.001	0.00	0.000	0.0	0.000
10.0	0.018	0.88	0.076	1978.2	0.066
12.0	0.033	1.07	0.110	2368.5	0.078
20.0	0.162	1.87	0.314	3896.2	0.129
30.0	0.437	2.93	0.735	5695.9	0.189
40.0	0.004	3.21	1.283	7322.5	0.243
50.0	-0.794	3.02	1.830	8726.6	0.289
60.0	-1.764	2.55	2.321	9865.6	0.327
70.0	-2.829	1.85	2.708	10704.8	0.355

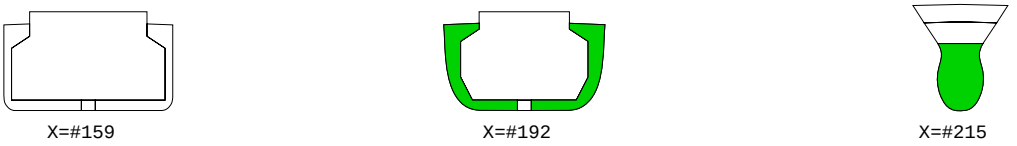


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	275	177	70.73	0.45
5.94	9.00	2874	742	98.29	18.86
18.60	25.00	15149	1125	122.72	103.42
25.80	34.00	23672	1237	102.20	153.41
40.20	52.00	34478	157	115.34	210.44
53.80	69.00	27355	-1128	198.51	223.74
68.20	87.00	945	-2597	118.67	222.74
82.60	105.00	-46906	-3914	214.84	218.92
97.00	123.00	-73573	248	510.17	214.85
111.40	141.00	-39719	4333	294.10	210.78
125.80	159.00	6245	1984	39.94	206.21
140.20	177.00	17992	-220	125.51	195.64
152.20	192.00	12502	-678	141.72	163.87
165.00	208.00	2978	-717	100.69	80.39
169.20	214.00	895	-313	125.45	50.03
173.40	220.00	118	-91	56.15	22.83

LOADING CASE 'LD03D -- HEAVY BALLAST CONDITION, ARRIVAL







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	10154.9	96.41	0.00	5.19	10748.87
Diesel Oil	DO	12.4	11.39	-2.61	7.47	149.80
Fresh Water	FW	51.6	4.32	-0.00	8.25	159.36
Grain Load	GRAIN	10223.4	96.78	0.00	8.67	0.00
Heavy Fuel Oil	HFO	156.7	43.37	-0.16	10.03	233.92
Lubricating Oil	LO	50.8	16.95	2.45	5.88	56.99
	MASS	27.6	75.87	0.00	20.42	0.00
Miscellaneous	MIS	55.8	19.16	-1.81	2.67	42.86
Deadweight		20733.3	95.48	-0.00	6.97	11391.80
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		20733.3	95.48	-0.00	6.97	
Total weight		29383.3	90.72	-0.00	7.92	

F L O A T I N G P O S I T I O N

Draught moulded	7.223	m	KM	13.43 m
Trim	-0.876	m	KG	7.92 m
Heel, PS=+	0.0	deg		
TA	7.661	m	GM0	5.51 m
TF	6.785	m	GMCORR	-0.39 m
Trimming moment	-45845	tonm	GM	5.13 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.756	mrad	OK
AREA40	Area under GZ curve .	0.090	1.320	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.565	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	3.313	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	38.649	deg	OK
GM0.15	GM > 0.15 m	0.150	5.125	m	OK
IMOWEATH.	IMO weather criterion	1.000	4.670		OK
VISIBILI.		359.760	166.629		OK
PROPIMMER	Propeller should imm.	50.000	84.553	%	OK

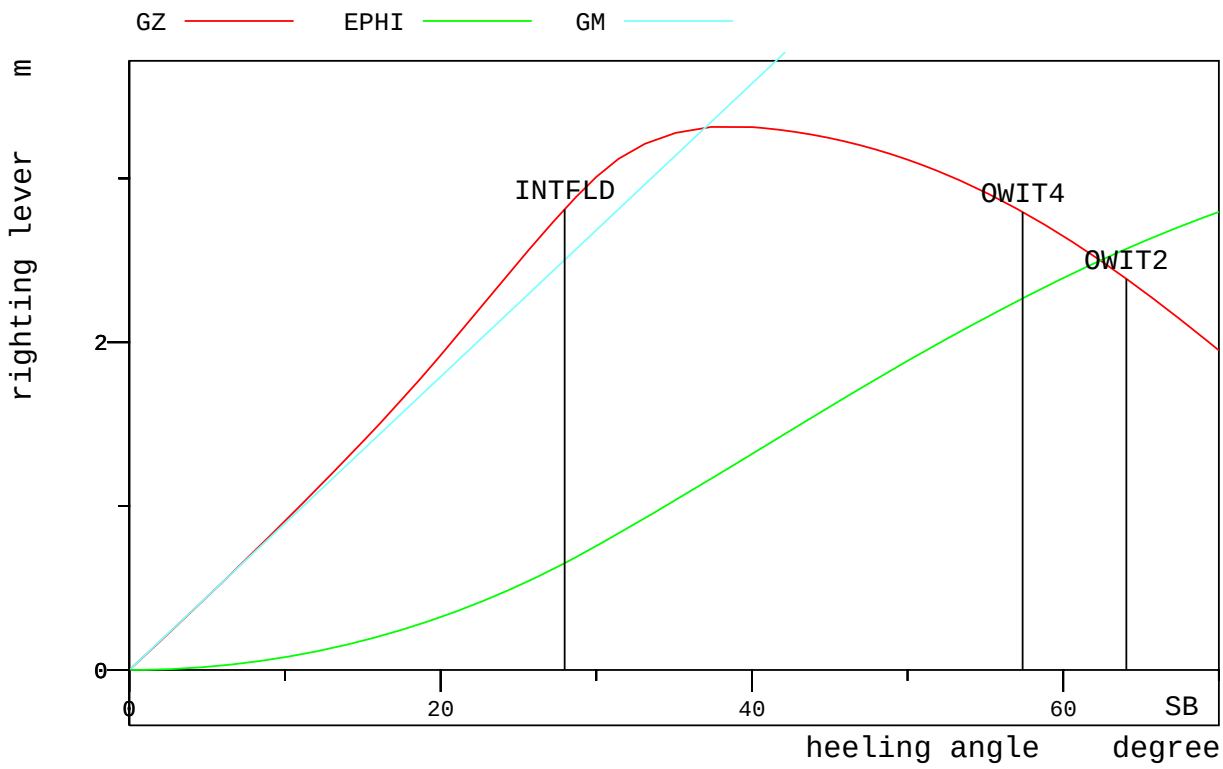
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-3933.8 t	(65.6%)	POSITION:	82.6 m	105
SHEAR FORCE (MAX)	4421.7 t	(73.7%)		111.4 m	141
SAGGING MOMENT	-78708.9 tm	(81.3%)		96.0 m	122
HOGGING MOMENT	30587.5 tm	(34.5%)		41.4 m	53

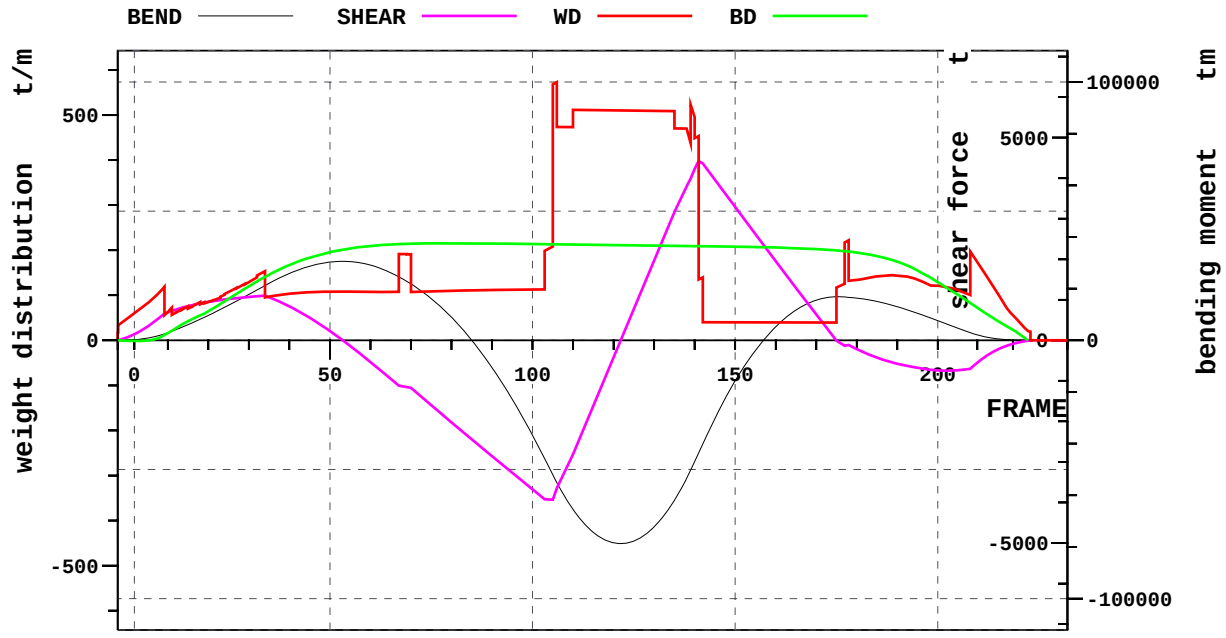
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.740)						
NO.1 CARGO HOLD	0.0	0.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	10223.4	100.0	96.78	0.00	8.67	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
SUBTOTAL	10223.4		96.78	0.00	8.67	0.0
, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 10%	2.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	99.00	0.00	19.20	0.0
SUBTOTAL	27.6		75.87	0.00	20.42	0.0
TOTAL	10251.0		96.73	0.00	8.70	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	339.3	100.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	774.2	100.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	0.00	0.00	0.00	5374.44
NO.2 WATER BALLAST T.	0.0	0.0	0.00	0.00	0.00	5374.44
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	929.2	100.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	929.2	100.0	39.87	-9.38	3.43	0.00
SUBTOTAL	10154.9		96.41	0.00	5.19	10748.87
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	2.1	10.0	8.28	-5.84	7.59	22.22
M.D.O. SETTL. TK.	2.5	10.0	9.88	-5.95	7.33	29.35
M.D.O. STOR. TK. (P)	3.5	10.0	12.81	5.91	7.45	43.37
M.D.O. STOR. TK. (S)	4.4	10.0	12.58	-5.91	7.52	54.86
SUBTOTAL	12.4		11.39	-2.61	7.47	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	4.1	10.0	4.98	7.15	11.13	31.97
FRESH WATER TK. (P)	9.4	10.0	1.14	6.32	11.15	44.74
FRESH WATER TK. (S)	13.4	10.0	2.40	-6.59	11.14	82.65
SUBTOTAL	51.6		4.32	-0.00	8.25	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	26.4	10.0	68.20	12.64	11.22	68.67
NO.1 FUEL OIL TK. (S)	26.4	10.0	68.20	-12.64	11.22	68.67
NO.2 FUEL OIL TK. (P)	25.5	10.0	40.23	12.62	11.23	0.00
NO.2 FUEL OIL TK. (S)	25.5	10.0	40.23	-12.62	11.23	0.00
H.F.O. TK. (P)	25.3	10.0	23.08	10.63	7.06	44.35
H.F.O. TK. (S)	18.0	10.0	23.44	-10.57	6.80	33.83
NO.1 H.F.O. SERV. TK.	2.5	10.0	15.49	-10.82	11.07	7.35
NO.1 H.F.O. SETTL. T.	3.0	10.0	18.65	-11.14	11.06	11.05
NO.2 H.F.O. SERV. TK.	1.9	10.0	12.65	-10.87	11.07	0.00
NO.2 H.F.O. SETTL. T.	2.2	10.0	9.70	-10.42	11.09	0.00
SUBTOTAL	156.7		43.37	-0.16	10.03	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	1.4	20.0	10.03	5.53	7.22	0.36
G/E L.O. STOR. TK.	2.6	20.0	10.04	7.48	7.96	10.28
M/E L.O. SETTL. TK.	3.5	20.0	8.83	6.21	7.80	19.67
M/E L.O. STOR. TK.	5.4	20.0	7.19	6.05	8.05	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	50.8		16.95	2.45	5.88	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	4.5	80.0	18.81	-2.29	1.40	1.69
BILGE TK.	12.9	80.0	11.52	0.00	1.14	9.85
F.O. DRAIN TK.	5.2	80.0	20.70	2.51	1.34	3.33
F.O. OVERFLOW TK.	7.2	80.0	23.04	2.98	1.31	7.97
L.O. DRAIN TK.	3.1	80.0	18.32	2.24	1.42	0.91
SEWAGE STOR. TK.	10.9	80.0	22.37	-2.83	1.31	10.07
SLUDGE TK.	12.0	80.0	21.80	-8.40	7.72	9.03
SUBTOTAL	55.8		19.16	-1.81	2.67	42.86
TOTAL	10482.3		94.27	-0.00	5.27	11391.80

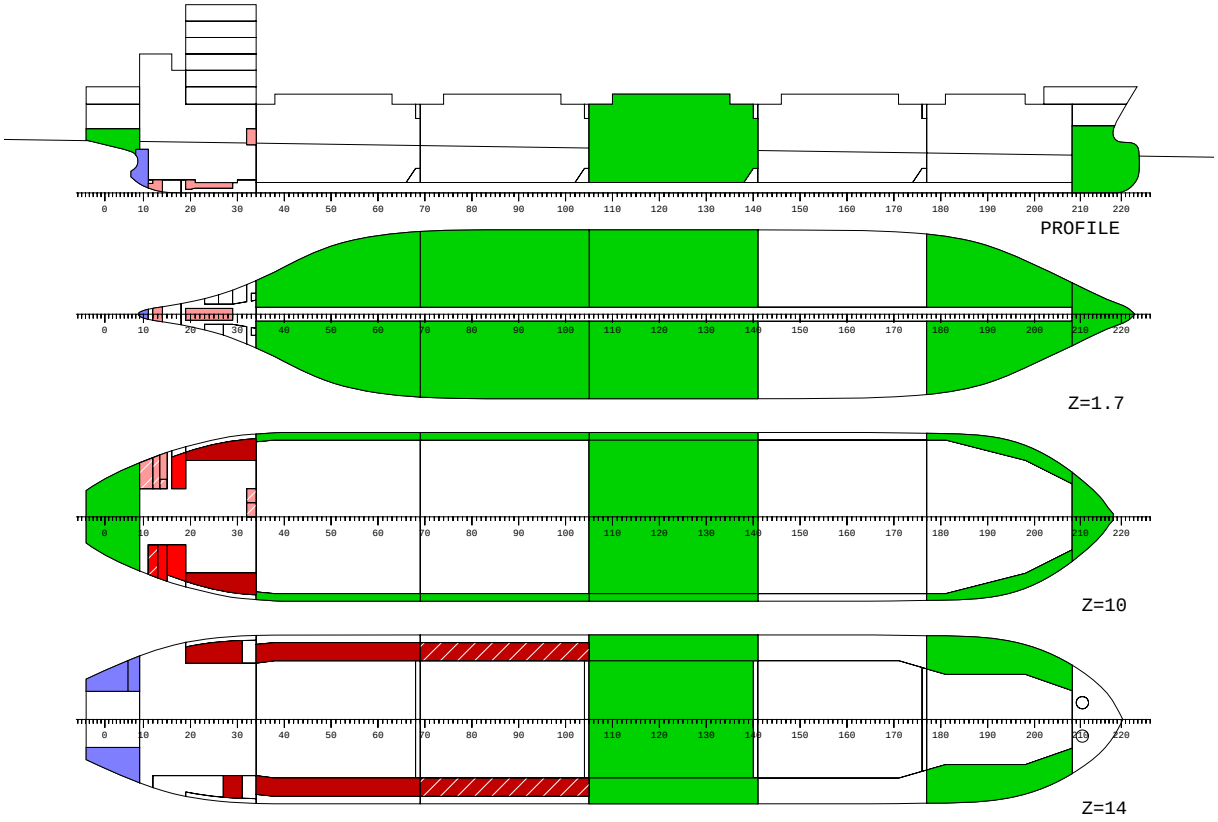


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.001	-0.00	0.000	0.0	0.000
10.0	0.019	0.91	0.078	1978.2	0.067
12.0	0.034	1.10	0.113	2368.5	0.081
20.0	0.168	1.92	0.323	3896.2	0.133
30.0	0.446	3.01	0.756	5695.9	0.194
40.0	0.018	3.31	1.320	7322.5	0.249
50.0	-0.813	3.11	1.885	8726.6	0.297
60.0	-1.791	2.65	2.391	9865.6	0.336
70.0	-2.866	1.95	2.795	10704.8	0.364

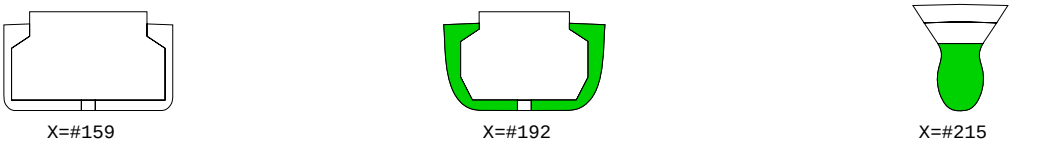


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	232	153	60.78	0.02
5.94	9.00	2509	660	87.29	11.83
18.60	25.00	13578	1018	106.94	91.73
25.80	34.00	21220	1100	95.11	140.98
40.20	52.00	30517	85	108.02	199.18
53.80	69.00	22715	-1155	191.74	214.28
68.20	87.00	-3996	-2611	111.34	215.19
82.60	105.00	-52098	-3934	208.11	213.26
97.00	123.00	-78563	295	510.17	211.10
111.40	141.00	-43724	4422	294.10	208.93
125.80	159.00	3607	2085	39.94	206.26
140.20	177.00	16712	-133	125.51	197.56
152.20	192.00	12065	-622	141.72	167.08
165.00	208.00	2958	-702	100.70	82.88
169.20	214.00	911	-306	125.45	51.73
173.40	220.00	140	-89	56.15	23.65

LOADING CASE 'LD03E -- HEAVY BALLAST CONDITION, FOR IACS, DEPARTURE







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	10154.9	96.41	0.00	5.19	10748.87
Diesel Oil	DO	121.6	11.23	-3.17	9.29	149.80
Fresh Water	FW	293.6	2.53	0.00	12.34	159.36
Grain Load	GRAIN	10223.4	96.78	0.00	8.67	0.00
Heavy Fuel Oil	HFO	1535.7	42.70	-0.15	12.05	233.92
Lubricating Oil	LO	101.1	12.74	4.87	7.77	56.99
	MASS	43.6	53.17	0.00	20.19	0.00
Miscellaneous	MIS	0.0	0.00	0.00	0.00	42.86
Deadweight		22474.0	90.76	-0.01	7.40	11391.80
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		22474.0	90.76	-0.01	7.40	
Total weight		31124.0	87.58	-0.00	8.18	

F L O A T I N G P O S I T I O N

Draught moulded	7.605	m	KM	13.31 m
Trim	-2.524	m	KG	8.18 m
Heel, PS=+	-0.0	deg		
TA	8.866	m	GM0	5.14 m
TF	6.343	m	GMCORR	-0.37 m
Trimming moment	-142898	tonm	GM	4.77 m
Heel	-0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.704	mrad	OK
AREA40	Area under GZ curve .	0.090	1.226	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.523	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	3.058	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	38.306	deg	OK
GM0.15	GM > 0.15 m	0.150	4.772	m	OK
IMOWEATH.	IMO weather criterion	1.000	4.800		OK
VISIBILI.		359.760	198.945		OK
PROPIMMER	Propeller should imm.	50.000	107.290	%	OK

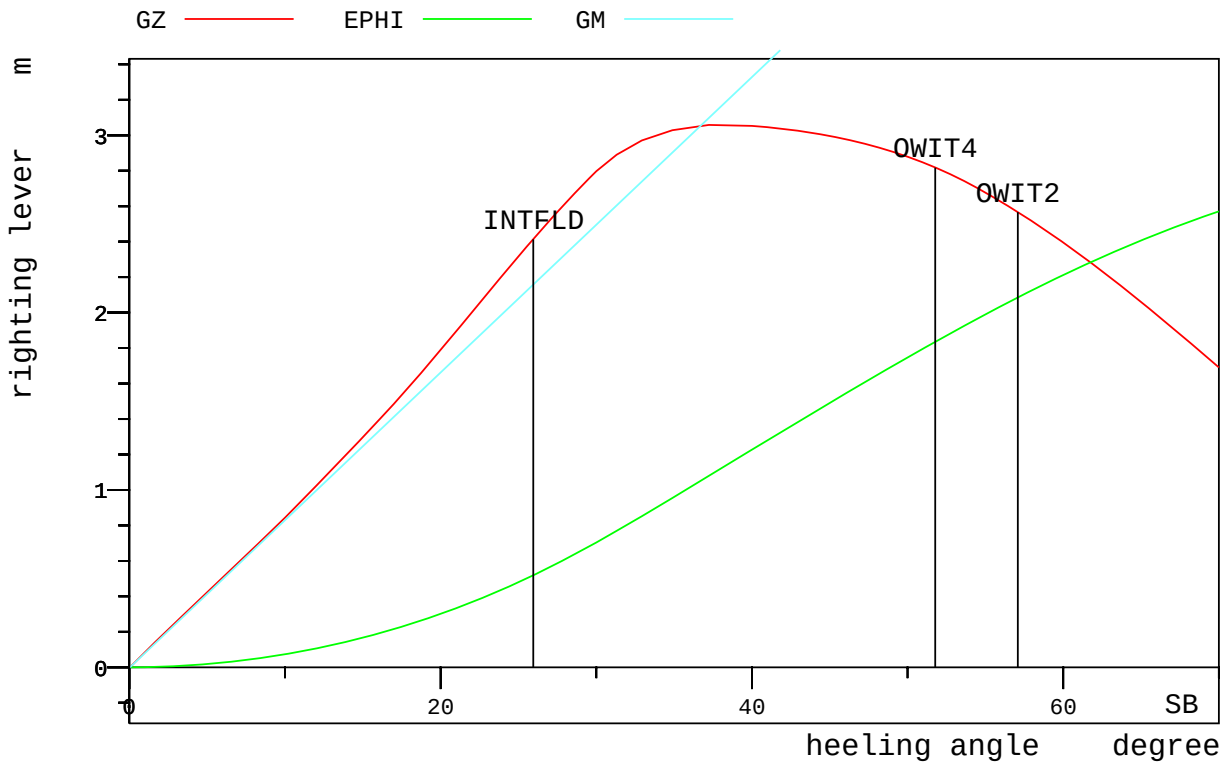
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-3884.2 t	(64.7%)	POSITION:	82.6 m	105
SHEAR FORCE (MAX)	4236.2 t	(70.6%)		111.4 m	141
SAGGING MOMENT	-68328.8 tm	(70.6%)		96.3 m	122
HOGGING MOMENT	38834.3 tm	(42.1%)		43.2 m	56

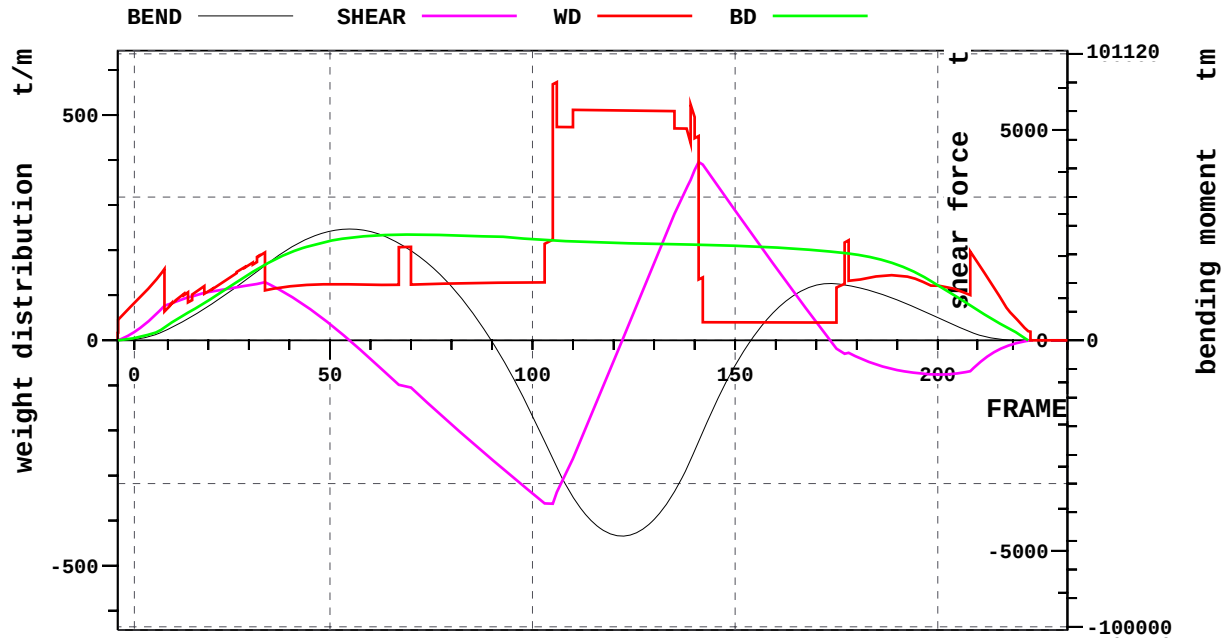
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.740)						
NO.1 CARGO HOLD	0.0	0.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	10223.4	100.0	96.78	0.00	8.67	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
SUBTOTAL	10223.4		96.78	0.00	8.67	0.0
, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	18.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	43.6		53.17	0.00	20.19	0.0
TOTAL	10267.0		96.60	0.00	8.72	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	339.3	100.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	774.2	100.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	0.00	0.00	0.00	5374.44
NO.2 WATER BALLAST T.	0.0	0.0	0.00	0.00	0.00	5374.44
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	929.2	100.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	929.2	100.0	39.87	-9.38	3.43	0.00
SUBTOTAL	10154.9		96.41	0.00	5.19	10748.87
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	20.2	98.0	8.23	-7.08	9.34	22.22
M.D.O. SETTL. TK.	24.2	98.0	9.82	-7.32	9.21	29.35
M.D.O. STOR. TK. (P)	33.9	98.0	12.66	7.26	9.29	43.37
M.D.O. STOR. TK. (S)	43.2	98.0	12.30	-7.22	9.32	54.86
SUBTOTAL	121.6		11.23	-3.17	9.29	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	40.7	100.0	4.97	7.43	12.96	31.97
FRESH WATER TK. (P)	93.7	100.0	0.94	6.60	13.03	44.74
FRESH WATER TK. (S)	134.4	100.0	2.17	-6.85	13.01	82.65
SUBTOTAL	293.6		2.53	0.00	12.34	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	258.4	98.0	68.20	11.81	12.57	68.67
NO.1 FUEL OIL TK. (S)	258.4	98.0	68.20	-11.81	12.57	68.67
NO.2 FUEL OIL TK. (P)	250.2	98.0	39.90	11.80	12.58	0.00
NO.2 FUEL OIL TK. (S)	250.2	98.0	39.90	-11.80	12.58	0.00
H.F.O. TK. (P)	248.1	98.0	20.62	11.15	11.01	44.35
H.F.O. TK. (S)	176.3	98.0	21.94	-11.14	10.32	33.83
NO.1 H.F.O. SERV. TK.	24.3	98.0	15.46	-11.00	12.38	7.35
NO.1 H.F.O. SETTL. T.	29.0	98.0	18.63	-11.27	12.36	11.05
NO.2 H.F.O. SERV. TK.	19.1	98.0	12.64	-11.07	12.39	0.00
NO.2 H.F.O. SETTL. T.	21.5	98.0	9.62	-10.64	12.44	0.00
SUBTOTAL	1535.7		42.70	-0.15	12.05	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	6.8	98.0	10.01	5.59	8.82	0.36
G/E L.O. STOR. TK.	12.9	98.0	10.02	8.28	9.39	10.28
M/E L.O. SETTL. TK.	17.2	98.0	8.81	7.17	9.29	19.67
M/E L.O. STOR. TK.	26.4	98.0	7.12	6.90	9.42	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.56	0.00	1.28	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	101.1		12.74	4.87	7.77	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	0.0	0.0	0.00	0.00	0.00	1.69
BILGE TK.	0.0	0.0	0.00	0.00	0.00	9.85
F.O. DRAIN TK.	0.0	0.0	0.00	0.00	0.00	3.33
F.O. OVERFLOW TK.	0.0	0.0	0.00	0.00	0.00	7.97
L.O. DRAIN TK.	0.0	0.0	0.00	0.00	0.00	0.91
SEWAGE STOR. TK.	0.0	0.0	0.00	0.00	0.00	10.07
SLUDGE TK.	0.0	0.0	21.80	-8.40	7.80	9.03
SUBTOTAL	0.0		0.00	0.00	0.00	42.86
TOTAL	12207.0		85.86	-0.01	6.29	11391.80

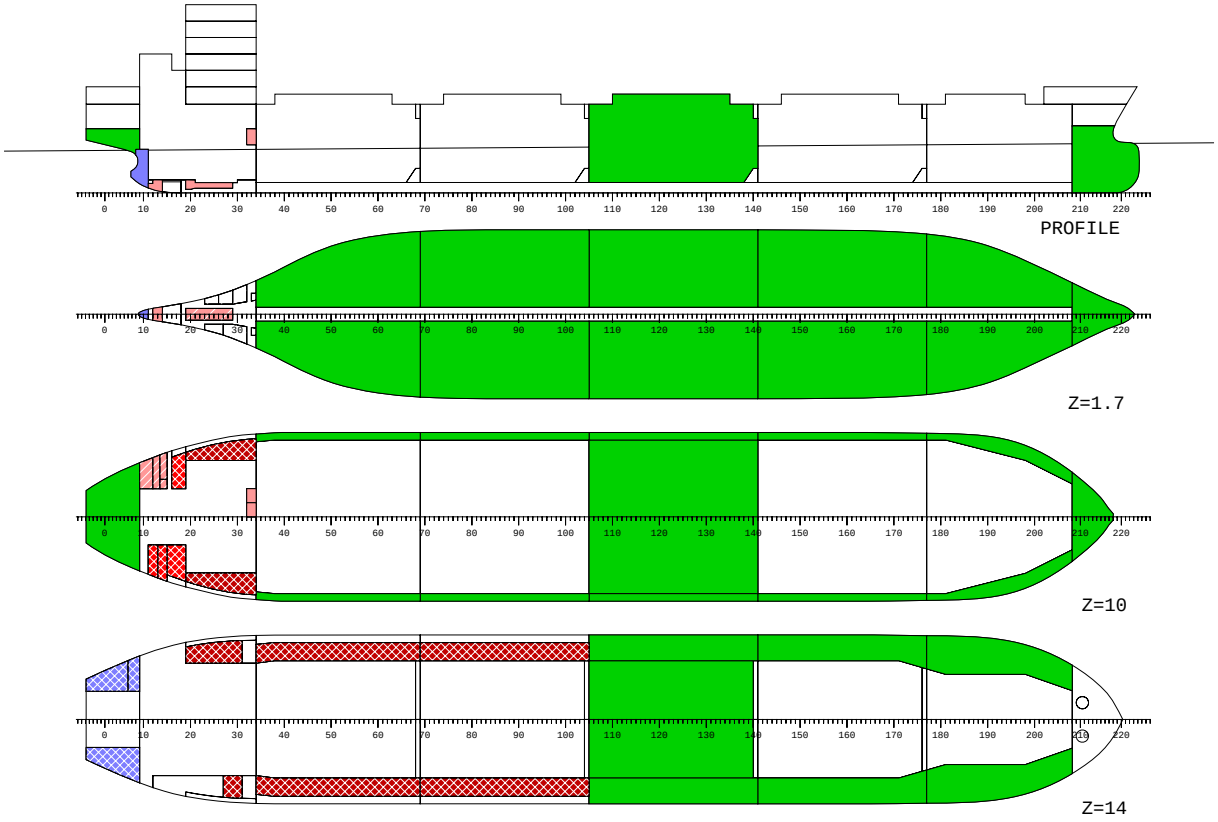


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.004	-0.00	0.000	0.0	0.000
10.0	0.015	0.84	0.074	1978.2	0.064
12.0	0.030	1.02	0.106	2368.5	0.076
20.0	0.156	1.79	0.301	3896.2	0.125
30.0	0.411	2.80	0.704	5695.9	0.183
40.0	-0.015	3.05	1.226	7322.5	0.235
50.0	-0.776	2.88	1.747	8726.6	0.280
60.0	-1.738	2.39	2.212	9865.6	0.317
70.0	-2.793	1.69	2.571	10704.8	0.344

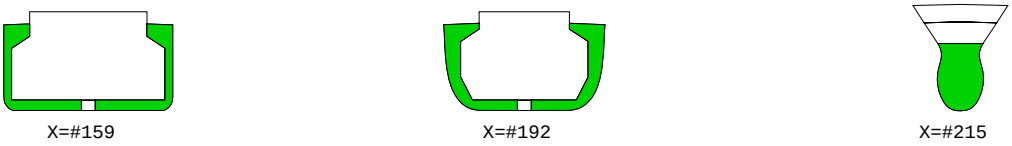


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	330	201	83.32	5.02
5.94	9.00	3262	824	64.06	28.39
18.60	25.00	16725	1227	137.66	117.28
25.80	34.00	26151	1382	110.86	167.88
40.20	52.00	38544	228	124.28	223.40
53.80	69.00	32133	-1101	207.09	234.65
68.20	87.00	6030	-2585	127.69	231.48
82.60	105.00	-41479	-3884	223.48	225.48
97.00	123.00	-68256	198	510.17	219.23
111.40	141.00	-35423	4236	294.10	212.98
125.80	159.00	9063	1871	39.94	206.24
140.20	177.00	19328	-317	125.51	193.52
152.20	192.00	12919	-740	141.72	160.28
165.00	208.00	2956	-735	100.69	77.59
169.20	214.00	842	-321	125.45	48.09
173.40	220.00	67	-92	56.15	21.88

LOADING CASE 'LD03F -- HEAVY BALLAST CONDITION, FOR IACS, ARRIVAL







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	12944.8	102.73	0.00	5.34	10748.87
Diesel Oil	DO	12.4	11.39	-2.61	7.47	149.80
Fresh Water	FW	51.6	4.32	-0.00	8.25	159.36
Grain Load	GRAIN	10223.4	96.78	0.00	8.67	0.00
Heavy Fuel Oil	HFO	156.7	43.37	-0.16	10.03	233.92
Lubricating Oil	LO	50.8	16.95	2.45	5.88	56.99
	MASS	27.6	75.87	0.00	20.42	0.00
Miscellaneous	MIS	55.8	19.16	-1.81	2.67	42.86
Deadweight		23523.2	99.07	-0.00	6.84	11391.80
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		23523.2	99.07	-0.00	6.84	
Total weight		32173.2	93.75	-0.00	7.74	

F L O A T I N G P O S I T I O N

Draught moulded	7.823	m	KM	12.92 m
Trim	1.182	m	KG	7.74 m
Heel, PS=+	0.0	deg		
TA	7.232	m	GM0	5.18 m
TF	8.414	m	GMCORR	-0.35 m
Trimming moment	60248	tonm	GM	4.82 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.712	mrad	OK
AREA40	Area under GZ curve .	0.090	1.245	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.534	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	3.161	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	40.776	deg	OK
GM0.15	GM > 0.15 m	0.150	4.822	m	OK
IMOWEATH.	IMO weather criterion	1.000	5.259		OK
VISIBILI.		359.760	123.859		OK
PROPIMMER	Propeller should imm.	50.000	76.454	%	OK

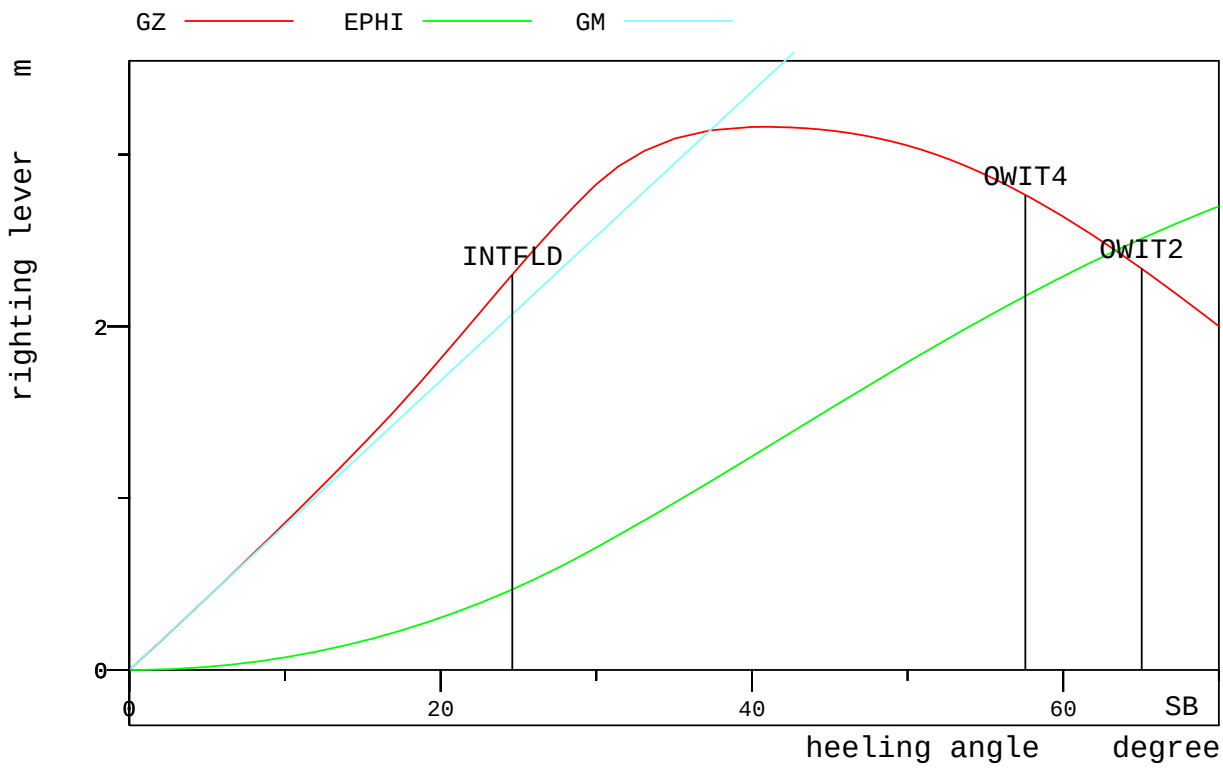
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-4204.0 t	(70.1%)	POSITION:	82.6 m	105
SHEAR FORCE (MAX)	3527.9 t	(58.8%)		111.4 m	141
SAGGING MOMENT	-83573.4 tm	(86.3%)		98.0 m	124
HOGGING MOMENT	33125.5 tm	(37.3%)		41.4 m	53

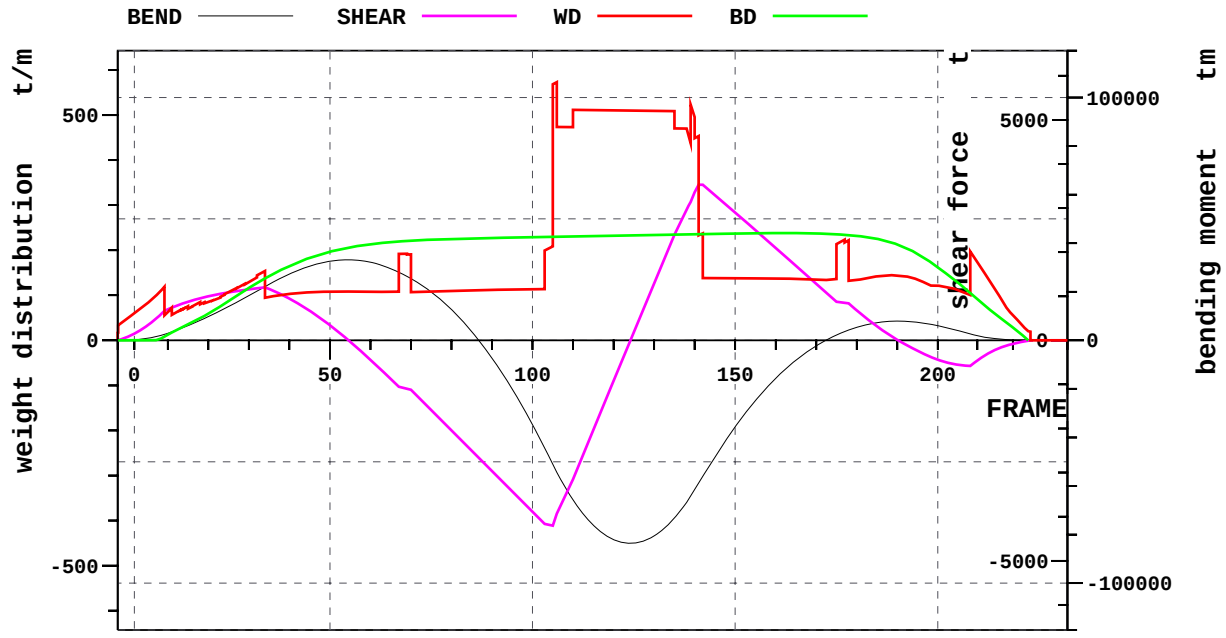
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.740)						
NO.1 CARGO HOLD	0.0	0.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	10223.4	100.0	96.78	0.00	8.67	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
SUBTOTAL	10223.4		96.78	0.00	8.67	0.0
, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 10%	2.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	99.00	0.00	19.20	0.0
SUBTOTAL	27.6		75.87	0.00	20.42	0.0
TOTAL	10251.0		96.73	0.00	8.70	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	339.3	100.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	774.2	100.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	10.47	5.87	5374.44
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	-10.47	5.87	5374.44
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	1409.3	100.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	972.8	100.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	929.2	100.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	929.2	100.0	39.87	-9.38	3.43	0.00
SUBTOTAL	12944.8		102.73	0.00	5.34	10748.87
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	2.1	10.0	8.28	-5.84	7.59	22.22
M.D.O. SETTL. TK.	2.5	10.0	9.88	-5.95	7.33	29.35
M.D.O. STOR. TK. (P)	3.5	10.0	12.81	5.91	7.45	43.37
M.D.O. STOR. TK. (S)	4.4	10.0	12.58	-5.91	7.52	54.86
SUBTOTAL	12.4		11.39	-2.61	7.47	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	4.1	10.0	4.98	7.15	11.13	31.97
FRESH WATER TK. (P)	9.4	10.0	1.14	6.32	11.15	44.74
FRESH WATER TK. (S)	13.4	10.0	2.40	-6.59	11.14	82.65
SUBTOTAL	51.6		4.32	-0.00	8.25	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	26.4	10.0	68.20	12.64	11.22	68.67
NO.1 FUEL OIL TK. (S)	26.4	10.0	68.20	-12.64	11.22	68.67
NO.2 FUEL OIL TK. (P)	25.5	10.0	40.23	12.62	11.23	0.00
NO.2 FUEL OIL TK. (S)	25.5	10.0	40.23	-12.62	11.23	0.00
H.F.O. TK. (P)	25.3	10.0	23.08	10.63	7.06	44.35
H.F.O. TK. (S)	18.0	10.0	23.44	-10.57	6.80	33.83
NO.1 H.F.O. SERV. TK.	2.5	10.0	15.49	-10.82	11.07	7.35
NO.1 H.F.O. SETTL. T.	3.0	10.0	18.65	-11.14	11.06	11.05
NO.2 H.F.O. SERV. TK.	1.9	10.0	12.65	-10.87	11.07	0.00
NO.2 H.F.O. SETTL. T.	2.2	10.0	9.70	-10.42	11.09	0.00
SUBTOTAL	156.7		43.37	-0.16	10.03	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	1.4	20.0	10.03	5.53	7.22	0.36
G/E L.O. STOR. TK.	2.6	20.0	10.04	7.48	7.96	10.28
M/E L.O. SETTL. TK.	3.5	20.0	8.83	6.21	7.80	19.67
M/E L.O. STOR. TK.	5.4	20.0	7.19	6.05	8.05	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	50.8		16.95	2.45	5.88	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	4.5	80.0	18.81	-2.29	1.40	1.69
BILGE TK.	12.9	80.0	11.52	0.00	1.14	9.85
F.O. DRAIN TK.	5.2	80.0	20.70	2.51	1.34	3.33
F.O. OVERFLOW TK.	7.2	80.0	23.04	2.98	1.31	7.97
L.O. DRAIN TK.	3.1	80.0	18.32	2.24	1.42	0.91
SEWAGE STOR. TK.	10.9	80.0	22.37	-2.83	1.31	10.07
SLUDGE TK.	12.0	80.0	21.80	-8.40	7.72	9.03
SUBTOTAL	55.8		19.16	-1.81	2.67	42.86
TOTAL	13272.2		100.88	-0.00	5.40	11391.80

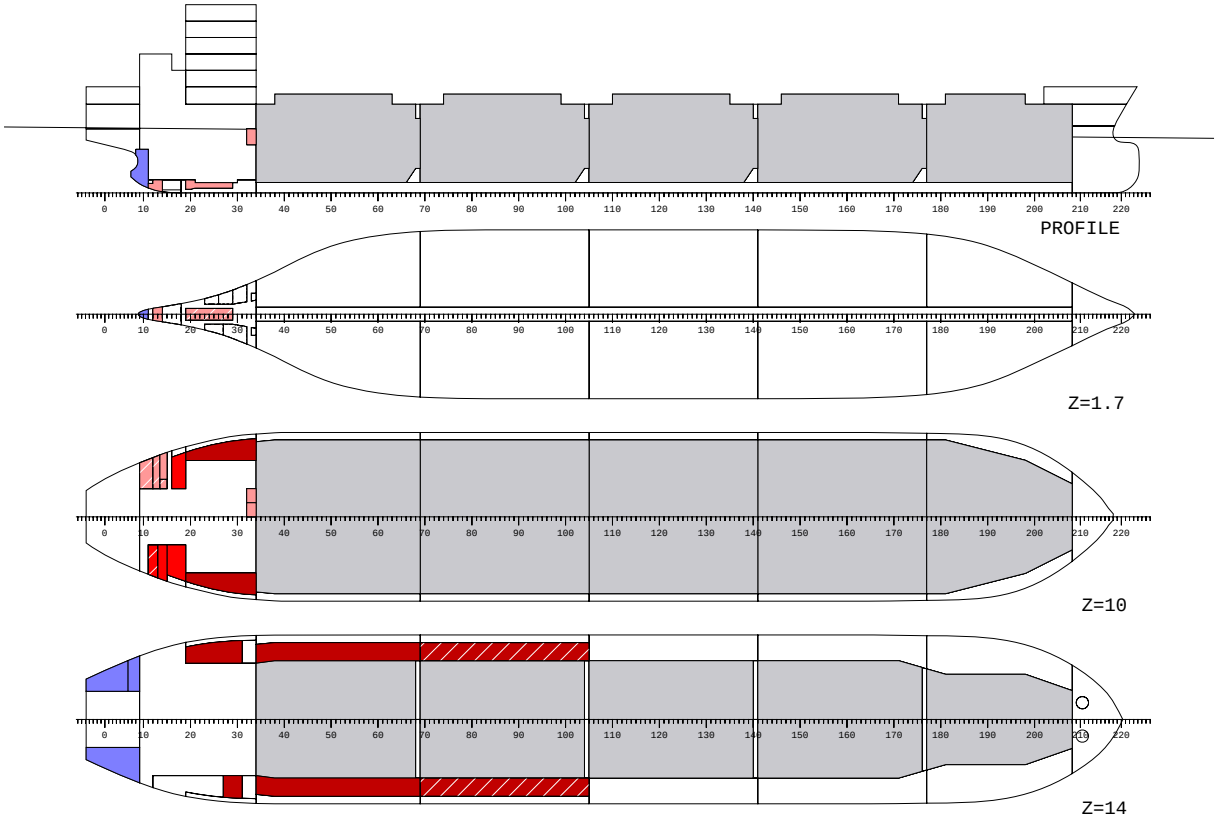


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.001	-0.00	0.000	0.0	0.000
10.0	0.018	0.86	0.074	1978.2	0.061
12.0	0.033	1.04	0.107	2368.5	0.074
20.0	0.161	1.81	0.304	3896.2	0.121
30.0	0.416	2.83	0.712	5695.9	0.177
40.0	0.061	3.16	1.245	7322.5	0.228
50.0	-0.642	3.05	1.791	8726.6	0.271
60.0	-1.538	2.64	2.292	9865.6	0.307
70.0	-2.531	2.00	2.699	10704.8	0.333

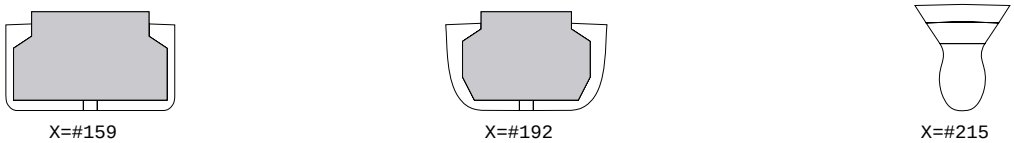


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	212	152	60.69	0.00
5.94	9.00	2462	668	87.41	8.18
18.60	25.00	13928	1087	106.74	86.82
25.80	34.00	22139	1200	124.32	137.65
40.20	52.00	32934	195	108.02	200.73
53.80	69.00	26268	-1095	192.28	220.64
68.20	87.00	-543	-2683	111.34	226.64
82.60	105.00	-51122	-4204	208.67	229.82
97.00	123.00	-83457	-250	510.17	232.74
111.40	141.00	-58978	3528	343.03	235.67
125.80	159.00	-17394	2178	137.33	238.10
140.20	177.00	3459	851	223.55	234.05
152.20	192.00	7843	-91	141.72	204.85
165.00	208.00	2622	-580	100.70	105.00
169.20	214.00	889	-259	125.45	65.13
173.40	220.00	197	-82	56.15	28.89

LOADING CASE 'LD04A -- HOMOGENEOUSLY LIGHT CARGO CONDITION, DEPARTURE







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	0.0	0.00	0.00	0.00	0.00
Diesel Oil	DO	121.6	11.23	-3.17	9.29	149.80
Fresh Water	FW	293.6	2.53	0.00	12.34	159.36
Grain Load	GRAIN	32851.6	93.51	0.00	8.76	0.00
Heavy Fuel Oil	HFO	1535.7	42.68	-0.15	12.40	233.92
Lubricating Oil	LO	101.1	12.73	4.87	7.77	56.99
	MASS	43.6	53.17	0.00	20.19	0.00
Miscellaneous	MIS	7.0	19.24	-1.81	1.99	42.86
Deadweight		34954.2	89.93	-0.00	8.96	642.92
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		34954.2	89.93	-0.00	8.96	
Total weight		43604.2	87.82	-0.00	9.20	

F L O A T I N G P O S I T I O N

Draught moulded	10.267	m	KM	12.31 m
Trim	-1.640	m	KG	9.20 m
Heel, PS=+	-0.1	deg		
TA	11.087	m	GM0	3.11 m
TF	9.447	m	GMCORR	-0.01 m
Trimming moment	-97996	tonm	GM	3.09 m
Heel	-0.1	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.424	mrad	OK
AREA40	Area under GZ curve .	0.090	0.655	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.231	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	1.652	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	41.446	deg	OK
GM0.15	GM > 0.15 m	0.150	3.092	m	OK
IMOWEATH.	IMO weather criterion	1.000	2.834		OK
VISIBILI.		359.760	139.328		OK
PROPIMMER	Propeller should imm.	50.000	149.194	%	OK

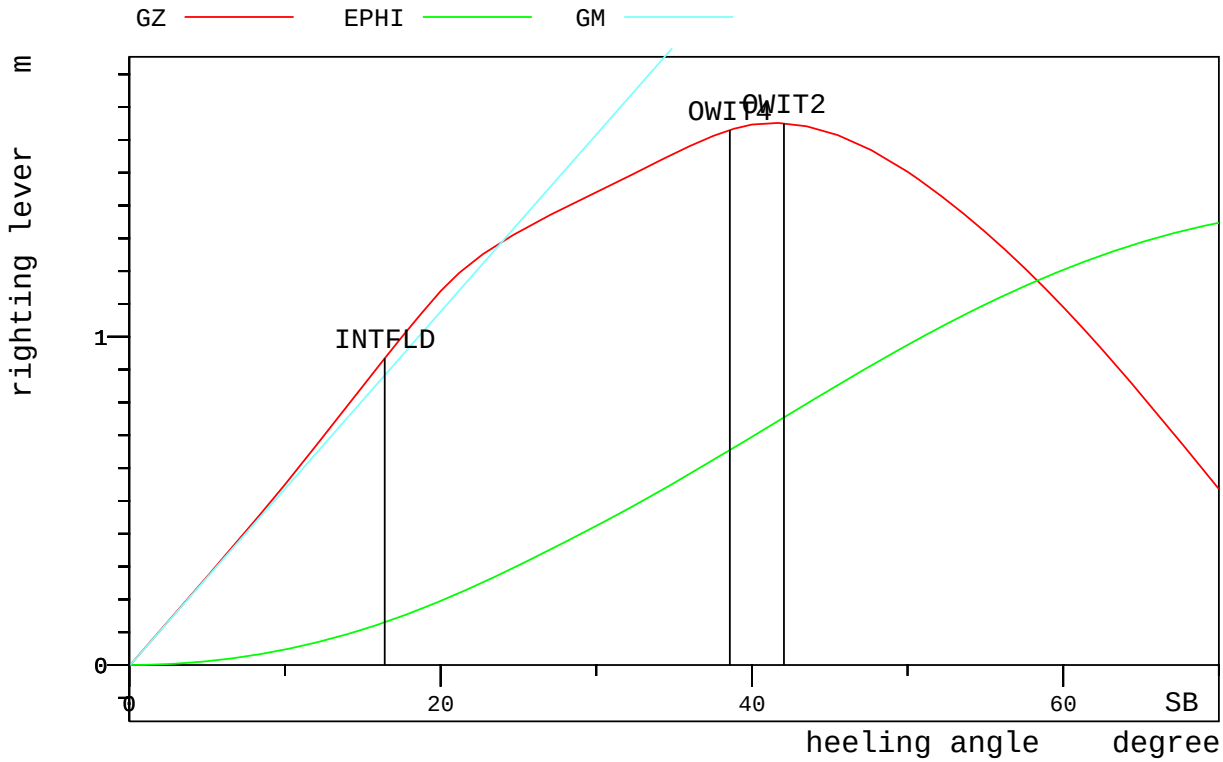
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-396.7 t	(6.6%)	POSITION:	29.0 m	38
SHEAR FORCE (MAX)	266.6 t	(4.4%)		165.0 m	208
SAGGING MOMENT	-9812.1 tm	(10.1%)		75.3 m	96
HOGGING MOMENT	2446.1 tm	(6.1%)		17.0 m	23

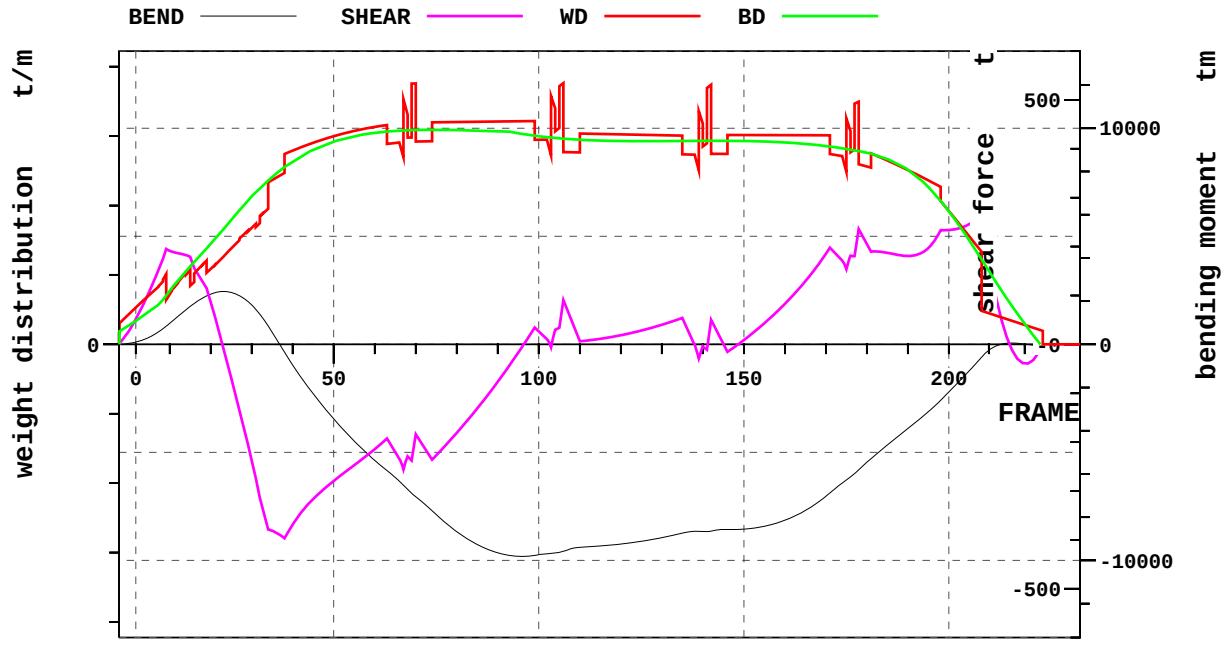
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.721)						
NO.1 CARGO HOLD	4797.4	100.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	7183.4	100.0	125.57	0.00	8.67	0.0
NO.3 CARGO HOLD	7192.3	100.0	96.78	0.00	8.67	0.0
NO.4 CARGO HOLD	7192.1	100.0	67.98	0.00	8.67	0.0
NO.5 CARGO HOLD	6486.4	100.0	40.05	0.00	9.09	0.0
SUBTOTAL	32851.6		93.51	0.00	8.76	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	18.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	43.6		53.17	0.00	20.19	0.0
TOTAL	32895.2		93.46	0.00	8.77	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	20.2	98.0	8.23	-7.08	9.34	22.22
M.D.O. SETTL. TK.	24.2	98.0	9.82	-7.32	9.21	29.35
M.D.O. STOR. TK. (P)	33.9	98.0	12.66	7.26	9.29	43.37
M.D.O. STOR. TK. (S)	43.2	98.0	12.30	-7.22	9.32	54.86
SUBTOTAL	121.6		11.23	-3.17	9.29	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	40.7	100.0	4.97	7.43	12.96	31.97
FRESH WATER TK. (P)	93.7	100.0	0.94	6.60	13.03	44.74
FRESH WATER TK. (S)	134.4	100.0	2.17	-6.85	13.01	82.65
SUBTOTAL	293.6		2.53	0.00	12.34	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	258.4	98.0	68.20	11.73	13.08	68.67
NO.1 FUEL OIL TK. (S)	258.4	98.0	68.20	-11.73	13.08	68.67
NO.2 FUEL OIL TK. (P)	250.2	98.0	39.86	11.71	13.09	0.00
NO.2 FUEL OIL TK. (S)	250.2	98.0	39.86	-11.71	13.09	0.00
H.F.O. TK. (P)	248.1	98.0	20.60	11.15	11.04	44.35
H.F.O. TK. (S)	176.3	98.0	21.94	-11.14	10.32	33.83
NO.1 H.F.O. SERV. TK.	24.3	98.0	15.46	-11.00	12.38	7.35
NO.1 H.F.O. SETTL. T.	29.0	98.0	18.63	-11.27	12.36	11.05
NO.2 H.F.O. SERV. TK.	19.1	98.0	12.64	-11.07	12.39	0.00
NO.2 H.F.O. SETTL. T.	21.5	98.0	9.62	-10.64	12.44	0.00
SUBTOTAL	1535.7		42.68	-0.15	12.40	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	6.8	98.0	10.01	5.59	8.82	0.36
G/E L.O. STOR. TK.	12.9	98.0	10.02	8.28	9.39	10.28
M/E L.O. SETTL. TK.	17.2	98.0	8.81	7.17	9.29	19.67
M/E L.O. STOR. TK.	26.4	98.0	7.12	6.90	9.42	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	101.1		12.73	4.87	7.77	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	0.6	10.0	18.98	-1.96	0.68	1.69
BILGE TK.	1.6	10.0	11.66	0.00	0.32	9.85
F.O. DRAIN TK.	0.6	10.0	20.77	2.04	0.56	3.33
F.O. OVERFLOW TK.	0.9	10.0	23.05	2.35	0.50	7.97
L.O. DRAIN TK.	0.4	10.0	18.43	1.94	0.72	0.91
SEWAGE STOR. TK.	1.4	10.0	22.44	-2.26	0.52	10.07
SLUDGE TK.	1.5	10.0	21.80	-8.40	7.44	9.03
SUBTOTAL	7.0		19.24	-1.81	1.99	42.86
TOTAL	2059.0		33.55	-0.07	11.94	642.92

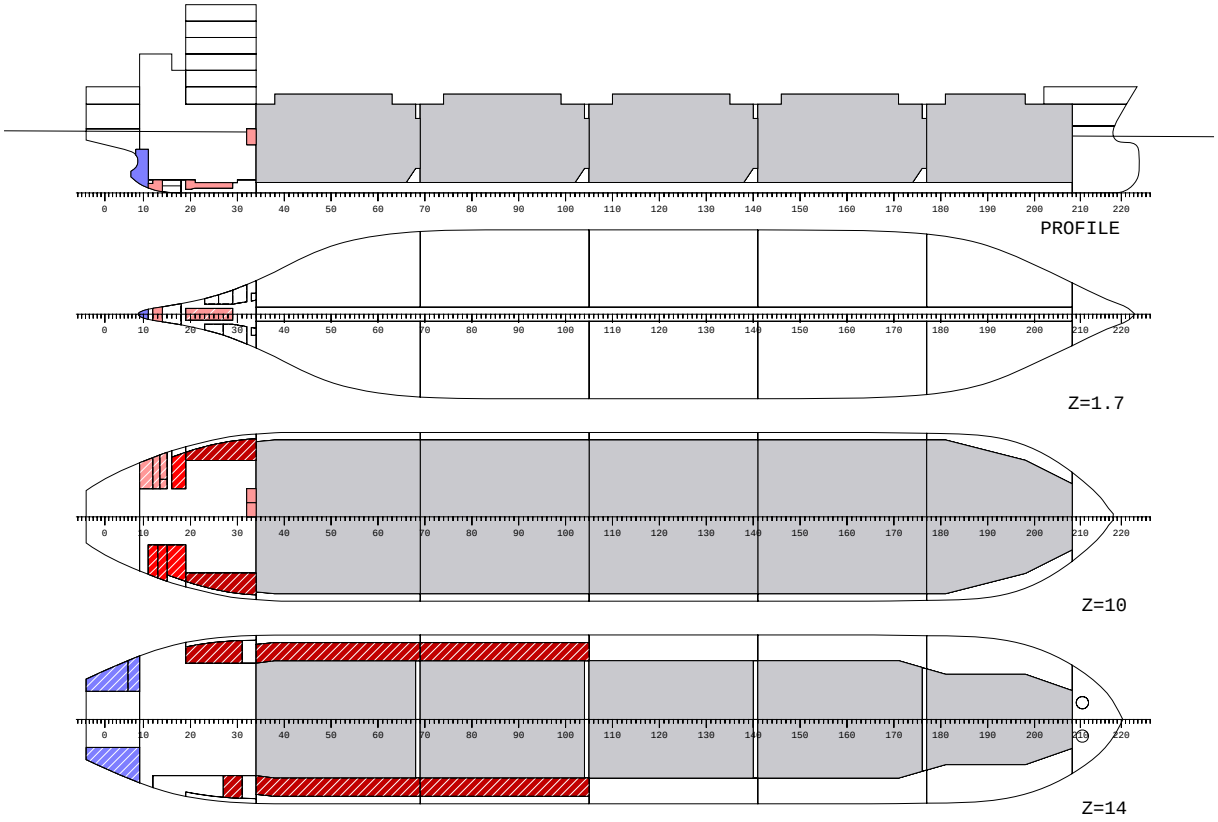


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.003	-0.00	0.000	0.0	0.000
10.0	0.013	0.55	0.047	111.6	0.003
12.0	0.025	0.67	0.068	133.7	0.003
20.0	0.082	1.14	0.195	219.9	0.005
30.0	-0.106	1.44	0.424	321.5	0.007
40.0	-0.341	1.65	0.695	413.3	0.009
50.0	-0.866	1.50	0.975	492.5	0.011
60.0	-1.588	1.09	1.204	556.8	0.013
70.0	-2.370	0.54	1.347	604.2	0.014

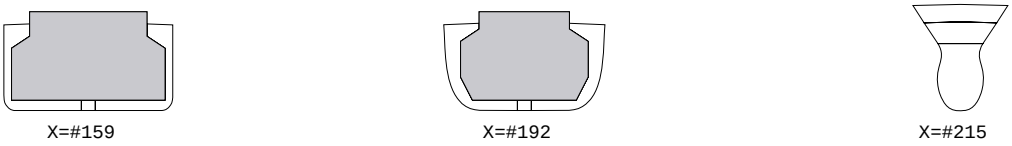


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	106	52	53.14	33.85
5.94	9.00	854	195	64.03	70.43
18.60	25.00	2400	-69	138.27	179.15
25.80	34.00	849	-379	233.35	235.85
40.20	52.00	-3876	-267	303.16	295.22
53.80	69.00	-6892	-238	297.96	308.54
68.20	87.00	-9445	-110	321.03	307.55
82.60	105.00	-9629	34	311.64	303.74
97.00	123.00	-9170	24	302.12	299.68
111.40	141.00	-8664	-11	289.51	295.62
125.80	159.00	-8207	74	301.39	291.06
140.20	177.00	-5983	180	280.74	279.50
152.20	192.00	-3586	183	245.19	242.14
165.00	208.00	-585	267	133.32	122.12
169.20	214.00	28	38	38.08	73.77
173.40	220.00	11	-39	27.85	29.28

LOADING CASE 'LD04B -- HOMOGENEOUSLY LIGHT CARGO CONDITION, HALFWAY







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	0.0	0.00	0.00	0.00	0.00
Diesel Oil	DO	62.0	11.26	-2.98	8.46	149.80
Fresh Water	FW	159.2	2.90	-0.00	10.95	159.36
Grain Load	GRAIN	32851.6	93.51	0.00	8.76	0.00
Heavy Fuel Oil	HFO	783.5	42.77	-0.15	11.35	233.92
Lubricating Oil	LO	76.6	14.10	4.02	6.96	56.99
	MASS	34.6	63.09	0.00	20.29	0.00
Miscellaneous	MIS	27.9	19.18	-1.81	2.34	42.86
Deadweight		33995.5	91.50	-0.00	8.83	642.92
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		33995.5	91.50	-0.00	8.83	
Total weight		42645.5	89.02	-0.00	9.11	

F L O A T I N G P O S I T I O N

Draught moulded	10.075	m	KM	12.32 m
Trim	-0.844	m	KG	9.11 m
Heel, PS=+	0.0	deg		
TA	10.497	m	GM0	3.21 m
TF	9.653	m	GMCORR	-0.02 m
Trimming moment	-50041	tonm	GM	3.20 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.447	mrad	OK
AREA40	Area under GZ curve .	0.090	0.738	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.291	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	1.779	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	41.764	deg	OK
GM0.15	GM > 0.15 m	0.150	3.198	m	OK
IMOWEATH.	IMO weather criterion	1.000	3.192		OK
VISIBILI.		359.760	127.571		OK
PROPIMMER	Propeller should imm.	50.000	138.061	%	OK

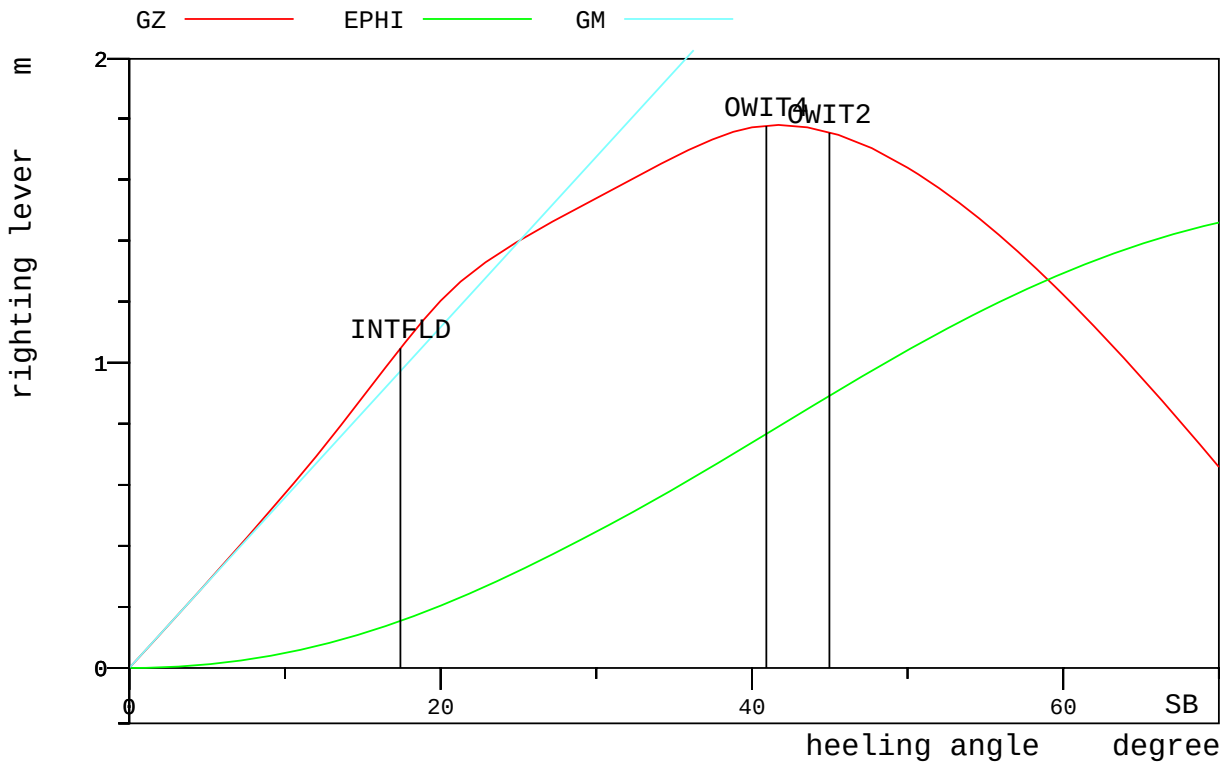
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-495.6 t	(8.3%)	POSITION:	29.0 m	38
SHEAR FORCE (MAX)	317.6 t	(5.3%)		141.0 m	178
SAGGING MOMENT	-13822.6 tm	(14.3%)		82.6 m	105
HOGGING MOMENT	1694.7 tm	(4.6%)		15.4 m	21

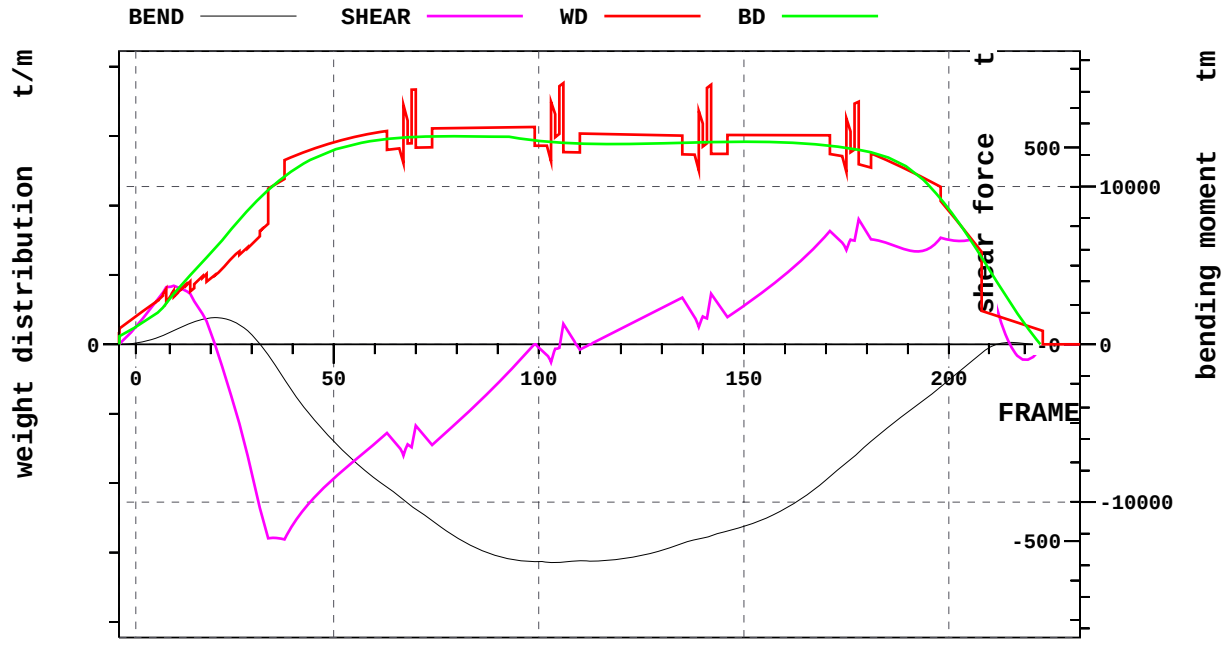
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.721)						
NO.1 CARGO HOLD	4797.4	100.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	7183.4	100.0	125.57	0.00	8.67	0.0
NO.3 CARGO HOLD	7192.3	100.0	96.78	0.00	8.67	0.0
NO.4 CARGO HOLD	7192.1	100.0	67.98	0.00	8.67	0.0
NO.5 CARGO HOLD	6486.4	100.0	40.05	0.00	9.09	0.0
SUBTOTAL	32851.6		93.51	0.00	8.76	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	9.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	34.6		63.09	0.00	20.29	0.0
TOTAL	32886.2		93.48	0.00	8.77	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	10.3	50.0	8.24	-6.66	8.53	22.22
M.D.O. SETTL. TK.	12.4	50.0	9.83	-6.86	8.34	29.35
M.D.O. STOR. TK. (P)	17.3	50.0	12.68	6.80	8.45	43.37
M.D.O. STOR. TK. (S)	22.0	50.0	12.35	-6.76	8.50	54.86
SUBTOTAL	62.0		11.26	-2.98	8.46	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	20.4	50.0	4.98	7.32	11.97	31.97
FRESH WATER TK. (P)	46.9	50.0	1.02	6.48	12.05	44.74
FRESH WATER TK. (S)	67.2	50.0	2.26	-6.75	12.02	82.65
SUBTOTAL	159.2		2.90	-0.00	10.95	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	131.9	50.0	68.20	11.90	12.29	68.67
NO.1 FUEL OIL TK. (S)	131.9	50.0	68.20	-11.90	12.29	68.67
NO.2 FUEL OIL TK. (P)	127.7	50.0	39.94	11.88	12.30	0.00
NO.2 FUEL OIL TK. (S)	127.7	50.0	39.94	-11.88	12.30	0.00
H.F.O. TK. (P)	126.6	50.0	21.26	10.98	9.26	44.35
H.F.O. TK. (S)	90.0	50.0	21.60	-10.89	8.63	33.83
NO.1 H.F.O. SERV. TK.	12.4	50.0	15.47	-10.92	11.69	7.35
NO.1 H.F.O. SETTL. T.	14.8	50.0	18.64	-11.21	11.66	11.05
NO.2 H.F.O. SERV. TK.	9.7	50.0	12.65	-10.98	11.70	0.00
NO.2 H.F.O. SETTL. T.	11.0	50.0	9.65	-10.54	11.75	0.00
SUBTOTAL	783.5		42.77	-0.15	11.35	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	4.1	60.0	10.01	5.58	8.05	0.36
G/E L.O. STOR. TK.	7.9	60.0	10.02	8.01	8.78	10.28
M/E L.O. SETTL. TK.	10.5	60.0	8.82	6.85	8.65	19.67
M/E L.O. STOR. TK.	16.1	60.0	7.14	6.62	8.83	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	76.6		14.10	4.02	6.96	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	2.3	40.0	18.84	-2.15	1.07	1.69
BILGE TK.	6.4	40.0	11.55	0.00	0.73	9.85
F.O. DRAIN TK.	2.6	40.0	20.71	2.30	0.98	3.33
F.O. OVERFLOW TK.	3.6	40.0	23.05	2.70	0.93	7.97
L.O. DRAIN TK.	1.5	40.0	18.35	2.11	1.09	0.91
SEWAGE STOR. TK.	5.4	40.0	22.38	-2.58	0.94	10.07
SLUDGE TK.	6.0	40.0	21.80	-8.40	7.56	9.03
SUBTOTAL	27.9		19.18	-1.81	2.34	42.86
TOTAL	1109.3		32.71	-0.04	10.60	642.92

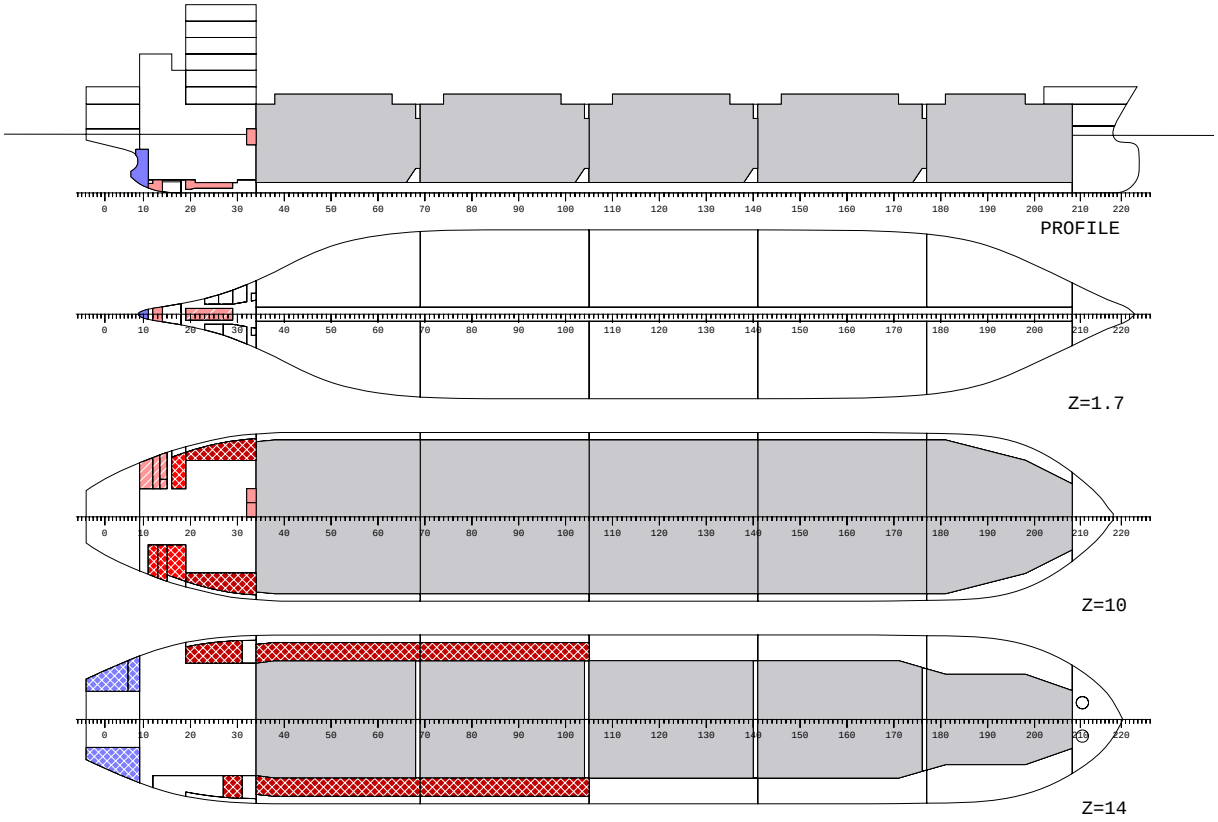


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.001	-0.00	0.000	0.0	0.000
10.0	0.015	0.57	0.049	111.6	0.003
12.0	0.027	0.69	0.071	133.7	0.003
20.0	0.109	1.20	0.204	219.9	0.005
30.0	-0.060	1.54	0.447	321.5	0.008
40.0	-0.285	1.77	0.738	413.3	0.010
50.0	-0.811	1.64	1.041	492.5	0.012
60.0	-1.547	1.22	1.293	556.8	0.013
70.0	-2.347	0.66	1.459	604.2	0.014

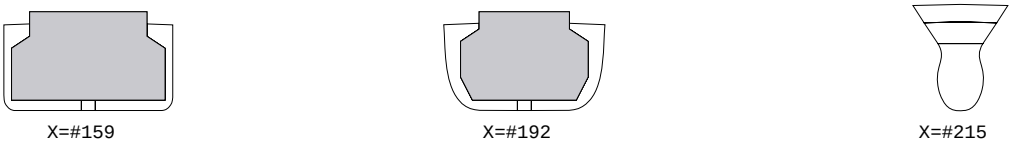


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	82	44	40.51	25.24
5.94	9.00	653	145	59.87	59.22
18.60	25.00	1500	-131	122.64	165.37
25.80	34.00	-693	-493	224.30	222.14
40.20	52.00	-6668	-323	294.35	283.28
53.80	69.00	-10256	-261	366.90	298.46
68.20	87.00	-13148	-130	312.13	299.45
82.60	105.00	-13823	-9	303.14	297.60
97.00	123.00	-13478	55	302.12	295.51
111.40	141.00	-12221	66	289.51	293.42
125.80	159.00	-10560	169	301.39	290.84
140.20	177.00	-7085	264	280.74	281.23
152.20	192.00	-3900	236	245.19	245.30
165.00	208.00	-540	279	133.32	124.77
169.20	214.00	85	41	38.08	75.32
173.40	220.00	56	-39	27.85	29.28

LOADING CASE 'LD04C -- HOMOGENEOUSLY LIGHT CARGO CONDITION, ARRIVAL







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	0.0	0.00	0.00	0.00	0.00
Diesel Oil	DO	12.4	11.39	-2.61	7.47	149.80
Fresh Water	FW	51.6	4.32	-0.00	8.25	159.36
Grain Load	GRAIN	32851.6	93.51	0.00	8.76	0.00
Heavy Fuel Oil	HFO	156.7	43.37	-0.16	10.03	233.92
Lubricating Oil	LO	50.8	16.95	2.45	5.88	56.99
	MASS	27.6	75.29	0.00	20.42	0.00
Miscellaneous	MIS	55.8	19.16	-1.81	2.67	42.86
Deadweight		33206.6	92.85	-0.00	8.75	642.92
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		33206.6	92.85	-0.00	8.75	
Total weight		41856.6	90.05	-0.00	9.05	

F L O A T I N G P O S I T I O N

Draught moulded	9.916	m	KM	12.33 m
Trim	-0.184	m	KG	9.05 m
Heel, PS=+	0.0	deg		
TA	10.008	m	GM0	3.27 m
TF	9.824	m	GMCORR	-0.02 m
Trimming moment	-10827	tonm	GM	3.26 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.461	mrad	OK
AREA40	Area under GZ curve .	0.090	0.765	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.304	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	1.853	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	41.808	deg	OK
GM0.15	GM > 0.15 m	0.150	3.258	m	OK
IMOWEATH.	IMO weather criterion	1.000	3.494		OK
VISIBILI.		359.760	118.886		OK
PROPIMMER	Propeller should imm.	50.000	128.828	%	OK

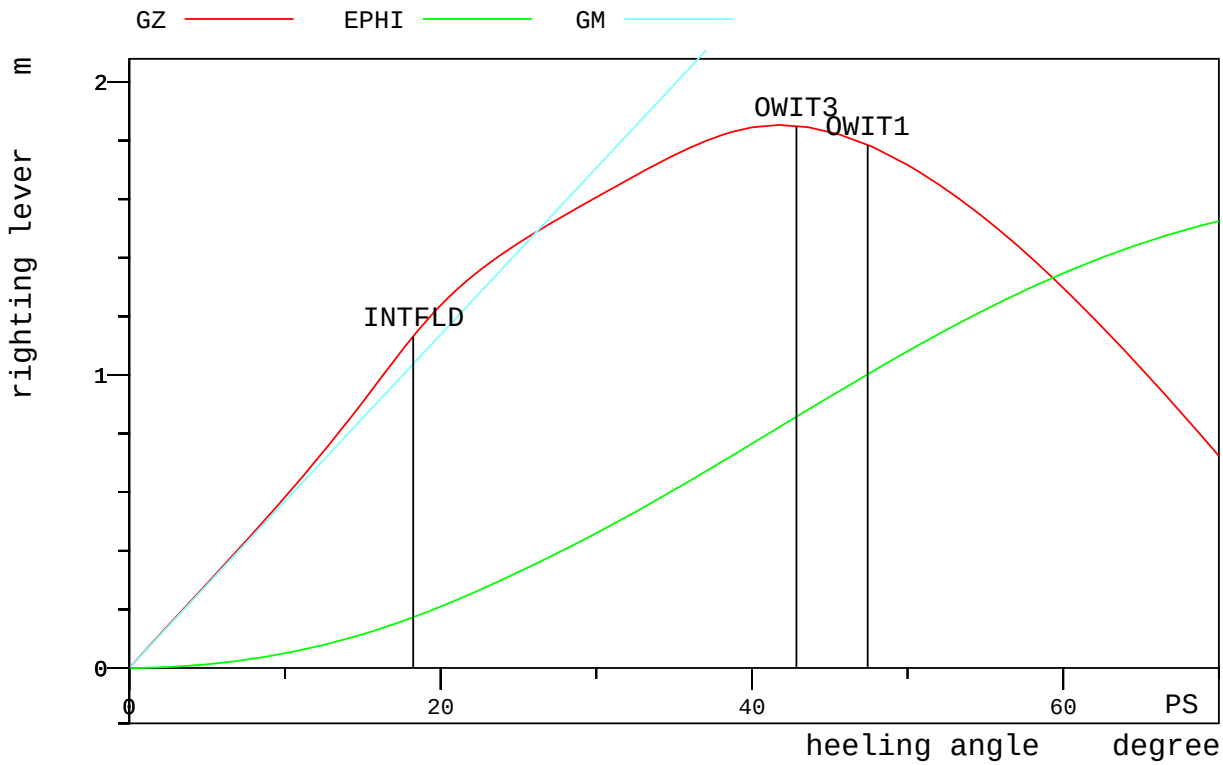
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-585.7 t	(9.8%)	POSITION:	25.8 m	34
SHEAR FORCE (MAX)	385.2 t	(6.4%)		141.0 m	178
SAGGING MOMENT	-17394.6 tm	(18.0%)		89.4 m	113
HOGGING MOMENT	1132.4 tm	(3.3%)		14.2 m	20

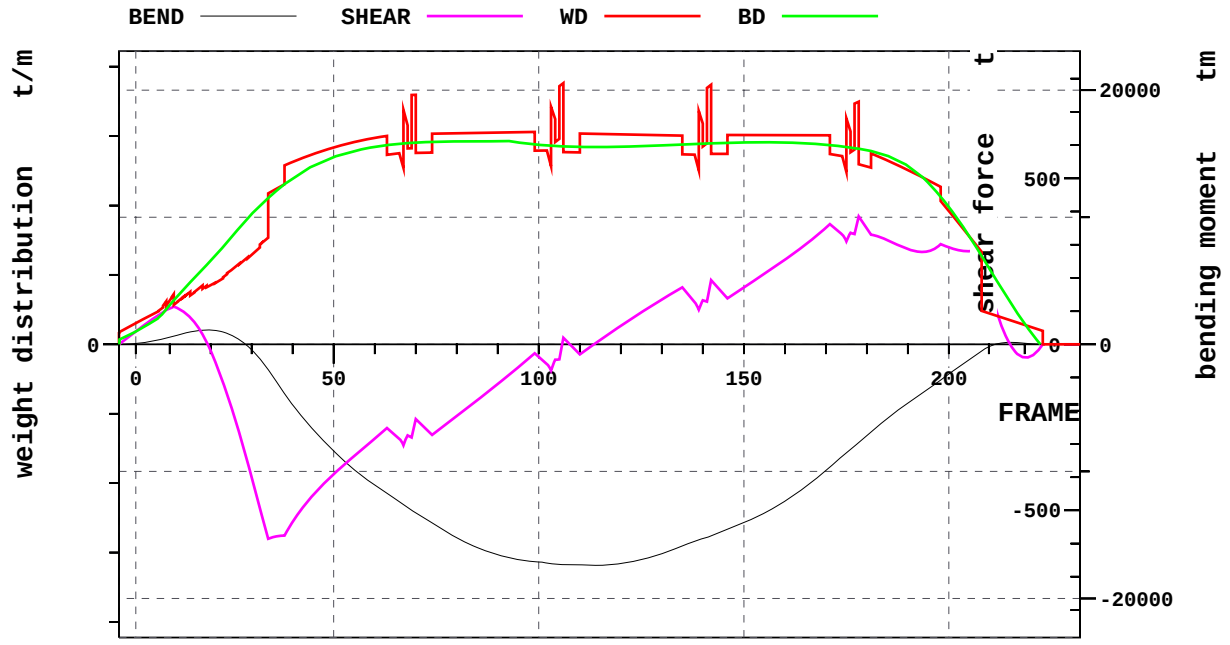
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.721)						
NO.1 CARGO HOLD	4797.4	100.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	7183.4	100.0	125.57	0.00	8.67	0.0
NO.3 CARGO HOLD	7192.3	100.0	96.78	0.00	8.67	0.0
NO.4 CARGO HOLD	7192.1	100.0	67.98	0.00	8.67	0.0
NO.5 CARGO HOLD	6486.4	100.0	40.05	0.00	9.09	0.0
SUBTOTAL	32851.6		93.51	0.00	8.76	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	2.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	27.6		75.29	0.00	20.42	0.0
TOTAL	32879.2		93.50	0.00	8.76	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	2.1	10.0	8.28	-5.84	7.59	22.22
M.D.O. SETTL. TK.	2.5	10.0	9.88	-5.95	7.33	29.35
M.D.O. STOR. TK. (P)	3.5	10.0	12.81	5.91	7.45	43.37
M.D.O. STOR. TK. (S)	4.4	10.0	12.58	-5.91	7.52	54.86
SUBTOTAL	12.4		11.39	-2.61	7.47	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	4.1	10.0	4.98	7.15	11.13	31.97
FRESH WATER TK. (P)	9.4	10.0	1.14	6.32	11.15	44.74
FRESH WATER TK. (S)	13.4	10.0	2.40	-6.59	11.14	82.65
SUBTOTAL	51.6		4.32	-0.00	8.25	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	26.4	10.0	68.20	12.64	11.22	68.67
NO.1 FUEL OIL TK. (S)	26.4	10.0	68.20	-12.64	11.22	68.67
NO.2 FUEL OIL TK. (P)	25.5	10.0	40.23	12.62	11.23	0.00
NO.2 FUEL OIL TK. (S)	25.5	10.0	40.23	-12.62	11.23	0.00
H.F.O. TK. (P)	25.3	10.0	23.08	10.63	7.06	44.35
H.F.O. TK. (S)	18.0	10.0	23.44	-10.57	6.80	33.83
NO.1 H.F.O. SERV. TK.	2.5	10.0	15.49	-10.82	11.07	7.35
NO.1 H.F.O. SETTL. T.	3.0	10.0	18.65	-11.14	11.06	11.05
NO.2 H.F.O. SERV. TK.	1.9	10.0	12.65	-10.87	11.07	0.00
NO.2 H.F.O. SETTL. T.	2.2	10.0	9.70	-10.42	11.09	0.00
SUBTOTAL	156.7		43.37	-0.16	10.03	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	1.4	20.0	10.03	5.53	7.22	0.36
G/E L.O. STOR. TK.	2.6	20.0	10.04	7.48	7.96	10.28
M/E L.O. SETTL. TK.	3.5	20.0	8.83	6.21	7.80	19.67
M/E L.O. STOR. TK.	5.4	20.0	7.19	6.05	8.05	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	50.8		16.95	2.45	5.88	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	4.5	80.0	18.81	-2.29	1.40	1.69
BILGE TK.	12.9	80.0	11.52	0.00	1.14	9.85
F.O. DRAIN TK.	5.2	80.0	20.70	2.51	1.34	3.33
F.O. OVERFLOW TK.	7.2	80.0	23.04	2.98	1.31	7.97
L.O. DRAIN TK.	3.1	80.0	18.32	2.24	1.42	0.91
SEWAGE STOR. TK.	10.9	80.0	22.37	-2.83	1.31	10.07
SLUDGE TK.	12.0	80.0	21.80	-8.40	7.72	9.03
SUBTOTAL	55.8		19.16	-1.81	2.67	42.86
TOTAL	327.4		27.78	-0.10	7.76	642.92

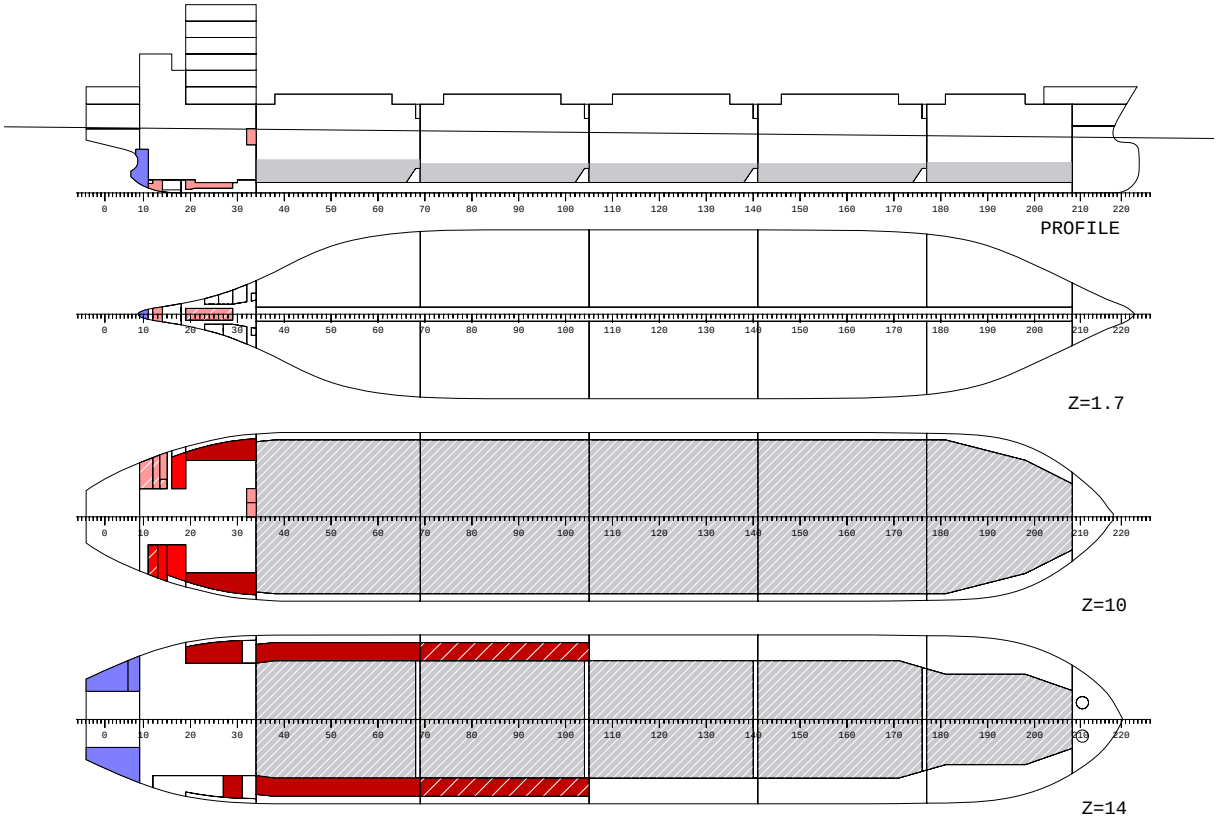


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	0.000	0.00	0.000	0.0	0.000
10.0	0.017	0.58	0.050	111.6	0.003
12.0	0.029	0.71	0.073	133.7	0.003
20.0	0.124	1.24	0.209	219.9	0.005
30.0	-0.023	1.61	0.461	321.5	0.008
40.0	-0.249	1.85	0.765	413.3	0.010
50.0	-0.779	1.72	1.081	492.5	0.012
60.0	-1.525	1.30	1.347	556.8	0.013
70.0	-2.338	0.72	1.525	604.2	0.014

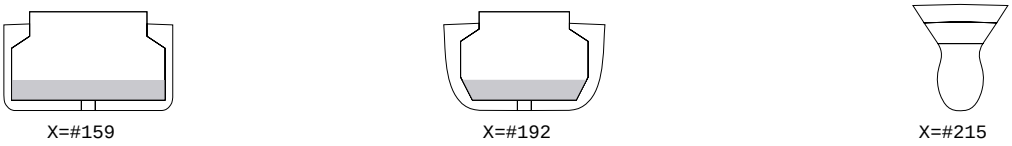


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	67	39	30.57	18.51
5.94	9.00	499	104	55.64	50.22
18.60	25.00	728	-189	106.87	154.05
25.80	34.00	-2012	-586	217.47	210.81
40.20	52.00	-9003	-369	287.02	273.38
53.80	69.00	-13055	-280	282.53	290.10
68.20	87.00	-16239	-149	304.81	292.72
82.60	105.00	-17330	-45	296.19	292.51
97.00	123.00	-17078	80	302.12	292.05
111.40	141.00	-15198	129	289.51	291.60
125.80	159.00	-12535	246	301.39	290.65
140.20	177.00	-8018	332	280.74	282.66
152.20	192.00	-4176	280	245.19	247.91
165.00	208.00	-512	289	133.32	126.98
169.20	214.00	125	44	38.08	76.62
173.40	220.00	87	-39	27.85	29.28

LOADING CASE 'LD05A -- HOMOGENEOUSLY HEAVY CARGO CONDITION, DEPARTURE







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	0.0	0.00	0.00	0.00	0.00
Diesel Oil	DO	121.6	11.23	-3.17	9.29	149.80
Fresh Water	FW	293.6	2.53	0.00	12.34	159.36
Grain Load	GRAIN	32856.1	93.42	0.00	3.55	0.00
Heavy Fuel Oil	HFO	1535.7	42.68	-0.15	12.40	233.92
Lubricating Oil	LO	101.1	12.73	4.87	7.77	56.99
	MASS	43.6	53.17	0.00	20.19	0.00
Miscellaneous	MIS	7.0	19.24	-1.81	1.99	42.86
Deadweight		34958.8	89.85	-0.00	4.06	642.92
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		34958.8	89.85	-0.00	4.06	
Total weight		43608.8	87.75	-0.00	5.28	

F L O A T I N G P O S I T I O N

Draught moulded	10.268	m	KM	12.31 m
Trim	-1.661	m	KG	5.28 m
Heel, PS=+	-0.0	deg		
TA	11.098	m	GM0	7.03 m
TF	9.437	m	GMCORR	-0.01 m
Trimming moment	-100884	tonm	GM	7.02 m
Heel	-0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.951	mrad	OK
AREA40	Area under GZ curve .	0.090	1.508	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.557	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.536	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	54.230	deg	OK
GM0.15	GM > 0.15 m	0.150	7.018	m	OK
IMOWEATH.	IMO weather criterion	1.000	3.239		OK
VISIBILI.		359.760	139.712		OK
PROPIMMER	Propeller should imm.	50.000	149.400	%	OK

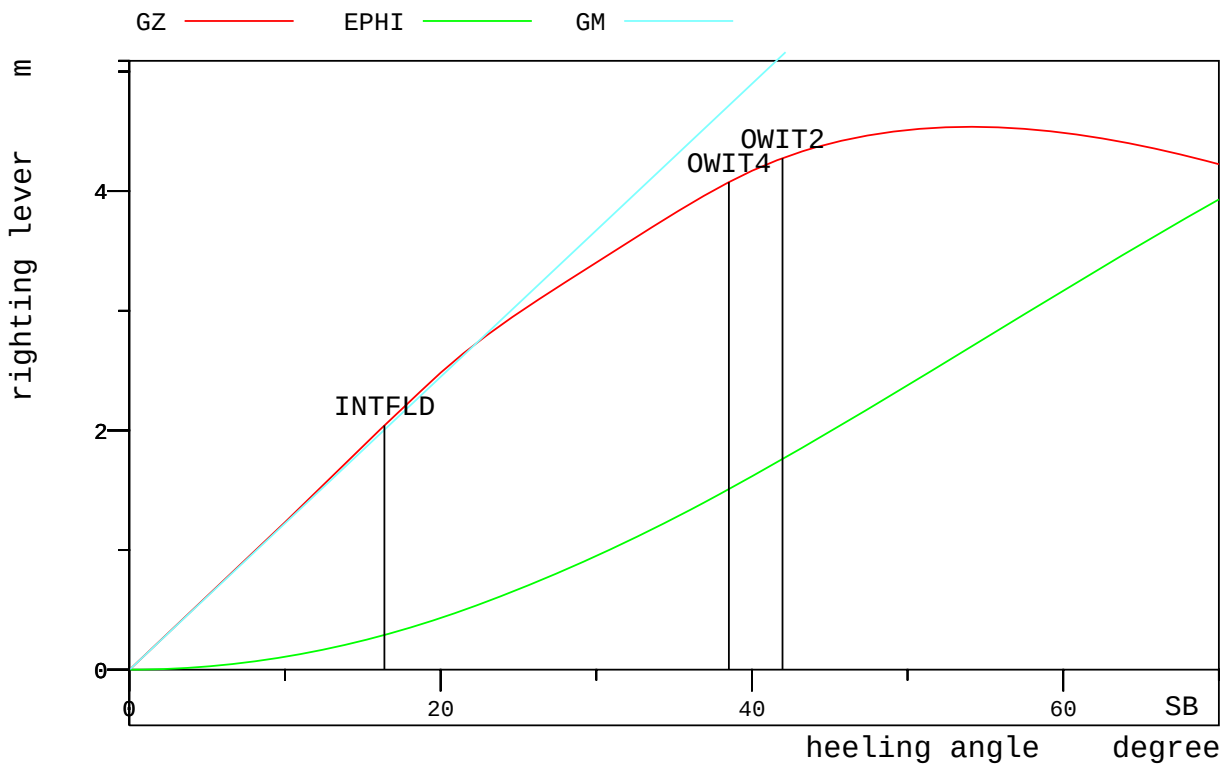
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-661.1 t	(11.0%)	POSITION:	37.6 m	49
SHEAR FORCE (MAX)	301.7 t	(5.0%)		138.0 m	174
SAGGING MOMENT	-15364.9 tm	(15.9%)		70.9 m	90
HOGGING MOMENT	2232.8 tm	(5.6%)		16.8 m	23

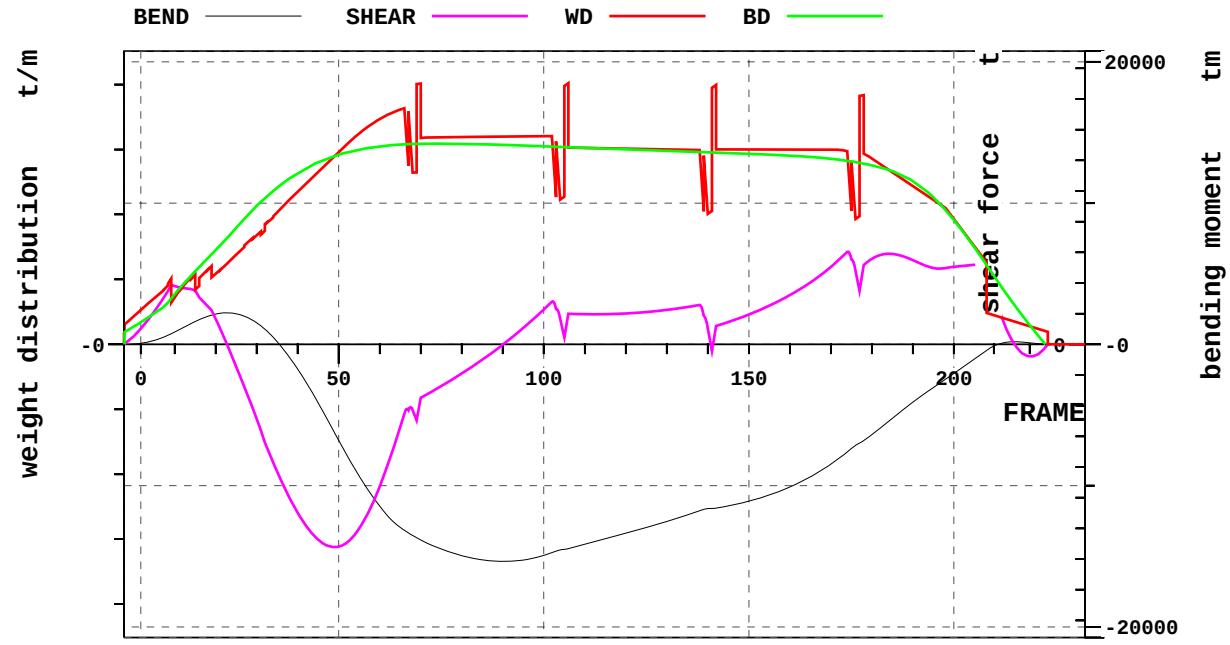
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 3.000)						
NO.1 CARGO HOLD	4798.0	24.0	150.87	0.00	3.54	0.0
NO.2 CARGO HOLD	7184.4	24.0	125.23	0.00	3.44	0.0
NO.3 CARGO HOLD	7193.3	24.0	96.43	0.00	3.44	0.0
NO.4 CARGO HOLD	7193.1	24.0	67.64	0.00	3.44	0.0
NO.5 CARGO HOLD	6487.3	24.0	40.97	0.00	3.91	0.0
SUBTOTAL	32856.1		93.42	0.00	3.55	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	18.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	43.6		53.17	0.00	20.19	0.0
TOTAL	32899.8		93.37	0.00	3.57	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	0.00	0.00	0.00	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	20.2	98.0	8.23	-7.08	9.34	22.22
M.D.O. SETTL. TK.	24.2	98.0	9.82	-7.32	9.21	29.35
M.D.O. STOR. TK. (P)	33.9	98.0	12.66	7.26	9.29	43.37
M.D.O. STOR. TK. (S)	43.2	98.0	12.30	-7.22	9.32	54.86
SUBTOTAL	121.6		11.23	-3.17	9.29	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	40.7	100.0	4.97	7.43	12.96	31.97
FRESH WATER TK. (P)	93.7	100.0	0.94	6.60	13.03	44.74
FRESH WATER TK. (S)	134.4	100.0	2.17	-6.85	13.01	82.65
SUBTOTAL	293.6		2.53	0.00	12.34	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	258.4	98.0	68.20	11.73	13.08	68.67
NO.1 FUEL OIL TK. (S)	258.4	98.0	68.20	-11.73	13.08	68.67
NO.2 FUEL OIL TK. (P)	250.2	98.0	39.86	11.71	13.09	0.00
NO.2 FUEL OIL TK. (S)	250.2	98.0	39.86	-11.71	13.09	0.00
H.F.O. TK. (P)	248.1	98.0	20.60	11.15	11.04	44.35
H.F.O. TK. (S)	176.3	98.0	21.94	-11.14	10.32	33.83
NO.1 H.F.O. SERV. TK.	24.3	98.0	15.46	-11.00	12.38	7.35
NO.1 H.F.O. SETTL. T.	29.0	98.0	18.63	-11.27	12.36	11.05
NO.2 H.F.O. SERV. TK.	19.1	98.0	12.64	-11.07	12.39	0.00
NO.2 H.F.O. SETTL. T.	21.5	98.0	9.62	-10.64	12.44	0.00
SUBTOTAL	1535.7		42.68	-0.15	12.40	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	6.8	98.0	10.01	5.59	8.82	0.36
G/E L.O. STOR. TK.	12.9	98.0	10.02	8.28	9.39	10.28
M/E L.O. SETTL. TK.	17.2	98.0	8.81	7.17	9.29	19.67
M/E L.O. STOR. TK.	26.4	98.0	7.12	6.90	9.42	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	101.1		12.73	4.87	7.77	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	0.6	10.0	18.98	-1.96	0.68	1.69
BILGE TK.	1.6	10.0	11.66	0.00	0.32	9.85
F.O. DRAIN TK.	0.6	10.0	20.77	2.04	0.56	3.33
F.O. OVERFLOW TK.	0.9	10.0	23.05	2.35	0.50	7.97
L.O. DRAIN TK.	0.4	10.0	18.43	1.94	0.72	0.91
SEWAGE STOR. TK.	1.4	10.0	22.44	-2.26	0.52	10.07
SLUDGE TK.	1.5	10.0	21.80	-8.40	7.44	9.03
SUBTOTAL	7.0		19.24	-1.81	1.99	42.86
TOTAL	2059.0		33.55	-0.07	11.94	642.92

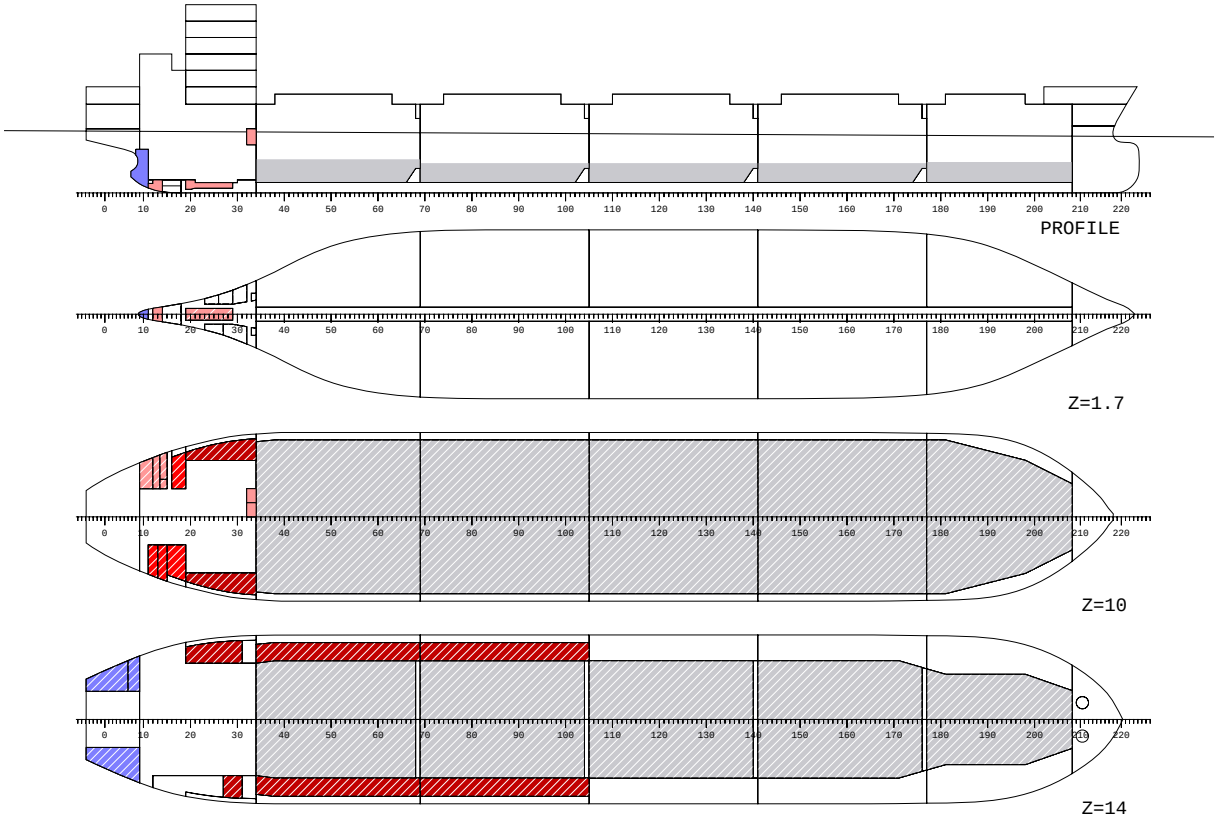


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.003	-0.00	0.000	0.0	0.000
10.0	0.013	1.23	0.107	111.6	0.003
12.0	0.025	1.48	0.155	133.7	0.003
20.0	0.081	2.48	0.432	219.9	0.005
30.0	-0.106	3.40	0.951	321.5	0.007
40.0	-0.341	4.17	1.614	413.3	0.009
50.0	-0.866	4.51	2.379	492.5	0.011
60.0	-1.589	4.49	3.168	556.8	0.013
70.0	-2.370	4.22	3.931	604.2	0.014

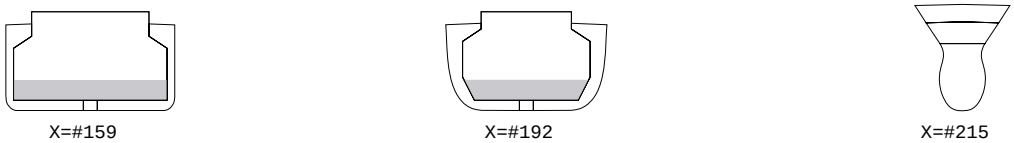


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	76	51	53.14	34.02
5.94	9.00	767	194	64.03	70.64
18.60	25.00	2165	-73	138.27	179.39
25.80	34.00	513	-385	196.03	236.07
40.20	52.00	-7824	-646	308.17	295.41
53.80	69.00	-13693	-248	332.66	308.67
68.20	87.00	-15323	-31	319.83	307.63
82.60	105.00	-14509	22	227.54	303.77
97.00	123.00	-13129	103	300.89	299.66
111.40	141.00	-11609	-21	205.39	295.55
125.80	159.00	-10222	153	299.94	290.94
140.20	177.00	-6991	175	197.25	279.33
152.20	192.00	-3662	263	235.18	241.94
165.00	208.00	-416	266	125.98	121.99
169.20	214.00	158	38	38.08	73.70
173.40	220.00	105	-39	27.85	29.28

LOADING CASE 'LD05B -- HOMOGENEOUSLY HEAVY CARGO CONDITION, HALFWAY







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	0.0	0.00	0.00	0.00	0.00
Diesel Oil	DO	62.0	11.26	-2.98	8.46	149.80
Fresh Water	FW	159.2	2.90	-0.00	10.95	159.36
Grain Load	GRAIN	32856.1	93.42	0.00	3.55	0.00
Heavy Fuel Oil	HFO	783.5	42.77	-0.15	11.35	233.92
Lubricating Oil	LO	76.6	14.10	4.02	6.96	56.99
	MASS	34.6	63.09	0.00	20.29	0.00
Miscellaneous	MIS	27.9	19.18	-1.81	2.34	42.86
Deadweight		34000.0	91.41	-0.00	3.79	642.92
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		34000.0	91.41	-0.00	3.79	
Total weight		42650.0	88.96	-0.00	5.09	

F L O A T I N G P O S I T I O N

Draught moulded	10.076	m	KM	12.32 m
Trim	-0.878	m	KG	5.09 m
Heel, PS=+	0.0	deg		
TA	10.515	m	GM0	7.23 m
TF	9.637	m	GMCORR	-0.02 m
Trimming moment	-52930	tonm	GM	7.21 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.986	mrad	OK
AREA40	Area under GZ curve .	0.090	1.678	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.692	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.739	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	54.418	deg	OK
GM0.15	GM > 0.15 m	0.150	7.212	m	OK
IMOWEATH.	IMO weather criterion	1.000	3.738		OK
VISIBILI.		359.760	128.143		OK
PROPIMMER	Propeller should imm.	50.000	138.390	%	OK

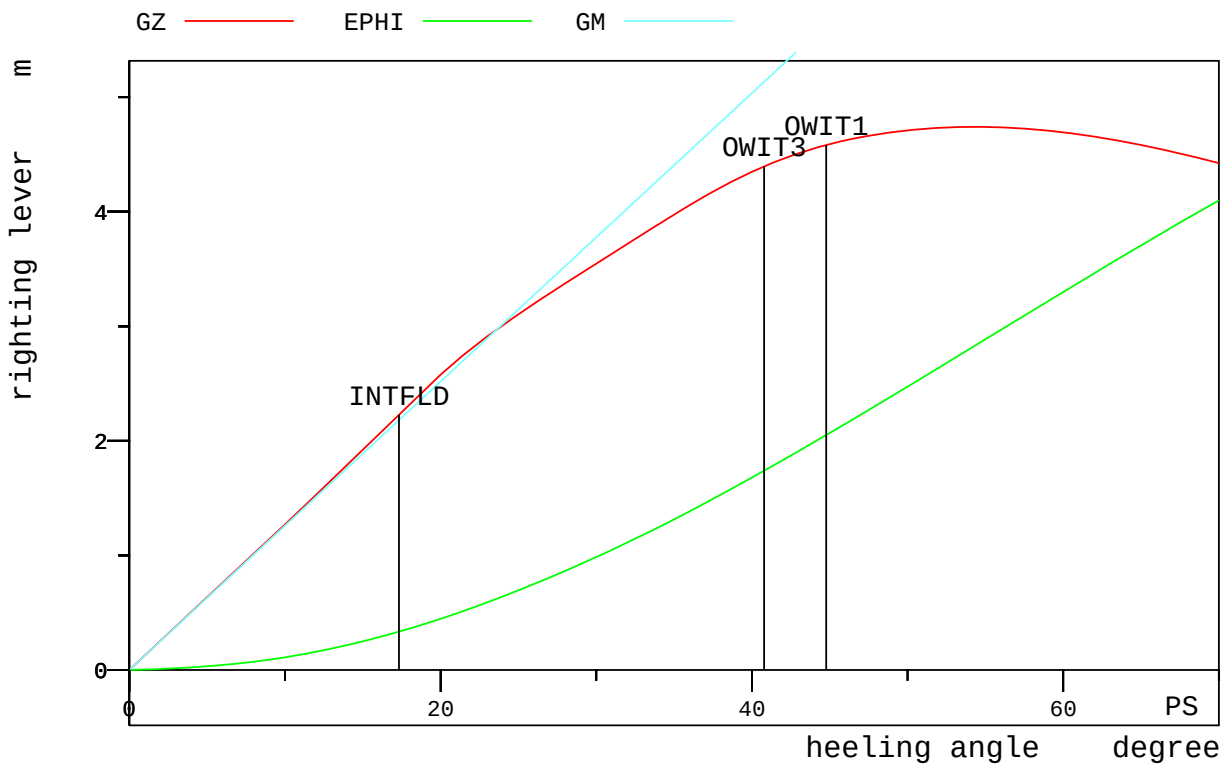
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-730.8 t	(12.2%)	POSITION:	37.1 m	48
SHEAR FORCE (MAX)	384.4 t	(6.4%)		138.0 m	174
SAGGING MOMENT	-19155.6 tm	(19.8%)		73.1 m	93
HOGGING MOMENT	1562.7 tm	(4.2%)		15.4 m	21

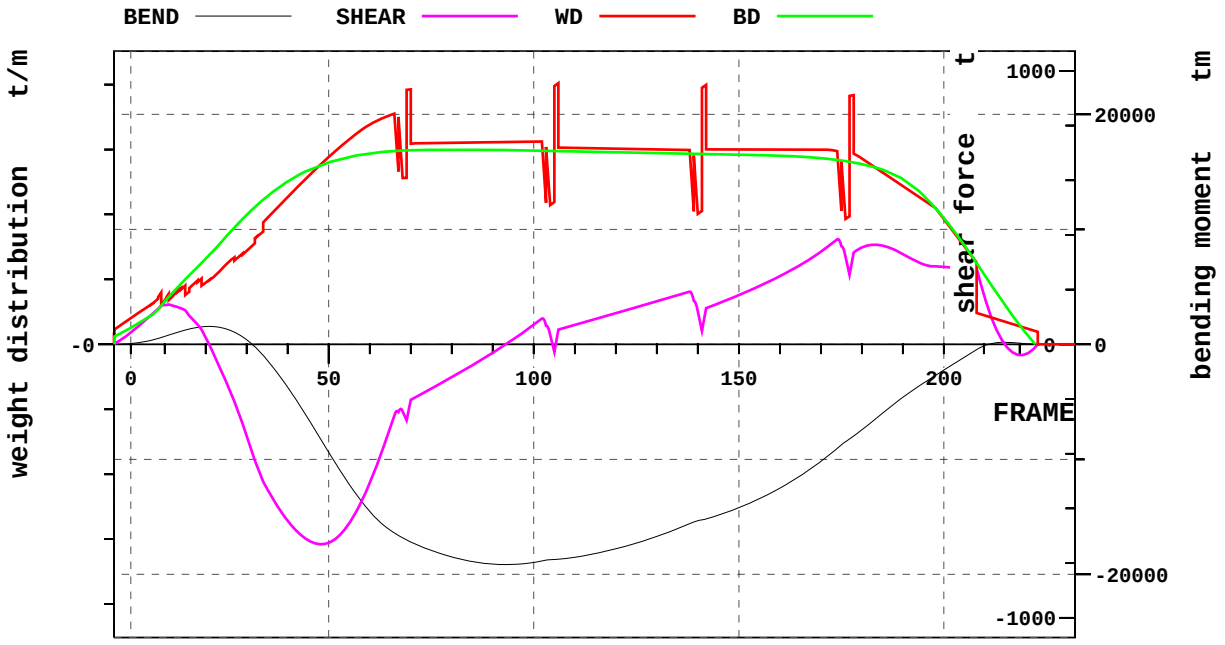
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 3.000)						
NO.1 CARGO HOLD	4798.0	24.0	150.87	0.00	3.54	0.0
NO.2 CARGO HOLD	7184.4	24.0	125.23	0.00	3.44	0.0
NO.3 CARGO HOLD	7193.3	24.0	96.43	0.00	3.44	0.0
NO.4 CARGO HOLD	7193.1	24.0	67.64	0.00	3.44	0.0
NO.5 CARGO HOLD	6487.3	24.0	40.97	0.00	3.91	0.0
SUBTOTAL	32856.1		93.42	0.00	3.55	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	9.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	34.6		63.09	0.00	20.29	0.0
TOTAL	32890.8		93.39	0.00	3.56	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	0.00	0.00	0.00	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	10.3	50.0	8.24	-6.66	8.53	22.22
M.D.O. SETTL. TK.	12.4	50.0	9.83	-6.86	8.34	29.35
M.D.O. STOR. TK. (P)	17.3	50.0	12.68	6.80	8.45	43.37
M.D.O. STOR. TK. (S)	22.0	50.0	12.35	-6.76	8.50	54.86
SUBTOTAL	62.0		11.26	-2.98	8.46	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	20.4	50.0	4.98	7.32	11.97	31.97
FRESH WATER TK. (P)	46.9	50.0	1.02	6.48	12.05	44.74
FRESH WATER TK. (S)	67.2	50.0	2.26	-6.75	12.02	82.65
SUBTOTAL	159.2		2.90	-0.00	10.95	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	131.9	50.0	68.20	11.90	12.29	68.67
NO.1 FUEL OIL TK. (S)	131.9	50.0	68.20	-11.90	12.29	68.67
NO.2 FUEL OIL TK. (P)	127.7	50.0	39.94	11.88	12.30	0.00
NO.2 FUEL OIL TK. (S)	127.7	50.0	39.94	-11.88	12.30	0.00
H.F.O. TK. (P)	126.6	50.0	21.26	10.98	9.26	44.35
H.F.O. TK. (S)	90.0	50.0	21.60	-10.89	8.63	33.83
NO.1 H.F.O. SERV. TK.	12.4	50.0	15.47	-10.92	11.69	7.35
NO.1 H.F.O. SETTL. T.	14.8	50.0	18.64	-11.21	11.66	11.05
NO.2 H.F.O. SERV. TK.	9.7	50.0	12.65	-10.98	11.70	0.00
NO.2 H.F.O. SETTL. T.	11.0	50.0	9.65	-10.54	11.75	0.00
SUBTOTAL	783.5		42.77	-0.15	11.35	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	4.1	60.0	10.01	5.58	8.05	0.36
G/E L.O. STOR. TK.	7.9	60.0	10.02	8.01	8.78	10.28
M/E L.O. SETTL. TK.	10.5	60.0	8.82	6.85	8.65	19.67
M/E L.O. STOR. TK.	16.1	60.0	7.14	6.62	8.83	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	76.6		14.10	4.02	6.96	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	2.3	40.0	18.84	-2.15	1.07	1.69
BILGE TK.	6.4	40.0	11.55	0.00	0.73	9.85
F.O. DRAIN TK.	2.6	40.0	20.71	2.30	0.98	3.33
F.O. OVERFLOW TK.	3.6	40.0	23.05	2.70	0.93	7.97
L.O. DRAIN TK.	1.5	40.0	18.35	2.11	1.09	0.91
SEWAGE STOR. TK.	5.4	40.0	22.38	-2.58	0.94	10.07
SLUDGE TK.	6.0	40.0	21.80	-8.40	7.56	9.03
SUBTOTAL	27.9		19.18	-1.81	2.34	42.86
TOTAL	1109.3		32.71	-0.04	10.60	642.92

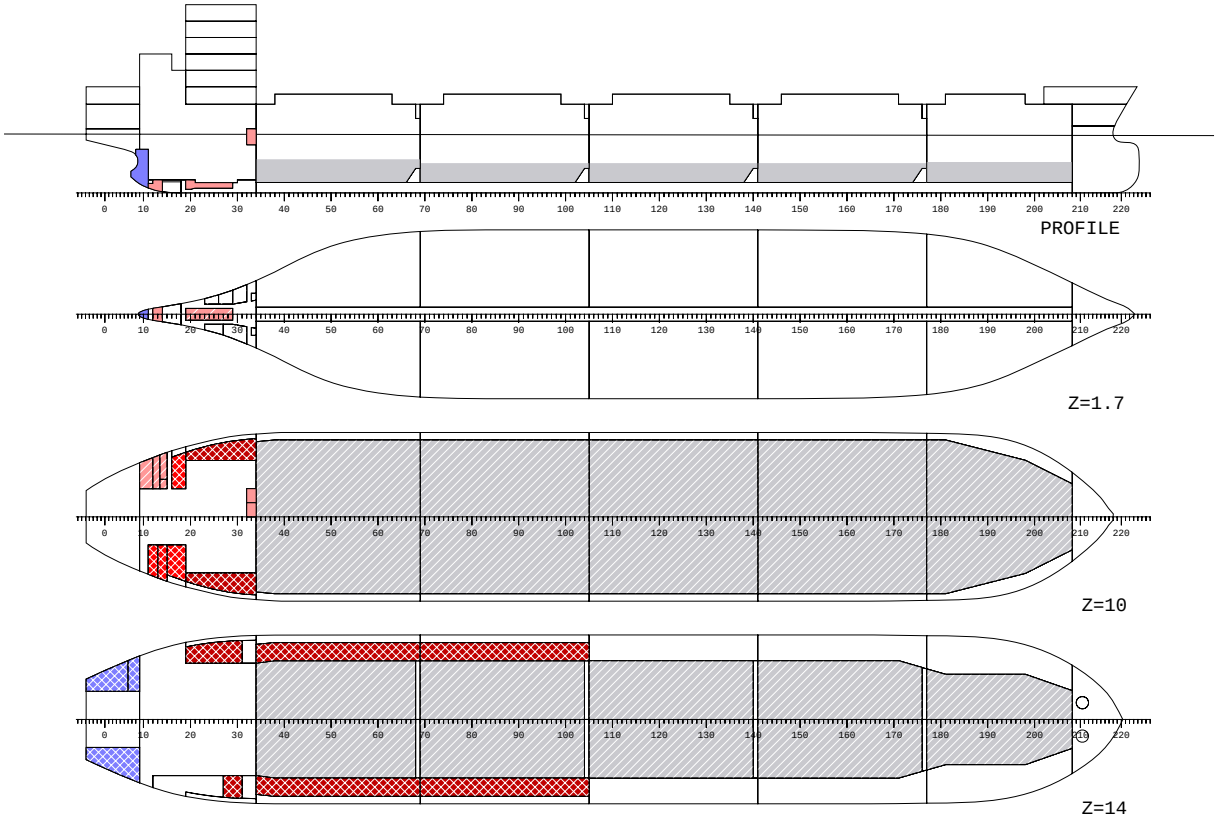


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	0.000	0.00	0.000	0.0	0.000
10.0	0.017	1.27	0.111	111.6	0.003
12.0	0.029	1.53	0.160	133.7	0.003
20.0	0.111	2.58	0.446	219.9	0.005
30.0	-0.060	3.55	0.986	321.5	0.008
40.0	-0.287	4.35	1.678	413.3	0.010
50.0	-0.816	4.71	2.475	492.5	0.012
60.0	-1.553	4.69	3.300	556.8	0.013
70.0	-2.353	4.42	4.098	604.2	0.014

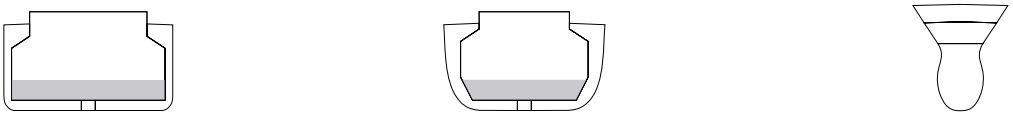


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	67	43	40.51	25.48
5.94	9.00	602	143	59.87	59.54
18.60	25.00	1333	-138	122.65	165.74
25.80	34.00	-951	-502	180.68	222.50
40.20	52.00	-10539	-707	299.35	283.57
53.80	69.00	-17001	-278	324.07	298.66
68.20	87.00	-19004	-59	310.94	299.56
82.60	105.00	-18722	-28	219.02	297.64
97.00	123.00	-17499	127	300.89	295.46
111.40	141.00	-15268	49	205.39	293.29
125.80	159.00	-12701	242	299.94	290.62
140.20	177.00	-8224	254	197.25	280.94
152.20	192.00	-4094	314	235.18	244.95
165.00	208.00	-451	278	125.98	124.54
169.20	214.00	153	41	38.08	75.20
173.40	220.00	104	-39	27.85	29.28

LOADING CASE 'LD05C -- HOMOGENEOUSLY HEAVY CARGO CONDITION, ARRIVAL







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	0.0	0.00	0.00	0.00	0.00
Diesel Oil	DO	12.4	11.39	-2.61	7.47	149.80
Fresh Water	FW	51.6	4.32	-0.00	8.25	159.36
Grain Load	GRAIN	32856.1	93.42	0.00	3.55	0.00
Heavy Fuel Oil	HFO	156.7	43.37	-0.16	10.03	233.92
Lubricating Oil	LO	50.8	16.95	2.45	5.88	56.99
	MASS	27.6	75.29	0.00	20.42	0.00
Miscellaneous	MIS	55.8	19.16	-1.81	2.67	42.86
Deadweight		33211.1	92.76	-0.00	3.60	642.92
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		33211.1	92.76	-0.00	3.60	
Total weight		41861.1	89.98	-0.00	4.96	

F L O A T I N G P O S I T I O N

Draught moulded	9.916	m	KM	12.33 m
Trim	-0.229	m	KG	4.96 m
Heel, PS=+	0.0	deg		
TA	10.031	m	GM0	7.36 m
TF	9.802	m	GMCORR	-0.02 m
Trimming moment	-13711	tonm	GM	7.35 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	1.010	mrad	OK
AREA40	Area under GZ curve .	0.090	1.723	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.713	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.884	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	54.587	deg	OK
GM0.15	GM > 0.15 m	0.150	7.349	m	OK
IMOWEATH.	IMO weather criterion	1.000	4.188		OK
VISIBILI.		359.760	119.586		OK
PROPIMMER	Propeller should imm.	50.000	129.262	%	OK

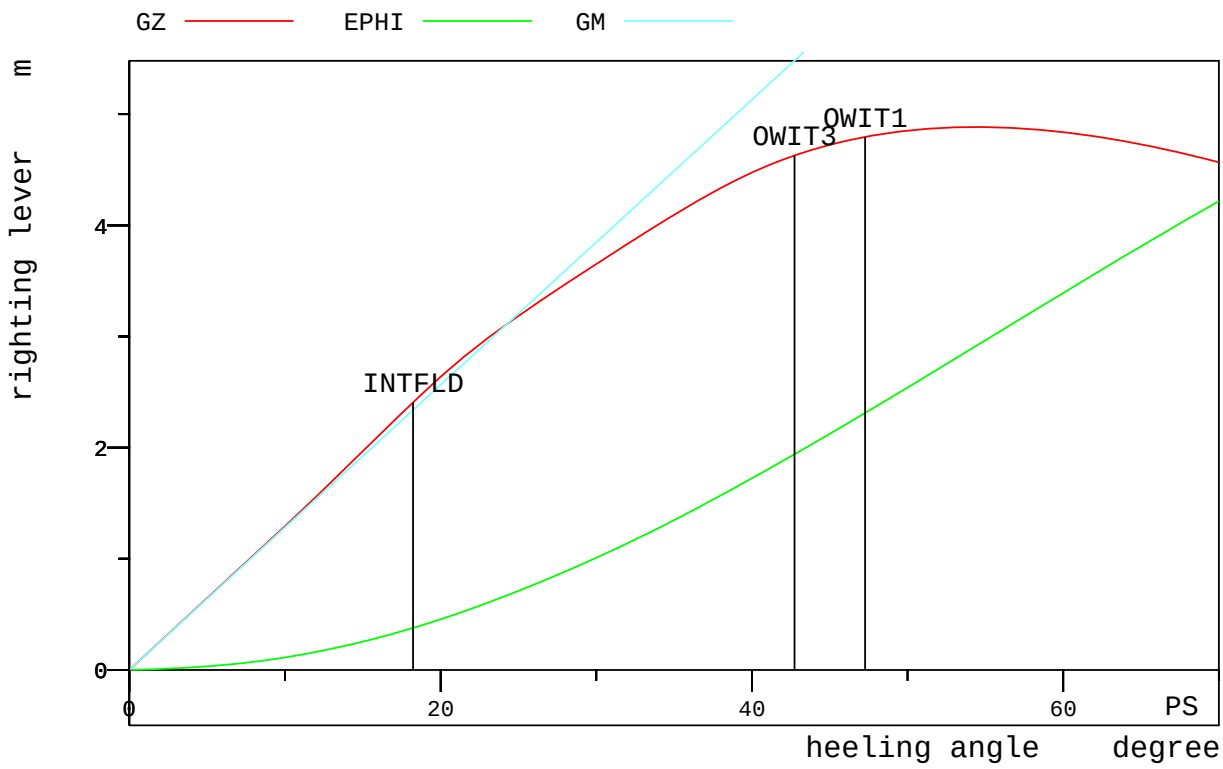
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-790.1 t	(13.2%)	POSITION:	36.6 m	47
SHEAR FORCE (MAX)	452.7 t	(7.5%)		138.0 m	174
SAGGING MOMENT	-22391.5 tm	(23.1%)		75.6 m	96
HOGGING MOMENT	1065.2 tm	(3.1%)		13.8 m	19

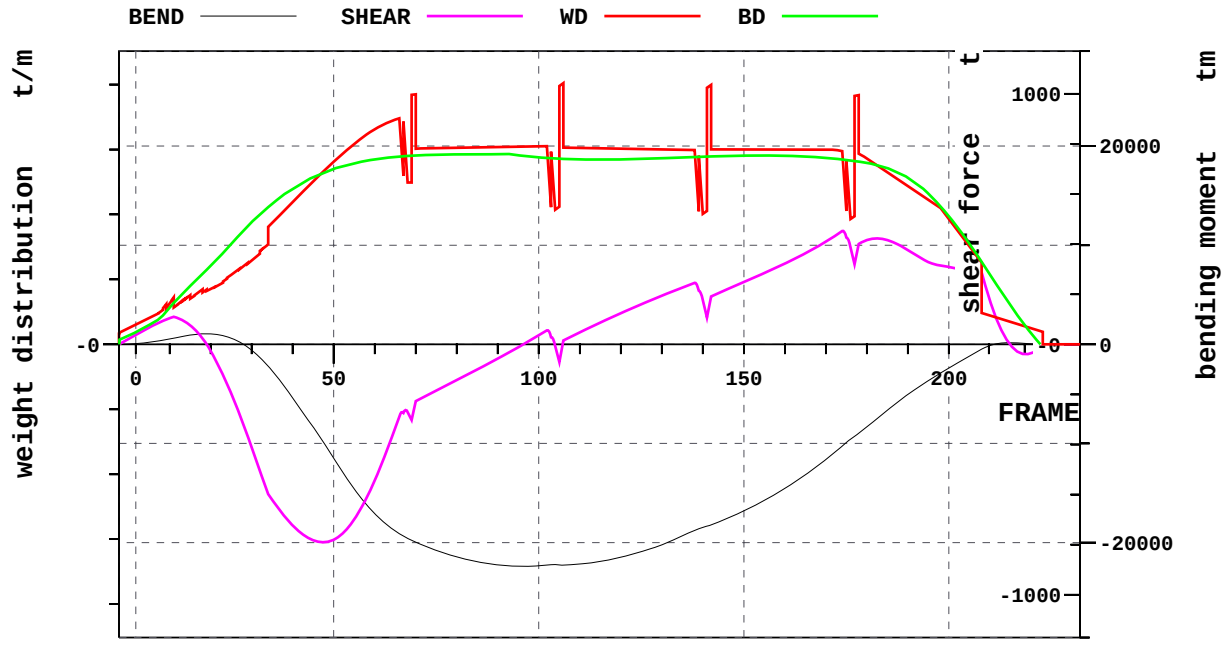
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 3.000)						
NO.1 CARGO HOLD	4798.0	24.0	150.87	0.00	3.54	0.0
NO.2 CARGO HOLD	7184.4	24.0	125.23	0.00	3.44	0.0
NO.3 CARGO HOLD	7193.3	24.0	96.43	0.00	3.44	0.0
NO.4 CARGO HOLD	7193.1	24.0	67.64	0.00	3.44	0.0
NO.5 CARGO HOLD	6487.3	24.0	40.97	0.00	3.91	0.0
SUBTOTAL	32856.1		93.42	0.00	3.55	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	2.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	27.6		75.29	0.00	20.42	0.0
TOTAL	32883.8		93.41	0.00	3.56	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	0.00	0.00	0.00	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	2.1	10.0	8.28	-5.84	7.59	22.22
M.D.O. SETTL. TK.	2.5	10.0	9.88	-5.95	7.33	29.35
M.D.O. STOR. TK. (P)	3.5	10.0	12.81	5.91	7.45	43.37
M.D.O. STOR. TK. (S)	4.4	10.0	12.58	-5.91	7.52	54.86
SUBTOTAL	12.4		11.39	-2.61	7.47	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	4.1	10.0	4.98	7.15	11.13	31.97
FRESH WATER TK. (P)	9.4	10.0	1.14	6.32	11.15	44.74
FRESH WATER TK. (S)	13.4	10.0	2.40	-6.59	11.14	82.65
SUBTOTAL	51.6		4.32	-0.00	8.25	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	26.4	10.0	68.20	12.64	11.22	68.67
NO.1 FUEL OIL TK. (S)	26.4	10.0	68.20	-12.64	11.22	68.67
NO.2 FUEL OIL TK. (P)	25.5	10.0	40.23	12.62	11.23	0.00
NO.2 FUEL OIL TK. (S)	25.5	10.0	40.23	-12.62	11.23	0.00
H.F.O. TK. (P)	25.3	10.0	23.08	10.63	7.06	44.35
H.F.O. TK. (S)	18.0	10.0	23.44	-10.57	6.80	33.83
NO.1 H.F.O. SERV. TK.	2.5	10.0	15.49	-10.82	11.07	7.35
NO.1 H.F.O. SETTL. T.	3.0	10.0	18.65	-11.14	11.06	11.05
NO.2 H.F.O. SERV. TK.	1.9	10.0	12.65	-10.87	11.07	0.00
NO.2 H.F.O. SETTL. T.	2.2	10.0	9.70	-10.42	11.09	0.00
SUBTOTAL	156.7		43.37	-0.16	10.03	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	1.4	20.0	10.03	5.53	7.22	0.36
G/E L.O. STOR. TK.	2.6	20.0	10.04	7.48	7.96	10.28
M/E L.O. SETTL. TK.	3.5	20.0	8.83	6.21	7.80	19.67
M/E L.O. STOR. TK.	5.4	20.0	7.19	6.05	8.05	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	50.8		16.95	2.45	5.88	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	4.5	80.0	18.81	-2.29	1.40	1.69
BILGE TK.	12.9	80.0	11.52	0.00	1.14	9.85
F.O. DRAIN TK.	5.2	80.0	20.70	2.51	1.34	3.33
F.O. OVERFLOW TK.	7.2	80.0	23.04	2.98	1.31	7.97
L.O. DRAIN TK.	3.1	80.0	18.32	2.24	1.42	0.91
SEWAGE STOR. TK.	10.9	80.0	22.37	-2.83	1.31	10.07
SLUDGE TK.	12.0	80.0	21.80	-8.40	7.72	9.03
SUBTOTAL	55.8		19.16	-1.81	2.67	42.86
TOTAL	327.4		27.78	-0.10	7.76	642.92

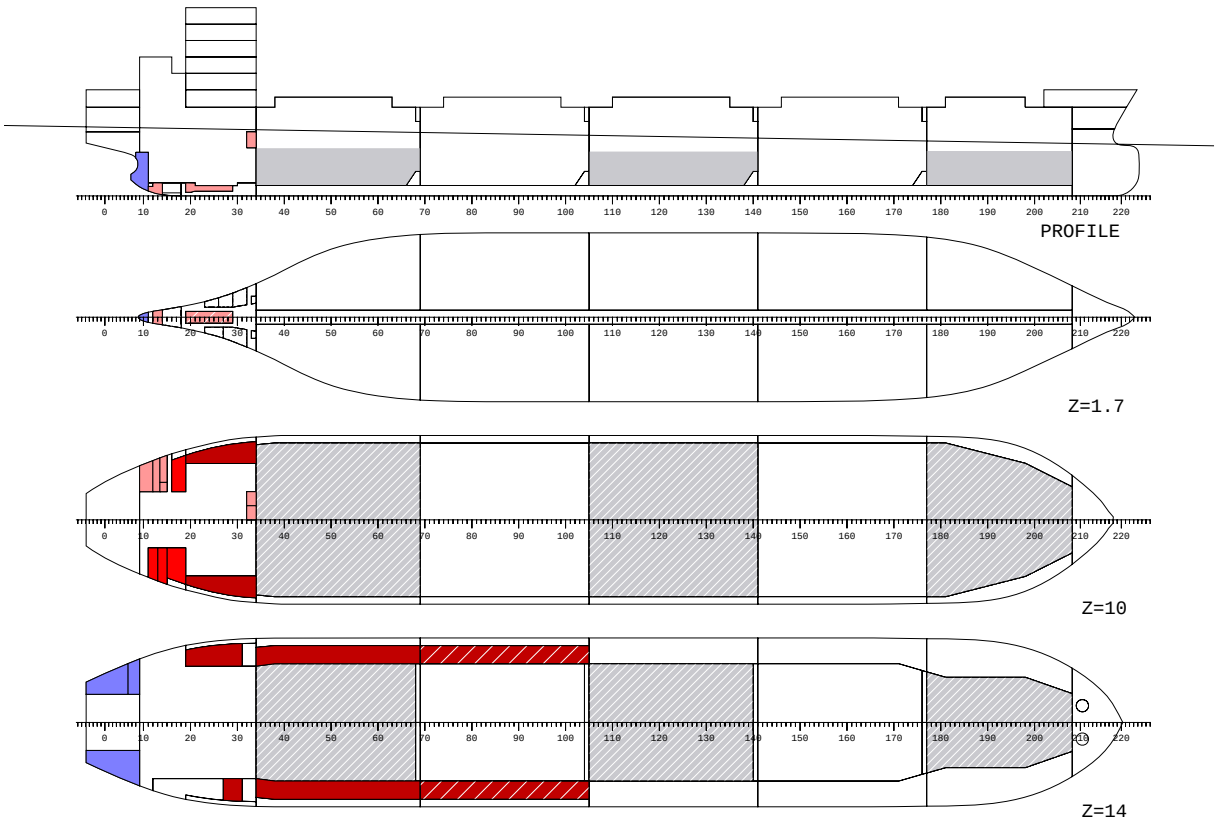


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	0.000	0.00	0.000	0.0	0.000
10.0	0.017	1.29	0.113	111.6	0.003
12.0	0.029	1.56	0.163	133.7	0.003
20.0	0.124	2.64	0.456	219.9	0.005
30.0	-0.022	3.65	1.010	321.5	0.008
40.0	-0.248	4.48	1.723	413.3	0.010
50.0	-0.779	4.85	2.543	492.5	0.012
60.0	-1.525	4.84	3.393	556.8	0.013
70.0	-2.338	4.57	4.217	604.2	0.014

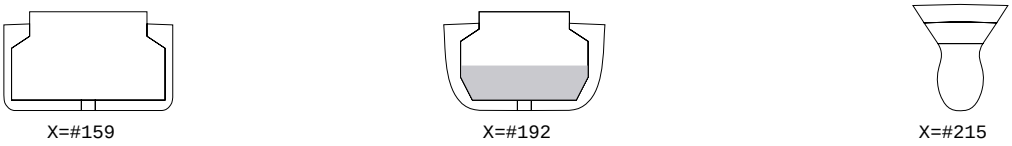


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	62	38	30.58	18.82
5.94	9.00	477	101	55.64	50.63
18.60	25.00	620	-198	106.88	154.54
25.80	34.00	-2203	-598	167.31	211.28
40.20	52.00	-12807	-758	292.02	273.75
53.80	69.00	-19750	-302	316.73	290.36
68.20	87.00	-22075	-83	303.61	292.87
82.60	105.00	-22246	-71	212.07	292.55
97.00	123.00	-21154	146	300.89	291.98
111.40	141.00	-18331	107	205.39	291.41
125.80	159.00	-14783	315	299.94	290.35
140.20	177.00	-9269	320	197.25	282.26
152.20	192.00	-4469	356	235.18	247.44
165.00	208.00	-489	287	125.98	126.66
169.20	214.00	141	44	38.08	76.44
173.40	220.00	98	-39	27.85	29.28

LOADING CASE 'LD06A -- ALTERNATIVE HEAVY CARGO CONDITION, DEPARTURE







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	0.0	0.00	0.00	0.00	0.00
Diesel Oil	DO	121.6	11.23	-3.17	9.29	149.80
Fresh Water	FW	293.6	2.53	0.00	12.34	159.36
Grain Load	GRAIN	32944.2	91.11	0.00	4.92	0.00
Heavy Fuel Oil	HFO	1535.7	42.68	-0.15	12.40	233.92
Lubricating Oil	LO	101.1	12.73	4.87	7.77	56.99
	MASS	43.6	53.17	0.00	20.19	0.00
Miscellaneous	MIS	7.0	19.24	-1.81	1.99	42.86
Deadweight		35046.8	87.68	-0.00	5.35	642.92
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		35046.8	87.68	-0.00	5.35	
Total weight		43696.8	86.02	-0.00	6.31	

F L O A T I N G P O S I T I O N

Draught moulded	10.268	m	KM	12.35 m
Trim	-2.911	m	KG	6.31 m
Heel, PS=+	-0.0	deg		
TA	11.724	m	GM0	6.04 m
TF	8.813	m	GMCORR	-0.01 m
Trimming moment	-177120	tonm	GM	6.02 m
Heel	-0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.813	mrad	OK
AREA40	Area under GZ curve .	0.090	1.162	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.349	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	3.723	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	50.541	deg	OK
GM0.15	GM > 0.15 m	0.150	6.025	m	OK
IMOWEATH.	IMO weather criterion	1.000	2.671		OK
VISIBILI.		359.760	166.620		OK
PROPIMMER	Propeller should imm.	50.000	161.203	%	OK

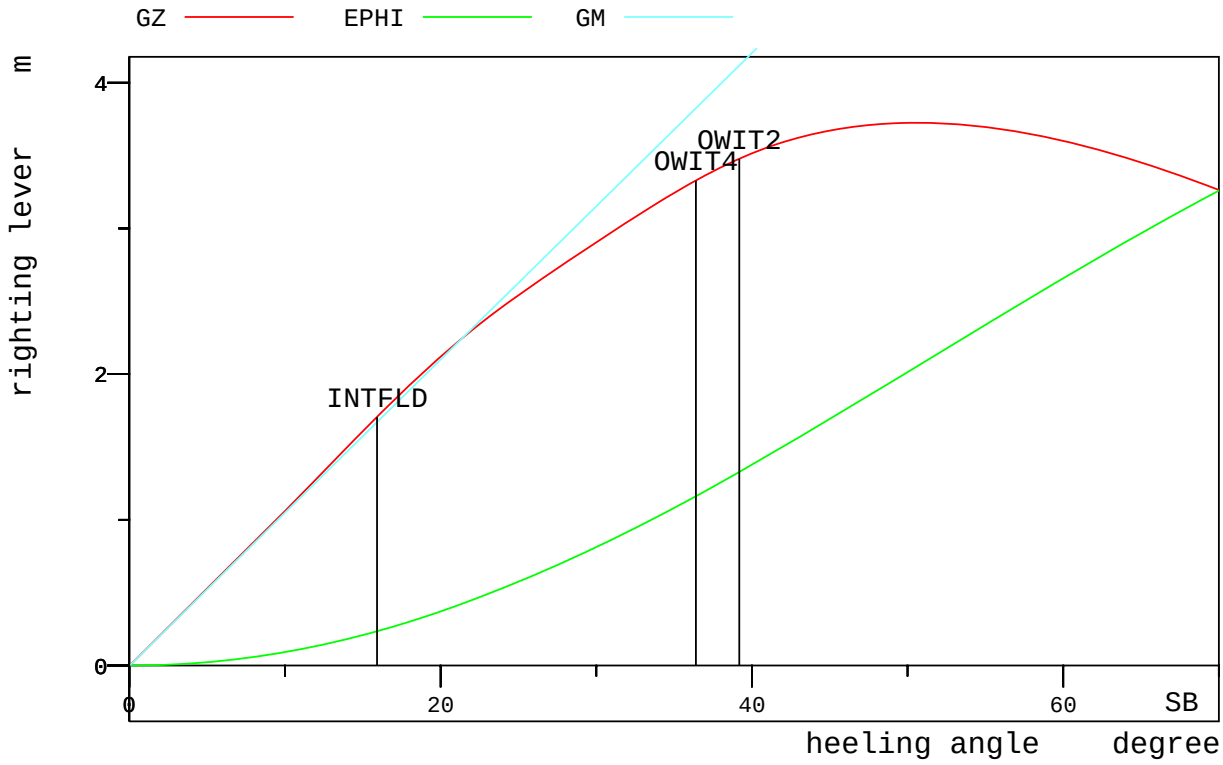
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN,CORR)	-3921.7 t	(65.4%)	POSITION:	140.2 m	177
SHEAR FORCE (MAX,CORR)	4197.9 t	(70.0%)		53.8 m	69
SAGGING MOMENT	-6674.8 tm	(10.7%)		32.1 m	42
HOGGING MOMENT	73338.3 tm	(70.5%)		123.1 m	156

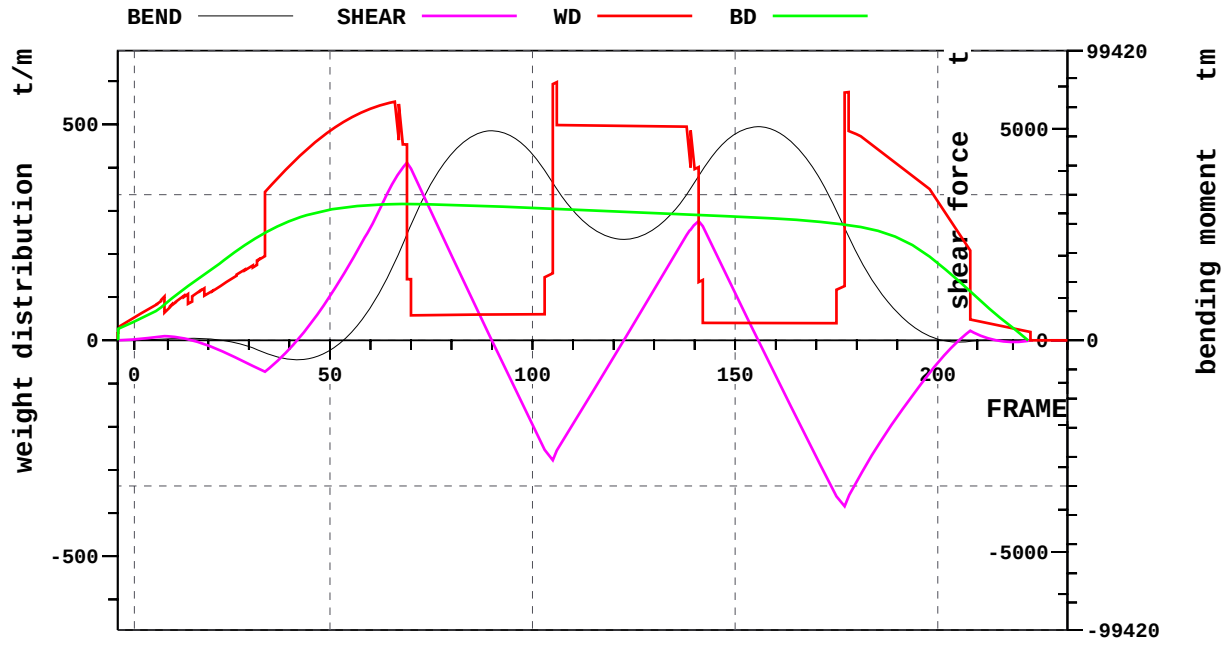
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 3.000)						
NO.1 CARGO HOLD	8554.0	42.9	150.97	0.00	4.81	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	12824.4	42.9	96.68	0.00	4.70	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	11565.8	42.9	40.65	0.00	5.24	0.0
SUBTOTAL	32944.2		91.11	0.00	4.92	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	18.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	43.6		53.17	0.00	20.19	0.0
TOTAL	32987.8		91.06	0.00	4.94	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	0.00	0.00	0.00	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	20.2	98.0	8.23	-7.08	9.34	22.22
M.D.O. SETTL. TK.	24.2	98.0	9.82	-7.32	9.21	29.35
M.D.O. STOR. TK. (P)	33.9	98.0	12.66	7.26	9.29	43.37
M.D.O. STOR. TK. (S)	43.2	98.0	12.30	-7.22	9.32	54.86
SUBTOTAL	121.6		11.23	-3.17	9.29	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	40.7	100.0	4.97	7.43	12.96	31.97
FRESH WATER TK. (P)	93.7	100.0	0.94	6.60	13.03	44.74
FRESH WATER TK. (S)	134.4	100.0	2.17	-6.85	13.01	82.65
SUBTOTAL	293.6		2.53	0.00	12.34	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	258.4	98.0	68.20	11.73	13.08	68.67
NO.1 FUEL OIL TK. (S)	258.4	98.0	68.20	-11.73	13.08	68.67
NO.2 FUEL OIL TK. (P)	250.2	98.0	39.86	11.71	13.09	0.00
NO.2 FUEL OIL TK. (S)	250.2	98.0	39.86	-11.71	13.09	0.00
H.F.O. TK. (P)	248.1	98.0	20.60	11.15	11.04	44.35
H.F.O. TK. (S)	176.3	98.0	21.94	-11.14	10.32	33.83
NO.1 H.F.O. SERV. TK.	24.3	98.0	15.46	-11.00	12.38	7.35
NO.1 H.F.O. SETTL. T.	29.0	98.0	18.63	-11.27	12.36	11.05
NO.2 H.F.O. SERV. TK.	19.1	98.0	12.64	-11.07	12.39	0.00
NO.2 H.F.O. SETTL. T.	21.5	98.0	9.62	-10.64	12.43	0.00
SUBTOTAL	1535.7		42.68	-0.15	12.40	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	6.8	98.0	10.01	5.59	8.82	0.36
G/E L.O. STOR. TK.	12.9	98.0	10.02	8.28	9.39	10.28
M/E L.O. SETTL. TK.	17.2	98.0	8.81	7.17	9.29	19.67
M/E L.O. STOR. TK.	26.4	98.0	7.12	6.90	9.42	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	101.1		12.73	4.87	7.77	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	0.6	10.0	18.98	-1.96	0.68	1.69
BILGE TK.	1.6	10.0	11.66	0.00	0.32	9.85
F.O. DRAIN TK.	0.6	10.0	20.77	2.04	0.56	3.33
F.O. OVERFLOW TK.	0.9	10.0	23.05	2.35	0.50	7.97
L.O. DRAIN TK.	0.4	10.0	18.43	1.94	0.72	0.91
SEWAGE STOR. TK.	1.4	10.0	22.44	-2.26	0.52	10.07
SLUDGE TK.	1.5	10.0	21.80	-8.40	7.44	9.03
SUBTOTAL	7.0		19.24	-1.81	1.99	42.86
TOTAL	2059.0		33.55	-0.07	11.94	642.92

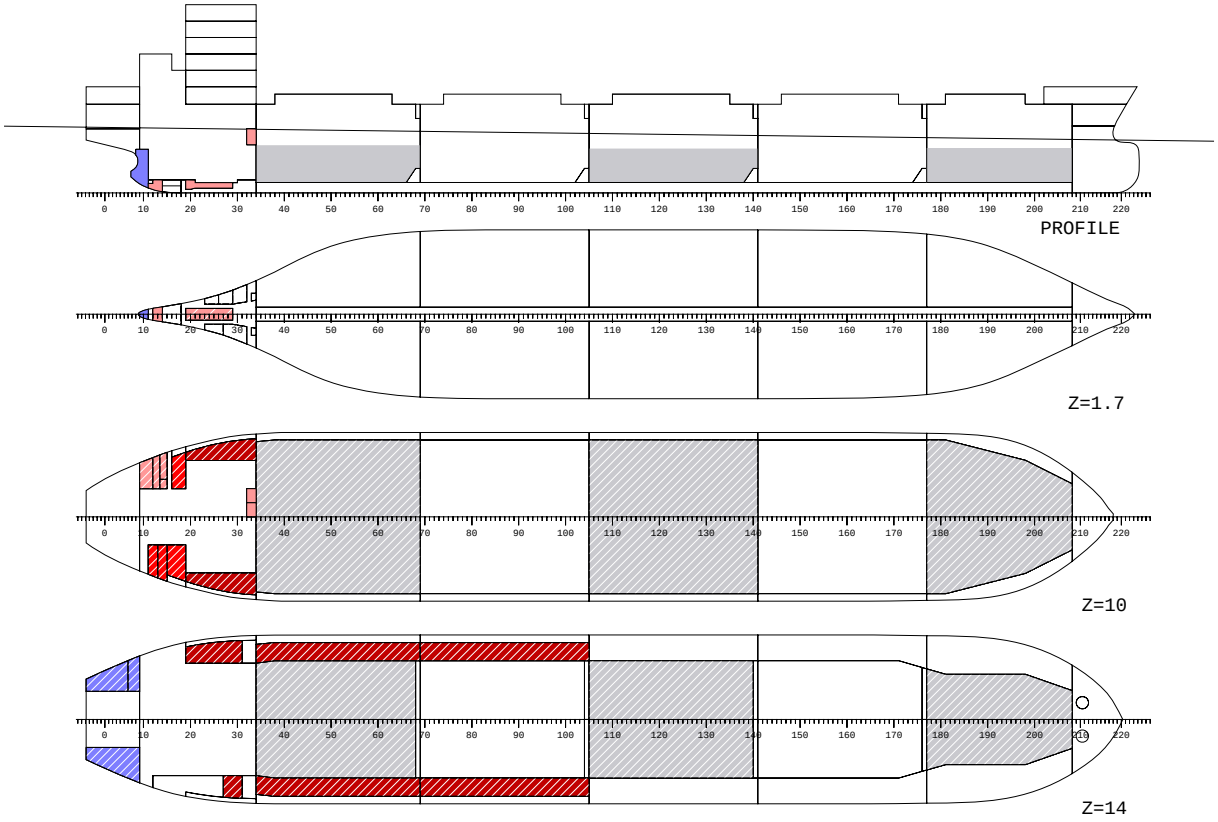


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.003	-0.00	0.000	0.0	0.000
10.0	0.013	1.06	0.092	111.6	0.003
12.0	0.026	1.28	0.133	133.7	0.003
20.0	0.059	2.12	0.372	219.9	0.005
30.0	-0.107	2.91	0.813	321.5	0.007
40.0	-0.357	3.52	1.378	413.3	0.009
50.0	-0.892	3.72	2.015	492.5	0.011
60.0	-1.616	3.60	2.658	556.8	0.013
70.0	-2.398	3.26	3.259	604.2	0.014

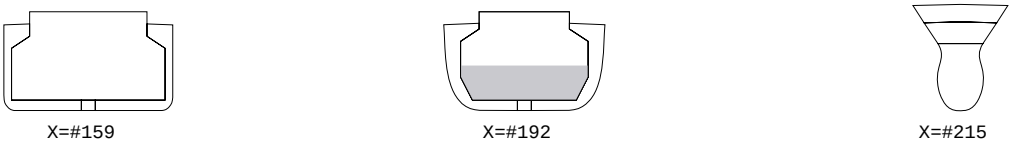


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	43	23	53.14	43.61
5.94	9.00	402	101	64.07	82.58
18.60	25.00	-359	-331	138.34	192.93
25.80	34.00	-4193	-739	343.57	248.86
40.20	52.00	-1532	1343	497.45	305.26
53.80	69.00	36364	4198	453.50	315.60
68.20	87.00	71229	592	59.75	311.47
82.60	105.00	54060	-2834	155.18	304.51
97.00	123.00	34653	73	496.41	297.30
111.40	141.00	56408	2820	400.91	290.09
125.80	159.00	72459	-648	39.94	282.39
140.20	177.00	38531	-3922	125.51	267.82
152.20	192.00	6956	-1535	393.53	228.70
165.00	208.00	-367	227	207.80	113.37
169.20	214.00	124	28	38.08	68.88
173.40	220.00	70	-39	27.85	29.28

LOADING CASE 'LD06B -- ALTERNATIVE HEAVY CARGO CONDITION, HALFWAY







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	0.0	0.00	0.00	0.00	0.00
Diesel Oil	DO	62.0	11.26	-2.98	8.46	149.80
Fresh Water	FW	159.2	2.90	-0.00	10.95	159.36
Grain Load	GRAIN	32944.2	91.11	0.00	4.92	0.00
Heavy Fuel Oil	HFO	783.5	42.77	-0.15	11.35	233.92
Lubricating Oil	LO	76.6	14.10	4.02	6.96	56.99
	MASS	34.6	63.09	0.00	20.29	0.00
Miscellaneous	MIS	27.9	19.18	-1.81	2.34	42.86
Deadweight		34088.0	89.18	-0.00	5.12	642.92
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		34088.0	89.18	-0.00	5.12	
Total weight		42738.0	87.18	-0.00	6.15	

F L O A T I N G P O S I T I O N

Draught moulded	10.078	m	KM	12.36 m
Trim	-2.134	m	KG	6.15 m
Heel, PS=+	0.0	deg		
TA	11.144	m	GM0	6.21 m
TF	9.011	m	GMCORR	-0.02 m
Trimming moment	-129067	tonm	GM	6.19 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.847	mrad	OK
AREA40	Area under GZ curve .	0.090	1.348	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.501	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	3.912	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	50.636	deg	OK
GM0.15	GM > 0.15 m	0.150	6.192	m	OK
IMOWEATH.	IMO weather criterion	1.000	3.063		OK
VISIBILI.		359.760	152.054		OK
PROPIMMER	Propeller should imm.	50.000	150.273	%	OK

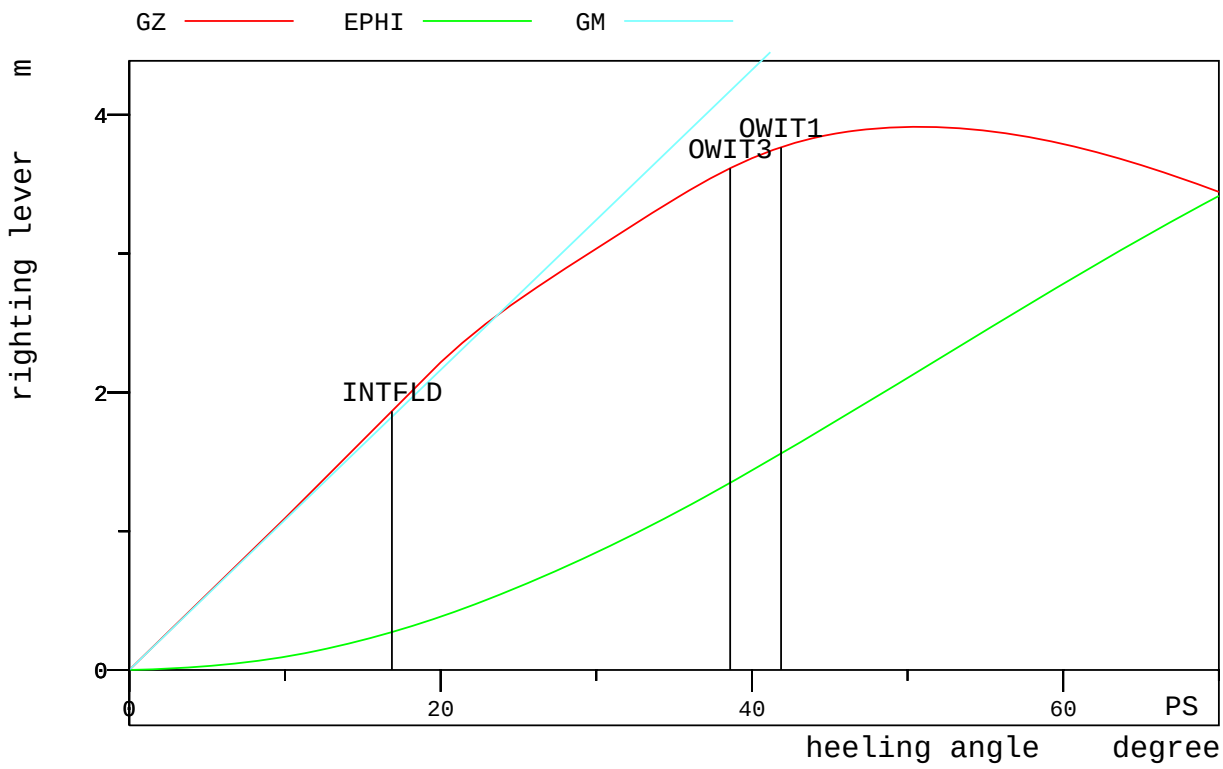
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN,CORR)	-3843.2 t	(64.1%)	POSITION:	140.2 m	177
SHEAR FORCE (MAX,CORR)	4170.6 t	(69.5%)		53.8 m	69
SAGGING MOMENT	-8730.6 tm	(13.7%)		32.8 m	43
HOGGING MOMENT	70757.7 tm	(68.0%)		123.1 m	156

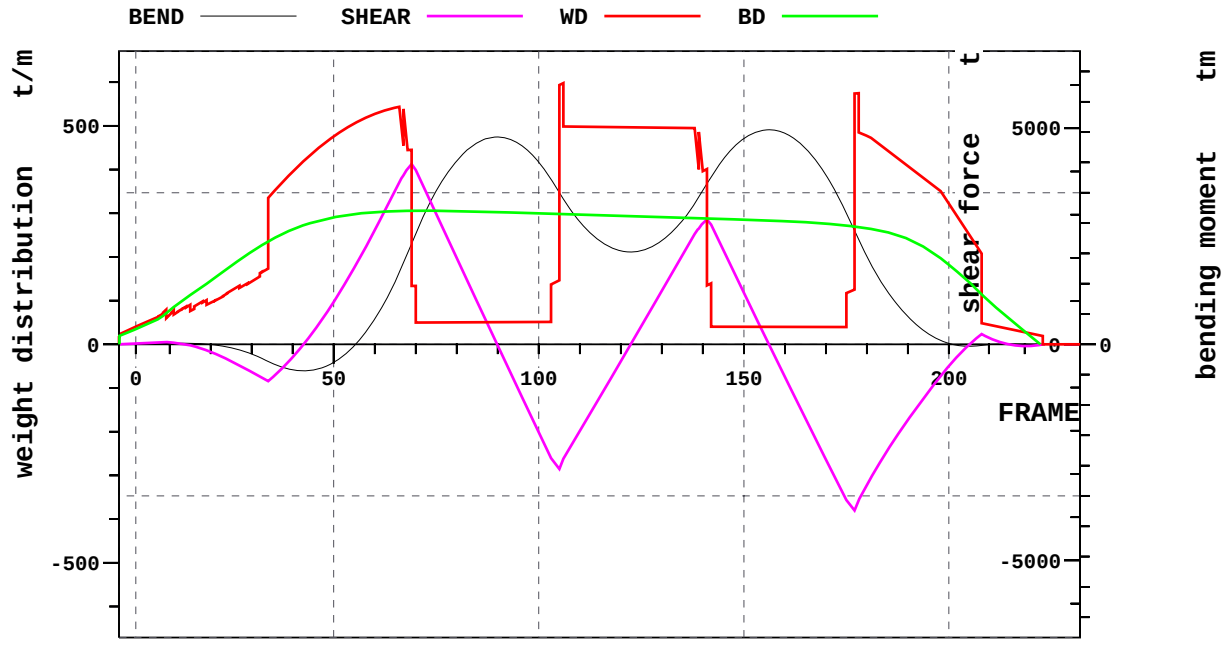
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 3.000)						
NO.1 CARGO HOLD	8554.0	42.9	150.97	0.00	4.81	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	12824.4	42.9	96.68	0.00	4.70	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	11565.8	42.9	40.65	0.00	5.24	0.0
SUBTOTAL	32944.2		91.11	0.00	4.92	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	9.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	34.6		63.09	0.00	20.29	0.0
TOTAL	32978.8		91.08	0.00	4.94	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	0.00	0.00	0.00	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	0.00	0.00	0.00	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	10.3	50.0	8.24	-6.66	8.53	22.22
M.D.O. SETTL. TK.	12.4	50.0	9.83	-6.86	8.34	29.35
M.D.O. STOR. TK. (P)	17.3	50.0	12.68	6.80	8.45	43.37
M.D.O. STOR. TK. (S)	22.0	50.0	12.35	-6.76	8.50	54.86
SUBTOTAL	62.0		11.26	-2.98	8.46	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	20.4	50.0	4.98	7.32	11.97	31.97
FRESH WATER TK. (P)	46.9	50.0	1.02	6.48	12.05	44.74
FRESH WATER TK. (S)	67.2	50.0	2.26	-6.75	12.02	82.65
SUBTOTAL	159.2		2.90	-0.00	10.95	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	131.9	50.0	68.20	11.90	12.29	68.67
NO.1 FUEL OIL TK. (S)	131.9	50.0	68.20	-11.90	12.29	68.67
NO.2 FUEL OIL TK. (P)	127.7	50.0	39.94	11.88	12.30	0.00
NO.2 FUEL OIL TK. (S)	127.7	50.0	39.94	-11.88	12.30	0.00
H.F.O. TK. (P)	126.6	50.0	21.26	10.98	9.26	44.35
H.F.O. TK. (S)	90.0	50.0	21.60	-10.89	8.63	33.83
NO.1 H.F.O. SERV. TK.	12.4	50.0	15.47	-10.92	11.69	7.35
NO.1 H.F.O. SETTL. T.	14.8	50.0	18.64	-11.21	11.66	11.05
NO.2 H.F.O. SERV. TK.	9.7	50.0	12.65	-10.98	11.70	0.00
NO.2 H.F.O. SETTL. T.	11.0	50.0	9.65	-10.54	11.75	0.00
SUBTOTAL	783.5		42.77	-0.15	11.35	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	4.1	60.0	10.01	5.58	8.05	0.36
G/E L.O. STOR. TK.	7.9	60.0	10.02	8.01	8.78	10.28
M/E L.O. SETTL. TK.	10.5	60.0	8.82	6.85	8.65	19.67
M/E L.O. STOR. TK.	16.1	60.0	7.14	6.62	8.83	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	76.6		14.10	4.02	6.96	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	2.3	40.0	18.84	-2.15	1.07	1.69
BILGE TK.	6.4	40.0	11.55	0.00	0.73	9.85
F.O. DRAIN TK.	2.6	40.0	20.71	2.30	0.98	3.33
F.O. OVERFLOW TK.	3.6	40.0	23.05	2.70	0.93	7.97
L.O. DRAIN TK.	1.5	40.0	18.35	2.11	1.09	0.91
SEWAGE STOR. TK.	5.4	40.0	22.38	-2.58	0.94	10.07
SLUDGE TK.	6.0	40.0	21.80	-8.40	7.56	9.03
SUBTOTAL	27.9		19.18	-1.81	2.34	42.86
TOTAL	1109.3		32.71	-0.04	10.60	642.92

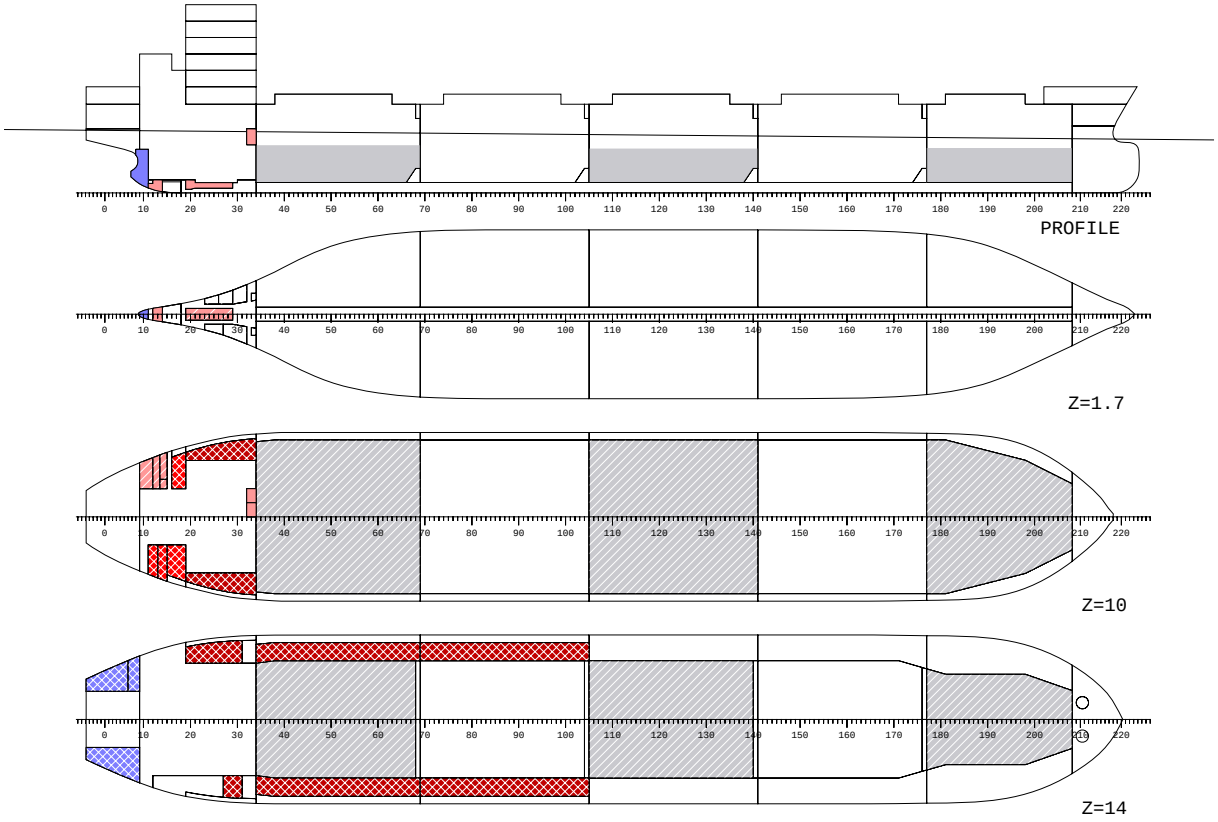


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	0.000	0.00	0.000	0.0	0.000
10.0	0.018	1.09	0.095	111.6	0.003
12.0	0.030	1.32	0.137	133.7	0.003
20.0	0.098	2.22	0.384	219.9	0.005
30.0	-0.060	3.04	0.847	321.5	0.008
40.0	-0.293	3.69	1.437	413.3	0.010
50.0	-0.831	3.91	2.107	492.5	0.012
60.0	-1.572	3.79	2.783	556.8	0.013
70.0	-2.375	3.44	3.416	604.2	0.014

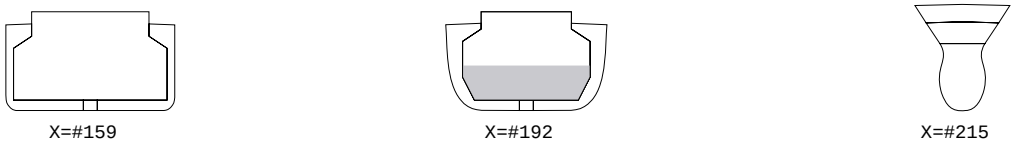


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	30	17	40.58	34.71
5.94	9.00	239	53	69.75	71.25
18.60	25.00	-1157	-391	122.78	179.26
25.80	34.00	-5600	-851	335.16	235.36
40.20	52.00	-4119	1289	488.45	293.51
53.80	69.00	33224	4171	444.78	305.67
68.20	87.00	67757	569	50.68	303.46
82.60	105.00	50068	-2884	146.40	298.43
97.00	123.00	30478	97	496.41	293.14
111.40	141.00	52916	2890	400.91	287.86
125.80	159.00	70112	-560	39.94	282.08
140.20	177.00	37390	-3843	125.51	269.41
152.20	192.00	6585	-1484	393.53	231.69
165.00	208.00	-368	239	207.80	115.84
169.20	214.00	146	31	38.08	70.36
173.40	220.00	89	-39	27.85	29.28

LOADING CASE 'LD06C -- ALTERNATIVE HEAVY CARGO CONDITION, ARRIVAL







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	0.0	0.00	0.00	0.00	0.00
Diesel Oil	DO	12.4	11.39	-2.61	7.47	149.80
Fresh Water	FW	51.6	4.32	-0.00	8.25	159.36
Grain Load	GRAIN	32944.2	91.11	0.00	4.92	0.00
Heavy Fuel Oil	HFO	156.7	43.37	-0.16	10.03	233.92
Lubricating Oil	LO	50.8	16.95	2.45	5.88	56.99
	MASS	27.6	75.29	0.00	20.42	0.00
Miscellaneous	MIS	55.8	19.16	-1.81	2.67	42.86
Deadweight		33299.1	90.47	-0.00	4.96	642.92
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		33299.1	90.47	-0.00	4.96	
Total weight		41949.1	88.17	-0.00	6.04	

F L O A T I N G P O S I T I O N

Draught moulded	9.920	m	KM	12.36 m
Trim	-1.492	m	KG	6.04 m
Heel, PS=+	0.0	deg		
TA	10.666	m	GM0	6.32 m
TF	9.174	m	GMCORR	-0.02 m
Trimming moment	-89658	tonm	GM	6.31 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.869	mrad	OK
AREA40	Area under GZ curve .	0.090	1.478	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.609	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.046	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	50.800	deg	OK
GM0.15	GM > 0.15 m	0.150	6.308	m	OK
IMOWEATH.	IMO weather criterion	1.000	3.411		OK
VISIBILI.		359.760	141.435		OK
PROPIMMER	Propeller should imm.	50.000	141.236	%	OK

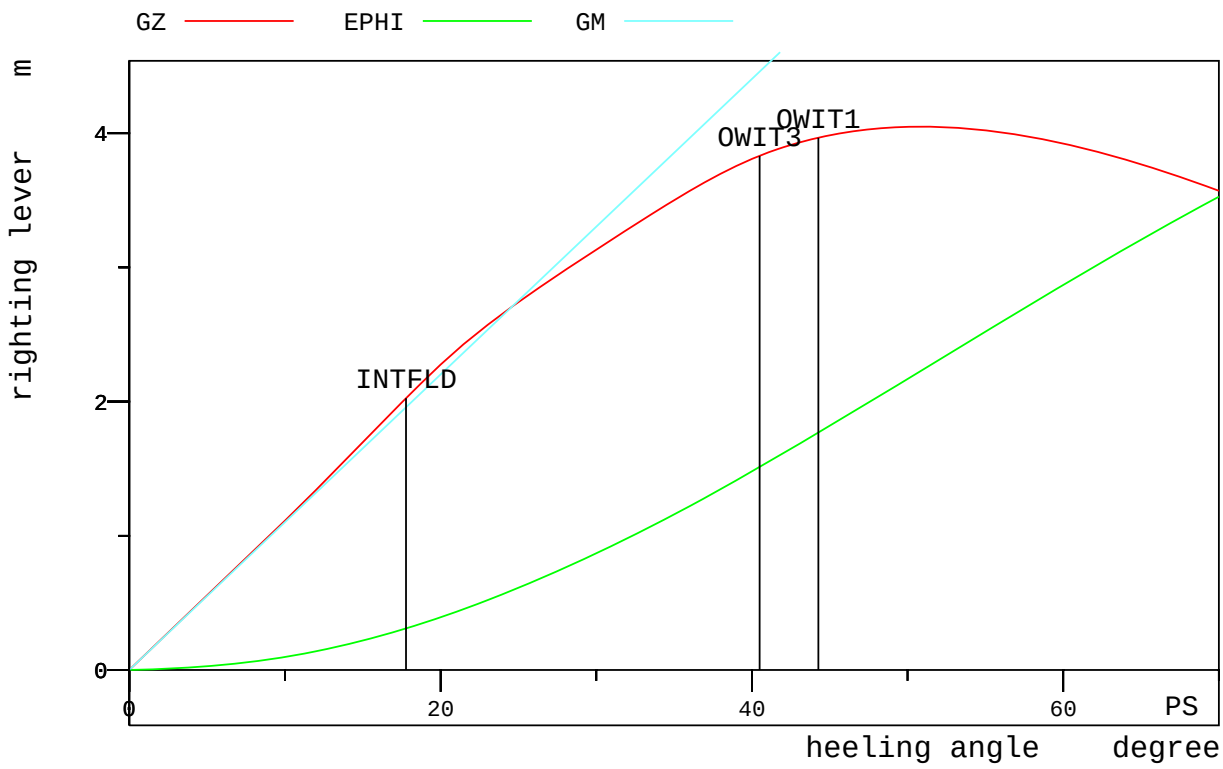
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN,CORR)	-3778.6 t	(63.0%)	POSITION:	140.2 m	177
SHEAR FORCE (MAX,CORR)	4148.5 t	(69.1%)		53.8 m	69
SAGGING MOMENT	-10455.2 tm	(16.3%)		33.1 m	43
HOGGING MOMENT	68538.7 tm	(65.8%)		123.1 m	156

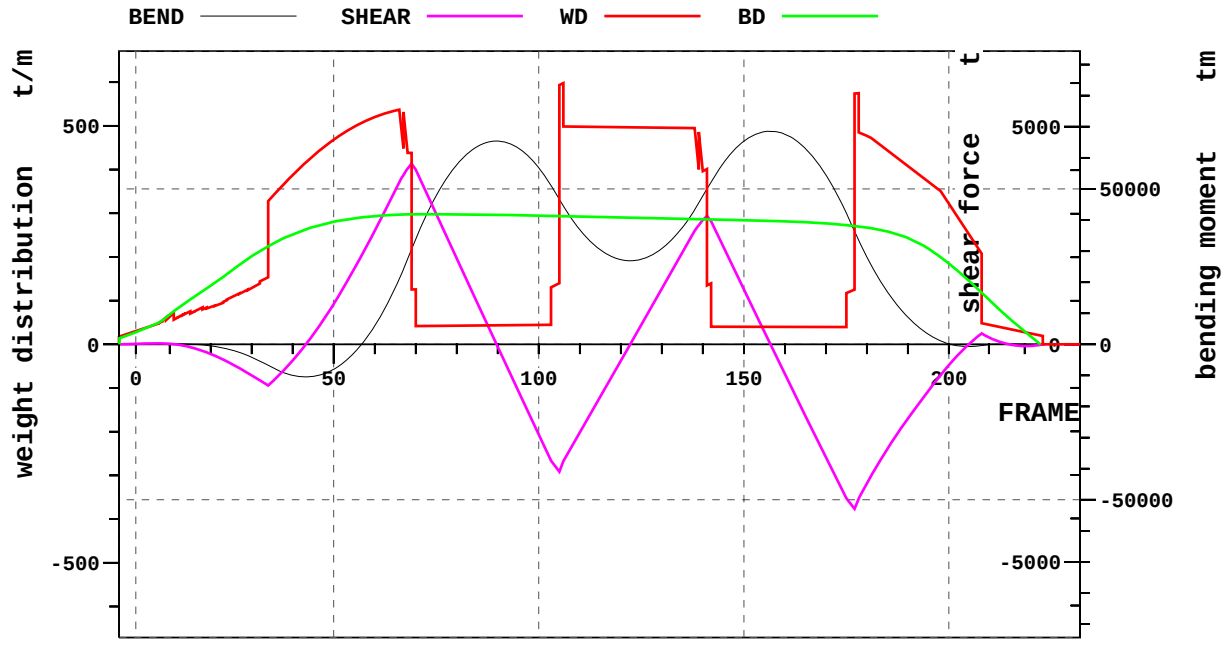
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 3.000)						
NO.1 CARGO HOLD	8554.0	42.9	150.97	0.00	4.81	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	12824.4	42.9	96.68	0.00	4.70	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	11565.8	42.9	40.65	0.00	5.24	0.0
SUBTOTAL	32944.2		91.11	0.00	4.92	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	2.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	27.6		75.29	0.00	20.42	0.0
TOTAL	32971.8		91.10	0.00	4.93	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	0.00	0.00	0.00	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	2.1	10.0	8.28	-5.84	7.59	22.22
M.D.O. SETTL. TK.	2.5	10.0	9.88	-5.95	7.33	29.35
M.D.O. STOR. TK. (P)	3.5	10.0	12.81	5.91	7.45	43.37
M.D.O. STOR. TK. (S)	4.4	10.0	12.58	-5.91	7.52	54.86
SUBTOTAL	12.4		11.39	-2.61	7.47	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	4.1	10.0	4.98	7.15	11.13	31.97
FRESH WATER TK. (P)	9.4	10.0	1.14	6.32	11.15	44.74
FRESH WATER TK. (S)	13.4	10.0	2.40	-6.59	11.14	82.65
SUBTOTAL	51.6		4.32	-0.00	8.25	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	26.4	10.0	68.20	12.64	11.22	68.67
NO.1 FUEL OIL TK. (S)	26.4	10.0	68.20	-12.64	11.22	68.67
NO.2 FUEL OIL TK. (P)	25.5	10.0	40.23	12.62	11.23	0.00
NO.2 FUEL OIL TK. (S)	25.5	10.0	40.23	-12.62	11.23	0.00
H.F.O. TK. (P)	25.3	10.0	23.08	10.63	7.06	44.35
H.F.O. TK. (S)	18.0	10.0	23.44	-10.57	6.80	33.83
NO.1 H.F.O. SERV. TK.	2.5	10.0	15.49	-10.82	11.07	7.35
NO.1 H.F.O. SETTL. T.	3.0	10.0	18.65	-11.14	11.06	11.05
NO.2 H.F.O. SERV. TK.	1.9	10.0	12.65	-10.87	11.07	0.00
NO.2 H.F.O. SETTL. T.	2.2	10.0	9.70	-10.42	11.09	0.00
SUBTOTAL	156.7		43.37	-0.16	10.03	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	1.4	20.0	10.03	5.53	7.22	0.36
G/E L.O. STOR. TK.	2.6	20.0	10.04	7.48	7.96	10.28
M/E L.O. SETTL. TK.	3.5	20.0	8.83	6.21	7.80	19.67
M/E L.O. STOR. TK.	5.4	20.0	7.19	6.05	8.05	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	50.8		16.95	2.45	5.88	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	4.5	80.0	18.81	-2.29	1.40	1.69
BILGE TK.	12.9	80.0	11.52	0.00	1.14	9.85
F.O. DRAIN TK.	5.2	80.0	20.70	2.51	1.34	3.33
F.O. OVERFLOW TK.	7.2	80.0	23.04	2.98	1.31	7.97
L.O. DRAIN TK.	3.1	80.0	18.32	2.24	1.42	0.91
SEWAGE STOR. TK.	10.9	80.0	22.37	-2.83	1.31	10.07
SLUDGE TK.	12.0	80.0	21.80	-8.40	7.72	9.03
SUBTOTAL	55.8		19.16	-1.81	2.67	42.86
TOTAL	327.4		27.78	-0.10	7.76	642.92

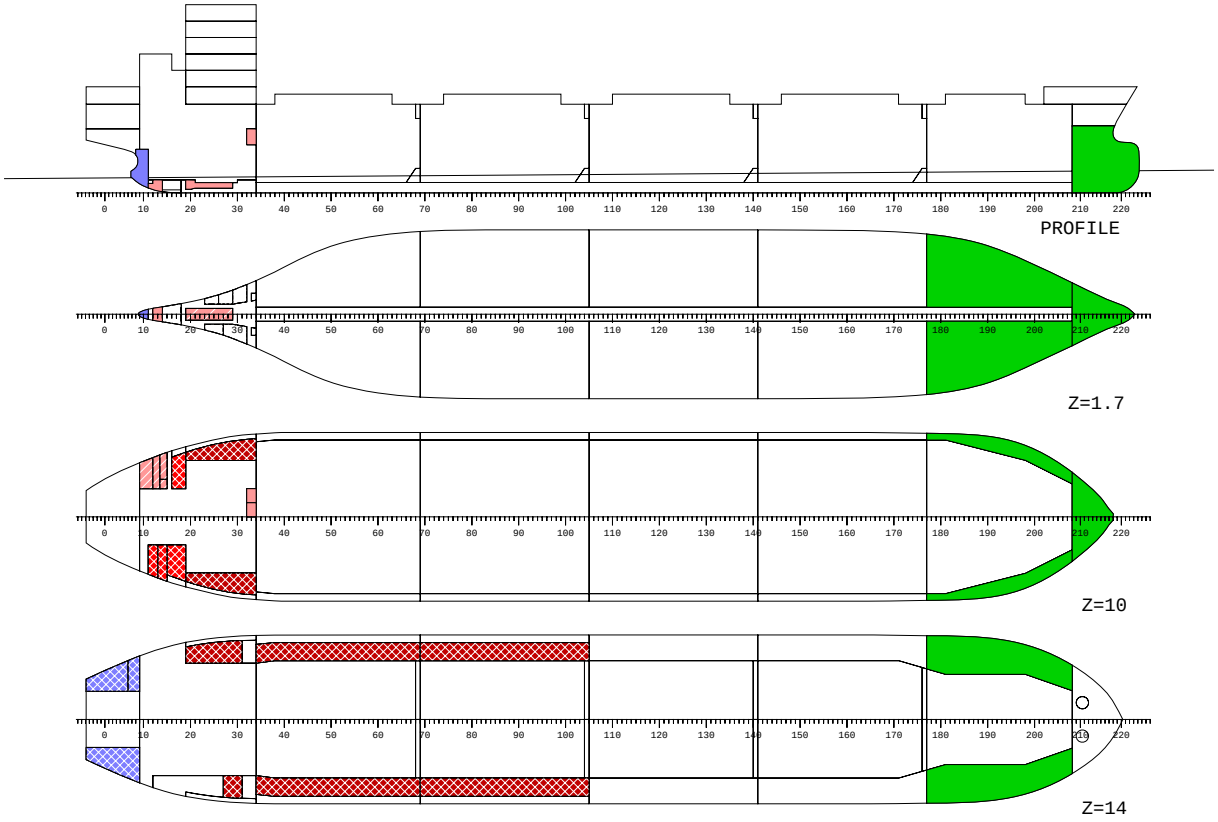


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	0.000	0.00	0.000	0.0	0.000
10.0	0.017	1.11	0.097	111.6	0.003
12.0	0.030	1.34	0.140	133.7	0.003
20.0	0.118	2.28	0.393	219.9	0.005
30.0	-0.022	3.13	0.869	321.5	0.008
40.0	-0.247	3.81	1.478	413.3	0.010
50.0	-0.786	4.05	2.170	492.5	0.012
60.0	-1.539	3.92	2.870	556.8	0.013
70.0	-2.357	3.57	3.526	604.2	0.014

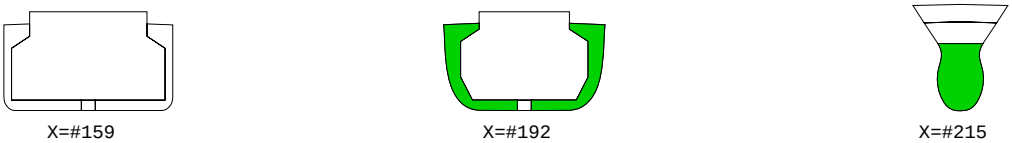


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	26	13	30.63	27.64
5.94	9.00	127	15	55.67	62.10
18.60	25.00	-1810	-446	107.00	168.05
25.80	34.00	-6762	-943	327.97	224.22
40.20	52.00	-6267	1240	481.13	283.79
53.80	69.00	30608	4148	438.09	297.45
68.20	87.00	64814	544	43.35	296.83
82.60	105.00	46652	-2927	139.75	293.39
97.00	123.00	26921	116	496.41	289.69
111.40	141.00	49933	2947	400.91	286.00
125.80	159.00	68089	-488	39.94	281.81
140.20	177.00	36381	-3779	125.51	270.72
152.20	192.00	6229	-1443	393.53	234.15
165.00	208.00	-399	249	207.80	117.89
169.20	214.00	140	34	38.08	71.57
173.40	220.00	88	-39	27.85	29.28

LOADING CASE 'LD07 -- PROPELLER SHAFT SEAL MAINTENANCE WITH 10% BUNKER,DOCKING







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	3193.0	155.95	0.00	6.75	0.00
Diesel Oil	DO	12.4	11.39	-2.61	7.47	149.80
Fresh Water	FW	51.6	4.32	-0.00	8.25	159.36
Grain Load	GRAIN	0.0	0.00	0.00	0.00	0.00
Heavy Fuel Oil	HFO	156.7	43.35	-0.16	10.12	380.41
Lubricating Oil	LO	50.8	16.95	2.45	5.88	56.99
	MASS	27.6	75.29	0.00	20.42	0.00
Miscellaneous	MIS	7.0	19.24	-1.81	1.99	42.86
Deadweight		3499.2	145.23	0.02	7.01	789.41
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		3499.2	145.23	0.02	7.01	
Total weight		12149.2	98.29	0.00	9.28	

F L O A T I N G P O S I T I O N

Draught moulded	3.143	m	KM	22.08 m
Trim	1.202	m	KG	9.28 m
Heel, PS=+	0.0	deg		
TA	2.542	m	GM0	12.80 m
TF	3.744	m	GMCORR	-0.06 m
Trimming moment	47273	tonm	GM	12.73 m
Heel	0.0	deg (PS)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	1.401	mrad	OK
AREA40	Area under GZ curve .	0.090	2.065	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.665	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	3.924	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	28.055	deg	OK
GM0.15	GM > 0.15 m	0.150	12.731	m	OK
IMOWEATH.	IMO weather criterion	1.000	1.319		OK
VISIBILI.		359.760	177.916		OK
PROPIMMER	Propeller should imm.	50.000	0.000	%	NOT MET

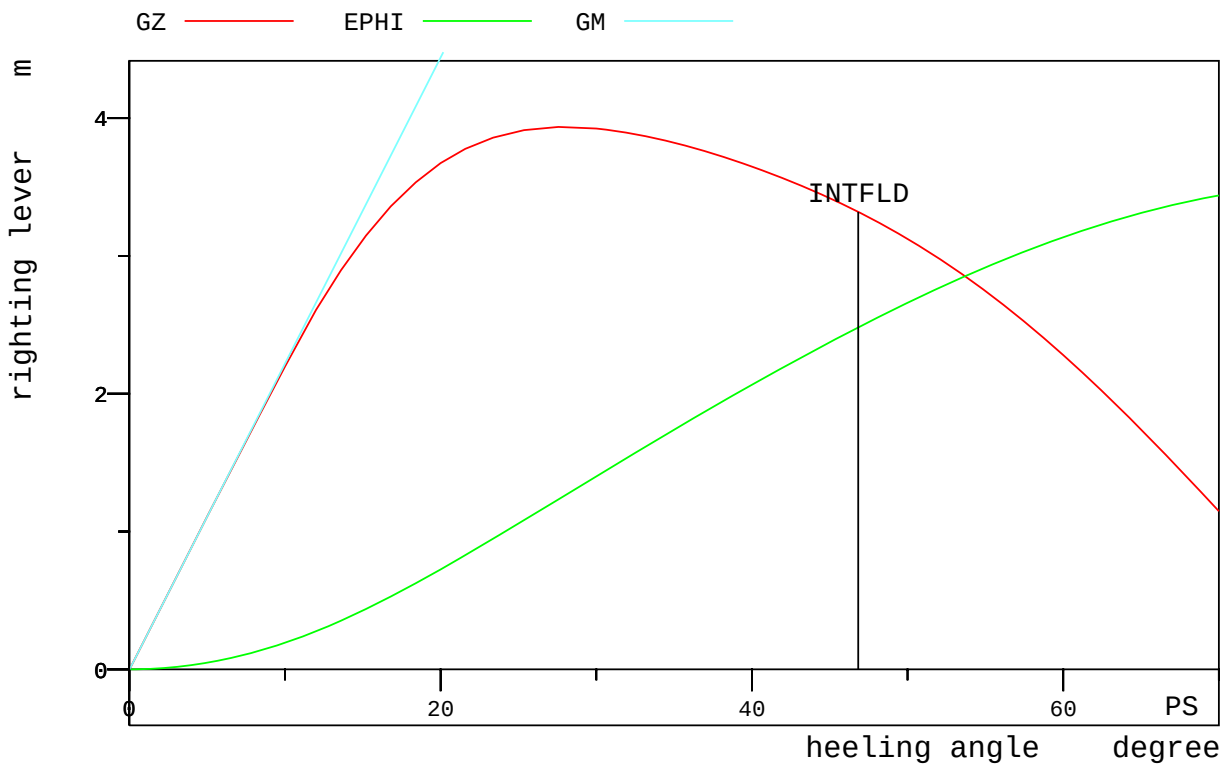
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-2402.8 t	(40.0%)	POSITION:	138.6 m	175
SHEAR FORCE (MAX)	1972.6 t	(32.9%)		29.0 m	38
SAGGING MOMENT	-0.4 tm	(0.2%)		-3.2 m	-5
HOGGING MOMENT	102141.0 tm	(93.6%)		93.4 m	118

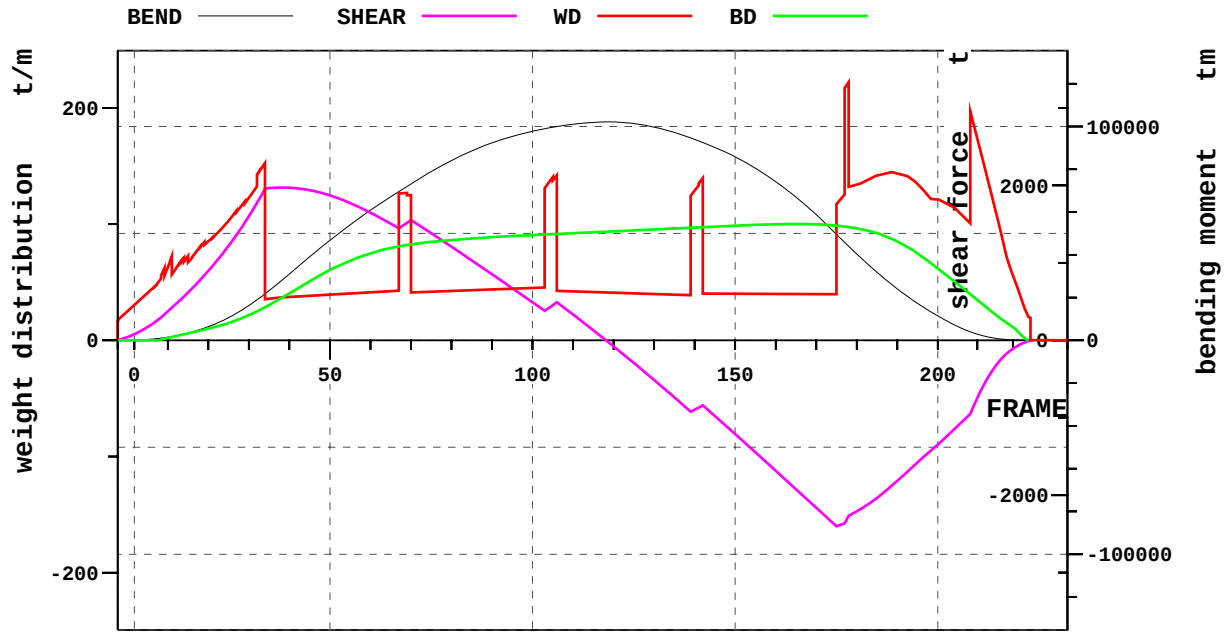
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.740)						
NO.1 CARGO HOLD	0.0	0.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
SUBTOTAL	0.0		0.00	0.00	0.00	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	2.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	27.6		75.29	0.00	20.42	0.0
TOTAL	27.6		75.29	0.00	20.42	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	774.2	100.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	1209.4	100.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	3193.0		155.95	0.00	6.75	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	2.1	10.0	8.28	-5.84	7.59	22.22
M.D.O. SETTL. TK.	2.5	10.0	9.88	-5.95	7.33	29.35
M.D.O. STOR. TK. (P)	3.5	10.0	12.81	5.91	7.45	43.37
M.D.O. STOR. TK. (S)	4.4	10.0	12.58	-5.91	7.52	54.86
SUBTOTAL	12.4		11.39	-2.61	7.47	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	4.1	10.0	4.98	7.15	11.13	31.97
FRESH WATER TK. (P)	9.4	10.0	1.14	6.32	11.15	44.74
FRESH WATER TK. (S)	13.4	10.0	2.40	-6.59	11.14	82.65
SUBTOTAL	51.6		4.32	-0.00	8.25	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	26.4	10.0	68.20	12.54	11.35	68.67
NO.1 FUEL OIL TK. (S)	26.4	10.0	68.20	-12.54	11.35	68.67
NO.2 FUEL OIL TK. (P)	25.5	10.0	40.16	12.52	11.37	67.49
NO.2 FUEL OIL TK. (S)	25.5	10.0	40.16	-12.52	11.37	67.49
H.F.O. TK. (P)	25.3	10.0	23.07	10.63	7.07	44.35
H.F.O. TK. (S)	18.0	10.0	23.44	-10.57	6.80	33.83
NO.1 H.F.O. SERV. TK.	2.5	10.0	15.49	-10.82	11.07	7.35
NO.1 H.F.O. SETTL. T.	3.0	10.0	18.65	-11.14	11.06	11.05
NO.2 H.F.O. SERV. TK.	1.9	10.0	12.65	-10.87	11.07	6.67
NO.2 H.F.O. SETTL. T.	2.2	10.0	9.70	-10.42	11.09	4.84
SUBTOTAL	156.7		43.35	-0.16	10.12	380.41
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	1.4	20.0	10.03	5.53	7.22	0.36
G/E L.O. STOR. TK.	2.6	20.0	10.04	7.48	7.96	10.28
M/E L.O. SETTL. TK.	3.5	20.0	8.83	6.21	7.80	19.67
M/E L.O. STOR. TK.	5.4	20.0	7.19	6.05	8.05	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	50.8		16.95	2.45	5.88	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	0.6	10.0	18.98	-1.96	0.68	1.69
BILGE TK.	1.6	10.0	11.66	0.00	0.32	9.85
F.O. DRAIN TK.	0.6	10.0	20.77	2.04	0.56	3.33
F.O. OVERFLOW TK.	0.9	10.0	23.05	2.35	0.50	7.97
L.O. DRAIN TK.	0.4	10.0	18.43	1.94	0.72	0.91
SEWAGE STOR. TK.	1.4	10.0	22.44	-2.26	0.52	10.07
SLUDGE TK.	1.5	10.0	21.80	-8.40	7.44	9.03
SUBTOTAL	7.0		19.24	-1.81	1.99	42.86
TOTAL	3471.6		145.78	0.02	6.90	789.41

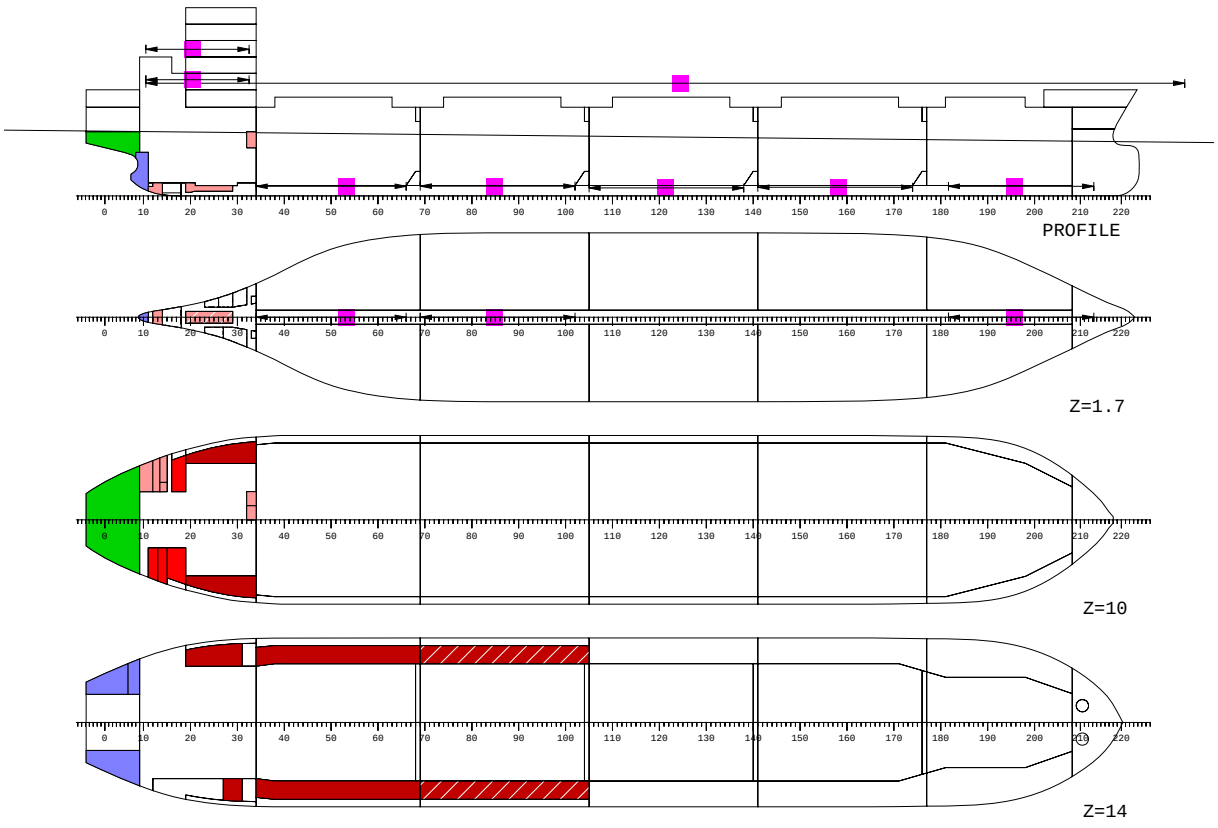


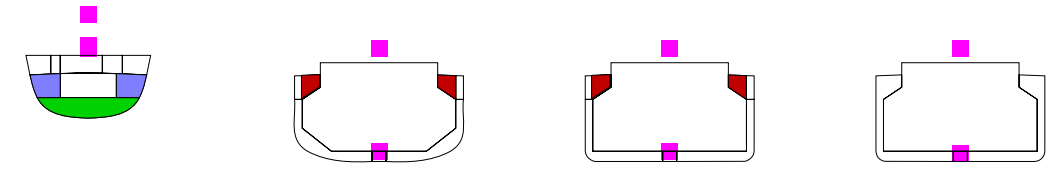
HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.004	-0.00	0.000	0.0	0.000
10.0	-0.012	2.20	0.193	137.1	0.011
12.0	-0.038	2.61	0.277	164.1	0.014
20.0	-0.681	3.67	0.726	270.0	0.022
30.0	-2.442	3.92	1.400	394.7	0.032
40.0	-4.536	3.65	2.065	507.4	0.042
50.0	-6.629	3.12	2.660	604.7	0.050
60.0	-8.746	2.28	3.136	683.6	0.056
70.0	-10.815	1.15	3.439	741.8	0.061

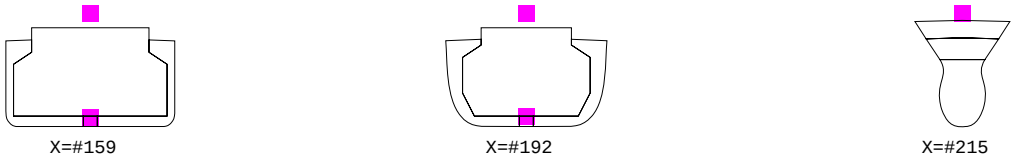


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	107	79	30.51	0.00
5.94	9.00	1226	334	55.62	1.40
18.60	25.00	10613	1222	102.97	14.99
25.80	34.00	21884	1960	35.41	28.70
40.20	52.00	49693	1835	39.70	63.90
53.80	69.00	71872	1519	126.81	82.10
68.20	87.00	90183	962	43.35	88.14
82.60	105.00	99397	453	140.48	91.37
97.00	123.00	101714	-206	40.66	94.34
111.40	141.00	93076	-870	133.42	97.32
125.80	159.00	75530	-1637	39.94	99.80
140.20	177.00	45788	-2366	125.51	97.80
152.20	192.00	21242	-1722	141.72	81.18
165.00	208.00	4024	-950	100.69	39.75
169.20	214.00	1246	-409	125.45	24.76
173.40	220.00	217	-109	56.15	11.41

LOADING CASE 'LD08A -- STEEL COIL CONDITION, DEPARTURE







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	339.3	2.28	0.00	9.53	0.00
Diesel Oil	DO	121.6	11.23	-3.17	9.29	149.80
Fresh Water	FW	293.6	2.53	0.00	12.34	159.36
Solid cargo	GRAIN	0.0	0.00	0.00	0.00	0.00
Heavy Fuel Oil	HFO	1535.7	42.68	-0.15	12.40	233.92
Lubricating Oil	LO	101.1	12.73	4.87	7.77	56.99
	MASS	32023.6	94.25	0.00	1.55	0.00
Miscellaneous	MIS	7.0	19.24	-1.81	1.99	42.86
Deadweight		34421.9	89.71	-0.00	2.25	642.92
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		34421.9	89.71	-0.00	2.25	
Total weight		43071.9	87.62	-0.00	3.85	

F L O A T I N G P O S I T I O N

Draught moulded	10.153	m	KM	12.33 m
Trim	-1.781	m	KG	3.85 m
Heel, PS=+	0.0	deg		
TA	11.043	m	GM0	8.48 m
TF	9.262	m	GMCORR	-0.01 m
Trimming moment	-108673	tonm	GM	8.47 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	1.148	mrad	OK
AREA40	Area under GZ curve .	0.090	1.861	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.713	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	5.765	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	58.411	deg	OK
GM0.15	GM > 0.15 m	0.150	8.466	m	OK
IMOWEATH.	IMO weather criterion	1.000	4.007		OK
VISIBILI.		359.760	143.722		OK
PROPIMMER	Propeller should imm.	50.000	148.366	%	OK

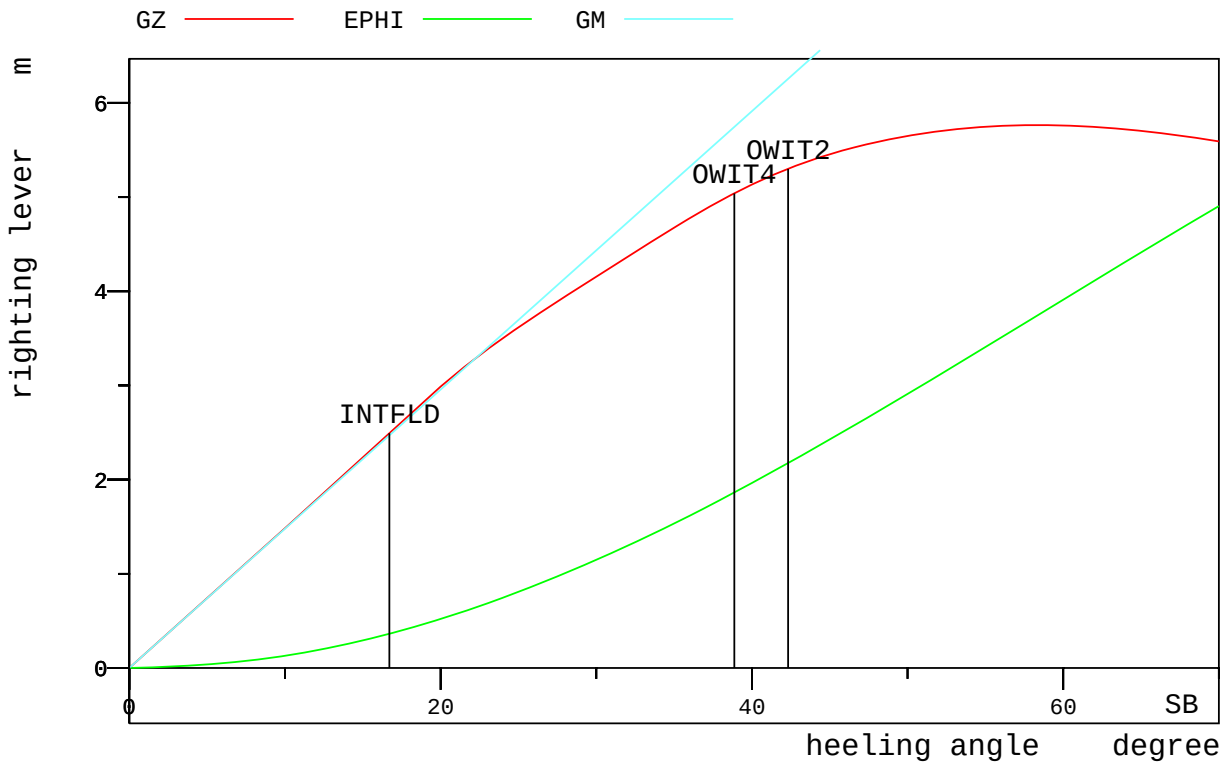
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN,CORR)	-744.7 t	(12.4%)	POSITION:	40.8 m	53
SHEAR FORCE (MAX,CORR)	1197.0 t	(19.9%)		137.8 m	174
SAGGING MOMENT	-8016.4 tm	(8.3%)		64.6 m	82
HOGGING MOMENT	8898.5 tm	(15.4%)		25.8 m	34

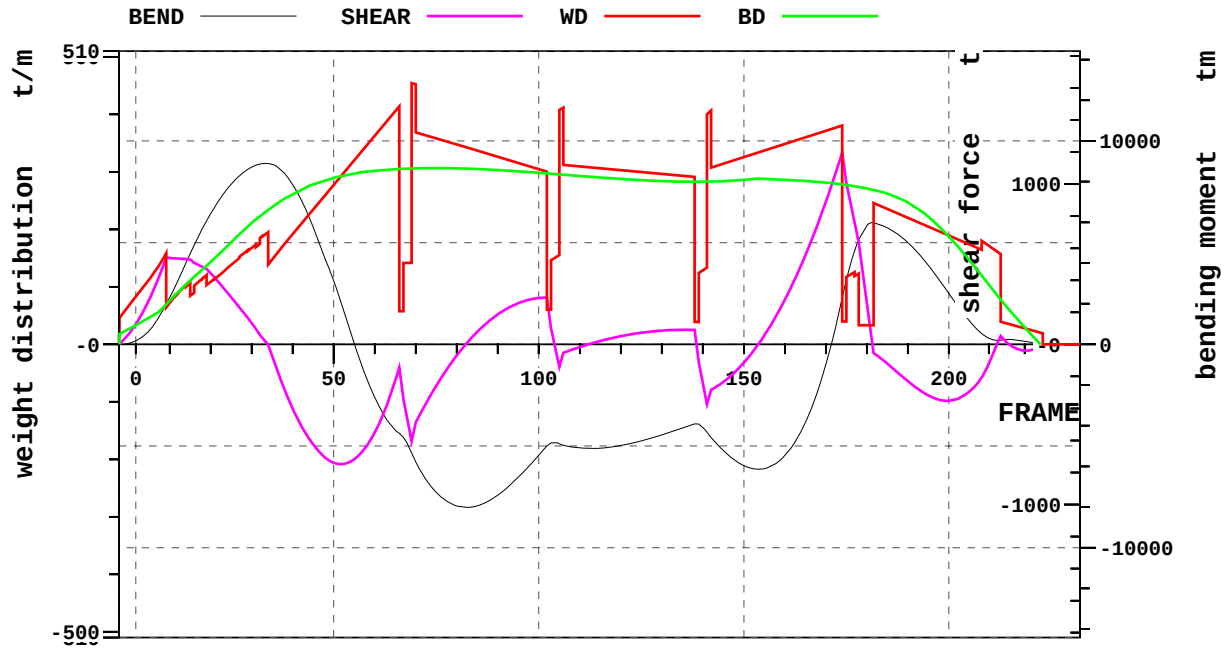
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Steel Coil						
NO.1 CARGO HOLD	4100.0	0.0	155.11	0.00	1.62	0.0
NO.2 CARGO HOLD	8000.0	0.0	125.15	0.00	1.47	0.0
NO.3 CARGO HOLD	6900.0	0.0	95.65	0.00	1.33	0.0
NO.4 CARGO HOLD	7300.0	0.0	66.42	0.00	1.65	0.0
NO.5 CARGO HOLD	5680.0	0.0	41.19	0.00	1.64	0.0
SUBTOTAL	31980.0		94.31	0.00	1.53	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	18.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	43.6		53.17	0.00	20.19	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	339.3	100.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	0.00	0.00	0.00	0.00
NO.5 WATER BALLAST T.	0.0	0.0	0.00	0.00	0.00	0.00
SUBTOTAL	339.3		2.28	0.00	9.53	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	20.2	98.0	8.23	-7.08	9.34	22.22
M.D.O. SETTL. TK.	24.2	98.0	9.82	-7.32	9.21	29.35
M.D.O. STOR. TK. (P)	33.9	98.0	12.66	7.26	9.29	43.37
M.D.O. STOR. TK. (S)	43.2	98.0	12.30	-7.22	9.32	54.86
SUBTOTAL	121.6		11.23	-3.17	9.29	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	40.7	100.0	4.97	7.43	12.96	31.97
FRESH WATER TK. (P)	93.7	100.0	0.94	6.60	13.03	44.74
FRESH WATER TK. (S)	134.4	100.0	2.17	-6.85	13.01	82.65
SUBTOTAL	293.6		2.53	0.00	12.34	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	258.4	98.0	68.20	11.73	13.08	68.67
NO.1 FUEL OIL TK. (S)	258.4	98.0	68.20	-11.73	13.08	68.67
NO.2 FUEL OIL TK. (P)	250.2	98.0	39.86	11.71	13.09	0.00
NO.2 FUEL OIL TK. (S)	250.2	98.0	39.86	-11.71	13.09	0.00
H.F.O. TK. (P)	248.1	98.0	20.60	11.15	11.04	44.35
H.F.O. TK. (S)	176.3	98.0	21.94	-11.14	10.32	33.83
NO.1 H.F.O. SERV. TK.	24.3	98.0	15.46	-11.00	12.38	7.35
NO.1 H.F.O. SETTL. T.	29.0	98.0	18.63	-11.27	12.36	11.05
NO.2 H.F.O. SERV. TK.	19.1	98.0	12.64	-11.07	12.39	0.00
NO.2 H.F.O. SETTL. T.	21.5	98.0	9.62	-10.64	12.43	0.00
SUBTOTAL	1535.7		42.68	-0.15	12.40	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	6.8	98.0	10.01	5.59	8.82	0.36
G/E L.O. STOR. TK.	12.9	98.0	10.02	8.28	9.39	10.28
M/E L.O. SETTL. TK.	17.2	98.0	8.81	7.17	9.29	19.67
M/E L.O. STOR. TK.	26.4	98.0	7.12	6.90	9.42	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	101.1		12.73	4.87	7.77	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	0.6	10.0	18.98	-1.96	0.68	1.69
BILGE TK.	1.6	10.0	11.66	0.00	0.32	9.85
F.O. DRAIN TK.	0.6	10.0	20.77	2.04	0.56	3.33
F.O. OVERFLOW TK.	0.9	10.0	23.05	2.35	0.50	7.97
L.O. DRAIN TK.	0.4	10.0	18.43	1.94	0.72	0.91
SEWAGE STOR. TK.	1.4	10.0	22.44	-2.26	0.52	10.07
SLUDGE TK.	1.5	10.0	21.80	-8.40	7.44	9.03
SUBTOTAL	7.0		19.24	-1.81	1.99	42.86
TOTAL	2398.3		29.12	-0.06	11.60	642.92

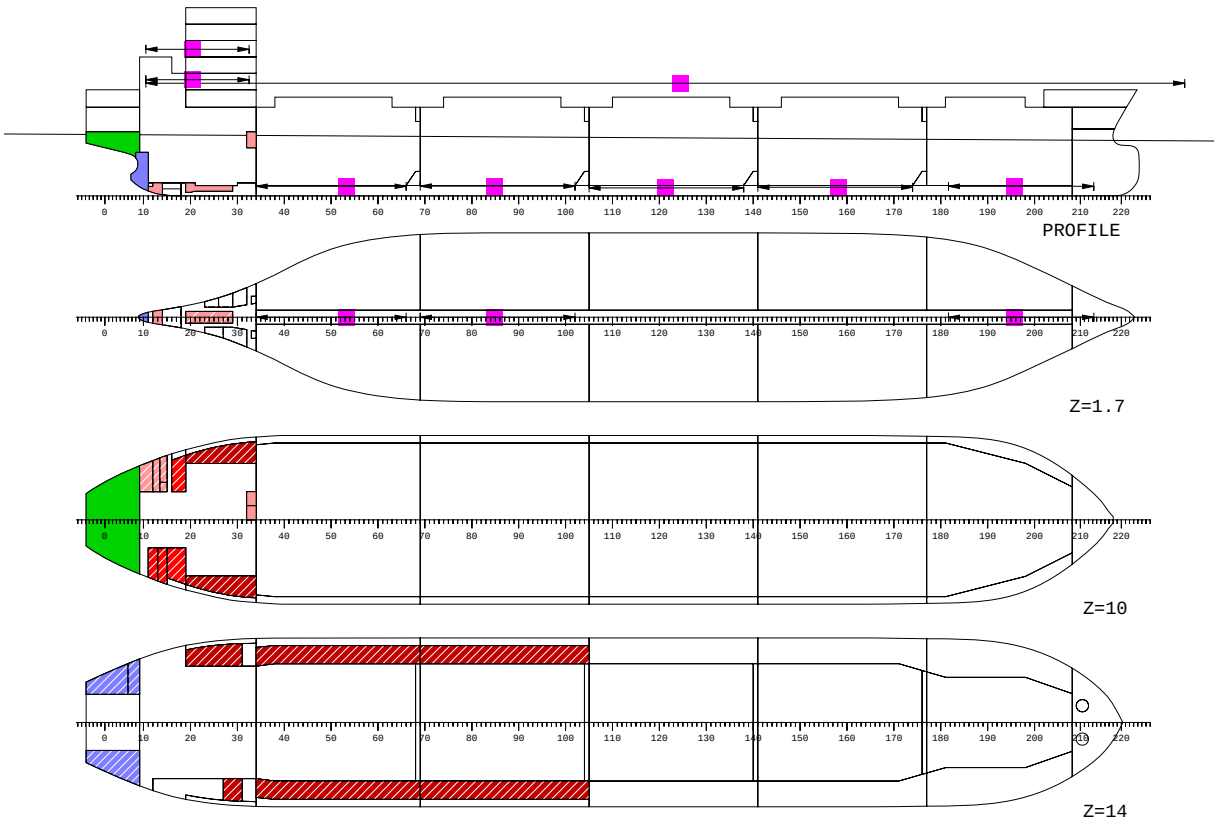


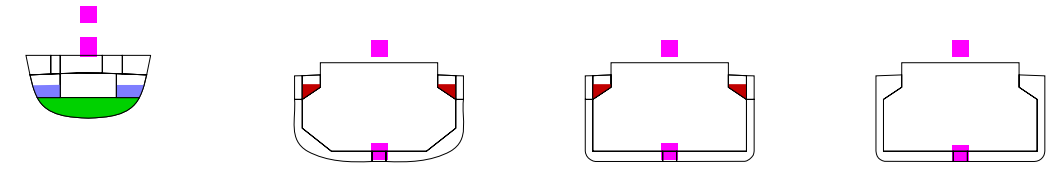
HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.003	-0.00	0.000	0.0	0.000
10.0	0.013	1.48	0.129	111.6	0.003
12.0	0.026	1.79	0.187	133.7	0.003
20.0	0.092	2.99	0.520	219.9	0.005
30.0	-0.079	4.15	1.148	321.5	0.007
40.0	-0.309	5.13	1.962	413.3	0.010
50.0	-0.838	5.65	2.910	492.5	0.011
60.0	-1.572	5.76	3.910	556.8	0.013
70.0	-2.365	5.59	4.904	604.2	0.014

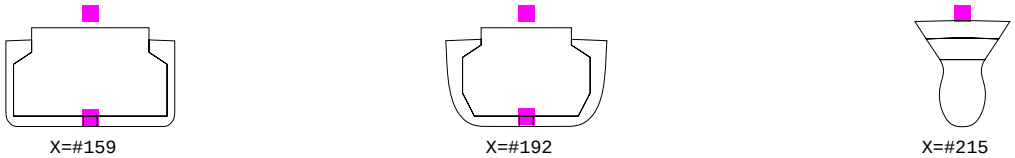


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	175	126	83.32	33.19
5.94	9.00	2012	542	111.27	69.45
18.60	25.00	7881	294	138.28	177.53
25.80	34.00	8899	-3	138.80	233.95
40.20	52.00	1775	-744	294.57	292.96
53.80	69.00	-5342	-605	141.82	305.94
68.20	87.00	-7782	127	332.80	304.61
82.60	105.00	-4885	-141	155.50	300.45
97.00	123.00	-4849	65	301.22	296.04
111.40	141.00	-4320	-373	133.42	291.63
125.80	159.00	-5674	231	346.43	286.72
140.20	177.00	4598	759	125.51	274.87
152.20	192.00	4598	-270	213.95	237.50
165.00	208.00	531	-207	164.62	119.41
169.20	214.00	221	35	38.08	72.32
173.40	220.00	148	-39	27.85	29.28

LOADING CASE 'LD08B -- STEEL COIL CONDITION, HALFWAY







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	339.3	2.28	0.00	9.53	0.00
Diesel Oil	DO	62.0	11.26	-2.98	8.46	149.80
Fresh Water	FW	159.2	2.90	-0.00	10.95	159.36
Solid cargo	GRAIN	0.0	0.00	0.00	0.00	0.00
Heavy Fuel Oil	HFO	783.5	42.77	-0.15	11.35	233.92
Lubricating Oil	LO	76.6	14.10	4.02	6.96	56.99
	MASS	32014.6	94.27	0.00	1.55	0.00
Miscellaneous	MIS	27.9	19.18	-1.81	2.34	42.86
Deadweight		33463.1	91.30	-0.00	1.93	642.92
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		33463.1	91.30	-0.00	1.93	
Total weight		42113.1	88.84	-0.00	3.63	

F L O A T I N G P O S I T I O N

Draught moulded	9.961	m	KM	12.34 m
Trim	-1.001	m	KG	3.63 m
Heel, PS=+	0.0	deg		
TA	10.461	m	GM0	8.71 m
TF	9.460	m	GMCORR	-0.02 m
Trimming moment	-60626	tonm	GM	8.70 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	1.188	mrad	OK
AREA40	Area under GZ curve .	0.090	2.034	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.846	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	5.998	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	58.524	deg	OK
GM0.15	GM > 0.15 m	0.150	8.697	m	OK
IMOWEATH.	IMO weather criterion	1.000	4.677		OK
VISIBILI.		359.760	131.854		OK
PROPIMMER	Propeller should imm.	50.000	137.383	%	OK

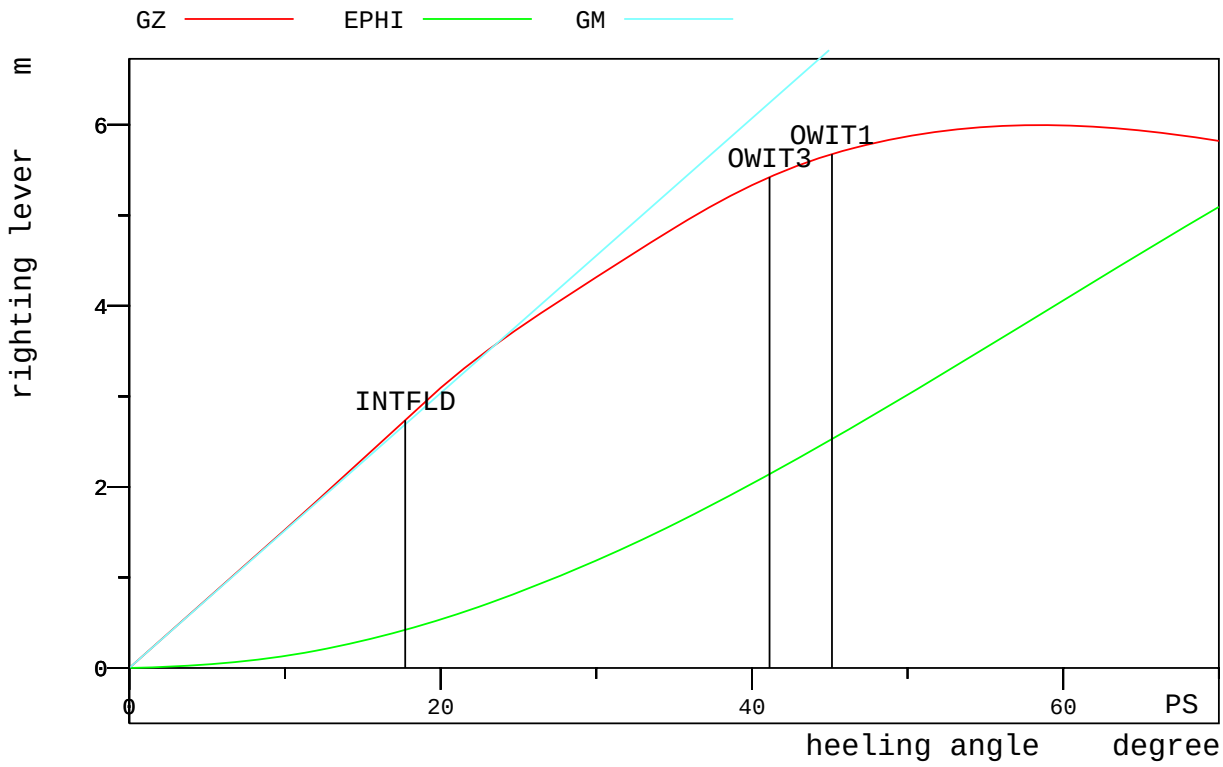
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN,CORR)	-808.8 t	(13.5%)	POSITION:	39.0 m	51
SHEAR FORCE (MAX,CORR)	1278.7 t	(21.3%)		137.8 m	174
SAGGING MOMENT	-11607.7 tm	(12.0%)		64.6 m	83
HOGGING MOMENT	7613.1 tm	(14.4%)		23.4 m	31

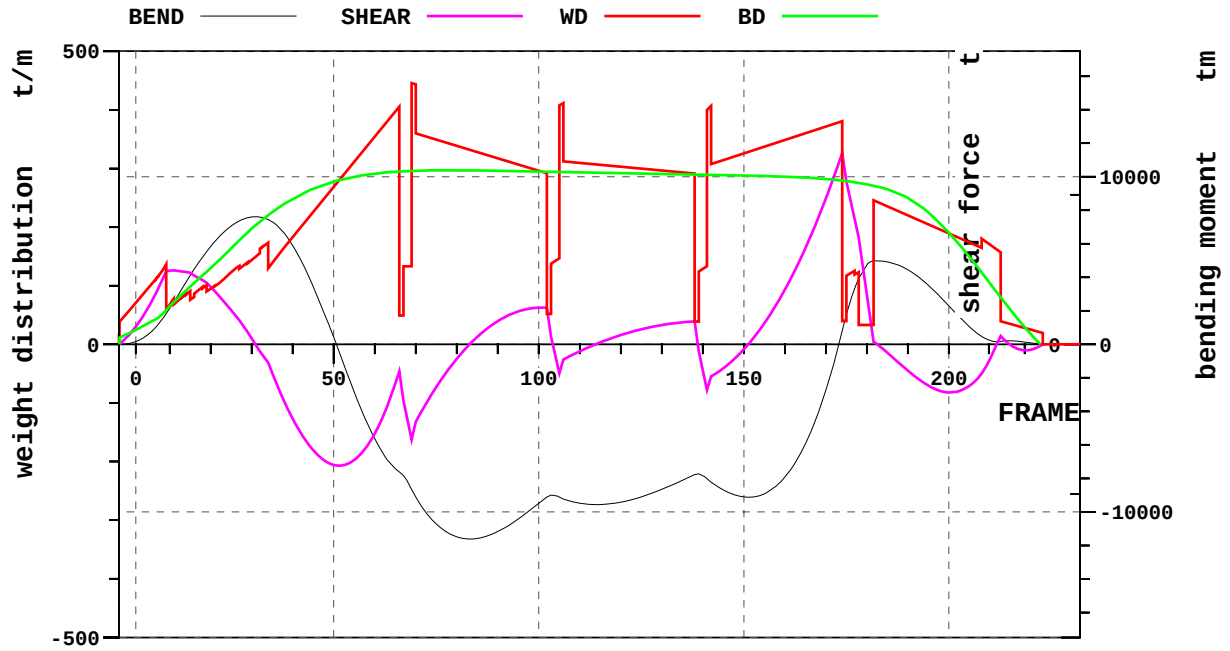
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Steel Coil						
NO.1 CARGO HOLD	4100.0	0.0	155.11	0.00	1.62	0.0
NO.2 CARGO HOLD	8000.0	0.0	125.15	0.00	1.47	0.0
NO.3 CARGO HOLD	6900.0	0.0	95.65	0.00	1.33	0.0
NO.4 CARGO HOLD	7300.0	0.0	66.42	0.00	1.65	0.0
NO.5 CARGO HOLD	5680.0	0.0	41.19	0.00	1.64	0.0
SUBTOTAL	31980.0		94.31	0.00	1.53	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	9.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	34.6		63.09	0.00	20.29	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	339.3	100.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	0.00	0.00	0.00	0.00
NO.5 WATER BALLAST T.	0.0	0.0	0.00	0.00	0.00	0.00
SUBTOTAL	339.3		2.28	0.00	9.53	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	10.3	50.0	8.24	-6.66	8.53	22.22
M.D.O. SETTL. TK.	12.4	50.0	9.83	-6.86	8.34	29.35
M.D.O. STOR. TK. (P)	17.3	50.0	12.68	6.80	8.45	43.37
M.D.O. STOR. TK. (S)	22.0	50.0	12.35	-6.76	8.50	54.86
SUBTOTAL	62.0		11.26	-2.98	8.46	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	20.4	50.0	4.98	7.32	11.97	31.97
FRESH WATER TK. (P)	46.9	50.0	1.02	6.48	12.05	44.74
FRESH WATER TK. (S)	67.2	50.0	2.26	-6.75	12.02	82.65
SUBTOTAL	159.2		2.90	-0.00	10.95	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	131.9	50.0	68.20	11.90	12.29	68.67
NO.1 FUEL OIL TK. (S)	131.9	50.0	68.20	-11.90	12.29	68.67
NO.2 FUEL OIL TK. (P)	127.7	50.0	39.94	11.88	12.30	0.00
NO.2 FUEL OIL TK. (S)	127.7	50.0	39.94	-11.88	12.30	0.00
H.F.O. TK. (P)	126.6	50.0	21.26	10.98	9.26	44.35
H.F.O. TK. (S)	90.0	50.0	21.60	-10.89	8.63	33.83
NO.1 H.F.O. SERV. TK.	12.4	50.0	15.47	-10.92	11.69	7.35
NO.1 H.F.O. SETTL. T.	14.8	50.0	18.64	-11.21	11.66	11.05
NO.2 H.F.O. SERV. TK.	9.7	50.0	12.65	-10.98	11.70	0.00
NO.2 H.F.O. SETTL. T.	11.0	50.0	9.65	-10.54	11.75	0.00
SUBTOTAL	783.5		42.77	-0.15	11.35	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	4.1	60.0	10.01	5.58	8.05	0.36
G/E L.O. STOR. TK.	7.9	60.0	10.02	8.01	8.78	10.28
M/E L.O. SETTL. TK.	10.5	60.0	8.82	6.85	8.65	19.67
M/E L.O. STOR. TK.	16.1	60.0	7.14	6.62	8.83	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	76.6		14.10	4.02	6.96	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	2.3	40.0	18.84	-2.15	1.07	1.69
BILGE TK.	6.4	40.0	11.55	0.00	0.73	9.85
F.O. DRAIN TK.	2.6	40.0	20.71	2.30	0.98	3.33
F.O. OVERFLOW TK.	3.6	40.0	23.05	2.70	0.93	7.97
L.O. DRAIN TK.	1.5	40.0	18.35	2.11	1.09	0.91
SEWAGE STOR. TK.	5.4	40.0	22.38	-2.58	0.94	10.07
SLUDGE TK.	6.0	40.0	21.80	-8.40	7.56	9.03
SUBTOTAL	27.9		19.18	-1.81	2.34	42.86
TOTAL	1448.5		25.59	-0.03	10.35	642.92

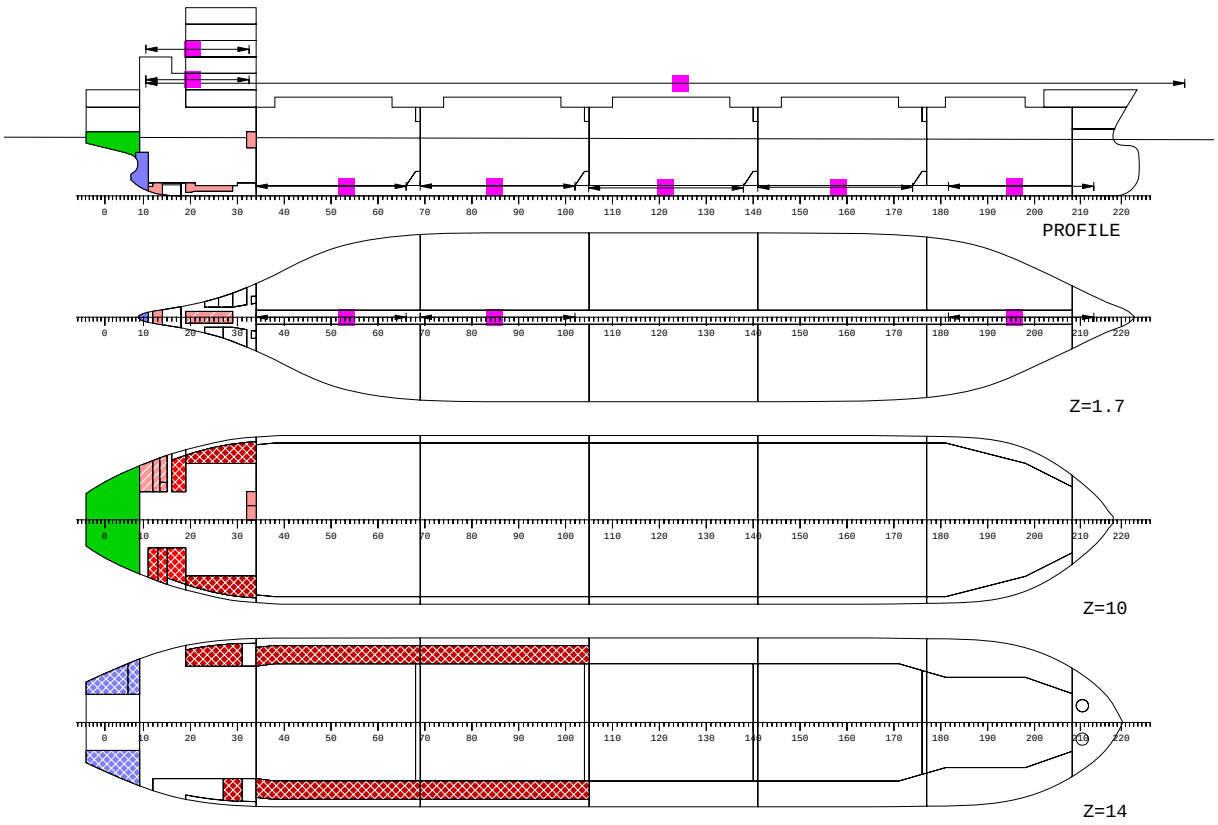


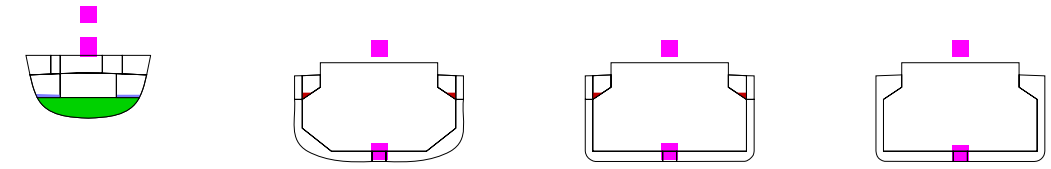
HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	0.000	0.00	0.000	0.0	0.000
10.0	0.017	1.53	0.133	111.6	0.003
12.0	0.030	1.84	0.192	133.7	0.003
20.0	0.119	3.09	0.536	219.9	0.005
30.0	-0.032	4.32	1.188	321.5	0.008
40.0	-0.256	5.33	2.034	413.3	0.010
50.0	-0.790	5.87	3.020	492.5	0.012
60.0	-1.538	5.99	4.060	556.8	0.013
70.0	-2.350	5.82	5.095	604.2	0.014

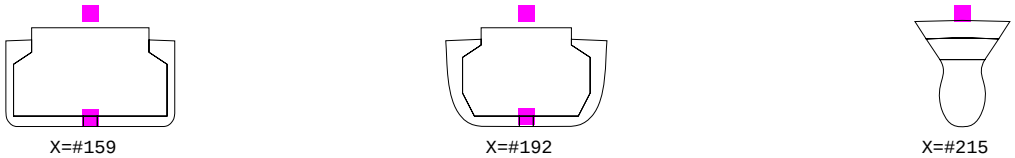


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	169	118	70.70	24.73
5.94	9.00	1857	490	98.33	58.41
18.60	25.00	7064	229	122.66	163.93
25.80	34.00	7448	-122	129.83	220.42
40.20	52.00	-927	-807	285.73	281.14
53.80	69.00	-8646	-637	445.86	295.95
68.20	87.00	-11470	97	323.89	296.54
82.60	105.00	-9116	-193	146.97	294.31
97.00	123.00	-9247	86	301.22	291.83
111.40	141.00	-8017	-305	133.42	289.35
125.80	159.00	-8198	318	346.43	286.38
140.20	177.00	3320	838	125.51	276.44
152.20	192.00	4126	-219	213.95	240.47
165.00	208.00	470	-196	164.62	121.92
169.20	214.00	195	39	38.08	73.79
173.40	220.00	132	-39	27.85	29.28

LOADING CASE 'LD08C -- STEEL COIL CONDITION, ARRIVAL







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	339.3	2.28	0.00	9.53	0.00
Diesel Oil	DO	12.4	11.39	-2.61	7.47	149.80
Fresh Water	FW	51.6	4.32	-0.00	8.25	159.36
Solid cargo	GRAIN	0.0	0.00	0.00	0.00	0.00
Heavy Fuel Oil	HFO	156.7	43.35	-0.16	10.12	233.92
Lubricating Oil	LO	50.8	16.95	2.45	5.88	56.99
	MASS	32007.6	94.29	0.00	1.55	0.00
Miscellaneous	MIS	55.8	19.16	-1.81	2.67	42.86
Deadweight		32674.2	92.67	-0.00	1.69	642.92
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		32674.2	92.67	-0.00	1.69	
Total weight		41324.2	89.87	-0.00	3.47	

F L O A T I N G P O S I T I O N

Draught moulded	9.801	m	KM	12.35 m
Trim	-0.355	m	KG	3.47 m
Heel, PS=+	0.0	deg		
TA	9.979	m	GM0	8.88 m
TF	9.623	m	GMCORR	-0.02 m
Trimming moment	-21324	tonm	GM	8.86 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	1.216	mrad	OK
AREA40	Area under GZ curve .	0.090	2.085	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.870	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	6.169	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	58.617	deg	OK
GM0.15	GM > 0.15 m	0.150	8.864	m	OK
IMOWEATH.	IMO weather criterion	1.000	5.267		OK
VISIBILI.		359.760	123.078		OK
PROPIMMER	Propeller should imm.	50.000	128.275	%	OK

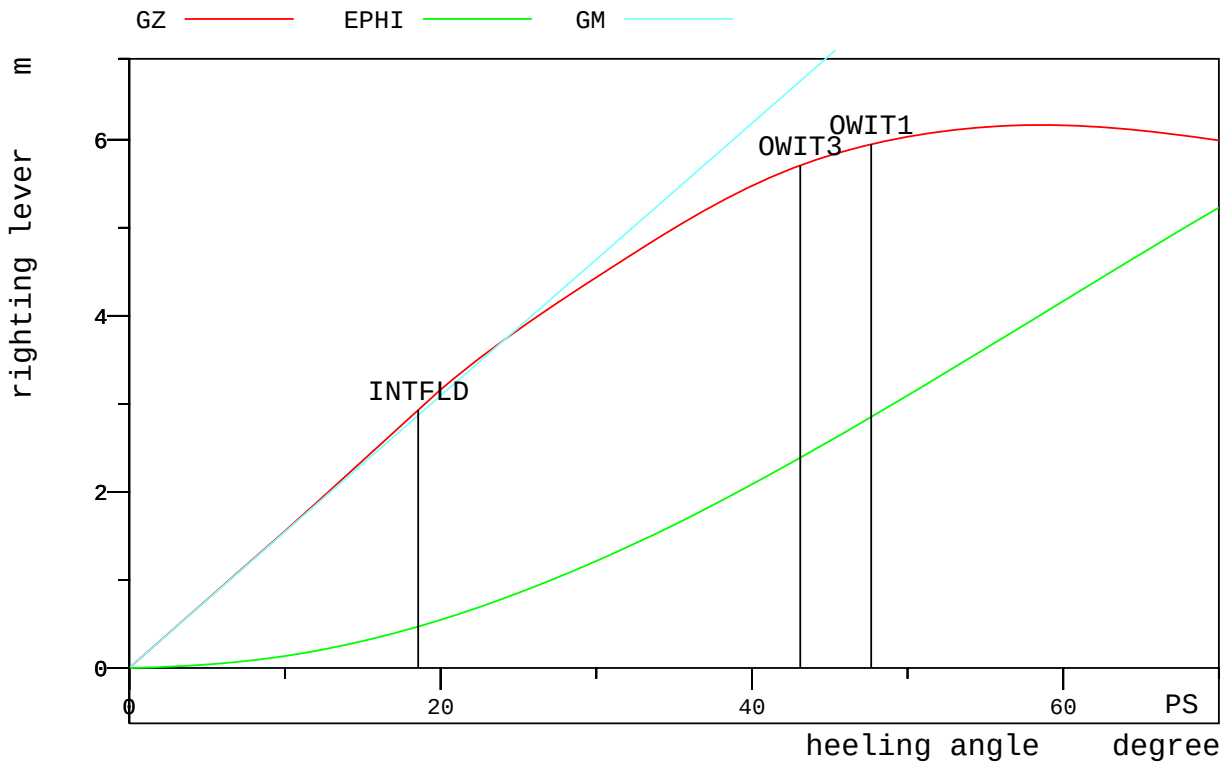
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN,CORR)	-864.0 t	(14.4%)	POSITION:	39.0 m	51
SHEAR FORCE (MAX,CORR)	1346.3 t	(22.4%)		137.8 m	174
SAGGING MOMENT	-14632.1 tm	(15.1%)		66.4 m	85
HOGGING MOMENT	6644.9 tm	(13.3%)		21.8 m	29

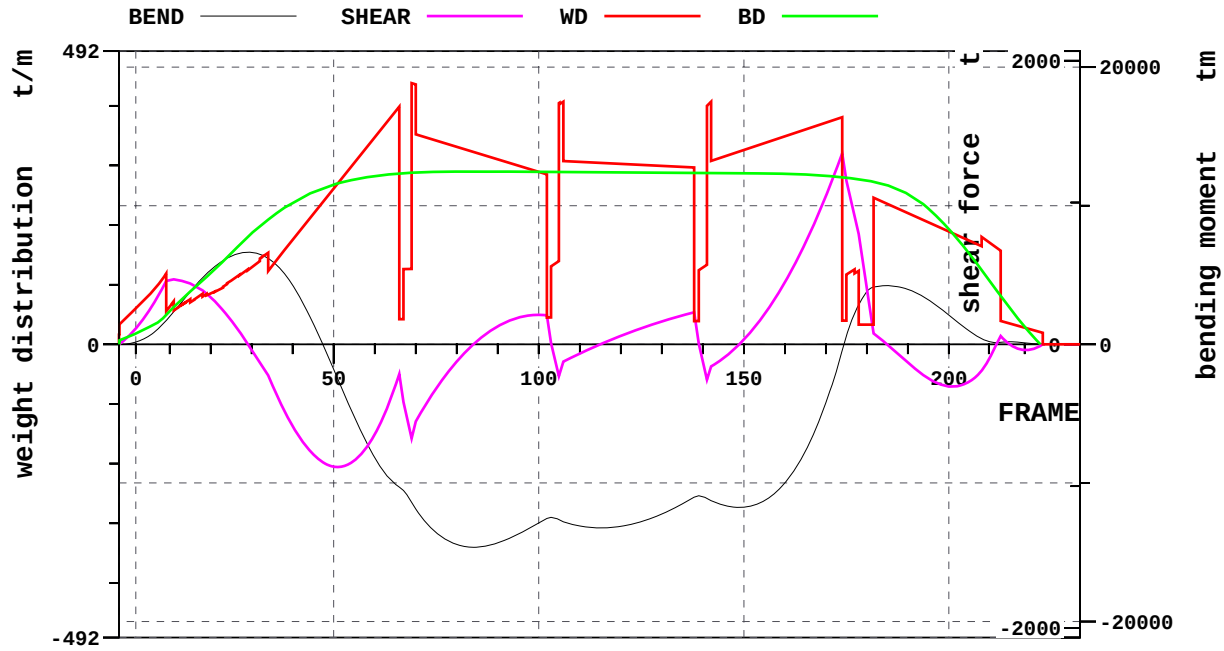
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Steel Coil						
NO.1 CARGO HOLD	4100.0	0.0	155.11	0.00	1.62	0.0
NO.2 CARGO HOLD	8000.0	0.0	125.15	0.00	1.47	0.0
NO.3 CARGO HOLD	6900.0	0.0	95.65	0.00	1.33	0.0
NO.4 CARGO HOLD	7300.0	0.0	66.42	0.00	1.65	0.0
NO.5 CARGO HOLD	5680.0	0.0	41.19	0.00	1.64	0.0
SUBTOTAL	31980.0		94.31	0.00	1.53	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	2.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	27.6		75.29	0.00	20.42	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	339.3	100.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	0.00	0.00	0.00	0.00
NO.5 W.B.T.(P)	0.0	0.0	0.00	0.00	0.00	0.00
NO.5 WATER BALLAST T.	0.0	0.0	0.00	0.00	0.00	0.00
SUBTOTAL	339.3		2.28	0.00	9.53	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	2.1	10.0	8.28	-5.84	7.59	22.22
M.D.O. SETTL. TK.	2.5	10.0	9.88	-5.95	7.33	29.35
M.D.O. STOR. TK. (P)	3.5	10.0	12.81	5.91	7.45	43.37
M.D.O. STOR. TK. (S)	4.4	10.0	12.58	-5.91	7.52	54.86
SUBTOTAL	12.4		11.39	-2.61	7.47	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	4.1	10.0	4.98	7.15	11.13	31.97
FRESH WATER TK. (P)	9.4	10.0	1.14	6.32	11.15	44.74
FRESH WATER TK. (S)	13.4	10.0	2.40	-6.59	11.14	82.65
SUBTOTAL	51.6		4.32	-0.00	8.25	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	26.4	10.0	68.20	12.54	11.35	68.67
NO.1 FUEL OIL TK. (S)	26.4	10.0	68.20	-12.54	11.35	68.67
NO.2 FUEL OIL TK. (P)	25.5	10.0	40.16	12.52	11.37	0.00
NO.2 FUEL OIL TK. (S)	25.5	10.0	40.16	-12.52	11.37	0.00
H.F.O. TK. (P)	25.3	10.0	23.07	10.63	7.07	44.35
H.F.O. TK. (S)	18.0	10.0	23.44	-10.57	6.80	33.83
NO.1 H.F.O. SERV. TK.	2.5	10.0	15.49	-10.82	11.07	7.35
NO.1 H.F.O. SETTL. T.	3.0	10.0	18.65	-11.14	11.06	11.05
NO.2 H.F.O. SERV. TK.	1.9	10.0	12.65	-10.87	11.07	0.00
NO.2 H.F.O. SETTL. T.	2.2	10.0	9.70	-10.42	11.09	0.00
SUBTOTAL	156.7		43.35	-0.16	10.12	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	1.4	20.0	10.03	5.53	7.22	0.36
G/E L.O. STOR. TK.	2.6	20.0	10.04	7.48	7.96	10.28
M/E L.O. SETTL. TK.	3.5	20.0	8.83	6.21	7.80	19.67
M/E L.O. STOR. TK.	5.4	20.0	7.19	6.05	8.05	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	50.8		16.95	2.45	5.88	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	4.5	80.0	18.81	-2.29	1.40	1.69
BILGE TK.	12.9	80.0	11.52	0.00	1.14	9.85
F.O. DRAIN TK.	5.2	80.0	20.70	2.51	1.34	3.33
F.O. OVERFLOW TK.	7.2	80.0	23.04	2.98	1.31	7.97
L.O. DRAIN TK.	3.1	80.0	18.32	2.24	1.42	0.91
SEWAGE STOR. TK.	10.9	80.0	22.37	-2.83	1.31	10.07
SLUDGE TK.	12.0	80.0	21.80	-8.40	7.72	9.03
SUBTOTAL	55.8		19.16	-1.81	2.67	42.86
TOTAL	666.6		14.80	-0.05	8.68	642.92

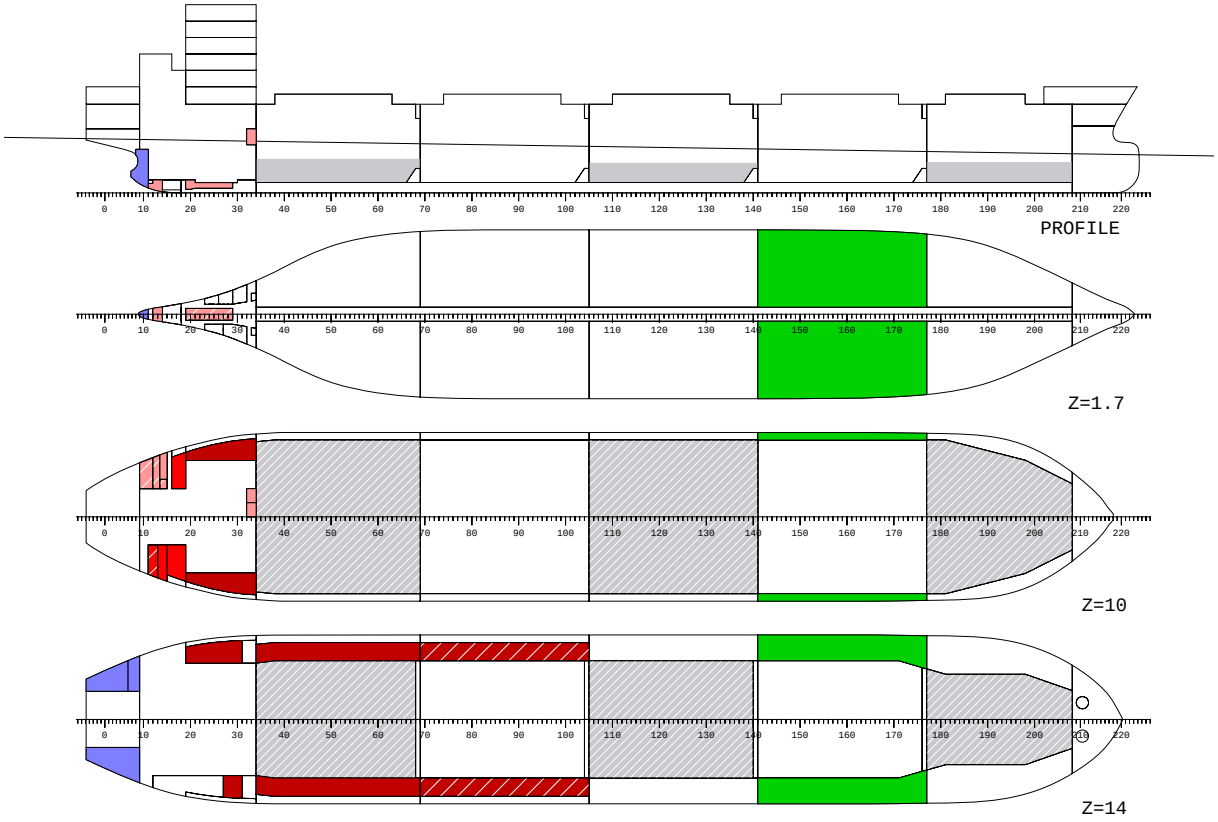


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	0.000	0.00	0.000	0.0	0.000
10.0	0.017	1.56	0.136	111.6	0.003
12.0	0.030	1.87	0.196	133.7	0.003
20.0	0.131	3.16	0.547	219.9	0.005
30.0	0.007	4.44	1.216	321.5	0.008
40.0	-0.219	5.48	2.085	413.3	0.010
50.0	-0.755	6.04	3.098	492.5	0.012
60.0	-1.511	6.17	4.168	556.8	0.013
70.0	-2.336	5.99	5.232	604.2	0.015

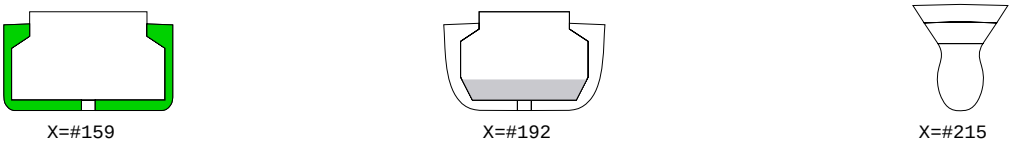


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	168	113	60.76	18.12
5.94	9.00	1741	448	87.32	49.55
18.60	25.00	6365	167	106.89	152.77
25.80	34.00	6209	-219	138.29	209.22
40.20	52.00	-3188	-859	278.40	271.33
53.80	69.00	-11398	-663	126.39	287.65
68.20	87.00	-14555	71	316.56	289.85
82.60	105.00	-12664	-215	405.13	289.22
97.00	123.00	-12944	93	301.66	288.34
111.40	141.00	-11119	-249	133.42	287.46
125.80	159.00	-10323	390	346.43	286.08
140.20	177.00	2234	902	125.51	277.73
152.20	192.00	3716	-177	213.95	242.93
165.00	208.00	407	-186	164.62	124.01
169.20	214.00	164	41	38.08	75.02
173.40	220.00	112	-39	27.85	29.28

LOADING CASE 'LD09A -- MULTI-PORT HEAVY CARGO LOADING, DEPARTURE







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	2789.9	125.72	0.00	5.87	10748.87
Diesel Oil	DO	121.6	11.23	-3.17	9.29	149.80
Fresh Water	FW	293.6	2.53	0.00	12.34	159.36
Grain Load	GRAIN	18870.7	91.10	0.00	3.67	0.00
Heavy Fuel Oil	HFO	1535.7	42.68	-0.15	12.40	233.92
Lubricating Oil	LO	101.1	12.73	4.87	7.77	56.99
	MASS	43.6	53.17	0.00	20.19	0.00
Miscellaneous	MIS	7.0	19.24	-1.81	1.99	42.86
Deadweight		23763.2	90.11	-0.01	4.67	11391.80
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		23763.2	90.11	-0.01	4.67	
Total weight		32413.2	87.22	-0.00	6.15	

F L O A T I N G P O S I T I O N

Draught moulded	7.881	m	KM	13.15 m
Trim	-2.647	m	KG	6.15 m
Heel, PS=+	-0.0	deg		
TA	9.204	m	GM0	7.01 m
TF	6.557	m	GMCORR	-0.35 m
Trimming moment	-153589	tonm	GM	6.66 m
Heel	-0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.953	mrad	OK
AREA40	Area under GZ curve .	0.090	1.662	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.709	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.376	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	48.556	deg	OK
GM0.15	GM > 0.15 m	0.150	6.656	m	OK
IMOWEATH.	IMO weather criterion	1.000	5.514		OK
VISIBILI.		359.760	197.879		OK
PROPIMMER	Propeller should imm.	50.000	113.667	%	OK

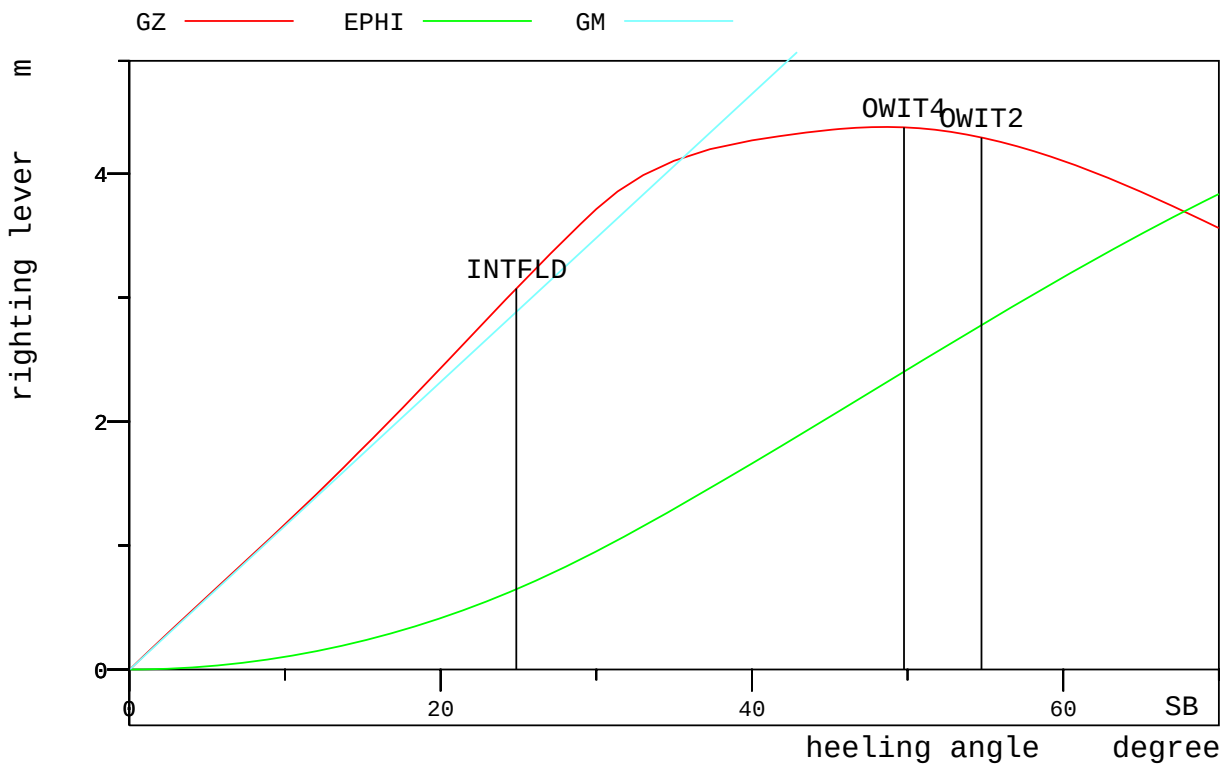
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN,CORR)	-2089.6 t	(34.8%)	POSITION:	82.6 m	105
SHEAR FORCE (MAX,CORR)	2886.9 t	(48.1%)		53.8 m	69
SAGGING MOMENT	-33.5 tm	(1.0%)		176.9 m	225
HOGGING MOMENT	86418.9 tm	(79.2%)		70.0 m	89

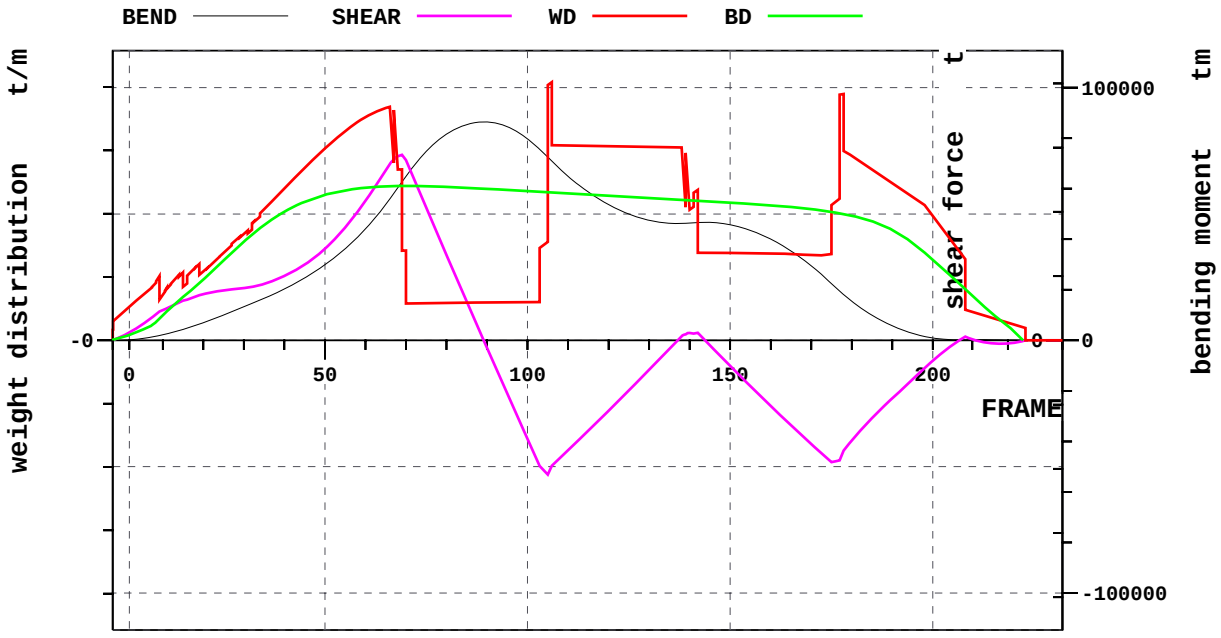
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 3.000)						
NO.1 CARGO HOLD	4899.8	24.6	150.87	0.00	3.58	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	7345.9	24.6	96.45	0.00	3.47	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	6625.0	24.5	40.97	0.00	3.95	0.0
SUBTOTAL	18870.7		91.10	0.00	3.67	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	18.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	43.6		53.17	0.00	20.19	0.0
TOTAL	18914.3		91.01	0.00	3.71	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	0.00	0.00	0.00	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	10.47	5.87	5374.44
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	-10.47	5.87	5374.44
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	2789.9		125.72	0.00	5.87	10748.87
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	20.2	98.0	8.23	-7.08	9.34	22.22
M.D.O. SETTL. TK.	24.2	98.0	9.82	-7.32	9.21	29.35
M.D.O. STOR. TK. (P)	33.9	98.0	12.66	7.26	9.29	43.37
M.D.O. STOR. TK. (S)	43.2	98.0	12.30	-7.22	9.32	54.86
SUBTOTAL	121.6		11.23	-3.17	9.29	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	40.7	100.0	4.97	7.43	12.96	31.97
FRESH WATER TK. (P)	93.7	100.0	0.94	6.60	13.03	44.74
FRESH WATER TK. (S)	134.4	100.0	2.17	-6.85	13.01	82.65
SUBTOTAL	293.6		2.53	0.00	12.34	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	258.4	98.0	68.20	11.73	13.08	68.67
NO.1 FUEL OIL TK. (S)	258.4	98.0	68.20	-11.73	13.08	68.67
NO.2 FUEL OIL TK. (P)	250.2	98.0	39.86	11.71	13.09	0.00
NO.2 FUEL OIL TK. (S)	250.2	98.0	39.86	-11.71	13.09	0.00
H.F.O. TK. (P)	248.1	98.0	20.60	11.15	11.04	44.35
H.F.O. TK. (S)	176.3	98.0	21.94	-11.14	10.32	33.83
NO.1 H.F.O. SERV. TK.	24.3	98.0	15.46	-11.00	12.38	7.35
NO.1 H.F.O. SETTL. T.	29.0	98.0	18.63	-11.27	12.36	11.05
NO.2 H.F.O. SERV. TK.	19.1	98.0	12.64	-11.07	12.39	0.00
NO.2 H.F.O. SETTL. T.	21.5	98.0	9.62	-10.64	12.44	0.00
SUBTOTAL	1535.7		42.68	-0.15	12.40	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	6.8	98.0	10.01	5.59	8.82	0.36
G/E L.O. STOR. TK.	12.9	98.0	10.02	8.28	9.39	10.28
M/E L.O. SETTL. TK.	17.2	98.0	8.81	7.17	9.29	19.67
M/E L.O. STOR. TK.	26.4	98.0	7.12	6.90	9.42	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	101.1		12.73	4.87	7.77	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	0.6	10.0	18.98	-1.96	0.68	1.69
BILGE TK.	1.6	10.0	11.66	0.00	0.32	9.85
F.O. DRAIN TK.	0.6	10.0	20.77	2.04	0.56	3.33
F.O. OVERFLOW TK.	0.9	10.0	23.05	2.35	0.50	7.97
L.O. DRAIN TK.	0.4	10.0	18.43	1.94	0.72	0.91
SEWAGE STOR. TK.	1.4	10.0	22.44	-2.26	0.52	10.07
SLUDGE TK.	1.5	10.0	21.80	-8.40	7.44	9.03
SUBTOTAL	7.0		19.24	-1.81	1.99	42.86
TOTAL	4848.9		86.58	-0.03	8.45	11391.80

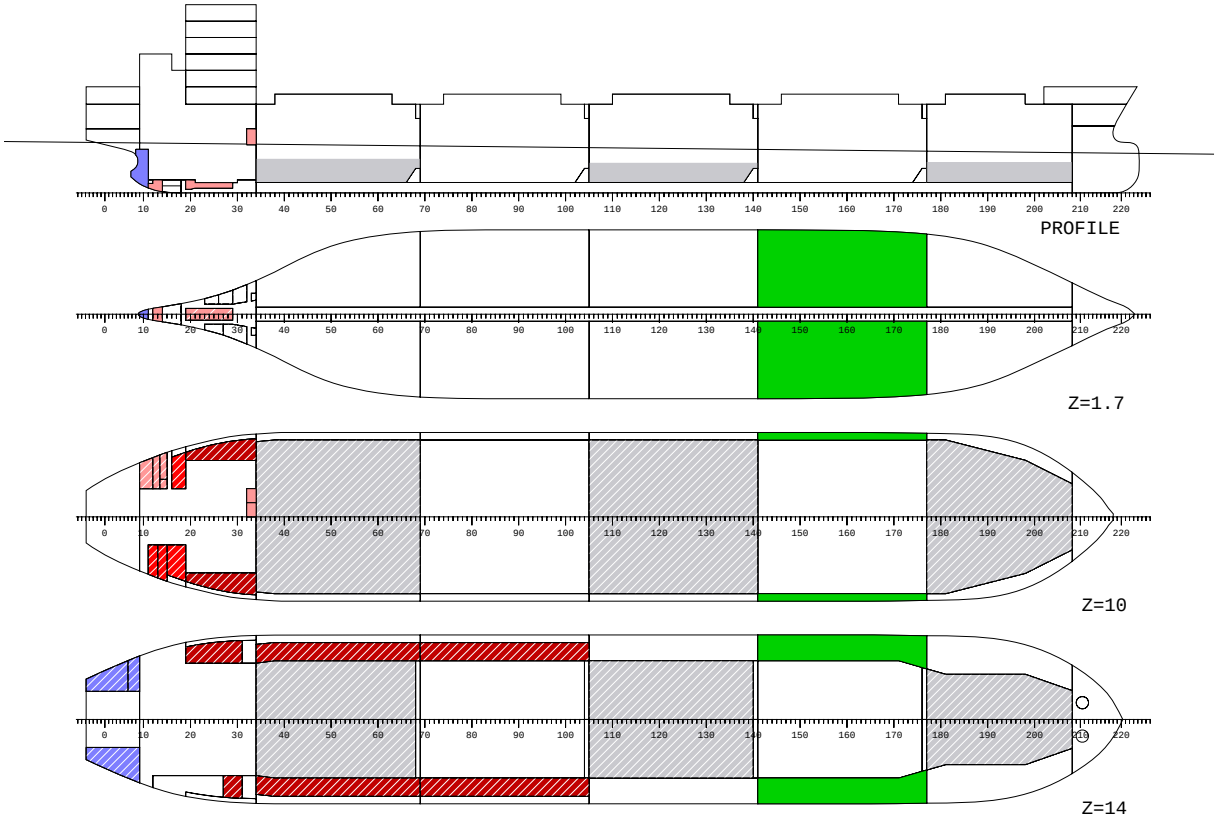


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.004	-0.00	0.000	0.0	0.000
10.0	0.015	1.17	0.102	1978.2	0.061
12.0	0.029	1.41	0.147	2368.5	0.073
20.0	0.153	2.43	0.415	3896.2	0.120
30.0	0.386	3.71	0.953	5695.9	0.176
40.0	-0.010	4.27	1.662	7322.5	0.226
50.0	-0.727	4.37	2.420	8726.6	0.269
60.0	-1.662	4.10	3.165	9865.6	0.304
70.0	-2.693	3.56	3.836	10704.8	0.330

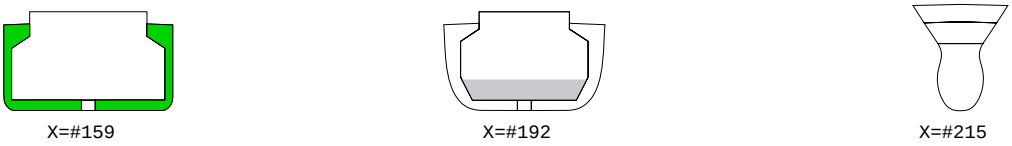


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	202	120	53.14	8.58
5.94	9.00	1855	448	64.06	34.04
18.60	25.00	10003	763	138.32	125.69
25.80	34.00	15861	859	200.41	177.03
40.20	52.00	32261	1547	313.67	232.60
53.80	69.00	62671	2887	269.81	243.56
68.20	87.00	86138	313	59.72	240.08
82.60	105.00	72369	-2090	155.25	233.78
97.00	123.00	51309	-907	306.19	227.22
111.40	141.00	46504	105	221.79	220.67
125.80	159.00	40832	-961	137.33	213.62
140.20	177.00	19502	-1869	223.55	200.49
152.20	192.00	4413	-783	239.47	166.46
165.00	208.00	200	56	128.15	80.71
169.20	214.00	233	-35	38.08	50.00
173.40	220.00	54	-48	27.85	22.73

LOADING CASE 'LD09B -- MULTI-PORT HEAVY CARGO LOADING, WBT2 FULL, FOLLOWED







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	2789.9	125.72	0.00	5.87	10748.87
Diesel Oil	DO	62.0	11.26	-2.98	8.46	149.80
Fresh Water	FW	159.2	2.90	-0.00	10.95	159.36
Grain Load	GRAIN	18870.7	91.10	0.00	3.67	0.00
Heavy Fuel Oil	HFO	783.5	42.77	-0.15	11.35	233.92
Lubricating Oil	LO	76.6	14.10	4.02	6.96	56.99
	MASS	34.6	63.09	0.00	20.29	0.00
Miscellaneous	MIS	27.9	19.18	-1.81	2.34	42.86
Deadweight		22804.4	92.45	-0.00	4.30	11391.80
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		22804.4	92.45	-0.00	4.30	
Total weight		31454.4	88.84	-0.00	5.92	

F L O A T I N G P O S I T I O N

Draught moulded	7.678	m	KM	13.21 m
Trim	-1.794	m	KG	5.92 m
Heel, PS=+	0.0	deg		
TA	8.576	m	GM0	7.29 m
TF	6.781	m	GMCORR	-0.36 m
Trimming moment	-100895	tonm	GM	6.93 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.994	mrad	OK
AREA40	Area under GZ curve .	0.090	1.735	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.742	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.574	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	48.770	deg	OK
GM0.15	GM > 0.15 m	0.150	6.927	m	OK
IMOWEATH.	IMO weather criterion	1.000	5.585		OK
VISIBILI.		359.760	179.997		OK
PROPIMMER	Propeller should imm.	50.000	101.804	%	OK

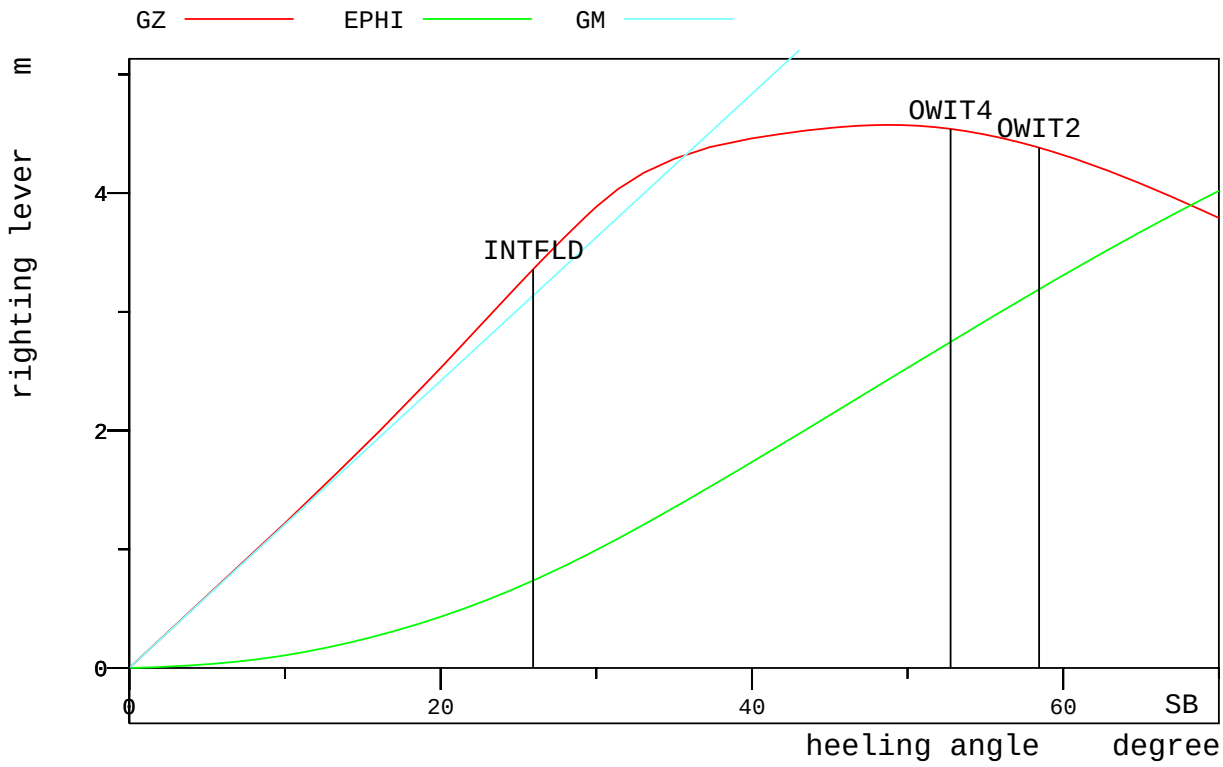
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN,CORR)	-2127.6 t	(35.5%)	POSITION:	82.6 m	105
SHEAR FORCE (MAX,CORR)	2854.9 t	(47.6%)		53.8 m	69
SAGGING MOMENT	-19.3 tm	(0.6%)		176.9 m	225
HOGGING MOMENT	81622.8 tm	(74.8%)		70.0 m	89

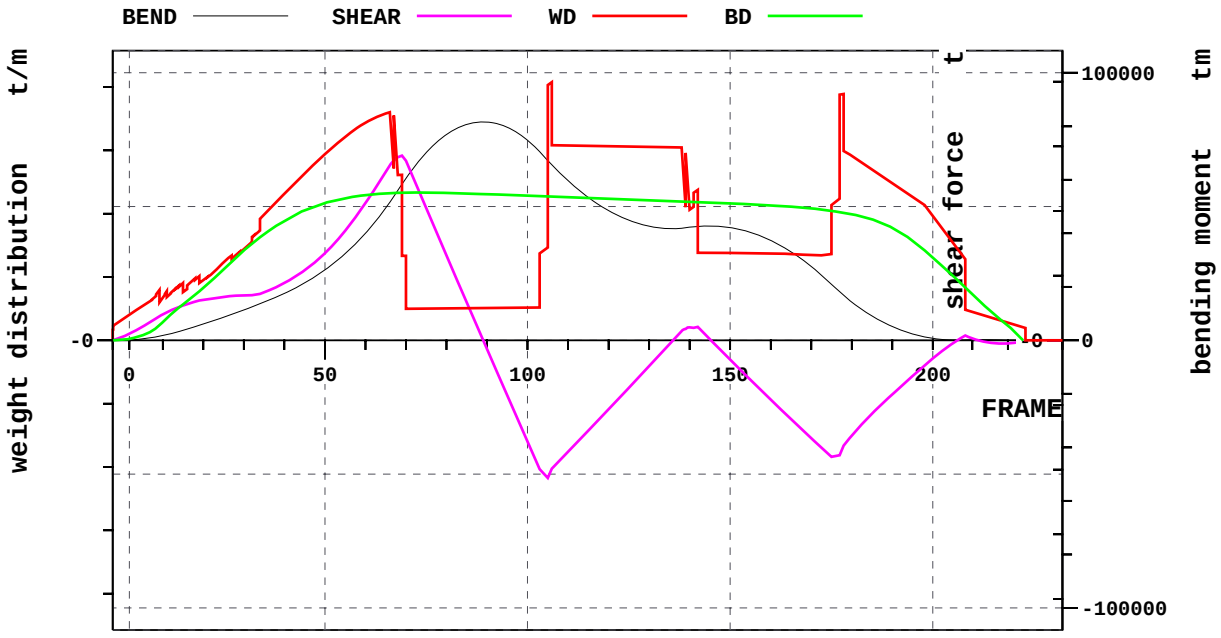
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 3.000)						
NO.1 CARGO HOLD	4899.8	24.6	150.87	0.00	3.58	0.0
NO.2 CARGO HOLD	0.0	0.0	125.57	0.00	8.67	0.0
NO.3 CARGO HOLD	7345.9	24.6	96.45	0.00	3.47	0.0
NO.4 CARGO HOLD	0.0	0.0	67.98	0.00	8.67	0.0
NO.5 CARGO HOLD	6625.0	24.5	40.97	0.00	3.95	0.0
SUBTOTAL	18870.7		91.10	0.00	3.67	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	9.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	34.6		63.09	0.00	20.29	0.0
TOTAL	18905.3		91.05	0.00	3.70	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	10.47	5.87	5374.44
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	-10.47	5.87	5374.44
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	2789.9		125.72	0.00	5.87	10748.87
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	10.3	50.0	8.24	-6.66	8.53	22.22
M.D.O. SETTL. TK.	12.4	50.0	9.83	-6.86	8.34	29.35
M.D.O. STOR. TK. (P)	17.3	50.0	12.68	6.80	8.45	43.37
M.D.O. STOR. TK. (S)	22.0	50.0	12.35	-6.76	8.50	54.86
SUBTOTAL	62.0		11.26	-2.98	8.46	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	20.4	50.0	4.98	7.32	11.97	31.97
FRESH WATER TK. (P)	46.9	50.0	1.02	6.48	12.05	44.74
FRESH WATER TK. (S)	67.2	50.0	2.26	-6.75	12.02	82.65
SUBTOTAL	159.2		2.90	-0.00	10.95	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	131.9	50.0	68.20	11.90	12.29	68.67
NO.1 FUEL OIL TK. (S)	131.9	50.0	68.20	-11.90	12.29	68.67
NO.2 FUEL OIL TK. (P)	127.7	50.0	39.94	11.88	12.30	0.00
NO.2 FUEL OIL TK. (S)	127.7	50.0	39.94	-11.88	12.30	0.00
H.F.O. TK. (P)	126.6	50.0	21.26	10.98	9.26	44.35
H.F.O. TK. (S)	90.0	50.0	21.60	-10.89	8.63	33.83
NO.1 H.F.O. SERV. TK.	12.4	50.0	15.47	-10.92	11.69	7.35
NO.1 H.F.O. SETTL. T.	14.8	50.0	18.64	-11.21	11.66	11.05
NO.2 H.F.O. SERV. TK.	9.7	50.0	12.65	-10.98	11.70	0.00
NO.2 H.F.O. SETTL. T.	11.0	50.0	9.65	-10.54	11.75	0.00
SUBTOTAL	783.5		42.77	-0.15	11.35	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	4.1	60.0	10.01	5.58	8.05	0.36
G/E L.O. STOR. TK.	7.9	60.0	10.02	8.01	8.78	10.28
M/E L.O. SETTL. TK.	10.5	60.0	8.82	6.85	8.65	19.67
M/E L.O. STOR. TK.	16.1	60.0	7.14	6.62	8.83	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	76.6		14.10	4.02	6.96	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	2.3	40.0	18.84	-2.15	1.07	1.69
BILGE TK.	6.4	40.0	11.55	0.00	0.73	9.85
F.O. DRAIN TK.	2.6	40.0	20.71	2.30	0.98	3.33
F.O. OVERFLOW TK.	3.6	40.0	23.05	2.70	0.93	7.97
L.O. DRAIN TK.	1.5	40.0	18.35	2.11	1.09	0.91
SEWAGE STOR. TK.	5.4	40.0	22.38	-2.58	0.94	10.07
SLUDGE TK.	6.0	40.0	21.80	-8.40	7.56	9.03
SUBTOTAL	27.9		19.18	-1.81	2.34	42.86
TOTAL	3899.1		99.26	-0.01	7.21	11391.80

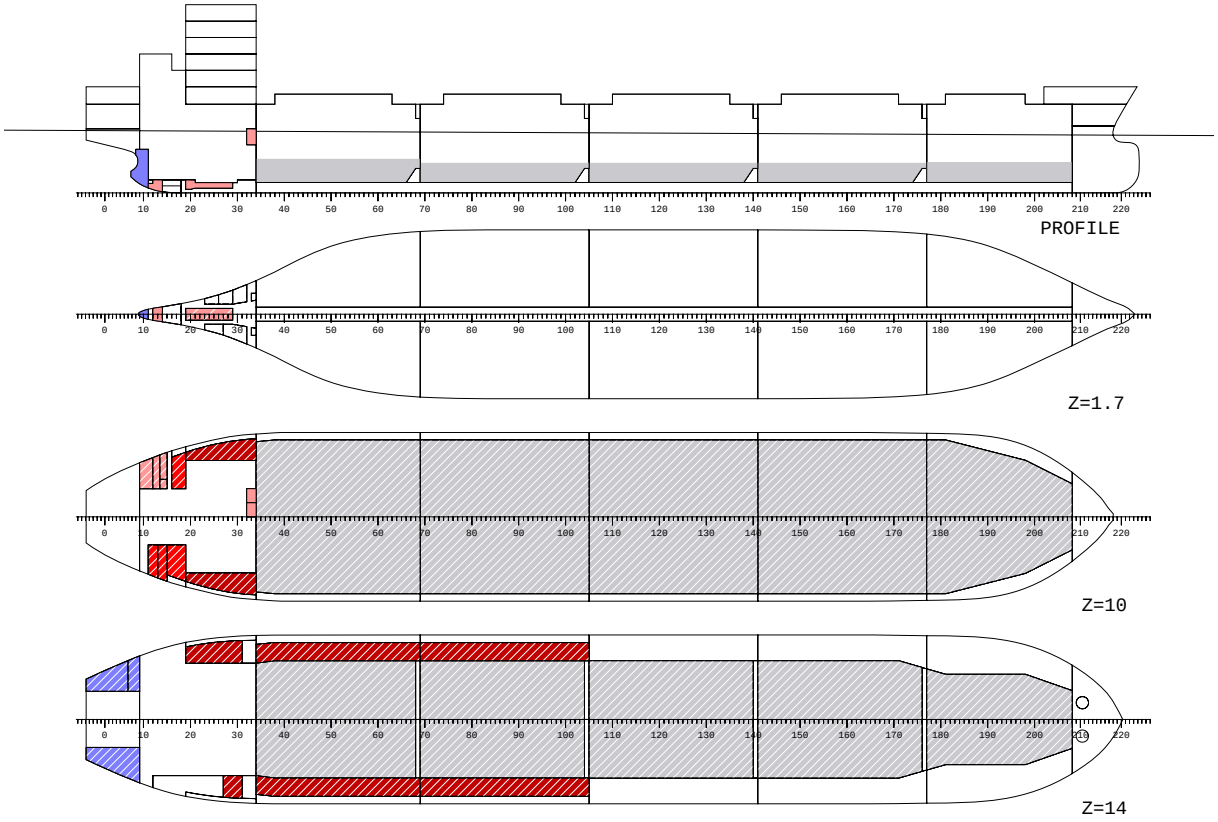


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.001	-0.00	0.000	0.0	0.000
10.0	0.018	1.22	0.107	1978.2	0.063
12.0	0.032	1.47	0.154	2368.5	0.075
20.0	0.159	2.53	0.432	3896.2	0.124
30.0	0.423	3.89	0.994	5695.9	0.181
40.0	0.009	4.46	1.735	7322.5	0.233
50.0	-0.735	4.57	2.527	8726.6	0.277
60.0	-1.678	4.32	3.309	9865.6	0.314
70.0	-2.718	3.79	4.019	10704.8	0.340

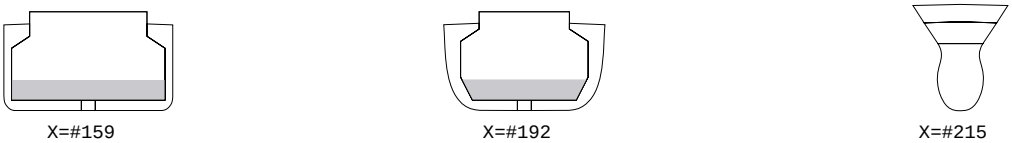


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	163	101	40.56	2.40
5.94	9.00	1545	375	59.90	24.12
18.60	25.00	8663	670	122.74	111.92
25.80	34.00	13678	717	191.83	162.78
40.20	52.00	28548	1475	304.71	219.90
53.80	69.00	58254	2855	261.20	232.86
68.20	87.00	81380	295	50.68	231.49
82.60	105.00	67206	-2128	146.57	227.30
97.00	123.00	46170	-867	306.19	222.85
111.40	141.00	42295	193	221.79	218.41
125.80	159.00	38022	-855	137.33	213.47
140.20	177.00	18119	-1777	223.55	202.44
152.20	192.00	3932	-724	239.47	169.85
165.00	208.00	170	73	128.15	83.37
169.20	214.00	244	-28	38.08	51.84
173.40	220.00	73	-47	27.85	23.61

LOADING CASE 'LD09C -- MULTI-PORT HEAVY CARGO LOADING, WBT2 EMPTY, FOLLOWED







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	0.0	0.00	0.00	0.00	10748.87
Diesel Oil	DO	62.0	11.26	-2.98	8.46	149.80
Fresh Water	FW	159.2	2.90	-0.00	10.95	159.36
Grain Load	GRAIN	33553.2	93.43	0.00	3.58	0.00
Heavy Fuel Oil	HFO	783.5	42.77	-0.15	11.35	233.92
Lubricating Oil	LO	76.6	14.10	4.02	6.96	56.99
	MASS	34.6	63.09	0.00	20.29	0.00
Miscellaneous	MIS	27.9	19.18	-1.81	2.34	42.86
Deadweight		34697.1	91.46	-0.00	3.82	11391.80
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		34697.1	91.46	-0.00	3.82	
Total weight		43347.1	89.03	-0.00	5.09	

F L O A T I N G P O S I T I O N

Draught moulded	10.224	m	KM	12.29 m
Trim	-0.763	m	KG	5.09 m
Heel, PS=+	0.0	deg		
TA	10.606	m	GM0	7.20 m
TF	9.843	m	GMCORR	-0.26 m
Trimming moment	-46123	tonm	GM	6.94 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.944	mrad	OK
AREA40	Area under GZ curve .	0.090	1.601	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.657	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.485	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	53.899	deg	OK
GM0.15	GM > 0.15 m	0.150	6.937	m	OK
IMOWEATH.	IMO weather criterion	1.000	3.620		OK
VISIBILI.		359.760	124.197		OK
PROPIMMER	Propeller should imm.	50.000	140.114	%	OK

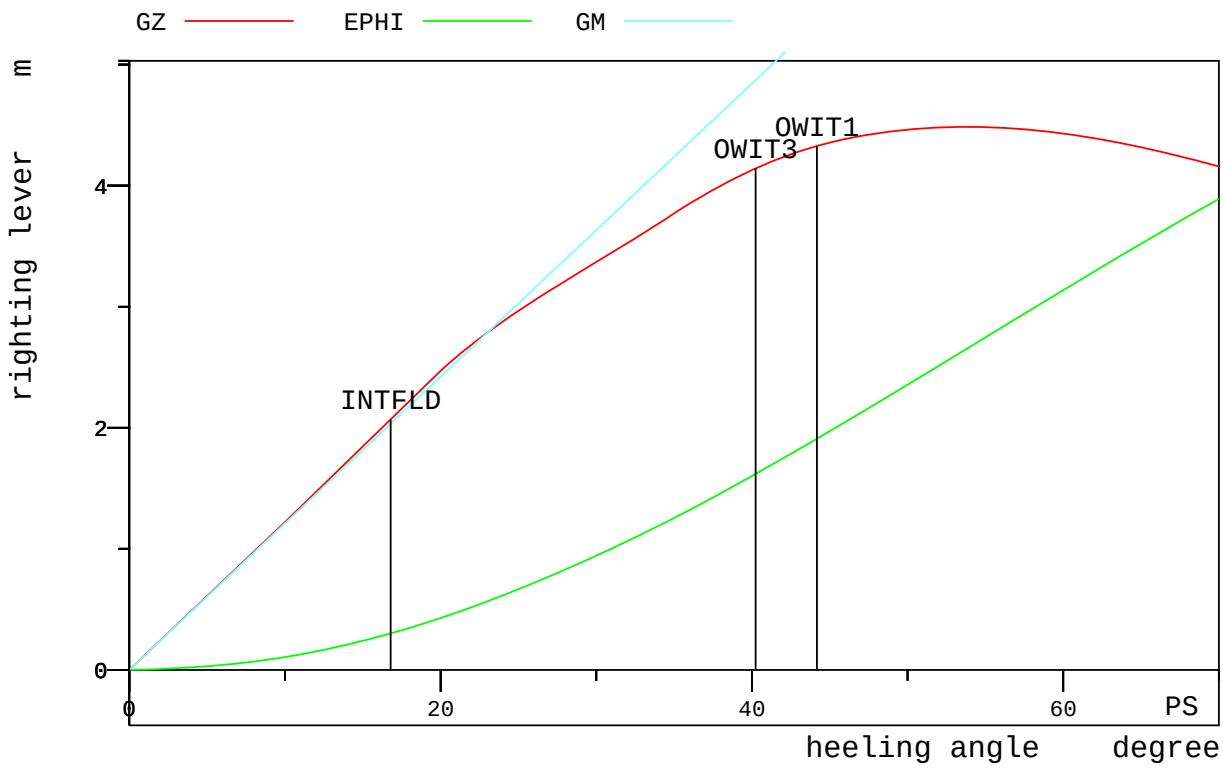
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN,CORR)	-785.5 t	(13.1%)	POSITION:	36.8 m	48
SHEAR FORCE (MAX,CORR)	418.7 t	(7.0%)		138.0 m	174
SAGGING MOMENT	-21510.0 tm	(22.2%)		73.1 m	93
HOGGING MOMENT	1310.3 tm	(3.8%)		14.2 m	20

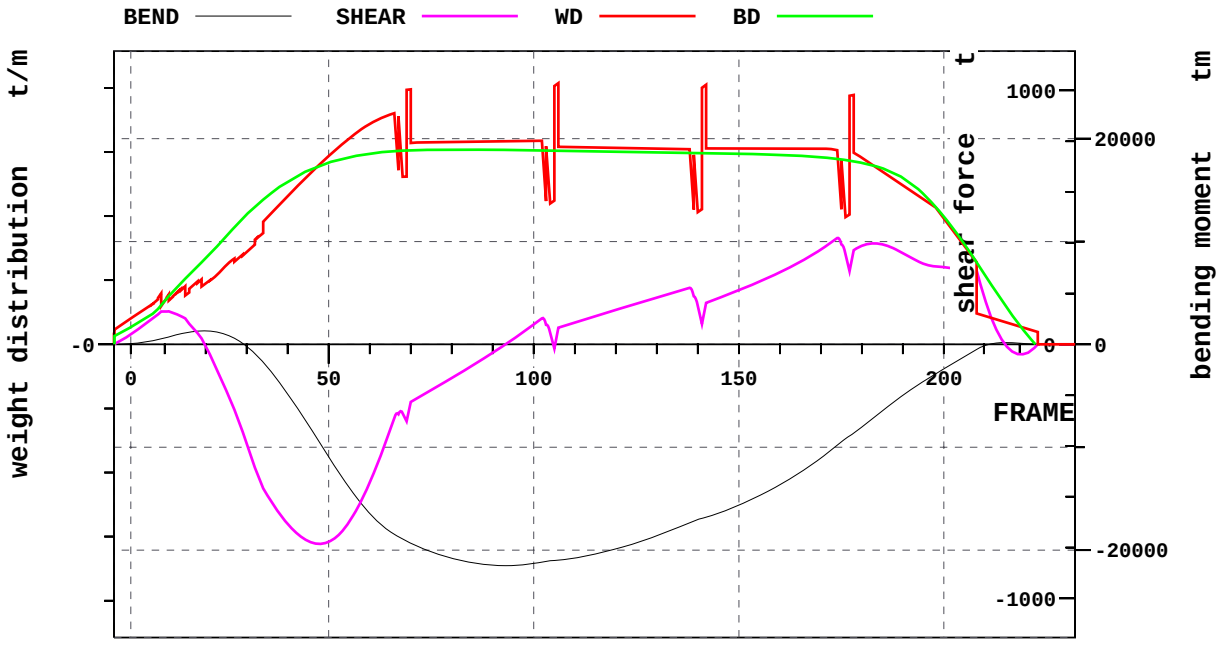
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 3.000)						
NO.1 CARGO HOLD	4899.8	24.6	150.87	0.00	3.58	0.0
NO.2 CARGO HOLD	7336.8	24.6	125.24	0.00	3.47	0.0
NO.3 CARGO HOLD	7345.9	24.6	96.45	0.00	3.47	0.0
NO.4 CARGO HOLD	7345.7	24.5	67.65	0.00	3.47	0.0
NO.5 CARGO HOLD	6625.0	24.5	40.97	0.00	3.95	0.0
SUBTOTAL	33553.2		93.43	0.00	3.58	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	9.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	34.6		63.09	0.00	20.29	0.0
TOTAL	33587.8		93.40	0.00	3.60	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	5374.44
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	5374.44
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	10748.87
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	10.3	50.0	8.24	-6.66	8.53	22.22
M.D.O. SETTL. TK.	12.4	50.0	9.83	-6.86	8.34	29.35
M.D.O. STOR. TK. (P)	17.3	50.0	12.68	6.80	8.45	43.37
M.D.O. STOR. TK. (S)	22.0	50.0	12.35	-6.76	8.50	54.86
SUBTOTAL	62.0		11.26	-2.98	8.46	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	20.4	50.0	4.98	7.32	11.97	31.97
FRESH WATER TK. (P)	46.9	50.0	1.02	6.48	12.05	44.74
FRESH WATER TK. (S)	67.2	50.0	2.26	-6.75	12.02	82.65
SUBTOTAL	159.2		2.90	-0.00	10.95	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	131.9	50.0	68.20	11.90	12.29	68.67
NO.1 FUEL OIL TK. (S)	131.9	50.0	68.20	-11.90	12.29	68.67
NO.2 FUEL OIL TK. (P)	127.7	50.0	39.94	11.88	12.30	0.00
NO.2 FUEL OIL TK. (S)	127.7	50.0	39.94	-11.88	12.30	0.00
H.F.O. TK. (P)	126.6	50.0	21.26	10.98	9.26	44.35
H.F.O. TK. (S)	90.0	50.0	21.60	-10.89	8.63	33.83
NO.1 H.F.O. SERV. TK.	12.4	50.0	15.47	-10.92	11.69	7.35
NO.1 H.F.O. SETTL. T.	14.8	50.0	18.64	-11.21	11.66	11.05
NO.2 H.F.O. SERV. TK.	9.7	50.0	12.65	-10.98	11.70	0.00
NO.2 H.F.O. SETTL. T.	11.0	50.0	9.65	-10.54	11.75	0.00
SUBTOTAL	783.5		42.77	-0.15	11.35	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	4.1	60.0	10.01	5.58	8.05	0.36
G/E L.O. STOR. TK.	7.9	60.0	10.02	8.01	8.78	10.28
M/E L.O. SETTL. TK.	10.5	60.0	8.82	6.85	8.65	19.67
M/E L.O. STOR. TK.	16.1	60.0	7.14	6.62	8.83	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	76.6		14.10	4.02	6.96	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	2.3	40.0	18.84	-2.15	1.07	1.69
BILGE TK.	6.4	40.0	11.55	0.00	0.73	9.85
F.O. DRAIN TK.	2.6	40.0	20.71	2.30	0.98	3.33
F.O. OVERFLOW TK.	3.6	40.0	23.05	2.70	0.93	7.97
L.O. DRAIN TK.	1.5	40.0	18.35	2.11	1.09	0.91
SEWAGE STOR. TK.	5.4	40.0	22.38	-2.58	0.94	10.07
SLUDGE TK.	6.0	40.0	21.80	-8.40	7.56	9.03
SUBTOTAL	27.9		19.18	-1.81	2.34	42.86
TOTAL	1109.3		32.71	-0.04	10.60	11391.80

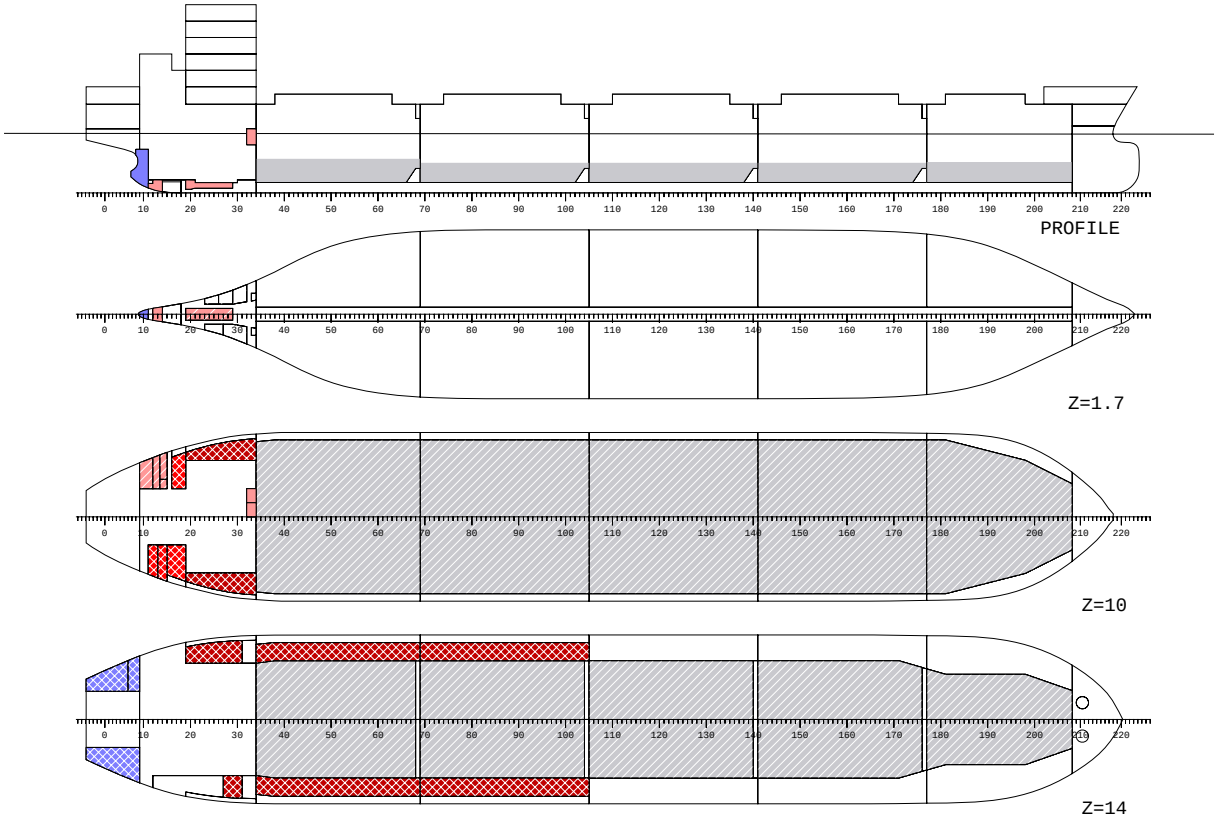


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	0.000	0.00	0.000	0.0	0.000
10.0	0.017	1.22	0.107	1978.2	0.046
12.0	0.029	1.47	0.154	2368.5	0.055
20.0	0.097	2.47	0.429	3896.2	0.090
30.0	-0.096	3.37	0.944	5695.9	0.131
40.0	-0.329	4.13	1.601	7322.5	0.169
50.0	-0.852	4.46	2.358	8726.6	0.201
60.0	-1.576	4.43	3.138	9865.6	0.228
70.0	-2.360	4.16	3.890	10704.8	0.247

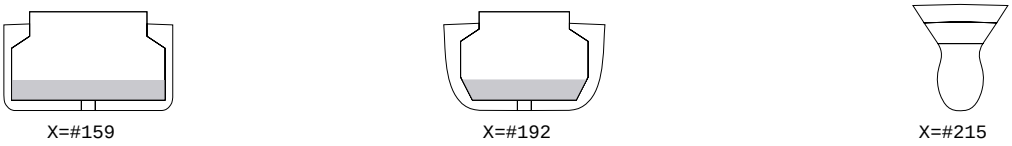


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	61	40	40.51	26.79
5.94	9.00	549	129	59.87	61.42
18.60	25.00	937	-181	122.63	168.57
25.80	34.00	-1737	-567	182.45	225.66
40.20	52.00	-12172	-756	304.72	287.06
53.80	69.00	-19132	-303	261.70	302.43
68.20	87.00	-21347	-64	316.24	303.61
82.60	105.00	-21019	-17	224.38	301.97
97.00	123.00	-19544	150	306.19	300.08
111.40	141.00	-16920	80	210.68	298.19
125.80	159.00	-13878	276	305.23	295.81
140.20	177.00	-8903	288	202.54	286.35
152.20	192.00	-4390	341	239.47	250.28
165.00	208.00	-488	291	128.15	127.64
169.20	214.00	148	44	38.08	76.85
173.40	220.00	104	-39	27.85	29.28

LOADING CASE 'LD09D -- MULTI-PORT HEAVY CARGO LOADING, ARRIVAL







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	0.0	0.00	0.00	0.00	10748.87
Diesel Oil	DO	12.4	11.39	-2.61	7.47	149.80
Fresh Water	FW	51.6	4.32	-0.00	8.25	159.36
Grain Load	GRAIN	33553.2	93.43	0.00	3.58	0.00
Heavy Fuel Oil	HFO	156.7	43.35	-0.16	10.12	233.92
Lubricating Oil	LO	50.8	16.95	2.45	5.88	56.99
	MASS	27.6	75.29	0.00	20.42	0.00
Miscellaneous	MIS	55.8	19.16	-1.81	2.67	42.86
Deadweight		33908.1	92.78	-0.00	3.64	11391.80
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		33908.1	92.78	-0.00	3.64	
Total weight		42558.1	90.04	-0.00	4.97	

F L O A T I N G P O S I T I O N

Draught moulded	10.065	m	KM	12.30 m
Trim	-0.117	m	KG	4.97 m
Heel, PS=+	0.0	deg		
TA	10.124	m	GM0	7.33 m
TF	10.007	m	GMCORR	-0.27 m
Trimming moment	-7001	tonm	GM	7.06 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.966	mrad	OK
AREA40	Area under GZ curve .	0.090	1.643	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.677	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.624	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	54.051	deg	OK
GM0.15	GM > 0.15 m	0.150	7.065	m	OK
IMOWEATH.	IMO weather criterion	1.000	4.062		OK
VISIBILI.		359.760	115.933		OK
PROPIMMER	Propeller should imm.	50.000	131.012	%	OK

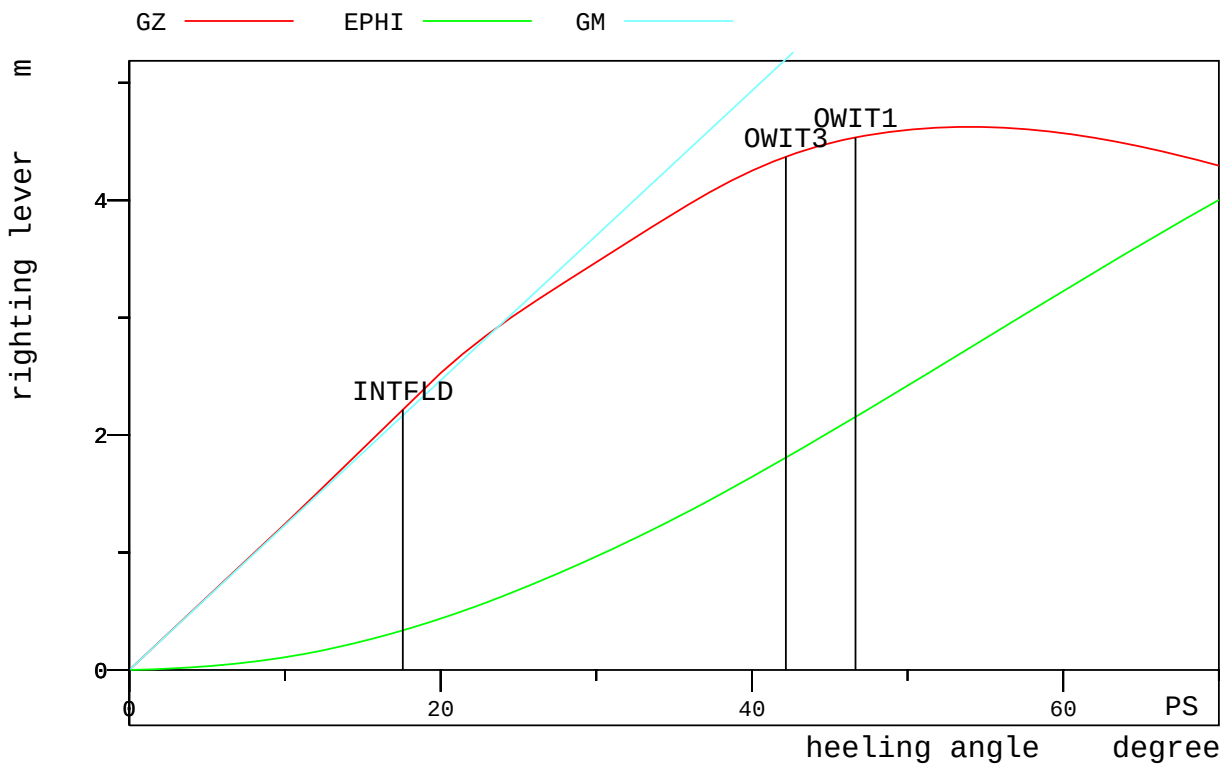
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN,CORR)	-845.3 t	(14.1%)	POSITION:	36.0 m	47
SHEAR FORCE (MAX,CORR)	486.7 t	(8.1%)		138.0 m	174
SAGGING MOMENT	-24711.6 tm	(25.5%)		75.4 m	96
HOGGING MOMENT	866.2 tm	(2.7%)		13.0 m	18

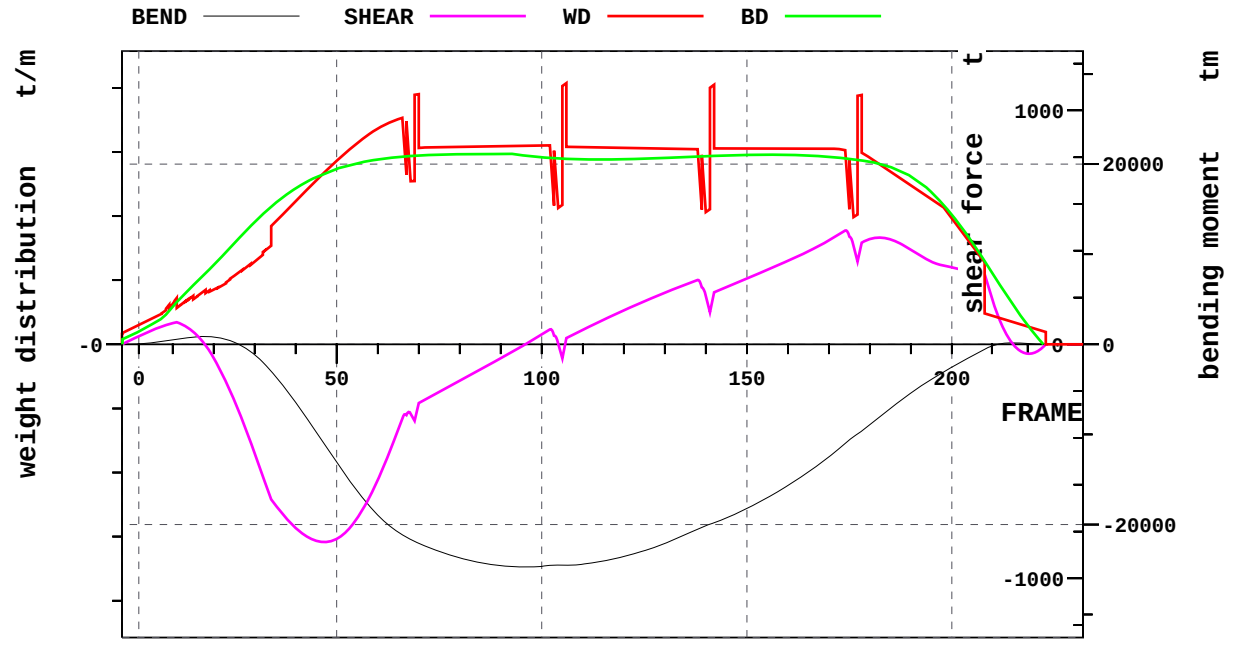
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 3.000)						
NO.1 CARGO HOLD	4899.8	24.6	150.87	0.00	3.58	0.0
NO.2 CARGO HOLD	7336.8	24.6	125.24	0.00	3.47	0.0
NO.3 CARGO HOLD	7345.9	24.6	96.45	0.00	3.47	0.0
NO.4 CARGO HOLD	7345.7	24.5	67.65	0.00	3.47	0.0
NO.5 CARGO HOLD	6625.0	24.5	40.97	0.00	3.95	0.0
SUBTOTAL	33553.2		93.43	0.00	3.58	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	2.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	27.6		75.29	0.00	20.42	0.0
TOTAL	33580.8		93.42	0.00	3.59	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	0.00	0.00	0.00	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	5374.44
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	5374.44
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	10748.87
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	2.1	10.0	8.28	-5.84	7.59	22.22
M.D.O. SETTL. TK.	2.5	10.0	9.88	-5.95	7.33	29.35
M.D.O. STOR. TK. (P)	3.5	10.0	12.81	5.91	7.45	43.37
M.D.O. STOR. TK. (S)	4.4	10.0	12.58	-5.91	7.52	54.86
SUBTOTAL	12.4		11.39	-2.61	7.47	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	4.1	10.0	4.98	7.15	11.13	31.97
FRESH WATER TK. (P)	9.4	10.0	1.14	6.32	11.15	44.74
FRESH WATER TK. (S)	13.4	10.0	2.40	-6.59	11.14	82.65
SUBTOTAL	51.6		4.32	-0.00	8.25	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	26.4	10.0	68.20	12.54	11.35	68.67
NO.1 FUEL OIL TK. (S)	26.4	10.0	68.20	-12.54	11.35	68.67
NO.2 FUEL OIL TK. (P)	25.5	10.0	40.16	12.52	11.37	0.00
NO.2 FUEL OIL TK. (S)	25.5	10.0	40.16	-12.52	11.37	0.00
H.F.O. TK. (P)	25.3	10.0	23.07	10.63	7.07	44.35
H.F.O. TK. (S)	18.0	10.0	23.44	-10.57	6.80	33.83
NO.1 H.F.O. SERV. TK.	2.5	10.0	15.49	-10.82	11.07	7.35
NO.1 H.F.O. SETTL. T.	3.0	10.0	18.65	-11.14	11.06	11.05
NO.2 H.F.O. SERV. TK.	1.9	10.0	12.65	-10.87	11.07	0.00
NO.2 H.F.O. SETTL. T.	2.2	10.0	9.70	-10.42	11.09	0.00
SUBTOTAL	156.7		43.35	-0.16	10.12	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	1.4	20.0	10.03	5.53	7.22	0.36
G/E L.O. STOR. TK.	2.6	20.0	10.04	7.48	7.96	10.28
M/E L.O. SETTL. TK.	3.5	20.0	8.83	6.21	7.80	19.67
M/E L.O. STOR. TK.	5.4	20.0	7.19	6.05	8.05	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	50.8		16.95	2.45	5.88	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	4.5	80.0	18.81	-2.29	1.40	1.69
BILGE TK.	12.9	80.0	11.52	0.00	1.14	9.85
F.O. DRAIN TK.	5.2	80.0	20.70	2.51	1.34	3.33
F.O. OVERFLOW TK.	7.2	80.0	23.04	2.98	1.31	7.97
L.O. DRAIN TK.	3.1	80.0	18.32	2.24	1.42	0.91
SEWAGE STOR. TK.	10.9	80.0	22.37	-2.83	1.31	10.07
SLUDGE TK.	12.0	80.0	21.80	-8.40	7.72	9.03
SUBTOTAL	55.8		19.16	-1.81	2.67	42.86
TOTAL	327.4		27.76	-0.10	7.80	11391.80

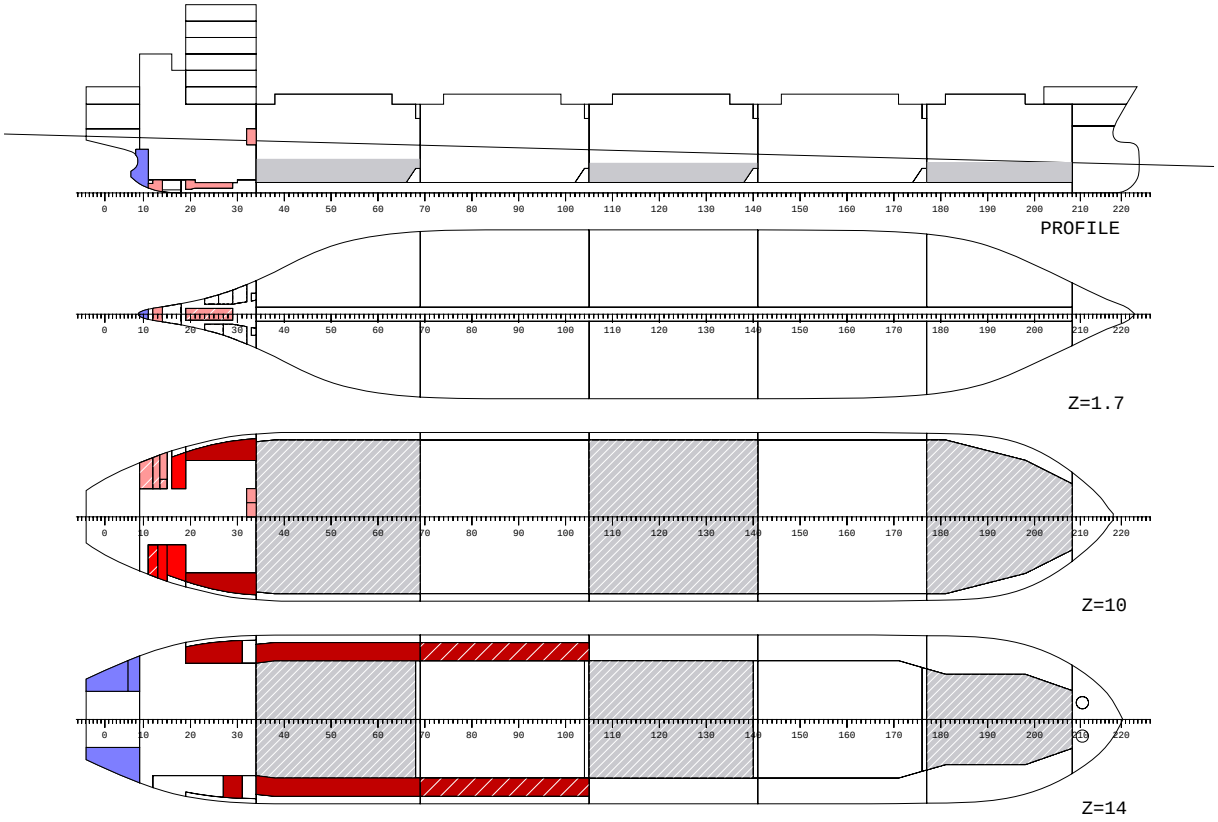


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	0.000	0.00	0.000	0.0	0.000
10.0	0.017	1.24	0.109	1978.2	0.046
12.0	0.029	1.50	0.156	2368.5	0.056
20.0	0.113	2.53	0.438	3896.2	0.092
30.0	-0.059	3.47	0.966	5695.9	0.134
40.0	-0.289	4.25	1.643	7322.5	0.172
50.0	-0.813	4.60	2.423	8726.6	0.205
60.0	-1.546	4.57	3.227	9865.6	0.232
70.0	-2.344	4.29	4.004	10704.8	0.252

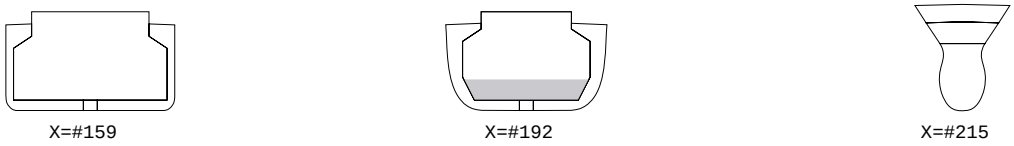


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	57	35	30.57	20.06
5.94	9.00	427	88	55.64	52.47
18.60	25.00	236	-240	106.87	157.37
25.80	34.00	-2971	-662	184.51	214.46
40.20	52.00	-14414	-806	297.39	277.27
53.80	69.00	-21851	-327	254.70	294.15
68.20	87.00	-24389	-89	308.91	296.93
82.60	105.00	-24516	-60	217.40	296.89
97.00	123.00	-23179	169	306.19	296.60
111.40	141.00	-19970	137	210.68	296.31
125.80	159.00	-15954	349	305.23	295.53
140.20	177.00	-9947	353	202.54	287.65
152.20	192.00	-4766	383	239.47	252.76
165.00	208.00	-527	300	128.15	129.78
169.20	214.00	135	46	38.08	78.10
173.40	220.00	96	-39	27.85	29.28

LOADING CASE 'LD09E -- MULTI-PORT HEAVY CARGO LOADING, FOR IACS, DEPARTURE







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	0.0	0.00	0.00	0.00	10748.87
Diesel Oil	DO	121.6	11.23	-3.17	9.29	149.80
Fresh Water	FW	293.6	2.53	0.00	12.34	159.36
Grain Load	GRAIN	18870.7	91.10	0.00	3.67	0.00
Heavy Fuel Oil	HFO	1535.7	42.68	-0.15	12.40	233.92
Lubricating Oil	LO	101.1	12.73	4.87	7.77	56.99
	MASS	43.6	53.17	0.00	20.19	0.00
Miscellaneous	MIS	7.0	19.24	-1.81	1.99	42.86
Deadweight		20973.3	85.37	-0.01	4.51	11391.80
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		20973.3	85.37	-0.01	4.51	
Total weight		29623.3	83.60	-0.00	6.17	

F L O A T I N G P O S I T I O N

Draught moulded	7.267	m	KM	13.74 m
Trim	-4.622	m	KG	6.17 m
Heel, PS=+	-0.0	deg		
TA	9.578	m	GM0	7.57 m
TF	4.956	m	GMCORR	-0.38 m
Trimming moment	-269498	tonm	GM	7.18 m
Heel	-0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	1.023	mrad	OK
AREA40	Area under GZ curve .	0.090	1.772	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.749	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.562	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	46.484	deg	OK
GM0.15	GM > 0.15 m	0.150	7.184	m	OK
IMOWEATH.	IMO weather criterion	1.000	4.987		OK
VISIBILI.		359.760	272.951		OK
PROPIMMER	Propeller should imm.	50.000	120.714	%	OK

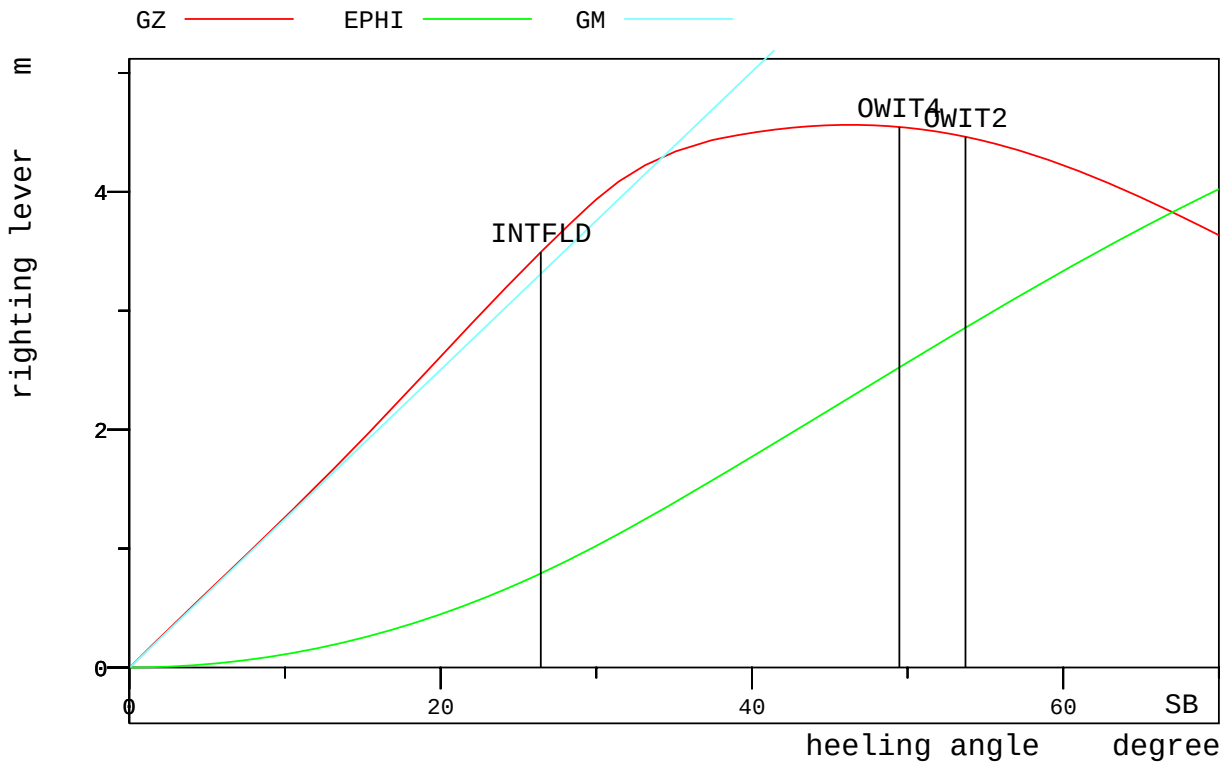
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN,CORR)	-2829.6 t	(47.2%)	POSITION:	140.2 m	177
SHEAR FORCE (MAX,CORR)	2827.0 t	(47.1%)		53.8 m	69
SAGGING MOMENT	-56.5 tm	(1.3%)		176.5 m	224
HOGGING MOMENT	81642.6 tm	(74.9%)		70.0 m	89

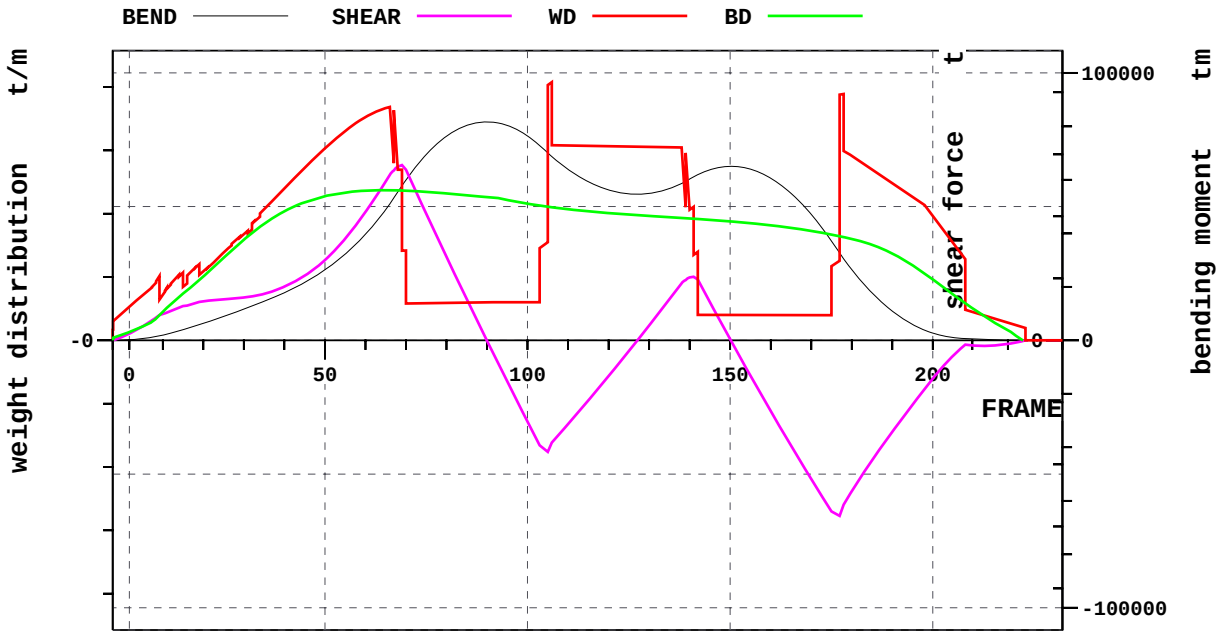
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 3.000)						
NO.1 CARGO HOLD	4899.8	24.6	150.87	0.00	3.58	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	7345.9	24.6	96.45	0.00	3.47	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	6625.0	24.5	40.97	0.00	3.95	0.0
SUBTOTAL	18870.7		91.10	0.00	3.67	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	18.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	43.6		53.17	0.00	20.19	0.0
TOTAL	18914.3		91.01	0.00	3.71	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	0.00	0.00	0.00	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	5374.44
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	5374.44
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	10748.87
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	20.2	98.0	8.23	-7.08	9.34	22.22
M.D.O. SETTL. TK.	24.2	98.0	9.82	-7.32	9.21	29.35
M.D.O. STOR. TK. (P)	33.9	98.0	12.66	7.26	9.29	43.37
M.D.O. STOR. TK. (S)	43.2	98.0	12.30	-7.22	9.32	54.86
SUBTOTAL	121.6		11.23	-3.17	9.29	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	40.7	100.0	4.97	7.43	12.96	31.97
FRESH WATER TK. (P)	93.7	100.0	0.94	6.60	13.03	44.74
FRESH WATER TK. (S)	134.4	100.0	2.17	-6.85	13.01	82.65
SUBTOTAL	293.6		2.53	0.00	12.34	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	258.4	98.0	68.20	11.73	13.08	68.67
NO.1 FUEL OIL TK. (S)	258.4	98.0	68.20	-11.73	13.08	68.67
NO.2 FUEL OIL TK. (P)	250.2	98.0	39.86	11.71	13.09	0.00
NO.2 FUEL OIL TK. (S)	250.2	98.0	39.86	-11.71	13.09	0.00
H.F.O. TK. (P)	248.1	98.0	20.60	11.15	11.04	44.35
H.F.O. TK. (S)	176.3	98.0	21.94	-11.14	10.32	33.83
NO.1 H.F.O. SERV. TK.	24.3	98.0	15.46	-11.00	12.38	7.35
NO.1 H.F.O. SETTL. T.	29.0	98.0	18.63	-11.27	12.36	11.05
NO.2 H.F.O. SERV. TK.	19.1	98.0	12.64	-11.07	12.39	0.00
NO.2 H.F.O. SETTL. T.	21.5	98.0	9.62	-10.64	12.44	0.00
SUBTOTAL	1535.7		42.68	-0.15	12.40	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	6.8	98.0	10.01	5.59	8.82	0.36
G/E L.O. STOR. TK.	12.9	98.0	10.02	8.28	9.39	10.28
M/E L.O. SETTL. TK.	17.2	98.0	8.81	7.17	9.29	19.67
M/E L.O. STOR. TK.	26.4	98.0	7.12	6.90	9.42	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	101.1		12.73	4.87	7.77	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	0.6	10.0	18.98	-1.96	0.68	1.69
BILGE TK.	1.6	10.0	11.66	0.00	0.32	9.85
F.O. DRAIN TK.	0.6	10.0	20.77	2.04	0.56	3.33
F.O. OVERFLOW TK.	0.9	10.0	23.05	2.35	0.50	7.97
L.O. DRAIN TK.	0.4	10.0	18.43	1.94	0.72	0.91
SEWAGE STOR. TK.	1.4	10.0	22.44	-2.26	0.52	10.07
SLUDGE TK.	1.5	10.0	21.80	-8.40	7.44	9.03
SUBTOTAL	7.0		19.24	-1.81	1.99	42.86
TOTAL	2059.0		33.55	-0.07	11.94	11391.80

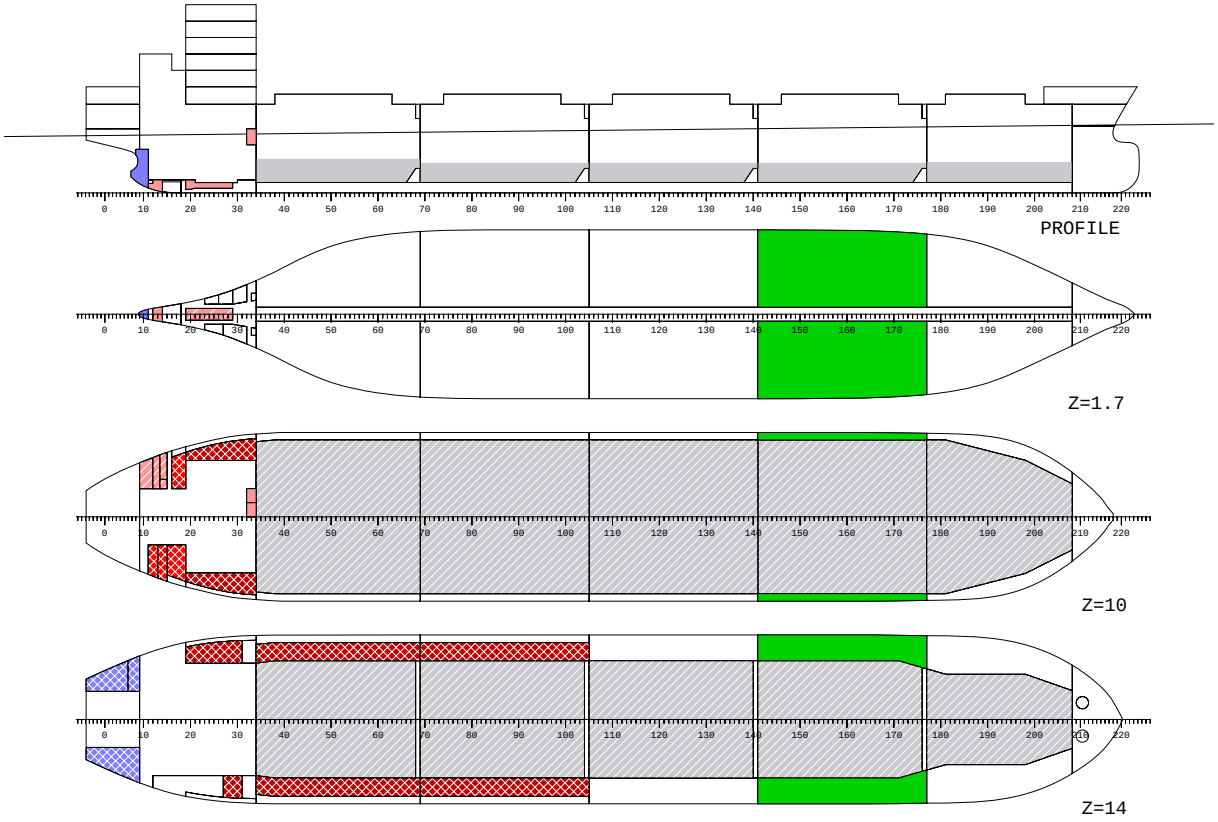


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.005	-0.00	0.000	0.0	0.000
10.0	0.015	1.26	0.110	1978.2	0.067
12.0	0.030	1.52	0.159	2368.5	0.080
20.0	0.157	2.61	0.448	3896.2	0.132
30.0	0.344	3.94	1.023	5695.9	0.192
40.0	-0.119	4.50	1.772	7322.5	0.247
50.0	-0.963	4.54	2.565	8726.6	0.295
60.0	-1.999	4.22	3.335	9865.6	0.333
70.0	-3.116	3.64	4.023	10704.8	0.361

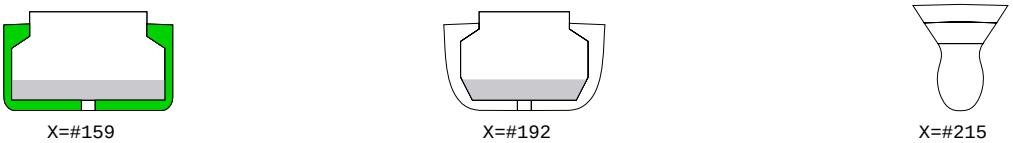


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	194	109	53.14	12.98
5.94	9.00	1709	407	82.80	39.43
18.60	25.00	8948	658	138.41	129.84
25.80	34.00	13976	730	200.41	179.23
40.20	52.00	28497	1420	313.82	229.99
53.80	69.00	57621	2827	269.37	236.33
68.20	87.00	81192	393	59.84	227.96
82.60	105.00	70061	-1800	154.80	216.77
97.00	123.00	55150	-337	306.19	205.32
111.40	141.00	61029	1025	210.68	193.88
125.80	159.00	61443	-1027	39.94	181.94
140.20	177.00	32670	-2830	125.51	164.43
152.20	192.00	8736	-1300	239.47	129.86
165.00	208.00	650	-71	128.15	59.47
169.20	214.00	327	-89	38.08	36.14
173.40	220.00	34	-59	27.85	16.19

LOADING CASE 'LD09F -- MULTI-PORT HEAVY CARGO LOADING, FOR IACS, ARRIVAL







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	2789.9	125.72	0.00	5.87	10748.87
Diesel Oil	DO	12.4	11.39	-2.61	7.47	149.80
Fresh Water	FW	51.6	4.32	-0.00	8.25	159.36
Grain Load	GRAIN	33553.2	93.43	0.00	3.58	0.00
Heavy Fuel Oil	HFO	156.7	43.35	-0.16	10.12	233.92
Lubricating Oil	LO	50.8	16.95	2.45	5.88	56.99
	MASS	27.6	75.29	0.00	20.42	0.00
Miscellaneous	MIS	55.8	19.16	-1.81	2.67	42.86
Deadweight		36698.0	95.29	-0.00	3.81	11391.80
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		36698.0	95.29	-0.00	3.81	
Total weight		45348.0	92.24	-0.00	5.02	

F L O A T I N G P O S I T I O N

Draught moulded	10.675	m	KM	12.21 m
Trim	1.806	m	KG	5.02 m
Heel, PS=+	0.0	deg		
TA	9.772	m	GM0	7.18 m
TF	11.578	m	GMCORR	-0.25 m
Trimming moment	108972	tonm	GM	6.93 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.921	mrad	OK
AREA40	Area under GZ curve .	0.090	1.553	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.631	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.350	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	55.603	deg	OK
GM0.15	GM > 0.15 m	0.150	6.932	m	OK
IMOWEATH.	IMO weather criterion	1.000	3.900		OK
VISIBILI.		359.760	83.360		OK
PROPIMMER	Propeller should imm.	50.000	124.383	%	OK

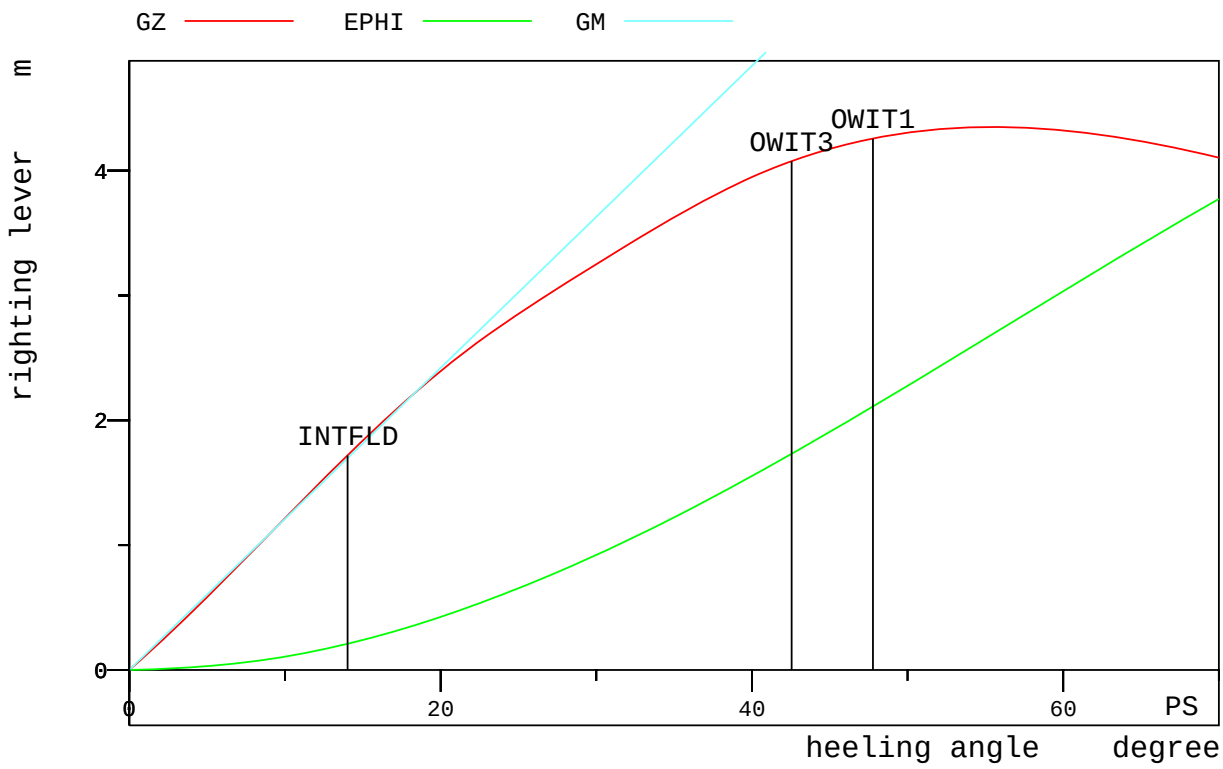
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN,CORR)	-789.3 t	(13.2%)	POSITION:	111.4 m	141
SHEAR FORCE (MAX,CORR)	1374.2 t	(22.9%)		141.0 m	178
SAGGING MOMENT	-37920.6 tm	(39.2%)		120.4 m	152
HOGGING MOMENT	1645.6 tm	(4.4%)		15.4 m	21

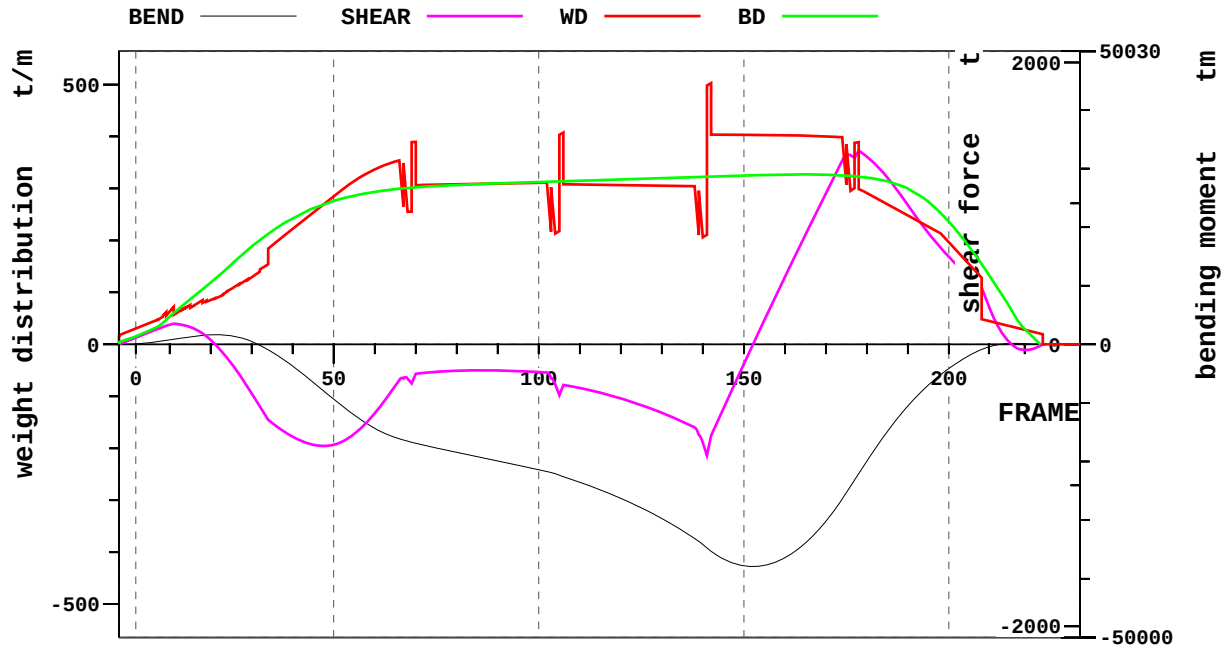
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 3.000)						
NO.1 CARGO HOLD	4899.8	24.6	150.87	0.00	3.58	0.0
NO.2 CARGO HOLD	7336.8	24.6	125.24	0.00	3.47	0.0
NO.3 CARGO HOLD	7345.9	24.6	96.45	0.00	3.47	0.0
NO.4 CARGO HOLD	7345.7	24.5	67.65	0.00	3.47	0.0
NO.5 CARGO HOLD	6625.0	24.5	40.97	0.00	3.95	0.0
SUBTOTAL	33553.2		93.43	0.00	3.58	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	2.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	27.6		75.29	0.00	20.42	0.0
TOTAL	33580.8		93.42	0.00	3.59	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	0.00	0.00	0.00	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	10.47	5.87	5374.44
NO.2 WATER BALLAST T.	1394.9	100.0	125.72	-10.47	5.87	5374.44
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	2789.9		125.72	0.00	5.87	10748.87
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	2.1	10.0	8.28	-5.84	7.59	22.22
M.D.O. SETTL. TK.	2.5	10.0	9.88	-5.95	7.33	29.35
M.D.O. STOR. TK. (P)	3.5	10.0	12.81	5.91	7.45	43.37
M.D.O. STOR. TK. (S)	4.4	10.0	12.58	-5.91	7.52	54.86
SUBTOTAL	12.4		11.39	-2.61	7.47	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	4.1	10.0	4.98	7.15	11.13	31.97
FRESH WATER TK. (P)	9.4	10.0	1.14	6.32	11.15	44.74
FRESH WATER TK. (S)	13.4	10.0	2.40	-6.59	11.14	82.65
SUBTOTAL	51.6		4.32	-0.00	8.25	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	26.4	10.0	68.20	12.54	11.35	68.67
NO.1 FUEL OIL TK. (S)	26.4	10.0	68.20	-12.54	11.35	68.67
NO.2 FUEL OIL TK. (P)	25.5	10.0	40.16	12.52	11.37	0.00
NO.2 FUEL OIL TK. (S)	25.5	10.0	40.16	-12.52	11.37	0.00
H.F.O. TK. (P)	25.3	10.0	23.07	10.63	7.07	44.35
H.F.O. TK. (S)	18.0	10.0	23.44	-10.57	6.80	33.83
NO.1 H.F.O. SERV. TK.	2.5	10.0	15.49	-10.82	11.07	7.35
NO.1 H.F.O. SETTL. T.	3.0	10.0	18.65	-11.14	11.06	11.05
NO.2 H.F.O. SERV. TK.	1.9	10.0	12.65	-10.87	11.07	0.00
NO.2 H.F.O. SETTL. T.	2.2	10.0	9.70	-10.42	11.09	0.00
SUBTOTAL	156.7		43.35	-0.16	10.12	233.92
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	1.4	20.0	10.03	5.53	7.22	0.36
G/E L.O. STOR. TK.	2.6	20.0	10.04	7.48	7.96	10.28
M/E L.O. SETTL. TK.	3.5	20.0	8.83	6.21	7.80	19.67
M/E L.O. STOR. TK.	5.4	20.0	7.19	6.05	8.05	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	50.8		16.95	2.45	5.88	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	4.5	80.0	18.81	-2.29	1.40	1.69
BILGE TK.	12.9	80.0	11.52	0.00	1.14	9.85
F.O. DRAIN TK.	5.2	80.0	20.70	2.51	1.34	3.33
F.O. OVERFLOW TK.	7.2	80.0	23.04	2.98	1.31	7.97
L.O. DRAIN TK.	3.1	80.0	18.32	2.24	1.42	0.91
SEWAGE STOR. TK.	10.9	80.0	22.37	-2.83	1.31	10.07
SLUDGE TK.	12.0	80.0	21.80	-8.40	7.72	9.03
SUBTOTAL	55.8		19.16	-1.81	2.67	42.86
TOTAL	3117.2		115.43	-0.01	6.07	11391.80

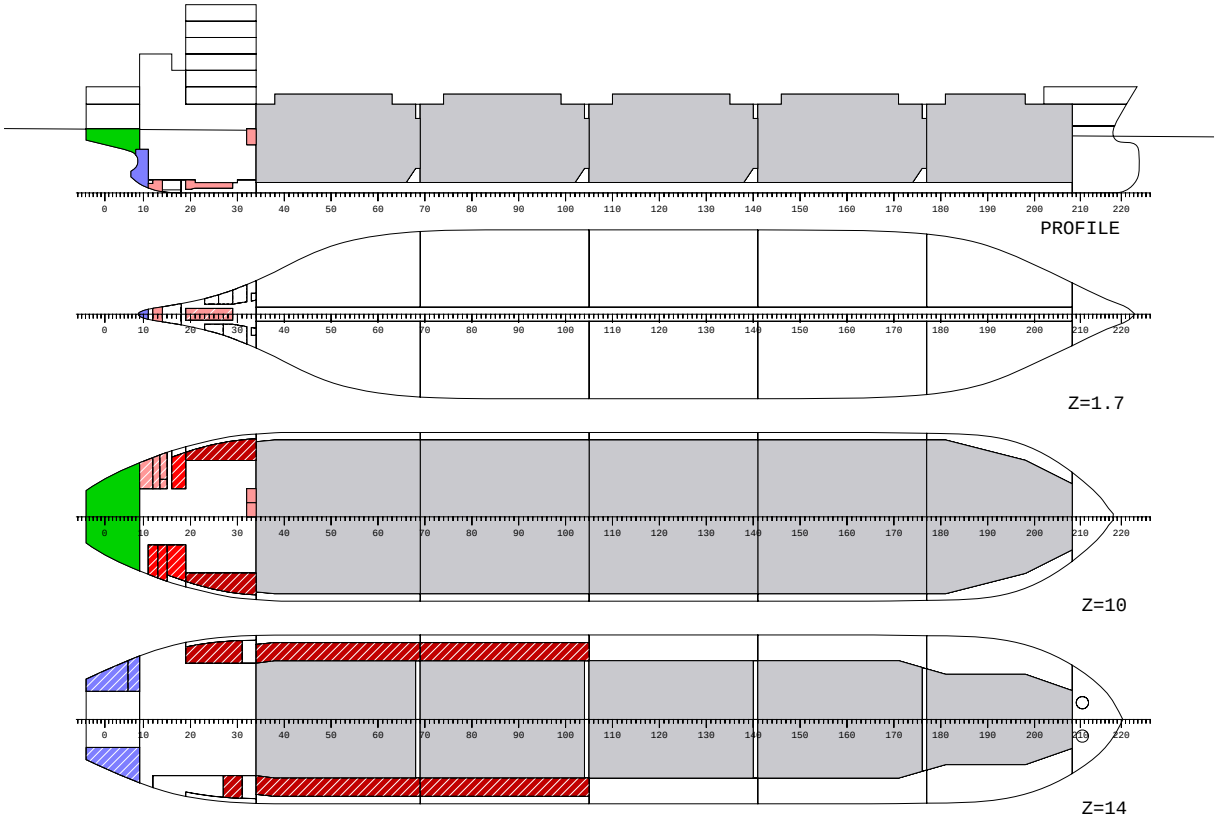


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	0.000	0.00	0.000	0.0	0.000
10.0	0.016	1.22	0.106	1978.2	0.044
12.0	0.027	1.47	0.153	2368.5	0.052
20.0	0.025	2.40	0.425	3896.2	0.086
30.0	-0.215	3.25	0.921	5695.9	0.126
40.0	-0.507	3.95	1.553	7322.5	0.161
50.0	-1.007	4.30	2.278	8726.6	0.192
60.0	-1.680	4.32	3.035	9865.6	0.218
70.0	-2.410	4.10	3.773	10704.8	0.236

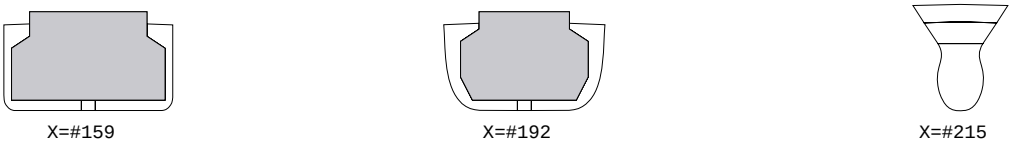


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	87	47	30.48	15.45
5.94	9.00	638	132	55.61	47.08
18.60	25.00	1448	-135	106.68	153.52
25.80	34.00	-896	-536	184.15	212.64
40.20	52.00	-10480	-691	297.39	280.18
53.80	69.00	-16701	-279	322.08	301.55
68.20	87.00	-19463	-185	308.90	309.10
82.60	105.00	-22338	-362	217.94	313.82
97.00	123.00	-27224	-411	306.19	318.29
111.40	141.00	-34734	-789	354.44	322.76
125.80	159.00	-36778	415	402.62	326.74
140.20	177.00	-23178	1322	300.58	323.26
152.20	192.00	-8979	908	239.47	290.21
165.00	208.00	-863	404	128.15	153.93
169.20	214.00	59	69	38.08	91.72
173.40	220.00	64	-39	27.85	29.28

LOADING CASE 'LD10A -- HOMOGENEOUSLY LIGHT CARGO CONDITION,SHORT VOYAGE, DEPARTURE







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	339.3	2.28	0.00	9.53	0.00
Diesel Oil	DO	62.0	11.26	-2.98	8.46	149.80
Fresh Water	FW	159.2	2.90	-0.00	10.95	159.36
Grain Load	GRAIN	33462.1	93.51	0.00	8.76	0.00
Heavy Fuel Oil	HFO	783.5	42.77	-0.15	11.35	380.41
Lubricating Oil	LO	76.6	14.10	4.02	6.96	56.99
	MASS	34.6	63.09	0.00	20.29	0.00
Miscellaneous	MIS	7.0	19.24	-1.81	1.99	42.86
Deadweight		34924.3	90.71	-0.00	8.84	789.41
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		34924.3	90.71	-0.00	8.84	
Total weight		43574.3	88.44	-0.00	9.11	

F L O A T I N G P O S I T I O N

Draught moulded	10.267	m	KM	12.30 m
Trim	-1.187	m	KG	9.11 m
Heel, PS=+	0.0	deg		
TA	10.860	m	GM0	3.19 m
TF	9.674	m	GMCORR	-0.02 m
Trimming moment	-70743	tonm	GM	3.17 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.436	mrad	OK
AREA40	Area under GZ curve .	0.090	0.696	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.259	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	1.706	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	41.744	deg	OK
GM0.15	GM > 0.15 m	0.150	3.174	m	OK
IMOWEATH.	IMO weather criterion	1.000	2.916		OK
VISIBILI.		359.760	130.930		OK
PROPIMMER	Propeller should imm.	50.000	144.910	%	OK

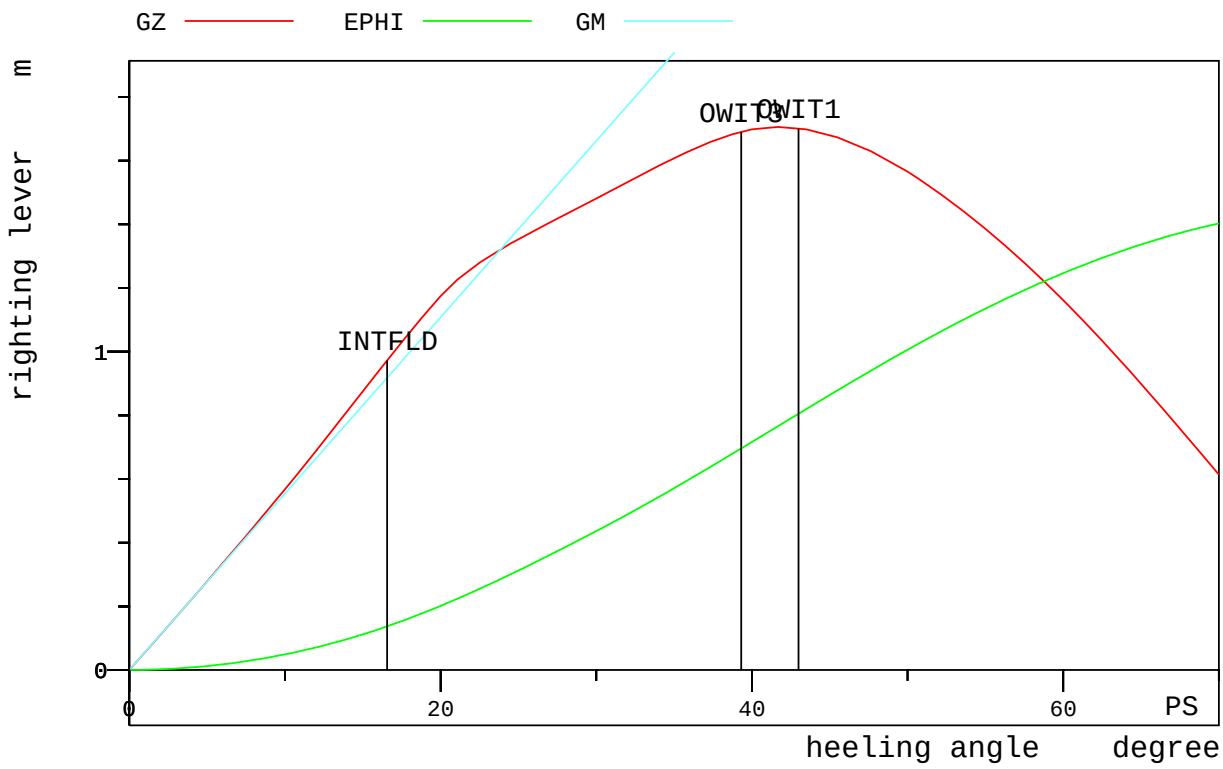
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-413.5 t	(6.9%)	POSITION:	29.0 m	38
SHEAR FORCE (MAX)	433.3 t	(7.2%)		5.9 m	9
SAGGING MOMENT	-9845.6 tm	(10.2%)		100.8 m	128
HOGGING MOMENT	5573.4 tm	(12.4%)		19.4 m	26

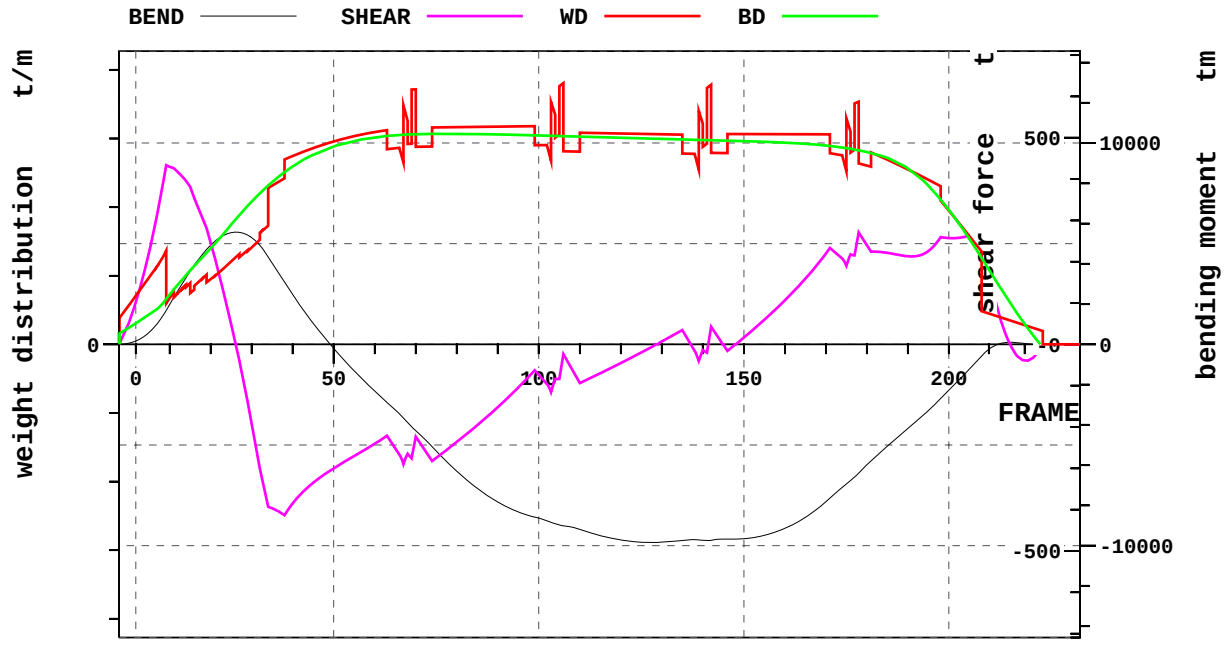
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.734)						
NO.1 CARGO HOLD	4886.5	100.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	7316.9	100.0	125.57	0.00	8.67	0.0
NO.3 CARGO HOLD	7326.0	100.0	96.78	0.00	8.67	0.0
NO.4 CARGO HOLD	7325.7	100.0	67.98	0.00	8.67	0.0
NO.5 CARGO HOLD	6607.0	100.0	40.05	0.00	9.09	0.0
SUBTOTAL	33462.1		93.51	0.00	8.76	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	9.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	34.6		63.09	0.00	20.29	0.0
TOTAL	33496.7		93.48	0.00	8.77	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	339.3	100.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	339.3		2.28	0.00	9.53	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	10.3	50.0	8.24	-6.66	8.53	22.22
M.D.O. SETTL. TK.	12.4	50.0	9.83	-6.86	8.34	29.35
M.D.O. STOR. TK. (P)	17.3	50.0	12.68	6.80	8.45	43.37
M.D.O. STOR. TK. (S)	22.0	50.0	12.35	-6.76	8.50	54.86
SUBTOTAL	62.0		11.26	-2.98	8.46	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	20.4	50.0	4.98	7.32	11.97	31.97
FRESH WATER TK. (P)	46.9	50.0	1.02	6.48	12.05	44.74
FRESH WATER TK. (S)	67.2	50.0	2.26	-6.75	12.02	82.65
SUBTOTAL	159.2		2.90	-0.00	10.95	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	131.9	50.0	68.20	11.90	12.29	68.67
NO.1 FUEL OIL TK. (S)	131.9	50.0	68.20	-11.90	12.29	68.67
NO.2 FUEL OIL TK. (P)	127.7	50.0	39.94	11.88	12.30	67.49
NO.2 FUEL OIL TK. (S)	127.7	50.0	39.94	-11.88	12.30	67.49
H.F.O. TK. (P)	126.6	50.0	21.26	10.98	9.26	44.35
H.F.O. TK. (S)	90.0	50.0	21.60	-10.89	8.63	33.83
NO.1 H.F.O. SERV. TK.	12.4	50.0	15.47	-10.92	11.69	7.35
NO.1 H.F.O. SETTL. T.	14.8	50.0	18.64	-11.21	11.66	11.05
NO.2 H.F.O. SERV. TK.	9.7	50.0	12.65	-10.98	11.70	6.67
NO.2 H.F.O. SETTL. T.	11.0	50.0	9.65	-10.54	11.75	4.84
SUBTOTAL	783.5		42.77	-0.15	11.35	380.41
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	4.1	60.0	10.01	5.58	8.05	0.36
G/E L.O. STOR. TK.	7.9	60.0	10.02	8.01	8.78	10.28
M/E L.O. SETTL. TK.	10.5	60.0	8.82	6.85	8.65	19.67
M/E L.O. STOR. TK.	16.1	60.0	7.14	6.62	8.83	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	76.6		14.10	4.02	6.96	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	0.6	10.0	18.98	-1.96	0.68	1.69
BILGE TK.	1.6	10.0	11.66	0.00	0.32	9.85
F.O. DRAIN TK.	0.6	10.0	20.77	2.04	0.56	3.33
F.O. OVERFLOW TK.	0.9	10.0	23.05	2.35	0.50	7.97
L.O. DRAIN TK.	0.4	10.0	18.43	1.94	0.72	0.91
SEWAGE STOR. TK.	1.4	10.0	22.44	-2.26	0.52	10.07
SLUDGE TK.	1.5	10.0	21.80	-8.40	7.44	9.03
SUBTOTAL	7.0		19.24	-1.81	1.99	42.86
TOTAL	1427.6		25.68	-0.00	10.46	789.41

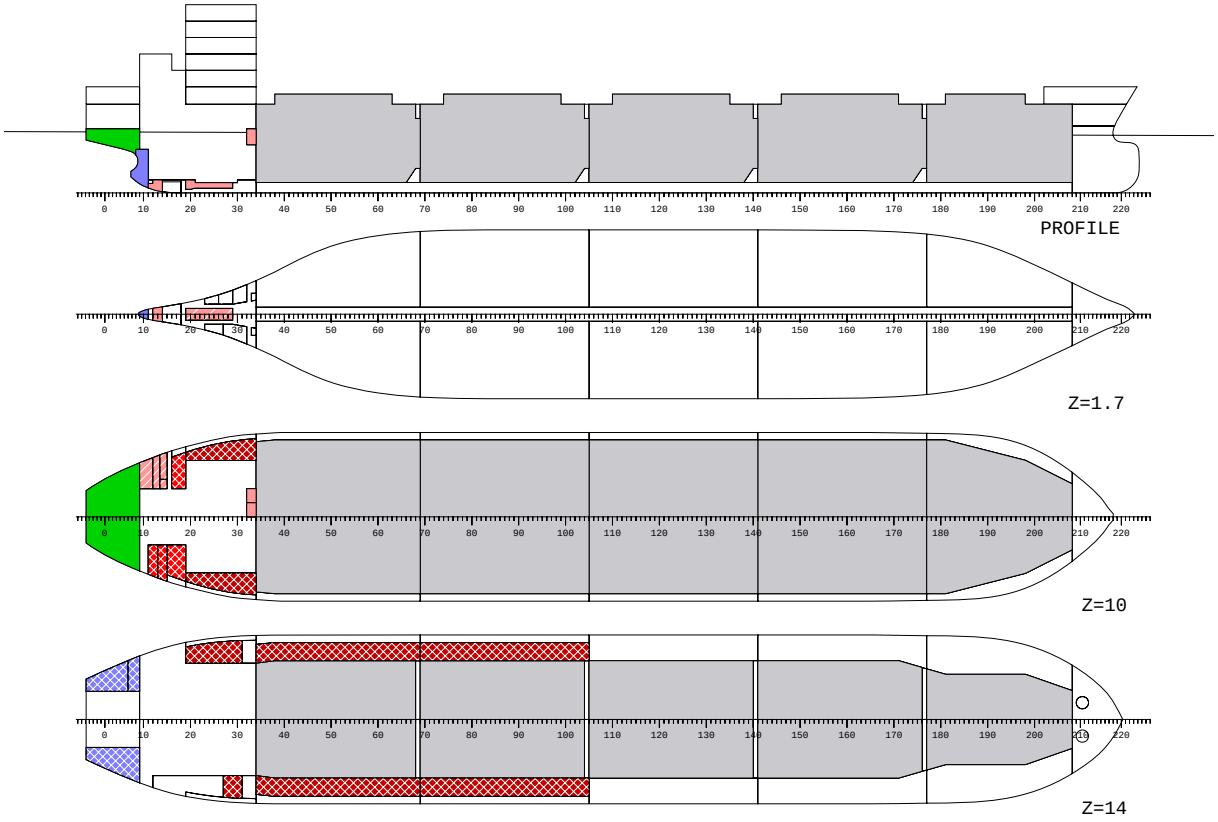


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	0.000	0.00	0.000	0.0	0.000
10.0	0.016	0.57	0.049	137.1	0.003
12.0	0.028	0.69	0.071	164.1	0.004
20.0	0.089	1.17	0.201	270.0	0.006
30.0	-0.106	1.48	0.436	394.7	0.009
40.0	-0.342	1.70	0.716	507.4	0.012
50.0	-0.866	1.57	1.006	604.7	0.014
60.0	-1.588	1.16	1.247	683.6	0.016
70.0	-2.368	0.61	1.403	741.8	0.017

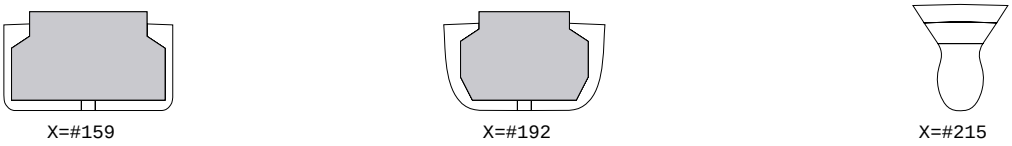


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	171	102	70.71	30.48
5.94	9.00	1697	433	98.32	66.18
18.60	25.00	5548	48	121.06	174.27
25.80	34.00	4363	-393	227.84	231.22
40.20	52.00	-664	-288	298.94	291.66
53.80	69.00	-4095	-276	371.42	306.03
68.20	87.00	-7448	-179	316.99	306.17
82.60	105.00	-8981	-84	305.87	303.48
97.00	123.00	-9795	-30	306.97	300.54
111.40	141.00	-9745	-20	292.42	297.60
125.80	159.00	-9229	97	306.25	294.17
140.20	177.00	-6593	215	283.62	283.69
152.20	192.00	-3825	214	249.13	246.96
165.00	208.00	-567	281	135.20	125.30
169.20	214.00	67	42	38.08	75.54
173.40	220.00	43	-39	27.85	29.28

LOADING CASE 'LD10B -- HOMOGENEOUSLY LIGHT CARGO CONDITION,SHORT VOYAGE, ARRIVAL







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	339.3	2.28	0.00	9.53	0.00
Diesel Oil	DO	12.4	11.39	-2.61	7.47	149.80
Fresh Water	FW	51.6	4.32	-0.00	8.25	159.36
Grain Load	GRAIN	33462.1	93.51	0.00	8.76	0.00
Heavy Fuel Oil	HFO	156.7	43.35	-0.16	10.12	380.41
Lubricating Oil	LO	50.8	16.95	2.45	5.88	56.99
	MASS	27.6	75.29	0.00	20.42	0.00
Miscellaneous	MIS	55.8	19.16	-1.81	2.67	42.86
Deadweight		34156.3	91.96	-0.00	8.76	789.41
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		34156.3	91.96	-0.00	8.76	
Total weight		42806.3	89.40	-0.00	9.05	

F L O A T I N G P O S I T I O N

Draught moulded	10.113	m	KM	12.31 m
Trim	-0.556	m	KG	9.05 m
Heel, PS=+	0.0	deg		
TA	10.391	m	GM0	3.25 m
TF	9.835	m	GMCORR	-0.02 m
Trimming moment	-32927	tonm	GM	3.23 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.450	mrad	OK
AREA40	Area under GZ curve .	0.090	0.743	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.293	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	1.792	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	41.931	deg	OK
GM0.15	GM > 0.15 m	0.150	3.234	m	OK
IMOWEATH.	IMO weather criterion	1.000	3.212		OK
VISIBILI.		359.760	122.260		OK
PROPIMMER	Propeller should imm.	50.000	136.051	%	OK

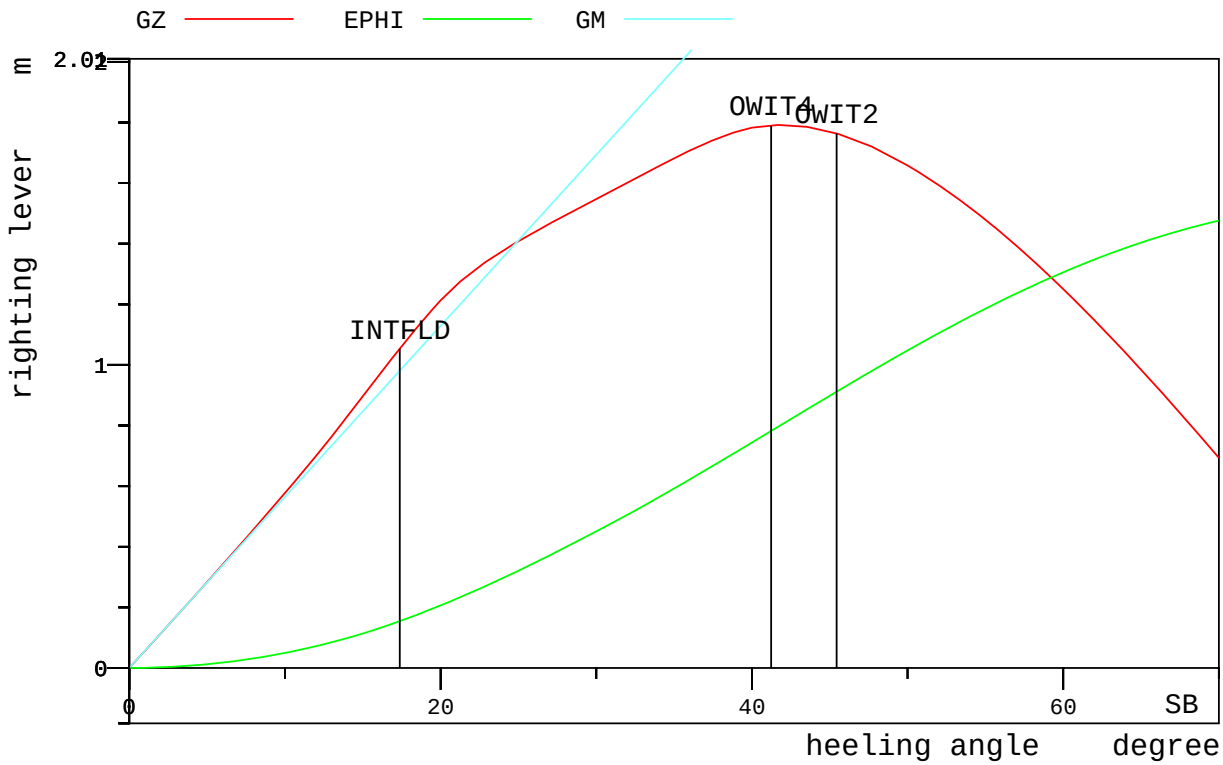
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-480.9 t	(8.0%)	POSITION:	29.0 m	38
SHEAR FORCE (MAX)	391.8 t	(6.5%)		5.9 m	9
SAGGING MOMENT	-13120.8 tm	(13.5%)		97.7 m	124
HOGGING MOMENT	4798.6 tm	(11.0%)		18.6 m	25

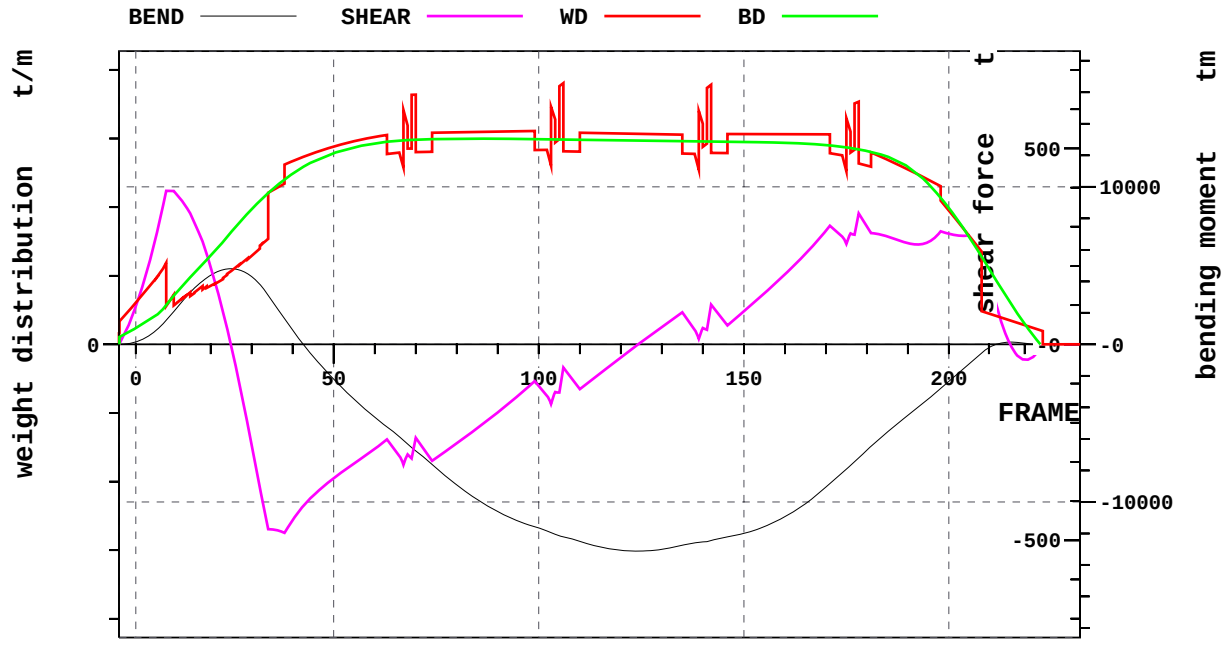
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 0.734)						
NO.1 CARGO HOLD	4886.5	100.0	151.18	0.00	8.68	0.0
NO.2 CARGO HOLD	7316.9	100.0	125.57	0.00	8.67	0.0
NO.3 CARGO HOLD	7326.0	100.0	96.78	0.00	8.67	0.0
NO.4 CARGO HOLD	7325.7	100.0	67.98	0.00	8.67	0.0
NO.5 CARGO HOLD	6607.0	100.0	40.05	0.00	9.09	0.0
SUBTOTAL	33462.1		93.51	0.00	8.76	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	2.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	27.6		75.29	0.00	20.42	0.0
TOTAL	33489.7		93.50	0.00	8.76	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	339.3	100.0	2.28	0.00	9.53	0.00
FORE PEAK TK.	0.0	0.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	339.3		2.28	0.00	9.53	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	2.1	10.0	8.28	-5.84	7.59	22.22
M.D.O. SETTL. TK.	2.5	10.0	9.88	-5.95	7.33	29.35
M.D.O. STOR. TK. (P)	3.5	10.0	12.81	5.91	7.45	43.37
M.D.O. STOR. TK. (S)	4.4	10.0	12.58	-5.91	7.52	54.86
SUBTOTAL	12.4		11.39	-2.61	7.47	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	4.1	10.0	4.98	7.15	11.13	31.97
FRESH WATER TK. (P)	9.4	10.0	1.14	6.32	11.15	44.74
FRESH WATER TK. (S)	13.4	10.0	2.40	-6.59	11.14	82.65
SUBTOTAL	51.6		4.32	-0.00	8.25	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	26.4	10.0	68.20	12.54	11.35	68.67
NO.1 FUEL OIL TK. (S)	26.4	10.0	68.20	-12.54	11.35	68.67
NO.2 FUEL OIL TK. (P)	25.5	10.0	40.16	12.52	11.37	67.49
NO.2 FUEL OIL TK. (S)	25.5	10.0	40.16	-12.52	11.37	67.49
H.F.O. TK. (P)	25.3	10.0	23.07	10.63	7.07	44.35
H.F.O. TK. (S)	18.0	10.0	23.44	-10.57	6.80	33.83
NO.1 H.F.O. SERV. TK.	2.5	10.0	15.49	-10.82	11.07	7.35
NO.1 H.F.O. SETTL. T.	3.0	10.0	18.65	-11.14	11.06	11.05
NO.2 H.F.O. SERV. TK.	1.9	10.0	12.65	-10.87	11.07	6.67
NO.2 H.F.O. SETTL. T.	2.2	10.0	9.70	-10.42	11.09	4.84
SUBTOTAL	156.7		43.35	-0.16	10.12	380.41
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	1.4	20.0	10.03	5.53	7.22	0.36
G/E L.O. STOR. TK.	2.6	20.0	10.04	7.48	7.96	10.28
M/E L.O. SETTL. TK.	3.5	20.0	8.83	6.21	7.80	19.67
M/E L.O. STOR. TK.	5.4	20.0	7.19	6.05	8.05	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	50.8		16.95	2.45	5.88	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	4.5	80.0	18.81	-2.29	1.40	1.69
BILGE TK.	12.9	80.0	11.52	0.00	1.14	9.85
F.O. DRAIN TK.	5.2	80.0	20.70	2.51	1.34	3.33
F.O. OVERFLOW TK.	7.2	80.0	23.04	2.98	1.31	7.97
L.O. DRAIN TK.	3.1	80.0	18.32	2.24	1.42	0.91
SEWAGE STOR. TK.	10.9	80.0	22.37	-2.83	1.31	10.07
SLUDGE TK.	12.0	80.0	21.80	-8.40	7.72	9.03
SUBTOTAL	55.8		19.16	-1.81	2.67	42.86
TOTAL	666.6		14.80	-0.05	8.68	789.41

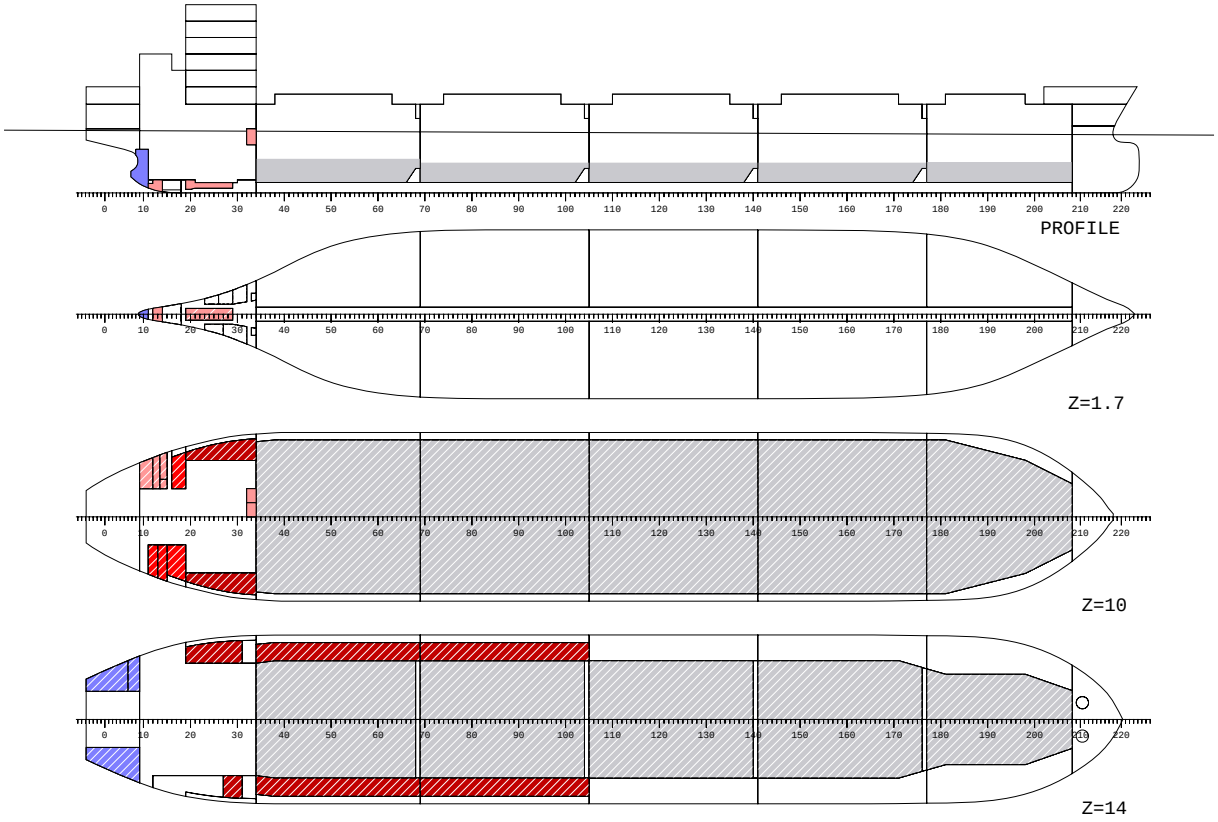


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	-0.001	-0.00	0.000	0.0	0.000
10.0	0.015	0.58	0.050	137.1	0.003
12.0	0.027	0.70	0.072	164.1	0.004
20.0	0.107	1.21	0.206	270.0	0.006
30.0	-0.070	1.55	0.450	394.7	0.009
40.0	-0.296	1.78	0.743	507.4	0.012
50.0	-0.819	1.66	1.049	604.7	0.014
60.0	-1.550	1.25	1.305	683.6	0.016
70.0	-2.345	0.69	1.476	741.8	0.017

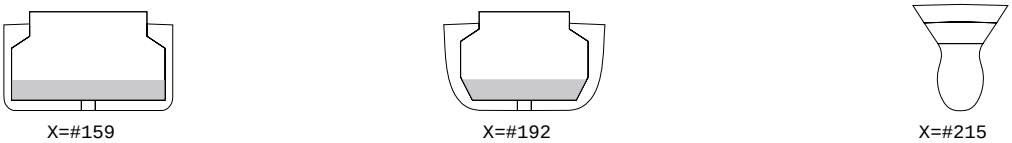


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	157	97	60.77	23.74
5.94	9.00	1546	392	87.31	57.34
18.60	25.00	4799	-7	106.91	163.32
25.80	34.00	3135	-472	220.92	220.31
40.20	52.00	-2744	-326	291.61	282.13
53.80	69.00	-6557	-291	285.33	297.98
68.20	87.00	-10178	-198	309.67	299.68
82.60	105.00	-12153	-123	376.21	298.55
97.00	123.00	-13117	-10	306.97	297.17
111.40	141.00	-12519	37	292.42	295.80
125.80	159.00	-11081	169	306.25	293.93
140.20	177.00	-7473	280	283.62	285.00
152.20	192.00	-4088	255	249.13	249.40
165.00	208.00	-542	290	135.20	127.39
169.20	214.00	104	44	38.08	76.76
173.40	220.00	72	-39	27.85	29.28

LOADING CASE 'LD11A -- HOMOGENEOUSLY HEAVY CARGO CONDITION,SHORT VOYAGE, DEPARTURE







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	0.0	0.00	0.00	0.00	0.00
Diesel Oil	DO	62.0	11.26	-2.98	8.46	149.80
Fresh Water	FW	159.2	2.90	-0.00	10.95	159.36
Grain Load	GRAIN	33786.9	93.43	0.00	3.59	0.00
Heavy Fuel Oil	HFO	783.5	42.77	-0.15	11.35	380.41
Lubricating Oil	LO	76.6	14.10	4.02	6.96	56.99
	MASS	34.6	63.09	0.00	20.29	0.00
Miscellaneous	MIS	7.0	19.24	-1.81	1.99	42.86
Deadweight		34909.8	91.52	-0.00	3.83	789.41
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		34909.8	91.52	-0.00	3.83	
Total weight		43559.8	89.09	-0.00	5.10	

F L O A T I N G P O S I T I O N

Draught moulded	10.270	m	KM	12.29 m
Trim	-0.702	m	KG	5.10 m
Heel, PS=+	0.0	deg		
TA	10.621	m	GM0	7.19 m
TF	9.919	m	GMCORR	-0.02 m
Trimming moment	-42472	tonm	GM	7.17 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.973	mrad	OK
AREA40	Area under GZ curve .	0.090	1.652	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.679	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.666	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	54.853	deg	OK
GM0.15	GM > 0.15 m	0.150	7.172	m	OK
IMOWEATH.	IMO weather criterion	1.000	3.568		OK
VISIBILI.		359.760	122.575		OK
PROPIMMER	Propeller should imm.	50.000	140.403	%	OK

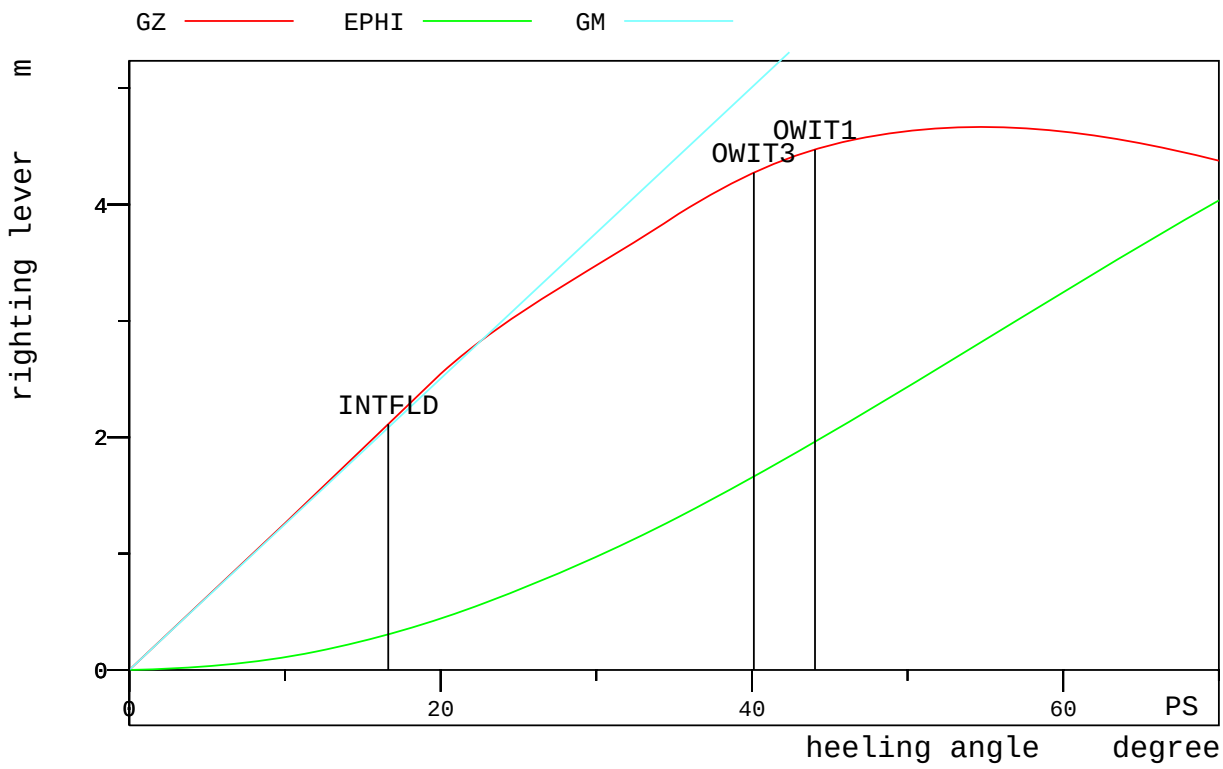
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-812.6 t	(13.5%)	POSITION:	36.6 m	47
SHEAR FORCE (MAX)	433.7 t	(7.2%)		138.0 m	174
SAGGING MOMENT	-22568.1 tm	(23.3%)		73.1 m	93
HOGGING MOMENT	1256.9 tm	(3.6%)		14.2 m	20

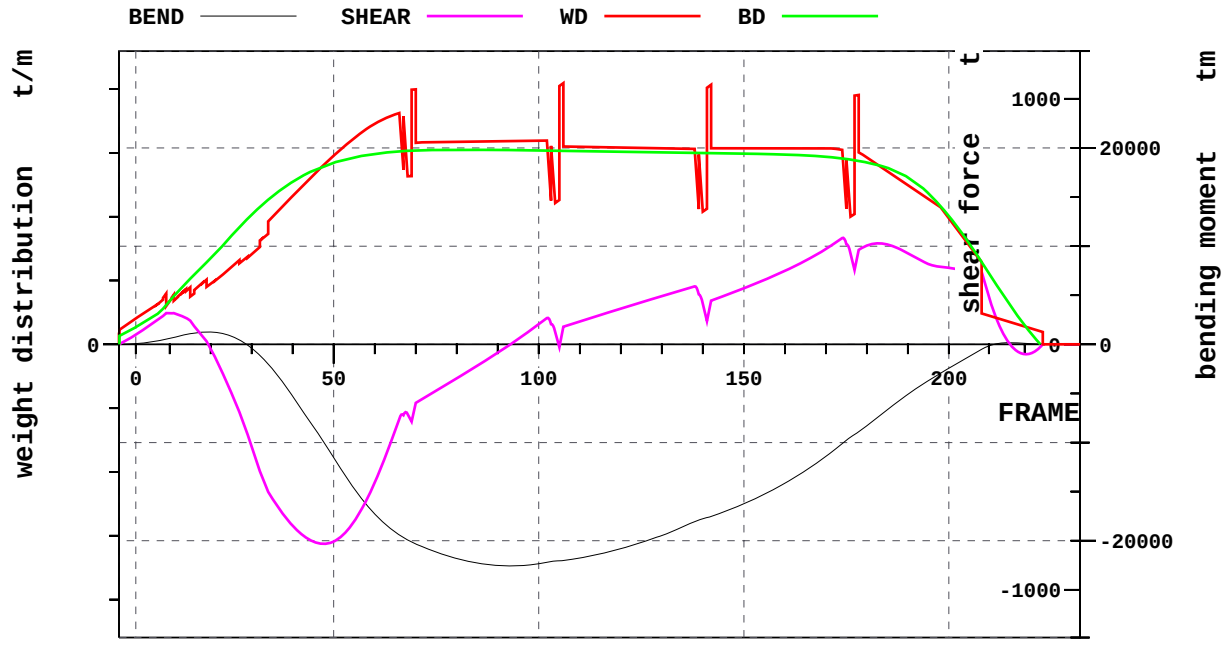
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 3.000)						
NO.1 CARGO HOLD	4933.9	24.7	150.88	0.00	3.59	0.0
NO.2 CARGO HOLD	7387.9	24.7	125.24	0.00	3.48	0.0
NO.3 CARGO HOLD	7397.1	24.7	96.45	0.00	3.48	0.0
NO.4 CARGO HOLD	7396.8	24.7	67.65	0.00	3.48	0.0
NO.5 CARGO HOLD	6671.1	24.7	40.96	0.00	3.97	0.0
SUBTOTAL	33786.9		93.43	0.00	3.59	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	9.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	34.6		63.09	0.00	20.29	0.0
TOTAL	33821.5		93.40	0.00	3.61	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	0.00	0.00	0.00	0.00
FORE PEAK TK.	0.0	0.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	10.3	50.0	8.24	-6.66	8.53	22.22
M.D.O. SETTL. TK.	12.4	50.0	9.83	-6.86	8.34	29.35
M.D.O. STOR. TK. (P)	17.3	50.0	12.68	6.80	8.45	43.37
M.D.O. STOR. TK. (S)	22.0	50.0	12.35	-6.76	8.50	54.86
SUBTOTAL	62.0		11.26	-2.98	8.46	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	20.4	50.0	4.98	7.32	11.97	31.97
FRESH WATER TK. (P)	46.9	50.0	1.02	6.48	12.05	44.74
FRESH WATER TK. (S)	67.2	50.0	2.26	-6.75	12.02	82.65
SUBTOTAL	159.2		2.90	-0.00	10.95	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	131.9	50.0	68.20	11.90	12.29	68.67
NO.1 FUEL OIL TK. (S)	131.9	50.0	68.20	-11.90	12.29	68.67
NO.2 FUEL OIL TK. (P)	127.7	50.0	39.94	11.88	12.30	67.49
NO.2 FUEL OIL TK. (S)	127.7	50.0	39.94	-11.88	12.30	67.49
H.F.O. TK. (P)	126.6	50.0	21.26	10.98	9.26	44.35
H.F.O. TK. (S)	90.0	50.0	21.60	-10.89	8.63	33.83
NO.1 H.F.O. SERV. TK.	12.4	50.0	15.47	-10.92	11.69	7.35
NO.1 H.F.O. SETTL. T.	14.8	50.0	18.64	-11.21	11.66	11.05
NO.2 H.F.O. SERV. TK.	9.7	50.0	12.65	-10.98	11.70	6.67
NO.2 H.F.O. SETTL. T.	11.0	50.0	9.65	-10.54	11.75	4.84
SUBTOTAL	783.5		42.77	-0.15	11.35	380.41
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	4.1	60.0	10.01	5.58	8.05	0.36
G/E L.O. STOR. TK.	7.9	60.0	10.02	8.01	8.78	10.28
M/E L.O. SETTL. TK.	10.5	60.0	8.82	6.85	8.65	19.67
M/E L.O. STOR. TK.	16.1	60.0	7.14	6.62	8.83	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	76.6		14.10	4.02	6.96	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	0.6	10.0	18.98	-1.96	0.68	1.69
BILGE TK.	1.6	10.0	11.66	0.00	0.32	9.85
F.O. DRAIN TK.	0.6	10.0	20.77	2.04	0.56	3.33
F.O. OVERFLOW TK.	0.9	10.0	23.05	2.35	0.50	7.97
L.O. DRAIN TK.	0.4	10.0	18.43	1.94	0.72	0.91
SEWAGE STOR. TK.	1.4	10.0	22.44	-2.26	0.52	10.07
SLUDGE TK.	1.5	10.0	21.80	-8.40	7.44	9.03
SUBTOTAL	7.0		19.24	-1.81	1.99	42.86
TOTAL	1088.3		32.98	-0.00	10.76	789.41

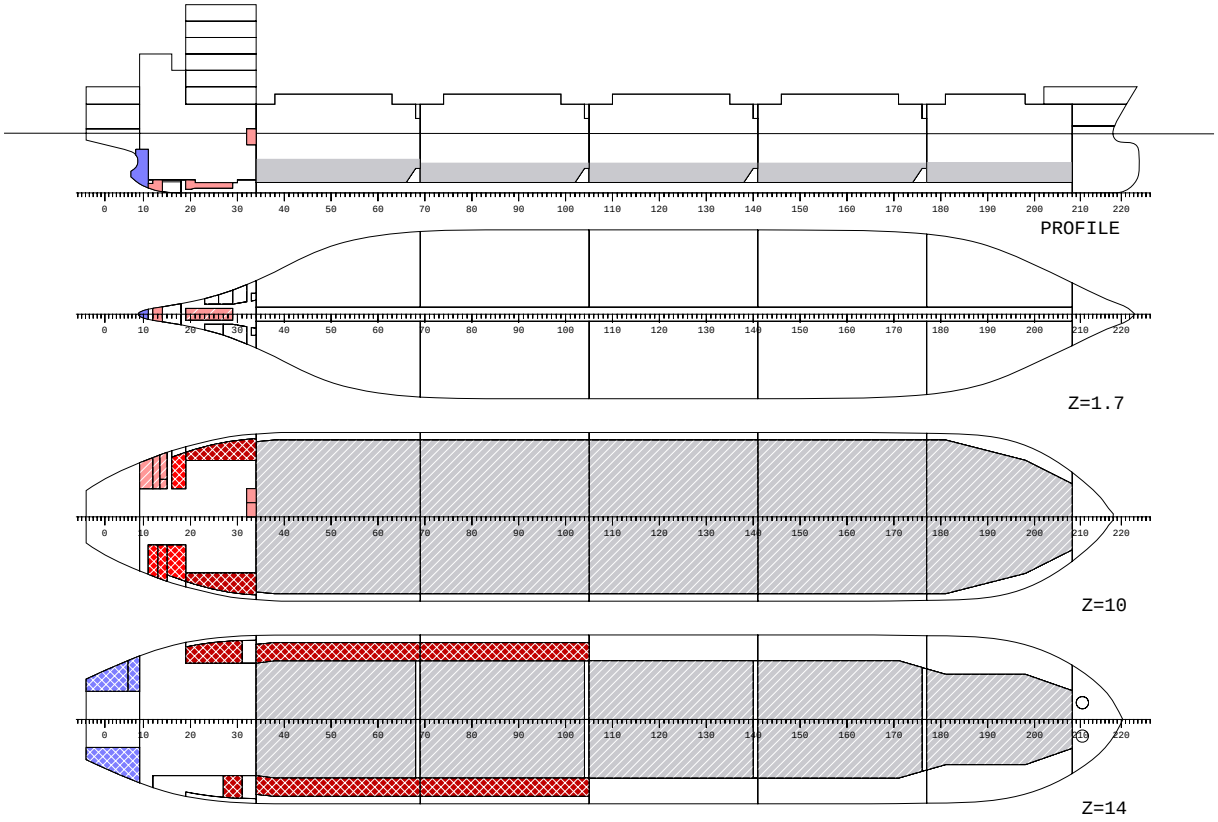


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	0.000	0.00	0.000	0.0	0.000
10.0	0.016	1.26	0.110	137.1	0.003
12.0	0.028	1.52	0.159	164.1	0.004
20.0	0.092	2.54	0.442	270.0	0.006
30.0	-0.107	3.48	0.973	394.7	0.009
40.0	-0.343	4.27	1.652	507.4	0.012
50.0	-0.864	4.63	2.435	604.7	0.014
60.0	-1.584	4.63	3.247	683.6	0.016
70.0	-2.363	4.38	4.035	741.8	0.017

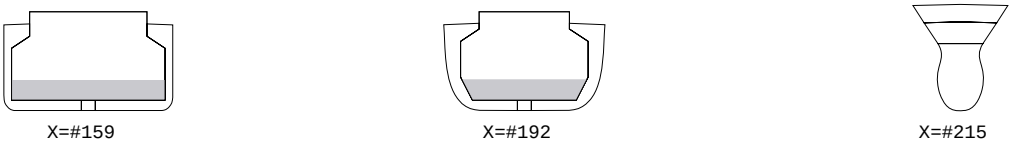


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	60	39	40.51	27.01
5.94	9.00	540	127	69.84	61.76
18.60	25.00	830	-196	121.01	169.17
25.80	34.00	-2023	-601	192.48	226.37
40.20	52.00	-12882	-780	306.52	287.93
53.80	69.00	-20084	-315	263.52	303.44
68.20	87.00	-22401	-66	318.01	304.78
82.60	105.00	-22047	-11	226.18	303.29
97.00	123.00	-20450	161	307.97	301.55
111.40	141.00	-17639	94	402.10	299.81
125.80	159.00	-14378	292	307.01	297.58
140.20	177.00	-9181	303	204.32	288.25
152.20	192.00	-4508	352	240.90	252.19
165.00	208.00	-502	296	128.88	128.78
169.20	214.00	146	45	38.08	77.46
173.40	220.00	103	-39	27.85	29.28

LOADING CASE 'LD11B -- HOMOGENEOUSLY HEAVY CARGO CONDITION,SHORT VOYAGE, ARRIVAL







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	0.0	0.00	0.00	0.00	0.00
Diesel Oil	DO	12.4	11.39	-2.61	7.47	149.80
Fresh Water	FW	51.6	4.32	-0.00	8.25	159.36
Grain Load	GRAIN	33786.9	93.43	0.00	3.59	0.00
Heavy Fuel Oil	HFO	156.7	43.35	-0.16	10.12	380.41
Lubricating Oil	LO	50.8	16.95	2.45	5.88	56.99
	MASS	27.6	75.29	0.00	20.42	0.00
Miscellaneous	MIS	55.8	19.16	-1.81	2.67	42.86
Deadweight		34141.9	92.79	-0.00	3.65	789.41
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		34141.9	92.79	-0.00	3.65	
Total weight		42791.9	90.06	-0.00	4.97	

F L O A T I N G P O S I T I O N

Draught moulded	10.115	m	KM	12.29 m
Trim	-0.079	m	KG	4.97 m
Heel, PS=+	0.0	deg		
TA	10.155	m	GM0	7.32 m
TF	10.076	m	GMCORR	-0.02 m
Trimming moment	-4743	tonm	GM	7.30 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.997	mrad	OK
AREA40	Area under GZ curve .	0.090	1.695	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.699	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	4.809	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	54.982	deg	OK
GM0.15	GM > 0.15 m	0.150	7.304	m	OK
IMOWEATH.	IMO weather criterion	1.000	3.990		OK
VISIBILI.		359.760	114.722		OK
PROPIMMER	Propeller should imm.	50.000	131.598	%	OK

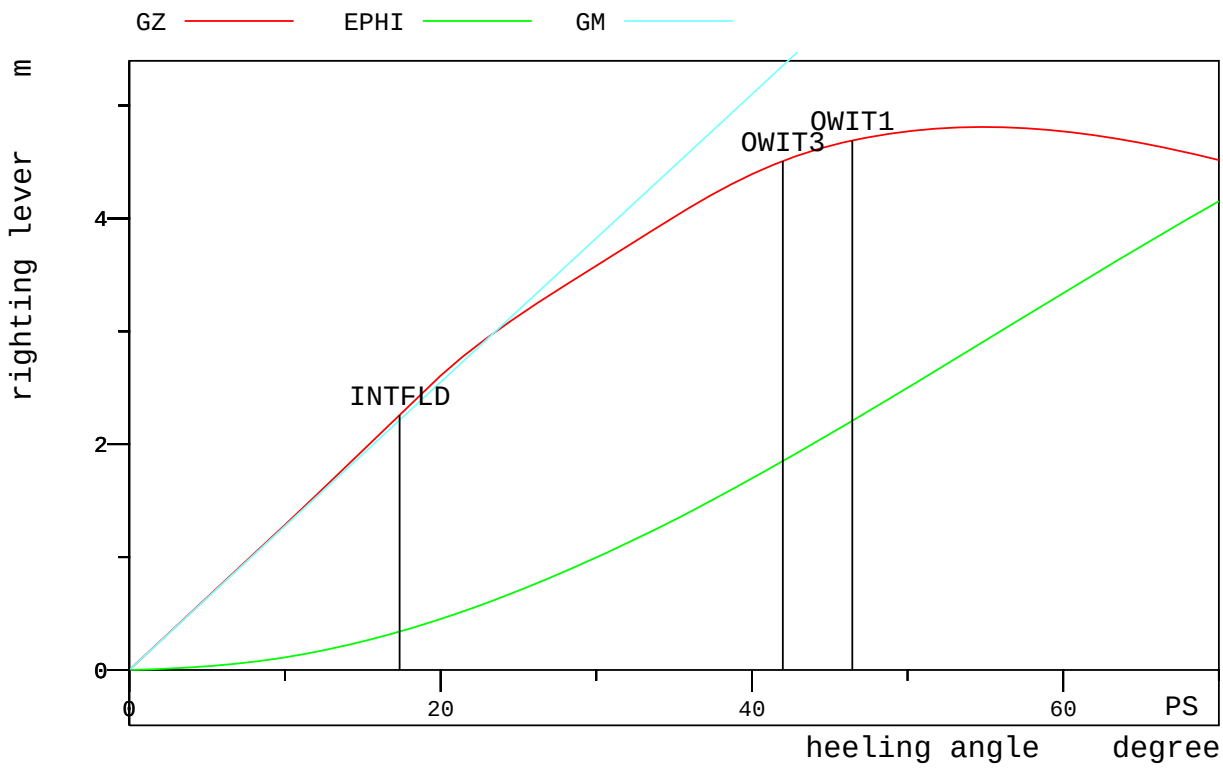
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-863.9 t	(14.4%)	POSITION:	36.0 m	47
SHEAR FORCE (MAX)	498.1 t	(8.3%)		138.0 m	174
SAGGING MOMENT	-25490.6 tm	(26.3%)		75.4 m	96
HOGGING MOMENT	805.8 tm	(2.6%)		12.6 m	18

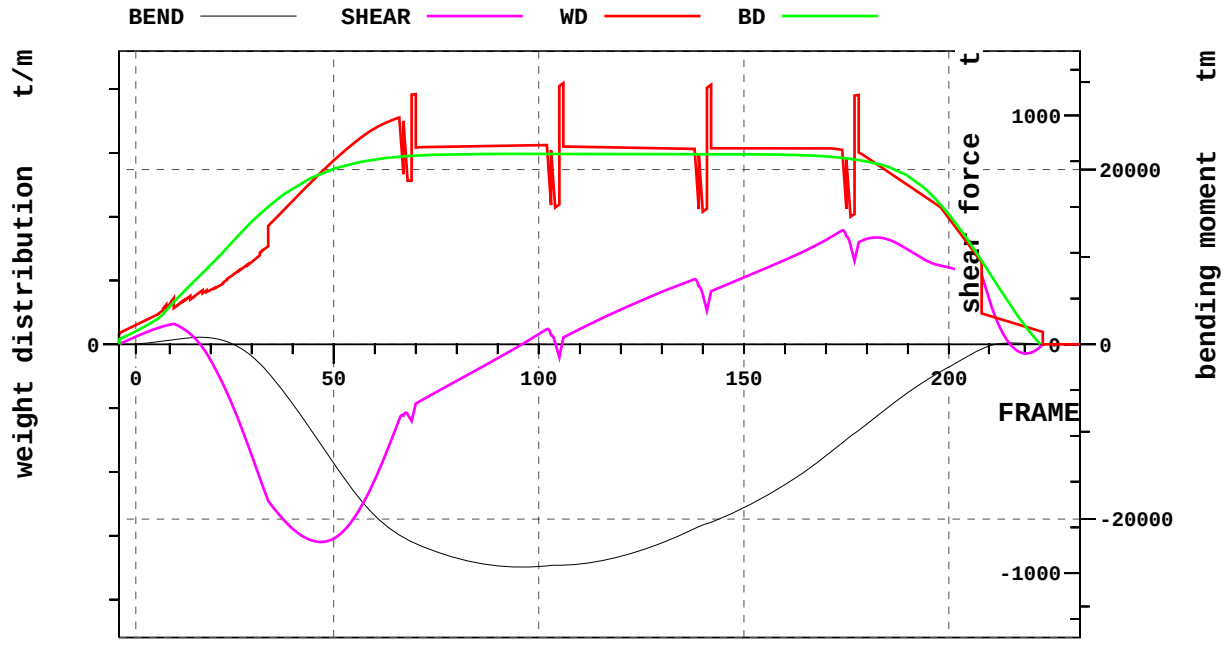
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 3.000)						
NO.1 CARGO HOLD	4933.9	24.7	150.88	0.00	3.59	0.0
NO.2 CARGO HOLD	7387.9	24.7	125.24	0.00	3.48	0.0
NO.3 CARGO HOLD	7397.1	24.7	96.45	0.00	3.48	0.0
NO.4 CARGO HOLD	7396.8	24.7	67.65	0.00	3.48	0.0
NO.5 CARGO HOLD	6671.1	24.7	40.96	0.00	3.97	0.0
SUBTOTAL	33786.9		93.43	0.00	3.59	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	2.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	27.6		75.29	0.00	20.42	0.0
TOTAL	33814.5		93.42	0.00	3.61	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	0.00	0.00	0.00	0.00
FORE PEAK TK.	0.0	0.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	2.1	10.0	8.28	-5.84	7.59	22.22
M.D.O. SETTL. TK.	2.5	10.0	9.88	-5.95	7.33	29.35
M.D.O. STOR. TK. (P)	3.5	10.0	12.81	5.91	7.45	43.37
M.D.O. STOR. TK. (S)	4.4	10.0	12.58	-5.91	7.52	54.86
SUBTOTAL	12.4		11.39	-2.61	7.47	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	4.1	10.0	4.98	7.15	11.13	31.97
FRESH WATER TK. (P)	9.4	10.0	1.14	6.32	11.15	44.74
FRESH WATER TK. (S)	13.4	10.0	2.40	-6.59	11.14	82.65
SUBTOTAL	51.6		4.32	-0.00	8.25	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	26.4	10.0	68.20	12.54	11.35	68.67
NO.1 FUEL OIL TK. (S)	26.4	10.0	68.20	-12.54	11.35	68.67
NO.2 FUEL OIL TK. (P)	25.5	10.0	40.16	12.52	11.37	67.49
NO.2 FUEL OIL TK. (S)	25.5	10.0	40.16	-12.52	11.37	67.49
H.F.O. TK. (P)	25.3	10.0	23.07	10.63	7.07	44.35
H.F.O. TK. (S)	18.0	10.0	23.44	-10.57	6.80	33.83
NO.1 H.F.O. SERV. TK.	2.5	10.0	15.49	-10.82	11.07	7.35
NO.1 H.F.O. SETTL. T.	3.0	10.0	18.65	-11.14	11.06	11.05
NO.2 H.F.O. SERV. TK.	1.9	10.0	12.65	-10.87	11.07	6.67
NO.2 H.F.O. SETTL. T.	2.2	10.0	9.70	-10.42	11.09	4.84
SUBTOTAL	156.7		43.35	-0.16	10.12	380.41
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	1.4	20.0	10.03	5.53	7.22	0.36
G/E L.O. STOR. TK.	2.6	20.0	10.04	7.48	7.96	10.28
M/E L.O. SETTL. TK.	3.5	20.0	8.83	6.21	7.80	19.67
M/E L.O. STOR. TK.	5.4	20.0	7.19	6.05	8.05	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	50.8		16.95	2.45	5.88	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	4.5	80.0	18.81	-2.29	1.40	1.69
BILGE TK.	12.9	80.0	11.52	0.00	1.14	9.85
F.O. DRAIN TK.	5.2	80.0	20.70	2.51	1.34	3.33
F.O. OVERFLOW TK.	7.2	80.0	23.04	2.98	1.31	7.97
L.O. DRAIN TK.	3.1	80.0	18.32	2.24	1.42	0.91
SEWAGE STOR. TK.	10.9	80.0	22.37	-2.83	1.31	10.07
SLUDGE TK.	12.0	80.0	21.80	-8.40	7.72	9.03
SUBTOTAL	55.8		19.16	-1.81	2.67	42.86
TOTAL	327.4		27.76	-0.10	7.80	789.41

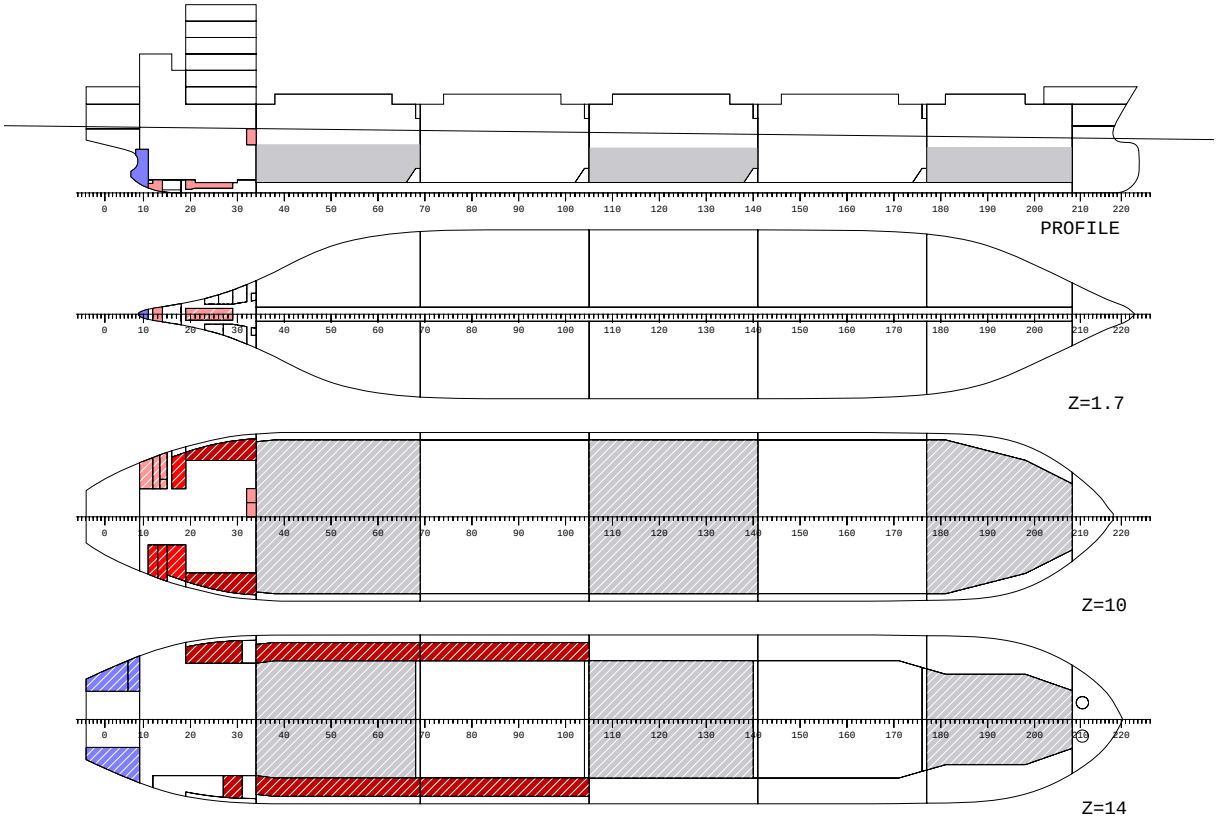


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	0.000	0.00	0.000	0.0	0.000
10.0	0.017	1.29	0.112	137.1	0.003
12.0	0.029	1.55	0.162	164.1	0.004
20.0	0.109	2.61	0.452	270.0	0.006
30.0	-0.072	3.58	0.997	394.7	0.009
40.0	-0.303	4.39	1.695	507.4	0.012
50.0	-0.825	4.77	2.502	604.7	0.014
60.0	-1.554	4.77	3.339	683.6	0.016
70.0	-2.347	4.52	4.152	741.8	0.017

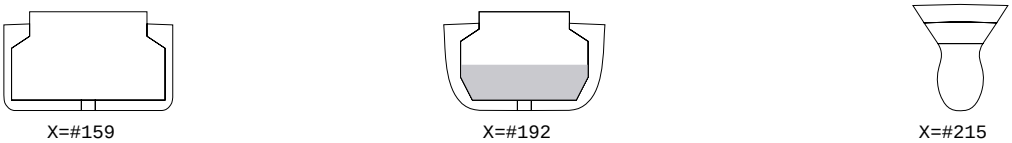


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	56	33	30.57	20.48
5.94	9.00	410	84	55.64	53.09
18.60	25.00	106	-255	106.86	158.33
25.80	34.00	-3229	-684	185.70	215.53
40.20	52.00	-14953	-822	299.19	278.45
53.80	69.00	-22556	-336	256.50	295.41
68.20	87.00	-25165	-90	310.69	298.29
82.60	105.00	-25278	-56	219.19	298.34
97.00	123.00	-23859	176	307.97	298.15
111.40	141.00	-20521	147	402.10	297.95
125.80	159.00	-16348	360	307.01	297.26
140.20	177.00	-10175	365	204.32	289.46
152.20	192.00	-4866	392	240.90	254.54
165.00	208.00	-540	305	128.88	130.83
169.20	214.00	133	47	38.08	78.66
173.40	220.00	96	-39	27.85	29.28

LOADING CASE 'LD12A -- ALTERNATIVE HEAVY CARGO CONDITION,SHORT VOYAGE, DEPARTURE







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	0.0	0.00	0.00	0.00	0.00
Diesel Oil	DO	62.0	11.26	-2.98	8.46	149.80
Fresh Water	FW	159.2	2.90	-0.00	10.95	159.36
Grain Load	GRAIN	33859.7	91.11	0.00	5.00	0.00
Heavy Fuel Oil	HFO	783.5	42.77	-0.15	11.35	380.41
Lubricating Oil	LO	76.6	14.10	4.02	6.96	56.99
	MASS	34.6	63.09	0.00	20.29	0.00
Miscellaneous	MIS	7.0	19.24	-1.81	1.99	42.86
Deadweight		34982.6	89.27	-0.00	5.19	789.41
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		34982.6	89.27	-0.00	5.19	
Total weight		43632.6	87.29	-0.00	6.19	

F L O A T I N G P O S I T I O N

Draught moulded	10.268	m	KM	12.32 m
Trim	-2.000	m	KG	6.19 m
Heel, PS=+	0.0	deg		
TA	11.268	m	GM0	6.14 m
TF	9.268	m	GMCORR	-0.02 m
Trimming moment	-121253	tonm	GM	6.12 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.830	mrad	OK
AREA40	Area under GZ curve .	0.090	1.274	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.444	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	3.810	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	50.854	deg	OK
GM0.15	GM > 0.15 m	0.150	6.117	m	OK
IMOWEATH.	IMO weather criterion	1.000	2.903		OK
VISIBILI.		359.760	146.440		OK
PROPIMMER	Propeller should imm.	50.000	152.606	%	OK

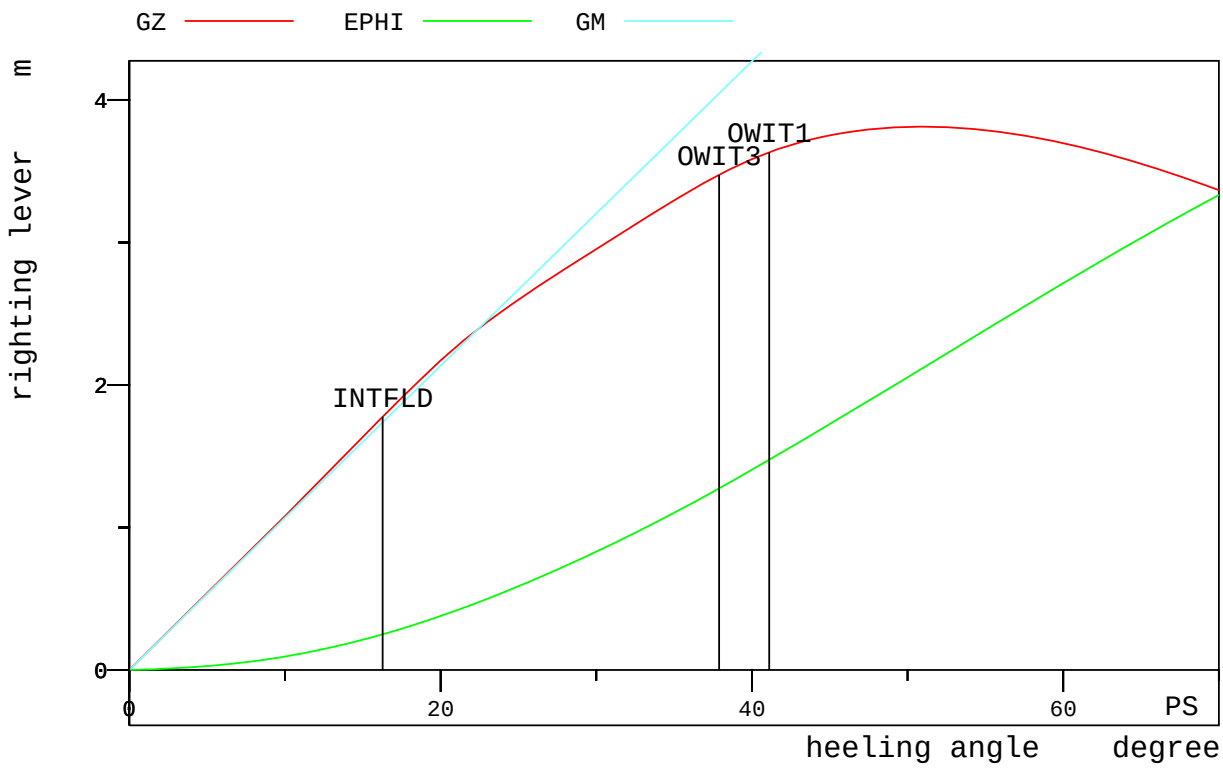
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-3911.2 t	(65.2%)	POSITION:	140.2 m	177
SHEAR FORCE (MAX)	4254.4 t	(70.9%)		53.8 m	69
SAGGING MOMENT	-10561.2 tm	(16.5%)		33.1 m	43
HOGGING MOMENT	71286.0 tm	(68.5%)		123.1 m	156

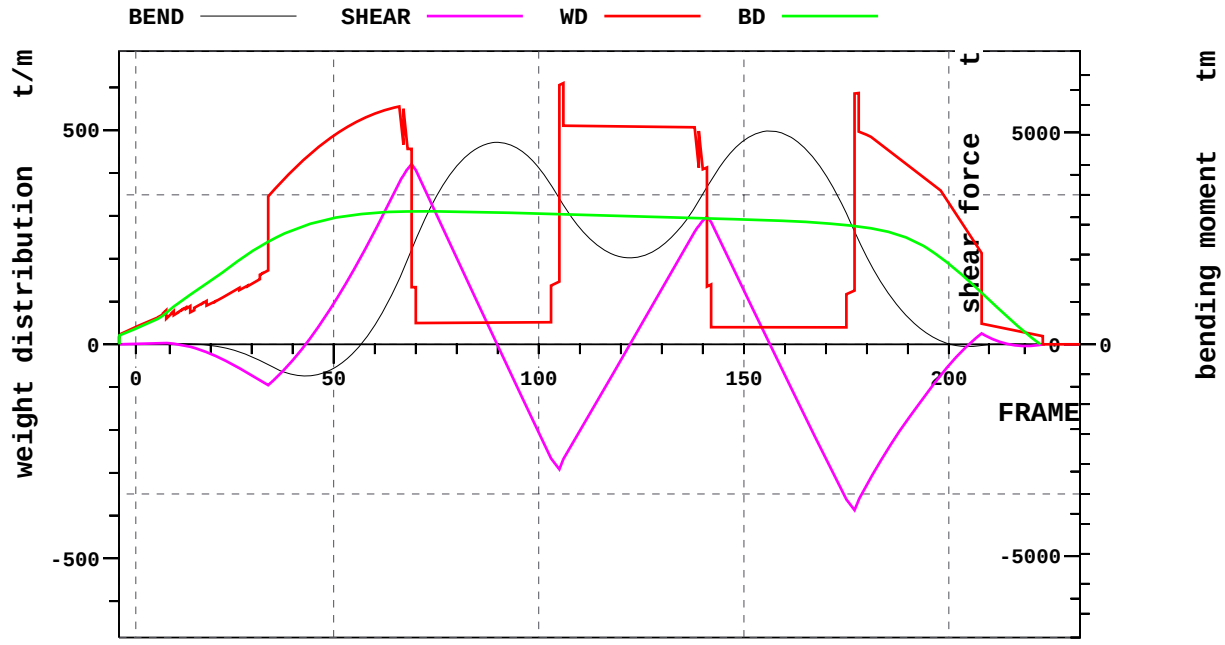
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 3.000)						
NO.1 CARGO HOLD	8791.7	44.1	150.98	0.00	4.89	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	13180.8	44.0	96.69	0.00	4.78	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	11887.2	44.0	40.63	0.00	5.32	0.0
SUBTOTAL	33859.7		91.11	0.00	5.00	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	9.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	34.6		63.09	0.00	20.29	0.0
TOTAL	33894.3		91.08	0.00	5.01	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	0.00	0.00	0.00	0.00
FORE PEAK TK.	0.0	0.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	10.3	50.0	8.24	-6.66	8.53	22.22
M.D.O. SETTL. TK.	12.4	50.0	9.83	-6.86	8.34	29.35
M.D.O. STOR. TK. (P)	17.3	50.0	12.68	6.80	8.45	43.37
M.D.O. STOR. TK. (S)	22.0	50.0	12.35	-6.76	8.50	54.86
SUBTOTAL	62.0		11.26	-2.98	8.46	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	20.4	50.0	4.98	7.32	11.97	31.97
FRESH WATER TK. (P)	46.9	50.0	1.02	6.48	12.05	44.74
FRESH WATER TK. (S)	67.2	50.0	2.26	-6.75	12.02	82.65
SUBTOTAL	159.2		2.90	-0.00	10.95	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	131.9	50.0	68.20	11.90	12.29	68.67
NO.1 FUEL OIL TK. (S)	131.9	50.0	68.20	-11.90	12.29	68.67
NO.2 FUEL OIL TK. (P)	127.7	50.0	39.94	11.88	12.30	67.49
NO.2 FUEL OIL TK. (S)	127.7	50.0	39.94	-11.88	12.30	67.49
H.F.O. TK. (P)	126.6	50.0	21.26	10.98	9.26	44.35
H.F.O. TK. (S)	90.0	50.0	21.60	-10.89	8.63	33.83
NO.1 H.F.O. SERV. TK.	12.4	50.0	15.47	-10.92	11.69	7.35
NO.1 H.F.O. SETTL. T.	14.8	50.0	18.64	-11.21	11.66	11.05
NO.2 H.F.O. SERV. TK.	9.7	50.0	12.65	-10.98	11.70	6.67
NO.2 H.F.O. SETTL. T.	11.0	50.0	9.65	-10.54	11.75	4.84
SUBTOTAL	783.5		42.77	-0.15	11.35	380.41
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	4.1	60.0	10.01	5.58	8.05	0.36
G/E L.O. STOR. TK.	7.9	60.0	10.02	8.01	8.78	10.28
M/E L.O. SETTL. TK.	10.5	60.0	8.82	6.85	8.65	19.67
M/E L.O. STOR. TK.	16.1	60.0	7.14	6.62	8.83	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	76.6		14.10	4.02	6.96	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	0.6	10.0	18.98	-1.96	0.68	1.69
BILGE TK.	1.6	10.0	11.66	0.00	0.32	9.85
F.O. DRAIN TK.	0.6	10.0	20.77	2.04	0.56	3.33
F.O. OVERFLOW TK.	0.9	10.0	23.05	2.35	0.50	7.97
L.O. DRAIN TK.	0.4	10.0	18.43	1.94	0.72	0.91
SEWAGE STOR. TK.	1.4	10.0	22.44	-2.26	0.52	10.07
SLUDGE TK.	1.5	10.0	21.80	-8.40	7.44	9.03
SUBTOTAL	7.0		19.24	-1.81	1.99	42.86
TOTAL	1088.3		32.98	-0.00	10.76	789.41

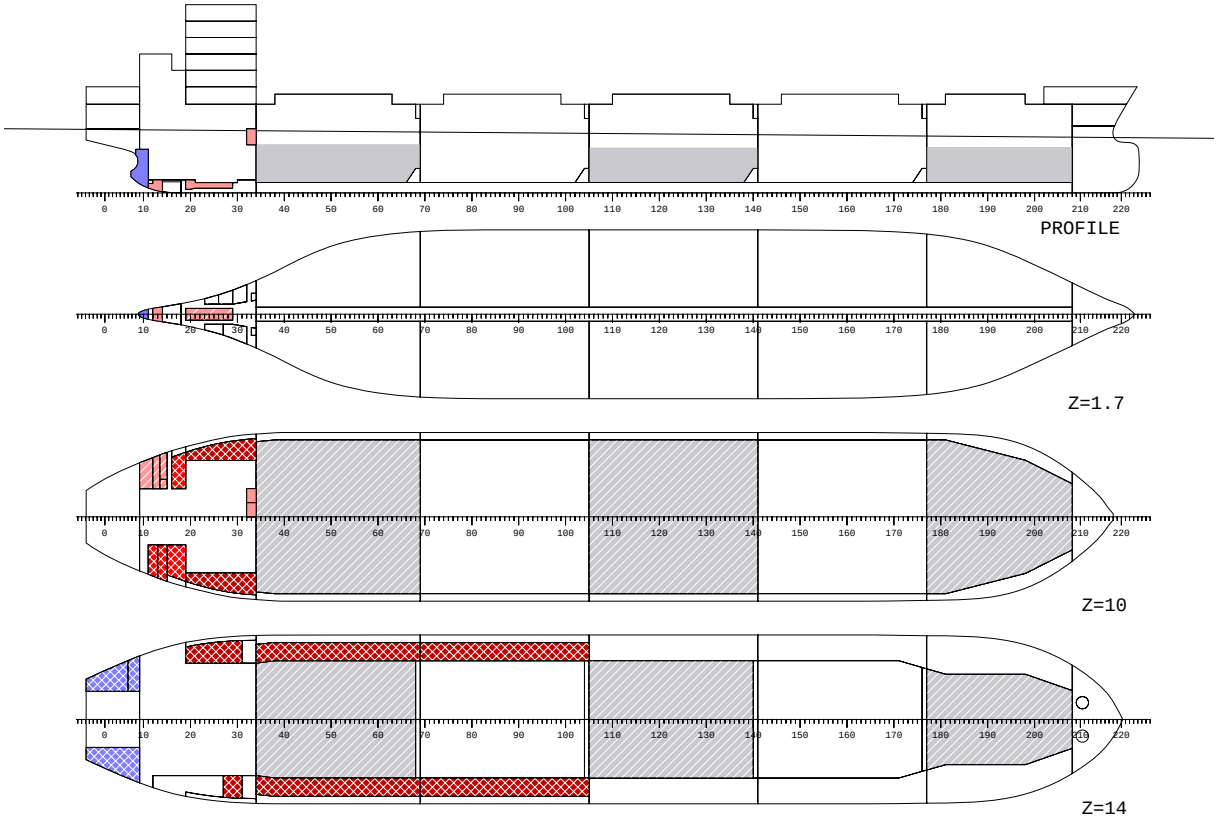


HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	0.000	0.00	0.000	0.0	0.000
10.0	0.017	1.08	0.094	137.1	0.003
12.0	0.029	1.30	0.136	164.1	0.004
20.0	0.079	2.17	0.379	270.0	0.006
30.0	-0.105	2.95	0.830	394.7	0.009
40.0	-0.347	3.59	1.404	507.4	0.012
50.0	-0.877	3.81	2.056	604.7	0.014
60.0	-1.600	3.70	2.714	683.6	0.016
70.0	-2.382	3.37	3.333	741.8	0.017

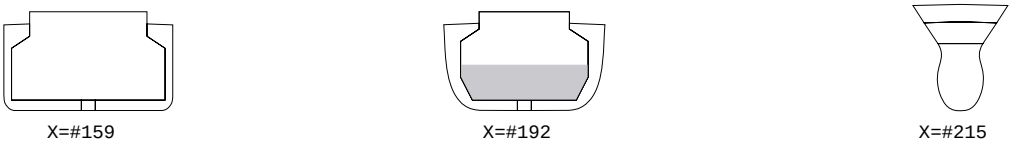


X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	23	12	40.57	36.58
5.94	9.00	165	35	69.76	73.85
18.60	25.00	-1744	-457	121.14	183.06
25.80	34.00	-6822	-961	259.08	239.55
40.20	52.00	-6204	1279	500.02	298.09
53.80	69.00	31654	4254	456.41	310.56
68.20	87.00	66875	580	50.68	308.68
82.60	105.00	48791	-2951	146.47	303.98
97.00	123.00	28942	126	508.79	299.03
111.40	141.00	52466	3010	135.03	294.08
125.80	159.00	70743	-532	39.94	288.63
140.20	177.00	37745	-3911	125.51	276.22
152.20	192.00	6479	-1498	403.55	238.35
165.00	208.00	-422	256	212.99	119.63
169.20	214.00	134	36	38.08	72.39
173.40	220.00	85	-39	27.85	29.28

LOADING CASE 'LD12B -- ALTERNATIVE HEAVY CARGO CONDITION,SHORT VOYAGE, ARRIVAL







NAME	PURP	MASS t	XM m	YM m	ZM m	FRSM tm
Ballast Water	BW	0.0	0.00	0.00	0.00	0.00
Diesel Oil	DO	12.4	11.39	-2.61	7.47	149.80
Fresh Water	FW	51.6	4.32	-0.00	8.25	159.36
Grain Load	GRAIN	33859.7	91.11	0.00	5.00	0.00
Heavy Fuel Oil	HFO	156.7	43.35	-0.16	10.12	380.41
Lubricating Oil	LO	50.8	16.95	2.45	5.88	56.99
	MASS	27.6	75.29	0.00	20.42	0.00
Miscellaneous	MIS	55.8	19.16	-1.81	2.67	42.86
Deadweight		34214.6	90.49	-0.00	5.04	789.41
Lightweight		8650.0	79.30	0.00	10.20	
Deadweight		34214.6	90.49	-0.00	5.04	
Total weight		42864.6	88.23	-0.00	6.08	

F L O A T I N G P O S I T I O N

Draught moulded	10.115	m	KM	12.33 m
Trim	-1.381	m	KG	6.08 m
Heel, PS=+	0.0	deg		
TA	10.805	m	GM0	6.25 m
TF	9.424	m	GMCORR	-0.02 m
Trimming moment	-83249	tonm	GM	6.23 m
Heel	0.0	deg (SB)		

Stability Criteria

RCR	TEXT	REQ	ATTV	UNIT	STAT
AREA30	Area under GZ curve .	0.055	0.853	mrad	OK
AREA40	Area under GZ curve .	0.090	1.426	mrad	OK
AREA3040	Area under GZ curve .	0.030	0.573	mrad	OK
GZ0.2	Max GZ > 0.2	0.200	3.943	m	OK
MAXGZ25	Max. GZ at an angle .	25.000	50.943	deg	OK
GM0.15	GM > 0.15 m	0.150	6.229	m	OK
IMOWEATH.	IMO weather criterion	1.000	3.229		OK
VISIBILI.		359.760	136.597		OK
PROPIMMER	Propeller should imm.	50.000	143.870	%	OK

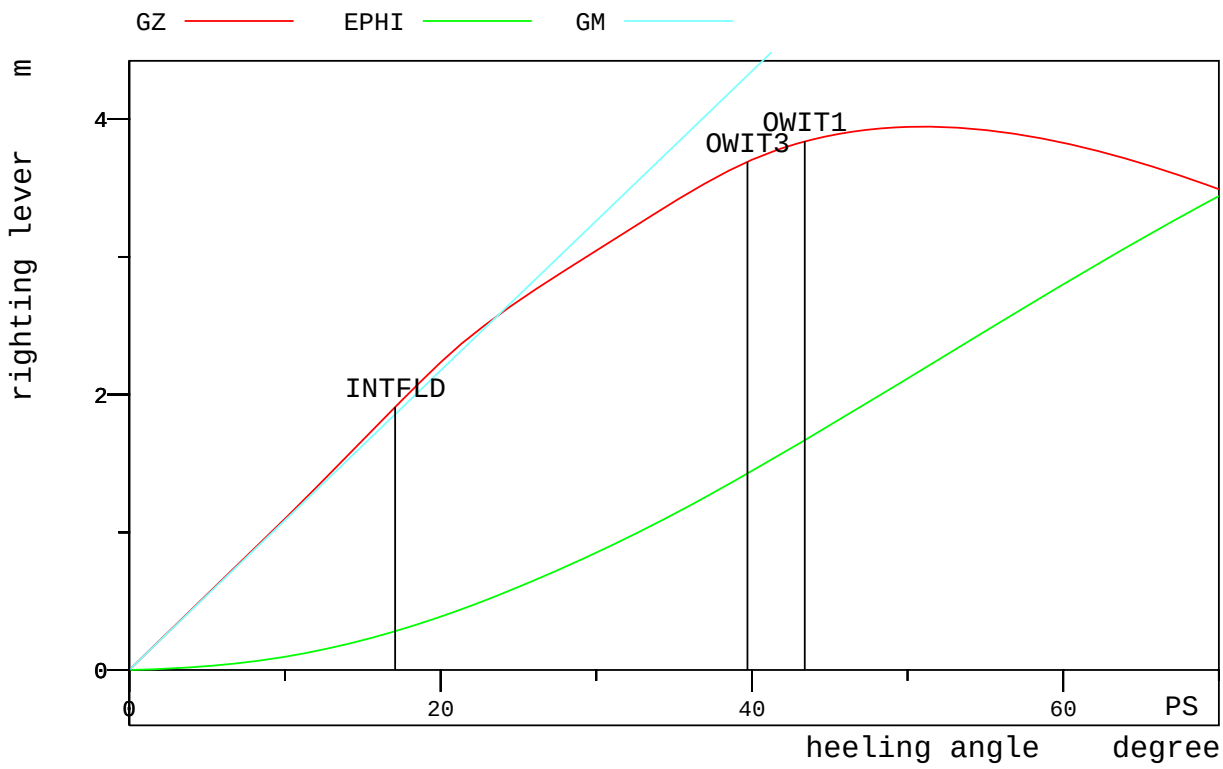
Longitudinal Strength

				X	FRAME
SHEAR FORCE (MIN)	-3849.7 t	(64.2%)	POSITION:	140.2 m	177
SHEAR FORCE (MAX)	4236.6 t	(70.6%)		53.8 m	69
SAGGING MOMENT	-12192.1 tm	(18.9%)		33.3 m	43
HOGGING MOMENT	69204.6 tm	(66.5%)		123.1 m	156

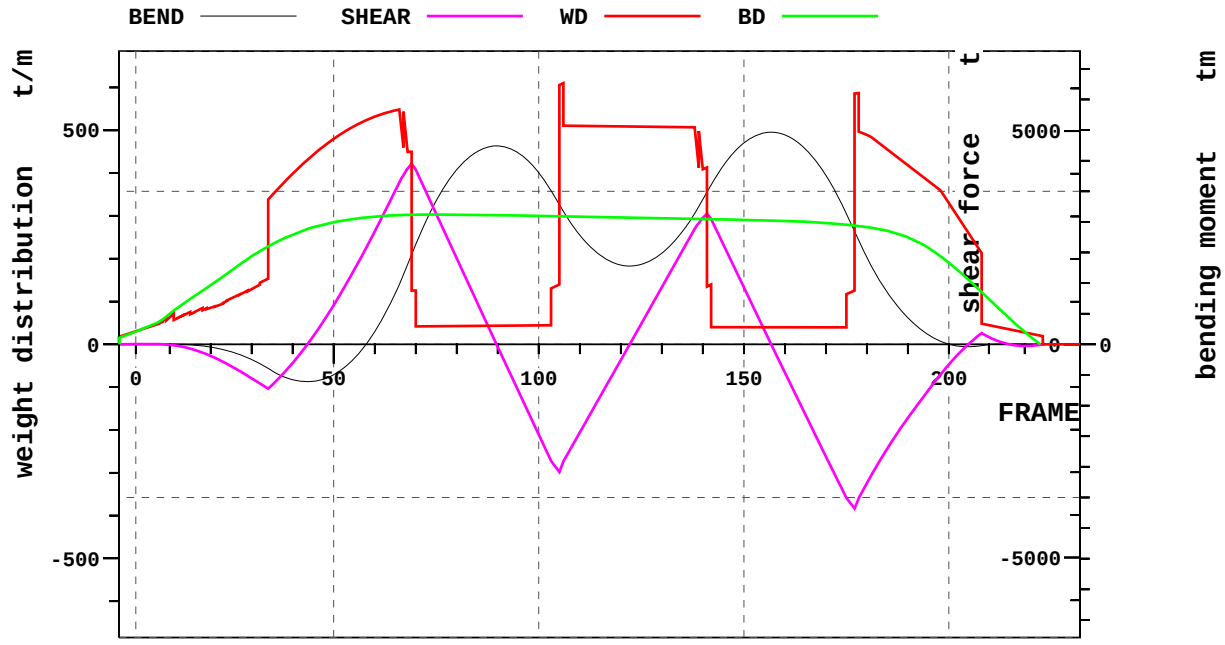
DES	MASS t	FILL %	CGX m	CGY m	CGZ m	GRM tm
Grain Load, (Dens = 3.000)						
NO.1 CARGO HOLD	8791.7	44.1	150.98	0.00	4.89	0.0
NO.2 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.3 CARGO HOLD	13180.8	44.0	96.69	0.00	4.78	0.0
NO.4 CARGO HOLD	0.0	0.0	0.00	0.00	0.00	0.0
NO.5 CARGO HOLD	11887.2	44.0	40.63	0.00	5.32	0.0
SUBTOTAL	33859.7		91.11	0.00	5.00	0.0
Mass Load, (Dens = 1.000)						
CREW & EFFECT	5.6	0.0	15.00	0.00	25.00	0.0
PROVISION 100%	2.0	0.0	15.00	0.00	19.80	0.0
SPARE	20.0	0.0	98.20	0.00	19.20	0.0
SUBTOTAL	27.6		75.29	0.00	20.42	0.0
TOTAL	33887.3		91.09	0.00	5.01	0.0

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Ballast Water, (Dens = 1.025)						
AFT PEAK TK.	0.0	0.0	0.00	0.00	0.00	0.00
FORE PEAK TK.	0.0	0.0	168.53	0.00	5.89	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	9.30	7.03	0.00
NO.1 WATER BALLAST T.	0.0	0.0	151.92	-9.30	7.03	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	10.47	5.87	0.00
NO.2 WATER BALLAST T.	0.0	0.0	125.72	-10.47	5.87	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	10.53	5.79	0.00
NO.3 WATER BALLAST T.	0.0	0.0	97.00	-10.53	5.79	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	9.66	2.70	0.00
NO.4 WATER BALLAST T.	0.0	0.0	68.29	-9.66	2.70	0.00
NO.5 W.B.T.(P)	0.0	0.0	39.87	9.38	3.43	0.00
NO.5 WATER BALLAST T.	0.0	0.0	39.87	-9.38	3.43	0.00
SUBTOTAL	0.0		0.00	0.00	0.00	0.00
Diesel Oil, (Dens = 0.850)						
M.D.O. SERV. TK.	2.1	10.0	8.28	-5.84	7.59	22.22
M.D.O. SETTL. TK.	2.5	10.0	9.88	-5.95	7.33	29.35
M.D.O. STOR. TK. (P)	3.5	10.0	12.81	5.91	7.45	43.37
M.D.O. STOR. TK. (S)	4.4	10.0	12.58	-5.91	7.52	54.86
SUBTOTAL	12.4		11.39	-2.61	7.47	149.80

DES	MASS t	FILL %	CGX m	CGY m	CGZ m	FRSM tm
Fresh Water, (Dens = 1.000)						
COOLING WATER TK.	24.7	100.0	6.47	0.00	5.10	0.00
DISTILLED WATER TK.	4.1	10.0	4.98	7.15	11.13	31.97
FRESH WATER TK. (P)	9.4	10.0	1.14	6.32	11.15	44.74
FRESH WATER TK. (S)	13.4	10.0	2.40	-6.59	11.14	82.65
SUBTOTAL	51.6		4.32	-0.00	8.25	159.36
Heavy Fuel Oil, (Dens = 0.980)						
NO.1 FUEL OIL TK. (P)	26.4	10.0	68.20	12.54	11.35	68.67
NO.1 FUEL OIL TK. (S)	26.4	10.0	68.20	-12.54	11.35	68.67
NO.2 FUEL OIL TK. (P)	25.5	10.0	40.16	12.52	11.37	67.49
NO.2 FUEL OIL TK. (S)	25.5	10.0	40.16	-12.52	11.37	67.49
H.F.O. TK. (P)	25.3	10.0	23.07	10.63	7.07	44.35
H.F.O. TK. (S)	18.0	10.0	23.44	-10.57	6.80	33.83
NO.1 H.F.O. SERV. TK.	2.5	10.0	15.49	-10.82	11.07	7.35
NO.1 H.F.O. SETTL. T.	3.0	10.0	18.65	-11.14	11.06	11.05
NO.2 H.F.O. SERV. TK.	1.9	10.0	12.65	-10.87	11.07	6.67
NO.2 H.F.O. SETTL. T.	2.2	10.0	9.70	-10.42	11.09	4.84
SUBTOTAL	156.7		43.35	-0.16	10.12	380.41
Lubricating Oil, (Dens = 0.900)						
G/E L.O. SETTL. TK.	1.4	20.0	10.03	5.53	7.22	0.36
G/E L.O. STOR. TK.	2.6	20.0	10.04	7.48	7.96	10.28
M/E L.O. SETTL. TK.	3.5	20.0	8.83	6.21	7.80	19.67
M/E L.O. STOR. TK.	5.4	20.0	7.19	6.05	8.05	26.68
M/E L.O. SUMP TK.	14.6	98.0	17.48	0.00	1.30	0.00
NO.1 CYL L.O. TK.	9.0	98.0	25.00	3.60	9.52	0.00
NO.2 CYL L.O. TK.	9.0	98.0	25.00	1.20	9.52	0.00
S/T L.O. SUMP TK.	5.4	98.0	8.88	-0.00	1.40	0.00
SUBTOTAL	50.8		16.95	2.45	5.88	56.99
Miscellaneous, (Dens = 1.000)						
BILGE SEP. OIL TK.	4.5	80.0	18.81	-2.29	1.40	1.69
BILGE TK.	12.9	80.0	11.52	0.00	1.14	9.85
F.O. DRAIN TK.	5.2	80.0	20.70	2.51	1.34	3.33
F.O. OVERFLOW TK.	7.2	80.0	23.04	2.98	1.31	7.97
L.O. DRAIN TK.	3.1	80.0	18.32	2.24	1.42	0.91
SEWAGE STOR. TK.	10.9	80.0	22.37	-2.83	1.31	10.07
SLUDGE TK.	12.0	80.0	21.80	-8.40	7.72	9.03
SUBTOTAL	55.8		19.16	-1.81	2.67	42.86
TOTAL	327.4		27.76	-0.10	7.80	789.41



HEEL degree	MS m	HPHI m	EPHI rad*m	FSMOM tm	DGZ m
0.0	0.000	0.00	0.000	0.0	0.000
10.0	0.017	1.10	0.096	137.1	0.003
12.0	0.029	1.32	0.138	164.1	0.004
20.0	0.104	2.23	0.387	270.0	0.006
30.0	-0.069	3.05	0.853	394.7	0.009
40.0	-0.298	3.71	1.445	507.4	0.012
50.0	-0.829	3.94	2.119	604.7	0.014
60.0	-1.565	3.83	2.801	683.6	0.016
70.0	-2.362	3.49	3.442	741.8	0.017



X m	FR #	BEND tm	SHEAR t	WD t/m	BD t/m
0.00	0.00	17	8	30.63	29.67
5.94	9.00	43	-6	55.66	64.94
18.60	25.00	-2419	-511	106.99	172.19
25.80	34.00	-7953	-1039	338.35	228.76
40.20	52.00	-8172	1239	492.69	288.68
53.80	69.00	29298	4237	449.69	302.60
68.20	87.00	64226	556	43.35	302.26
82.60	105.00	45662	-2995	139.78	299.08
97.00	123.00	25631	142	508.79	295.66
111.40	141.00	49672	3064	135.03	292.25
125.80	159.00	68845	-463	39.94	288.33
140.20	177.00	36803	-3850	125.51	277.44
152.20	192.00	6155	-1459	403.55	240.69
165.00	208.00	-442	265	212.99	121.61
169.20	214.00	136	38	38.08	73.56
173.40	220.00	89	-39	27.85	29.28

6. APPENDIX

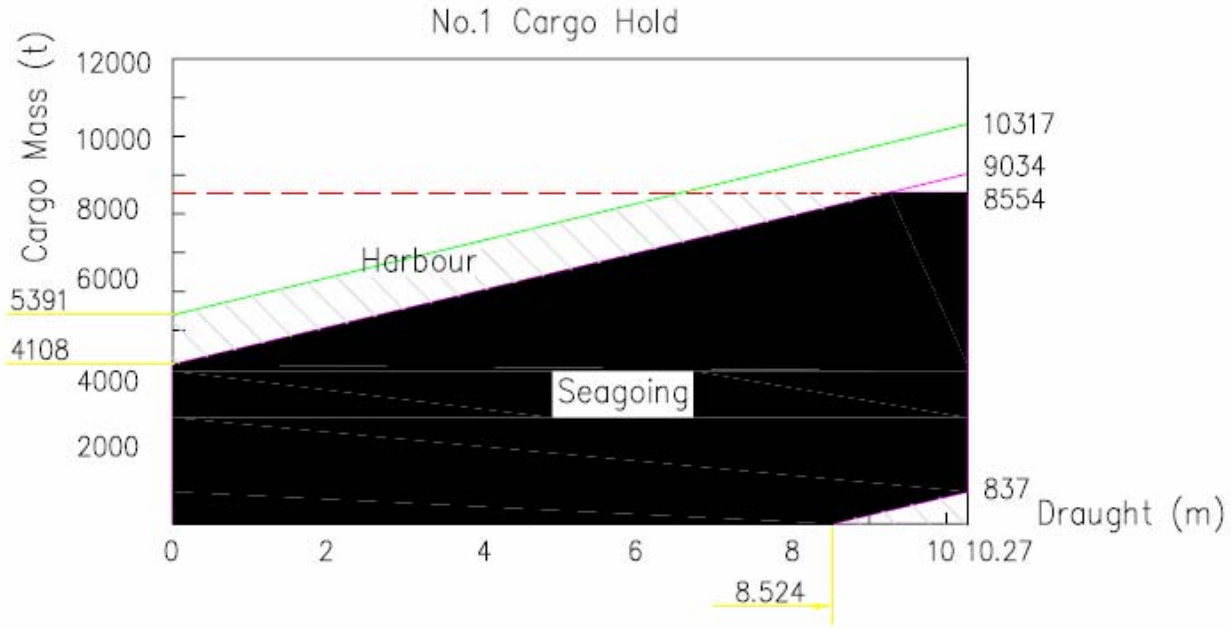
6.1. Lightship Weight Distribution

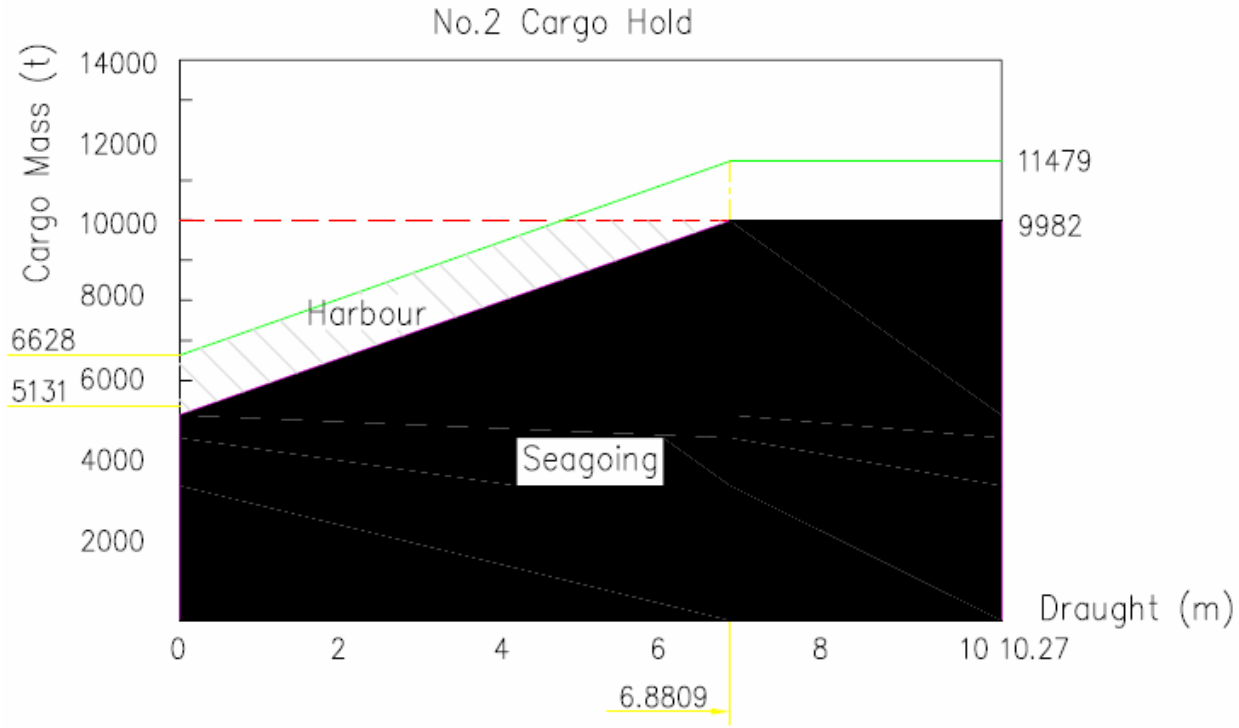
Item	Weight(t)	XCG (m)	YCG (m)	ZCG (m)
STERN	269.0	2.17	0	10.2
ER1	555.0	17.94	0	10.2
ER2	400.0	15.40	0	10.2
HOLD5	885.0	40.30	0	10.2
BHD2	201.0	53.40	0	10.2
HOLD4	1035.0	68.50	0	10.2
BHD3	222.0	82.23	0	10.2
HOLD3	1026.0	96.76	0	10.2
BHD4	222.0	111.03	0	10.2
HOLD2	1021.0	125.80	0	10.2
BHD5	201.0	139.83	0	10.2
HOLD1	717.0	152.60	0	10.2
FORE	360.0	170.00	0	10.2
SUPER	700.0	18.40	0	10.2
PIPE	200.0	75.00	0	10.2
MIS	386	86.00	0	10.2
ADD	250	86.00	0	10.2
$\sum$	8650	79.30	0	10.2

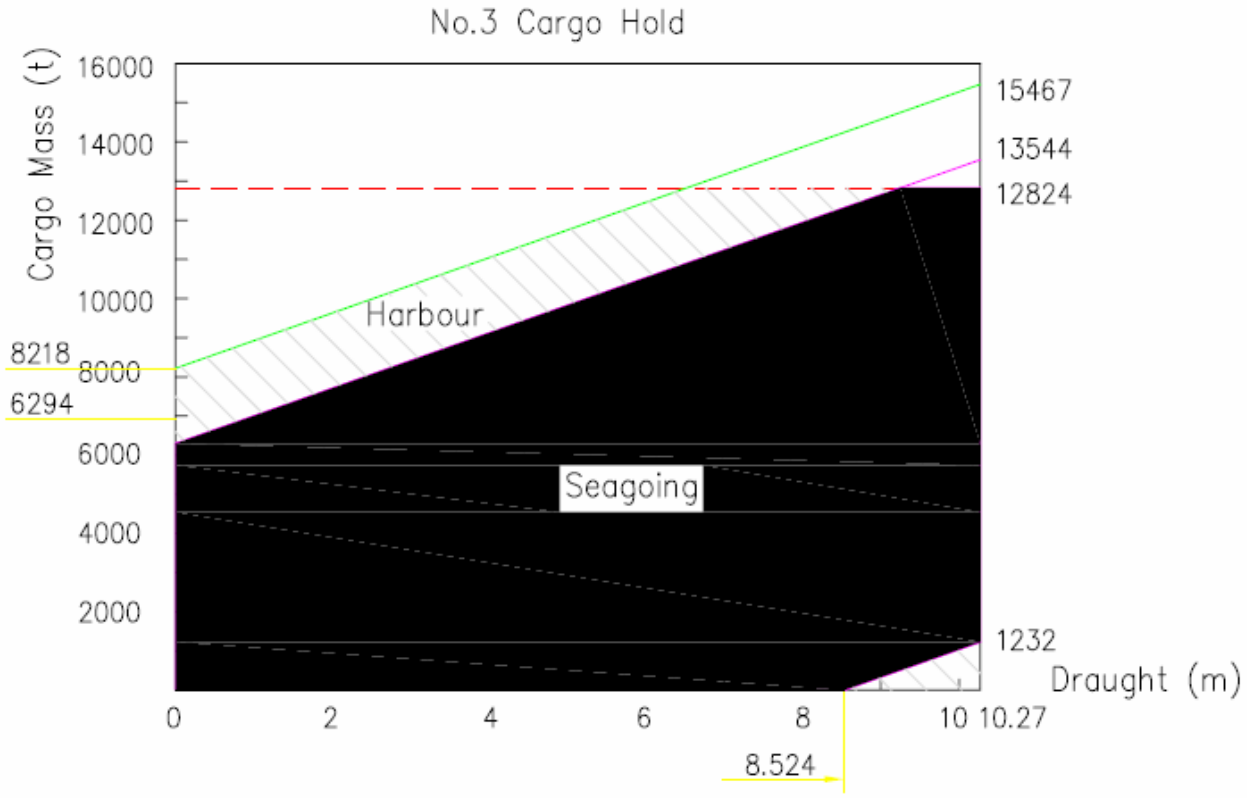
Light Ship Weight	8650 t
CGX	79.30 m from AP
CGY	0 m
CGZ	10.20 m from BL

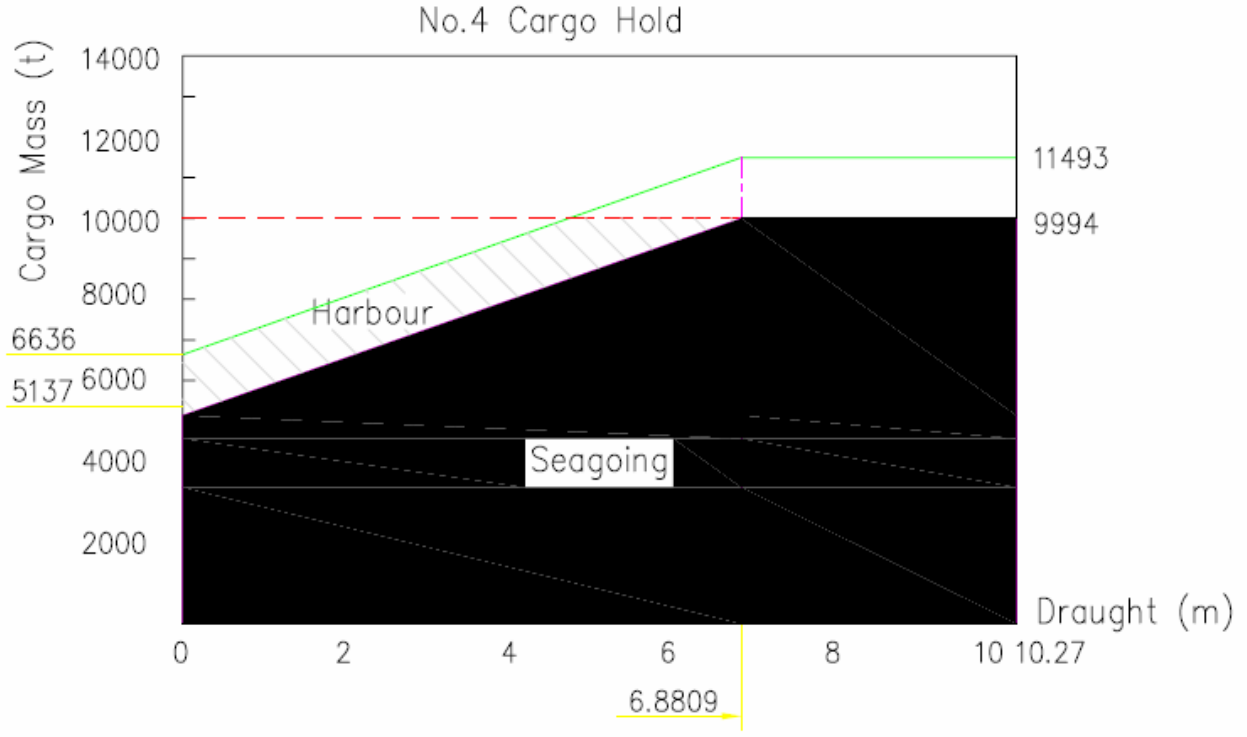
6.2. Allowable Cargo Mass Curve

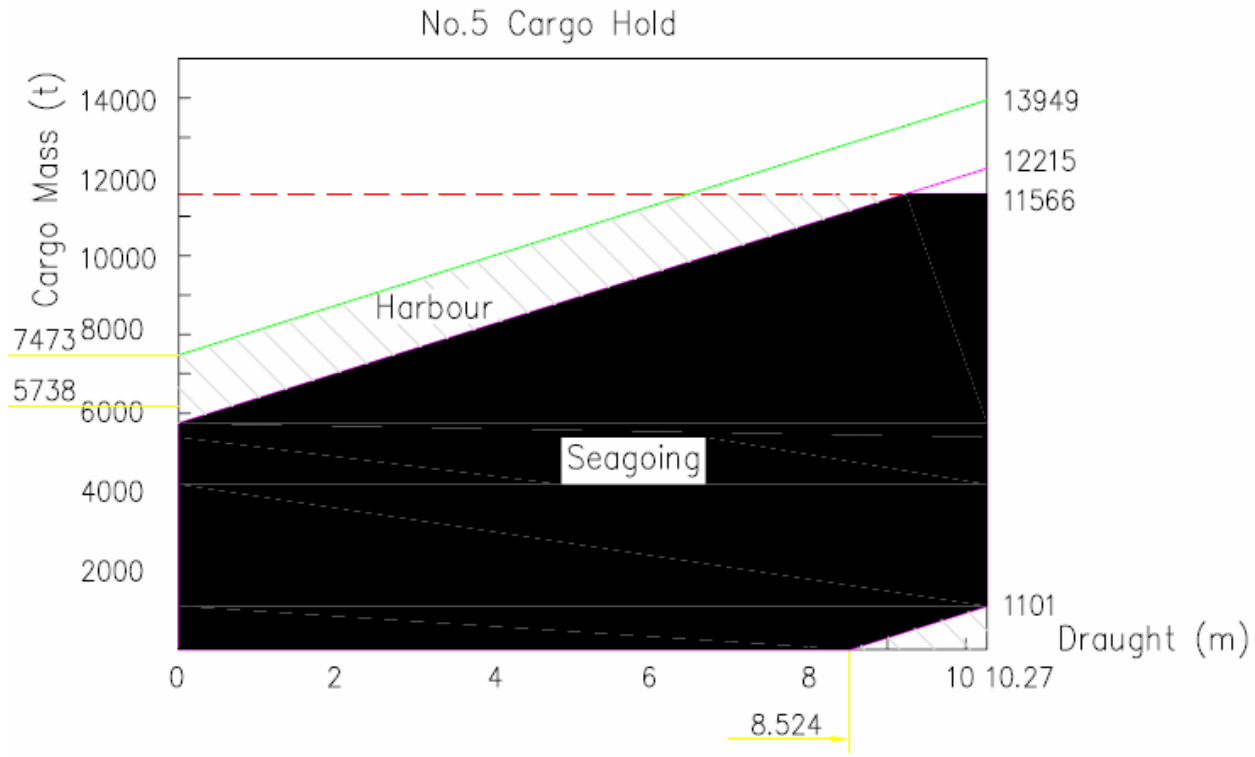
1). Maximum permissible mass and minimum required masses of single cargo hold in seagoing condition and harbour condition.



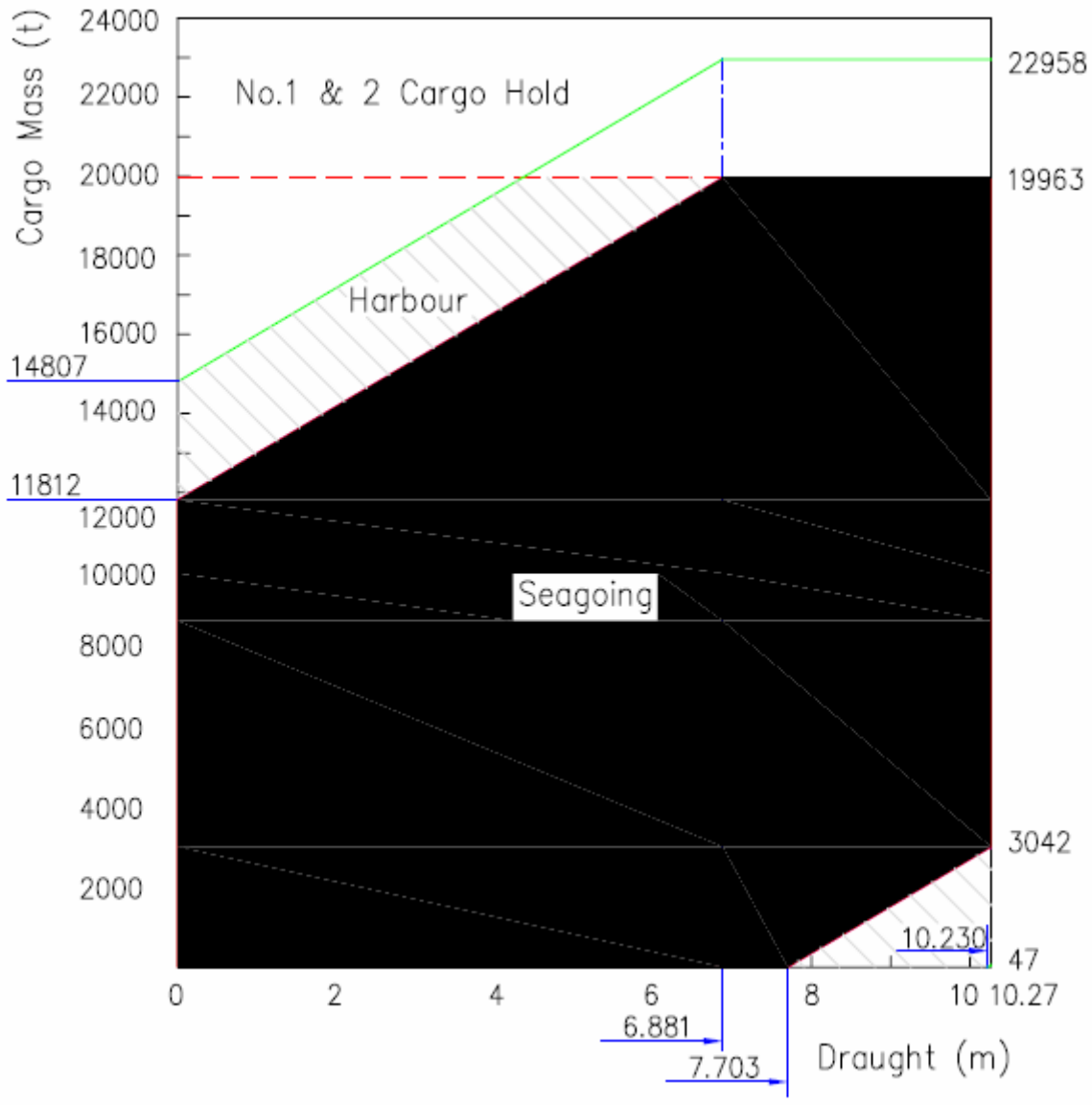


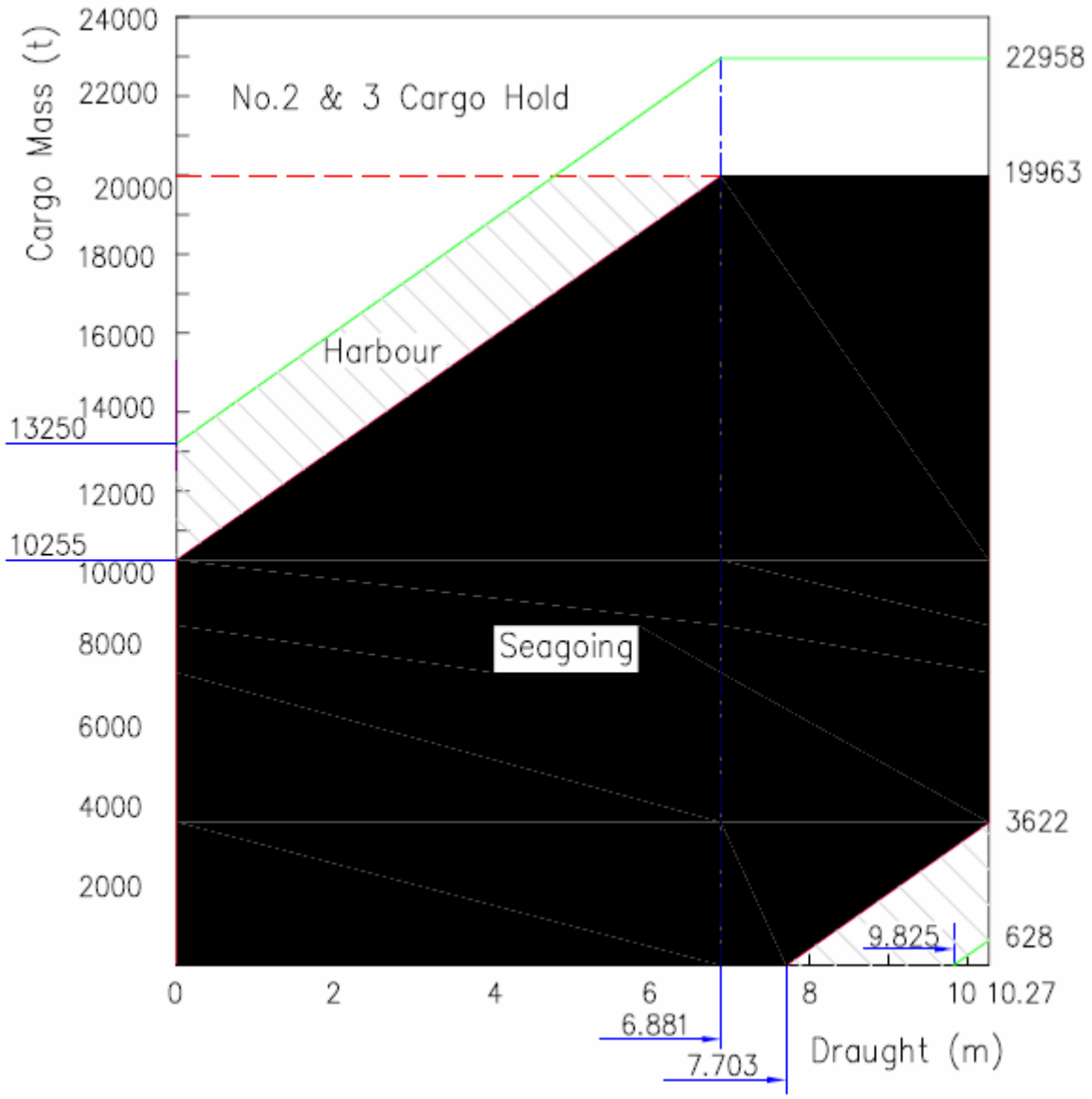

No.3 Cargo Hold
Cargo Mass (t)
16000
14000
12000
10000
8000
6000
4000
2000
8218
6294
15467
13544
12824
1232
Harbour
Seagoing
0
2
4
6
8
10
10.27
8.524
Draught (m)

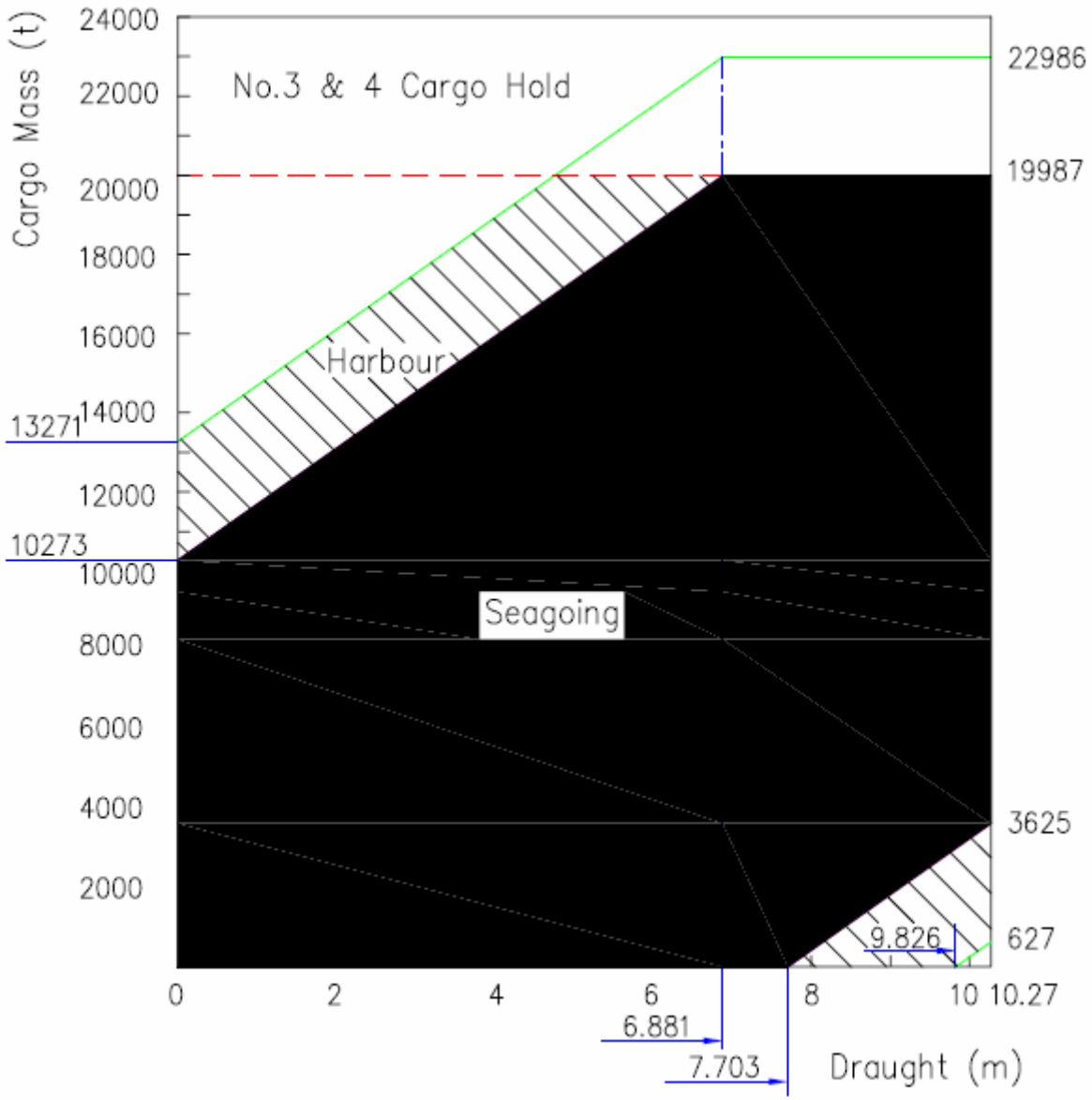

No.4 Cargo Hold
Cargo Mass (t)
14000
12000
10000
8000
6000
4000
2000
6636
5137
11493
9994
Harbour
Seagoing
0
2
4
6
8
10
10.27
6.8809
Draught (m)

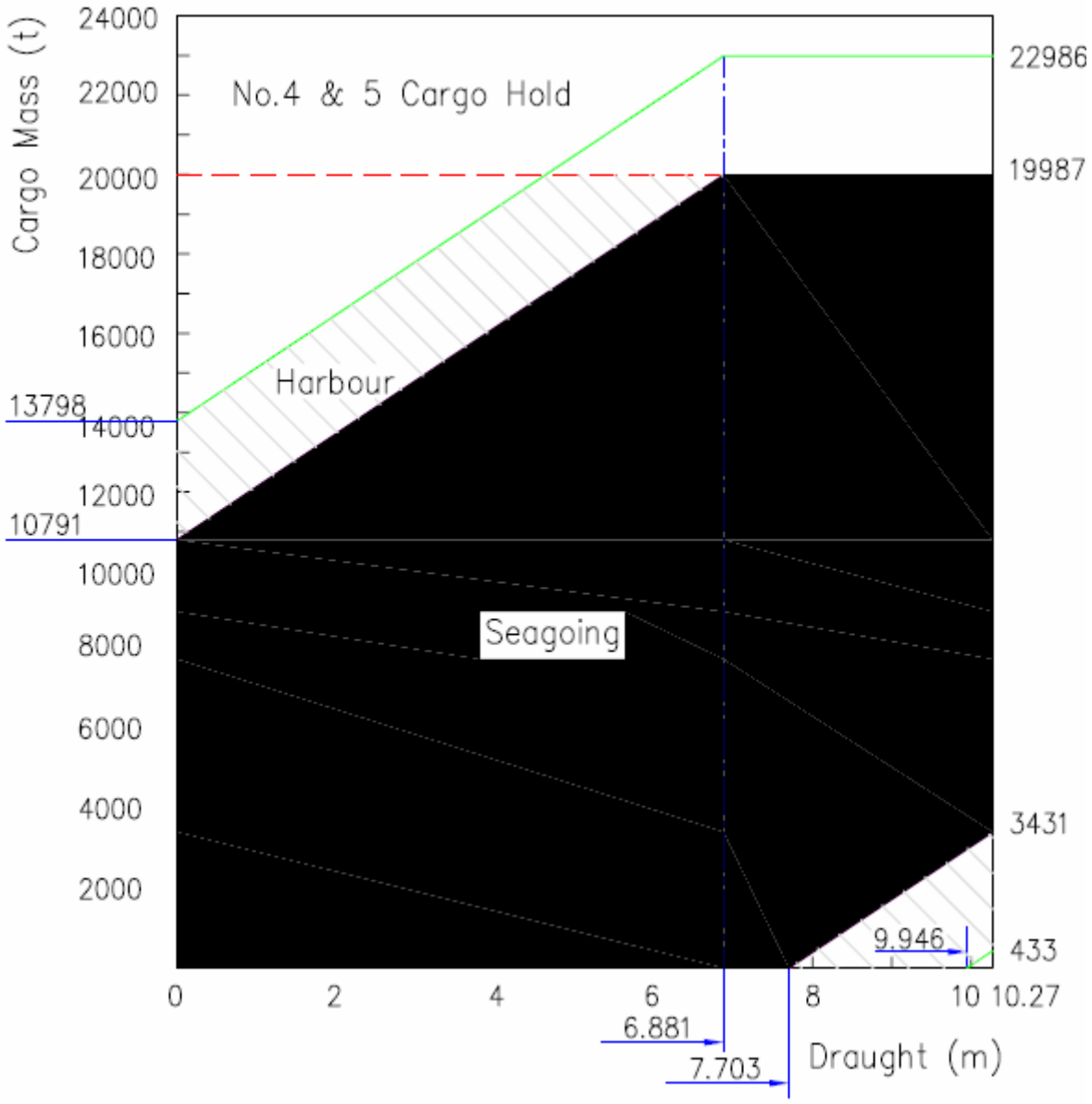

No.5 Cargo Hold
Cargo Mass (t)
14000
12000
10000
8000
6000
4000
2000
7473
5738
13949
12215
11566
1101
Harbour
Seagoing
0
2
4
6
8
10
10.27
8.524
Draught (m)

2). Maximum permissible mass and minimum required masses of two adjacent cargo holds in seagoing condition and harbour condition.




Cargo Mass (t)
24000
22000
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
No.2 & 3 Cargo Hold
Harbour
Seagoing
22958
19963
13250
10255
3622
628
9.825
0
2
4
6
8
10
10.27
6.881
7.703
Draught (m)


Cargo Mass (t)
24000
22000
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
No.3 & 4 Cargo Hold
Harbour
Seagoing
22986
19987
13271
10273
3625
627
9.826
0
2
4
6
8
10
10.27
6.881
7.703
Draught (m)


Cargo Mass (t)
24000
22000
20000
18000
16000
14000
12000
10000
8000
6000
4000
2000
No.4 & 5 Cargo Hold
Harbour
Seagoing
22986
19987
13798
10791
3431
433
9.946
0
2
4
6
8
10
10.27
6.881
7.703
Draught (m)